Preliminary

MCP Specification

254FBGA, 11.5x13x1.0mmt
64GB e.MMC + 32Gb(16Gb*2) DDP LPDDR4X SDRAM

datasheet

SAMSUNG

Revision History

Revision No.	History	Draft Date	Remark	Editor
0.0	- Initial issue.	13th Oct, 2017	Target	S.H.Kim
0.5	- None	30th Oct, 2017	Preliminary	S.H.Kim

1.0 FEATURES

<Common>
- Operating Temperature : -25°C ~ 85°C
- Package : 254Ball FBGA Type - 11.5mm x 13mm x 1.0mmt
  0.5mm ball pitch

<e.MMC>
- embedded MultiMediaCard Ver. 5.1 compatible.
- SAMSUNG eMMC supports features of eMMC5.1 which are defined in JEDEC Standard
  - Supported Features : Packed command, Cache, Discard, Sanitize, Power Off Notification, Data Tag, Partition types, Context ID, Real Time Clock, Dynamic Device Capacity, Command Queuing, Enhanced Strobe Mode, Secure Write Protection, HS200, HS400, Field Firmware Update.
  - Non-supported Features : Large Sector Size (4KB)
- Full backward compatibility with previous MultiMediaCard system specification (1bit data bus, multi-eMMC systems)
- Data bus width : 1bit (Default), 4bit and 8bit
- MMC I/F Clock Frequency : 0 ~ 200MHz
  MMC I/F Boot Frequency : 0 ~ 52MHz
- Power : Interface power → VDD(VCCQ) (1.70V ~ 1.95V or 2.7V ~ 3.6V)
, Memory power → VDDF(VCC) (2.7V ~ 3.6V)

<LPDDR4X SDRAM>
- Double-data rate architecture; two data transfers per clock cycle
- Bidirectional data strobes (DQS_t, DQS_c), These are transmitted/ received with data to be used in capturing data at the receiver
- Differential clock inputs (CK_t and CK_c)
- Differential data strobes (DQS_t and DQS_c)
- Commands & addresses entered positive CK edges; data and data mask referenced to both edges of DQS
- 2channel composition per die
- 8 internal banks for each channel
- DMI Pin : DBI (Data Bus Inversion) when normal write and read operation, Data mask (DM) for masked write when DBI off
  - Counting # of DQ's 1 for masked write when DBI on
- Burst Length: 16, 32 (OTF)
- Burst Type: Sequential
- Read & Write latency : Refer to Table 49 LPDDR4X AC Timing Table
- Auto Precharge option for each burst access
- Configurable Drive Strength
- Refresh and Self Refresh Modes
- Partial Array Self Refresh and Temperature Compensated Self Refresh
- Write Leveling
- CA Calibration
- Internal VREF and VREF training
- FIFO based write/read training
- MPC (Multi Purpose Command)
- LVSTLE (Low Voltage Swing Terminated Logic Extension) IO
- VDD1/VDD2/VDDQ : 1.8V/1.1V/0.6V
- VSSQ Termination
- No DLL : CK to DQS is not synchronized
- Edge aligned data output, write training for data input center align

Refresh rate : 3.9us

[Table 1] LPDDR4X SDRAM Addressing

Memory Density (per Die)		16Gb
Memory Density (per channel)		8Gb
Configuration		64Mb x 16DQ x 8 banks x 2 channels
Number of Channels (per die)		2
Number of Banks (per channel)		8
Array Pre-Fetch (bits, per channel)		256
Number of Rows (per Channel)		65,536
Number of Columns (fetch boundaries)		64
Page Size (Bytes)		2048
Channel Density (Bits per channel)		8,589,934,592
Total Density (Bits per die)		17,179,869,184
Bank Addresses		BA0-BA2
x16	Row Addresses 2)	R0-R15
	Column Addresses 1), 2)	C0-C9
Burst Starting Address Boundary		64 - bit

NOTE :
1) The lower two column addresses (C0-C1) are assumed to be "zero" and are not transmitted on the CA bus.
2) Row and Column Address values on the CA bus that are not used for a particular density be at valid logic levels.

2.0 GENERAL DESCRIPTION

The KMDH6001DA is a Multi Chip Package Memory which combines 64GB e.MMC and 32Gb(16Gb*2) DDP LPDDR4X SDRAM.

SAMSUNG eMMC is an embedded MMC solution designed in a BGA package form. eMMC operation is identical to a MMC device and therefore is a simple read and write to memory using MMC protocol v5.1 which is a industry standard.
eMMC consists of NAND flash and a MMC controller. 3V supply voltage is required for the NAND area (VDDF or VCC) whereas 1.8V or 3V dual supply voltage (VDD or VCCQ) is supported for the MMC controller. SAMSUNG eMMC supports HS400 in order to improve sequential bandwidth, especially sequential read performance.
There are several advantages of using eMMC. It is easy to use as the MMC interface allows easy integration with any microprocessor with MMC host. Any revision or amendment of NAND is invisible to the host as the embedded MMC controller insulates NAND technology from the host. This leads to faster product development as well as faster times to market.
The embedded flash management software or FTL(Flash Transition Layer) of eMMC manages Wear Leveling, Bad Block Management and ECC. The FTL supports all features of the Samsung NAND flash and achieves optimal performance.

LPDDR4X devices use a 2 or 4 clocks architecture on the Command/Address (CA) bus to reduce the number of input pins in the system. The 6-bit CA bus contains command, address, and bank information. Each command uses 1, 2 or 4 clock cycle, during which command information is transferred on the positive edge of the clock. See command truth table for details.

These devices use a double data rate architecture on the DQ pins to achieve high speed operation. The double data rate architecture is essentially an 16n prefetch architecture with an interface designed to transfer two data bits per DQ every clock cycle at the I/O pins. A single read or write access for the LPDDR4X SDRAM effectively consists of a single 16n-bit wide, one clock cycle data transfer at the internal DRAM core and eight corresponding n-bit wide, one-half-clock-cycle data transfers at the I/O pins. Read and write accesses to the LPDDR4X SDRAMs are burst oriented; accesses start at a selected location and continue for a programmed number of locations in a programmed sequence. Accesses begin with the registration of an Activate command, which is then followed by a Read, Write or Mask Write command. The address and BA bits registered coincident with the Activate command are used to select the row and the Bank to be accessed. The address bits registered coincident with the Read, Write or Mask Write command are used to select the Bank and the starting column location for the burst access.

Prior to normal operation, the LPDDR4X SDRAM must be initialized. The following section provides detailed information covering device initialization, register definition, command description and device operation.

The KMDH6001DA suitable for use in data memory of mobile communication system to reduce not only mount area but also power consumption. This device is available in 254-ball FBGA Type.

3.0 PIN CONFIGURATION

254Ball

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
A	DNU	DNU	DQ0_A	VDD1	VDD2	VDDQ	VDDQ	VDD2	VDD1				VDDQ	VDDQ	VDD1	VDD1	DNU	DNU
B	DNU		DQ1_A	VSS	VDDQ	VSS	DQ4_A	VSS	VDD2				VDD2	VDD2	VDD1	ZQ0_A		DNU
C			DQ2_A	VSS	VSS	DQ5_A	VSS	DQ7_A	DQS0_t_A				CA2_A	VSS	CA5_A	ZQ1_A		
D			DQ3_A	VSS	DMI0_A	VSS	DQ6_A	VSS	DQS0_c_A				CA3_A	VSS	VSS	NC		
E													CA4_A	VSS	CS0_A	CKE0_A		
F													CA1_A	VSS	CS1_A	CKE1_A		
G			DQ13_A	VSS	VSS	VSS	VDD2	VDD2	VDD2				VSS	CA0_A	VSS	CK_c_A		
H			DMI1_A	VSS	VDDQ	DQ14_A	VSS	DQ15_A	VDDQ				VSS	NC	VSS	CK_t_A		
J			DQ11_A	VDDQ	VDDQ	VSS	DQ12_A	VDDQ	DQS1_c_A				ODT_A[1), 2)]	NC	VCCQ	VCCQ	VCCQ	
K		VDD2	DQ10_A	VSS	DQ8_A	DQ9_A	VSS	VSS	DQS1_t_A				VSS_m	VSS_m	VCCQ	VSS_m	NC	
L							VDD2	VDD2	VDD2			VSS_m	DAT7	DAT6	VSS_m	VSS_m	VDDI	
M			VSF1	VSF3	VSF5	VSF7	VSF9	VSS_m	CMD			DS	VSS_m	VSS_m	DAT1	DAT4	VCC	
N			VSF2	VSF4	VSF6	VSF8	NC	VSS_m	RST_n			VSS_m	DAT2	DAT5	VSS_m	VSS_m	VCC	
P							VDD2	VDD2	VDD2			CLK	VSS_m	VSS_m	DAT3	DAT0	VCC	
R		VDD2	DQ10_B	VSS	DQ8_B	DQ9_B	VSS	VSS	DQS1_t_B				VCCQ	VCCQ	VSS_m	VSS_m	VSS_m	
T			DQ11_B	VDDQ	VDDQ	VSS	DQ12_B	VDDQ	DQS1_c_B				ODT_B[1), 2)]	NC	VCCQ	VCCQ	NC	
U			DMI1_B	VSS	VDDQ	DQ14_B	VSS	DQ15_B	VDDQ				VSS	NC	VSS	CK_t_B		
V			DQ13_B	VSS	VSS	VSS	VDD2	VDD2	VDD2				VSS	CA0_B	VSS	CK_c_B		
W													CA1_B	VSS	CS1_B	CKE1_B		
Y													CA4_B	VSS	CS0_B	CKE0_B		
AA			DQ3_B	VSS	DMI0_B	VSS	DQ6_B	VSS	DQS0_c_B				CA3_B	VSS	VSS	RESET_n		
AB			DQ2_B	VSS	VSS	DQ5_B	VSS	DQ7_B	DQS0_t_B				CA2_B	VSS	CA5_B	NC		
AC	DNU		DQ1_B	VSS	VDDQ	VSS	DQ4_B	VSS	VDD2				VDD2	VDD2	VDD1	NC		DNU
AD	DNU	DNU	DQ0_B	VDD1	VDD2	VDDQ	VDDQ	VDD2	VDD1				VDDQ	VDDQ	VDD1	VDD1	DNU	DNU

254FBGA: Top View (Ball Down)

(dark blue)	LPDDR4X Channel A	(salmon)	LPDDR4X Channel B
(red)	DRAM Power	(brown)	e.MMC
(cyan)	ODT	(olive)	Ground
(yellow)	ZQ	(white)	NC, RFU DNU, VSF

NOTE :

1) ODT(CA)_[x] balls are wired to ODT(CA)_[x] pads of Rank 0 DRAM die. ODT(CA)_[x] pads for other ranks (if present) are disabled in the package. .

2) For the LPDDR4X, ODT_Bond pad is ignored. ODT pin should connect to VDD2 or VSS.

4.0 PIN DESCRIPTION

Pin Name	Pin Function(e.MMC)
DAT0 ~ DAT7	Data Input/Output
CLK	Clock
DS	Data Strobe
CMD	Command
RST_n	Reset
VCC	Power Supply for Flash
VCCQ	Power Supply for Controller
VDDI	External Capacitance for internal power stability
VSS_m	Ground for Controller / Flash

Pin Name	Pin Function Channel-A~B (LPDDR4X-Per Channel A~B)
CK_c_A, CK_t_A CK_c_B, CK_t_B	System Differential Clock
CKE[1:0]_A, CKE[1:0]_B	Clock Enable
CS[1:0]_A, CS[1:0]_B	Chip Select
CA[5:0]_A, CA[5:0]_B	DDR Command / Address Inputs
DMI[1:0]_A, DMI[1:0]_B	Input Data Inversion
DQS[1:0]_t_A, DQS[1:0]_t_B	Data Strobe Bi-directional
DQS[1:0]_c_A, DQS[1:0]_c_B	Data Strobe Complementary
DQ[15:0]_A, DQ[15:0]_B	Data Input / Output
VDD1	Core Power Supply 1
VDD2	Core Power Supply 2
VDDQ	I/O Power Supply
VSS	Ground
ZQ[1:0]_A	Reference Pin for Output Drive Strength Calibration
ODT_A, ODT_B	On die termination[1), 2)]
RESET_n	RESET

NOTE :
1) ODT(CA)_[x] balls are wired to ODT(CA)_[x] pads of Rank 0 DRAM die. ODT(CA)_[x] pads for other ranks (if present) are disabled in the package. .
2) For the LPDDR4X, ODT_Bond pad is ignored. ODT pin should connect to VDD2 or VSS.

Pin Name	Pin Function
NC	Not Connected
DNU	Do Not Use
VSF	Vendor Specific Function
RFU	Reserved For Use

5.0 ORDERING INFORMATION

KM D H 6 0 0 1D A - B 422

KM : Memory eMCP

Device Type
D : e.MMC + LPDDR4X + LPDDR4X

e.MMC Density & VCC & Org.
H : 64GByte, 256G*2, MLC, 3.3V, 1.8V~3.3V, x8

LSI
6 : Vincent

NOR Flash Density & VCC & Org.
0 : NONE

NAND/OneNAND Density & VCC & Org
0 : NONE

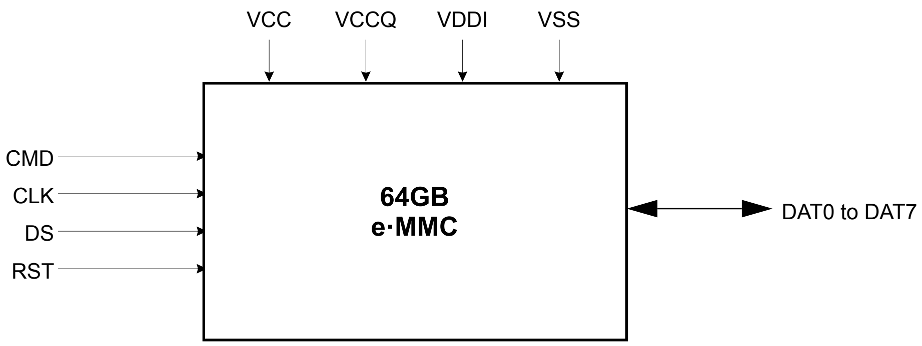
DRAM I/F & Density & VCC & Org.
1D : LPDDR4X*2, 16G*2, 1.8V, 0.6V, x32, 2CS/2CKE

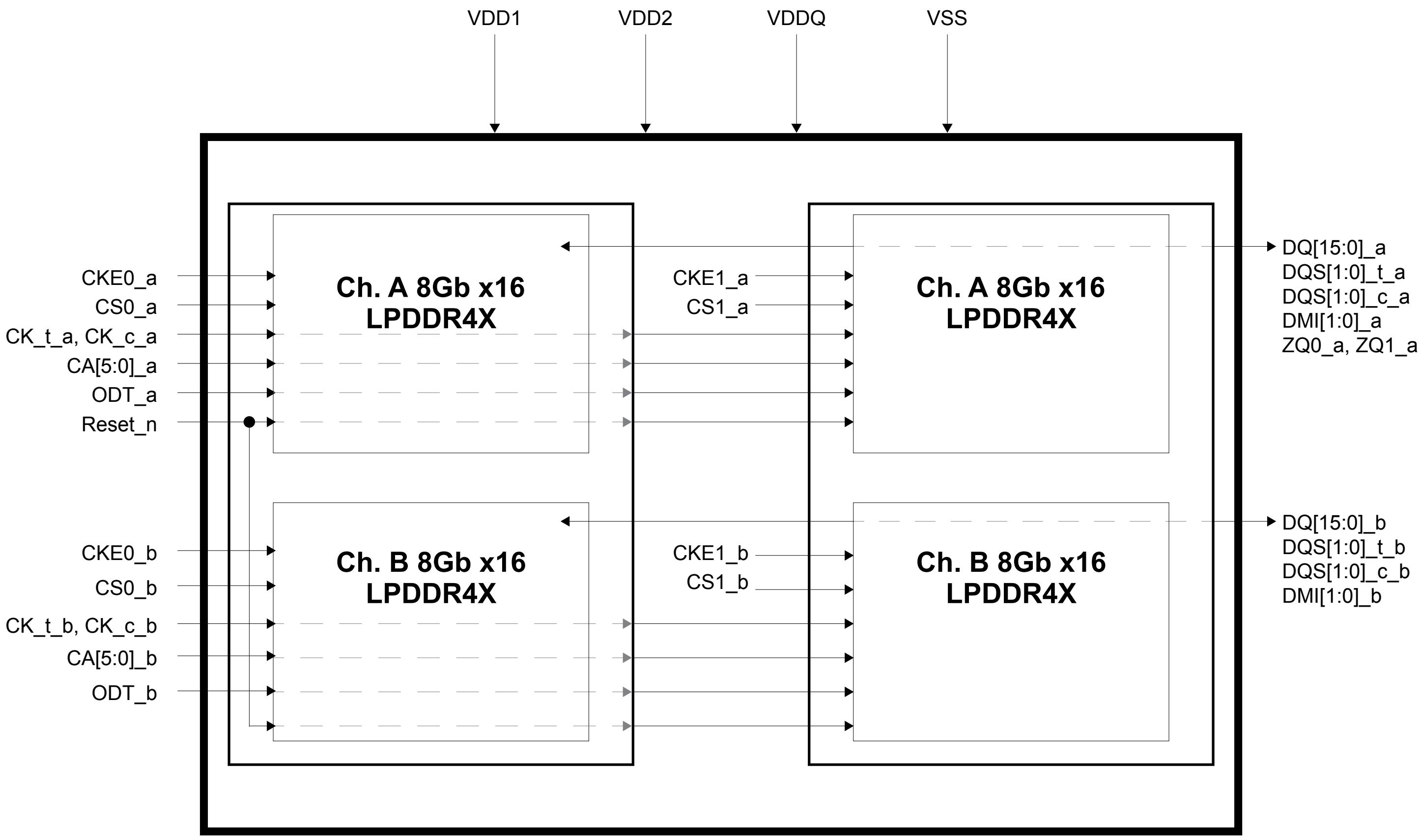
Generation
A : 2nd Generation

Package
B : FBGA(OSP)

422 :
e.MMC = 5ns(2)
LPDDR4X = 0.53ns, 0.53ns

6.0 FUNCTIONAL BLOCK DIAGRAM

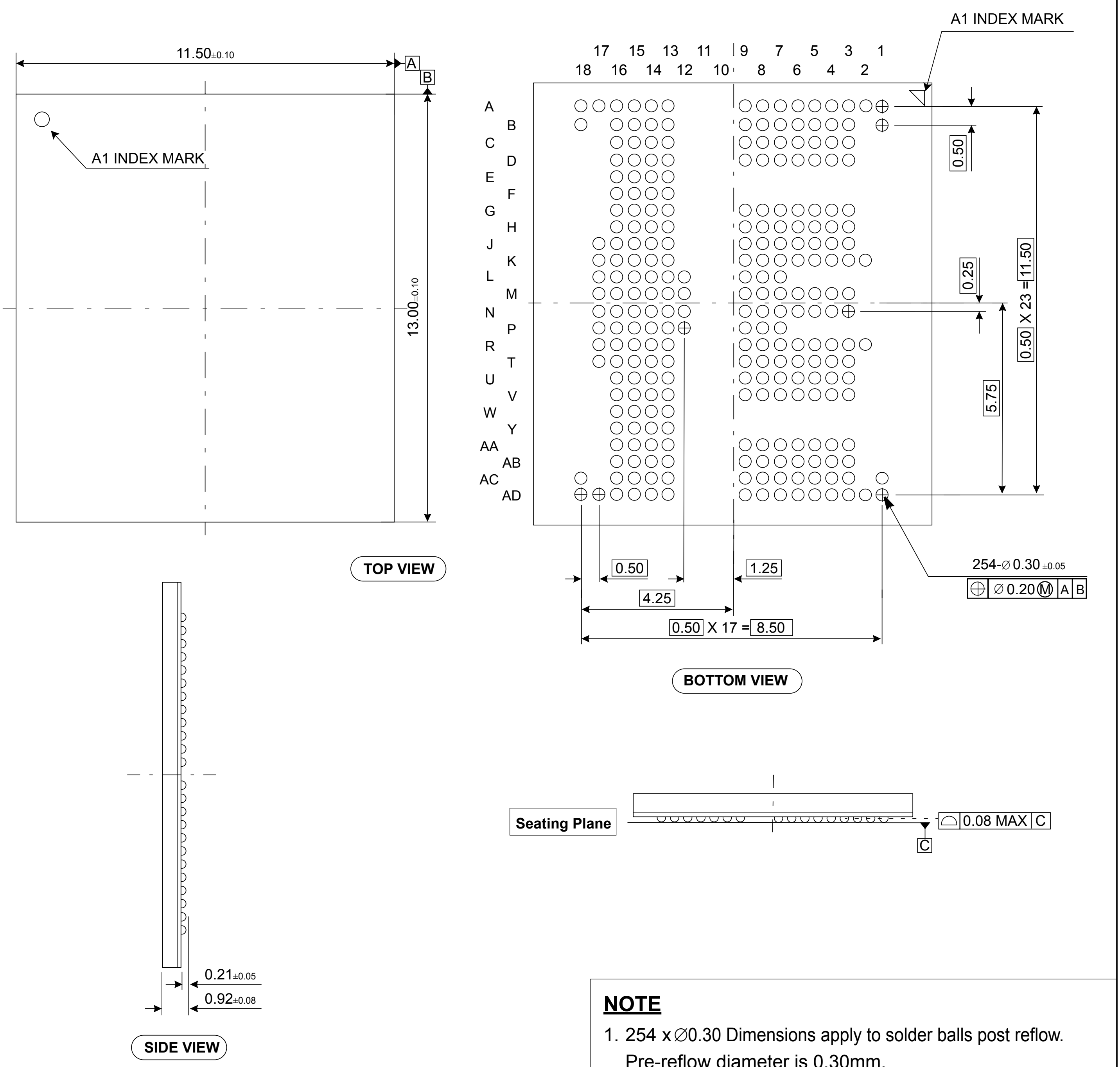




7.0 PACKAGE DIMENSION

254-Ball Fine pitch Ball Grid Array Package (measured in millimeters)

Units:millimeters



NOTE

1. 254 x∅0.30 Dimensions apply to solder balls post reflow.
   Pre-reflow diameter is 0.30mm.

64GB e.MMC

1.0 Product Architecture

- eMMC consists of NAND Flash and Controller. V_{CCQ} is for Controller power and V_{CC} is for flash power

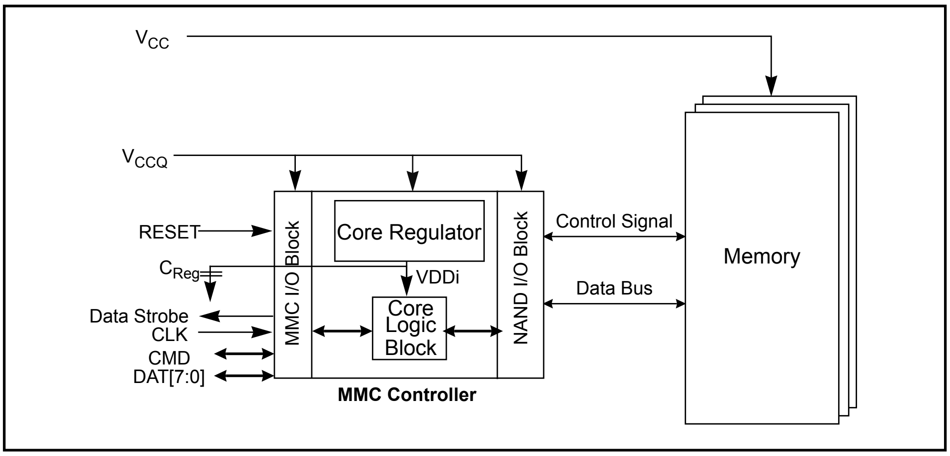


Figure 1. eMMC Block Diagram

2.0 HS400 mode

eMMC5.0 product supports high speed DDR interface timing mode up to 400MB/s at 200MHz with 1.8V I/O supply.
HS400 mode supports the following features :
- DDR Data sampling method
- CLK frequency up to 200MHz DDR (up to 400Mbps)
- Only 8-bits bus width available
- Signaling levels of 1.8V
- Six selectable Drive Strength (refer to the table below)

[Table 1] I/O driver strength types

Driver Type	HS200 & HS400 Support	Nominal Impedance	Approximated driving capability compared to Type-0	Remark
0	Default	50Ω	x1	Default Driver Type. Supports up to 200MHz operation.
1	Optional	33Ω	x1.5	Supports up to 200MHz Operation.
2	Optional	66Ω	x0.75	The weakest driver that supports up to 200MHz operation.
3	Optional	100Ω	x0.5	For low noise and low EMI systems. Maximal operating frequency is decided by Host design.
4	Optional	40Ω	x1.2	Supports up to 200MHz DDR operation

NOTE:
1) Support of Driver Type-0 is default for HS200 & HS400 Device, while supporting Driver types 1~4 are optional for HS200 & HS400 Device.

[Table 2] Device type values (EXT_CSD register : DEVICE_TYPE [196])

Bit	Device Type	Supportability
7	HS400 Dual Data Rate eMMC @ 200 MHz - 1.2V I/O	Not support
6	HS400 Dual Data Rate eMMC @ 200 MHz - 1.8V I/O	Support
5	HS200 Single Data Rate *eMMC* @ 200 MHz - 1.2V I/O	Not support
4	HS200 Single Data Rate *eMMC* @ 200 MHz - 1.8V I/O	Support
3	High-Speed Dual Data Rate *eMMC* @ 52MHz - 1.2V I/O	Not support
2	High-Speed Dual Data Rate *eMMC* @ 52MHz - 1.8V or 3V I/O	Support
1	High-Speed *eMMC* @ 52MHz - at rated device voltage(s)	Support
0	High-Speed *eMMC* @ 26MHz - at rated device voltage(s)	Support

[Table 3] Extended CSD revisions (EXT_CSD register : EXT_CSD_REV [192])

Value	Timing Interface	EXT_CSD Register Value
255-8	Reserved	-
8	Revision 1.8 (for MMC V5.1)	0x08
7	Revision 1.7 (for MMC V5.0)	-
6	Revision 1.6 (for MMC V4.5, V4.51)	-
5	Revision 1.5 (for MMC V4.41)	-
4	Revision 1.4 (Obsolete)	-
3	Revision 1.3 (for MMC V4.3)	-
2	Revision 1.2 (for MMC V4.2)	-
1	Revision 1.1 (for MMC V4.1)	-
0	Revision 1.0 (for MMC V4.0)	-

[Table 4] High speed timing values (EXT_CSD register : HS_TIMING [185])

Value	Timing Interface	Supportability
0x0	Selecting backwards compatibility interface timing	Support
0x1	High Speed	Support
0x2	HS200	Support
0x3	HS400	Support

3.0 New eMMC5.1 Features

3.1 Overview

New Feature	JEDEC	Support
Cache Flushing Report	Mandatory	Yes
Background operation control	Mandatory	Yes
Command Queuing	Optional	Yes
Enhanced Strobe	Optional	Yes
RPMB Throughput improve	Optional	Yes
Secure Write Protection	Optional	Yes

3.2 Command Queuing

To facilitate command queuing in eMMC, the device manages an internal task queue that the host can queue during data transfer tasks.

Every task is issued by the host and initially queued as pending. The device works to prepare pending tasks for execution. When a task is ready for execution, its state changes to “ready for execution”.

The host tracks the state of all queued tasks and may order the execution of any task, marked as “ready for execution”, by sending a command indicating its task ID. The device executes the data transfer transaction after receiving the execute command(CMD46/CMD47)

3.2.1 CMD Set Description

[Table 5] CMD Set Description and Details

CMD	Type	Argument	Abbreviation	Purpose
CMD44	ac/R1	[31] Reliable Write Request [30] DAT_DIR - "0" write / "1" read [29] tag request [28:25] context ID [24] forced programming [23] Priority: “0” simple / “1” high [20:16] TASK ID [15:0] number of blocks	QUEUED_TASK _PARAMS	Define direction of operation (Read or Write) and Set high priority CMD Queue with task ID
CMD45	ac/R1	[31:0] Start block address	QUEUED_TASK_ADDRESS	Indicate data address for Queued CMD
CMD46	adtc/R1	[20:16] TASK ID	EXECUTE_READ_TASK	(Read) Transmit the requested number of data blocks
CMD47	adtc/R1	[20:16] TASK ID	EXECUTE_WRITE_TASK	(Write) Transmit the requested number of data blocks
CMD48	ac/R1b	[20:16] Task ID [3:0] TM op-code	CMDQ_TASK _MGMT	Reset a specific task or entire queue. [20:16] when TM op-code = 2h these bits represent TaskID. When TM op-code = 1h these bits are reserved."

3.2.2 New Response : QSR (Queue Status Register)

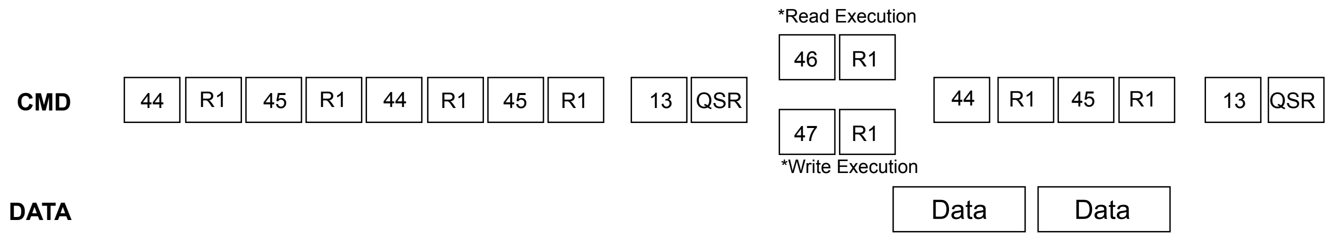
The 32-bit Queue Status Register (QSR) carries the state of tasks in the queue at a specific point in time. The host has read access to this register through device response to SEND_STATUS command (CMD13 with bit[15]=”1”), R1’s argument will be the 32- bit Queue Status Register (QSR). Every bit in the QSR represents the task who’s ID corresponds to the bit index. If bit QSR[i] = “0”, then the queued task with a Task ID i is not ready for execution. The task may be queued and pending, or the Task ID is unused. If bit QSR[i] = “1”, then the queued task with Task ID i is ready for execution.

3.2.3 Send Status : CMD13

CMD13 for reading the Queue Status Register (QSR) by the host. If bit[15] in CMD13’s argument is set to 1, then the device shall send an R1 Response with the QSR instead of the Device Status. * There is still legacy CMD13 with R` response

3.2.4 Mechanism of CMD Queue operation

Host issues CMD44 with Task ID number, Sector, Count, Direction, Priority to the device followed by CMD45 and host checks the Queue Status check with CMD13 [15]bits to 1 . After that host issues CMD46 for Read or CMD47 for write During CMD queue operation, CMD44/CMD45 is able to be issued at anytime when the CMD line is not in use



3.2.5 CMD Queue Register description

Configuration and capability structures shall be added to the EXT_CSD register, as described below

[Table 6] CMD Queuing Support (EXT_CSD register : CMDQ_SUPPORT [308])

Bit7	Bit6	Bit5	Bit4	Bit 3	Bit2	Bit1	Bit0
Reserved							CMD Queue supportability

This field indicates whether the device supports command queuing or not

0x0: CMD Queue function is not supported
0x1: CMD Queue function is supported

[Table 7] Command Queue Mode Enable(EXT_CSD register : CMDQ_MODE_EN [15])

Bit7	Bit6	Bit5	Bit4	Bit 3	Bit2	Bit1	Bit0
Reserved							0x00

This field is used by the host enable command queuing

0x0: Queue function is not enabled
0x1: Queue function is enabled

[Table 8] CMD Queuing Depth(EXT_CSD register : CMDQ_DEPTH [307])

Bit7	Bit6	Bit5	Bit4	Bit 3	Bit2	Bit1	Bit0
Reserved			0x0F				

This field is used to calculate the depth of the queue supported by the device

Bit encoding:
[7:5]: Reserved
[4:0]: N,a parameter used to calculate the Queue Depth of task queue in the device.
Queue Depth = N+1.

3.3 Enhanced Strobe Mode

This product supports Enhanced Strobe in HS400 mode and refer to the details as described in eMMC5.1 JEDEC standard

3.4 RPMB Throughput improve

[Table 9] Related parameter register in EXT_CSD : WR_REL_PARAM [166]

Name	Field	Bit	Type
Enhanced RPMB Reliable Write	EN_RPMB_REL_WR	4	R

Bit[4]: EN_RPMB_REL_WR(R)
    0x0: RPMB transfer size is either 256B (single 512B frame) or 512B (Two 512B frame).
    0 x1: RPMB transfer size is either 256B (single 512B frame), 512B (Two 512B frame), or 8KB(Thirthy two 512B frames).

3.5 Secure Write Protection

Configuration and capability structures shall be added to the EXT_CSD register and Authenticated Device Configuration Area as described below

[Table 10] Parameter register in EXT_CSD : SECURE_WP_INFO [211]

Bit7	Bit6	Bit5	Bit4	Bit	Bit2	Bit1	Bit0
Reserved						SECURE_WP_EN_STATUS	SECURE_WP_SUPPORT

Bit[7:2]: Reserved
Bit[1]: SECURE_WP_EN_STATUS(R)
 0x0: Legacy Write Protection mode.
 0x1: Secure Write Protection mode.
Bit[0]: SECURE_WP_SUPPORT(R)
 0x0: Secure Write Protection is NOT supported by this device
 0x1: Secure Write Protection is supported by this device

[Table 11] Authenticated Device Configuration Area[1] : SECURE_WP_MODE_ENABLE

Bit7	Bit6	Bit5	Bit4	Bit	Bit2	Bit1	Bit0
Reserved							0x00

Bit[7:1] : Reserved
Bit[0] : SECURE_WP_EN (R/W/E)
The default value of this field is 0x0.

- 0x0 : Legacy Write Protection mode, i.e., TMP_WRITE_PROTECT[12] , PERM_WRITE_PROTECT[13] is updated by CMD27. USER_WP[171], BOOT_WP[173] and BOOT_WP_STATUS[174] are updated by CMD6.

- 0x1 : Secure Write Protection mode. The access to the write protection related EXT_CSD and CSD fields depends on the value of SECURE_WP_MASK bit in SECURE_WP_MODE_CONFIG field.

[Table 12] Authenticated Device Configuration Area[2] : SECURE_WP_MODE_CONFIG

Bit7	Bit6	Bit5	Bit4	Bit	Bit2	Bit1	Bit0
Reserved							0x00

Bit[7:1] : Reserved
Bit[0] : SECURE_WP_MASK (R/W/E_P)
The default value of this field is 0x0.

- 0x0: Disabling updating WP related EXT_CSD and CSD fields. CMD27 (Program CSD) will generate generic error for setting TMP_WRITE_PROTECT[12] , PERM_WRITE_PROTECT[13]. CMD6 for updating USER_WP[171], BOOT_WP[173] and BOOT_WP_STATUS[174] generates SWITCH_ERROR. If a force erase command is issued, the command will fail (Device stays locked) and the LOCK_UNLOCK_FAILED error bit will be set in the status register. If CMD28 or CMD29 is issued, then generic error will be occurred. Power-on Write Protected boot partitions will keep protected mode after power failure, H/W reset assertion and any CMD0 reset. The device keeps the current value of BOOT_WP_STATUS in the EXT_CSD register to be same after power cycle, H/W reset assertion, and any CMD0 reset.

- 0x1: Enabling updating WP related EXT_CSD and CSD fields. I.e TMP_WRITE_PROTECT[12] , PERM_WRITE_PROTECT[13] , USER_WP[171], BOOT_WP[173] and BOOT_WP_STATUS[174] are accessed using CMD6, CMD8 and CMD27.
 If a force erase command is issued and accepted, then ALL THE DEVICE CONTENT WILL BE ERASED including the PWD and PWD_LEN register content and the locked Device will get unlocked. If a force erase command is issued and power-on protected or a permanently-write-protected write protect groups exist on the device, the command will fail (Device stays locked) and the LOCK_UNLOCK_FAILED error bit will be set in the status register. An attempt to force erase on an unlocked Device will fail and LOCK_UNLOCK_FAILED error bit will be set in the status register. Write Protection is applied to the WPG indicated by CMD28 with the WP type indicated by the bit[2] and bit[0] of USER_WP[171]. All temporary WP Groups and power-on Write Protected boot partitions become writable/erasable temporarily which means write protect type is not changed. All power-on and permanent WP Groups in user area will not become writable/erasable temporarily. Those temporarily writable/erasable area will become write protected when this bit is cleared to 0x0 by the host or when there is power failure, H/W reset assertion and any CMD0 reset. The device keeps the current value of BOOT_WP_STATUS in the EXT_CSD register to be same after power cycle, H/W reset assertion, and any CMD0 reset.

4.0 Technical Notes

4.1 S/W Algorithm

4.1.1 Partition Management

The device initially consists of two Boot Partitions and RPMB Partition and User Data Area.

The User Data Area can be divided into four General Purpose Area Partitions and User Data Area partition. Each of the General Purpose Area partitions and a section of User Data Area partition can be configured as enhanced partition.

4.1.1.1 Enhanced Partition (Area)

SAMSUNG eMMC adopts Enhanced User Data Area as SLC Mode. Therefore when master adopts some portion as enhanced user data area in User Data Area, that area occupies double size of original set up size. (ex> if master set 1MB for enhanced mode, total 3MB user data area is needed to generate 1MB enhanced area)

Max Enhanced User Data Area size is defined as (MAX_ENH_SIZE_MULT x HC_WP_GRP_SIZE x HC_ERASE_GRP_SIZE x 512kBytes)

4.1.2 Boot operation

Device supports not only boot mode but also alternative boot mode. Device supports high speed timing and dual data rate during boot.

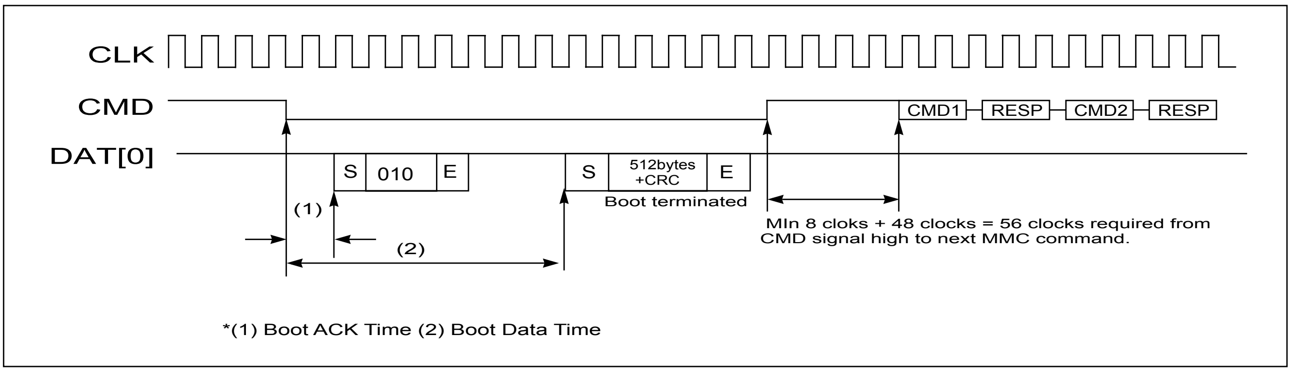


Figure 2. embedded MultiMediaCard state diagram (boot mode)

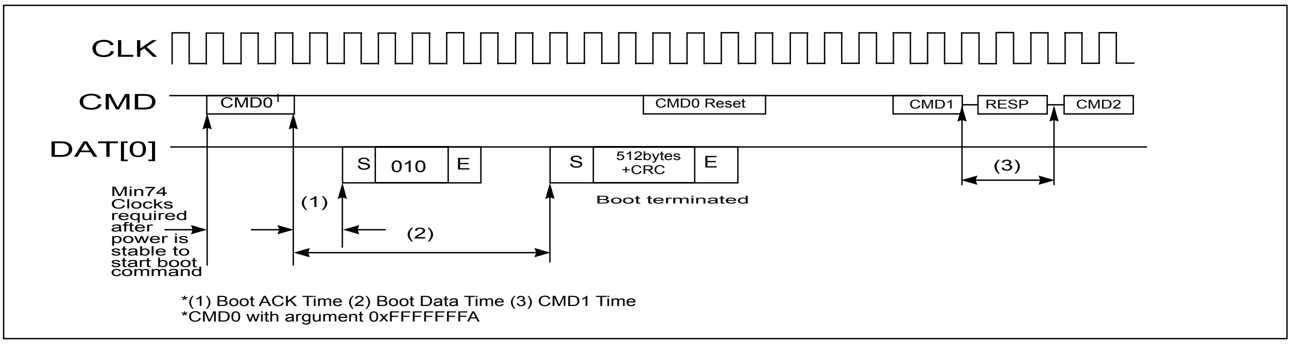


Figure 3. embedded MultiMediaCard state diagram (alternative boot mode)

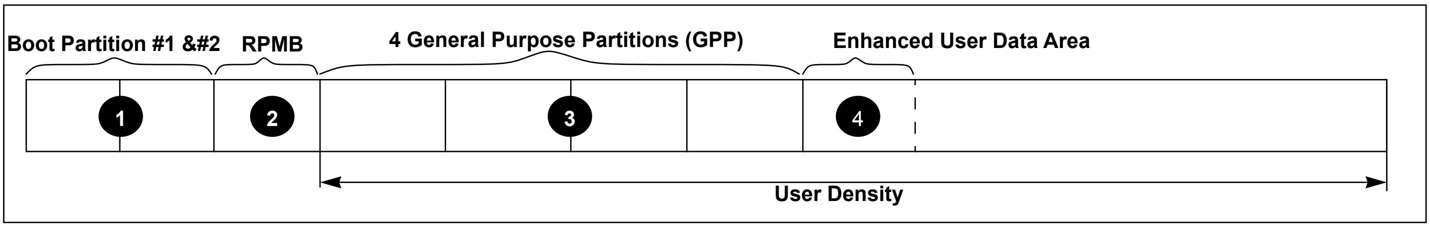
[Table 13] Boot ack, boot data and initialization Time

Timing Factor	Value
(1) Boot ACK Time	< 50 ms
(2) Boot Data Time	< 100 ms
(3) Initialization Time[1)]	< 3 secs

NOTE:
1) This initialization time includes partition setting, Please refer to INI_TIMEOUT_AP in 6.4 Extended CSD Register.
Normal initialization time (without partition setting) is completed within 1sec

4.1.3 User Density

Total User Density depends on device type.
For example, 32MB in the SLC Mode requires 64MB in TLC.
This results in decreasing of user density



[Table 14] Capacity according to partition

	Boot partition 1	Boot partition 2	RPMB
Default.	4,096KB	4,096KB	16,384KB
Max.	4,096KB	4,096KB	16,384KB

[Table 15] Maximum Enhanced Partition Size

Device	Max. Enhanced Partition Size
64 GB	TBD Byte

[Table 16] User Density Size

Device	User Density Size
64 GB	TBD Byte

4.1.4 Auto Power Saving Mode

If host does not issue any command during a certain duration (1ms), after previously issued command is completed, the device enters "Power Saving mode" to reduce power consumption.

At this time, commands arriving at the device while it is in power saving mode will be serviced in normal fashion

[Table 17] Auto Power Saving Mode and exit

Mode	Enter Condition	Escape Condition
Auto Power Saving Mode	When previous operation which came from Host is completed and no command is issued during a certain time.	If Host issues any command

[Table 18] Auto Power Saving Mode and Sleep Mode

	Auto Power Saving Mode	Sleep Mode
NAND Power	ON	OFF
Goto Sleep Time	< 1ms	< 1ms

4.1.5 Performance

[Table 19] Performance

Density	Sequential Read (MB/s)	Sequential Write (MB/s)
64 GB	TBD	TBD

* Test Condition : Bus width x8, HS400, 512KB data transfer, w/o file system overhead, measured on Samsung's internal board

5.0 REGISTER VALUE

5.1 OCR Register

The 32-bit operation conditions register stores the V_{CCQ} voltage profile of the eMMC. In addition, this register includes a status information bit. This status bit is set if the eMMC power up procedure has been finished. The OCR register shall be implemented by all eMMCs.

[Table 20] OCR Register

OCR bit	VCCQ voltage window[2]	Register Value
[6:0]	Reserved	00 00000b
[7]	1.70 - 1.95	1b
[14:8]	2.0-2.6	000 0000b
[23:15]	2.7-3.6	1 1111 1111b
[28:24]	Reserved	0 0000b
[30:29]	Access Mode	00b (byte mode) 10b (sector mode) -[*Higher than 2GB only]
[31]	eMMC power up status bit (busy)[1]	

NOTE :
1) This bit is set to LOW if the eMMC has not finished the power up routine
2) The voltage for internal flash memory(V_{CC}) should be 2.7-3.6v regardless of OCR Register value.

5.2 CID Register

[Table 21] CID Register

Name	Field	Width	CID-slice	CID Value
Manufacturer ID	MID	8	[127:120]	0x15
Reserved		6	[119:114]	---
Card/BGA	CBX	2	[113:112]	01
OEM/Application ID	OID	8	[111:104]	---[1]
Product name	PNM	48	[103:56]	See Product name table
Product revision	PRV	8	[55:48]	---[2]
Product serial number	PSN	32	[47:16]	---[3]
Manufacturing date	MDT	8	[15:8]	---[4]
CRC7 checksum	CRC	7	[7:1]	---[5]
not used, always '1'	-	1	[0:0]	---

NOTE :
1),4),5) description are same as eMMC JEDEC standard
2) PRV is composed of the revision count of controller and the revision count of F/W patch
3) A 32 bits unsigned binary integer. (Random Number)

5.2.1 Product name table (In CID Register)

[Table 22] Product name table

Part Number	Density	Product Name in CID Register (PNM)
KMDH6001DA-B422	64 GB	0 x 444836444142

5.3 CSD Register

The Card-Specific Data register provides information on how to access the eMMC contents. The CSD defines the data format, error correction type, maximum data access time, data transfer speed, whether the DSR register can be used etc. The programmable part of the register (entries marked by W or E, see below) can be changed by CMD27. The type of the entries in the table below is coded as follows:
R: Read only
W: One time programmable and not readable.
R/W: One time programmable and readable.
W/E: Multiple writable with value kept after power failure, H/W reset assertion and any CMD0 reset and not readable.
R/W/E: Multiple writable with value kept after power failure, H/W reset assertion and any CMD0 reset and readable.
R/W/C_P: Writable after value cleared by power failure and HW/ rest assertion (the value not cleared by CMD0 reset) and readable.
R/W/E_P: Multiple writable with value reset after power failure, H/W reset assertion and any CMD0 reset and readable.

W/E/_P: Multiple writable with value reset after power failure, H/W reset assertion and any CMD0 reset and not readable.

[Table 23] CSD Register

Name	Field	Width	Cell Type	CSD-slice	CSD Value
					64GB
CSD structure	CSD_STRUCTURE	2	R	[127:126]	TBD
System specification version	SPEC_VERS	4	R	[125:122]	TBD
Reserved	-	2	R	[121:120]	-
Data read access-time 1	TAAC	8	R	[119:112]	TBD
Data read access-time 2 in CLK cycles (NSAC*100)	NSAC	8	R	[111:104]	TBD
Max. bus clock frequency	TRAN_SPEED	8	R	[103:96]	TBD
Device command classes	CCC	12	R	[95:84]	TBD
Max. read data block length	READ_BL_LEN	4	R	[83:80]	TBD
Partial blocks for read allowed	READ_BL_PARTIAL	1	R	[79:79]	TBD
Write block misalignment	WRITE_BLK_MISALIGN	1	R	[78:78]	TBD
Read block misalignment	READ_BLK_MISALIGN	1	R	[77:77]	TBD
DSR implemented	DSR_IMP	1	R	[76:76]	TBD
Reserved	-	2	R	[75:74]	-
Device size	C_SIZE	12	R	[73:62]	TBD
Max. read current @ VDD min	VDD_R_CURR_MIN	3	R	[61:59]	TBD
Max. read current @ VDD max	VDD_R_CURR_MAX	3	R	[58:56]	TBD
Max. write current @ VDD min	VDD_W_CURR_MIN	3	R	[55:53]	TBD
Max. write current @ VDD max	VDD_W_CURR_MAX	3	R	[52:50]	TBD
Device size multiplier	C_SIZE_MULT	3	R	[49:47]	TBD
Erase group size	ERASE_GRP_SIZE	5	R	[46:42]	TBD
Erase group size multiplier	ERASE_GRP_MULT	5	R	[41:37]	TBD
Write protect group size	WP_GRP_SIZE	5	R	[36:32]	TBD
Write protect group enable	WP_GRP_ENABLE	1	R	[31:31]	TBD
Manufacturer default ECC	DEFAULT_ECC	2	R	[30:29]	TBD
Write speed factor	R2W_FACTOR	3	R	[28:26]	TBD
Max. write data block length	WRITE_BL_LEN	4	R	[25:22]	TBD
Partial blocks for write allowed	WRITE_BL_PARTIAL	1	R	[21:21]	TBD
Reserved	-	4	R	[20:17]	-
Content protection application	CONTENT_PROT_APP	1	R	[16:16]	TBD
File format group	FILE_FORMAT_GRP	1	R/W	[15:15]	TBD
Copy flag (OTP)	COPY	1	R/W	[14:14]	TBD
Permanent write protection	PERM_WRITE_PROTECT	1	R/W	[13:13]	TBD
Temporary write protection	TMP_WRITE_PROTECT	1	R/W/E	[12:12]	TBD
File format	FILE_FORMAT	2	R/W	[11:10]	TBD
ECC code	ECC	2	R/W/E	[9:8]	TBD
CRC	CRC	7	R/W/E	[7:1]	-
Not used, always'1'	-	1	—	[0:0]	-

5.4 Extended CSD Register

The Extended CSD register defines the eMMC properties and selected modes. It is 512 bytes long.
The most significant 320 bytes are the Properties segment, which defines the eMMC capabilities and cannot be modified by the host. The lower 192 bytes are the Modes segment, which defines the configuration the eMMC is working in. These modes can be changed by the host by means of the SWITCH command.

R: Read only
W: One time programmable and not readable.
R/W: One time programmable and readable.
W/E: Multiple writable with value kept after power failure, H/W reset assertion and any CMD0 reset and not readable.
R/W/E: Multiple writable with value kept after power failure, H/W reset assertion and any CMD0 reset and readable.
R/W/C_P: Writable after value cleared by power failure and HW/ rest assertion (the value not cleared by CMD0 reset) and readable.
R/W/E_P: Multiple writable with value reset after power failure, H/W reset assertion and any CMD0 reset and readable.
W/E/_P: Multiple writable with value reset after power failure, H/W reset assertion and any CMD0 reset and not readable

[Table 24] Extended CSD Register

Name	Field	Size (Bytes)	Cell Type	CSD slice	CSD Value
					64 GB
Properties Segment					
Reserved[1]		6	-	[511:506]	-
Extended Security Commands Error	EXT_SECURITY_ERR	1	R	[505]	TBD
Supported Command Sets	S_CMD_SET	1	R	[504]	TBD
HPI features	HPI_FEATURES	1	R	[503]	TBD
Background operations support	BKOPS_SUPPORT	1	R	[502]	TBD
Max packed read commands	MAX_PACKED_READS	1	R	[501]	TBD
Max packed write commands	MAX_PACKED_WRITES	1	R	[500]	TBD
Data Tag Support	DATA_TAG_SUPPORT	1	R	[499]	TBD
Tag Unit Size	TAG_UNIT_SIZE	1	R	[498]	TBD
Tag Resources Size	TAG_RES_SIZE	1	R	[497]	TBD
Context management capabilities	CONTEXT_CAPABILITIES	1	R	[496]	TBD
Large Unit size	LARGE_UNIT_SIZE_M1	1	R	[495]	TBD
Extended partitions attribute support	EXT_SUPPORT	1	R	[494]	TBD
Supported modes	SUPPORTED_MODES	1	R	[493]	TBD
FFU features	FFU_FEATURES	1	R	[492]	TBD
Operation codes timeout	OPERATION_CODE_TIMEOUT	1	R	[491]	TBD
FFU Argument	FFU_ARG	4	R	[490:487]	TBD
Barrier support	BARRIER_SUPPORT	1	R	[486]	TBD
Reserved[1]		177	-	[485:309]	-
CMD Queuing Support	CMDQ_SUPPORT	1	R	[308]	TBD
CMD Queuing Depth	CMDQ_DEPTH	1	R	[307]	TBD
Reserved[1]		1	-	[306]	-
Number of FW sectors correctly programmed	NUMBER_OF_FW_SEC-TORS_CORRECTLY_PRO-GRAMMED	4	R	[305:302]	TBD
Vendor proprietary health report	VENDOR_PROPRI-ETARY_HEALTH_REPORT	32	R	[301:270]	TBD
Device life time estimation type B	DEVICE_LIFE_TIME_EST_TYP_B	1	R	[269]	TBD
Device life time estimation type A	DEVICE_LIFE_TIME_EST_TYP_A	1	R	[268]	TBD
Pre EOL information	PRE_EOL_INFO	1	R	[267]	TBD
Optimal read size	OPTIMAL_READ_SIZE	1	R	[266]	TBD
Optimal write size	OPTIMAL_WRITE_SIZE	1	R	[265]	TBD
Optimal trim unit size	OPTIMAL_TRIM_UNIT_SIZE	1	R	[264]	TBD
Device version	DEVICE_VERSION	2	R	[263:262]	TBD

Firmware version	FIRMWARE_VERSION	3	R	[261:254]	Patch Version
Power class for 200MHz, DDR at VCC=3.6V	PWR_CL_DDR_200_360	1	R	[253]	TBD
Cache size	CACHE_SIZE	4	R	[252:249]	TBD
Generic CMD6 timeout	GENERIC_CMD6_TIME	1	R	[248]	TBD
Power off notification (long) timeout	POWER_OFF_LONG_TIME	1	R	[247]	TBD
Background operations status	BKOPS_STATUS	1	R	[246]	TBD
Number of correctly programmed sectors	CORRECTLY_PRG_SECTORS_-NUM	4	R	[245:242]	TBD
1st initialization time after partitioning	INI_TIMEOUT_AP	1	R	[241]	TBD
Cache flush policy	CACHE_FLUSH_POLICY	1	R	[240]	TBD
Power class for 52MHz, DDR at 3.6V	PWR_CL_DDR_52_360	1	R	[239]	TBD
Power class for 52MHz, DDR at 1.95V	PWR_CL_DDR_52_195	1	R	[238]	TBD
Power class for 200MHz at V_{CCQ}=1.95V, V_{CC}=3.6V	PWR_CL_200_360	1	R	[237]	TBD
Power class for 200MHz, at V_{CCQ}=1.3V, V_{CC}=3.6V	PWR_CL_200_195	1	R	[236]	TBD
Minimum Write Performance for 8bit at 52MHz in DDR mode	MIN_PERF_DDR_W_8_52	1	R	[235]	TBD
Minimum Read Performance for 8bit at 52MHz in DDR mode	MIN_PERF_DDR_R_8_52	1	R	[234]	TBD
Reserved[1]		1	-	[233]	-
TRIM Multiplier	TRIM_MULT	1	R	[232]	TBD
Secure Feature support	SEC_FEATURE_SUPPORT	1	R	[231]	TBD
Secure Erase Multiplier	SEC_ERASE_MULT	1	R	[230]	TBD
Secure TRIM Multiplier	SEC_TRIM_MULT	1	R	[229]	TBD
Boot information	BOOT_INFO	1	R	[228]	TBD
Reserved[1]		1	-	[227]	-
Boot partition size	BOOT_SIZE_MULT	1	R	[226]	TBD
Access size	ACC_SIZE	1	R	[225]	TBD
High-capacity erase unit size	HC_ERASE_GRP_SIZE	1	R	[224]	TBD
High-capacity erase timeout	ERASE_TIMEOUT_MULT	1	R	[223]	TBD
Reliable write sector count	REL_WR_SEC_C	1	R	[222]	TBD
High-capacity write protect group size	HC_WP_GRP_SIZE	1	R	[221]	TBD
Sleep current (V_{CC})	S_C_VCC	1	R	[220]	TBD
Sleep current (V_{CCQ})	S_C_VCCQ	1	R	[219]	TBD
Production state awareness timeout	PRODUCTION_STATE_AWARE-NESS_TIMEOUT	1	R	[218]	TBD
Sleep/awake timeout	S_A_TIMEOUT	1	R	[217]	TBD
Sleep Notification Timeout	SLEEP_NOTIFICATION_TIME	1	R	[216]	TBD
Sector Count	SEC_COUNT	4	R	[215:212]	TBD
Secure Write Protect Information	SECURE_WP_INFO	1	R	[211]	TBD
Minimum Write Performance for 8bit at 52MHz	MIN_PERF_W_8_52	1	R	[210]	TBD
Minimum Read Performance for 8bit at 52MHz	MIN_PERF_R_8_52	1	R	[209]	TBD
Minimum Write Performance for 8bit at 26MHz, for 4bit at 52MHz	MIN_PERF_W_8_26_4_52	1	R	[208]	TBD
Minimum Read Performance for 8bit at 26MHz, for 4bit at 52MHz	MIN_PERF_R_8_26_4_52	1	R	[207]	TBD
Minimum Write Performance for 4bit at 26MHz	MIN_PERF_W_4_26	1	R	[206]	TBD
Minimum Read Performance for 4bit at 26MHz	MIN_PERF_R_4_26	1	R	[205]	0x00

Reserved[1]		1	-	[204]	-
Power class for 26MHz at 3.6V 1 R	PWR_CL_26_360	1	R	[203]	TBD
Power class for 52MHz at 3.6V 1 R	PWR_CL_52_360	1	R	[202]	TBD
Power class for 26MHz at 1.95V 1 R	PWR_CL_26_195	1	R	[201]	TBD
Power class for 52MHz at 1.95V 1 R	PWR_CL_52_195	1	R	[200]	TBD
Partition switching timing	PARTITION_SWITCH_TIME	1	R	[199]	TBD
Out-of-interrupt busy timing	OUT_OF_INTERRUPT_TIME	1	R	[198]	TBD
I/O Driver Strength	DRIVER_STRENGTH	1	R	[197]	TBD
Device type	DEVICE_TYPE	1	R	[196]	TBD
Reserved[1]		1	-	[195]	-
CSD structure	CSD_STRUCTURE	1	R	[194]	TBD
Reserved[1]		1	-	[193]	-
Extended CSD revision	EXT_CSD_REV	1	R	[192]	TBD
Modes Segment					
Command set	CMD_SET	1	R/W/E_P	[191]	TBD
Reserved[1]		1	-	[190]	-
Command set revision	CMD_SET_REV	1	R	[189]	TBD
Reserved[1]		1	-	[188]	-
Power class	POWER_CLASS	1	R/W/E_P	[187]	TBD
Reserved[1]		1	-	[186]	-
High-speed interface timing	HS_TIMING	1	R/W/E_P	[185]	TBD
Strobe Support	STROBE_SUPPORT	1	R	[184]	TBD
Bus width mode	BUS_WIDTH	1	W/E_P	[183]	TBD
Reserved[1]		1	-	[182]	-
Erased memory content	ERASED_MEM_CONT	1	R	[181]	TBD
Reserved[1]		1	-	[180]	-
Partition configuration	PARTITION_CONFIG	1	R/W/E & R/W/E_P	[179]	TBD
Boot config protection	BOOT_CONFIG_PROT	1	R/W & R/W/C_P	[178]	TBD
Boot bus Conditions	BOOT_BUS_CONDITIONS	1	R/W/E	[177]	TBD
Reserved[1]		1	-	[176]	-
High-density erase group definition	ERASE_GROUP_DEF	1	R/W/E_P	[175]	TBD
Boot write protection status registers	BOOT_WP_STATUS	1	R	[174]	TBD
Boot area write protection register	BOOT_WP	1	R/W & R/W/C_P	[173]	TBD
Reserved[1]		1	-	[172]	-
User area write protection register	USER_WP	1	R/W, R/W/C_P &R/W/E_P	[171]	TBD
Reserved[1]		1	-	[170]	-
FW configuration	FW_CONFIG	1	R/W	[169]	TBD
RPMB Size	RPMB_SIZE_MULT	1	R	[168]	TBD
Write reliability setting register	WR_REL_SET	1	R/W	[167]	TBD
Write reliability parameter register	WR_REL_PARAM	1	R	[166]	TBD
Start Sanitize operation	SANITIZE_START	1	W/E_P	[165]	TBD
Manually start background operations	BKOPS_START	1	W/E_P	[164]	TBD
Enable background operations handshake	BKOPS_EN	1	R/W&R/W/E	[163]	TBD

H/W reset function	RST_n_FUNCTION	1	R/W	[162]	TBD
HPI management	HPI_MGMT	1	R/W/E_P	[161]	TBD
Partitioning Support	PARTITIONING_SUPPORT	1	R	[160]	TBD
Max Enhanced Area Size	MAX_ENH_SIZE_MULT	3	R	[159:157]	TBD
Partitions attribute	PARTITIONS_ATTRIBUTE	1	R/W	[156]	TBD
Partitioning Setting	PARTITION_SETTING_ COMPLETED	1	R/W	[155]	TBD
General Purpose Partition Size	GP_SIZE_MULT	12	R/W	[154:143]	TBD
Enhanced User Data Area Size	ENH_SIZE_MULT	3	R/W	[142:140]	TBD
Enhanced User Data Start Address	ENH_START_ADDR	4	R/W	[139:136]	TBD
Reserved[1]		1	-	[135]	-
Bad Block Management mode	SEC_BAD_BLK_MGMNT	1	R/W	[134]	TBD
Production state awareness	PRODUCTION_STATE_AWARE-NESS	1	W/E_P	[133]	TBD
Package Case Temperature is controlled	TCASE_SUPPORT	1	W/E_P	[132]	TBD
Periodic Wake-up	PERIODIC_WAKEUP	1	R/W/E	[131]	TBD
Program CID/CSD in DDR mode support	PROGRAM_CID_CSD_DDR_SUP-PORT	1	R	[130]	TBD
Reserved[1]		2	-	[129:128]	-
Vendor Mode Configuration	VENDOR_MODE_CONFIG	1	W/E_P	[127]	TBD
Reserved[1]		1	-	[126]	-
SMT mode config	SMT_MODE_CONFIG	1	R/W	[125]	TBD
Vendor Feature Support	VENDOR_FEATURE_SUPPORT	1	R	[124]	TBD
Status of Burst Write	BURST_WRITE_BUFFER_STATU S	1	R	[123]	TBD
Reserved[1]		1	-	[122]	-
Burst Write Buffer Size	BURST_WRITE_BUFFER_SIZE	1	R/W/E	[121]	TBD
Burst Write Performance	BURST_WRITE_PERF	1	R/W/E	[120]	TBD
Burst Write Disable Count	BURST_WRITE_DISABLE_CNT	1	R	[119]	TBD
Burst Write Enable Count	BURST_WRITE_ENABLE_CNT	1	R	[118]	TBD
Set up drive strength manual mode	SETUP_DS_MANUAL_MODE_R	1	R/W/E_P	[117]	TBD
Set up drive strength manual mode	SETUP_DS_MANUAL_MODE_W	1	R/W/E_P	[116]	TBD
Set up Async Auto Busy Clear	SETUP_ASYNC_AUTO_BUSY_-CLEAR	1	W/E_P	[115]	TBD
Package Information	PACKAGE_INFO	1	R	[114]	TBD
Cumulative Written Data Size	TOTAL_WRITTEN_DATA	4	R	[113:110]	TBD
Reserved[1]		45	-	[109:69]	-
Modify HS200 data output delay	MODIFY_HS200_DAT_OUT-PUT_DELAY	1	R/W/E	[68]	TBD
Modify HS200 cmd output delay	MODIFY_HS200_CMD_OUT-PUT_DELAY	1	R/W/E	[67]	TBD
Reserved[1]		1	-	[66]	-
Optimized Features	OPTIMIZED_FEATURES	2	R	[65:64]	TBD
Native sector size	NATIVE_SECTOR_SIZE	1	R	[63]	TBD
Sector size emulation	USE_NATIVE_SECTOR	1	R/W	[62]	TBD
Sector size	DATA_SECTOR_SIZE	1	R	[61]	TBD
1st initialization after disabling sector size emulation	INI_TIMEOUT_EMU	1	R	[60]	TBD
Class 6 commands control	CLASS_6_CTRL	1	R/W/E_P	[59]	TBD
Number of addressed group to be Released	DYNCAP_NEEDED	1	R	[58]	TBD
Exception events control	EXCEPTION_EVENTS_CTRL	2	R/W/E_P	[57:56]	TBD

Exception events status	EXCEPTION_EVENTS_STATUS	2	R	[55:54]	TBD
Extended Partitions Attribute	EXT_PARTITIONS_ATTRIBUTE	2	R/W	[53:52]	TBD
Context configuration	CONTEXT_CONF	15	R/W/E_P	[51:37]	TBD
Packed command status	PACKED_COMMAND_STATUS	1	R	[36]	TBD
Packed command failure index	PACKED_FAILURE_INDEX	1	R	[35]	TBD
Power Off Notification	POWER_OFF_NOTIFICATION	1	R/W/E_P	[34]	TBD
Control to turn the Cache ON/OFF	CACHE_CTRL	1	R/W/E_P	[33]	TBD
Flushing of the cache	FLUSH_CACHE	1	W/E_P	[32]	TBD
Control to turn the Barrier ON/OFF	BARRIER_CTRL	1	R	[31]	TBD
Mode config	MODE_CONFIG	1	R/W/E_P	[30]	TBD
Mode operation codes	MODE_OPERATION_CODES	1	W/E_P	[29]	TBD
Reserved[1]		2	-	[28:27]	-
FFU status	FFU_STATUS	1	R	[26]	TBD
Pre loading data size	PRE_LOADING_DATA_SIZE	4	R/W/E_P	[25:22]	TBD
Max pre loading data size	MAX_PRE_LOADING_DATA_ SIZE	4	R	[21:18]	TBD
Product state awareness enablement	PRODUCT_STATE_AWARENESS _ENABLEMENT	1	R/W/E & R	[17]	TBD
Secure Removal Type	SECURE_REMOVAL_TYPE	1	R/W & R	[16]	TBD
Command Queue Mode Enable	CMDQ_MODE_EN	1	R/W/E_P	[15]	TBD
Reserved[1]		15	-	[14:0]	-

NOTE :
1) Reserved bits should be read as “0.”

6.0 AC PARAMETER

6.1 Timing Parameter

[Table 25] Timing Parameter

Timing Parameter		Max. Value	Unit
Initialization Time (tINIT)	Normal [1)]	1	s
	After partition setting [2)]	3	s
Read Timeout		100	ms
Write Timeout		350	ms
Erase Timeout		20	ms
Force Erase Timeout		3	min
Secure Erase Timeout		8	s
Secure Trim step1 Timeout		5	s
Secure Trim step2 Timeout		3	s
Trim Timeout		600	ms
Partition Switching Timeout (after Init)		1	ms
Power Off Notification (Short) Timeout		100	ms
Power Off Notification (Long) Timeout		600	ms

NOTE:
1) Normal Initialization Time without partition setting
2) Initialization Time after partition setting, refer to INI_TIMEOUT_AP in Chapter 5.4 EXT_CSD register
3) Be advised Timeout Values specified in Table above are for testing purposes under Samsung test pattern only and actual timeout situations may vary
4) EXCEPTION_EVENT may occur and the actual timeout values may vary due to user environment

6.2 Previous Bus Timing Parameters for DDR52 and HS200 mode are defined by JEDEC standard

6.3 Bus Timing Specification in HS400 mode

6.3.1 HS400 Device Input Timing

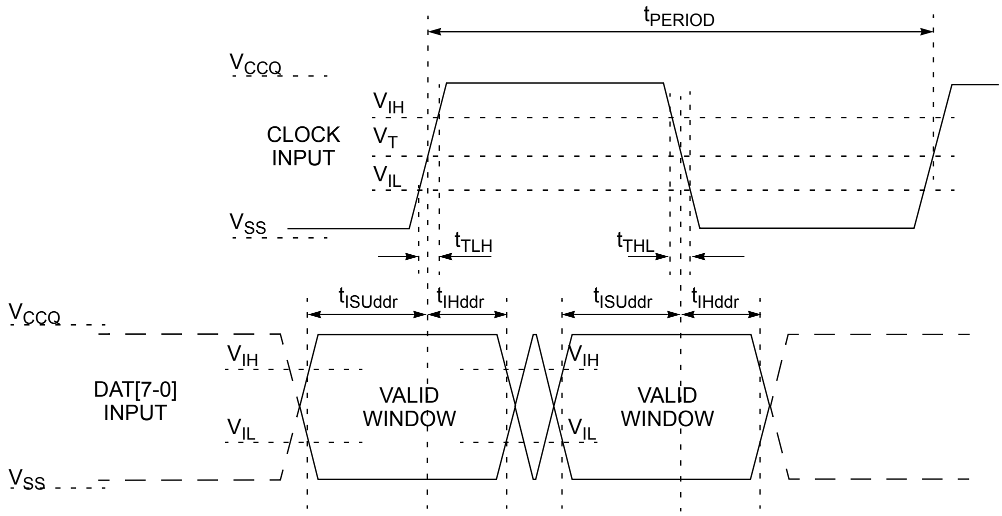


Figure 4. HS400 Device Input Timing

NOTE:
1) t_{ISU} and t_{IH} are measured at V_{IL}(max.) and V_{IH}(min).
2) V_{IH} denotes V_{IH}(min.) and V_{IL} denotes V_{IL}(max.)

[Table 26] HS400 Device input timing

Parameter	Symbol	Min	Max	Unit
Input CLK				
Cycle time data transfer mode	t_{PERIOD}	5		ns
Slew rate	SR	1.125		V/ns
Duty cycle distortion	t_{CKDCD}	0.0	0.3	ns
Minimum pulse width	t_{CKMPW}	2.2		ns
Input DAT (referenced to CLK)				
Input set-up time	t_{ISUddr}	0.4		ns
Input hold time	t_{IHddr}	0.4		ns
Slew rate	SR	1.125		V/ns

6.3.2 HS400 Device Output Timing

Data Strobe is used to read data (data read and CRC status response read) in HS400 mode.
The device output value of Data Strobe is “High-Z” when the device is not in outputting data (data read, CRC status response).
Data Strobe is toggled only during data read period.

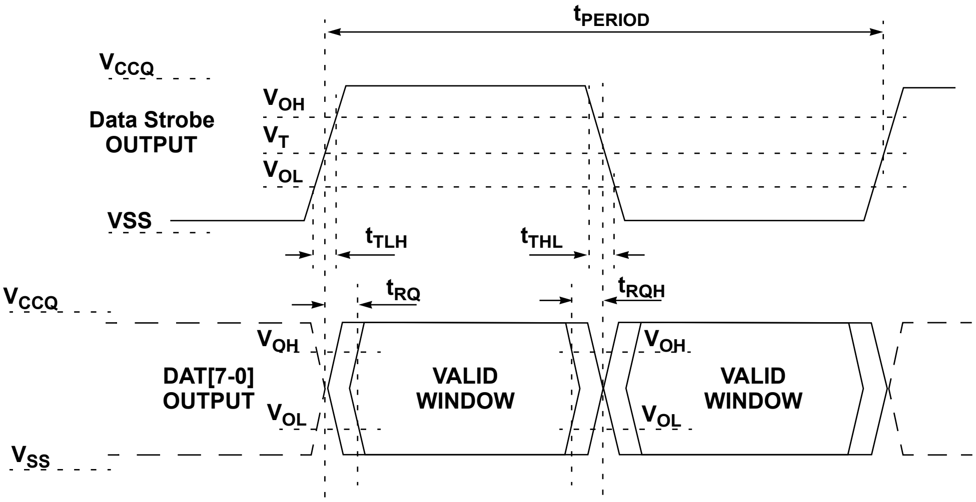


Figure 5. HS400 Device Output Timing

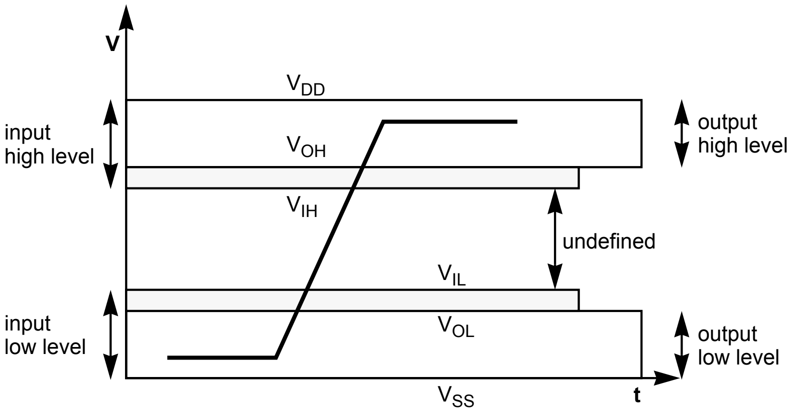
NOTE:
V_{OH} denotes V_{OH}(min.) and V_{OL} denotes V_{OL}(max.).

[Table 27] HS400 Device Output timing

Parameter	Symbol	Min	Max	Unit
Data Strobe				
Cycle time data transfer mode	t_{PERIOD}	5		ns
Slew rate	SR	1.125		V/ns
Duty cycle distortion	t_{DSDCD}	0.0	0.2	ns
Minimum pulse width	t_{DSMPW}	2.0		ns
Read pre-amble	t_{RPRE}	0.4	-	t_{PERIOD}
Read post-amble	t_{RPST}	0.4	-	t_{PERIOD}
Output DAT (referenced to Data Strobe)				
Output skew	t_{RQ}	-	0.4	ns
Output hold skew	t_{RQH}	-	0.4	ns
Slew rate	SR	1.125		V/ns

6.4 Bus signal levels

As the bus can be supplied with a variable supply voltage, all signal levels are related to the supply voltage.



6.4.1 Open-drain mode bus signal level

[Table 28] Open-drain bus signal level

Parameter	Symbol	Min	Max.	Unit	Conditions
Output HIGH voltage	V_{OH}	V_{DD} - 0.2	-	V	1)
Output LOW voltage	V_{OL}	-	0.3	V	I_{OL} = 2 mA

NOTE:
1) Because Voh depends on external resistance value (including outside the package), this value does not apply as device specification.
Host is responsible to choose the external pull-up and open drain resistance value to meet Voh Min value.

6.4.2 Push-pull mode bus signal level eMMC

The device input and output voltages shall be within the following specified ranges for any V_{CCQ} of the allowed voltage range

[Table 29] Push-pull signal level—high-voltage eMMC

Parameter	Symbol	Min	Max.	Unit	Conditions
Output HIGH voltage	V_{OH}	0.75*V_{DD}	-	V	I_{OH} = -100 uA@V_{CCQ} min
Output LOW voltage	V_{OL}	-	0.125*V_{DD}	V	I_{OL} = 100 uA@V_{CCQ} min
Input HIGH voltage	V_{IH}	0.625*V_{DD}	V_{DD} + 0.3	V	-
Input LOW voltage	V_{IL}	V_{SS} - 0.3	0.25*V_{DD}	V	-

[Table 30] Push-pull signal level—1.70 - 1.95 V_{CCQ} voltage Range

Parameter	Symbol	Min	Max.	Unit	Conditions
Output HIGH voltage	V_{OH}	V_{CCQ} - 0.45V	-	V	I_{OH} = -2mA
Output LOW voltage	V_{OL}	-	0.45V	V	I_{OL} = 2mA
Input HIGH voltage	V_{IH}	0.65*V_{CCQ} 1)	V_{CCQ} + 0.3	V	-
Input LOW voltage	V_{IL}	V_{SS} - 0.3	0.35*V_{CCQ} 2)	V	-

NOTE:
1) 0.7*V_{CCQ} for MMC4.3 and older revisions.
2) 0.3*V_{CCQ} for MMC4.3 and older revisions.

7.0 DC PARAMETER

7.1 Active Power Consumption during operation

[Table 31] Active Power Consumption during operation

Density	NAND Type	V_{CCQ}	V_{CC}	Unit
64 GB	256Gb x 2	TBD	TBD	mA

* Power Measurement conditions: Bus configuration =x8 @HS400
* The measurement for max RMS current is the average RMS current consumption over a period of 100ms.

7.2 Standby Power Consumption in auto power saving mode and standby state.

[Table 32] Standby Power Consumption in auto power saving mode and standby state

Density	NAND Type	V_{CCQ}		V_{CC}		Unit
		25°C(Typ)	85°C	25°C(Typ)	85°C	
64 GB	256Gb x 2	TBD	TBD	TBD	TBD	uA

NOTE:
Power Measurement conditions: Bus configuration =x8, No CLK
*Typical value is measured at V_{CC}=3.3V, TA=25°C. Not 100% tested.

7.3 Sleep Power Consumption in Sleep State

[Table 33] Sleep Power Consumption in Sleep State

Density	NAND Type	V_{CCQ}		V_{CC}	Unit
		25°C(Typ)	85°C		
64 GB	256Gb x 2	TBD	TBD	0 1)	uA

NOTE:
Power Measurement conditions: Bus configuration =x8, No CLK
1) In auto power saving mode, V_{CC} power can not be turned off. However in sleep mode V_{CC} power can be turned off.

7.4 Supply Voltage

[Table 34] Supply voltage

Item	Min	Max	Unit
V_{CCQ}	1.70	1.95	V
V_{CC}	2.7	3.6	V
V_{SS}	-0.5	0.5	V

7.5 Bus Signal Line Load

The total capacitance C_L of each line of the eMMC bus is the sum of the bus master capacitance C_{HOST}, the bus capacitance C_{BUS} itself and the capacitance C_{DEVICE} of the eMMC connected to this line:

$$C_L = C_{HOST} + C_{BUS} + C_{DEVICE}$$

The sum of the host and bus capacitances should be under 20pF.

[Table 35] Bus Signal Line Load

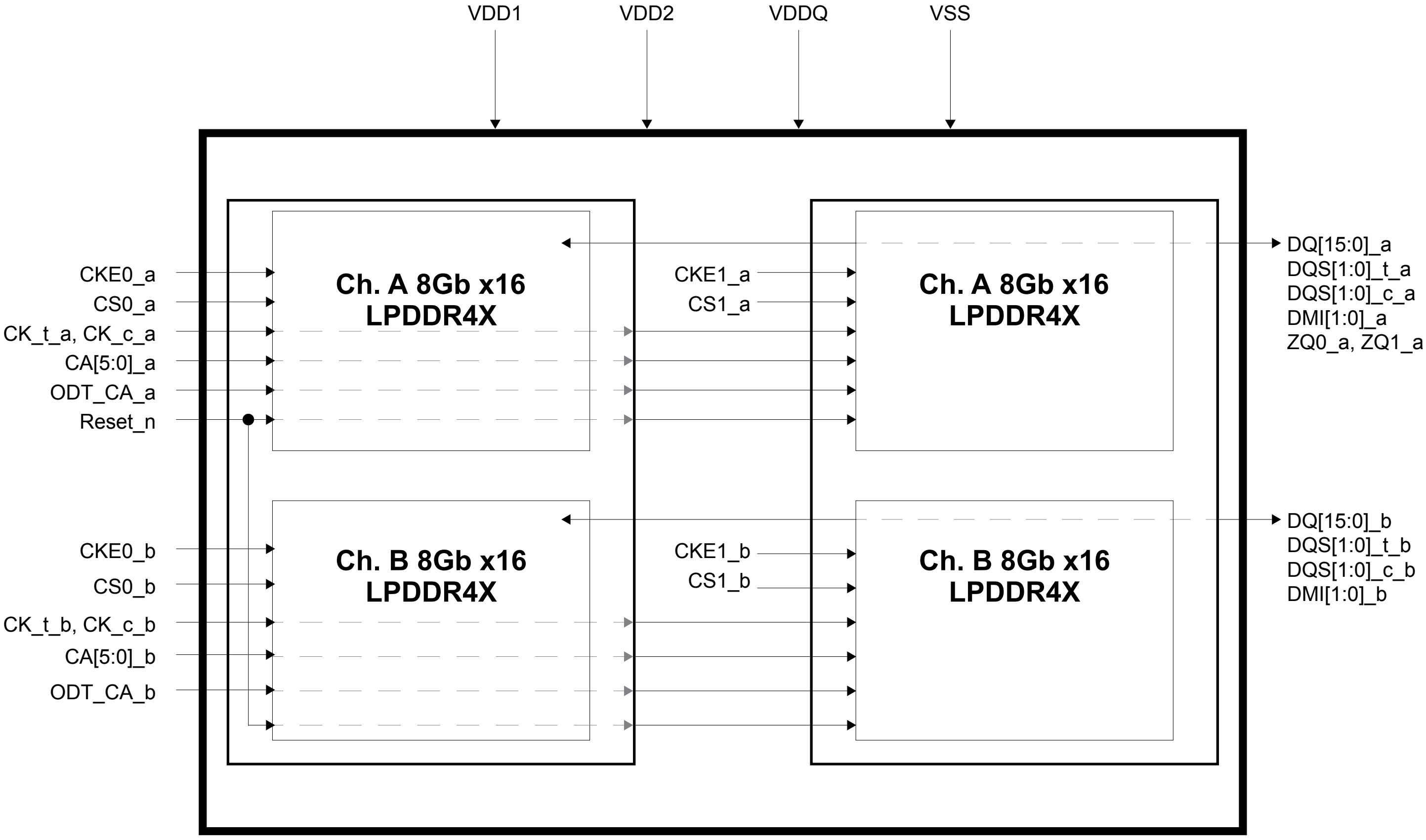
sParameter	Symbol	Min	Typ.	Max	Unit	Remark
Pull-up resistance for CMD	R_{CMD}	4.7		100	KOhm	to prevent bus floating
Pull-up resistance for DAT0-DAT7	R_{DAT}	10		100	KOhm	to prevent bus floating
Internal pull up resistance DAT1-DAT7	R_{int}	10		150	KOhm	to prevent unconnected lines floating
Single Device capacitance	C_{DEVICE}			12	pF	
Maximum signal line inductance				16	nH	f_{PP} <= 52 MHz

[Table 36] Capacitance and Resistance for HS400 mode

Parameter	Symbol	Min	Typ	Max	Unit	Remark
Bus signal line capacitance	C_L			13	pF	Single Device
Single Device capacitance	C_{DEVICE}			6	pF	
Pull-down resistance for Data Strobe	$R_{Data\ Strobe}$	10		100	KOhm	

32Gb(16Gb*2) DDP LPDDR4X SDRAM

1.0 Functional Block Diagram



1.1 LPDDR4X Pad Definition and Description

[Table 1] Pad Definition and Description

Name	Type	Description
CK_t_A CK_c_A CK_t_B CK_c_B	Input	**Clock:** CK_t and CK_c are differential clock inputs. All address, command, and control input signals are sampled on the crossing of the positive edge of CK_t and the negative edge of CK_c. AC timings for CA parameters are referenced to CK. Each channel (A & B) has its own clock pair.
CKE_A CKE_B	Input	**Clock Enable:** CKE HIGH activates and CKE LOW deactivates the internal clock circuits, input buffers, and output drivers. Power-saving modes are entered and exited via CKE transitions. CKE is part of the command code. Each channel (A & B) has its own CKE signal.
CS_A, CS_B	Input	**Chip Select:** CS is part of the command code. Each channel (A & B) has its own CS signal.
CA[5:0]_A CA[5:0]_B	Input	**Command/Address Inputs:** CA signals provide the Command and Address inputs according to the Command Truth Table. Each channel (A&B) has its own CA signals.
ODT_CA_A ODT_CA_B	Input	**CA ODT Control:** The ODT_CA pin is used in conjunction with the Mode Register to turn on/off the On-Die-Termination for CA pins.
DQ[15:0]_A DQ[15:0]_B	I/O	**Data Inputs/Outputs:** Bi-direction data bus
DQS[1:0]_t_A DQS[1:0]_c_A DQS[1:0]_t_B DQS[1:0]_c_B	I/O	**Data Strobe:** DQS_t and DQS_c are bi-directional differential output clock signals used to strobe data during a READ or WRITE. The Data Strobe is generated by the DRAM for a READ and is edge-aligned with Data. The Data Strobe is generated by the Memory Controller for a WRITE and must arrive prior to Data. Each byte of data has a Data Strobe signal pair. Each channel (A & B) has its own DQS strobes.
DMI[1:0]_A DMI[1:0]_B	I/O	**Data Mask Inversion:** DMI is a bi-directional signal which is driven HIGH when the data on the data bus is inverted, or driven LOW when the data is in its normal state. Data Inversion can be disabled via a mode register setting. Each byte of data has a DMI signal. Each channel (A & B) has its own DMI signals. This signal is also used along with the DQ signals to provide write data masking information to the DRAM. The DMI pin function - Data Inversion or Data mask - depends on Mode Register setting.
ZQ	Reference	**Calibration Reference:** Used to calibrate the output drive strength and the termination resistance. There is one ZQ pin per die. The ZQ pin shall be connected to VDDQ through a 240Ω ± 1% resistor.
VDDQ,VDD1, VDD2	Supply	**Power Supplies**: Isolated on the die for improved noise immunity.
V_{SS}, V_{SSQ}	GND	**Ground Reference:** Power supply ground reference.
RESET_n	Input	**RESET:** When asserted LOW, the RESET_n signal resets all channels of the die. There is one RESET_n pad per die.

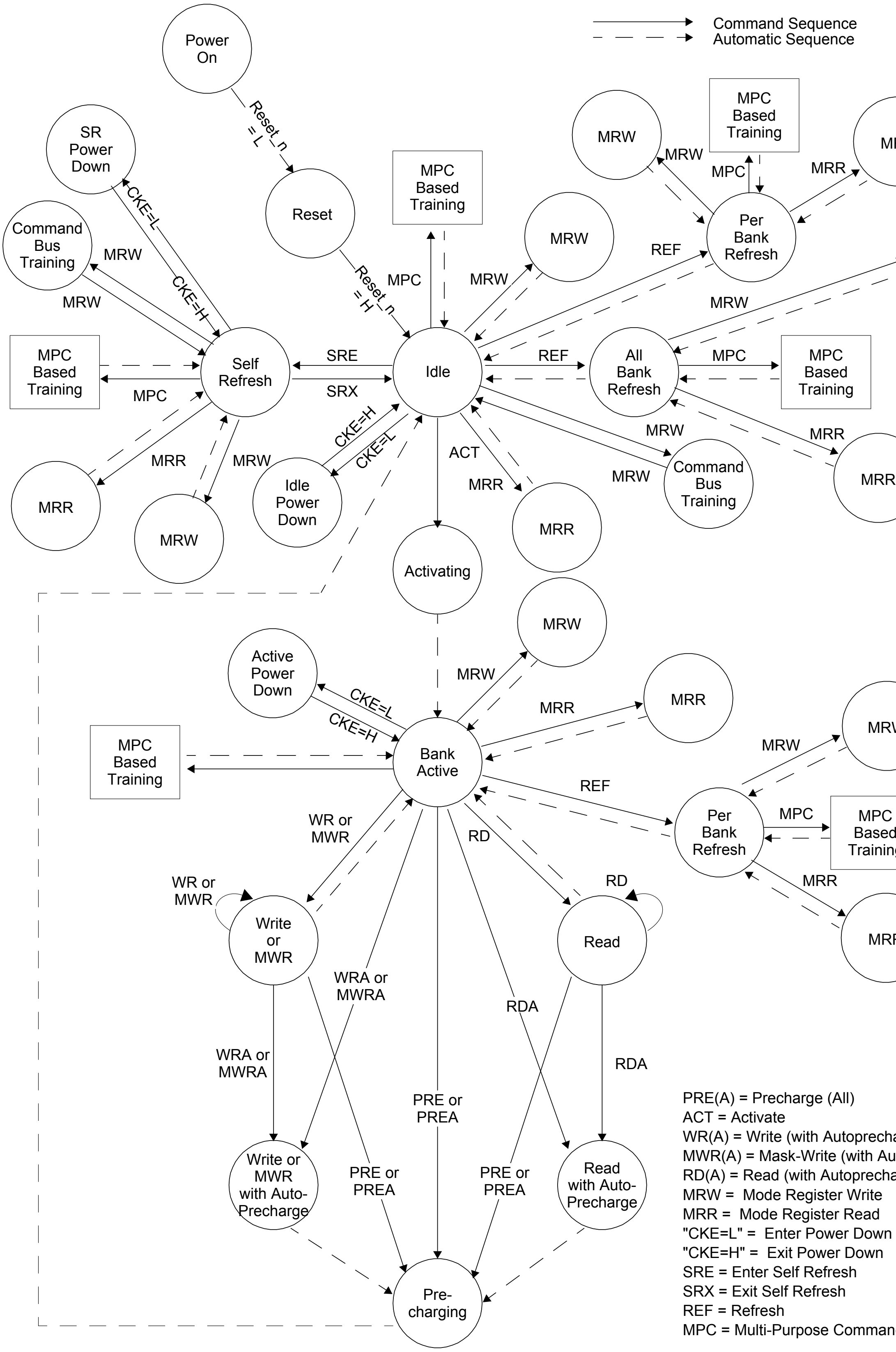
NOTE :
1) "_A" and "_B" indicate DRAM channel "_A" pads are present in all devices. "_B" pads are present in dual channel SDRAM devices only.

2.0 Simplified LPDDR4X State Diagram

LPDDR4X-SDRAM state diagram provides a simplified illustration of allowed state transitions and the related commands to control them. For a complete definition of the device behavior, the information provided by the state diagram should be integrated with the truth tables and timing specification.

The truth tables provide complementary information to the state diagram, they clarify the device behavior and the applied restrictions when considering the actual state of all the banks.

For the command definition, see datasheet of [Command Definition & Timing Diagram].



PRE(A) = Precharge (All)
ACT = Activate
WR(A) = Write (with Autoprecharge)
MWR(A) = Mask-Write (with Autoprecharge)
RD(A) = Read (with Autoprecharge)
MRW = Mode Register Write
MRR = Mode Register Read
"CKE=L" = Enter Power Down
"CKE=H" = Exit Power Down
SRE = Enter Self Refresh
SRX = Exit Self Refresh
REF = Refresh
MPC = Multi-Purpose Command (w/NOP)

Figure 1. LPDDR4X: Simplified Bus Interface State Diagram-1

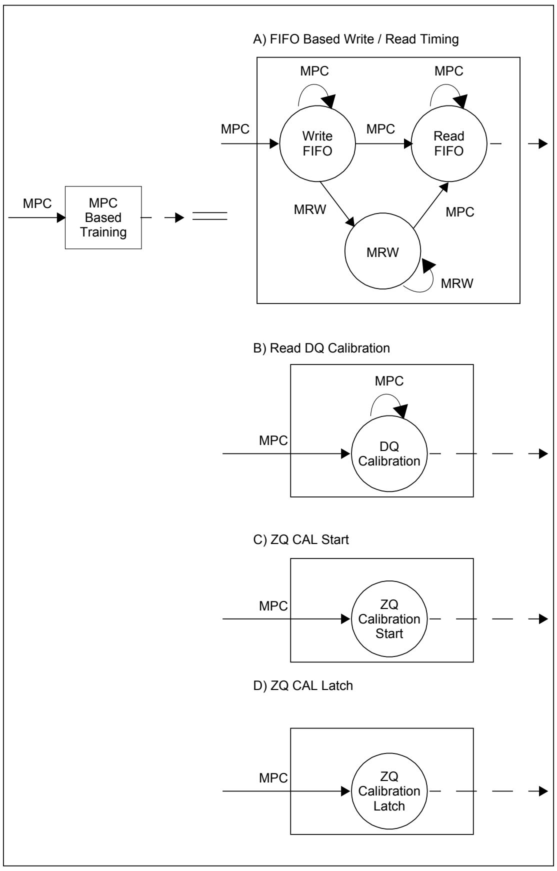


Figure 2. LPDDR4X: Simplified Bus Interface State Diagram -2

NOTE :
1) From the Self-Refresh state the device can enter Power-Down, MRR, MRW, or MPC states. See the section on Self-Refresh for more information.
2) In IDLE state, all banks are precharged.
3) In the case of a MRW command to enter a training mode, the state machine will not automatically return to the IDLE state at the conclusion of training. See the applicable training section for more information.
4) In the case of a MPC command to enter a training mode, the state machine may not automatically return to the IDLE state at the conclusion of training. See the applicable training section for more information.
5) This simplified State Diagram is intended to provide an overview of the possible state transitions and the commands to control them. In particular, situations involving more than one bank, the enabling or disabling of on-die termination, and some other events are not captured in full detail.
6) States that have an “automatic return” and can be accessed from more than one prior state (Ex. MRW from either Idle or Active states) will return to the state from when they were initiated (Ex. MRW from Idle will return to Idle).
7) The RESET_n pin can be asserted from any state, and will cause the SDRAM to go to the Reset State. The diagram shows RESET applied from the Power-On as an example, but the Diagram should not be construed as a restriction on RESET_n.

2.1 Mode Register Definition

2.1.1 Mode Register Assignment and Definition in LPDDR4X SDRAM

shows the mode registers for LPDDR4X SDRAM. Each register is denoted as “R” if it can be read but not written, “W” if it can be written but not read, and “R/W” if it can be read and written. A Mode Register Read command is used to read a mode register. A Mode Register Write command is used to write a mode register.

[Table 2] Mode Register Assignment in LPDDR4X SDRAM

MR#	MA <7:0>	Function	Access	OP7	OP6	OP5	OP4	OP3	OP2	OP1	OP0
0	00_H	Device Info.	R	CATR	(RFU)	PPR	RZQI		(RFU)		Refresh mode
1	01_H	Device Feature 1	W	RPST0	nWR0			RD-PRE0	WR-PRE0	BL	
				RPST1	nWR1			RD-PRE1	WR-PRE1		
2	02_H	Device Feature 2	W	WR Lev	WL Select0	WL0			RL0		
					WL Select1	WL1			RL1		
3	03_H	I/O Configuration-1	W	DBI-WR0	DBI-RD0	PDDS0			PPR Protection	WR PST	PU-CAL0
				DBI-WR1	DBI-RD1	PDDS1				WR PST	PU-CAL1
4	04_H	Refresh Rate	R/W	TUF	Thermal Offset		PPRE	SR Abort	Refresh Rate		
5	05_H	Basic Configuration-1	R	LPDDR4X Manufacturer ID							
6	06_H	Basic Configuration-2	R	Revision ID-1							
7	07_H	Basic Configuration-3	R	Revision ID-2							
8	08_H	Basic Configuration-4	R	I/O width		Density				Type	
9	09_H	Test Mode	W	Vendor Specific Test Register							
10	$0A_H$	IO Calibration	W	(RFU)							ZQ-RESET
11	$0B_H$	ODT Feature	W	(RFU)	CA ODT0			(RFU)	DQ ODT0		
					CA ODT1				DQ ODT1		
12	$0C_H$	VREF(ca) Setting/Range	R/W	(RFU)	VR-CA0	VREF0(ca)					
				(RFU)	VR-CA1	VREF1(ca)					
13	$0D_H$	CBT,RPT,VRO,VRCG, RRO, DM_DIS,FSP-WR,FSP-OP	W	FSP-OP	FSP-WR	DM_DIS	RRO	VRCG	VRO	RPT	CBT
14	$0E_H$	VREF(dq) Setting/Range	R/W	(RFU)	VR-DQ0	$V_{REF0(DQ)}$					
				(RFU)	VR-DQ1	$V_{REF1(DQ)}$					
15	$0F_H$	Lower-Byte Invert for DQ Calibration	W	Lower-Byte Invert Register for DQ Calibration							
16	10_H	PASR_Bank	W	PASR Bank Mask							
17	11_H	PASR_Segment	W	PASR Segment Mask							
18	12_H	IT-LSB	R	DQS Oscillator Count-LSB							
19	13_H	IT-MSB	R	DQS Oscillator Count-MSB							
20	14_H	Upper-Byte Invert for DQ Calibration	W	Upper-Byte Invert Register for DQ Calibration							
21	15_H	RFU	N/A	(RFU)							
22	16_H	ODT Feature	W	(RFU)		ODT-CA0	ODT-CS0	ODT-CK0	SoC ODT0		
				(RFU)		ODT-CA1	ODT-CS1	ODT-CK1	SoC ODT1		
23	17_H	DQS interval timer run time	W	DQS interval timer run time setting							

[Table 2] Mode Register Assignment in LPDDR4X SDRAM

MR#	MA <7:0>	Function	Access	OP7	OP6	OP5	OP4	OP3	OP2	OP1	OP0
24	18_H	TRR	R/W	TRR Mode	TRR Mode BAn			Unlim-ited MAC	MAC Value		
25	19_H	PPR Resource	R	Bank 7	Bank 6	Bank 5	Bank 4	Bank 3	Bank 2	Bank 1	Bank 0
26:29	$1A_H$: $1D_H$	RFU	N/A	Reserved for Future Use							
30	$1E_H$	Reserved for Testing	N/A	Reserved for Testing-SDRAM will ignore							
31	$1F_H$	RFU	N/A	Reserved for Future Use							
32	20_H	DQ Calibration Pattern A	W	DQ Calibration Pattern "A" (default = 5AH)							
33:38	21_H~26_H	(Do Not Use)	NA	Do Not Use							
39	27_H	Reserved for Testing	N/A	Reserved for Testing-SDRAM will ignore							
40	28_H	DQ Calibration Pattern B	W	DQ Calibration Pattern "B" (default = 3CH)							
41:47	29_H~$2F_H$	(Do Not Use)	NA	Do Not Use							
48:63	30_H~$3F_H$	RFU	NA	(RFU)							

	Applied when FSP = 0
	Applied when FSP = 1

NOTE :
1) RFU bits shall be set to '0' during writes.
2) RFU bits shall be read as '0' during reads.
3) All mode registers that are specified as RFU or write-only shall return undefined data when read and DQS_t, DQS_c shall be toggled.
4) All mode registers that are specified as RFU shall not be written.
5) Writes to read-only registers shall have no impact on the functionality of the device.

MR0_Device Information (MA<7:0> = 00_H) :

OP7	OP6	OP5	OP4	OP3	OP2	OP1	OP0
CATR	(RFU)	PPR	RZQI		(RFU)		Refresh mode

Function	Register Type	Operand	Data	Notes
Refresh mode	Read-only	OP[0]	**0_B**: Both legacy & modified refresh mode supported **1_B**: Only modified refresh mode supported	
RZQI (Built-in Self-Test for RZQ)		OP[4:3]	**00_B:** RZQ self-test not supported **01_B:** ZQ-pin may connect to V_{SSQ} or float **10_B:** ZQ-pin may short to V_{DDQ} **11_B:** ZQ-pin self test completed, no error condition detected (ZQ-pin may not connect to V_{SSQ} or float, nor short to V_{DDQ})	1,2,3,4
PPR (Post-Package Repair)		OP[5]	**0_B**: PPR not supported **1_B**: PPR supported	6,7
CATR (CA Terminating Rank)		OP[7]	**0_B**: CA for this rank is not terminated **1_B**: Vendor specific	5

NOTE :
1) RZQI MR value, if supported, will be valid after the following sequence:
a. Completion of MPC ZQCAL Start command to either channel.
b. Completion of MPC ZQCAL Latch command to either channel then t_{ZQLAT} is satisfied.
RZQI value will be lost after Reset.
2) If the ZQ-pin is connected to VSSQ to set default calibration, OP[4:3] shall be set to 01_B. If the ZQ-pin is not connected to VSSQ, either OP[4:3] = 01_B or OP[4:3] =10_B might indicate a ZQ-pin assembly error. It is recommended that the assembly error is corrected.
3) In the case of possible assembly error, the LPDDR4X-SDRAM device will default to factory trim settings for RON, and will ignore ZQ calibration commands. In either case, the device may not function as intended.
4) In ZQ self-test returns OP[4:3] = 11_B, the device has detected a resistor connected to the ZQ pin. However, this result cannot be used to validate the ZQ resistor value or that the ZQ resistor tolerance meets the specified limits (i.e 240-Ω +/- 1%).
5) CATR functionality is Vendor specific. CATR can either indicate the connection status of the ODTCA pad for the die or whether CA for the rank is terminated. Consult the vendor device datasheet for details.
6) If post-package repair is supported, then the LPDDR4X-SDRAM can be put into repair mode by setting the PPR entry/exit bit in MR4.
7) PPR (Post-Package Repair) function is JEDEC optional spec. Please contact Samsung for more information.

MR1_Device Feature 1 (MA<7:0> = 01_H) :

OP7	OP6	OP5	OP4	OP3	OP2	OP1	OP0
RPST	nWR (for AP)			RD-PRE	WR-PRE	BL	

Function	Register Type	Operand	Data	Notes
BL (Burst Length)	Write-only	OP[1:0]	**00_B :** BL=16 Sequential (default) **01_B :** BL=32 Sequential **10_B :** BL=16 or 32 Sequential (on-the-fly) **All others:** Reserved	1
WR-PRE (WR Pre-amble Length)		OP[2]	**0_B** : Reserved **1_B** : WR Pre-amble = 2×tCK	5,6
RD-PRE (RD Pre-amble Type)		OP[3]	**0_B** : RD Pre-amble = Static (default) **1_B** : RD Pre-amble = Toggle	3,5,6
nWR (Write-Recovery for Auto-Precharge commands)		OP[6:4]	**000_B:** nWR = 6 (default) **001_B:** nWR = 10 **010_B:** nWR = 16 **011_B:** nWR = 20 **100_B:** nWR = 24 **101_B:** nWR = 30 **110_B:** nWR = 34 **111_B:** nWR = 40	2,5,6
RPST (RD Post-Amble Length)		OP[7]	**0_B** : RD Post-amble = 0.5×tCK (default) **1_B** : RD Post-amble = 1.5×tCK	4,5,6

NOTE :
1) Burst length on-the-fly can be set to either BL=16 or BL=32 by setting the “BL” bit in the command operands. See the Command Truth Table.
2) The programmed value of nWR is the number of clock cycles the LPDDR4X-SDRAM device uses to determine the starting point of an internal Precharge operation after a Write burst with AP (auto-precharge) enabled. See , "Frequency Ranges for RL, WL, and nWR Settings" later in this section.
3) For Read operations this bit must be set to select between a “toggling” pre-amble and a “Non-toggling” pre-amble. See the Read Preamble and Postamble section in Operation timing for a drawing of each type of pre-amble.
4) OP[7] provides an optional READ post-amble with an additional rising and falling edge of DQS_t. The optional postamble cycle is provided for the benefit of certain memory controllers.
5) There are two physical registers assigned to each bit of this MR parameter, designated set point 0 and set point 1. Only the registers for the set point determined by the state of the FSP-WR bit (MR13 OP[6]) will be written to with an MRW command to this MR address, or read from with an MRR command to this address.
6) There are two physical registers assigned to each bit of this MR parameter, designated set point 0 and set point 1. The device will operate only according to the values stored in the registers for the active set point, i.e., the set point determined by the state of the FSP-OP bit (MR13 OP[7]). The values in the registers for the inactive set point will be ignored by the device, and may be changed without affecting device operation.

[Table 3] Read and Write Latencies

Read Latency		Write Latency		nWR	nRTP	Lower Clock Frequency Limit (Greater than)	Upper Clock Frequency Limit (Same or less than)	Units	Notes
No DBI	w/ DBI	Set "A"	Set "B"						
6	6	4	4	6	8	10	266	MHz	1,2,3,4,5,6
10	12	6	8	10	8	266	533		
14	16	8	12	16	8	533	800		
20	22	10	18	20	8	800	1066		
24	28	12	22	24	10	1066	1333		
28	32	14	26	30	12	1333	1600		
32	36	16	30	34	14	1600	1866		
36	40	18	34	40	16	1866	2133		

NOTE :
1) The LPDDR4X-SDRAM device should not be operated at a frequency above the Upper Frequency Limit, or below the Lower Frequency Limit, shown for each RL, WL, nRTP, or nWR value.
2) DBI for Read operations is enabled in MR3 OP[6]. When MR3 OP[6]=0_B, then the “No DBI” column should be used for Read Latency. When MR3 OP[6]=1_B, then the “w/DBI” column should be used for Read Latency.
3) Write Latency Set “A” and Set “B” is determined by MR2 OP[6]. When MR2 OP[6]=0_B, then Write Latency Set “A” should be used. When MR2 OP[6]=1_B, then Write Latency Set “B” should be used.
4) The programmed value of nWR is the number of clock cycles the LPDDR4X-SDRAM device uses to determine the starting point of an internal Precharge operation after a Write burst with AP (auto precharge). It is determined by RU(tWR/tCK).
5) The programmed value of nRTP is the number of clock cycles the LPDDR4X-SDRAM device uses to determine the starting point of an internal Precharge operation after a Read burst with AP (auto precharge). It is determined by RU(tRTP/tCK).
6) nRTP shown in this table is valid for BL16 only. For BL32, the SDRAM will add 8 clocks to the nRTP value before starting a precharge.

[Table 4] Burst Sequence for READ

BL	BT	C4	C3	C2	C1	C0	Burst Cycle Number and Burst Address Sequence																														
							1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32
16	seq	V	0_B	0_B	0_B	0_B	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F																
		V	0_B	1_B	0_B	0_B	4	5	6	7	8	9	A	B	C	D	E	F	0	1	2	3																
		V	1_B	0_B	0_B	0_B	8	9	A	B	C	D	E	F	0	1	2	3	4	5	6	7																
		V	1_B	1_B	0_B	0_B	C	D	E	F	0	1	2	3	4	5	6	7	8	9	A	B																
32	seq	0_B	0_B	0_B	0_B	0_B	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F	10	11	12	13	14	15	16	17	18	19	1A	1B	1C	1D	1E	1F
		0_B	0_B	1_B	0_B	0_B	4	5	6	7	8	9	A	B	C	D	E	F	0	1	2	3	14	15	16	17	18	19	1A	1B	1C	1D	1E	1F	10	11	12	13
		0_B	1_B	0_B	0_B	0_B	8	9	A	B	C	D	E	F	0	1	2	3	4	5	6	7	18	19	1A	1B	1C	1D	1E	1F	10	11	12	13	14	15	16	17
		0_B	1_B	1_B	0_B	0_B	C	D	E	F	0	1	2	3	4	5	6	7	8	9	A	B	1C	1D	1E	1F	10	11	12	13	14	15	16	17	18	19	1A	1B
		1_B	0_B	0_B	0_B	0_B	10	11	12	13	14	15	16	17	18	19	1A	1B	1C	1D	1E	1F	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
		1_B	0_B	1_B	0_B	0_B	14	15	16	17	18	19	1A	1B	1C	1D	1E	1F	10	11	12	13	4	5	6	7	8	9	A	B	C	D	E	F	0	1	2	3
		1_B	1_B	0_B	0_B	0_B	18	19	1A	1B	1C	1D	1E	1F	10	11	12	13	14	15	16	17	8	9	A	B	C	D	E	F	0	1	2	3	4	5	6	7
		1_B	1_B	1_B	0_B	0_B	1C	1D	1E	1F	10	11	12	13	14	15	16	17	18	19	1A	1B	C	D	E	F	0	1	2	3	4	5	6	7	8	9	A	B

NOTE :
1) C0-C1 are assumed to be '0', and are not transmitted on the command bus.
2) The starting burst address is on 64-bit (4n) boundaries.

[Table 5] Burst Sequence for Write

BL	BT	C4	C3	C2	C1	C0	Burst Cycle Number and Burst Address Sequence																														
							1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32
16	seq	V	0	0	0	0	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F																
32	seq	0	0	0	0	0	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F	10	11	12	13	14	15	16	17	18	19	1A	1B	1C	1D	1E	1F

NOTE :
1) C0-C1 are assumed to be '0', and are not transmitted on the command bus.
2) The starting address is on 256-bit (16n) boundaries for Burst length 16.
3) The starting address is on 512-bit (32n) boundaries for Burst length 32.
4) C2-C3 shall be set to '0' for all Write operations.

MR2_Device Feature 2 (MA<7:0> = 02_H):

OP7	OP6	OP5	OP4	OP3	OP2	OP1	OP0
WR Lev	WLS	WL			RL		

Function	Register Type	Operand	Data	Notes
RL (Read latency)	Write only	OP[2:0]	**RL & nRTP for DBI-RD Disabled (MR3 OP[6]=0_B)** **000_B**: RL=6, nRTP=8 (Default) **001_B**: RL=10, nRTP=8 **010_B**: RL=14, nRTP=8 **011_B**: RL=20, nRTP=8 **100_B**: RL=24, nRTP=10 **101_B**: RL=28, nRTP=12 **110_B**: RL=32, nRTP=14 **111_B**: RL=36, nRTP=16 **RL & nRTP for DBI-RD Enabled (MR3 OP[6]=1_B)** **000_B**: RL= 6, nRTP=8 **001_B**: RL= 12, nRTP=8 **010_B**: RL= 16, nRTP=8 **011_B**: RL= 22, nRTP=8 **100_B**: RL= 28, nRTP=10 **101_B**: RL= 32, nRTP=12 **110_B**: RL= 36, nRTP=14 **111_B**: RL= 40, nRTP=16	1,3,4
WL (Write latency)		OP[5:3]	**WL Set "A" (MR2 OP[6]=0_B)** **000_B**: WL=4 (Default) **001_B**: WL=6 **010_B**: WL=8 **011_B**: WL=10 **100_B**: WL=12 **101_B**: WL=14 **110_B**: WL=16 **111_B**: WL=18 **WL Set "B" (MR2 OP[6]=1_B)** **000_B**: WL=4 **001_B**: WL=8 **010_B**: WL=12 **011_B**: WL=18 **100_B**: WL=22 **101_B**: WL=26 **110_B**: WL=30 **111_B**: WL=34	1,3,4
WLS (Write latency set)		OP[6]	**0_B**: WL Set "A" (default) **1_B**: WL Set "B"	1,3,4
WR Leveling		OP[7]	**0_B**: Disabled (default) **1_B**: Enabled	2

NOTE :
1) See Latency Code Frequency Table for allowable frequency ranges for RL/WL/nWR/nRTP.
2) After a MRW to set the Write Leveling Enable bit (OP[7]=1_B), the LPDDR4X-SDRAM device remains in the MRW state until another MRW command clears the bit (OP[7]=0_B). No other commands are allowed until the Write Leveling Enable bit is cleared.
3) There are two physical registers assigned to each bit of this MR parameter, designated set point 0 and set point 1. Only the registers for the set point determined by the state of the FSP-WR bit (MR13 OP[6]) will be written to with an MRW command to this MR address, or read from with an MRR command to this address.
4) There are two physical registers assigned to each bit of this MR parameter, designated set point 0 and set point 1. The device will operate only according to the values stored in the registers for the active set point, i.e., the set point determined by the state of the FSP-OP bit (MR13 OP[7]). The values in the registers for the inactive set point will be ignored by the device, and may be changed without affecting device operation.

MR3_I/O Configuration 1 (MA<7:0> = 03_H):

OP7	OP6	OP5	OP4	OP3	OP2	OP1	OP0
DBI-WR	DBI-RD	PDDS			PPRP	WR PST	PU-CAL

Function	Register Type	Operand	Data	Notes
PU-Cal (Pull-up Calibration Point)	Write only	OP[0]	**0_B**: V_{DDQ}×0.6 **1_B**: V_{DDQ}×0.5 (default)	1,4
WR PST (WR Post-Amble Length)		OP[1]	**0_B**: WR Post-amble = 0.5×tCK (default) **1_B**: WR Post-amble = 1.5×tCK (Vendor specific function)	2,3,5
Post Package Repair Protection		OP[2]	**0_B**: PPR protection disabled (default) **1_B**: PPR protection enabled	6
PDDS (Pull-Down Drive Strength)		OP[5:3]	**000_B**: RFU **001_B**: RZQ/1 **010_B**: RZQ/2 **011_B**: RZQ/3 **100_B**: RZQ/4 **101_B**: RZQ/5 **110_B**: RZQ/6 (default) **111_B**: Reserved	1,2,3
DBI-RD (DBI-Read Enable)		OP[6]	**0_B**: Disabled (default) **1_B**: Enabled	2,3
DBI-WR (DBI-Write Enable)		OP[7]	**0_B**: Disabled (default) **1_B**: Enabled	2,3

NOTE :
1) All values are “typical”. The actual value after calibration will be within the specified tolerance for a given voltage and temperature. Re-calibration may be required as voltage and temperature vary.
2) There are two physical registers assigned to each bit of this MR parameter, designated set point 0 and set point 1. Only the registers for the set point determined by the state of the FSP-WR bit (MR13 OP[6]) will be written to with an MRW command to this MR address, or read from with an MRR command to this address.
3) There are two physical registers assigned to each bit of this MR parameter, designated set point 0 and set point 1. The device will operate only according to the values stored in the registers for the active set point, i.e., the set point determined by the state of the FSP-OP bit (MR13 OP[7]). The values in the registers for the inactive set point will be ignored by the device, and may be changed without affecting device operation.
4) For dual channel devices, PU-CAL setting is required as the same value for both Ch.A and Ch.B before issuing ZQ Cal start command.
5) Refer to the supplier data sheet for vender specific function. 1.5×tCK apply > 1.6GHz clock.
6) If MR3 OP[2] is set to 1b then PPR protection mode is enabled. The PPR Protection bit is a sticky bit and can only be set to 0b by a power on reset.
MR4 OP[4] controls entry to PPR Mode. If PPR protection is enabled then DRAM will not allow writing of 1 to MR4 OP[4].

MR4_Refresh rate (MA<7:0> = 04_H)

OP7	OP6	OP5	OP4	OP3	OP2	OP1	OP0
TUF	Thermal Offset		PPRE	SR Abort	Refresh Rate		

Function	Register Type	Operand	Data	Notes
Refresh Rate	Read-only	OP[2:0]	**000_B**: SDRAM Low temperature operating limit exceeded **001_B**: 4x refresh **010_B**: 2x refresh **011_B**: 1x refresh (default) **100_B**: 0.5x refresh **101_B**: 0.25x refresh, no de-rating **110_B**: 0.25x refresh, with de-rating **111_B**: SDRAM High temperature operating limit exceeded	1,2,3,4 7,8,9
SR Abort (Self Refresh Abort)	Write-only	OP[3]	**0_B** : Disable (default) **1_B** : Enable	9,11
PPRE (Post-package repair entry/exit)	Write-only	OP[4]	**0_B**: Exit PPR mode (default) **1_B**: Enter PPR mode	5,9
Thermal Offset (Vender Specific Function)	Write-only	OP[6:5]	**00_B**: No offset, 0~5°C gradient (default) **01_B**: 5°C offset, 5~10°C gradient **10_B**: 10°C offset, 10~15°C gradient **11_B**: Reserved	10
TUF (Temperature Update Flag)	Read-only	OP[7]	**0_B**: No change in OP[2:0] since last MR4 read (default) **1_B**: Change in OP[2:0] since last MR4 read	6,7,8

NOTE :
1) The refresh rate for each MR4 OP[2:0] setting applies to tREFI, tREFIpb and tREFW. OP[2:0]=011_B corresponds to a device temperature of 85°C. Other values require either a longer (2x, 4x) refresh interval at lower temperatures, or a shorter (0.5x, 0.25x) refresh interval at higher temperatures. If OP[2]=1_B, the device temperature is greater than 85°C.
2) At higher temperatures (>85°C), AC timing derating may be required. If derating is required the LPDDR4X-SDRAM will set OP[2:0]=110_B. See derating timing requirements in the AC Timing section.
3) DRAM vendors may or may not report all of the possible settings over the operating temperature range of the device. Each vendor guarantees that their device will work at any temperature within the range using the refresh interval requested by their device.
4) The device may not operate properly when OP[2:0]=000_B or 111_B.
5) Post-package repair can be entered or exited by writing to OP[4].
6) When OP[7]=1_B, the refresh rate reported in OP[2:0] has changed since the last MR4 read. A mode register read from MR4 will reset OP[7] to ‘0’.
7) OP[7]=0_B at power-up. OP[2:0] bits are valid after initialization sequence (Te).
8) See the section on “Temperature Sensor” for information on the recommended frequency of reading MR4.
9) OP[6:3] bits that can be written in this register. All other bits will be ignored by the DRAM during a MRW to this register.
10) Refer to the supplier data sheet for vender specific function.
11) Self refresh abort feature is available for higher density devices starting with 12Gb device.

MR5_Basic Configuration 1 (MA<7:0> = 05_H):

OP7	OP6	OP5	OP4	OP3	OP2	OP1	OP0
LPDDR4X Manufacturer ID							

Function	Register Type	Operand	Data	Notes
LPDDR4X Manufacturer ID	Read-only	OP[7:0]	**0000 0001_B** : Samsung	

MR6_Basic Configuration 2 (MA<7:0> = 06_H):

OP7	OP6	OP5	OP4	OP3	OP2	OP1	OP0
Revision ID-1							

Function	Register Type	Operand	Data	Notes
LPDDR4X Revision ID-1	Read-only	OP[7:0]	**0000 0110_B** : G-version	1

NOTE :
1) MR6 is vendor specific.

MR7_Basic Configuration 3 (MA<7:0> = 07_H):

OP7	OP6	OP5	OP4	OP3	OP2	OP1	OP0
Revision ID-2							

Function	Register Type	Operand	Data	Notes
LPDDR4X Revision ID-2	Read-only	OP[7:0]	**0001 0000_B : B-version**	1

NOTE :
1) MR7 is vendor specific.

MR8_Basic Configuration 4 (MA<7:0> = 08_H) :

OP7	OP6	OP5	OP4	OP3	OP2	OP1	OP0
I/O width		Density				Type	

Function	Register Type	Operand	Data	Notes
Type	Read-only	OP[1:0]	**00_B**: LPDDR4 S16 SDRAM (16n pre-fetch) Standard VDDQ(1.1V) only **10_B**: LPDDR4X S16 SDRAM (16n pre-fetch) LOW VDDQ(0.6V) support only **All Others**: Reserved	
Density		OP[5:2]	**0000_B**: 4Gb dual channel die / 2Gb single channel die **0001_B**: 6Gb dual channel die / 3Gb single channel die **0010_B**: 8Gb dual channel die / 4Gb single channel die **0011_B**: 12Gb dual channel die / 6Gb single channel die **0100_B**: 16Gb dual channel die / 8Gb single channel die **0101_B**: 24Gb dual channel die / 12Gb single channel die **0110_B**: 32Gb dual channel die / 16Gb single channel die **All Others**: Reserved	
I/O width		OP[7:6]	**00_B:** x16 (per channel) **All Others** : Reserved	

MR9_Test Mode (MA<7:0> = 09_H):

OP7	OP6	OP5	OP4	OP3	OP2	OP1	OP0
Vendor-specific Test Register							

NOTE :
1) Only 00_H should be written to this register.

MR10_IO Calibration (MA<7:0> = $0A_H$):

OP7	OP6	OP5	OP4	OP3	OP2	OP1	OP0
(RFU)							ZQ-Reset

Function	Register Type	Operand	Data	Notes
ZQ-Reset	Write-only	OP[0]	**0_B**: Normal Operation (Default) **1_B**: ZQ Reset	1,2

NOTE :
1) See the AC Timing tables for calibration latency and timing
2) If the ZQ-pin is connected to V_{DDQ} through RZQ, either the ZQ calibration function or default calibration (via ZQ-Reset) is supported. If the ZQ-pin is connected to V_{SS}, the device operates with default calibration, and ZQ calibration commands are ignored. In both cases, the ZQ connection shall not change after power is applied to the device.

MR11_ODT Feature (MA<7:0> = $0B_H$):

OP7	OP6	OP5	OP4	OP3	OP2	OP1	OP0
(RFU)	CA ODT			(RFU)	DQ ODT		

Function	Register Type	Operand	Data	Notes
DQ ODT (DQ Bus Receiver On-Die-Termination)	Write-only	OP[2:0]	**000_B**: Disable (Default) **001_B**: RZQ/1 **010_B**: RZQ/2 **011_B**: RZQ/3 **100_B**: RZQ/4 **101_B**: RZQ/5 **110_B**: RZQ/6 **111_B**: RFU	1,2,3
CA ODT (CA Bus Receiver On-Die-Termination)		OP[6:4]	**000_B**: Disable (Default) **001_B**: RZQ/1 **010_B**: RZQ/2 **011_B**: RZQ/3 **100_B**: RZQ/4 **101_B**: RZQ/5 **110_B**: RZQ/6 **111_B**: RFU	1,2,3

NOTE :
1) All values are “typical”. The actual value after calibration will be within the specified tolerance for a given voltage and temperature. Re.calibration may be required as voltage and temperature vary.
2) There are two physical registers assigned to each bit of this MR parameter, designated set point 0 and set point 1. Only the registers for the set point determined by the state of the FSP-WR bit (MR13 OP[6]) will be written to with an MRW command to this MR address, or read from with an MRR command to this address.
3) There are two physical registers assigned to each bit of this MR parameter, designated set point 0 and set point 1. The device will operate only according to the values stored in the registers for the active set point, i.e., the set point determined by the state of the FSP-OP bit (MR13 OP[7]). The values in the registers for the inactive set point will be ignored by the device, and may be changed without affecting device operation.

MR12_$V_{REF(CA)}$ Setting/Range (MA<7:0> = $0C_H$):

OP7	OP6	OP5	OP4	OP3	OP2	OP1	OP0
(RFU)	VR-CA	$V_{REF(CA)}$					

Function	Register Type	Operand	Data	Notes
$V_{REF(CA)}$ ($V_{REF(CA)}$ Setting)	Read/Write	OP[5:0]	**000000$_B$**: -- Thru . **110010$_B$**: See table below **All Others**: Reserved	1,2,3,5,6
VR-CA ($V_{REF(CA)}$ Range)		OP[6]	**0$_B$**: $V_{REF(CA)}$ Range[0] enabled **1$_B$**: $V_{REF(CA)}$ Range[1] enabled (default)	1,2,4,5,6

NOTE :
1) This register controls the $V_{REF(CA)}$ levels.
2) A read to this register places the contents of OP[7:0] on DQ[7:0]. Any RFU bits and unused DQ's shall be set to '0'. See the section on MRR Operation.
3) A write to OP[5:0] sets the internal $V_{REF(CA)}$ level for FSP[0] when MR13 OP[6]=0_B, or sets FSP[1] when MR13 OP[6]=1_B. The time required for $V_{REF(CA)}$ to reach the set level depends on the step size from the current level to the new level. See the section on $V_{REF(CA)}$ training for more information.
4) A write to OP[6] switches the LPDDR4X-SDRAM between two internal $V_{REF(CA)}$ ranges. The range (Range[0] or Range[1]) must be selected when setting the $V_{REF(CA)}$ register. The value, once set, will be retained until overwritten, or until the next power-on or RESET event.
5) There are two physical registers assigned to each bit of this MR parameter, designated set point 0 and set point 1. Only the registers for the set point determined by the state of the FSP-WR bit (MR13 OP[6]) will be written to with an MRW command to this MR address, or read from with an MRR command to this address.
6) There are two physical registers assigned to each bit of this MR parameter, designated set point 0 and set point 1. The device will operate only according to the values stored in the registers for the active set point, i.e., the set point determined by the state of the FSP-OP bit (MR13 OP[7]). The values in the registers for the inactive set point will be ignored by the device, and may be changed without affecting device operation.

[Table 6] V_{REF} Settings for Range[0] and Range[1]

Function	Operand	Range[0] Values (%of V_{DDQ})		Range[1] Values (%of V_{DDQ})		Notes
V_{REF} Settings for MR12	OP[5:0]	**000000$_B$**: 15.0%	**011010$_B$**: 30.5%	**000000$_B$**: 32.9%	**011010$_B$**: 48.5%	1,2,3
		000001$_B$: 15.6%	**011011$_B$**: 31.1%	**000001$_B$**: 33.5%	**011011$_B$**: 49.1%	
		000010$_B$: 16.2%	**011100$_B$**: 31.7%	**000010$_B$**: 34.1%	**011100$_B$**: 49.7%	
		000011$_B$: 16.8%	**011101$_B$**: 32.3%	**000011$_B$**: 34.7%	**011101$_B$**: 50.3% (Default)	
		000100$_B$: 17.4%	**011110$_B$**: 32.9%	**000100$_B$**: 35.3%	**011110$_B$**: 50.9%	
		000101$_B$: 18.0%	**011111$_B$**: 33.5%	**000101$_B$**: 35.9%	**011111$_B$**: 51.5%	
		000110$_B$: 18.6%	**100000$_B$**: 34.1%	**000110$_B$**: 36.5%	**100000$_B$**: 52.1%	
		000111$_B$: 19.2%	**100001$_B$**: 34.7%	**000111$_B$**: 37.1%	**100001$_B$**: 52.7%	
		001000$_B$: 19.8%	**100010$_B$**: 35.3%	**001000$_B$**: 37.7%	**100010$_B$**: 53.3%	
		001001$_B$: 20.4%	**100011$_B$**: 35.9%	**001001$_B$**: 38.3%	**100011$_B$**: 53.9%	
		001010$_B$: 21.0%	**100100$_B$**: 36.5%	**001010$_B$**: 38.9%	**100100$_B$**: 54.5%	
		001011$_B$: 21.6%	**100101$_B$**: 37.1%	**001011$_B$**: 39.5%	**100101$_B$**: 55.1%	
		001100$_B$: 22.2%	**100110$_B$**: 37.7%	**001100$_B$**: 40.1%	**100110$_B$**: 55.7%	
		001101$_B$: 22.8%	**100111$_B$**: 38.3%	**001101$_B$**: 40.7%	**100111$_B$**: 56.3%	
		001110$_B$: 23.4%	**101000$_B$**: 38.9%	**001110$_B$**: 41.3%	**101000$_B$**: 56.9%	
		001111$_B$: 24.0%	**101001$_B$**: 39.5%	**001111$_B$**: 41.9%	**101001$_B$**: 57.5%	
		010000$_B$: 24.6%	**101010$_B$**: 40.1%	**010000$_B$**: 42.5%	**101010$_B$**: 58.1%	
		010001$_B$: 25.1%	**101011$_B$**: 40.7%	**010001$_B$**: 43.1%	**101011$_B$**: 58.7%	
		010010$_B$: 25.7%	**101100$_B$**: 41.3%	**010010$_B$**: 43.7%	**101100$_B$**: 59.3%	
		010011$_B$: 26.3%	**101101$_B$**: 41.9%	**010011$_B$**: 44.3%	**101101$_B$**: 59.9%	
		010100$_B$: 26.9%	**101110$_B$**: 42.5%	**010100$_B$**: 44.9%	**101110$_B$**: 60.5%	
		010101$_B$: 27.5%	**101111$_B$**: 43.1%	**010101$_B$**: 45.5%	**101111$_B$**: 61.1%	
		010110$_B$: 28.1%	**110000$_B$**: 43.7%	**010110$_B$**: 46.1%	**110000$_B$**: 61.7%	
		010111$_B$: 28.7%	**110001$_B$**: 44.3%	**010111$_B$**: 46.7%	**110001$_B$**: 62.3%	
		011000$_B$: 29.3%	**110010$_B$**: 44.9%	**011000$_B$**: 47.3%	**110010$_B$**: 62.9%	
		011001$_B$: 29.9%	**All Others**: Reserved	**011001$_B$**: 47.9%	**All Others**: Reserved	

NOTE:
1) These values may be used for MR12 OP[5:0] to set the $V_{REF(CA)}$ levels in the LPDDR4X-SDRAM.
2) The range may be selected in the MR12 register by setting OP[6] appropriately.
3) The MR12 registers represents either FSP[0] or FSP[1]. Two frequency-set-points each for CA and DQ are provided to allow for faster switching between terminated and unterminated operation, or between different high-frequency setting which may use different terminations values.

MR13_CBT,RPT,VRO,VRCG,RRO,DM_DIS,FSP-WR, FSP-OP (MA<7:0> = $0D_H$):

OP7	OP6	OP5	OP4	OP3	OP2	OP1	OP0
FSP-OP	FSP-WR	DMD	RRO	VRCG	VRO	RPT	CBT

Function	Register Type	Operand	Data	Notes
CBT (Command Bus Training)	Write-only	OP[0]	0_B: Normal Operation (default) 1_B: Command Bus Training Mode Enabled	1
RPT (Read Preamble Training Mode)		OP[1]	0_B: Disable (default) 1_B: Enable	
VRO (V_{REF} Output)		OP[2]	0_B: Normal operation (default) 1_B: Output the $V_{REF(CA)}$ and $V_{REF(DQ)}$ values on DQ bits	2
VRCG (V_{REF} Current Generator)		OP[3]	0_B: Normal operation (default) 1_B: V_{REF} fast response (high current) mode	3
RRO (Refresh Rate Option)		OP[4]	0_B: Disable codes 001 and 010 in MR4 OP[2:0] 1_B: Enable all codes in MR4 OP[2:0]	4,5
DMD (Data Mask Disable)		OP[5]	0_B: Data Mask Operation Enabled (default) 1_B: Data Mask Operation Disabled	6
FSP-WR (Frequency Set Point Write/Read)		OP[6]	0_B: Frequency-Set-Point [0] (default) 1_B: Frequency-Set-Point [1]	7
FSP-OP (Frequency Set Point Operation Mode)		OP[7]	0_B: Frequency-Set-Point [0] (default) 1_B: Frequency-Set-Point [1]	8

NOTE :

1) A write to set OP[0]=1_B causes the LPDDR4X-SDRAM to enter the Command Bus training mode. When OP[0]=1_B and CKE goes LOW, commands are ignored and the contents of CA[5:0] are mapped to the DQ bus. CKE must be brought HIGH before doing a MRW to clear this bit (OP[0]=0_B) and return to normal operation. See the Command Bus Training section for more information.
2) When set, the LPDDR4X-SDRAM will output the $V_{REF(CA)}$ and $V_{REF(DQ)}$ voltages on DQ pins. Only the "active" frequency-set-point, as defined by MR13 OP[7], will be output on the DQ pins. This function allows an external test system to measure the internal VREF levels. The DQ pins used for V_{REF} output are vendor specific.
3) When OP[3]=1_B, the V_{REF} circuit uses a high-current mode to improve V_{REF} settling time.
4) MR13 OP[4] RRO bit is valid only when MR0 OP[0]= 1_B. For LPDDR4X devices with MR0 OP[0] = 0_B, MR4 OP[2:0] bits are not dependent on MR13 OP4.
5) When OP[4] = 0_B, only 001_B and 010_B in MR4 OP[2:0] are disabled. LPDDR4X devices must report 011_B instead of 001_B or 010_B in this case. Controller should follow the refresh mode reported by MR4 OP[2:0], regardless of RRO setting. TCSR function does not depend on RRO setting.
6) When enabled (OP[5]=0_B) data masking is enabled for the device. When disabled (OP[5]=1_B), masked write command is illegal. See LPDDR4X Data Mask (DM) and Data Bus Inversion (DBIdc) Function in operation timing datasheet.
7) FSP-WR determines which frequency-set-point registers are accessed with MRW commands for the following functions such as $V_{REF(CA)}$ Setting, $V_{REF(CA)}$ Range, $V_{REF(DQ)}$ Setting, $V_{REF(DQ)}$ Range. For more information, refer to Frequency Set Point section in operations and timing spec.
8) FSP-OP determines which frequency-set-point register values are currently used to specify device operation for the following functions such as $V_{REF(CA)}$ Setting, $V_{REF(CA)}$ Range, $V_{REF(DQ)}$ Setting, $V_{REF(DQ)}$ Range. For more information, refer to Frequency Set Point section in operations and timing spec.

MR14_$V_{REF(DQ)}$ Setting/Range (MA<7:0> = $0E_H$):

OP7	OP6	OP5	OP4	OP3	OP2	OP1	OP0
(RFU)	VR(DQ)	$V_{REF(DQ)}$					

Function	Register Type	Operand	Data	Notes
$V_{REF(DQ)}$ ($V_{REF(DQ)}$ Setting)	Read/Write	OP[5:0]	**000000$_B$**: -- Thru . **110010$_B$**: See table below **All Others**: Reserved	1,2,3,5,6
$V_{REF(DQ)}$ ($V_{REF(DQ)}$ Range)		OP[6]	**0$_B$**: $V_{REF(DQ)}$ Range [0] enabled **1$_B$**: $V_{REF(DQ)}$ Range [1] enabled (default)	1,2,4,5,6

NOTE :
1) This register controls the $V_{REF(DQ)}$ levels for Frequency-Set-Point[1:0]. Values from either VR(DQ)[0] or VR(DQ)[1] may be selected by setting OP[6] appropriately.
2) A read (MRR) to this register places the contents of OP[7:0] on DQ[7:0]. Any RFU bits and unused DQ's shall be set to '0'. See the section on MRR Operation.
3) A write to OP[5:0] sets the internal $V_{REF(DQ)}$ level for FSP[0] when MR13 OP[6]=0_B, or sets FSP[1] when MR13 OP[6]=1_B. The time required for $V_{REF(DQ)}$ to reach the set level depends on the step size from the current level to the new level. See the section on $V_{REF(DQ)}$ training for more information.
4) A write to OP[6] switches the LPDDR4X-SDRAM between two internal $V_{REF(DQ)}$ ranges. The range (Range[0] or Range[1]) must be selected when setting the $V_{REF(DQ)}$ register. The value, once set, will be retained until overwritten, or until the next power-on or RESET event.
5) There are two physical registers assigned to each bit of this MR parameter, designated set point 0 and set point 1. Only the registers for the set point determined by the state of the FSP-WR bit (MR13 OP[6]) will be written to with an MRW command to this MR address, or read from with an MRR command to this address.
6) There are two physical registers assigned to each bit of this MR parameter, designated set point 0 and set point 1. The device will operate only according to the values stored in the registers for the active set point, i.e., the set point determined by the state of the FSP-OP bit (MR13 OP[7]). The values in the registers for the inactive set point will be ignored by the device, and may be changed without affecting device operation.

[Table 7] VREF Settings for Range[0] and Range[1]

Function	Operand	Range[0] Values (%of VDDQ)		Range[1] Values (%of VDDQ)		Notes
VREF Settings for MR14	OP[5:0]	**000000$_B$**: 15.0%	**011010$_B$**: 30.5%	**000000$_B$**: 32.9%	**011010$_B$**: 48.5%	1,2,3
		000001$_B$: 15.6%	**011011$_B$**: 31.1%	**000001$_B$**: 33.5%	**011011$_B$**: 49.1%	
		000010$_B$: 16.2%	**011100$_B$**: 31.7%	**000010$_B$**: 34.1%	**011100$_B$**: 49.7%	
		000011$_B$: 16.8%	**011101$_B$**: 32.3%	**000011$_B$**: 34.7%	**011101$_B$**: 50.3% (Default)	
		000100$_B$: 17.4%	**011110$_B$**: 32.9%	**000100$_B$**: 35.3%	**011110$_B$**: 50.9%	
		000101$_B$: 18.0%	**011111$_B$**: 33.5%	**000101$_B$**: 35.9%	**011111$_B$**: 51.5%	
		000110$_B$: 18.6%	**100000$_B$**: 34.1%	**000110$_B$**: 36.5%	**100000$_B$**: 52.1%	
		000111$_B$: 19.2%	**100001$_B$**: 34.7%	**000111$_B$**: 37.1%	**100001$_B$**: 52.7%	
		001000$_B$: 19.8%	**100010$_B$**: 35.3%	**001000$_B$**: 37.7%	**100010$_B$**: 53.3%	
		001001$_B$: 20.4%	**100011$_B$**: 35.9%	**001001$_B$**: 38.3%	**100011$_B$**: 53.9%	
		001010$_B$: 21.0%	**100100$_B$**: 36.5%	**001010$_B$**: 38.9%	**100100$_B$**: 54.5%	
		001011$_B$: 21.6%	**100101$_B$**: 37.1%	**001011$_B$**: 39.5%	**100101$_B$**: 55.1%	
		001100$_B$: 22.2%	**100110$_B$**: 37.7%	**001100$_B$**: 40.1%	**100110$_B$**: 55.7%	
		001101$_B$: 22.8%	**100111$_B$**: 38.3%	**001101$_B$**: 40.7%	**100111$_B$**: 56.3%	
		001110$_B$: 23.4%	**101000$_B$**: 38.9%	**001110$_B$**: 41.3%	**101000$_B$**: 56.9%	
		001111$_B$: 24.0%	**101001$_B$**: 39.5%	**001111$_B$**: 41.9%	**101001$_B$**: 57.5%	
		010000$_B$: 24.6%	**101010$_B$**: 40.1%	**010000$_B$**: 42.5%	**101010$_B$**: 58.1%	
		010001$_B$: 25.1%	**101011$_B$**: 40.7%	**010001$_B$**: 43.1%	**101011$_B$**: 58.7%	
		010010$_B$: 25.7%	**101100$_B$**: 41.3%	**010010$_B$**: 43.7%	**101100$_B$**: 59.3%	
		010011$_B$: 26.3%	**101101$_B$**: 41.9%	**010011$_B$**: 44.3%	**101101$_B$**: 59.9%	
		010100$_B$: 26.9%	**101110$_B$**: 42.5%	**010100$_B$**: 44.9%	**101110$_B$**: 60.5%	
		010101$_B$: 27.5%	**101111$_B$**: 43.1%	**010101$_B$**: 45.5%	**101111$_B$**: 61.1%	
		010110$_B$: 28.1%	**110000$_B$**: 43.7%	**010110$_B$**: 46.1%	**110000$_B$**: 61.7%	
		010111$_B$: 28.7%	**110001$_B$**: 44.3%	**010111$_B$**: 46.7%	**110001$_B$**: 62.3%	
		011000$_B$: 29.3%	**110010$_B$**: 44.9%	**011000$_B$**: 47.3%	**110010$_B$**: 62.9%	
		011001$_B$: 29.9%	**All Others**: Reserved	**011001$_B$**: 47.9%	**All Others**: Reserved	

NOTE:
1) These values may be used for MR14 OP[5:0] to set the $V_{REF(DQ)}$ levels in the LPDDR4X-SDRAM.
2) The range may be selected in the MR14 register by setting OP[6] appropriately.
3) The MR14 registers represents either FSP[0] or FSP[1]. Two frequency-set-points each for CA and DQ are provided to allow for faster switching between terminated and unterminated operation, or between different high-frequency setting which may use different terminations values.

MR15_Lower-Byte Invert for DQ Calibration (MA<7:0> = $0F_H$):

OP7	OP6	OP5	OP4	OP3	OP2	OP1	OP0
Lower-Byte Invert Register for DQ Calibration							

Function	Register Type	Operand	Data	Notes
Lower-Byte Invert for DQ Calibration	Write-only	OP[7:0]	The following values may be written for any operand OP[7:0], and will be applied to the corresponding DQ locations DQ[7:0] within a byte lane: **0_B**: Do not invert **1_B**: Invert the DQ Calibration patterns in MR32 and MR40 Default value for OP[7:0]=55_H	1,2,3

NOTE :
1) This register will invert the DQ Calibration pattern found in MR32 and MR40 for any single DQ, or any combination of DQ's. Example: If MR15 OP[7:0]=00010101_B, then the DQ Calibration patterns transmitted on DQ[7,6,5,3,1] will not be inverted, but the DQ Calibration patterns transmitted on DQ[4,2,0] will be inverted.
2) DMI[0] is not inverted, and always transmits the "true" data contained in MR32/MR40.
3) No Data Bus Inversion (DBI) function is enacted during DQ Read Calibration, even if DBI is enabled in MR3-OP[6].

[Table 8] MR15 Invert Register Pin Mapping

PIN	DQ0	DQ1	DQ2	DQ3	DMI0	DQ4	DQ5	DQ6	DQ7
MR15	OP0	OP1	OP2	OP3	NO-Invert	OP4	OP5	OP6	OP7

MR16_PASR_Bank Mask (MA<7:0> = 010_H):

OP7	OP6	OP5	OP4	OP3	OP2	OP1	OP0
PASR Bank Mask							

Function	Register Type	Operand	Data	Notes
Bank [7:0] Mask	Write-only	OP[7:0]	**0_B**: Bank Refresh Enabled (default) : Unmasked **1_B**: Bank Refresh disabled : Masked	1

OP	Bank Mask	8-Bank SDRAM
0	XXXXXXX1	Bank 0
1	XXXXXX1X	Bank 1
2	XXXXX1XX	Bank 2
3	XXXX1XXX	Bank 3
4	XXX1XXXX	Bank 4
5	XX1XXXXX	Bank 5
6	X1XXXXXX	Bank 6
7	1XXXXXXX	Bank 7

NOTE :
1) When a mask bit is asserted (OP[n]=1), refresh to that bank is disabled.
2) PASR bank-masking is on a per-channel basis. The two channels on the die may have different bank masking in dual channel devices.

MR17_PASR Segment Mask (MA<7:0> = 011_H):

OP7	OP6	OP5	OP4	OP3	OP2	OP1	OP0
PASR Segment Mask							

Function	Register Type	Operand	Data	Notes
PASR Segment Mask	Write-only	OP[7:0]	0_B: Segment Refresh enabled (default) 1_B: Segment Refresh disabled	

<table>
<tr><th rowspan="2">Segment</th><th rowspan="2">OP</th><th rowspan="2">Segment Mask</th><th>2Gb per channel</th><th>3Gb per channel</th><th>4Gb per channel</th><th>6Gb per channel</th><th>8Gb per channel</th><th>12Gb per channel</th><th>16Gb per channel</th></tr>
<tr><th>R13:11</th><th>R14:12</th><th>R14:12</th><th>R15:13</th><th>R15:13</th><th>R16:14</th><th>R16:14</th></tr>
<tr><td>0</td><td>0</td><td>XXXXXXX1</td><td colspan="7">000_B</td></tr>
<tr><td>1</td><td>1</td><td>XXXXXX1X</td><td colspan="7">001_B</td></tr>
<tr><td>2</td><td>2</td><td>XXXXX1XX</td><td colspan="7">010_B</td></tr>
<tr><td>3</td><td>3</td><td>XXXX1XXX</td><td colspan="7">011_B</td></tr>
<tr><td>4</td><td>4</td><td>XXX1XXXX</td><td colspan="7">100_B</td></tr>
<tr><td>5</td><td>5</td><td>XX1XXXXX</td><td colspan="7">101_B</td></tr>
<tr><td>6</td><td>6</td><td>X1XXXXXX</td><td>110_B</td><td rowspan="2">Not Allowed</td><td>110_B</td><td rowspan="2">Not Allowed</td><td>110_B</td><td rowspan="2">Not Allowed</td><td>110_B</td></tr>
<tr><td>7</td><td>7</td><td>1XXXXXXX</td><td>111_B</td><td>111_B</td><td>111_B</td><td>111_B</td></tr>
</table>

NOTE :
1) This table indicates the range of row addresses in each masked segment. "X" is don't care for a particular segment.
2) PASR segment-masking is on a per-channel basis. The two channels on the die may have different segment masking in dual channel devices.
3) For 3Gb, 6Gb, and 12Gb per channel densities, OP[7:6] must always be LOW (=00B).

MR18_IT-LSB (MA<7:0> = 12_H) :

OP7	OP6	OP5	OP4	OP3	OP2	OP1	OP0
DQS Oscillator Count-LSB							

Function	Register Type	Operand	Data	Notes
DQS Oscillator (WR Training DQS Oscillator)	Read-only	OP[7:0]	0-255 LSB DRAM DQS Oscillator Count	1,2,3

NOTE :
1) MR18 reports the LSB bits of the DRAM DQS Oscillator count. The DRAM DQS Oscillator count value is used to train DQS to the DQ data valid window. The value reported by the DRAM in this mode register can be used by the memory controller to periodically adjust the phase of DQS relative to DQ.
2) Both MR18 and MR19 must be read (MRR) and combined to get the value of the DQS Oscillator count.
3) A new MPC [Start DQS Oscillator] should be issued to reset the contents of MR18/MR19.

MR19_IT-MSB (MA<7:0> = 13_H) :

OP7	OP6	OP5	OP4	OP3	OP2	OP1	OP0
DQS Oscillator Count-MSB							

Function	Register Type	Operand	Data	Notes
DQS Oscillator (WR Training DQS Oscillator)	Read-only	OP[7:0]	0-255 MSB DRAM DQS Oscillator Count	1,2,3

NOTE :
1) MR19 reports the MSB bits of the DRAM DQS Oscillator count. The DRAM DQS Oscillator count value is used to train DQS to the DQ data valid window. The value reported by the DRAM in this mode register can be used by the memory controller to periodically adjust the phase of DQS relative to DQ.
2) Both MR18 and MR19 must be read (MRR) and combined to get the value of the DQS Oscillator count.
3) A new MPC [Start DQS Oscillator] should be issued to reset the contents of MR18/MR19.

MR20_Upper-Byte Invert for DQ Calibration (MA<7:0> = 14_H):

OP7	OP6	OP5	OP4	OP3	OP2	OP1	OP0
Upper-Byte Invert Register for DQ Calibration							

Function	Register Type	Operand	Data	Notes
Upper-Byte Invert for DQ Calibration	Write-only	OP[7:0]	The following values may be written for any operand OP[7:0], and will be applied to the corresponding DQ locations DQ[15:8] within a byte lane: 0_B: Do not invert 1_B: Invert the DQ Calibration patterns in MR32 and MR40 Default value for OP[7:0] = 55_H	1,2

NOTE:
1) This register will invert the DQ Calibration pattern found in MR32 and MR40 for any single DQ, or any combination of DQ's. Example: If MR20 OP[7:0]=00010101B, then the DQ Calibration patterns transmitted on DQ[15,14,13,11,9] will not be inverted, but the DQ Calibration patterns transmitted on DQ[12,10,8] will be inverted.
2) DMI[1] is not inverted, and always transmits the "true" data contained in MR32/MR40.
3) No Data Bus Inversion (DBI) function is enacted during DQ Read Calibration, even if DBI is enabled in MR3-OP[6].

[Table 9] MR20 Invert Register Pin Mapping

PIN	DQ8	DQ9	DQ10	DQ11	DMI1	DQ12	DQ13	DQ14	DQ15
MR20	OP0	OP1	OP2	OP3	NO-Invert	OP4	OP5	OP6	OP7

MR21_(RFU) (MA<7:0> = 015_H):

MR22_ODT Feature (MA<7:0> = 16_H):

OP7	OP6	OP5	OP4	OP3	OP2	OP1	OP0
(RFU)		ODT-CA	ODT-CS	ODT-CK	SoC ODT		

Function	Register Type	Operand	Data	Notes
SoC ODT (Controller ODT Value for VOH calibration)	Write-only	OP[2:0]	**000_B**: Disable (Default) **001_B**: RZQ/1 (illegal if MR3 OP0=0_B) **010_B**: RZQ/2 **011_B**: RZQ/3 (illegal if MR3 OP0=0_B) **100_B**: RZQ/4 **101_B**: RZQ/5 (illegal if MR3 OP0=0_B) **110_B**: RZQ/6 (illegal if MR3 OP0=0_B) **111_B**: RFU	1,2,3
ODT-CK		OP[3]	ODT bond PAD is ignored **0_B** : ODT-CK Enable (Default) **1_B** : ODT-CK Disable	2,3,4
ODT-CS		OP[4]	ODT bond PAD is ignored **0_B**: ODT-CS Enable (Default) **1_B**: ODT-CS Disable	2,3,4
ODT-CA		OP[5]	ODT bond PAD is ignored **0_B**: ODT-CA Enable (default) **1_B**: ODT-CA Disabled	2,3,4

NOTE :
1) All values are “typical”.
2) There are two physical registers assigned to each bit of this MR parameter, designated set point 0 and set point 1. Only the registers for the set point determined by the state of the FSP-WR bit (MR13 OP[6]) will be written to with an MRW command to this MR address, or read from with an MRR command to this address.
3) There are two physical registers assigned to each bit of this MR parameter, designated set point 0 and set point 1. The device will operate only according to the values stored in the registers for the active set point, i.e., the set point determined by the state of the FSP-OP bit (MR13 OP[7]). The values in the registers for the inactive set point will be ignored by the device, and may be changed without affecting device operation.
4) LPDDR4X device ignore ODT bond PAD.

[Table 10] DRAM PU strength & SoC ODT relation

VOH CAL (MR3 OP0)		SoC ODT Value (MR22 OP[2:0])							
		000_B	001_B	010_B	011_B	100_B	101_B	110_B	111_B
0_B (VOH=VDDQ*3/5)	SoC ODT	Disable	illegal	RZQ/ 2=120ohm	illegal	RZQ/ 4=60ohm	illegal	illegal	RFU
	DRAM Pull-up	Default	N/A	RZQ/ 3=80ohm	N/A	RZQ/ 6=40ohm	N/A	N/A	RFU
1_B (VOH=VDDQ/2)	SoC ODT	Disable	RZQ/ 1=240ohm	RZQ/ 2=120ohm	RZQ/ 3=80ohm	RZQ/ 4=60ohm	RZQ/ 5=48ohm	RZQ/ 6=40ohm	RFU
	DRAM Pull-up	Default	RZQ/ 1=240ohm	RZQ/ 2=120ohm	RZQ/ 3=80ohm	RZQ/ 4=60ohm	RZQ/ 5=48ohm	RZQ/ 6=40ohm	RFU

A) There is no corresponding RZQ/x value for 001B,011B,101B,110B when MR3 OP0 = 0B to support 3/5 VDDQ VOH value.

MR23_DQS Interval Timer Run Time (MA<7:0> = 17_H):

OP7	OP6	OP5	OP4	OP3	OP2	OP1	OP0
DQS interval timer run time setting							

Function	Register Type	Operand	Data	Notes
DQS interval timer run time	Write-only	OP[7:0]	**00000000**$_B$: DQS interval timer stop via MPC Command (Default) **00000001**$_B$: DQS timer stops automatically at 16th clocks after timer start **00000010**$_B$: DQS timer stops automatically at 32nd clocks after timer start **00000011**$_B$: DQS timer stops automatically at 48th clocks after timer start **00000100**$_B$: DQS timer stops automatically at 64th clocks after timer start -------------- Thru ---------------------- **00111111**$_B$: DQS timer stops automatically at (63X16)th clocks after timer start **01XXXXXX**$_B$: DQS timer stops automatically at 2048th clocks after timer start **10XXXXXX**$_B$: DQS timer stops automatically at 4096th clocks after timer start **11XXXXXX**$_B$: DQS timer stops automatically at 8192nd clocks after timer start	1,2

NOTE :
1) MPC command with OP[6:0]=1001101_B (Stop DQS Interval Oscillator) stops DQS interval timer in case of MR23 OP[7:0] = 00000000_B.
2) MPC command with OP[6:0]=1001101_B (Stop DQS Interval Oscillator) is illegal with non-zero values in MR23 OP[7:0].

MR24_TRR (MA<7:0> = 18_H):

OP7	OP6	OP5	OP4	OP3	OP2	OP1	OP0
TRR Mode	TRR mode BAn			Unlim-ited MAC	MAC Value		

Function	Register Type	Operand	Data	Notes
MAC Value	Read-only	OP[2:0]	**000**$_B$: Unknown when bit OP3 =0 [1)] Unlimited when bit OP3=1 [2)] **001**$_B$: 700K **010**$_B$: 600K **011**$_B$: 500K **100**$_B$: 400K **101**$_B$: 300K **110**$_B$: 200K **111**$_B$: Reserved	
Unlimited MAC		OP[3]	**0**$_B$: OP[2:0] define MAC value **1**$_B$: Unlimited MAC value [2), 3)]	
TRR Mode BAn	Write-only	OP[6:4]	**000**$_B$: Bank 0 **001**$_B$: Bank 1 **010**$_B$: Bank 2 **011**$_B$: Bank 3 **100**$_B$: Bank 4 **101**$_B$: Bank 5 **110**$_B$: Bank 6 **111**$_B$: Bank 7	
TRR Mode		OP[7]	**0**$_B$: Disabled (default) **1**$_B$: Enabled	

NOTE :
1) Unknown means that the device is not tested for tMAC and pass/fail values are unknown.
2) There is no restriction to number of activates.
3) MR24 OP [2:0] is set to ZERO.

MR25_PPR Resources (MA<7:0> = 19_H):

Mode Register 25 contains one bit of readout per bank indicating that at least one resource is available for Post Package Repair programming.

OP7	OP6	OP5	OP4	OP3	OP2	OP1	OP0
Bank 7	Bank 6	Bank 5	Bank 4	Bank 3	Bank 2	Bank 1	Bank 0

Function	Register Type	Operand	Data	Notes
PPR Resource	Read-only	OP[7:0]	0_B : PPR Resource is not available 1_B : PPR Resource is available	

MR26-29_(RFU) (MA<7:0> = $1A_H$ - $1D_H$):

MR30_Reserved for Testing (MA<7:0> = $1E_H$):

OP7	OP6	OP5	OP4	OP3	OP2	OP1	OP0
Valid 0 or 1							

Function	Register Type	Operand	Data	Notes
SDRAM will ignore	Write-only	OP[7:0]	Don't care	1

NOTE :
1) This register is reserved for testing purposes. The logical data values written to OP[7:0] shall have no effect on SDRAM operation, however timings need to be observed as for any other MR access command.

MR31_(RFU) (MA<7:0> = $1F_H$):

MR32_DQ Calibration Pattern A (MA<7:0>=20_H):

OP7	OP6	OP5	OP4	OP3	OP2	OP1	OP0
DQ Calibration Pattern "A" (default = $5A_H$)							

Function	Register Type	Operand	Data	Notes
Return DQ Calibration Pattern MR32 + MR40	Write	OP[7:0]	X_B: An MPC command with OP[6:0]=1000011_B causes the device to return the DQ Calibration Pattern contained in this register and (followed by) the contents of MR40. A default pattern "$5A_H$" is loaded at power-up or RESET, or the pattern may be overwritten with a MRW to this register. The contents of MR15 and MR20 will invert the data pattern for a given DQ (See MR15 for more information)	

MR33:38_(Do Not Use) (MA<7:0> = 21_H-26_H):

MR39_Reserved for Testing (MA<7:0> = 27_H):

OP7	OP6	OP5	OP4	OP3	OP2	OP1	OP0
Valid 0 or 1							

Function	Register Type	Operand	Data	Notes
SDRAM will ignore	Write-only	OP[7:0]	Don't care	1

NOTE :
1) This register is reserved for testing purposes. The logical data values written to OP[7:0] shall have no effect on SDRAM operation, however timings need to be observed as for any other MR access command.

MR40_DQ Calibration Pattern B (MA<7:0>=28_H):

OP7	OP6	OP5	OP4	OP3	OP2	OP1	OP0
DQ Calibration Pattern “B” (default = $3C_H$)							

Function	Register Type	Operand	Data	Notes
Return DQ Calibration Pattern MR32 + MR40	Write-only	OP[7:0]	**X_B**: A default pattern “$3C_H$” is loaded at power-up or RESET, or the pattern may be overwritten with a MRW to this register. See MR32,for more information.	1,2,3

NOTE :

1) The pattern contained in MR40 is concatenated to the end of MR32 and transmitted on DQ[15:0] and DMI[1:0] when DQ Read Calibration is initiated via a MPC command. The pattern transmitted serially on each data lane, organized “little endian” such that the low-order bit in a byte is transmitted first. If the data pattern in MR40 is 27_H, then the first bit transmitted with be a ‘1’, followed by ‘1’, ‘1’, ‘0’, ‘0’, ‘1’, ‘0’, and ‘0’. The bit stream will be 00100111_B.
2) MR15 and MR20 may be used to invert the MR32/MR40 data patterns on the DQ pins. See MR15 and MR22 for more information. Data is never inverted on the DMI[1:0] pins.
3) The data pattern is not transmitted on the DMI[1:0] pins if DBI-RD is disabled via MR3 OP[6].
4) No Data Bus Inversion (DBI) function is enacted during DQ Read Calibration, even if DBI is enabled in MR3 OP[6].

MR41:47_ (Do Not Use)(MA<7:0> = 29_H-$2F_H$):

MR48:63_(RFU) (MA<7:0> = 30_H - $3F_H$) :

3.0 TRUTH TABLES

Operation or timing that is not specified is illegal, and after such an event, in order to guarantee proper operation, the LPDDR4X device must be reset or power-cycled and then restarted through the specified initialization sequence before normal operation can continue.
CKE signal has to be held High when the commands listed in the command truth table input.

[Table 11] Command truth table

	SDR Command Pins	SDR CA pins (6)							
SDRAM Command	**CS**	**CA0**	**CA1**	**CA2**	**CA3**	**CA4**	**CA5**	**CK_t edge**	**Notes**
Deselect (DES)	L	X						R1	1,2
Multi-Purpose Command (MPC)	H	L	L	L	L	L	OP6	R1	1,9
	L	OP0	OP1	OP2	OP3	OP4	OP5	R2	
Precharge (PRE) (Per Bank, All Bank)	H	L	L	L	L	H	AB	R1	1,2,3,4
	L	BA0	BA1	BA2	V	V	V	R2	
Refresh (REF) (Per Bank, All Bank)	H	L	L	L	H	L	AB	R1	1,2,3,4
	L	BA0	BA1	BA2	V	V	V	R2	
Self Refresh Entry (SRE)	H	L	L	L	H	H	V	R1	1,2
	L	V						R2	
Write-1 (WR-1)	H	L	L	H	L	L	BL	R1	1,2,3,6,7,9
	L	BA0	BA1	BA2	V	C9	AP	R2	
Self Refresh Exit (SRX)	H	L	L	H	L	H	V	R1	1,2
	L	V						R2	
Mask Write-1 (MWR-1)	H	L	L	H	H	L	L	R1	1,2,3,5,6,9
	L	BA0	BA1	BA2	V	C9	AP	R2	
RFU	H	L	L	H	H	H	V	R1	1,2
	L	V						R2	
Read-1 (RD-1)	H	L	H	L	L	L	BL	R1	1,2,3,6,7,9
	L	BA0	BA1	BA2	V	C9	AP	R2	
CAS-2 (Write-2, Mask Write-2, Read-2, MRR-2, MPC)	H	L	H	L	L	H	C8	R1	1,8,9
	L	C2	C3	C4	C5	C6	C7	R2	
RFU	H	L	H	L	H	L	V	R1	1,2
	L	V						R2	
RFU	H	L	H	L	H	H	V	R1	1,2
	L	V						R2	
Mode Register Write-1 (MRW-1)	H	L	H	H	L	L	OP7	R1	1,11
	L	MA0	MA1	MA2	MA3	MA4	MA5	R2	
Mode Register Write-2 (MRW-2)	H	L	H	H	L	H	OP6	R1	1,11
	L	OP0	OP1	OP2	OP3	OP4	OP5	R2	
Mode Register Read-1 (MRR-1)	H	L	H	H	H	L	V	R1	1,2,12
	L	MA0	MA1	MA2	MA3	MA4	MA5	R2	
RFU	H	L	H	H	H	H	V	R1	1,2
	L	V						R2	
Activate-1 (ACT-1)	H	H	L	R12	R13	R14	R15	R1	1,2,3,10
	L	BA0	BA1	BA2	V	R10	R11	R2	
Activate-2 (ACT-2)	H	H	H	R6	R7	R8	R9	R1	1,10
	L	R0	R1	R2	R3	R4	R5	R2	

NOTE:

1) All LPDDR4X commands except for Deselect are 2 clock cycle long and defined by states of CS and CA[5:0] at the first rising edge of clock. Deselect command is 1 clock cycle long.

2) “V” means “H” or “L” (a defined logic level). “X” means don’t care in which case CA[5:0] can be floated.

3) Bank addresses BA[2:0] determine which bank is to be operated upon.

4) AB “HIGH” during Precharge or Refresh command indicates that command must be applied to all banks and bank address is a don’t care.

5) Mask Write-1 command supports only BL 16. For Mark Write-1 command, CA5 must be driven LOW on first rising clock cycle (R1).

6) AP “HIGH” during Write-1, Mask Write-1 or Read-1 commands indicates that an auto-precharge will occur to the bank associated with the Write, Mask Write or Read command.

7) If Burst Length on-the-fly is enabled, BL “HIGH” during Write-1 or Read-1 command indicates that Burst Length should be set on-the-Fly to BL=32. BL “LOW” during Write-1 or Read-1 command indicates that Burst Length should be set on-the-fly to BL=16. If Burst Length on-the-fly is disabled, then BL must be driven to defined logic level “H” or “L”.

8) For CAS-2 commands (Write-2 or Mask Write-2 or Read-2 or MRR-2 or MPC (Only Write FIFO, Read FIFO & Read DQ Calibration), C[1:0] are not transmitted on the CA[5:0] bus and are assumed to be zero. Note that for CAS-2 Write-2 or CAS-2 Mask Write-2 command, C[3:2] must be driven LOW.

9) Write-1 or Mask Write-1 or Read-1 or Mode Register Read-1 or MPC (Only Write FIFO, Read FIFO & Read DQ Calibration) command must be immediately followed by CAS-2 command consecutively without any other command in between. Write-1 or Mask Write-1 or Read-1 or mode register Read-1 or MPC (Only Write FIFO, Read FIFO & Read DQ Calibration) command must be issued first before issuing CAS-2 command. MPC (Only Start & Stop DQS Oscillator, Start & Latch ZQ Calibration) commands do not require CAS-2 command; they require two additional DES or NOP commands consecutively before issuing any other commands.

10) Activate-1 command must be immediately followed by Activate-2 command consecutively without any other command in between. Activate-1 command must be issued first before issuing Activate-2 command. Once Activate-1 command is issued, Activate-2 command must be issued before issuing another Activate-1 command.

11) MRW-1 command must be immediately followed by MRW-2 command consecutively without any other command in between. MRW-1 command must be issued first before issuing MRW-2 command.

12) MRR-1 command must be immediately followed by CAS-2 command consecutively without any other command in between. MRR-1 command must be issued first before issuing CAS-2 command.

3.1 CKE Truth Tables

[Table 12] LPDDR4X : CKE Table 1), 2), 3), 4), 8)

Device Current State	CKE_{n-1}	CKE_n	Command n	Operation	Device Next State	Notes
Active Power Down	L	L	X	Maintain Active Power Down	Active Power Down	
	L	H	Deselect	Exit Active Power Down	Active	5,6
Idle Power Down	L	L	X	Maintain Idle Power Down	Idle Power Down	
	L	H	Deselect	Exit Idle Power Down	Idle	5,6
Self Refresh	L	L	X	Maintain power-down state within Self Refresh	Self Refresh	
	L	H	Deselect	Exit SREF power-down, enable command decode	Self Refresh	5,6,7
	H	L	Deselect	Enter SREF Power-Down, disable command decode	Self Refresh	5,7
	H	H	See Note 7	See Note 7	Self Refresh	7
Bank(s) Active	H	L	Deselect	Enter Active Power Down	Active Power Down	5
All Banks Idle	H	L	Deselect	Enter Idle Power Down	Idle Power Down	5, 8
Command Entry	H	H	Refer to the Command Truth Table			

NOTE :
1) CKE is a strictly asynchronous input, and as such, has no relationship to CK.
2) "X" means "don't care."
3) "Current State" is the state of the LPDDR4X-SDRAM prior to a toggle of CKE.
4) "CKEn-1" is the logic state of CKE prior to a CKE toggle event, and "CKEn" is the state of CKE after the toggle event.
5) "Deselect" is the only valid command that can be present on the bus when CKE is toggled.
6) Power-Down exit time (tXP) should elapse before a command other than Deselect is issued. The clock must toggle at least twice during the tXP period, and must be stable before issuing a command.
7) When the device is in Self.Refresh, only MRR, MRW, or MPC commands are allowed. Certain restrictions apply to changing register contents via a MRW command during SREF. See MRW section for more information.
8) In the case of ODT disabled, all DQ output shall be Hi-Z. In the case of ODT enabled, all DQ shall be terminated to VSSQ.

3.2 State Truth Table

The truth tables provide complementary information to the state diagram, they clarify the device behavior and the applied restrictions when considering the actual state of all banks.

[Table 13] Current State Bank n - Command to Bank n

Current State	Command	Operation	Next State	NOTES
Any	NOP	Continue previous operation	Current State	
Idle	ACTIVATE	Select and activate row	Active	
	Refresh (Per Bank)	Begin to refresh	Refreshing (Per Bank)	6
	Refresh (All Bank)	Begin to refresh	Refreshing (All Bank)	7
	MRW	Write value to	MR Writing	7
	MRR	Read value from	Idle MR Reading	
	Precharge	Deactivate row in bank or banks	Precharging	8, 13
Row Active	Read	Select column, and start read burst	Reading	10
	Write	Select column, and start write burst	Writing	10
	MRR	Read value from	Active MR Reading	
	Precharge	Deactivate row in bank or banks	Precharging	8
Reading	Read	Select column, and start new read burst	Reading	9, 10
	Write	Select column, and start write burst	Writing	9, 10, 11
Writing	Write	Select column, and start new write burst	Writing	9, 10
	Read	Select column, and start read burst	Reading	9, 10, 12

NOTE :
1) The table applies when both CKEn-1 and CKEn are HIGH, and after t_{XSR} or t_{XP} has been met if the previous state was Self Refresh or Power Down.
2) All states and sequences not shown are illegal or reserved.
3) Current State Definitions:
 - Idle: The bank or banks have been precharged, and tRP has been met.
 - Active: A row in the bank has been activated, and tRCD has been met. No data bursts / accesses and no register accesses are in progress.
 - Reading: A Read burst has been initiated, with Auto Precharge disabled.
 - Writing: A Write burst has been initiated, with Auto Precharge disabled.
4) The following states must not be interrupted by a command issued to the same bank. NOP commands or allowable commands to the other bank should be issued on any clock edge occurring during these states. Allowable commands to the other banks are determined by its current state and , and according to .
 - Precharging: starts with the registration of a Precharge command and ends when tRP is met. Once tRP is met, the bank will be in the idle state.
 - Row Activating: starts with registration of an Activate command and ends when tRCD is met. Once tRCD is met, the bank will be in the ‘Active’ state.
 - Read with AP Enabled: starts with the registration of the Read command with Auto Precharge enabled and ends when tRP has been met. Once tRP has been met, the bank will be in the idle state.
 - Write with AP Enabled: starts with registration of a Write command with Auto Precharge enabled and ends when tRP has been met. Once tRP is met, the bank will be in the idle state.
5) The following states must not be interrupted by any executable command; NOP commands must be applied to each positive clock edge during these states.
 - Refreshing (Per Bank): starts with registration of a Refresh (Per Bank) command and ends when tRFCpb is met. Once tRFCpb is met, the bank will be in an ‘idle’ state.
 - Refreshing (All Bank): starts with registration of an Refresh (All Bank) command and ends when tRFCab is met. Once tRFCab is met, the device will be in an ‘all banks idle’ state.
 - Idle MR Reading: starts with the registration of a MRR command and ends when tMRR has been met. Once tMRR has been met, the bank will be in the Idle state.
 - Active MR Reading: starts with the registration of a MRR command and ends when tMRR has been met. Once tMRR has been met, the bank will be in the Active state.
 - Idle MR Writing: starts with the registration of a MRW command and ends when tMRW has been met. Once tMRW has been met, the bank will be in the Idle state.
 - Active MR Writing: starts with the registration of a MRW command and ends when tMRW has been met. Once tMRW has been met, the bank will be in the Active state.
 - Precharging All: starts with the registration of a Precharge-All command and ends when tRP is met. Once tRP is met, the bank will be in the idle state.
6) Bank-specific; requires that the bank is idle and no bursts are in progress.
7) Not bank-specific; requires that all banks are idle and no bursts are in progress.
8) This command may or may not be bank specific. If all banks are being precharged, they must be in a valid state for precharging.
9) A command other than NOP should not be issued to the same bank while a Read or Write burst with Auto Precharge is enabled.
10) The new Read or Write command could be Auto Precharge enabled or Auto Precharge disabled.
11) A Write command may be applied after the completion of the Read burst; burst terminates are not permitted.
12) A Read command may be applied after the completion of the Write burst, burst terminates are not permitted.
13) If a Precharge command is issued to a bank in the Idle state, tRP shall still apply.

[Table 14] Current State Bank n - Command to Bank m

Current State of Bank n	Command for Bank m	Operation	Next State for Bank m	NOTES
Any	NOP	Continue previous operation	Current State of Bank m	
Idle	Any	Any command allowed to Bank m	-	
Row Activating, Active, or Precharging	Activate	Select and activate row in Bank m	Active	6
	Read	Select column, and start read burst from Bank m	Reading	7
	Write	Select column, and start write burst to Bank m	Writing	7
	Precharge	Deactivate row in bank or banks	Precharging	8
	MRR	Read value from	Idle MR Reading or Active MR Reading	9,10,
Reading (Autoprecharge disabled)	Read	Select column, and start read burst from Bank m	Reading	7
	Write	Select column, and start write burst to Bank m	Writing	7,12
	Activate	Select and activate row in Bank m	Active	
	Precharge	Deactivate row in bank or banks	Precharging	8
Writing/Masked Writing (Autoprecharge disabled)	Read	Select column, and start read burst from Bank m	Reading	7,14
	Write	Select column, and start write burst to Bank m	Writing	7
	Activate	Select and activate row in Bank m	Active	
	Precharge	Deactivate row in bank or banks	Precharging	8
Reading with Autoprecharge	Read	Select column, and start read burst from Bank m	Reading	7,13
	Write	Select column, and start write burst to Bank m	Writing	7,12,13
	Activate	Select and activate row in Bank m	Active	
	Precharge	Deactivate row in bank or banks	Precharging	8
Writing/Masked Writing with Autoprecharge	Read	Select column, and start read burst from Bank m	Reading	7,13,14
	Write	Select column, and start write burst to Bank m	Writing	7,13
	Activate	Select and activate row in Bank m	Active	
	Precharge	Deactivate row in bank or banks	Precharging	8

NOTE :

1) The table applies when both CKEn-1 and CKEn are HIGH, and after t_{XSR} or t_{XP} has been met if the previous state was Self Refresh or Power Down.
2) All states and sequences not shown are illegal or reserved.
3) Current State Definitions:
 - Idle: The bank has been precharged, and tRP has been met.
 - Active: A row in the bank has been activated, and tRCD has been met. No data bursts/accesses and no register accesses are in progress.
 - Reading: A Read burst has been initiated, with Auto Precharge disabled.
 - Writing: A Write burst has been initiated, with Auto Precharge disabled
4) Refresh, Self-Refresh, and Mode register Write commands may only be issued when all bank are idle.
5) The following states must not be interrupted by any executable command; NOP commands must be applied during each clock cycle while in these states:
 - Idle MR Reading: starts with the registration of a MRR command and ends when t_{MRR} has been met. Once t_{MRR} has been met, the bank will be in the Idle state.
 - Active MR Reading: starts with the registration of a MRR command and ends when t_{MRR} has been met. Once t_{MRR} has been met, the bank will be in the Active state.
 - Idle MR Writing: starts with the registration of a MRW command and ends when t_{MRW} has been met. Once t_{MRW} has been met, the bank will be in the Idle state.
 - Active MR Writing: starts with the registration of a MRW command and ends when tMRW has been met. Once tMRW has been met, the bank will be in the Active state.
6) t_{RRD} must be met between Activate command to Bank n and a subsequent Activate command to Bank m. Additionally, in the case of multiple banks activated, t_{FAW} must be satisfied.
7) Reads or Writes listed in the Command column include Reads and Writes with Auto Precharge enabled and Reads and Writes with Auto Precharge disabled.
8) This command may or may not be bank specific. If all banks are being precharged, they must be in a valid state for precharging.
9) MRR is allowed during the Row Activating state (Row Activating starts with registration of an Activate command and ends when t_{RCD} is met.)
10) MRR is allowed during the Precharging state. (Precharging starts with registration of a Precharge command and ends when t_{RP} is met.)
11) The next state for Bank m depends on the current state of Bank m (Idle, Row Activating, Precharging, or Active). The reader shall note that the state may be in transition when a MRR is issued. Therefore, if Bank m is in the Row Activating state and Precharging, the next state may be Active and Precharge dependent upon t_{RCD} and t_{RP} respectively.
12) A Write command may be applied after the completion of the Read burst, burst terminates are not permitted.
13) Read with auto precharge enabled or a Write with Auto Precharge enabled may be followed by any valid command to other banks provided that the timing restrictions described in the Precharge & Auto Precharge clarification table are followed.
14) A Read command may be applied after the completion of the Write burst, burst terminates are not permitted.

4.0 ABSOLUTE MAXIMUM DC RATINGS

Stresses greater than those listed may cause permanent damage to the device. This is a stress rating only, and functional operation of the device at these or any other conditions above those indicated in the operational sections of this specification is not implied. Exposure to absolute maximum rating conditions for extended periods may affect reliability.

[Table 15] Absolute Maximum DC Ratings

Parameter	Symbol	Min	Max	Units	Notes
V_{DD1} supply voltage relative to V_{SS}	V_{DD1}	-0.4	2.1	V	1
V_{DD2} supply voltage relative to V_{SS}	V_{DD2}	-0.4	1.5	V	1
V_{DDQ} supply voltage relative to V_{SSQ}	V_{DDQ}	-0.4	1.0	V	1
Voltage on any ball except V_{DD1} relative to V_{SS}	V_{IN}, V_{OUT}	-0.4	1.5	V	
Storage Temperature	T_{STG}	-55	125	°C	2

NOTE :
1) See Power Ramp for relationships between power supplies.
2) Storage Temperature is the case surface temperature on the center/top side of the LPDDR4X device. For the measurement conditions, please refer to JESD51-2 standard.

5.0 AC & DC OPERATING CONDITIONS

5.1 Recommended DC Operating Conditions

[Table 16] Recommended DC Operating Conditions

	DRAM	LPDDR4X			Unit	Notes
		Min	Typ	Max		
VDD1	Core 1 Power	1.70	1.80	1.95	V	1,2
VDD2	Core 2 Power / Input Buffer Power	1.06	1.10	1.17	V	1,2,3
VDDQ	I/O Buffer Power	0.57	0.60	0.65	V	2,3

NOTE :
1) VDD1 uses significantly less current than VDD2.
2) The voltage range is for DC voltage only. DC is defined as the voltage supplied at the DRAM and is inclusive of all noise up to 20MHz at the DRAM package ball.
3) The voltage noise tolerance from DC to 20MHz exceeding a pk-pk tolerance of 45mv at the DRAM ball is not included in the TdIVW.

5.2 Input Leakage Current

[Table 17] Input Leakage Current

	Symbol	Min	Max	Unit	Notes
Input Leakage current	I_L	-8	8	uA	1,2

NOTE :
1) For CK_t, CK_c, CKE, CS, CA, ODT_CA and RESET_n. Any input 0V ≤ VIN ≤ VDD2 (All other pins not under test = 0V).
2) CA ODT is disabled for CK_t, CK_c, CS, and CA.

5.3 Input/Output Leakage Current

Parameter/Condition	Symbol	Min.	Max.	Unit	Notes
Input/Output Leakage current	I_{OZ}	-10	10	uA	1,2

NOTE :
1) For DQ, DQS_t, DQS_c and DMI. Any I/O 0V ≤ V_{OUT} ≤ V_{DDQ}.
2) I/Os status are disabled: High Impedance and ODT Off.

5.4 Operating Temperature Range

[Table 18] Operating Temperature Range

	Symbol	Min	Max	Unit
Standard	T_{OPER}	-25	85	°C

NOTE :
1) Operating Temperature is the case surface temperature on the center top side of the LPDDR4X device. For the measurement conditions, please refer to JESD51-2.
2) Either the device case temperature rating or the temperature sensor (See "Temperature Sensor" on [Command Definition & Timing Diagram]) may be used to set an appropriate refresh rate, determine the need for AC timing de-rating and/or monitor the operating temperature. When using the temperature sensor, the actual device case temperature may be higher than the T_{OPER} rating that applies for the Standard or Extended Temperature Ranges. For example, T_{CASE} may be above 85°C when the temperature sensor indicates a temperature of less than 85°C.

6.0 AC AND DC INPUT MEASUREMENT LEVELS

6.1 1.1V High speed LVCMOS (HS_LLVCMOS)

6.1.1 Standard specifications

All voltages are referenced to ground except where noted.

6.1.2 DC electrical characteristics

6.1.2.1 LPDDR4X Input Level for CKE

This definition applies to CKE_A/B.

[Table 19] LPDDR4X Input Level for CKE

Parameter	Symbol	Min.	Max.	Unit	Note
Input high level (AC)	$V_{IH(AC)}$	0.75 × V_{DD2}	V_{DD2} + 0.2	V	1
Input low level (AC)	$V_{IL(AC)}$	-0.2	0.25 × V_{DD2}	V	1
Input high level (DC)	$V_{IH(DC)}$	0.65 × V_{DD2}	V_{DD2} + 0.2	V	
Input low level (DC)	$V_{IL(DC)}$	-0.2	0.35 × V_{DD2}	V	

NOTE :
1) Refer LPDDR4X AC Over/Undershoot section.

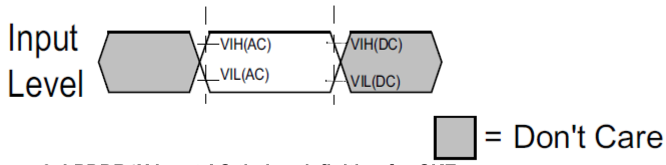


Figure 3. LPDDR4X Input AC timing definition for CKE
1-1). AC level is guaranteed transition point.
1-2). DC level is hysteresis.

6.1.2.2 LPDDR4X Input Level for Reset_n and ODT_CA

This definition applies to Reset_n and ODT_CA.

[Table 20] LPDDR4X Input Level for Reset_n and ODT_CA

Parameter	Symbol	Min.	Max.	Unit	Note
Input high level	VIH	0.80 × V_{DD2}	V_{DD2} + 0.2	V	1
Input low level	VIL	-0.2	0.20× V_{DD2}	V	1

NOTE :
1) Refer LPDDR4X AC Over/Undershoot section.

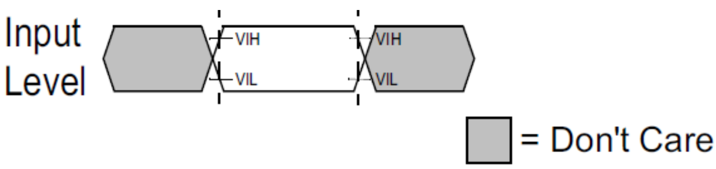


Figure 4. LPDDR4X Input AC timing definition for Reset_n and ODT_CA

6.1.3 AC Over/Undershoot

6.1.3.1 LPDDR4X AC Over/Undershoot

[Table 21] LPDDR4X AC Over/Undershoot

Parameter	Specification
Maximum peak amplitude allowed for overshoot area.	0.35V
Maximum peak amplitude allowed for undershoot area.	0.35V
Maximum overshoot area above V_{DD}/V_{DDQ}.	0.8V-ns
Maximum undershoot area below V_{SS}/V_{SSQ}.	0.8V-ns

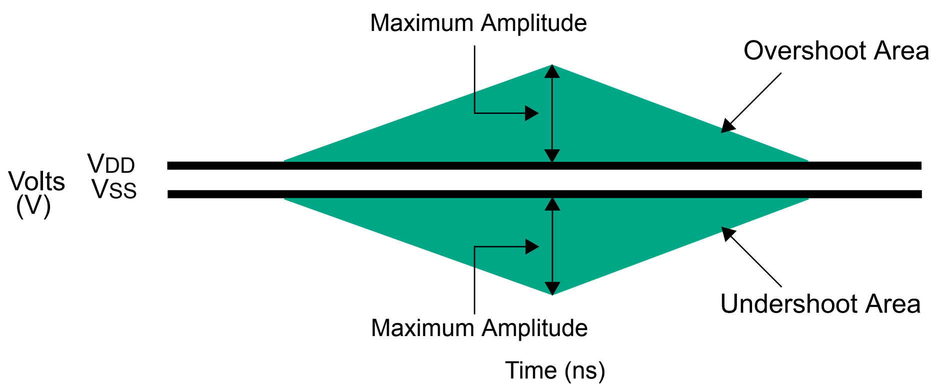


Figure 5. AC Overshoot and Undershoot Definition for Address and Control Pins

6.2 Differential Input Voltage

6.2.1 Differential Input Voltage for CK

The minimum input voltage need to satisfy both Vindiff_CK and Vindiff_CK /2 specification at input receiver and their measurement period is 1tCK. Vindiff_CK is the peak to peak voltage centered on 0 volts differential and Vindiff_CK /2 is max and min peak voltage from 0V.

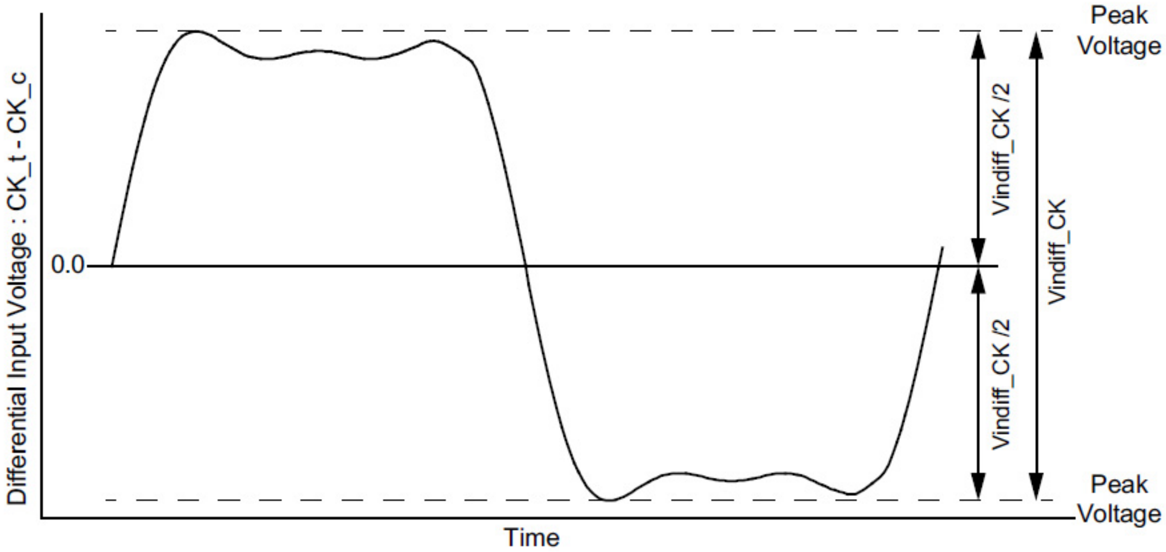


Figure 6. CK Differential Input Voltage

[Table 22] CK differential input voltage

Parameter	Symbol	Data Rate		Unit	Note
		3733			
		Min	Max		
CK differential input voltage	Vindiff_CK	360	-	mV	1

NOTE:
1) The peak voltage of Differential CK signals is calculated in a following equation.
Vindiff_CK = (Max Peak Voltage) - (Min Peak Voltage)
Max Peak Voltage = Max(f(t))
Min Peak Voltage = Min(f(t))
f(t) = VCK_t - VCK_c
a) The following requirements apply for DQ operating frequencies at or below 1333Gbps for all speed bins for the first column 1600/1866.

6.2.2 Peak voltage calculation method

The peak voltage of Differential Clock signals are calculated in a following equation.

VIH.DIFF.Peak Voltage = Max(f(t))
VIL.DIFF.Peak Voltage = Min(f(t))
f(t) = VCK_t - VCK_c

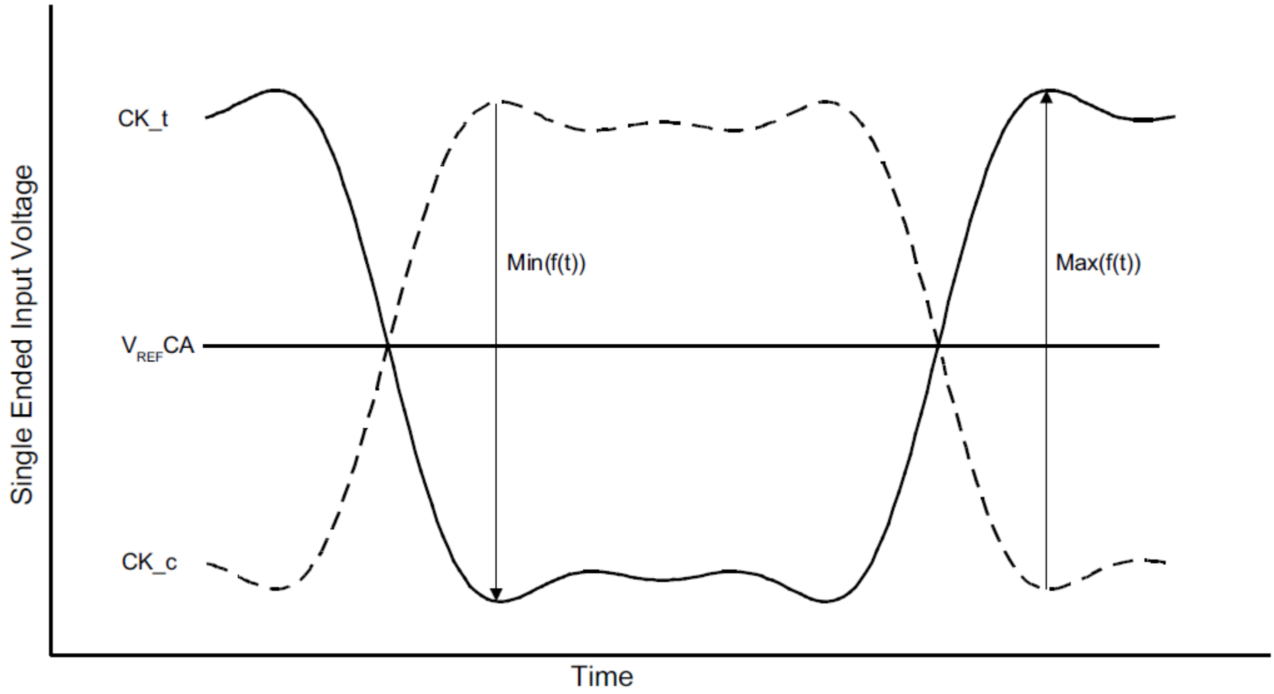


Figure 7. Definition of differential Clock Peak Voltage

NOTE:
1) VREFCA is LPDDR4X SDRAM internal setting value by VREF Training.

6.2.3 Single-Ended Input Voltage for Clock

The minimum input voltage need to satisfy both Vinse_CK, Vinse_CK_High/Low specification at input receiver.

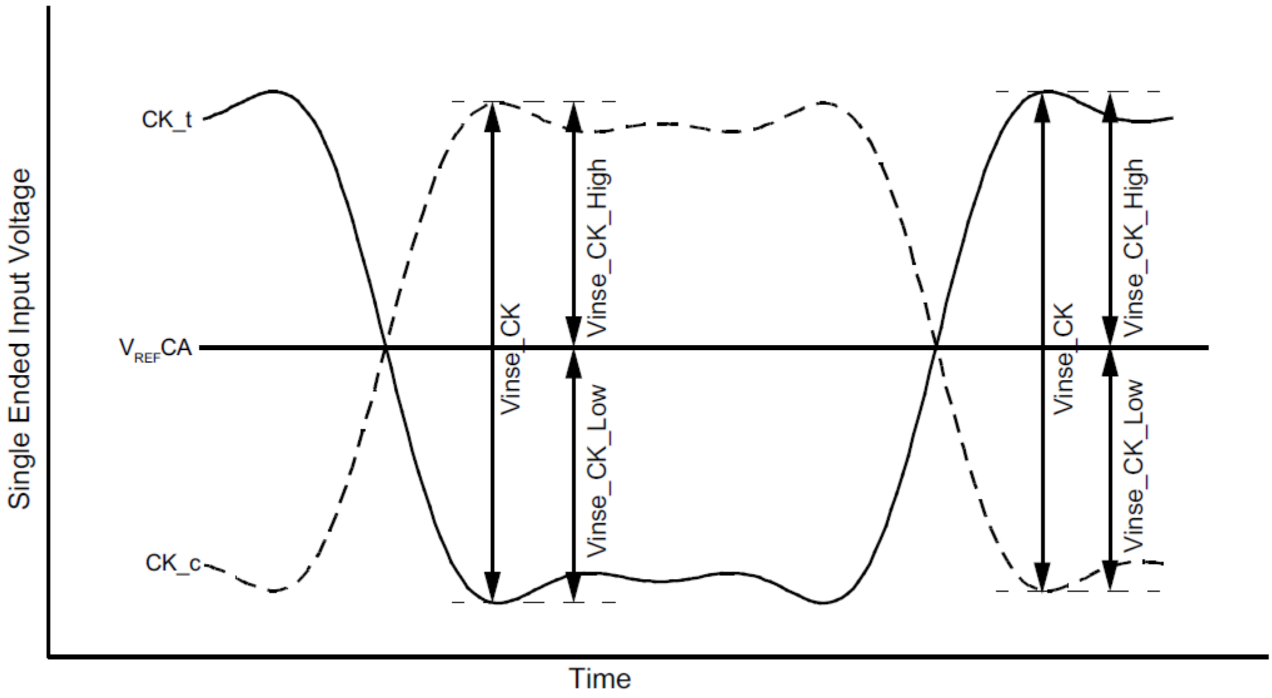


Figure 8. Clock Single-Ended Input Voltage

NOTE:
1) VREFCA is LPDDR4X SDRAM internal setting value by VREF Training.

[Table 23] Clock Single-Ended input voltage

Parameter	Symbol	Data Rate 3733 Min	Data Rate 3733 Max	Unit
Clock Single-Ended input voltage	Vinse_CK	180	-	mV
Clock Single-Ended input voltage High from VREFDQ	Vinse_CK_High	90	-	mV
Clock Single-Ended input voltage Low from VREFDQ	Vinse_CK_Low	90	-	mV

6.2.4 Differential Input Slew Rate Definition for Clock

Input slew rate for differential signals (CK_t, CK_c) are defined and measured as shown in Figure 9. and the following Tables.

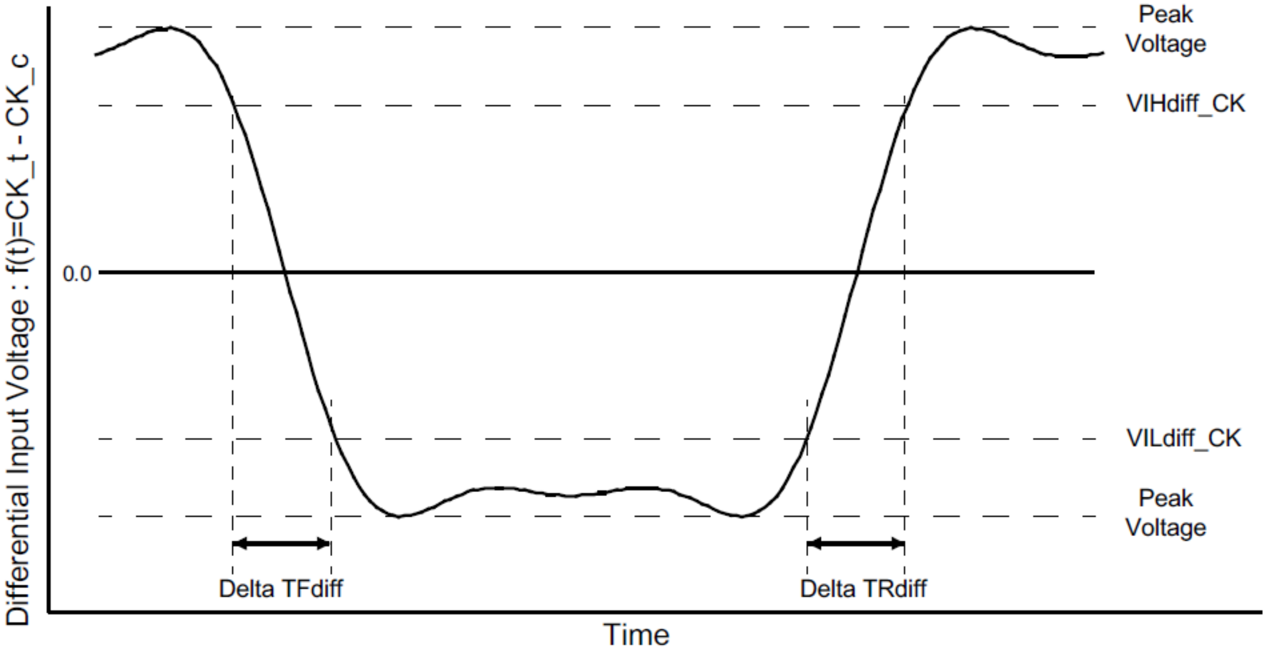


Figure 9. Differential Input Slew Rate Definition for CK_t, CK_c

NOTE :
1) Differential signal rising edge from VILdiff_CK to VIHdiff_CK must be monotonic slope.
2) Differential signal falling edge from VIHdiff_CK to VILdiff_CK must be monotonic slope.

[Table 24] Differential Input Slew Rate Definition for CK_t, CK_c

Description	From	To	Defined by
Differential input slew rate for rising edge(CK_t - CK_c)	VILdiff_CK	VIHdiff_CK	|VILdiff_CK - VIHdiff_CK|/DeltaTRdiff
Differential input slew rate for falling edge(CK_t - CK_c)	VIHdiff_CK	VILdiff_CK	|VILdiff_CK - VIHdiff_CK|/DeltaTFdiff

[Table 25] Differential Input Level for CK_t, CK_c

Parameter	Symbol	Data Rate 3733		Unit	Note
		Min	Max		
Differential Input High	VIHdiff_CK	145	-	mV	
Differential Input Low	VILdiff_CK	-	-145	mV	

[Table 26] Differential Input Slew Rate for CK_t, CK_c

Parameter	Symbol	Data Rate 3733		Unit	Note
		Min	Max		
Differential Input Slew Rate for Clock	SRIdiff_CK	2	14	V/ns	

6.2.5 Differential Input Cross Point Voltage for Clock

The cross point voltage of differential input signals (CK_t, CK_c) must meet the requirements in [Table 27]. The differential input cross point voltage VIX is measured from the actual cross point of true and complement signals to the mid level that is V_{REF}CA.

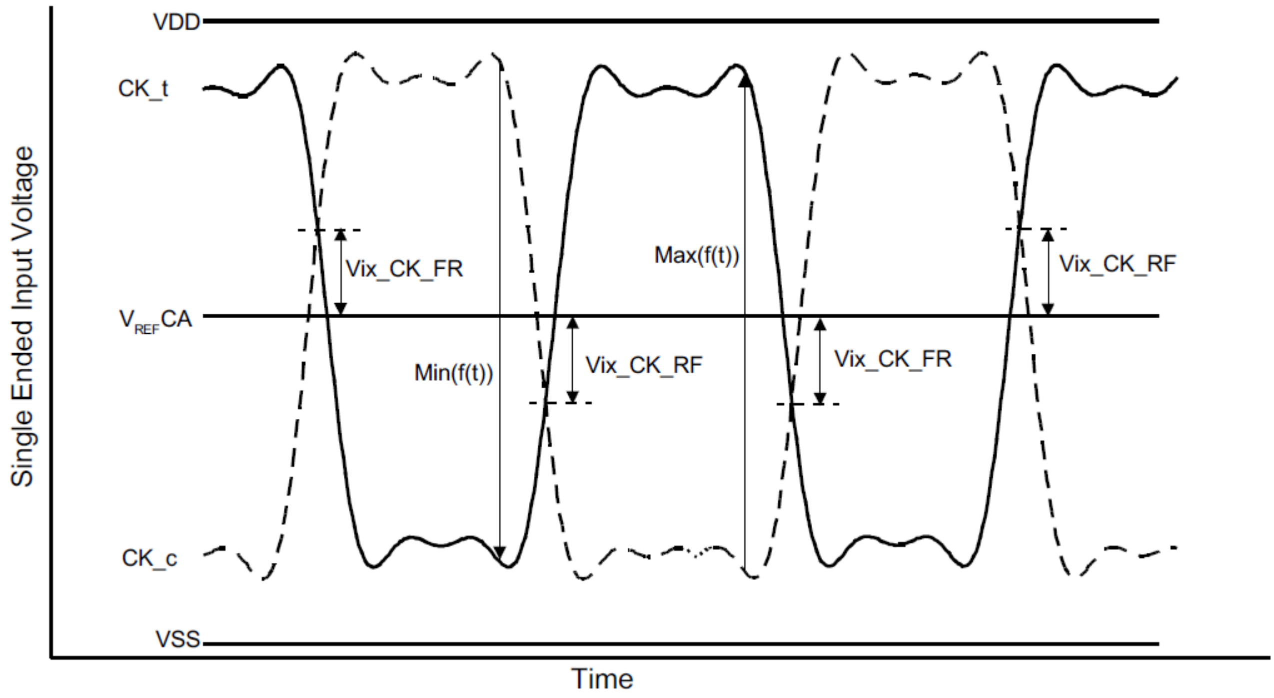


Figure 10. Vix Definition (Clock)

[Table 27] Cross point voltage for differential input signals (Clock)

Parameter	Symbol	Data Rate 3733		Unit	Note
		Min	Max		
Clock Differential input cross point voltage ratio	Vix_CK_ratio	-	25	%	1,2

NOTE :
1) Vix_CK_Ratio is defined by this equation: Vix_CK_Ratio = Vix_CK_FR/|Min(f(t))|
2) Vix_CK_Ratio is defined by this equation: Vix_CK_Ratio = Vix_CK_RF/Max(f(t))

6.2.6 AC/DC Input level for ODT input

[Table 28] LPDDR4X Input Level for ODT

Symbol		Min	Max	Unit	Note
$V_{IHODT(AC)}$	ODT Input High Level (AC)	0.75 × V_{DD}	V_{DD} + 0.2	V	1
$V_{ILODT(AC)}$	ODT Input Low Level (AC)	-0.2	0.25 × V_{DD}	V	1
$V_{IHODT(DC)}$	ODT Input High Level (DC)	0.65 × V_{DD}	V_{DD} + 0.2	V	
$V_{ILODT(DC)}$	ODT Input Low Level (DC)	-0.2	0.35 × V_{DD}	V	

NOTE :
1) See Overshoot and Undershoot Specifications.

6.2.7 Differential Input Voltage for DQS

The minimum input voltage need to satisfy both Vindiff_DQS and Vindiff_DQS /2 specification at input receiver and their measurement period is 1UI(tCK/2). Vindiff_DQS is the peak to peak voltage centered on 0 volts differential and Vindiff_DQS /2 is max and min peak voltage from 0V.

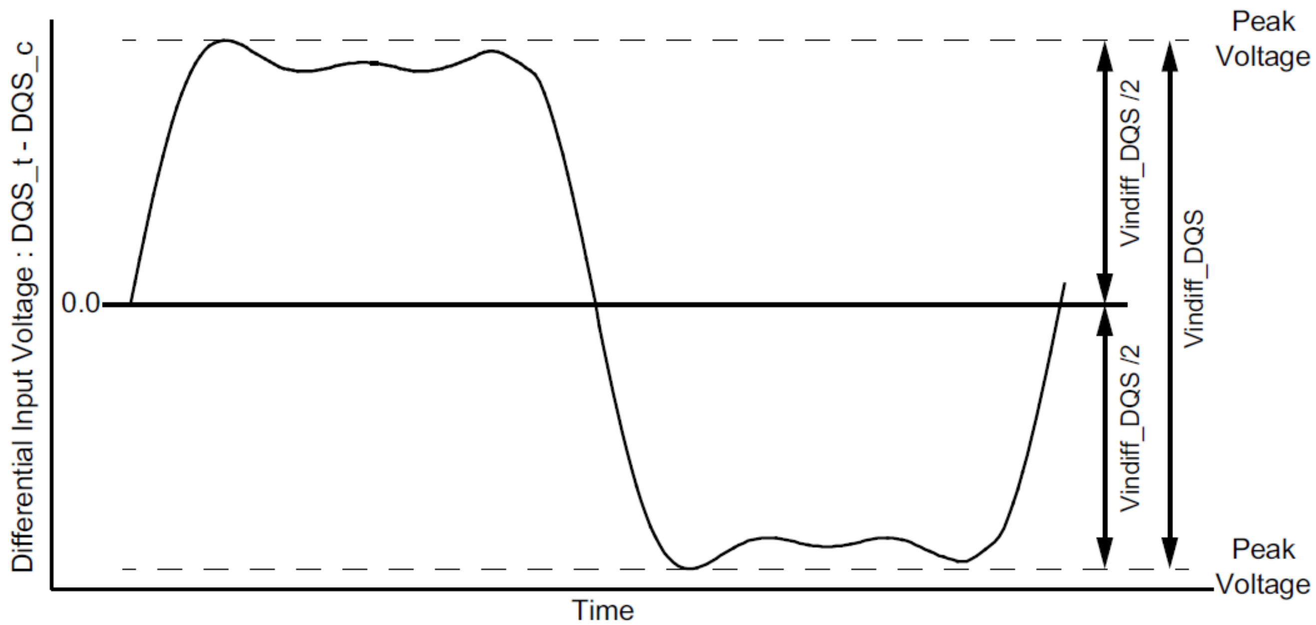


Figure 11. DQS Differential Input Voltage

[Table 29] DQS differential input voltage

Parameter	Symbol	Data Rate 3733		Unit	Note
		Min	Max		
DQS differential input	Vindiff_DQS	340	-	mV	1

NOTE :
1) The peak voltage of Differential DQS signals is calculated in a following equation.
Vindiff_DQS = (Max Peak Voltage) - (Min Peak Voltage)
Max Peak Voltage = Max(f(t))
Min Peak Voltage = Min(f(t))
f(t) = VDQS_t - VDQS_c

6.2.8 Peak voltage calculation method

The peak voltage of Differential DQS signals are calculated in a following equation.

VIH.DIFF.Peak Voltage = Max(f(t))
VIL.DIFF.Peak Voltage = Min(f(t))
f(t) = VDQS_t - VDQS_c

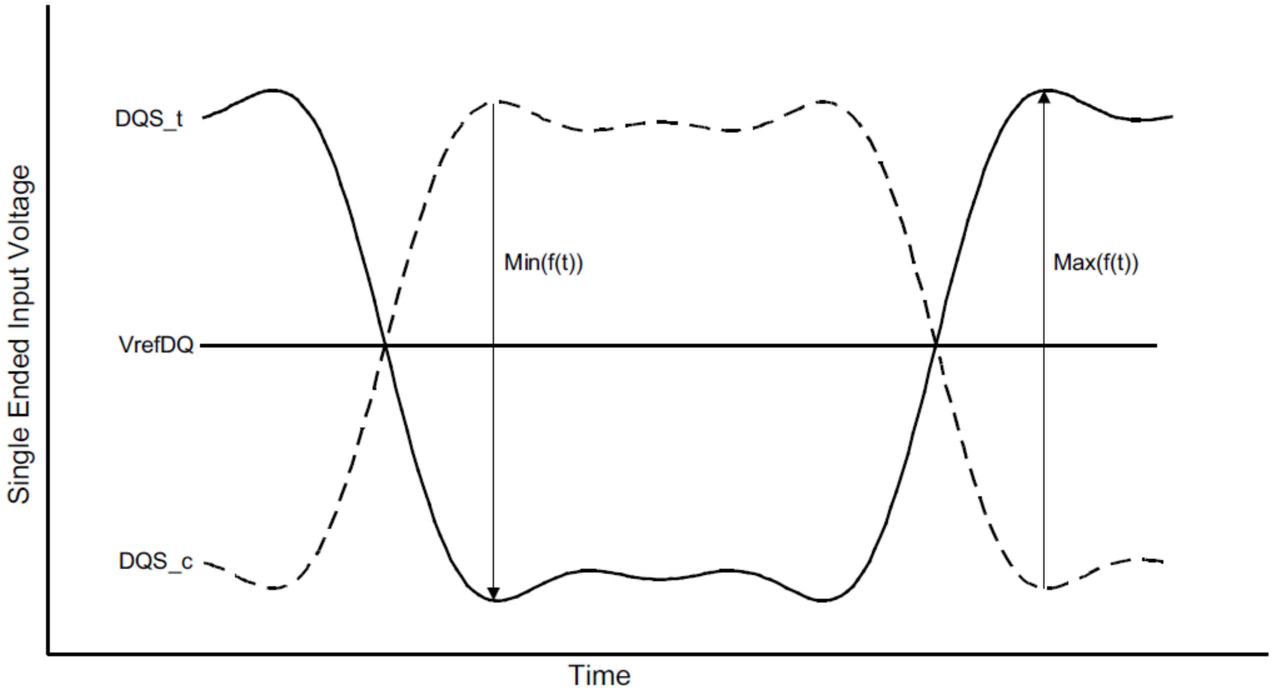


Figure 12. Definition of differential DQS Peak Voltage

6.2.9 Single-Ended Input Voltage for DQS

The minimum input voltage need to satisfy both Vinse_DQS, Vinse_DQS_High/Low specification at input receiver.

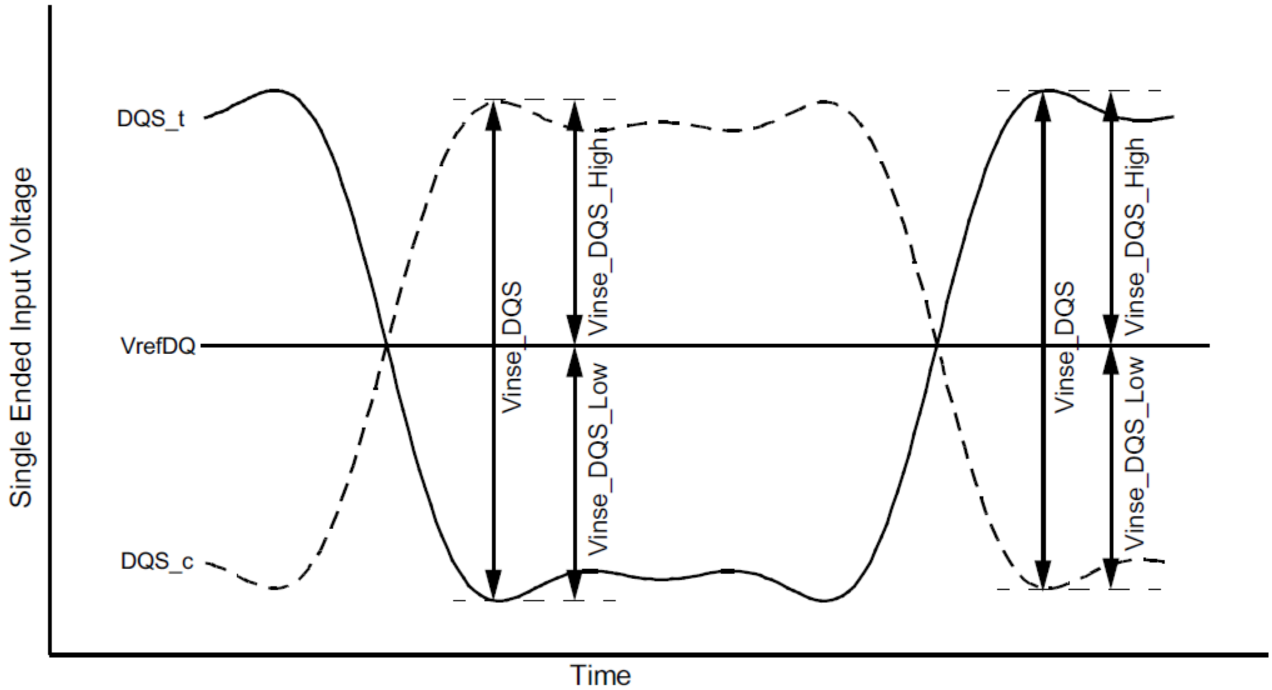


Figure 13. DQS Single-Ended Input Voltage

[Table 30] DQS Single-Ended input voltage

Parameter	Symbol	Data Rate 3733		Unit	Note
		Min	Max		
DQS Single-Ended input voltage	Vinse_DQS	170	-	mV	
DQS Single-Ended input voltage High from VrefDQ	Vinse_DQS_High	85	-	mV	
DQS Single-Ended input voltage Low from VrefDQ	Vinse_DQS_Low	85	-	mV	

6.2.10 Differential Input Slew Rate Definition for DQS

Input slew rate for differential signals (DQS_t, DQS_c) are defined and measured as shown in Figure 14. and [Table 31].

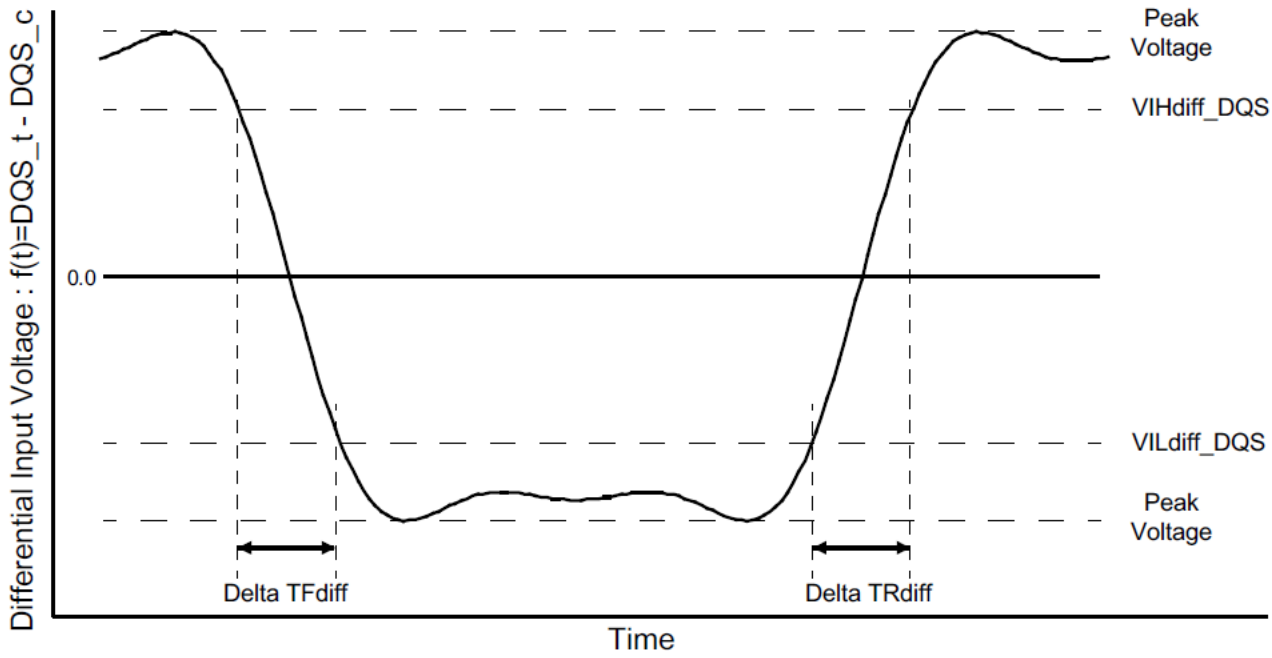


Figure 14. Differential Input Slew Rate Definition for DQS_t, DQS_c

[Table 31] Differential Input Slew Rate Definition for DQS_t, DQS_c

Description	From	To	Defined by
Differential input slew rate for rising edge(DQS_t - DQS_c)	VILdiff_DQS	VIHdiff_DQS	|VILdiff_DQS - VIHdiff_DQS|/DeltaTRdiff
Differential input slew rate for falling edge(DQS_t - DQS_c)	VIHdiff_DQS	VILdiff_DQS	|VILdiff_DQS - VIHdiff_DQS|/DeltaTFdiff

[Table 32] Differential Input Level for DQS_t, DQS_c

Parameter	Symbol	Data Rate 3733 Min	Data Rate 3733 Max	Unit	Note
Differential Input High	VIHdiff_DQS	120	-	mV	
Differential Input Low	VILdiff_DQS	-	-120	mV	

[Table 33] Differential Input Slew Rate for DQS_t, DQS_c

Parameter	Symbol	Data Rate 3733 Min	Data Rate 3733 Max	Unit	Note
Differential Input Slew Rate	SRIdiff	2	14	V/ns	

6.3 Differential Input Cross Point Voltage for DQS

The cross point voltage of differential input signals (DQS_t, DQS_c) must meet the requirements in [Table 35]. The differential input cross point voltage VIX is measured from the actual cross point of true and complement signals to the mid level that is $V_{REF}DQ$.

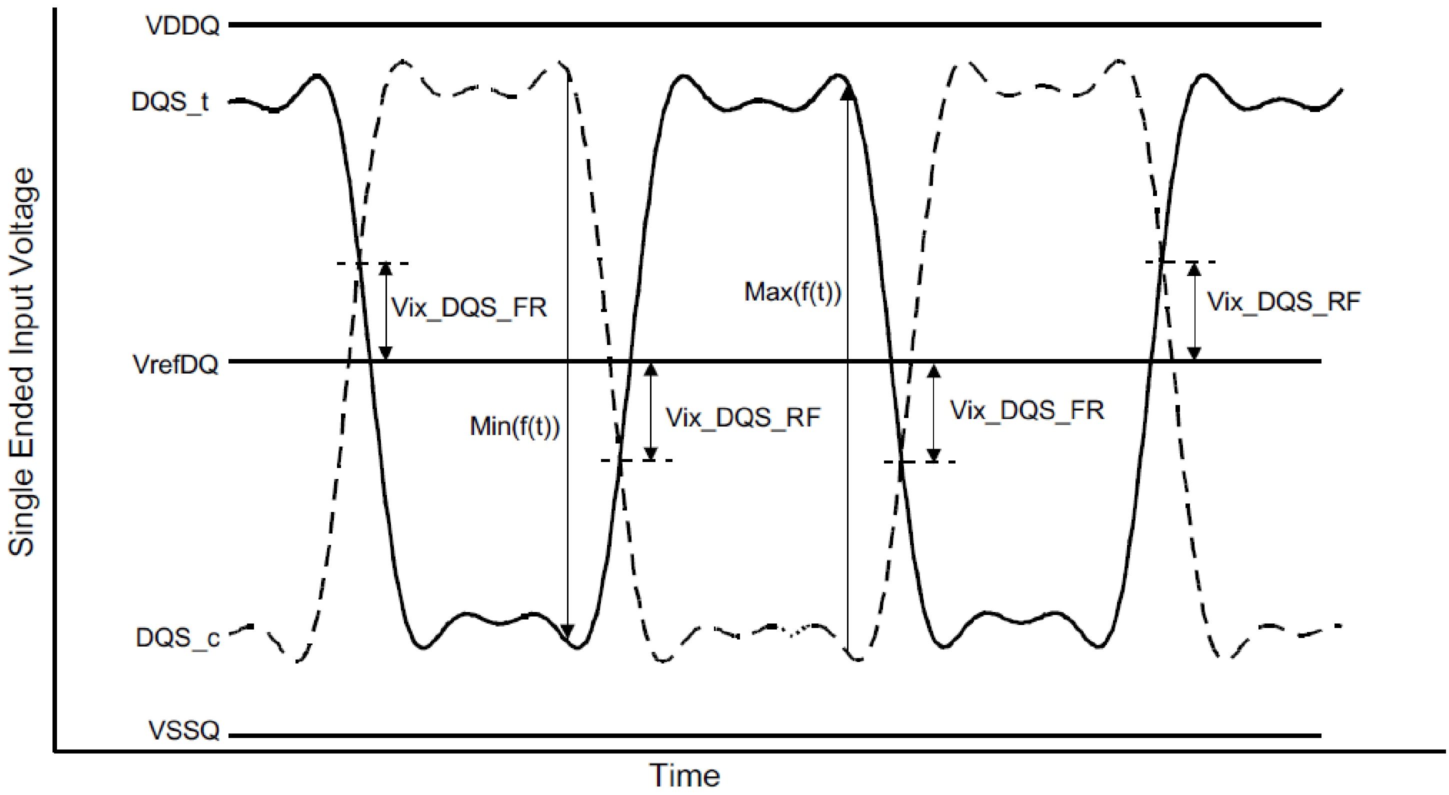


Figure 15. Vix Definition (DQS)

[Table 34] Cross point voltage for differential input signals (DQS)

Parameter	Symbol	Data Rate		Units	Notes
		3733			
		Min	Max		
DQS Differential input crosspoint voltage ratio	Vix_DQS_ratio	-	20	%	1,2

NOTE :
1) Vix_DQS_Ratio is defined by this equation: Vix_DQS_Ratio = Vix_DQS_FR/|Min(f(t))|
2) Vix_DQS_Ratio is defined by this equation: Vix_DQS_Ratio = Vix_DQS_RF/Max(f(t))

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6.4 Single Ended Output Slew Rate

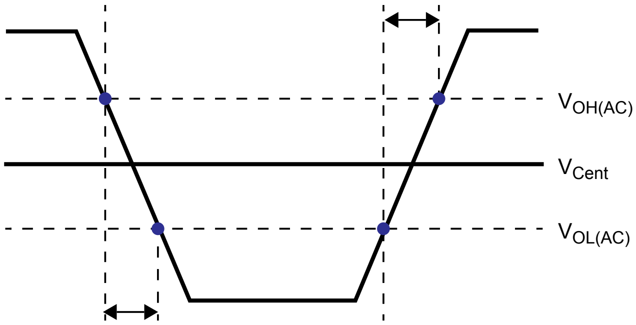


Figure 16. Single Ended Output Slew Rate Definition

[Table 35] Output Slew Rate (single-ended) for 0.6V VDDQ

Parameter	Symbol	Value		Units
		Min[1)]	Max[2)]	
Single-ended Output Slew Rate (V_{OH} = VDDQ×0.5)	S_{RQse}	3	9.0	V/ns
Output slew-rate matching Ratio (Rise to Fall)		0.8	1.2	-
Description: SR: Slew Rate Q: Query Output (like in DQ, which stands for Data-in, Query-Output) se: Single-ended Signals				

NOTE :
1) Measured with output reference load.
2) The ratio of pull-up to pull-down slew rate is specified for the same temperature and voltage, over the entire temperature and voltage range. For a given output, it represents the maximum difference between pull-up and pull-down drivers due to process variation.
3) The output slew rate for falling and rising edges is defined and measured between $V_{OL(AC)}$ = 0.2×$V_{OH(DC)}$ and $V_{OH(AC)}$=0.8×$V_{OH(DC)}$.
4) Slew rates are measured under average SSO conditions, with 50% of DQ signals per data byte switching.

6.5 Differential Output Slew Rate

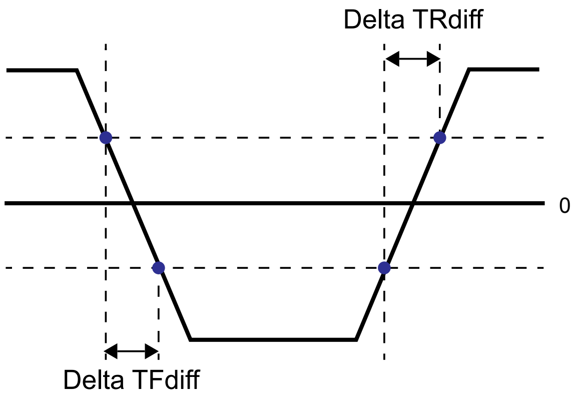


Figure 17. Differential Output Slew Rate Definition

[Table 36] Differential Output Slew Rate for 0.6V VDDQ

Parameter	Symbol	Value		Units
		Min	Max	
Differential Output Slew Rate (V_{OH} = VDDQ×0.5)	SRQdiff	6	18.0	V/ns
Description: SR: Slew Rate Q: Query Output (like in DQ, which stands for Data-in, Query-Output) diff: Differential Signals				

NOTE :
1) Measured with output reference load.
2) The output slew rate for falling and rising edges is defined and measured between $V_{OL(AC)}$=-0.8×$V_{OH(DC)}$ and $V_{OH(AC)}$=0.8×$V_{OH(DC)}$.
3) Slew rates are measured under average SSO conditions, with 50% of DQ signals per data byte switching.

6.6 Overshoot and Undershoot for LVSTL

[Table 37] AC Overshoot/Undershoot Specification

Parameter		Data Rate	Units
		3733	
Maximum peak amplitude allowed for overshoot area. (See Figure 18.)	Max	0.3	V
Maximum peak amplitude allowed for undershoot area. (See Figure 18.)	Max	0.3	V
Maximum overshoot area above V_{DD}. (See Figure 18.)	Max	0.1	V-ns
Maximum undershoot area below V_{SS}. (See Figure 18.)	Max	0.1	V-ns

NOTE :
1) V_{DD2} stands for V_{DD} for CA[5:0], CK_t, CK_c, CS_n, CKE and ODT. V_{DD} stands for V_{DDQ} for DQ, DMI, DQS_t and DQS_c.
2) V_{SS} stands for V_{SS} for CA[5:0], CK_t, CK_c, CS_n, CKE and ODT. V_{SS} stands for V_{SSQ} for DQ, DMI, DQS_t and DQS_c.
3) Maximum peak amplitude values are referenced from actual V_{DD} and V_{SS} values.
4) Maximum area values are referenced from maximum operating V_{DD} and V_{SS} values.

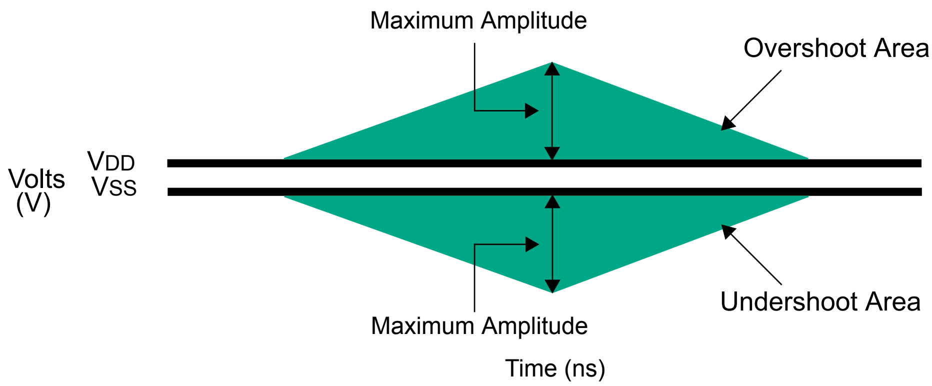


Figure 18. Overshoot and Undershoot Definition

7.0 OUTPUT BUFFER CHARACTERISTICS

7.1 LPDDR4X Driver Output Timing Reference Load

These 'Timing Reference Loads' are not intended as a precise representation of any particular system environment or a depiction of the actual load presented by a production tester. System designers should use IBIS or other simulation tools to correlate the timing reference load to a system environment. Manufacturers correlate to their production test conditions, generally one or more coaxial transmission lines terminated at the tester electronics.

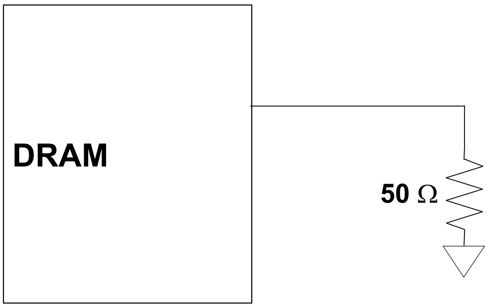


Figure 19. Driver Output Reference Load for Timing and Slew Rate

NOTE :
1) All output timing parameter values are reported with respect to this reference load. This reference load is also used to report slew rate.

7.2 LVSTL (Low Voltage Swing Terminated Logic) IO System

LVSTL I/O cell is comprised of pull-up, pull-down driver and a terminator. The basic cell is shown in Figure 20 .

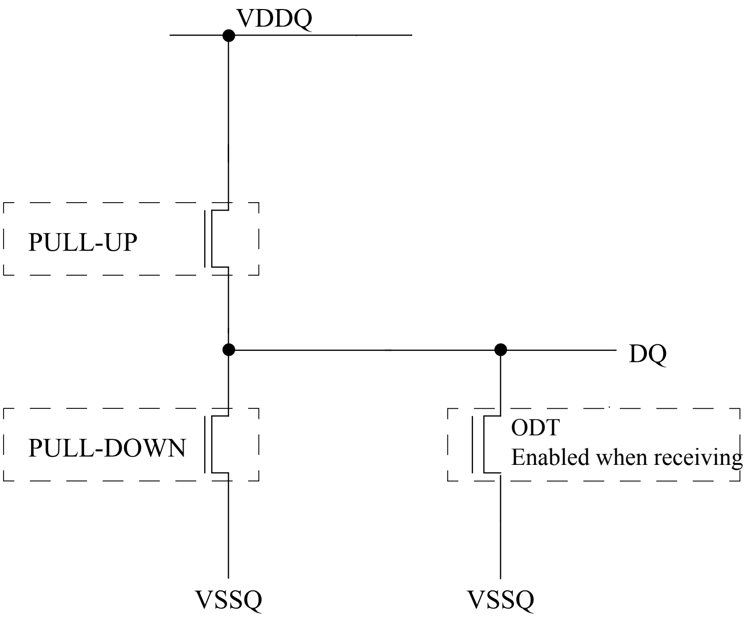


Figure 20. LVSTL I/O Cell

To ensure that the target impedance is achieved the LVSTL I/O cell is designed to calibrated as below procedure.

1) First calibrate the pull-down device against a 240 Ohm resister to VDDQ via the ZQ pin

Set Strength Control to minimum setting
Increase drive strength until comparator detects data bit is less than VDDQ/2.
NMOS pull-down device is calibrated to 240 Ohms

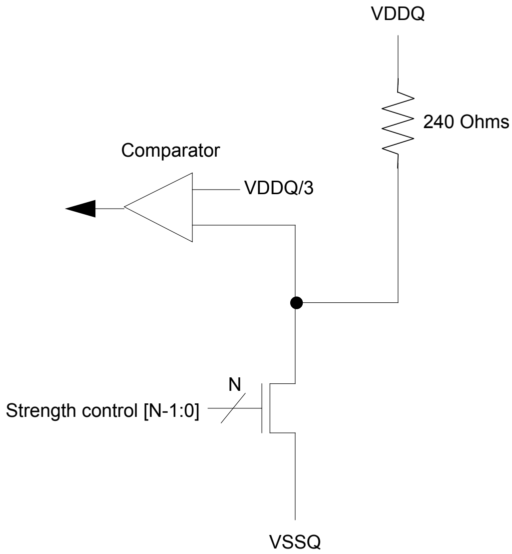


Figure 21. pull-down calibration

2) Then calibrate the pull-up device against the calibrated pull-down device.

Set VOH target and NMOS controller ODT replica via MRS (VOH can be automatically controlled by ODT MRS)
Set Strength Control to minimum setting
Increase drive strength until comparator detects data bit is greater than VOH target
NMOS pull-up device is now calibrated to VOH target

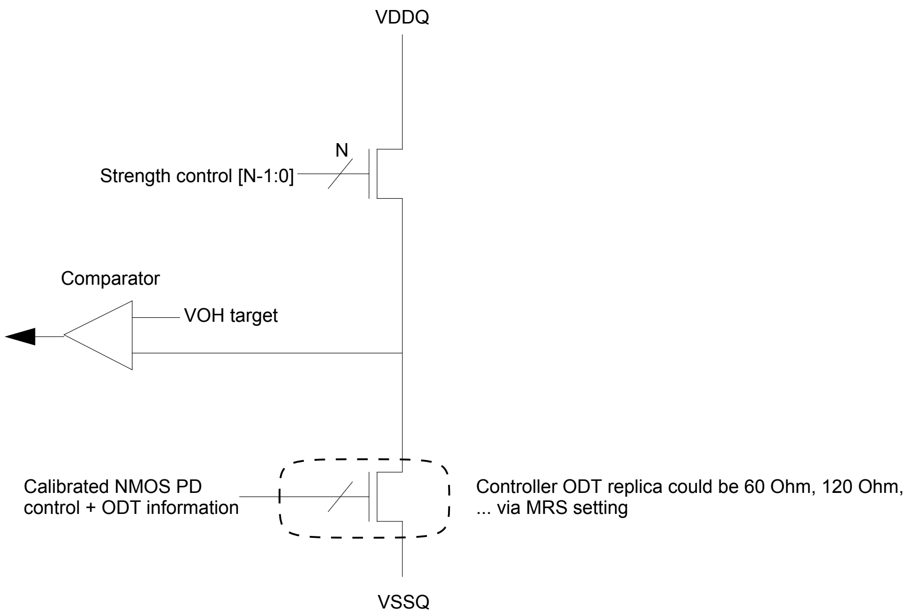


Figure 22. pull-up calibration

8.0 INPUT/OUTPUT CAPACITANCE

[Table 38] Input/Output Capacitance

Parameter	Symbol	Min/Max	Value	Unit	Notes
Input capacitance, CK_t and CK_c	CCK	Min	2.1	pF	1,2
		Max	4.0		
Input capacitance delta, CK_t and CK_c	CDCK	Min	0.0	pF	1,2,3
		Max	0.2		
Input capacitance, all other input-only pins except CS and CKE	CI1	Min	2.1	pF	1,2,4
		Max	4.0		
Cin, CS0 / CS1 and CKE0 / CKE1	CI2	Min	0.9	pF	1,2,4
		Max	2.8		
Input capacitance delta, all other input-only pins except CS and CKE	CDI1	Min	-0.3	pF	1,2,5
		Max	0.3		
Cdelta, CS0 / CS1 and CKE0 / CKE1	CDI2	Min	-1.6	pF	1,2,5,9
		Max	-0.8		
Input/output capacitance, DQ, DMI, DQS_t and DQS_c	CIO	Min	2.5	pF	1,2,6
		Max	5.0		
Input/output capacitance delta, DQS_t and DQS_c	CDDQS	Min	0.0	pF	1,2,7
		Max	0.3		
Input/output capacitance delta, DQ and DMI	CDIO	Min	-0.6	pF	1,2,8
		Max	0.6		
Input/output capacitance ZQ pin	CZQ	Min	8.5	pF	1,2
		Max	10.5		

NOTE :
1) This parameter applies to both die and package.
2) This parameter is not subject to production test. It is verified by design and characterization. The capacitance is measured according to JEP147 (Procedure for measuring input capacitance using a vector network analyzer (VNA) with VDD1, VDD2, VDDQ, VSS, VSSQ applied and all other pins floating.
3) Absolute value of CCK_t - CCK_c.
4) CI applies to CS_n, CKE, CA0-CA5.
5) CDI = CI - 0.5 × (CCK_t + CCK_c)
6) DMI loading matches DQ and DQS.
7) Absolute value of CDQS_t and CDQS_c.
8) CDIO = CIO - 0.5 × (CDQS_t + CDQS_c) in byte-lane.

9.0 IDD SPECIFICATION PARAMETERS AND TEST CONDITIONS

9.1 IDD Measurement Conditions

The following definitions are used within the IDD measurement tables unless stated otherwise:
LOW: $V_{IN} \le V_{IL}$(DC) MAX
HIGH: $V_{IN} \ge V_{IH}$(DC) MIN
STABLE: Inputs are stable at a HIGH or LOW level
SWITCHING: See Table 39 and Table 40.

[Table 39] Definition of Switching for CA Input Signals

Switching for CA								
CK_t edge	R1	R2	R3	R4	R5	R6	R7	R8
CKE	HIGH	HIGH	HIGH	HIGH	HIGH	HIGH	HIGH	HIGH
CS	LOW	LOW	LOW	LOW	LOW	LOW	LOW	LOW
CA0	HIGH	LOW	LOW	LOW	LOW	HIGH	HIGH	HIGH
CA1	HIGH	HIGH	HIGH	LOW	LOW	LOW	LOW	HIGH
CA2	HIGH	LOW	LOW	LOW	LOW	HIGH	HIGH	HIGH
CA3	HIGH	HIGH	HIGH	LOW	LOW	LOW	LOW	HIGH
CA4	HIGH	LOW	LOW	LOW	LOW	HIGH	HIGH	HIGH
CA5	HIGH	HIGH	HIGH	LOW	LOW	LOW	LOW	HIGH

NOTE :
1) CS must always be driven LOW.
2) 50% of CA bus is changing between HIGH and LOW once per clock for the CA bus.
3) The above pattern is used continuously during IDD measurement for IDD values that require switching on the CA bus.

[Table 40] CA pattern for IDD4R

Clock Cycle Number	CKE	CS	Command	CA0	CA1	CA2	CA3	CA4	CA5
N	HIGH	HIGH	Read-1	L	H	L	L	L	L
N+1	HIGH	LOW		L	H	L	L	L	L
N+2	HIGH	HIGH	CAS-2	L	H	L	L	H	L
N+3	HIGH	LOW		L	L	L	L	L	L
N+4	HIGH	LOW	DES	L	L	L	L	L	L
N+5	HIGH	LOW	DES	L	L	L	L	L	L
N+6	HIGH	LOW	DES	L	L	L	L	L	L
N+7	HIGH	LOW	DES	L	L	L	L	L	L
N+8	HIGH	HIGH	Read-1	L	H	L	L	L	L
N+9	HIGH	LOW		L	H	L	L	H	L
N+10	HIGH	HIGH	CAS-2	L	H	L	L	H	H
N+11	HIGH	LOW		H	H	H	H	H	H
N+12	HIGH	LOW	DES	L	L	L	L	L	L
N+13	HIGH	LOW	DES	L	L	L	L	L	L
N+14	HIGH	LOW	DES	L	L	L	L	L	L
N+15	HIGH	LOW	DES	L	L	L	L	L	L

NOTE :
1) BA[2:0] = 010_B, CA[9:4] = 000000_B or 111111_B, Burst Order CA[3:2] = 00_B or 11_B (Same as LPDDR3 IDD4R Spec)
2) Difference from LPDDR3 Spec : CA pins are kept low with DES CMD to reduce ODT current.

[Table 41] CA pattern for IDD4W

Clock Cycle Number	CKE	CS	Command	CA0	CA1	CA2	CA3	CA4	CA5
N	HIGH	HIGH	Write-1	L	L	H	L	L	L
N+1	HIGH	LOW		L	H	L	L	L	L
N+2	HIGH	HIGH	CAS-2	L	H	L	L	H	L
N+3	HIGH	LOW		L	L	L	L	L	L
N+4	HIGH	LOW	DES	L	L	L	L	L	L
N+5	HIGH	LOW	DES	L	L	L	L	L	L
N+6	HIGH	LOW	DES	L	L	L	L	L	L
N+7	HIGH	LOW	DES	L	L	L	L	L	L
N+8	HIGH	HIGH	Write-1	L	L	H	L	L	L
N+9	HIGH	LOW		L	H	L	L	H	L
N+10	HIGH	HIGH	CAS-2	L	H	L	L	H	H
N+11	HIGH	LOW		L	L	H	H	H	H
N+12	HIGH	LOW	DES	L	L	L	L	L	L
N+13	HIGH	LOW	DES	L	L	L	L	L	L
N+14	HIGH	LOW	DES	L	L	L	L	L	L
N+15	HIGH	LOW	DES	L	L	L	L	L	L

NOTE :
1) BA[2:0] = 010_B, CA[9:4] = 000000_B or 111111_B (Same as LPDDR3 IDD4W Spec.)
2) Difference from LPDDR3 Spec :
1-No burst ordering
2-CA pins are kept low with DES CMD to reduce ODT current.

[Table 42] Data Pattern for IDD4W (DBI off)

	DBI OFF Case									No. of 1's
	DQ[7]	DQ[6]	DQ[5]	DQ[4]	DQ[3]	DQ[2]	DQ[1]	DQ[0]	DBI	
BL0	1	1	1	1	1	1	1	1	0	8
BL1	1	1	1	1	0	0	0	0	0	4
BL2	0	0	0	0	0	0	0	0	0	0
BL3	0	0	0	0	1	1	1	1	0	4
BL4	0	0	0	0	0	0	1	1	0	2
BL5	0	0	0	0	1	1	1	1	0	4
BL6	1	1	1	1	1	1	0	0	0	6
BL7	1	1	1	1	0	0	0	0	0	4
BL8	1	1	1	1	1	1	1	1	0	8
BL9	1	1	1	1	0	0	0	0	0	4
BL10	0	0	0	0	0	0	0	0	0	0
BL11	0	0	0	0	1	1	1	1	0	4
BL12	0	0	0	0	0	0	1	1	0	2
BL13	0	0	0	0	1	1	1	1	0	4
BL14	1	1	1	1	1	1	0	0	0	6
BL15	1	1	1	1	0	0	0	0	0	4
BL16	1	1	1	1	1	1	0	0	0	6
BL17	1	1	1	1	0	0	0	0	0	4
BL18	0	0	0	0	0	0	1	1	0	2
BL19	0	0	0	0	1	1	1	1	0	4
BL20	0	0	0	0	0	0	0	0	0	0
BL21	0	0	0	0	1	1	1	1	0	4
BL22	1	1	1	1	1	1	1	1	0	8
BL23	1	1	1	1	0	0	0	0	0	4
BL24	0	0	0	0	0	0	1	1	0	2
BL25	0	0	0	0	1	1	1	1	0	4
BL26	1	1	1	1	1	1	0	0	0	6
BL27	1	1	1	1	0	0	0	0	0	4
BL28	1	1	1	1	1	1	1	1	0	8
BL29	1	1	1	1	0	0	0	0	0	4
BL30	0	0	0	0	0	0	0	0	0	0
BL31	0	0	0	0	1	1	1	1	0	4
No. of 1's	16	16	16	16	16	16	16	16		

NOTE:
1) Simplified pattern compared with last showing.
Same data pattern was applied to DQ[4], DQ[5], DQ[6], DQ[7] for reducing complexity for IDD4W/R pattern programming.

[Table 43] Data Pattern for IDD4R (DBI off)

	DBI OFF Case									No. of 1's
	DQ[7]	DQ[6]	DQ[5]	DQ[4]	DQ[3]	DQ[2]	DQ[1]	DQ[0]	DBI	
BL0	1	1	1	1	1	1	1	1	0	8
BL1	1	1	1	1	0	0	0	0	0	4
BL2	0	0	0	0	0	0	0	0	0	0
BL3	0	0	0	0	1	1	1	1	0	4
BL4	0	0	0	0	0	0	1	1	0	2
BL5	0	0	0	0	1	1	1	1	0	4
BL6	1	1	1	1	1	1	0	0	0	6
BL7	1	1	1	1	0	0	0	0	0	4
BL8	1	1	1	1	1	1	1	1	0	8
BL9	1	1	1	1	0	0	0	0	0	4
BL10	0	0	0	0	0	0	0	0	0	0
BL11	0	0	0	0	1	1	1	1	0	4
BL12	0	0	0	0	0	0	1	1	0	2
BL13	0	0	0	0	1	1	1	1	0	4
BL14	1	1	1	1	1	1	0	0	0	6
BL15	1	1	1	1	0	0	0	0	0	4
BL16	1	1	1	1	1	1	1	1	0	8
BL17	1	1	1	1	0	0	0	0	0	4
BL18	0	0	0	0	0	0	0	0	0	0
BL19	0	0	0	0	1	1	1	1	0	4
BL20	1	1	1	1	1	1	0	0	0	6
BL21	1	1	1	1	0	0	0	0	0	4
BL22	0	0	0	0	0	0	1	1	0	2
BL23	0	0	0	0	1	1	1	1	0	4
BL24	0	0	0	0	0	0	0	0	0	0
BL25	0	0	0	0	1	1	1	1	0	4
BL26	1	1	1	1	1	1	1	1	0	8
BL27	1	1	1	1	0	0	0	0	0	4
BL28	0	0	0	0	0	0	1	1	0	2
BL29	0	0	0	0	1	1	1	1	0	4
BL30	1	1	1	1	1	1	0	0	0	6
BL31	1	1	1	1	0	0	0	0	0	4
No. of 1's	16	16	16	16	16	16	16	16		

NOTE:
1) Same data pattern was applied to DQ[4], DQ[5], DQ[6], DQ[7] for reducing complexity for IDD4W/R pattern programming.

[Table 44] Data Pattern for IDD4W (DBI on)

	DBI ON Case									No. of 1's
	DQ[7]	DQ[6]	DQ[5]	DQ[4]	DQ[3]	DQ[2]	DQ[1]	DQ[0]	DBI	
BL0	0	0	0	0	0	0	0	0	1	1
BL1	1	1	1	1	0	0	0	0	0	4
BL2	0	0	0	0	0	0	0	0	0	0
BL3	0	0	0	0	1	1	1	1	0	4
BL4	0	0	0	0	0	0	1	1	0	2
BL5	0	0	0	0	1	1	1	1	0	4
BL6	0	0	0	0	0	0	1	1	1	3
BL7	1	1	1	1	0	0	0	0	0	4
BL8	0	0	0	0	0	0	0	0	1	1
BL9	1	1	1	1	0	0	0	0	0	4
BL10	0	0	0	0	0	0	0	0	0	0
BL11	0	0	0	0	1	1	1	1	0	4
BL12	0	0	0	0	0	0	1	1	0	2
BL13	0	0	0	0	1	1	1	1	0	4
BL14	0	0	0	0	0	0	1	1	1	3
BL15	1	1	1	1	0	0	0	0	0	4
BL16	0	0	0	0	0	0	1	1	1	3
BL17	1	1	1	1	0	0	0	0	0	4
BL18	0	0	0	0	0	0	1	1	0	2
BL19	0	0	0	0	1	1	1	1	0	4
BL20	0	0	0	0	0	0	0	0	0	0
BL21	0	0	0	0	1	1	1	1	0	4
BL22	0	0	0	0	0	0	0	0	1	1
BL23	1	1	1	1	0	0	0	0	0	4
BL24	0	0	0	0	0	0	1	1	0	2
BL25	0	0	0	0	1	1	1	1	0	4
BL26	0	0	0	0	0	0	1	1	1	3
BL27	1	1	1	1	0	0	0	0	0	4
BL28	0	0	0	0	0	0	0	0	1	1
BL29	1	1	1	1	0	0	0	0	0	4
BL30	0	0	0	0	0	0	0	0	0	0
BL31	0	0	0	0	1	1	1	1	0	4
No. of 1's	8	8	8	8	8	8	16	16	8	

DBI enabled burst

[Table 45] Data Pattern for IDD4R (DBI on)

	DBI ON Case									No. of 1's
	DQ[7]	DQ[6]	DQ[5]	DQ[4]	DQ[3]	DQ[2]	DQ[1]	DQ[0]	DBI	
BL0	0	0	0	0	0	0	0	0	1	1
BL1	1	1	1	1	0	0	0	0	0	4
BL2	0	0	0	0	0	0	0	0	0	0
BL3	0	0	0	0	1	1	1	1	0	4
BL4	0	0	0	0	0	0	1	1	0	2
BL5	0	0	0	0	1	1	1	1	0	4
BL6	0	0	0	0	0	0	1	1	1	3
BL7	1	1	1	1	0	0	0	0	0	4
BL8	0	0	0	0	0	0	0	0	1	1
BL9	1	1	1	1	0	0	0	0	0	4
BL10	0	0	0	0	0	0	0	0	0	0
BL11	0	0	0	0	1	1	1	1	0	4
BL12	0	0	0	0	0	0	1	1	0	2
BL13	0	0	0	0	1	1	1	1	0	4
BL14	0	0	0	0	0	0	1	1	1	3
BL15	1	1	1	1	0	0	0	0	0	4
BL16	0	0	0	0	0	0	0	0	1	1
BL17	1	1	1	1	0	0	0	0	0	4
BL18	0	0	0	0	0	0	0	0	0	0
BL19	0	0	0	0	1	1	1	1	0	4
BL20	0	0	0	0	0	0	1	1	1	3
BL21	1	1	1	1	0	0	0	0	0	4
BL22	0	0	0	0	0	0	1	1	0	2
BL23	0	0	0	0	1	1	1	1	0	4
BL24	0	0	0	0	0	0	0	0	0	0
BL25	0	0	0	0	1	1	1	1	0	4
BL26	0	0	0	0	0	0	0	0	1	1
BL27	1	1	1	1	0	0	0	0	0	4
BL28	0	0	0	0	0	0	1	1	0	2
BL29	0	0	0	0	1	1	1	1	0	4
BL30	0	0	0	0	0	0	1	1	1	3
BL31	1	1	1	1	0	0	0	0	0	4
No. of 1's	8	8	8	8	8	8	16	16	8	

9.2 IDD Specifications

IDD values are for the entire operating voltage range, and all of them are for the entire standard range, with the exception of IDD6ET which is for the entire elevated temperature range.

[Table 46] LPDDR4X IDD Specification Parameters and Operating Conditions

<table>
<tr><th>Parameter/Condition</th><th>Symbol</th><th>Power Supply</th><th>Notes</th></tr>
<tr><td rowspan="3">Operating one bank active-precharge current:
$t_{CK} = t_{CKmin}$; $t_{RC} = t_{RCmin}$;
CKE is HIGH;
CS is LOW between valid commands;
CA bus inputs are switching;
Data bus inputs are stable
ODT disabled</td><td>$IDD0_1$</td><td>VDD1</td><td>1,10,11</td></tr>
<tr><td>$IDD0_2$</td><td>VDD2</td><td>1,10,11</td></tr>
<tr><td>$IDD0_Q$</td><td>VDDQ</td><td>1,3,10,11</td></tr>
<tr><td rowspan="3">Idle power-down standby current:
$t_{CK} = t_{CKmin}$;
CKE is LOW;
CS is LOW;
All banks are idle;
CA bus inputs are switching;
Data bus inputs are stable
ODT disabled</td><td>$IDD2P_1$</td><td>VDD1</td><td>1,10,11</td></tr>
<tr><td>$IDD2P_2$</td><td>VDD2</td><td>1,10,11</td></tr>
<tr><td>$IDD2P_Q$</td><td>VDDQ</td><td>1,3,10,11</td></tr>
<tr><td rowspan="3">Idle power-down standby current with clock stop:
CK_t =LOW, CK_c =HIGH;
CKE is LOW;
CS is LOW;
All banks are idle;
CA bus inputs are stable;
Data bus inputs are stable
ODT disabled</td><td>$IDD2PS_1$</td><td>VDD1</td><td>1,10,11</td></tr>
<tr><td>$IDD2PS_2$</td><td>VDD2</td><td>1,10,11</td></tr>
<tr><td>$IDD2PS_Q$</td><td>VDDQ</td><td>1,3,10,11</td></tr>
<tr><td rowspan="3">Idle non power-down standby current:
$t_{CK} = t_{CKmin}$;
CKE is HIGH;
CS is LOW;
All banks are idle;
CA bus inputs are switching;
Data bus inputs are stable
ODT disabled</td><td>$IDD2N_1$</td><td>VDD1</td><td>1,10,11</td></tr>
<tr><td>$IDD2N_2$</td><td>VDD2</td><td>1,10,11</td></tr>
<tr><td>$IDD2N_Q$</td><td>VDDQ</td><td>1,3,10,11</td></tr>
<tr><td rowspan="3">Idle non power-down standby current with clock stopped:
CK_t=LOW; CK_c=HIGH;
CKE is HIGH;
CS is LOW;
All banks are idle;
CA bus inputs are stable;
Data bus inputs are stable
ODT disabled</td><td>$IDD2NS_1$</td><td>VDD1</td><td>1,10,11</td></tr>
<tr><td>$IDD2NS_2$</td><td>VDD2</td><td>1,10,11</td></tr>
<tr><td>$IDD2NS_Q$</td><td>VDDQ</td><td>1,3,10,11</td></tr>
<tr><td rowspan="3">Active power-down standby current:
$t_{CK} = t_{CKmin}$;
CKE is LOW;
CS is LOW;
One bank is active;
CA bus inputs are switching;
Data bus inputs are stable
ODT disabled</td><td>$IDD3P_1$</td><td>VDD1</td><td>1,10,11</td></tr>
<tr><td>$IDD3P_2$</td><td>VDD2</td><td>1,10,11</td></tr>
<tr><td>$IDD3P_Q$</td><td>VDDQ</td><td>1,3,10,11</td></tr>
<tr><td rowspan="3">Active power-down standby current with clock stop:
CK_t=LOW, CK_c=HIGH;
CKE is LOW;
CS is LOW;
One bank is active;
CA bus inputs are stable;
Data bus inputs are stable
ODT disabled</td><td>$IDD3PS_1$</td><td>VDD1</td><td>1,10,11</td></tr>
<tr><td>$IDD3PS_2$</td><td>VDD2</td><td>1,10,11</td></tr>
<tr><td>$IDD3PS_Q$</td><td>VDDQ</td><td>1,4,10,11</td></tr>
</table>

[Table 46] LPDDR4X IDD Specification Parameters and Operating Conditions

Parameter/Condition	Symbol	Power Supply	Notes
Active non-power-down standby current: $t_{CK} = t_{CKmin}$; CKE is HIGH; CS is LOW; One bank is active; CA bus inputs are switching; Data bus inputs are stable ODT disabled	$IDD3N_1$	VDD1	1,10,11
	$IDD3N_2$	VDD2	1,10,11
	$IDD3N_Q$	VDDQ	1,4,10,11
Active non-power-down standby current with clock stopped: CK_t=LOW, CK_c=HIGH; CKE is HIGH; CS is LOW; One bank is active; CA bus inputs are stable; Data bus inputs are stable ODT disabled	$IDD3NS_1$	VDD1	1,10,11
	$IDD3NS_2$	VDD2	1,10,11
	$IDD3NS_Q$	VDDQ	1,4,10,11
Operating burst READ current: $t_{CK} = t_{CKmin}$; CS is LOW between valid commands; One bank is active; BL = 16 or 32; RL = RL(MIN); CA bus inputs are switching; 50% data change each burst transfer ODT disabled	$IDD4R_1$	VDD1	1,10,11
	$IDD4R_2$	VDD2	1,10,11
	$IDD4R_Q$	VDDQ	1,5,10,11
Operating burst WRITE current: $t_{CK} = t_{CKmin}$; CS is LOW between valid commands; One bank is active; BL = 16 or 32; WL = WLmin; CA bus inputs are switching; 50% data change each burst transfer ODT disabled	$IDD4W_1$	VDD1	1,10,11
	$IDD4W_2$	VDD2	1,10,11
	$IDD4W_Q$	VDDQ	1,4,10,11
All-bank REFRESH Burst current: $t_{CK} = t_{CKmin}$; CKE is HIGH between valid commands; $t_{RC} = t_{RFCabmin}$; Burst refresh; CA bus inputs are switching; Data bus inputs are stable; ODT disabled	$IDD5_1$	VDD1	1,10,11
	$IDD5_2$	VDD2	1,10,11
	$IDD5_Q$	VDDQ	1,4,10,11
All-bank REFRESH Average current: $t_{CK} = t_{CKmin}$; CKE is HIGH between valid commands; $t_{RC} = t_{REFI}$; CA bus inputs are switching; Data bus inputs are stable; ODT disabled	$IDD5AB_1$	VDD1	1,10,11
	$IDD5AB_2$	VDD2	1,10,11
	$IDD5AB_Q$	VDDQ	1,4,10,11
Per-bank REFRESH Average current: $t_{CK} = t_{CKmin}$; CKE is HIGH between valid commands; $t_{RC} = t_{REFI}/8$; CA bus inputs are switching; Data bus inputs are stable; ODT disabled	$IDD5PB_1$	VDD1	1,10,11
	$IDD5PB_2$	VDD2	1,10,11
	$IDD5PB_Q$	VDDQ	1,4,10,11
Power Down Self refresh current (-25°C to +85°C): CK_t=LOW, CK_c=HIGH; CKE is LOW; CA bus inputs are stable; Data bus inputs are stable; Maximum 1x Self-Refresh Rate; ODT disabled	$IDD6_1$	VDD1	6,7,9,10,11
	$IDD6_2$	VDD2	6,7,9,10,11
	$IDD6_Q$	VDDQ	4,6,7,9,10,11

NOTE :
1) Published IDD values are the maximum of the distribution of the arithmetic mean.
2) ODT disabled: MR11[2:0] = 000_B.
3) IDD current specifications are tested after the device is properly initialized.
4) Measured currents are the summation of VDDQ and VDD2.
5) Guaranteed by design with output load = 5pF and RON = 40 ohm.
6) The 1x Self-Refresh Rate is the rate at which the LPDDR4X device is refreshed internally during Self-Refresh, before going into the elevated Temperature range.
7) This is the general definition that applies to full array Self Refresh.
8) For all IDD measurements, VIHCKE = 0.8 x VDD2, VILCKE = 0.2 x VDD2.
9) IDD6 25°C is guaranteed, IDD6 85°C is typical of the distribution of the arithmetic mean.
10) These specification values are the summation of all the channel current and both channels are under the same condition at the same time.
11) Dual Channel devices are specified in dual channel operation (both channels operating together).

9.3 IDD Spec Table

[Table 47] IDD Specification for 16Gb LPDDR4X (2Channel operation)

Symbol		Power Supply	16Gb (2Ch, x16/Ch) 3733Mbps	Units
IDD0	$IDD0_1$	VDD1	10	mA
	$IDD0_2$	VDD2	65	mA
	$IDD0_Q$	VDDQ	0.5	mA
IDD2P	$IDD2P_1$	VDD1	2	mA
	$IDD2P_2$	VDD2	6.25	mA
	$IDD2P_Q$	VDDQ	0.5	mA
IDD2PS	$IDD2PS_1$	VDD1	2	mA
	$IDD2PS_2$	VDD2	6.25	mA
	$IDD2PS_Q$	VDDQ	0.5	mA
IDD2N	$IDD2N_1$	VDD1	3	mA
	$IDD2N_2$	VDD2	26.5	mA
	$IDD2N_Q$	VDDQ	0.5	mA
IDD2NS	$IDD2NS_1$	VDD1	3	mA
	$IDD2NS_2$	VDD2	20	mA
	$IDD2NS_Q$	VDDQ	0.5	mA
IDD3P	$IDD3P_1$	VDD1	2.8	mA
	$IDD3P_2$	VDD2	13.5	mA
	$IDD3P_Q$	VDDQ	0.5	mA
IDD3PS	$IDD3PS_1$	VDD1	2.8	mA
	$IDD3PS_2$	VDD2	13.5	mA
	$IDD3PS_Q$	VDDQ	0.5	mA
IDD3N	$IDD3N_1$	VDD1	3	mA
	$IDD3N_2$	VDD2	32	mA
	$IDD3N_Q$	VDDQ	0.5	mA
IDD3NS	$IDD3NS_1$	VDD1	3	mA
	$IDD3NS_2$	VDD2	28	mA
	$IDD3NS_Q$	VDDQ	0.5	mA
IDD4R	$IDD4R_1$	VDD1	8.5	mA
	$IDD4R_2$	VDD2	450	mA
	$IDD4R_Q$	VDDQ	200	mA
IDD4W	$IDD4W_1$	VDD1	3	mA
	$IDD4W_2$	VDD2	434.5	mA
	$IDD4W_Q$	VDDQ	0.5	mA
IDD5	$IDD5_1$	VDD1	75	mA
	$IDD5_2$	VDD2	315	mA
	$IDD5_Q$	VDDQ	0.5	mA
IDD5AB	$IDD5AB_1$	VDD1	7	mA
	$IDD5AB_2$	VDD2	41	mA
	$IDD5AB_Q$	VDDQ	0.5	mA

Symbol			Power Supply	16Gb (2Ch, x16/Ch) 3733Mbps	Units
IDD5PB	$IDD5PB_1$		VDD1	7	mA
	$IDD5PB_2$		VDD2	42	mA
	$IDD5PB_Q$		VDDQ	0.5	mA
IDD6	$IDD6_1$	25°C	VDD1	1	mA
		85°C		5	
	$IDD6_2$	25°C	VDD2	2.7	mA
		85°C		22	
	$IDD6_Q$	25°C	VDDQ	0.4	mA
		85°C		0.5	

10.0 AC AND DC OUTPUT MEASUREMENT LEVELS

10.1 Single Ended AC and DC Output Levels

Table 48 shows the output levels used for measurements of single ended signals.

[Table 48] Single-ended AC and DC Output Levels

Symbol	Parameter	Value			Unit	Notes
		Under LPDDR4X-TBD Un-term	TBD to 3200 VSSQ term	3200 to 4266 VSSQ term		
V_{OH} (DC)	AC, DC output high measurement level	VDDQ	VDDQ/2	VDDQ/2	V	1
V_{OL} (DC)	AC, DC output low measurement level	VSSQ	VSSQ	VSSQ	V	

NOTE :
1) 60ohm ODT value is assumed.

10.2 Pull Up/Pull Down Driver Characteristics and Calibration

[Table 49] Pull-down Driver Characteristics, with ZQ Calibration

$R_{ONPD,NOM}$	Resistor	Min	Nom	Max	Unit
40.0 Ohm	R_{ON40PD}	0.90	1.00	1.10	$R_{ZQ/6}$
48.0 Ohm	R_{ON48PD}	0.90	1.00	1.10	$R_{ZQ/5}$
60.0 Ohm	R_{ON60PD}	0.90	1.00	1.10	$R_{ZQ/4}$
80.0 Ohm	R_{ON80PD}	0.90	1.00	1.10	$R_{ZQ/3}$
120.0 Ohm	$R_{ON120PD}$	0.90	1.00	1.10	$R_{ZQ/2}$
240.0 Ohm	$R_{ON240PD}$	0.90	1.00	1.10	$R_{ZQ/1}$

NOTE :
1) All value are after ZQ Calibration. Without ZQ Calibration RONPD values are ± 30%.

[Table 50] Terminated Pull-up Characteristics, with ZQ Calibration

VOH_{PU}, nom	VOH,nom (mV)	Min	Nor	Max	Unit
VDDQ×0.5	300	0.90	1.0	1.10	VOH,nom
VDDQ×0.6	360	0.90	1.0	1.10	VOH,nom

NOTE :
1) All values are after ZQ Calibration. Without ZQ Calibration VOH(nom) values are ± 30%
2) VOH,nom (mV) values are based on a nominal VDDQ = 0.6V.

[Table 51] Terminated Valid Calibration Points

VOH_{PU}, nom	ODT Value					
	240	120	80	60	48	40
VDDQ×0.5	VALID	VALID	VALID	VALID	VALID	VALID
VDDQ×0.6	DNU	VALID	DNU	VALID	DNU	DNU

NOTE :
1) Once the output is calibrated for a given VOH(nom) calibration point, the ODT value may be changed without recalibration.
2) If the VOH(nom) calibration point is changed, then re-calibration is required.
3) DNU = Do Not Use.

[Table 52] Pull-down Characteristics without ZQ Calibration

$R_{ONPD,NOM}$	Resistor	Vout	Min	Nom	Max	Unit	Notes
40.0Ω	R_{ON40PD}	0.5 × V_{OH}	0.70	1.00	1.30	$R_{ZQ/6}$	1
48.0Ω	R_{ON48PD}	0.5 × V_{OH}	0.70	1.00	1.30	$R_{ZQ/5}$	1

NOTE:
1) Across entire operating temperature range, without calibration.

[Table 53] Pull-up Characteristics without V_{OH} Calibration (Die to Die variation)

VOH_{PU}, (nom)	VOH(nom) (mV)	Variation			Unit	Notes
		Min	Nor	Max		
VDDQ/2.5	440	0.70	1.0	1.30	VOH(nom)	1
VDDQ/3	367	0.70	1.0	1.30	VOH(nom)	1

NOTE :
1) ODT value of Memory controller should be informed with MRW before V_{OH} calibration.

[Table 54] V_{OUT} level of un-terminated condition

Parameter	Symbol	Min	Max	Unit	Note
Output High voltage level when ODT of memory controller is turned off	V_{OH}_un-term	VDDQ-0.55	VDDQ-0.15	V	

11.0 ELECTRICAL CHARACTERISTICS AND AC TIMING

11.1 Clock Specification

The jitter specified is a random jitter meeting a Gaussian distribution. Input clocks violating the min/max values may result in malfunction of the LPDDR4X device.

11.1.1 Definition for tCK(avg) and nCK

tCK(avg) is calculated as the average clock period across any consecutive 200 cycle window, where each clock period is calculated from rising edge to rising edge.

$$tCK(avg) = \left(\sum_{j=1}^{N} tCK_j\right) / N$$
$$where \qquad N = 200$$

Unit 'tCK(avg)' represents the actual clock average tCK(avg) of the input clock under operation. Unit 'nCK' represents one clock cycle of the input clock, counting the actual clock edges.
tCK(avg) may change by up to +/-1% within a 100 clock cycle window, provided that all jitter and timing specs are met.

11.1.2 Definition for tCK(abs)

t_{CK}(abs) is defined as the absolute clock period, as measured from one rising edge to the next consecutive rising edge.
t_{CK}(abs) is not subject to production test.

11.1.3 Definition for tCH(avg) and tCL(avg)

t_{CH}(avg) is defined as the average high pulse width, as calculated across any consecutive 200 high pulses.

$$tCH(avg) = \left(\sum_{j=1}^{N} tCH_j\right) / (N \times tCK(avg))$$
$$where \qquad N = 200$$

t_{CL}(avg) is defined as the average low pulse width, as calculated across any consecutive 200 low pulses.

$$tCL(avg) = \left(\sum_{j=1}^{N} tCL_j\right) / (N \times tCK(avg))$$
$$where \qquad N = 200$$

11.1.4 Definition for tCH(abs) and tCL(abs)

tCH(abs) is the absolute instantaneous clock high pulse width, as measured from one rising edge to the following falling edge.
tCL(abs) is the absolute instantaneous clock low pulse width, as measured from one falling edge to the following rising edge.
Both tCH(abs) and tCL(abs) are not subject to production test.

11.1.5 Definition for tJIT(per)

t_{JIT}(per) is the single period jitter defined as the largest deviation of any signal tCK from tCK(avg).
t_{JIT}(per) = Min/max of {tCK_i - tCK(avg) where i = 1 to 200}.
t_{JIT}(per)$,_{act}$ is the actual clock jitter for a given system.
t_{JIT}(per)$,_{allowed}$ is the specified allowed clock period jitter.
t_{JIT}(per) is not subject to production test.

11.1.6 Definition for tJIT(cc)

tJIT(cc) is defined as the absolute difference in clock period between two consecutive clock cycles.
t_{JIT}(cc) = Max of |{tCK(i +1)- tCK(i)}|.
t_{JIT}(cc) defines the cycle to cycle jitter.
t_{JIT}(cc) is not subject to production test.

11.1.7 Definition for tERR(nper)

t_{ERR}(nper) is defined as the cumulative error across n multiple consecutive cycles from tCK(avg).
t_{ERR}(nper)$_{,act}$ is the actual clock jitter over n cycles for a given system.
t_{ERR}(nper)$_{,allowed}$ is the specified allowed clock period jitter over n cycles.
t_{ERR}(nper) is not subject to production test.

$$tERR(nper) = \left(\sum_{j=i}^{i+n-1} tCK_j \right) - n \times tCK(avg)$$

t_{ERR}(nper),min can be calculated by the formula shown below:

$$tERR(nper), min = (1 + 0.68LN(n)) \times tJIT(per), min$$

t_{ERR}(nper),max can be calculated by the formula shown below

$$tERR(nper), max = (1 + 0.68LN(n)) \times tJIT(per), max$$

Using these equations, t_{ERR}(nper) tables can be generated for each t_{JIT}(per),act value.

11.1.8 Definition for duty cycle jitter tJIT(duty)

t_{JIT}(duty) is defined with absolute and average specification of tCH / tCL.

$$tJIT(duty), min = MIN((tCH(abs), min - tCH(avg), min), (tCL(abs), min - tCL(avg), min)) \times tCK(avg)$$

$$tJIT(duty), max = MAX((tCH(abs), max - tCH(avg), max), (tCL(abs), max - tCL(avg), max)) \times tCK(avg)$$

11.1.9 Definition for tCK(abs), tCH(abs) and tCL(abs)

These parameters are specified per their average values, however it is understood that the following relationship between the average timing and the absolute instantaneous timing holds at all times.

[Table 55] Definition for tCK(abs), tCH(abs), and tCL(abs)

Parameter	Symbol	Min	Unit
Absolute Clock Period	t_{CK}(abs)	tCK(avg),min + tJIT(per),min	ps
Absolute Clock HIGH Pulse Width	t_{CH}(abs)	tCH(avg),min + tJIT(duty),min / tCK(avg)min	tCK(avg)
Absolute Clock LOW Pulse Width	t_{CL}(abs)	tCL(avg),min + tJIT(duty),min / tCK(avg)min	tCK(avg)

NOTE :
1) tCK(avg),min is expressed is ps for this table.
2) tJIT(duty),min is a negative value.

11.2 Period Clock Jitter

LPDDR4X devices can tolerate some clock period jitter without core timing parameter de-rating. This section describes device timing requirements in the presence of clock period jitter (tJIT(per)) in excess of the values found in Table 57, LPDDR4X AC Timing Table and how to determine cycle time de-rating and clock cycle de-rating.

11.2.1 Clock period jitter effects on core timing parameters

(tRCD, tRP, tRTP, tWR, tWRA, tWTR, tRC, tRAS, tRRD, tFAW)

Core timing parameters extend across multiple clock cycles. Period clock jitter will impact these parameters when measured in numbers of clock cycles. When the device is operated with clock jitter within the specification limits, the LPDDR4X device is characterized and verified to support tnPARAM = RU{tPARAM / tCK(avg)}.

When the device is operated with clock jitter outside specification limits, the number of clocks or tCK(avg) may need to be increased based on the values for each core timing parameter.

11.2.1.1 Cycle time de-rating for core timing parameters

For a given number of clocks (tnPARAM), for each core timing parameter, average clock period (tCK(avg)) and actual cumulative period error (tERR(tnPARAM),act) in excess of the allowed cumulative period error (tERR(tnPARAM),allowed), the equation below calculates the amount of cycle time de-rating (in ns) required if the equation results in a positive value for a core timing parameter.

$$CycleTimeDerating = MAX\left\{\left(\frac{tPARAM + tERR(tnPARAM), act - tERR(tnPARAM), allowed}{tnPARAM} - tCK(avg)\right), 0\right\}$$

A cycle time derating analysis should be conducted for each core timing parameter. The amount of cycle time derating required is the maximum of the cycle time de-ratings determined for each individual core timing parameter.

11.2.1.2 Clock Cycle de-rating for core timing parameters

For a given number of clocks (tnPARAM) for each core timing parameter, clock cycle de-rating should be specified with amount of period jitter (tJIT(per)).

For a given number of clocks (tnPARAM), for each core timing parameter, average clock period (tCK(avg)) and actual cumulative period error (tERR(tnPARAM),act) in excess of the allowed cumulative period error (tERR(tnPARAM),allowed), the equation below calculates the clock cycle derating (in clocks) required if the equation results in a positive value for a core timing parameter.

$$ClockCycleDerating = RU\left\{\frac{tPARAM + tERR(tnPARAM), act - tERR(tnPARAM), allowed}{tCK(avg)}\right\} - tnPARAM$$

A clock cycle de-rating analysis should be conducted for each core timing parameter.

11.2.2 Clock jitter effects on Command/Address timing parameters

Command/address timing parameters (t_{IS}, t_{IH}, t_{ISb}, t_{IHb}) are measured from a command/address signal (CS or CA[5:0]) transition edge to its respective clock signal (CK_t/ CK_c) crossing. The specification values are not affected by the tJIT(per) applied, because the setup and hold times are relative to the clock signal crossing that latches the command/address. Regardless of clock jitter values, these values must be met.

11.2.3 Clock jitter effects on Read timing parameters

11.2.3.1 tRPRE

When the device is operated with input clock jitter, tRPRE needs to be de-rated by the actual period jitter (tJIT(per),act,max) of the input clock in excess of the allowed period jitter (tJIT(per),allowed,max). Output de-ratings are relative to the input clock.

$$tRPRE(min, derated) = 0.9 - \left(\frac{tJIT(per), act,max - tJIT(per),allowed,max}{tCK(avg)}\right)$$

For example,

if the measured jitter into a LPDDR4X device has tCK(avg) = 625ps, tJIT(per),act,min = -xx, and tJIT(per),act,max = +xx ps, then tRPRE,min,derated = 0.9 - (tJIT(per),act,max - tJIT(per),allowed,max)/tCK(avg) = 0.9 - (xx - xx)/xx = yy tCK(avg).

11.2.3.2 tLZ(DQ), tHZ(DQ), tDQSCK, tLZ(DQS), tHZ(DQS)

These parameters are measured from a specific clock edge to a data signal (DMn, DQm.: n=0,1,2,3. m=0 –31) transition and will be met with respect to that clock edge. Therefore, they are not affected by the amount of clock jitter applied (i.e. tJIT(per).

11.2.3.3 tQSH, tQSL

These parameters are affected by duty cycle jitter which is represented by tCH(abs)min and tCL(abs)min.

These parameters determine absolute Data-Valid window(DVW) at the LPDDR4X device pin.

Absolute min DVW @LPDDR4X device pin = min { (tQSH(abs)min – tDQSQmax), (tQSL(abs)min – tDQSQmax) }

This minimum DVW shall be met at the target frequency regardless of clock jitter.

11.2.3.4 tRPST

tRPST is affected by duty cycle jitter which is represented by tCL(abs). Therefore tRPST(abs)min can be specified by tCL(abs)min.

tRPST(abs)min = tCL(abs)min – 0.05 = tQSL(abs)min

11.2.4 Clock jitter effects on Write timing parameters

11.2.4.1 tDS, tDH

These parameters are measured from a data signal (DMn, DQm.: n=0,1,2,3. m=0 –31) transition edge to its respective data strobe signal (DQSn_t, DQSn_c : n=0,1,2,3) crossing. The spec values are not affected by the amount of clock jitter applied (i.e. tJIT(per), as the setup and hold are relative to the data strobe signal crossing that latches the data. Regardless of clock jitter values, these values shall be met.

11.2.4.2 tDSS, tDSH

These parameters are measured from a data strobe signal (DQSx_t, DQSx_c) crossing to its respective clock signal (CK_t/CK_c) crossing. The spec values are not affected by the amount of clock jitter applied (i.e. tJIT(per)), as the setup and hold of the data strobes are relative to the corresponding clock signal crossing. Regardless of clock jitter values, these values shall be met.

11.2.4.3 tDQSS

This parameter is measured from a data strobe signal (DQSx_t, DQSx_c) crossing to the subsequent clock signal (CK_t/CK_c) crossing. When the device is operated with input clock jitter, this parameter needs to be de-rated by the actual period jitter tJIT(per),act of the input clock in excess of the allowed period jitter tJIT(per),allowed.

$$tDQSS(min, derated) = 0.75 - \frac{tJIT(per), act, min - tJIT(per), allowed, min}{tCK(avg)}$$

$$tDQSS(max, derated) = 1.25 - \frac{tJIT(per), act, max - tJIT(per), allowed, max}{tCK(avg)}$$

For example,

if the measured jitter into an LPDDR4X device has tCK(avg) = 625ps, tJIT(per),act,min = -xxps, and tJIT(per),act,max = +xx ps, then:

tDQSS,(min,derated) = 0.75 - (-xx + yy)/625 = xxxx tCK(avg)

tDQSS,(max,derated) = 1.25 - (xx . yy)/625 = xxxx tCK(avg)

11.3 LPDDR4X Refresh Requirements

[Table 56] LPDDR4X Refresh Requirement Parameters (per density)

Parameter		Symbol	16Gb	Unit
Density per Channel			8Gb	
Number of Banks per Channel			8	
Refresh Window Tcase ≤ 85°C		t_{REFW}	32	ms
Refresh Window 1/2-Rate Refresh		t_{REFW}	16	ms
Refresh Window 1/4-Rate Refresh		t_{REFW}	8	ms
Required number of REFRESH commands in a t_{REFW} window (min)		R	8,192	-
Average Refresh Internal	REFab	t_{REFI} 3)	3.904	us
	REFpb	t_{REFIpb}	488	ns
Refresh Cycle time (All Banks)		t_{RFCab}	280	ns
Refresh Cycle time (Per Bank)		t_{RFCpb}	140	ns
Per-bank Refresh to Per-bank Refresh different bank Time		$t_{pbR2pbR}$	90	ns

NOTE :
1) Refresh for each channel is independent of the other channel on the die, or other channels in a package. Power delivery in the user's system should be verified to make sure the DC operating conditions are maintained when multiple channels are refreshed simultaneously.
2) Self refresh abort feature is available for higher density devices starting with 12Gb dual channel device and 6Gb single channel device and tXSR_abort(min) is defined as tRFCpb + 17.5ns.
3) t_{REFI} values for all bank refresh is Tc = -25~85°C, Tc means Operating Case Temperature.

11.4 AC Timing

[Table 57] LPDDR4X AC Timing Table

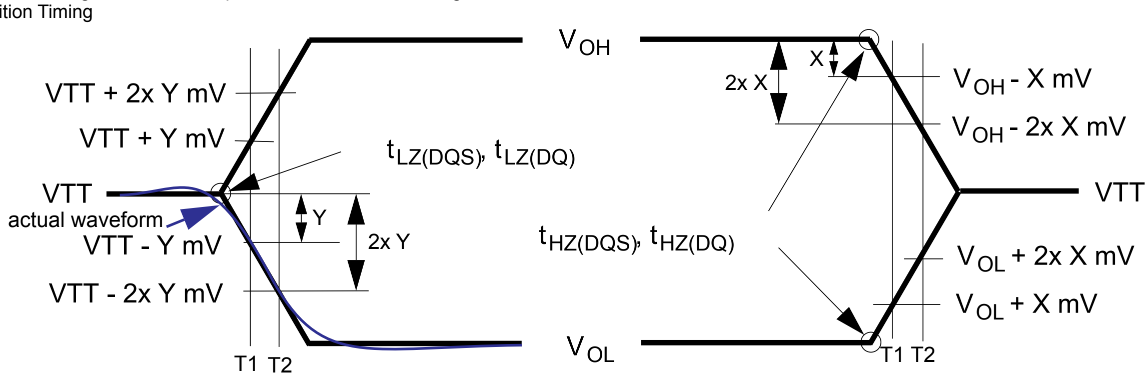
Parameter	Symbol	Min/Max	LPDDR4X 3733Mbps	Unit
Maximum clock frequency		~	1866	MHz
Clock Timing				
Average Clock Period	$t_{CK(avg)}$	MIN	0.536	ns
		MAX	100	
Average HIGH pulse width	$t_{CH(avg)}$	MIN	0.45	t_{CK}(avg)
		MAX	0.55	
Average LOW pulse width	$t_{CL(avg)}$	MIN	0.45	t_{CK}(avg)
		MAX	0.55	
Absolute clock period	$t_{CK(abs)}$	MIN	t_{CK}(avg) MIN + t_{JIT}(per) MIN	ns
Absolute HIGH clock pulse width	$t_{CH(abs)}$	MIN	0.43	t_{CK}(avg)
		MAX	0.57	
Absolute LOW clock pulse width	$t_{CL(abs)}$	MIN	0.43	t_{CK}(avg)
		MAX	0.57	
Clock period jitter	$t_{JIT(per)}$	MIN	-36	ps
		MAX	36	
Maximum Clock Jitter between two consecutive cycles	$t_{JIT(cc)}$	MAX	72	ps
Duty cycle jitter (with supported jitter)	$t_{JIT(duty)}$, allowed	MIN	min((t_{CH}(abs),min - t_{CH}(avg),min), (t_{CL}(abs),min - t_{CL}(avg),min)) × t_{CK}(avg)	ps
		MAX	max((t_{CH}(abs),max - t_{CH}(avg),max), (t_{CL}(abs),max - t_{CL}(avg),max)) × t_{CK}(avg)	
Core Parameters [17]				
READ latency (no DBI)	RL	MIN	32	t_{CK}(avg)
WRITE latency (set A)	WL	MIN	16	t_{CK}(avg)
ACTIVATE-to-ACTIVATE command period (same bank)	t_{RC}	MIN	t_{RAS} + t_{RPab} (with all-bank precharge) t_{RAS}+ t_{RPpb} (with per-bank precharge)	ns
Minimum Self-Refresh Time (Entry to Exit)	t_{SR}	MIN	max(15ns, 3tCK)	ns
SELF REFRESH exit to next valid command delay	t_{XSR}	MIN	Max (t_{RFCab} + 7.5ns, 2tCK)	ns
Exit power down to next valid command delay	t_{XP}	MIN	Max(7.5ns, 5tCK)	ns
CAS-to-CAS delay	t_{CCD}	MIN	BL/2	t_{CK}(avg)
CAS to CAS delay Masked Write	t_{CCDMW}[31]	MIN	4 × t_{CCD}	t_{CK}(avg)
Internal READ to PRECHARGE command delay	t_{RTP}	MIN	Max(7.5ns, 8tCK)	ns
RAS-to-CAS delay	t_{RCD}	MIN	Max (18ns, 4tCK)	ns
Row Precharge Time (single bank)	t_{RPpb}	MIN	Max (18ns, 4tCK)	ns
Row Precharge Time (all banks)	t_{RPab}	MIN	Max(21ns, 4tCK)	ns
Row active time	t_{RAS}	MIN	Max(42ns, 3tCK)	ns
		MAX	min (9 × t_{REFI} × Refresh Rate[19], 70.2)	us
WRITE recovery time	t_{WR}	MIN	Max(18ns, 6tCK)	ns
WRITE-to-READ delay	t_{WTR}	MIN	Max(10ns, 8tCK)	ns
Active bank-A to Active bank-B	t_{RRD}	MIN	Max(10ns, 4tCK)	ns
Precharge to Precharge Delay	t_{PPD}	MIN	4	tCK
Four-bank ACTIVATE Window	t_{FAW}	MIN	40	ns
CKE minimum pulse width during SELF REFRESH (low pulse width during SELF REFRESH)	t_{CKELPD}	MIN	Max(7.5ns, 3tCK)	ns
READ Parameters [4]				

Parameter	Symbol	Min/ Max	LPDDR4X 3733Mbps	Unit
Read preamble	t_{RPRE} 5), 8)	MIN	2.0	t_{CK}(avg)
0.5 tCK Read postamble	t_{RPST} 5), 9)	MIN	0.5	t_{CK}(avg)
1.5 tCK Read postamble	t_{RPST}	MIN	1.5	t_{CK}(avg)
DQ low-impedance time from CK_t, CK_c	$t_{LZ(DQ)}$ 5)	MIN	(RL × t_{CK}) + $t_{DQSCK(Min)}$ - 200ps	ps
DQ high impedance time from CK_t, CK_c	$t_{HZ(DQ)}$ 5)	MAX	(RL × t_{CK}) + $t_{DQSCK(Max)}$ + $t_{DQSQ(Max)}$ + (BL/2 × t_{CK}) - 100ps	ps
DQS_c low-impedance time from CK_t, CK_c	$t_{LZ(DQS)}$ 5)	MIN	(RL × t_{CK}) + $t_{DQSCK(Min)}$ - ($t_{PRE(Max)}$ × t_{CK}) - 200ps	ps
DQS_c high impedance time from CK_t, CK_c	$t_{HZ(DQS)}$ 5)	MAX	(RL × t_{CK}) + $t_{DQSCK(Max)}$+ (BL/2 × t_{CK}) - (RPST(Max) × t_{CK}) - 100ps	ps
DQS-DQ skew	t_{DQSQ}	MAX	0.18	UI
tDQSCK Parameters				
DQS output access time from CK_t/CK_c	t_{DQSCK} 14)	MIN	1500	ps
		MAX	3500	
DQS output access time from CK_t/CK_c temperature variation	t_{DQSCK_temp} 15)	MAX	4	ps/°C
DQS output access time from CK_t/CK_c voltage variation	t_{DQSCK_volt} 16)	MAX	7	ps/mV
CK to DQS rank to rank variation	$t_{DQSCK\_rank2rank}$ 22),23)	MAX	1.0	ns
Self Refresh Parameters				
Delay from SRE command to CKE Input low	t_{ESCKE} 24)	MIN	Max(1.75ns, 3nCK)	ns
Minimum Self Refresh Time	t_{SR} 24)	MIN	Max(15ns, 3tCK)	ns
Exit Self Refresh to Valid commands	t_{XSR} 24),25)	MIN	Max(tRFCab + 7.5ns, 2tCK)	ns
WRITE Parameters 4)				
Write command to 1^{st} DQS latching	t_{DQSS}	MIN	0.75	t_{CK}(avg)
		MAX	1.25	
DQS input high-level width	t_{DQSH}	MIN	0.4	t_{CK}(avg)
DQS input low-level width	t_{DQSL}	MIN	0.4	t_{CK}(avg)
DQS falling edge to CK setup time	t_{DSS}	MIN	0.2	t_{CK}(avg)
DQS falling edge hold time from CK	t_{DSH}	MIN	0.2	t_{CK}(avg)
Write preamble	t_{WPRE}	MIN	2.0	t_{CK}(avg)
0.5 tCK Write postamble	t_{WPST} 21)	MIN	0.5	t_{CK}(avg)
1.5 tCK Write postamble	t_{WPST} 21)	MIN	1.5	t_{CK}(avg)
ZQ Calibration Parameters				
ZQ Calibration	t_{ZQCAL}	MIN	1	us
ZQ Calibration Values Latch Time	t_{ZQLAT}	MN	Max (30ns, 8tCK)	ns
ZQ Calibration RESET time	$t_{ZQRESET}$	MIN	Max (50ns, 3tCK)	ns
Power Down Parameters				
CKE minimum pulse width (HIGH and LOW pulse width)	t_{CKE}	MIN	max(7.5ns, 4tCK)	ns
Delay from Valid command to CKE Input low	t_{CMDCKE} 26)	MIN	Max(1.75ns, 3tCK)	ns
Valid Clock Requirement after CKE Input Low	t_{CKELCK} 26)	MIN	Max(5ns, 5tCK)	-
Valid CS Requirement before CKE Input Low	t_{CSCKE}	MIN	1.75	ns
Valid CS Requirement after CKE Input Low	t_{CKELCS}	MIN	Max(5ns,5tCK)	ns
Valid Clock Requirement before CKE Input High	t_{CKCKEH} 26)	MIN	Max(1.75ns, 3tCK)	-ns

Parameter	Symbol	Min/ Max	LPDDR4X 3733Mbps	Unit
Exit power- down to next valid command delay	t_{XP} 26)	MIN	Max(7.5ns, 5tCK)	ns
Valid CS Requirement before CKE Input High	t_{CSCKEH}	MIN	1.75	ns
Valid CS Requirement after CKE Input High	t_{CKEHCS}	MIN	Max(7.5ns,5tCK)	ns
Valid Clock and CS Requirement after CKE Input low after MRW Command	$t_{MRWCKEL}$ 26)	MIN	Max(14ns,10tCK)	ns
Valid Clock and CS Requirement after CKE Input low after ZQ Calibration Start Command	t_{ZQCKE} 26)	MIN	Max(1.75ns,3tCK)	ns
Command Address Input Parameters 4)				
Rx Mask voltage - p-p	VcIVW	MAX	150	mV
Rx timing window	TcIVW	MAX	0.3	UI*
CA AC input pulse amplitude pk-pk	VIHL_AC	MIN	180	mV
CA input pulse width	TcIPW	MIN	0.6	UI*
Input Slew Rate over VcIVW	SRIN_cIVW	MIN	1	V/ns
		MAX	7	
Mode Register Read/Write AC Timing				
Additional time after tXP has expired until MRR command	t_{MRRI}	MIN	t_{RCD} + 3nCK	-
MODE REGISTER READ command period	t_{MRR}	MIN	8	nCK
MODE REGISTER WRITE command period	t_{MRW}	MIN	Max(10ns, 10nCK)	-
Mode register set command delay	t_{MRD}	MIN	Max(14ns, 10tCK)	-
Boot Parameters (10 MHz - 55 MHz) 11), 12), 13)				
Clock Cycle Time	t_{CKb}	max	100	ns
		MIN	18	
Address & Control Input Setup Time	t_{ISb}	MIN	1150	ps
Address & Control Input Hold Time	t_{IHb}	MIN	1150	ps
DQS Output Data Access Time from CK_t/CK_c	t_{DQSCKb}	MIN	2.0	ns
		MAX	10.0	
Data Strobe Edge to Output Data Edge	t_{DQSQb}	MAX	1.2	ns
Command Bus Training AC Parameters				
Valid Clock Requirement after CKE Input low	t_{CKELCK}	MIN	Max(5ns, 5nCK)	tCK
Data Setup for VREF Training Mode	$t_{DStrain}$	MIN	2	ns
Data Hold for VREF Training Mode	$t_{DHtrain}$	MIN	2	ns
Asynchronous Data Read	t_{ADR}	MAX	20	ns
CA Bus Training command to CA Bus Training command delay	t_{CACD} 29)	MIN	RU(t_{ADR}/t_{CK})	tCK
Valid Strobe Requirement before CKE Low	t_{DQSCKE} 30)	MIN	10	ns
First CA Bus Training Command Following CKE LOW	t_{CAENT}	MIN	250	ns
VREF Step Time-multiple steps	t_{VREFCA_LONG}	MAX	250	ns
VREF Step Time-one step	t_{VREFCA_SHORT}	MAX	80	ns
Valid Clock Requirement before CS High	$t_{CKPRECS}$	MIN	$2t_{CK} + t_{XP}$ (t_{XP} = max(7.5ns, 5nCK))	-
Valid Clock Requirement after CS High	$t_{CKPSTCS}$	MIN	max(7.5ns, 5nCK)	-
Minimum delay from CS to DQS toggle in command bus training	t_{CS_VREF}	MIN	2	tCK
Minimum delay from CKE High to Strobe High Impedance	$t_{CKEHDQS}$	-	10	ns
Valid Clock Requirement before CKE Input High	t_{CKCKEH}	MIN	Max(1.75ns, 3tCK)	
CA Bus Training CKE High to DQ Tri-state	t_{MRZ}	MIN	1.5	ns
ODT turn-on Latency from CKE	$t_{CKELODTon}$	MIN	20	ns
ODT turn-off Latency from CKE	$t_{CKELODToff}$	MIN	20	ns

Parameter	Symbol	Min/ Max	LPDDR4X 3733Mbps	Unit
Exit Command Bus Training Mode to next valid command delay [33)]	t_{XCBT_Short}	MIN	Max(5nCK, 200ns)	-
	t_{XCBT_Middle}	MIN	Max(5nCK, 200ns)	-
	t_{XCBT_Long}	MIN	Max(5nCK, 250ns)	-
Write Leveling Parameters				
DQS_t/DQS_c delay after write leveling mode is programmed	$t_{WLDQSEN}$	MIN	20	tCK
Write preamble for Write Leveling	t_{WLWPRE}	MIN	20	tCK
First DQS_t/DQS_c edge after write leveling mode is programmed	t_{WLMRD}	MIN	40	tCK
Write leveling output delay	t_{WLO}	MAX	20	ns
Mode register set command delay	t_{MRD}	MIN	Max(14ns, 10tCK)	ns
Valid Clock Requirement before DQS Toggle	$t_{CKPRDQS}$	MIN	Max(7.5ns, 4tCK)	-
Valid Clock Requirement after DQS Toggle	$t_{CKPSTDQS}$	MIN	Max(7.5ns, 4tCK)	-
Write leveling hold time	t_{WLH} [27)]	MIN	60	ps
Write leveling setup time	t_{WLS} [27)]	MIN	60	ps
Write leveling input valid window	t_{WLIVW} [28)]	MIN	100	ps
Temperature De-Rating AC Timing [20)]				
DQS output access time from CK_t/CK_c (derated)	t_{DQSCK}	MAX	3600	ps
RAS-to-CAS delay (derated)	t_{RCD}	MIN	t_{RCD} + 1.875	ns
ACTIVATE-to- ACTIVATE command period (derated)	t_{RC}	MIN	t_{RC} + 3.75	ns
Row active time (derated)	t_{RAS}	MIN	t_{RAS} + 1.875	ns
Row precharge time (derated)	t_{RP}	MIN	t_{RP} + 1.875	ns
Active bank A to active bank B (derated)	t_{RRD}	MIN	t_{RRD} + 1.875	ns

NOTE :
1) Frequency values are for reference only. Clock cycle time (tCK) is used to determine device capabilities.
2) All AC timings assume an input slew rate of TBDV/ns.
3) Measured with 4 V/ns differential CK_t/CK_c slew rate and nominal VIX.
4) READ, WRITE, and Input setup and hold values are referenced to V_{REF}.
5) For LOW-to-HIGH and HIGH-to-LOW transitions, the timing reference is at the point when the signal crosses the transition threshold (V_{TT}). tHZ and tLZ transitions occur in the same access time (with respect to clock) as valid data transitions. These parameters are not referenced to a specific voltage level but to the time when the device output is no longer driving (for tRPST, tHZ(DQS) and tHZ(DQ)), or begins driving (for tRPRE, tLZ(DQS), tLZ(DQ)). Operating and Timing [Burst Read:RL=12, BL=8, tDQSCK<tCK] shows a method to calculate the point when device is no longer driving tHZ(DQS) and tHZ(DQ), or begins driving tLZ(DQS), tLZ(DQ) by measuring the signal at two different voltages. The actual voltage measurement points are not critical as long as the calculation is consistent.
6) Output Transition Timing



Start driving point = 2 x T1 - T2 End driving point = 2 x T1 - T2

7)The parameters tLZ(DQS), tLZ(DQ), tHZ(DQS), and tHZ(DQ) are defined as single-ended. The timing parameters tRPRE and tRPST are determined from the differential signal DQS_t-DQS_c.
8) Measured from the point when DQS_t/DQS_c begins driving the signal to the point when DQS_t/DQS_c begins driving the first rising strobe edge. See Pre and Post-amble section in Operating & Timing spec
9) Measured from the last falling strobe edge of DQS_t/DQS_c to the point when DQS_t/DQS_c finishes driving the signal.
10) Input set-up/hold time for signal (CA[9:0], CS).
11) To ensure device operation before the device is configured, a number of AC boot-timing parameters are defined in this table. Boot parameter symbols have the letter b appended (for example, tCK during boot is tCKb).
12) The LPDDR4X device will set some default values upon receiving a RESET (MRW) command as specified in "Definition".
13) The output skew parameters are measured with default output impedance settings using the reference load.
14) Includes DRAM process, voltage and temperature variation. It includes the AC noise impact for frequencies > 20 MHz and max voltage of 45 mV pk-pk from DC-20 MHz at a fixed temperature on the package. The voltage supply noise must comply to the component Min-Max DC Operating conditions.
15) tDQSCK_temp max delay variation as a function of Temperature.

16) tDQSCK_volt max delay variation as a function of DC voltage variation for VDDQ and VDD2. t_{DQSCK_volt} should be used to calculate timing variation due to VDDQ and VDD2 noise < 20 MHz. Host controller do not need to account for any variation due to VDDQ and VDD2 noise > 20 MHz. The voltage supply noise must comply to the component Min-Max DC Operating conditions. The voltage variation is defined as the Max[abs{tDQSCKmin@V1-tDQSCKmax@V2}, abs{tDQSCKmax@V1-tDQSCKmin@V2}]/abs{V1-V2}.
17) Precharge to precharge timing restriction does not apply to Auto-Precharge commands.
18) tXSR/tXP/tZQLAT are defined as “to the first rising clock edge next valid command”.
19) Refresh Rate is specified by MR4, OP[2:0].
20) Timing derating applies for operation at 85°C to 105°C.
21) The length of Write Postamble depends on MR3 OP1 setting.
22) The same voltage and temperature are applied to $t_{DQS2CK\_rank2rank}$.
23) $t_{DQSCK\_rank2rank}$ parameter is applied to multi-ranks per byte lane within a package consisting of the same design dies.
24) Delay time has to satisfy both analog time(ns) and clock count(tCK).
It means that t_{ESCKE} will not expire until CK has toggled through at least 3 full cycles (3 *tCK) and 1.75ns has transpired. The case which 3tCK is applied to is shown below.

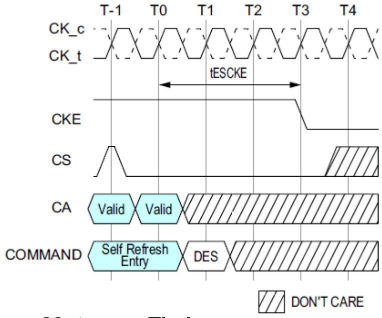


Figure 23. t_{ESCKE} Timing

25) MRR-1, CAS-2, DES, MPC, MRW-1 and MRW-2 except PASR Bank/Segment setting are only allowed during this period.
26) Delay time has to satisfy both analog time(ns) and clock count(nCK).
For example, t_{CMDCKE} will not expire until CK has toggled through at least 3 full cycles (3 *tCK) and 1.75ns has transpired.The case which 3nCK is applied to is shown below.

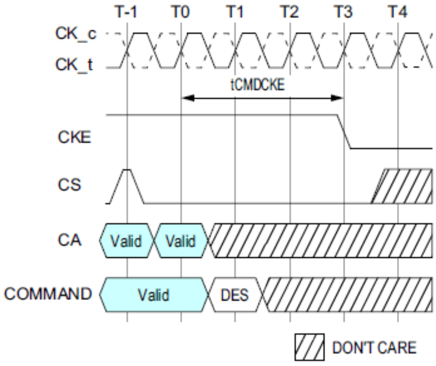


Figure 24. t_{CMDCKE} Timing

27) In addition to the traditional setup and hold time specifications above, there is value in a input valid window based specification for write-leveling training. As the training is based on each device, worst case process skews for setup and hold do not make sense to close timing between CK and DQS.
28) t_{WLIVW} is defined in a similar manner to tdIVW_Total, except that here it is a DQS input valid window with respect to CK. This would need to account for all VT (voltage and temperature) drift terms between CK and DQS within the DRAM that affect the write-leveling input valid window.
The DQS input mask for timing with respect to CK is shown in Figure 25. The "total" mask (tWLIVW) defines the time the input signal must not encroach in order for the DQS input to be successfully captured by CK with a BER of lower than tbd. The mask is a receiver property and it is not the valid data-eye.

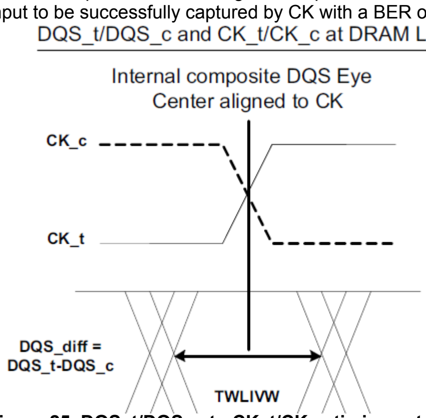


Figure 25. DQS_t/DQS_c to CK_t/CK_c timings at the DRAM pins referenced from the internal latch

29) If tCACD is violated, the data for samples which violate tCACD will not be available, except for the last sample (where tCACD after this sample is met). Valid data for the last sample will be available after tADR.
30) DQS_t has to retain a low level during t_{DQSCKE} period, as well as DQS_c has to retain a high level.
31) See Masked Write Operation for detail.
32) Devices supporting 4266Mbps specification shall support these timings at lower data rates.
33) Exit Command Bus Training Mode to next valid command delay Time depends on value of V_{REF}(CA) setting: MR12 OP[5:0] and V_{REF}(CA) Range: MR12 OP[6] of FSP-OP 0 and 1. The details are shown in tFC value mapping table. Additionally exit Command Bus Training Mode to next valid command delay Time may affect V_{REF}(DQ) setting. Settling time of V_{REF}(DQ) level is same as V_{REF}(CA) level.

11.5 CA Rx Voltage and Timing

The command and address(CA) including CS input receiver compliance mask for voltage and timing is shown in the figure below. All CA, CS signals apply the same compliance mask and operate in single data rate mode.
The CA input receiver mask for voltage and timing is shown in the figure below is applied across all CA pins. The receiver mask (Rx Mask) defines the area that the input signal must not encroach in order for the DRAM input receiver to be expected to be able to successfully capture a valid input signal; it is not the valid data-eye.

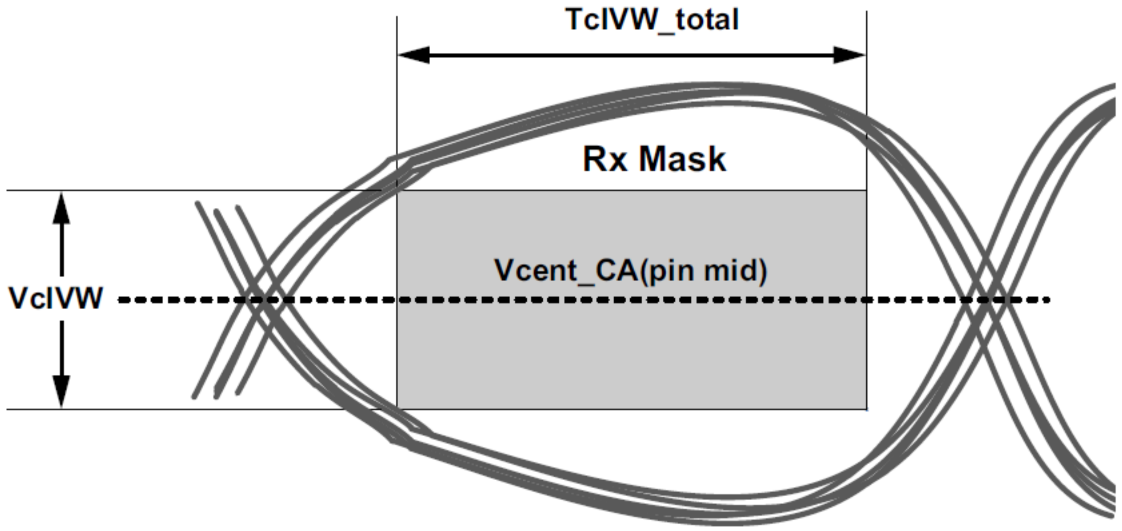


Figure 26. CA Receiver (Rx) mask

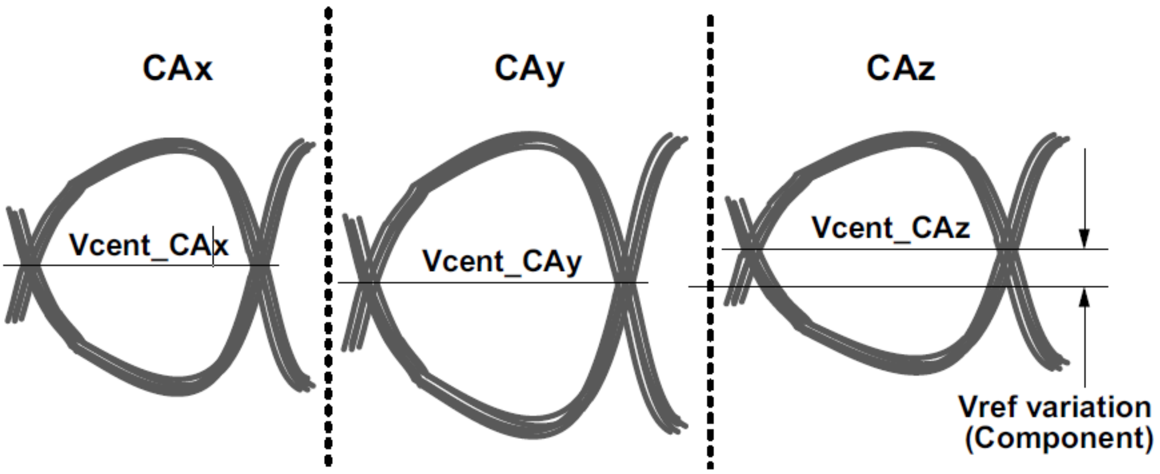


Figure 27. Across pin V_{REFCA} voltage variation

Vcent_CA(pin avg) is defined as the midpoint between the largest Vcent_CA voltage level and the smallest Vcent_CA voltage level across all CA and CS pins for a given DRAM component. Each CA Vcent level is defined by the center, i.e. widest opening, of the cumulative data input eye as depicted in Figure 27..This clarifies that any DRAM component level variation must be accounted for within the DRAM CA Rx mask. The component level Vref will be set by the system to account for Ron and ODT settings.

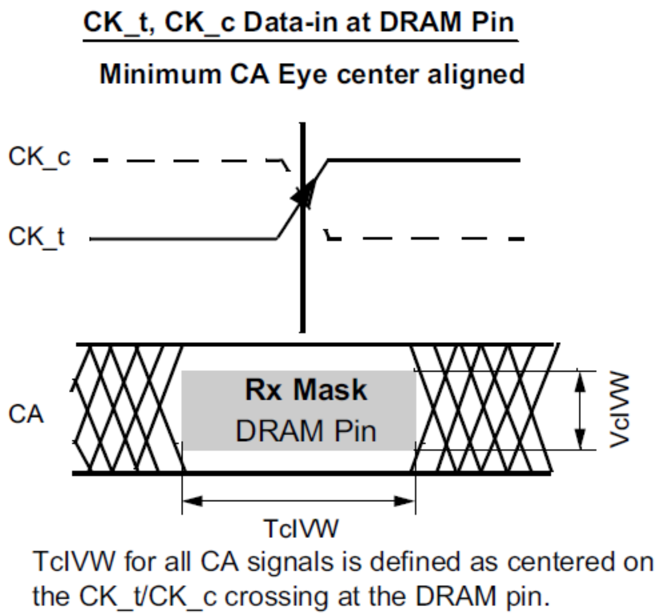


Figure 28. CA Timings at the DRAM pins

All of the timing terms in Figure 28. are measured from the CK_t/CK_c to the center(midpoint) of the TcIVW window taken at the VcIVW_total voltage levels centered around Vcent_CA(pin mid).

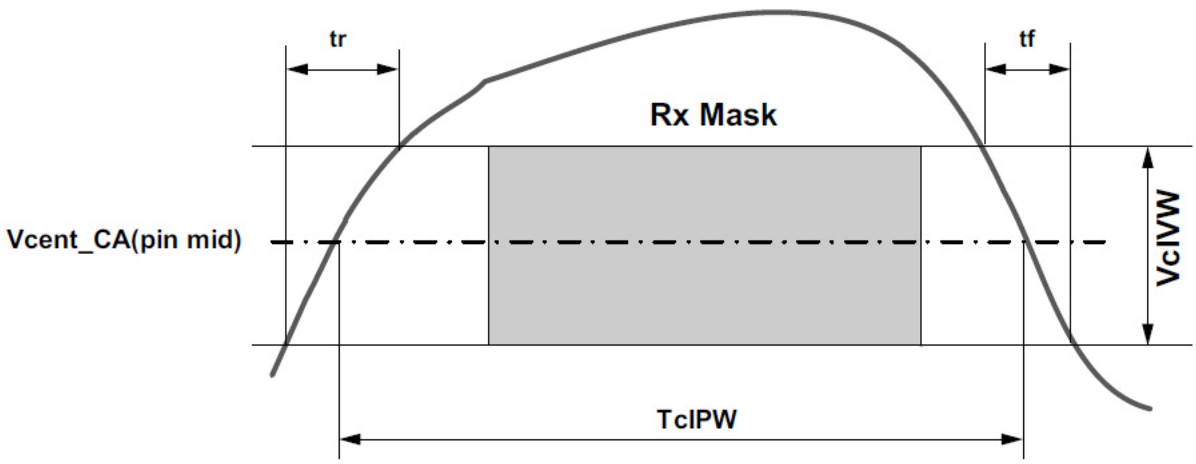


Figure 29. CA TcIPW and SRIN_cIVW definition (for each input pulse)

NOTE :
1) SRIN_cIVW=VcIVW_Total/(tr or tf), signal must be monotonic within tr and tf range.

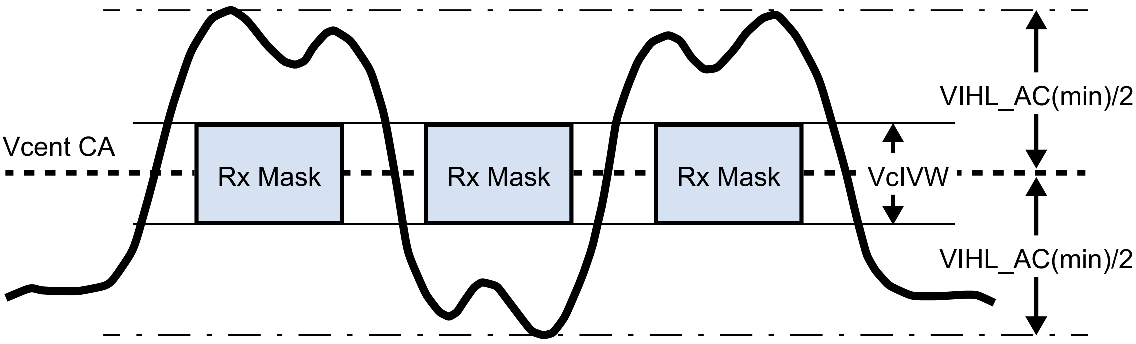


Figure 30. CA VIHL_AC definition (for each input pulse)

[Table 58] DRAM CMD/ADR, CS

Symbol	Parameter	DQ-3733		Unit	NOTE
		min	max		
VcIVW	Rx Mask voltage - p-p	-	150	mV	1,2,3
TcIVW	Rx timing window	-	0.3	UI*	1,2,3
VIHL_AC	CA AC input pulse amplitude pk-pk	180	-	mV	4,7
TcIPW	CA input pulse width	0.6	-	UI*	5
SRIN_cIVW	Input Slew Rate over VcIVW	1	7	V/ns	6

*** UI=tCK(avg)min**

NOTE :
1) CA Rx mask voltage and timing parameters at the pin including voltage and temperature drift.
2) Rx mask voltage VcIVW total(max) must be centered around Vcent_CA (pin_mid).
3) Vcent_CA must be within the adjustment range of the CA internal Vref.
4) CA only input pulse signal amplitude into the receiver must meet or exceed VIHL AC at any point over the total UI. No timing requirement above level. VIHL AC is the peak to peak voltage centered around Vcent_CA(pin mid) such that VIHL_AC/2 min must be met both above and below Vcent_CA.
5) CA only minimum input pulse width defined at the Vcent_CA (pin mid).
6) Input slew rate over VcIVW Mask centered at Vcent_CA (pin mid).
7) VIHL_AC does not have to be met when no transitions are occurring.

11.6 DRAM Data Timing

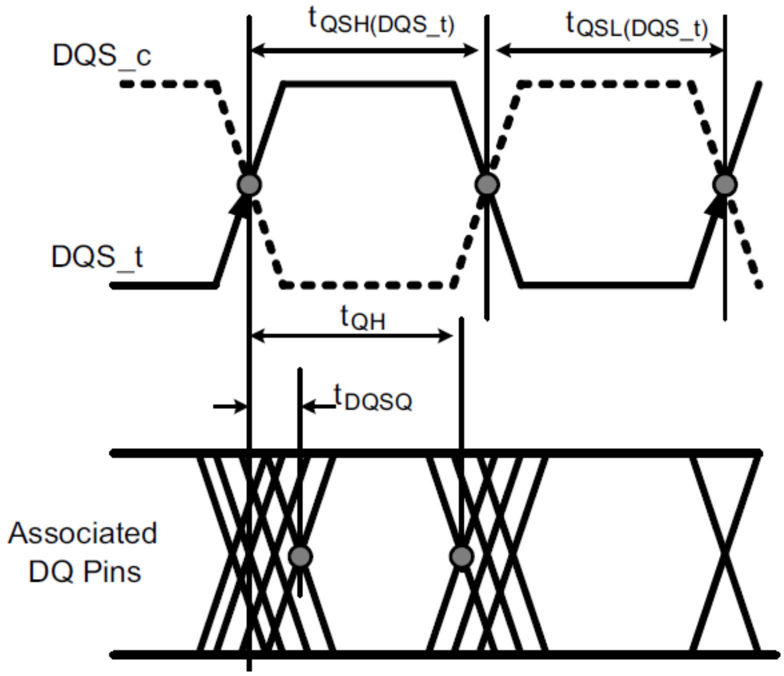


Figure 31. Read data timing definitions tQH and tDQSQ across on DQ signals per DQS group

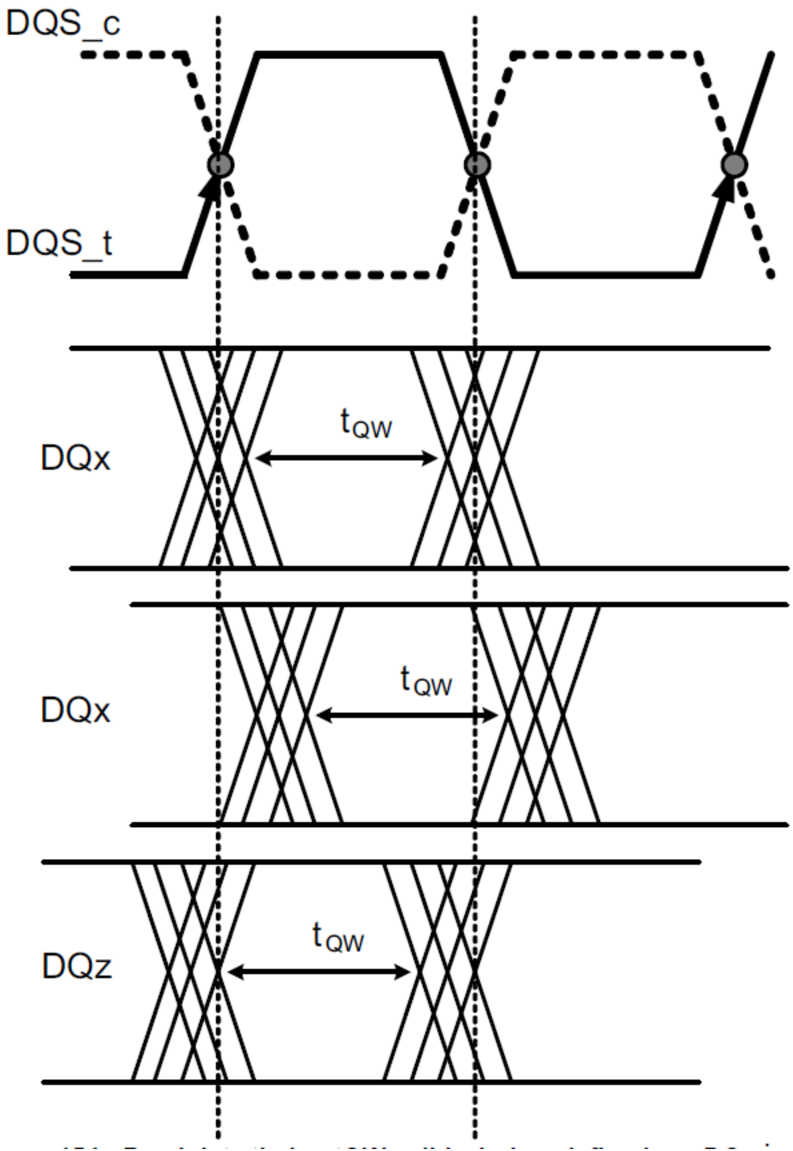


Figure 32. Read data timing tQW valid window defined per DQ signal

[Table 59] Read output timings

Parameter	Symbol	LPDDR4X-3733		Units	Notes
		Min	Max		
Data Timing					
DQS_t,DQS_c to DQ Skew total, per group, per access (DBI-Disabled)	t_{DQSQ}	-	0.18	UI	
DQ output hold time total from DQS_t, DQS_c (DBI-Disabled)	t_{QH}	min(t_{QSH}, t_{QSL})	-	UI	
DQ output window time total, per pin (DBI-Disabled)	t_{QW_total}	0.7	-	UI	3
DQ output window time deterministic, per pin (DBI-Disabled)	t_{QW_dj}	-	TBD	UI	2,3
DQS_t,DQS_c to DQ Skew total, per group, per access (DBI Enabled)	t_{DQSQ_DBI}	-	0.18	UI	
DQ output hold time total from DQS_t, DQS_c (DBI-Enabled)	t_{QH_DBI}	min(t_{QSH_DBI}, t_{QSL_DBI})	-	UI	
DQ output window time total, per pin (DBI-Enabled)	$t_{QW\_total\_DBI}$	0.7	-	UI	3
Data Strobe Timing					
DQS_t, DQS_c differential output low time (DBI-Disabled)	t_{QSL}	$t_{CL(abs)}$-0.05	-	$t_{CK(avg)}$	3,4
DQS_t, DQS_c differential output high time (DBI-Disabled)	t_{QSH}	$t_{CH(abs)}$-0.05	-	$t_{CK(avg)}$	3,5
DQS_t, DQS_c differential output low time (DBI-Enabled)	t_{QSL_DBI}	$t_{CL(abs)}$-0.045	-	$t_{CK(avg)}$	4,6
DQS_t, DQS_c differential output high time (DBI-Enabled)	t_{QSH_DBI}	$t_{CH(abs)}$-0.045	-	$t_{CK(avg)}$	5,6

Unit UI = tCK(avg)min/2

NOTE :
1) The deterministic component of the total timing. Measurement method tbd.
2) This parameter will be characterized and guaranteed by design.
3) This parameter is function of input clock jitter. These values assume the min tCH(abs) and tCL(abs). When the input clock jitter min tCH(abs) and tCL(abs) is 0.44 or greater of tck(avg) the min value of tQSL will be tCL(abs)-0.04 and tQSH will be tCH(abs) -0.04.
4) tQSL describes the instantaneous differential output low pulse width on DQS_t - DQS_c, as it measured the next rising edge from an arbitrary falling edge.
5) tQSH describes the instantaneous differential output high pulse width on DQS_t - DQS_c, as it measured the next rising edge from an arbitrary falling edge.
6) This parameter is function of input clock jitter. These values assume the min tCH(abs) and tCL(abs). When the input clock jitter min tCH(abs) and tCL(abs) is 0.44 or greater of tck(avg) the min value of tQSL will be tCL(abs)-0.04 and tQSH will be tCH(abs) -0.04.

11.7 DQ Rx Voltage And Timing

The DQ input receiver mask for voltage and timing is shown Figure 33. is applied per pin. The “total” mask (VdIVW_total, TdiVW_total) defines the area the input signal must not encroach in order for the DQ input receiver to successfully capture an input signal with a BER of lower than tbd. The mask is a receiver property and it is not the valid data-eye.

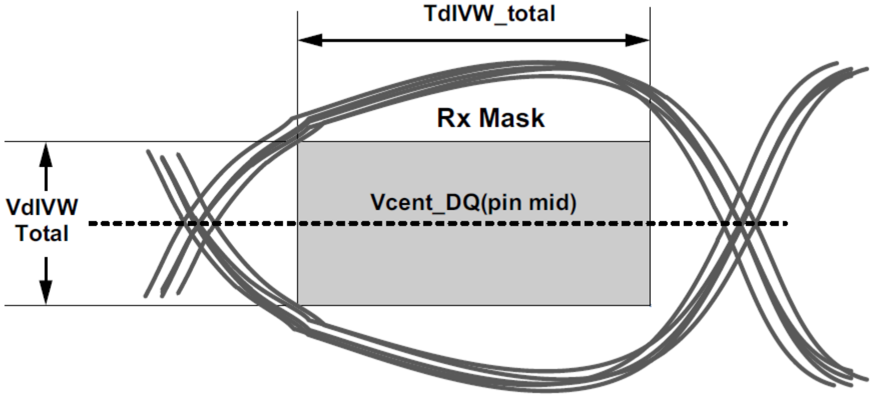


Figure 33. DQ Receiver(Rx) mask

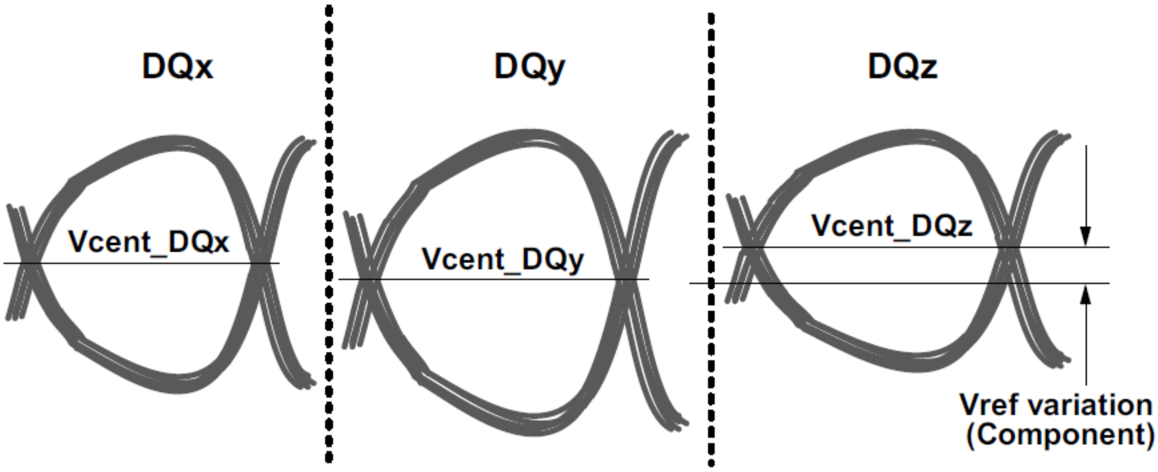


Figure 34. Across pin V_{REF}DQ voltage variation

Vcent_DQ(pin mid) is defined as the midpoint between the largest Vcent_DQ voltage level and the smallest Vcent_DQ voltage level across all DQ pins for a given DRAM component. Each DQ Vcent is defined by the center, i.e. widest opening, of the cumulative data input eye as depicted in Figure 34..This clarifies that any DRAM component level variation must be accounted for within the DRAM Rx mask. The component level Vref will be set by the system to account for Ron and ODT settings.

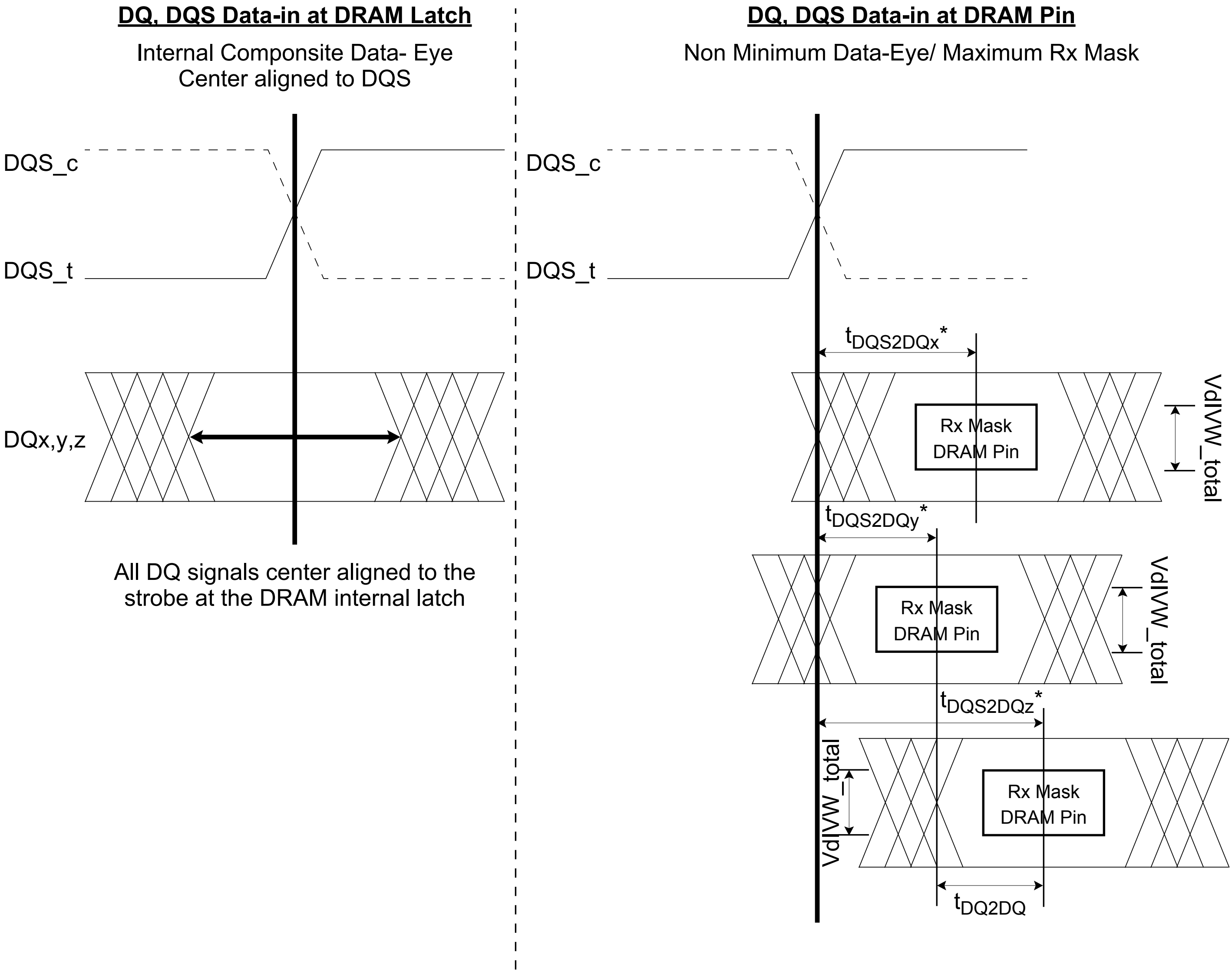


Figure 35. DQ to DQS t_{DQS2DQ} & t_{DQDQ} Timings at the DRAM pins referenced from the internal latch

NOTE :
1) t_{DQS2DQ} is measured at the center(midpoint) of the TdiVW window.
2) DQz represents the max t_{DQS2DQ} in this example.
3) DQy represents the min t_{DQS2DQ} in this example.

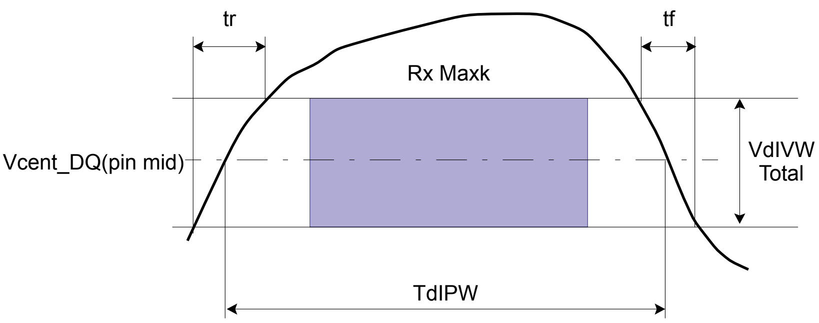


Figure 36. DQ TdIPW and SRIN_dIVW definition (for each input pulse)

NOTE :

1) SRIN_dIVW=VdIVW_Total/(tr or tf), signal must be monotonic within tr and tf range.

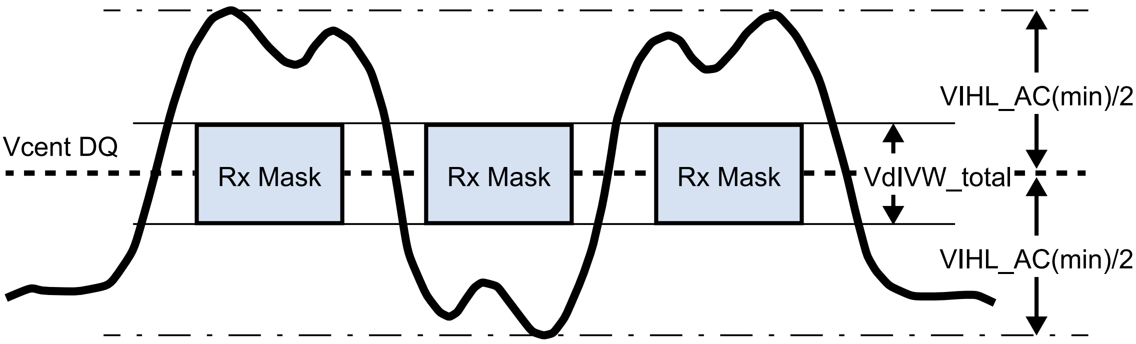


Figure 37. DQ VIHL_AC definition (for each input pulse)

[Table 60] DRAM DQs In Receive Mode;

Symbol	Parameter	3733		Unit	NOTE
		min	max		
VdIVW_total	Rx Mask voltage - p-p total	-	130	mV	1,2,3,4
TdIVW_total	Rx timing window total (At VdIVW voltage levels)	-	0.25	UI*	1,2,4
TdIVW_1bit	Rx timing window 1 bit toggle (At VdIVW voltage levels)	-	TBD	UI*	1,2,4,12
VIHL_AC	DQ AC input pulse amplitude pk-pk	180	-	mV	5,13
TdIPW DQ	Input pulse width (At Vcent_DQ)	0.45	-	UI*	6
t_{DQS2DQ}	DQ to DQS offset	250	700	ps	7
t_{DQ2DQ}	DQ to DQ offset	-	30	ps	8
t_{DQS2DQ_temp}	DQ to DQS offset temperature variation	-	0.4	ps/°C	9
t_{DQS2DQ_volt}	DQ to DQS offset voltage variation	-	25	ps/50mV	10
SRIN_dIVW	Input Slew Rate over VdIVW_total	1	7	V/ns	11
$t_{DQS2DQ\_rank2rank}$	DQ to DQS offset rank to rank variation	-	200	ps	14,15,16

*** UI=tck(avg)min/2**

NOTE :
1) Data Rx mask voltage and timing parameters are applied per pin and includes the DRAM DQ to DQS voltage AC noise impact for frequencies >20MHz and max voltage of 45mv pk-pk from DC-20MHz at a fixed temperature on the package. The voltage supply noise must comply to the component Min-Max DC operating conditions.
2) The design specification is a BER <tbd. The BER will be characterized and extrapolated if necessary using a dual dirac method.
3) Rx mask voltage VdIVW total(max) must be centered around Vcent_DQ(pin_mid).
4) Vcent_DQ must be within the adjustment range of the DQ internal Vref.
5) DQ only input pulse amplitude into the receiver must meet or exceed VIHL AC at any point over the total UI. No timing requirement above level. VIHL AC is the peak to peak voltage centered around Vcent_DQ(pin_mid) such that VIHL_AC/2 min must be met both above and below Vcent_DQ
6) DQ only minimum input pulse width defined at the Vcent_DQ(pin_mid).
7) DQ to DQS offset is within byte from DRAM pin to DRAM internal latch. Includes all DRAM process, voltage and temperature variation
8) DQ to DQ offset defined within byte from DRAM pin to DRAM internal latch for a given component.
9) TDQS2DQ max delay variation as a function of temperature.
10) TDQS2DQ max delay variation as a function of the DC voltage variation for VDDQ and VDD2. It includes the VDDQ and VDD2 AC noise impact for frequencies > 20MHz and max voltage of 45mv pk-pk from DC-20MHz at a fixed temperature on the package.
11) Input slew rate over VdIVW Mask centered at Vcent_DQ(pin mid).
12) Rx mask defined for a one pin toggling with other DQ signals in a steady state.
13) VIHL_AC does not have to be met when no transitions are occurring.
14) The same voltage and temperature are applied to $t_{DQS2DQ\_rank2rank}$.
15) $t_{DQS2DQ\_rank2rank}$ parameter is applied to multi-ranks per byte lane within a package consisting of the same design dies.
16) $t_{DQS2DQ\_rabk2rank}$ support was added to JESD209-4B, some older devices designed to support JESD209-4 and JESD209-4A may not support this parameter. Refer to vendor datasheet.

LPDDR4 SDRAM Command Definitions and Timing Diagrams

1.0 TRUTH TABLES

Operation or timing that is not specified is illegal, and after such an event, in order to guarantee proper operation, the LPDDR4 device must be reset or power-cycled and then restarted through the specified initialization sequence before normal operation can continue.
CKE signal has to be held High when the commands listed in the command truth table input.

[Table 1] Command truth table

	SDR Command Pins	SDR CA pins (6)							
SDRAM Command	**CS**	**CA0**	**CA1**	**CA2**	**CA3**	**CA4**	**CA5**	**CK_t edge**	**Notes**
Deselect (DES)	L	X						R1	1,2
Multi-Purpose Command (MPC)	H	L	L	L	L	L	OP6	R1	1,9
	L	OP0	OP1	OP2	OP3	OP4	OP5	R2	
Precharge (PRE) (Per Bank, All Bank)	H	L	L	L	L	H	AB	R1	1,2,3,4
	L	BA0	BA1	BA2	V	V	V	R2	
Refresh (REF) (Per Bank, All Bank)	H	L	L	L	H	L	AB	R1	1,2,3,4
	L	BA0	BA1	BA2	V	V	V	R2	
Self Refresh Entry (SRE)	H	L	L	L	H	H	V	R1	1,2
	L	V						R2	
Write-1 (WR-1)	H	L	L	H	L	L	BL	R1	1,2,3,6,7,9
	L	BA0	BA1	BA2	V	C9	AP	R2	
Self Refresh Exit (SRX)	H	L	L	H	L	H	V	R1	1,2
	L	V						R2	
Mask Write-1 (MWR-1)	H	L	L	H	H	L	L	R1	1,2,3,5,6,9
	L	BA0	BA1	BA2	V	C9	AP	R2	
RFU	H	L	L	H	H	H	V	R1	1,2
	L	V						R2	
Read-1 (RD-1)	H	L	H	L	L	L	BL	R1	1,2,3,6,7,9
	L	BA0	BA1	BA2	V	C9	AP	R2	
CAS-2 (Write-2, Mask Write-2, Read-2, MRR-2, MPC)	H	L	H	L	L	H	C8	R1	1,8,9
	L	C2	C3	C4	C5	C6	C7	R2	
RFU	H	L	H	L	H	L	V	R1	1,2
	L	V						R2	
RFU	H	L	H	L	H	H	V	R1	1,2
	L	V						R2	
Mode Register Write-1 (MRW-1)	H	L	H	H	L	L	OP7	R1	1,11
	L	MA0	MA1	MA2	MA3	MA4	MA5	R2	
Mode Register Write-2 (MRW-2)	H	L	H	H	L	H	OP6	R1	1,11
	L	OP0	OP1	OP2	OP3	OP4	OP5	R2	
Mode Register Read-1 (MRR-1)	H	L	H	H	H	L	V	R1	1,2,12
	L	MA0	MA1	MA2	MA3	MA4	MA5	R2	
RFU	H	L	H	H	H	H	V	R1	1,2
	L	V						R2	
Activate-1 (ACT-1)	H	H	L	R12	R13	R14	R15	R1	1,2,3,10
	L	BA0	BA1	BA2	V	R10	R11	R2	
Activate-2 (ACT-2)	H	H	H	R6	R7	R8	R9	R1	1,10
	L	R0	R1	R2	R3	R4	R5	R2	

NOTE:
1) All LPDDR4 commands except for Deselect are 2 clock cycle long and defined by states of CS and CA[5:0] at the first rising edge of clock. Deselect command is 1 clock cycle long.
2) “V” means “H” or “L” (a defined logic level). “X” means don’t care in which case CA[5:0] can be floated.
3) Bank addresses BA[2:0] determine which bank is to be operated upon.
4) AB “HIGH” during Precharge or Refresh command indicates that command must be applied to all banks and bank address is a don’t care.
5) Mask Write-1 command supports only BL 16. For Mark Write-1 command, CA5 must be driven LOW on first rising clock cycle (R1).
6) AP “HIGH” during Write-1, Mask Write-1 or Read-1 commands indicates that an auto-precharge will occur to the bank associated with the Write, Mask Write or Read command.
7) If Burst Length on-the-fly is enabled, BL “HIGH” during Write-1,or Read-1 command indicates that Burst Length should be set on-the-Fly to BL=32. BL “LOW” during Write-1 or Read-1 command indicates that Burst Length should be set on-the-fly to BL=16. If Burst Length on-the-fly is disabled, then BL must be driven to defined logic level “H” or “L”.
8) For CAS-2 commands (Write-2 or Mask Write-2 or Read-2 or MRR-2 or MPC (Only Write FIFO, Read FIFO & Read DQ Calibration)), C[1:0] are not transmitted on the CA[5:0] bus and are assumed to be zero. Note that for CAS-2 Write-2 or CAS-2 Mask Write-2 command, C[3:2] must be driven LOW.
9) Write-1 or Mask Write-1 or Read-1 or Mode Register Read-1 or MPC (Only Write FIFO, Read FIFO & Read DQ Calibration) command must be immediately followed by CAS-2 command consecutively without any other command in between. Write-1 or Mask Write-1 or Read-1 or mode register Read-1 or MPC (Only Write FIFO, Read FIFO & Read DQ Calibration) command must be issued first before issuing CAS-2 command. MPC (Only Start & Stop DQS Oscillator, Start & Latch ZQ Calibration) commands do not require CAS-2 command; they require two additional DES or NOP commands consecutively before issuing any other commands.
10) Activate-1 command must be immediately followed by Activate-2 command consecutively without any other command in between. Activate-1 command must be issued first before issuing Activate-2 command. Once Activate-1 command is issued, Activate-2 command must be issued before issuing another Activate-1 command.
11) MRW-1 command must be immediately followed by MRW-2 command consecutively without any other command in between. MRW-1 command must be issued first before issuing MRW-2 command.
12) MRR-1 command must be immediately followed by CAS-2 command consecutively without any other command in between. MRR-1 command must be issued first before issuing CAS-2 command.

2.0 POWER-UP, INITIALIZATION AND POWER-OFF PROCEDURE

For power-up and reset initialization, in order to prevent DRAM from functioning improperly, default values of the following MR settings are defined as Table 2.

[Table 2] MRS defaults settings

Item	MRS	Default Setting	Description
FSP-OP/WR	MR13 OP[7:6]	00_B	FSP-OP/WR[0] are enabled
WLS	MR2 OP[6]	0_B	Write Latency Set 0 is selected
WL	MR2 OP[5:3]	000_B	WL = 4
RL	MR2 OP[2:0]	000_B	RL = 6, nRTP=8
nWR	MR1 OP[6:4]	000_B	nWR = 6
DBI-WR/RD	MR3 OP[7:6]	00_B	Write & Read DBI are disabled
CA ODT	MR11 OP[6:4]	000_B	CA ODT is disabled
DQ ODT	MR11 OP[2:0]	000_B	DQ ODT is disabled
$V_{REF(CA)}$ Setting	MR12 OP[6]	1_B	$V_{REF(CA)}$ Range[1] enabled
$V_{REF(CA)}$ Value	MR12 OP[5:0]	001101_B	Range1 : 27.2% of V_{DD2}
$V_{REF(DQ)}$ Setting	MR14 OP[6]	1_B	$V_{REF(DQ)}$ Range[1] enabled
$V_{REF(DQ)}$ Value	MR14 OP[5:0]	001101_B	Range1 : 27.2% of V_{DDQ}

2.1 Voltage Ramp and Device Initialization

The following sequence shall be used to power up the LPDDR4 device. Unless specified otherwise, these steps are mandatory. Note that the power-up sequence of all channels must proceed simultaneously.

1. While applying power (after Ta), RESET_n is recommended to be LOW (≤0.2 × V_{DD2}) and all other inputs must be between VILmin and VIHmax. The device outputs remain at High-Z while RESET_n is held LOW. Power supply voltage ramp requirements are provided in Table . V_{DD1} must ramp at the same time or earlier than V_{DD2}. V_{DD2} must ramp at the same time or earlier than V_{DDQ}.

[Table 3] Voltage Ramp Conditions

After	Applicable Conditions
Ta is reached	V_{DD1} must be greater than V_{DD2}
	V_{DD2} must be greater than V_{DDQ} - 200mV

NOTE :
1) Ta is the point when any power supply first reaches 300mV.
2) Voltage ramp conditions in Table apply between Ta and power-off (controlled or uncontrolled).
3) Tb is the point at which all supply and reference voltages are within their defined ranges.
4) Power ramp duration tINIT0 (Tb-Ta) must not exceed 20ms.
5) The voltage difference between any of Vss and V_{SSQ} pins must not excess 100mV.

2. Following the completion of the voltage ramp (Tb), RESET_n must be maintained LOW. DQ, DMI, DQS_t and DQS_c voltage levels must be between V_{SSQ} and V_{DDQ} during voltage ramp to avoid latch-up. CKE, CK_t, CK_c, CS and CA input levels must be between V_{SS} and V_{DD2} during voltage ramp to avoid latch-up.

3. Beginning at Tb, RESET_n must remain LOW for at least tINIT1 (Tc), after which RESET_n can be de-asserted to HIGH (Tc). At least 10ns before RESET_n de-assertion, CKE is required to be set LOW. All other input signals are "Don't Care".

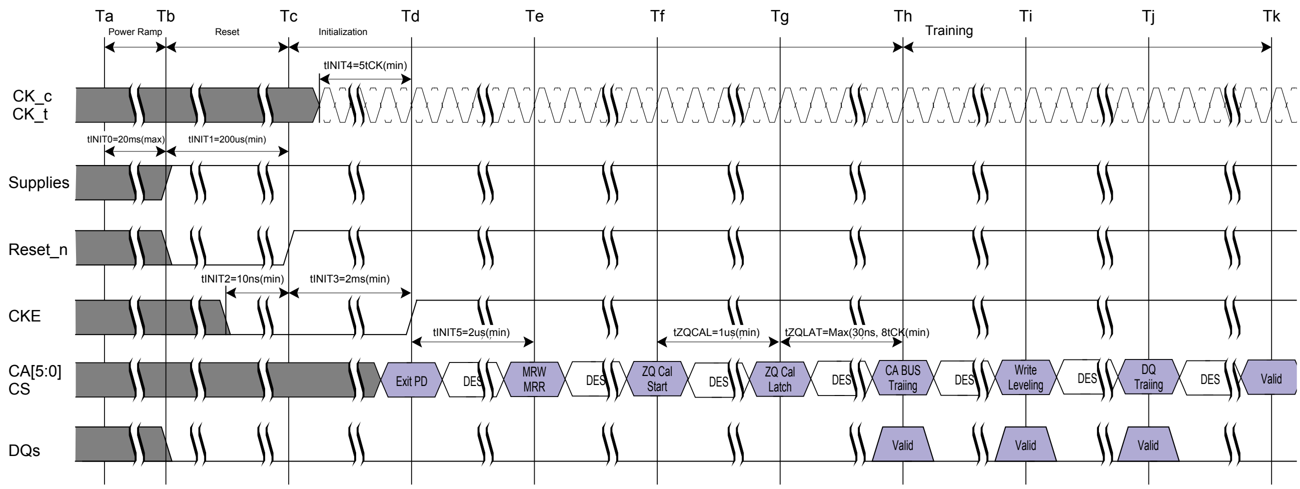


Figure 1: Power Ramp and Initialization Sequence

NOTE :
1) Training is optional and may be done at the system architects discretion. The training sequence after ZQ_CAL Latch (Th, Sequence7~9) in Figure 1 is simplified recommendation and actual training sequence may vary depending on systems.

4. After RESET_n is de-asserted (Tc), wait at least t_{INIT3} before activating CKE. Clock (CK_t, CK_c) is required to be started and stabilized for t_{INIT4} before CKE goes active (Td). CS is required to be maintained LOW when controller activates CKE.

5. After setting CKE high, wait minimum of t_{INIT5} to issue any MRR or MRW commands (Te). For both MRR and MRW commands, the clock frequency must be within the range defined for t_{CKb}. Some AC parameters (for example, t_{DQSCK}) could have relaxed timings (such as t_{DQSCKb}) before the system is appropriately configured.

6. After completing all MRW commands to set the Pull-up, Pull-down and Rx termination values, the DRAM controller can issue ZQCAL Start command to the memory (Tf). This command is used to calibrate VOH level and output impedance over process, voltage and temperature. In systems where more than one LPDDR4 DRAM devices share one external ZQ resistor, the controller must not overlap the ZQ calibration sequence of each LPDDR4 device. ZQ calibration sequence is completed after $t_{ZQCAL\ (Tg)}$ and the ZQCAL Latch command must be issued to update the DQ drivers and DQ+CA ODT to the calibrated values.

7. After t_{ZQLAT} is satisfied (Th) the command bus (internal $V_{REF(CA)}$, CS, and CA) should be trained for high-speed operation by issuing an MRW command (Command Bus Training Mode). This command is used to calibrate the device's internal VREF and align CS/CA with CK for high-speed operation. The LPDDR4 device will power-up with receivers configured for low-speed operations, and $V_{REF(CA)}$ set to a default factory setting. Normal device operation at clock speeds higher than t_{CKb} may not be possible until command bus training has been completed.

NOTE :
1) The command bus training MRW command uses the CA bus as inputs for the calibration data stream, and outputs the results asynchronously on the DQ bus. See Command Bus Training section (item 1.), MRW for information on how to enter/exit the training mode.

8. After command bus training, DRAM controller must perform write leveling. Write leveling mode is enabled when MR2 OP[7] is high (Ti). See Mode Register Write-WR leveling Mode section for detailed description of write leveling entry and exit sequence. In write leveling mode, the DRAM controller adjusts write DQS_t/_c timing to the point where the LPDDR4 device recognizes the start of write DQ data burst with desired write latency.

9. After write leveling, the DQ Bus (internal $V_{REF(DQ)}$, DQS, and DQ) should be trained for high-speed operation using the MPC training commands and by issuing MRW commands to adjust $V_{REF(DQ)}$(Tj). The LPDDR4 device will power-up with receivers configured for low-speed operations and $V_{REF(DQ)}$ set to a default factory setting. Normal device operation at clock speeds higher than tCKb should not be attempted until DQ Bus training has been completed.
The MPC Read Calibration command is used together with MPC FIFO Write/Read commands to train DQ bus without disturbing the memory array contents. See DQ Bus Training section for detailed DQ Bus Training sequence.

10. At Tk the LPDDR4 device is ready for normal operation, and is ready to accept any valid command. Any more registers that have not previously been set up for normal operation should be written at this time.

[Table 4] Initialization Timing Parameters

Parameter	Value		Unit	Comment
	Min	Max		
t_{INIT0}	-	20	ms	Maximum voltage-ramp time
t_{INIT1}	200	-	us	Minimum RESET_n LOW time after completion of voltage ramp
t_{INIT2}	10	-	ns	Minimum CKE low time before RESET_n high
t_{INIT3}	2	-	ms	Minimum CKE low time after RESET_n high
t_{INIT4}	5	-	t_{CK}	Minimum stable clock before first CKE high
t_{INIT5}	2	-	us	Minimum idle time before first MRW/MRR command
t_{ZQCAL}	1	-	us	ZQ calibration time
t_{ZQLAT}	Max(30ns, 8t_{CK})	-	ns	ZQCAL latch quiet time.
t_{CKb}	NOTE 1), 2)	NOTE 1), 2)	ns	Clock cycle time during boot

NOTE :
1) Min tCKb guaranteed by DRAM test is 18ns.
2) The system may boot at a higher frequency than dictated by min t_{CKb}. The higher boot frequency is system dependent.

2.2 Reset Initialization with Stable Power

The following sequence is required for RESET at no power interruption initialization.

1. Assert RESET_n below 0.2 × V_{DD2} anytime when reset is needed. RESET_n needs to be maintained for minimum t_{PW_RESET}. CKE must be pulled LOW at least 10 ns before de-asserting RESET_n.
2. Repeat steps 4 to 10 in "Voltage Ramp and Device Initialization" section.

[Table 5] Reset Timing Parameter

Parameter	Value		Unit	Comment
	Min	Max		
t_{PW_RESET}	100	-	ns	Minimum RESET_n low Time for Reset Initialization with stable power

2.3 Power-off Sequence

The following procedure is required to power off the device.

While powering off, CKE must be held LOW (≤0.2 × V_{DD2}) and all other inputs must be between V_{ILmin} and V_{IHmax}. The device outputs remain at High-Z while CKE is held LOW. DQ, DMI, DQS_t and DQS_c voltage levels must be between V_{SSQ} and V_{DDQ} during voltage ramp to avoid latch-up. RESET_n, CK_t, CK_c, CS and CA input levels must be between Vss and V_{DD2} during voltage ramp to avoid latch-up.

T_X is the point where any power supply drops below the minimum value specified.

T_Z is the point where all power supplies are below 300mV. After T_Z, the device is powered off.

[Table 6] Power Supply Conditions

Between	Applicable Conditions
Tx and Tz	V_{DD1} must be greater than V_{DD2}
	V_{DD2} must be greater than V_{DDQ} - 200mV

The voltage difference between any of V_{SS}, V_{SSQ} pins must not exceed 100mV.

2.3.1 Uncontrolled Power-Off Sequence

When an uncontrolled power-off occurs, the following conditions must be met:

At Tx, when the power supply drops below the minimum values specified, all power supplies must be turned off and all power supply current capacity must be at zero, except any static charge remaining in the system.

After Tz (the point at which all power supplies first reach 300mV), the device must power off. During this period the relative voltage between power supplies is uncontrolled. V_{DD1} and V_{DD2} must decrease with a slope lower than 0.5V/μs between Tx and Tz.

An uncontrolled power-off sequence can occur a maximum of 400 times over the life of the device.

[Table 7] Timing Parameters Power Off

Symbol	Value		Unit	Comment
	Min	Max		
t_{POFF}	-	2	s	Maximum Power-off ramp item

3.0 COMMAND DEFINITIONS AND TIMING DIAGRAMS

3.1 Activate Command

The ACTIVATE command is composed of two consecutive commands, Activate-1 command and Activate-2. Activate-1 command is issued by holding CS HIGH, CA0 HIGH and CA1 LOW at the first rising edge of the clock and Activate-2 command issued by holding CS HIGH, CA0 HIGH and CA1 HIGH at the first rising edge of the clock. The bank addresses BA0, BA1 and BA2 are used to select desired bank. Row addresses are used to determine which row to activate in the selected bank. The ACTIVATE command must be applied before any READ or WRITE operation can be executed. The device can accept a READ or WRITE command at t_{RCD} after the ACTIVATE command is issued. After a bank has been activated it must be precharged before another ACTIVATE command can be applied to the same bank. The bank active and precharge times are defined as t_{RAS} and t_{RP} respectively. The minimum time interval between ACTIVATE commands to the same bank is determined by the RAS cycle time of the device (t_{RC}). The minimum time interval between ACTIVATE commands to different banks is t_{RRD}.

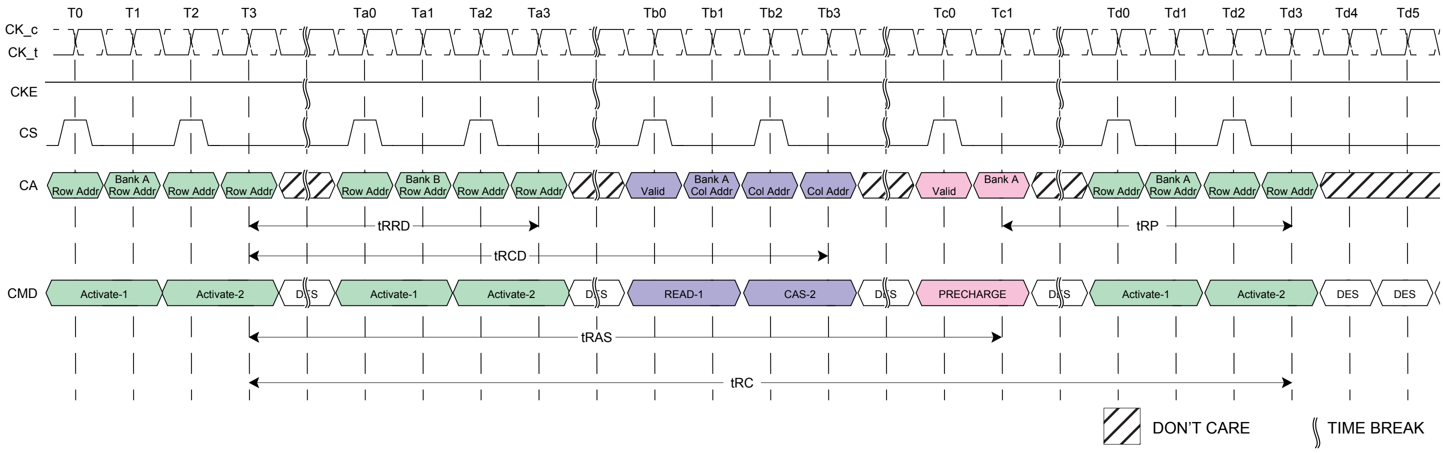


Figure 2: ACTIVATE command

NOTE :
1) A PRECHARGE command uses t_{RPab} timing for all-bank PRECHARGE and t_{RPpb} timing for single-bank PRECHARGE.
In this figure, t_{RP} is used to denote either all-bank PRECHARGE or a single-bank PRECHARGE.

3.2 8-Bank Device Operation

Certain restrictions on operation of the 8-bank LPDDR4 devices must be observed. There are two rules: One rule restricts the number of sequential ACTIVATE commands that can be issued; the other provides more time for RAS precharge for a PRECHARGE ALL command. The rules are as follows:

- **8-Bank Device Sequential Bank Activation Restriction :** No more than 4 banks may be activated (or refreshed, in the case of REFpb) in a rolling t_{FAW} window. The number of clocks in a t_{FAW} period is dependent upon the clock frequency, which may vary. If the clock frequency is not changed over this period, converting clocks is done by dividing t_{FAW} [ns] by t_{CK} [ns], and rounding up to the next integer value. As an example of the rolling window, if RU (t_{FAW}/t_{CK}) is 10 clocks, and an ACTIVATE command is issued in clock n, no more than three further ACTIVATE commands can be issued at or between clock n + 1 and n + 9. REFpb also counts as bank activation for purposes of t_{FAW}. If the clock frequency is changed during the t_{FAW} period, the rolling t_{FAW} window may be calculated in clock cycles by adding up the time spent in each clock period. The t_{FAW} requirement is met when the previous n clock cycles exceeds the t_{FAW} time.

- **The 8-Bank Device Precharge All Allowance :** t_{RP} for a PRECHARGE ALL command must equal t_{RPab}, which is greater than t_{RPpb}.

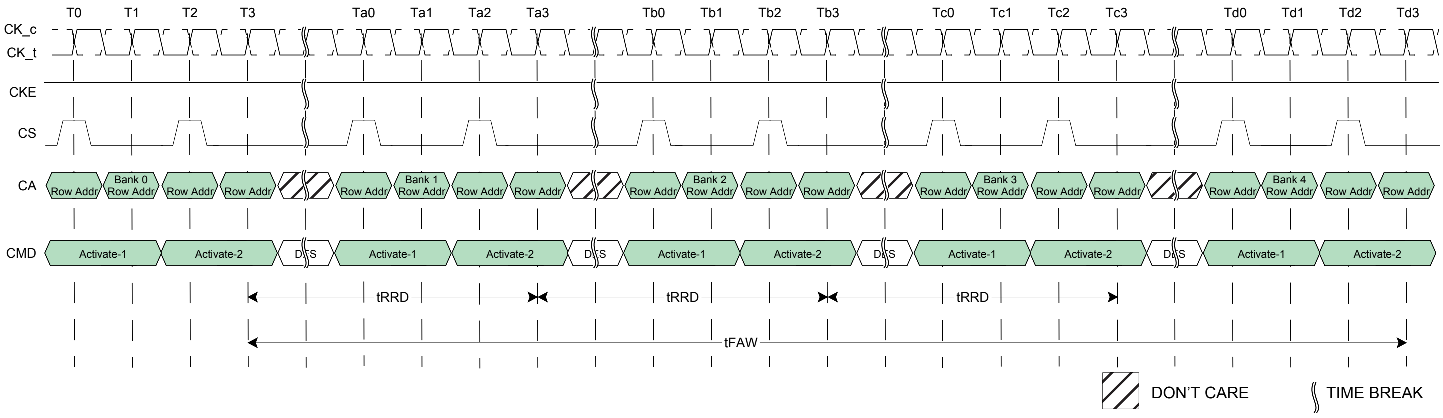


Figure 3: t_{FAW} Timing

3.3 LPDDR4 Command Input Signal Timing Definition

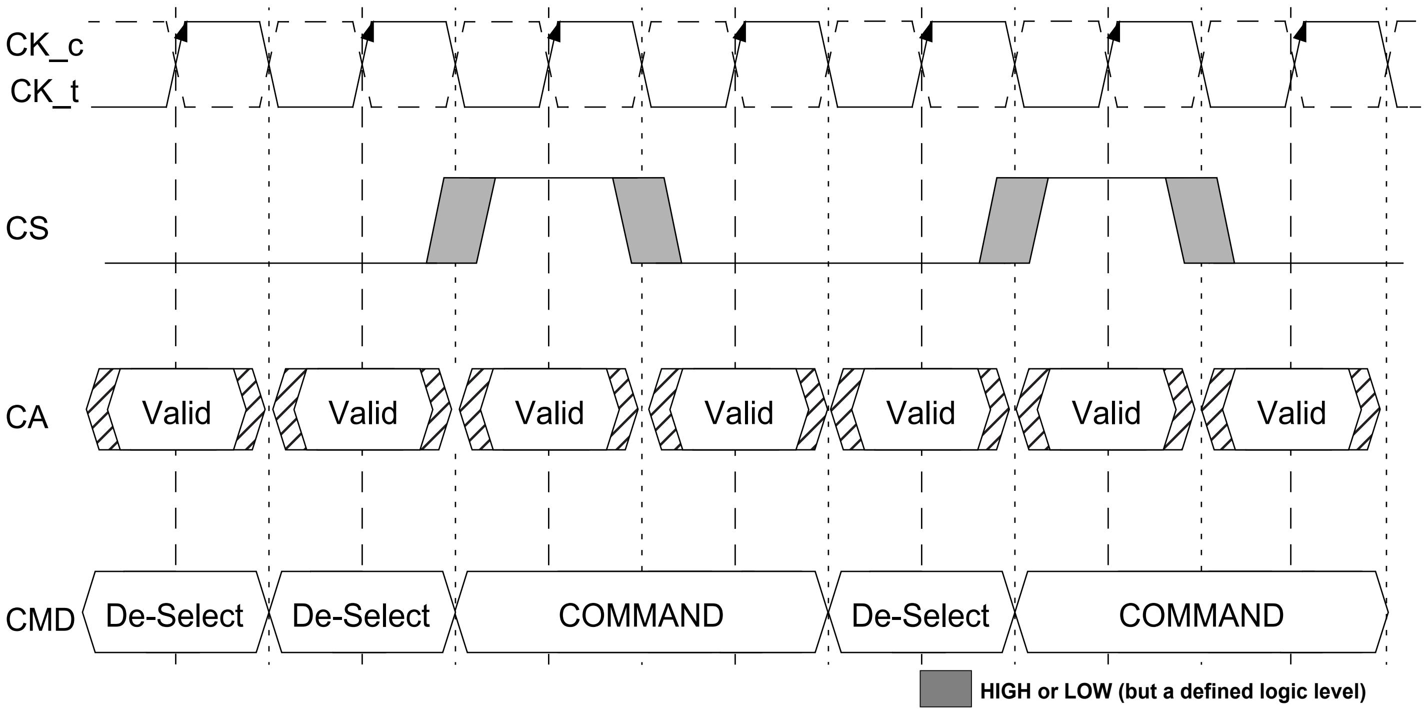


Figure 4: LPDDR4 : Command Input Setup and Hold Timing

NOTE :
1) Setup and hold conditions also apply to the CKE pin. See section related to power down for timing diagrams related to the CKE pin.

3.4 Read And Write Access Operations

After a bank has been activated, a read or write command can be executed. This is accomplished by asserting CKE asynchronously, with CS and CA[5:0] set to the proper state (see Command Truth Table) at a rising edge of CK.

The LPDDR4-SDRAM provides a fast column access operation. A single Read or Write command will initiate a burst read or write operation, where data is transferred to/from the DRAM on successive clock cycles. Burst interrupts are not allowed, but the optimal burst length may be set on the fly (see command truth table).

3.5 Read Preamble And Postamble

The DQS strobe for the LPDDR4-SDRAM requires a pre-amble prior to the first latching edge (the rising edge of DQS_t with DATA "valid"), and it requires a post-amble after the last latching edge. The pre-amble and post-amble lengths are set via mode register writes (MRW).

For READ operations the pre-amble is 2 × t_{CK}, but the pre-amble is static (no-toggle) or toggling, selectable via mode register.

LPDDR4 will have a DQS Read post-amble of 0.5 × t_{CK} (or extended to 1.5 × t_{CK}). Standard DQS post-amble will be 0.5 × t_{CK} driven by the DRAM for Reads. A mode register setting instructs the DRAM to drive an additional (extended) one cycle DQS Read post-amble. The drawings below show examples of DQS Read post-amble for both standard (t_{RPST}) and extended (t_{RPSTE}) post-amble operation.

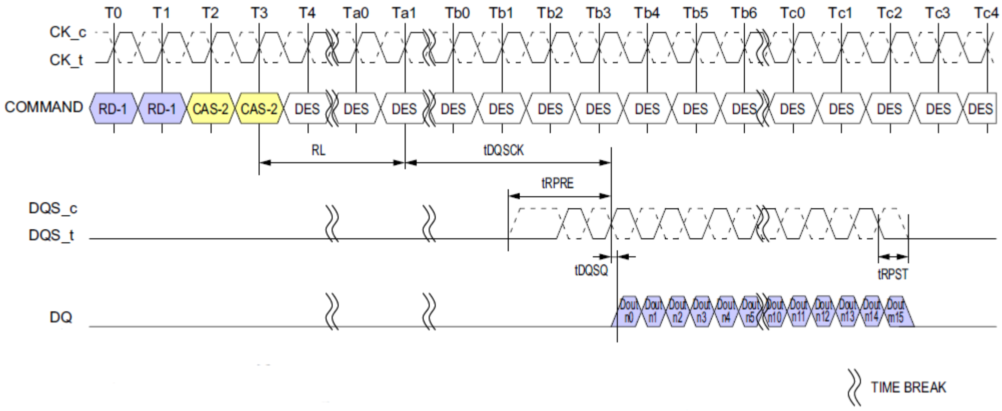


Figure 5: DQS READ Preamble and Postamble : Toggling Preamble and 0.5nCK Postamble

NOTE :
1) BL = 16, Preamble = Toggling, Postamble = 0.5nCK.
2) DQS and DQ terminated V_{SSQ}.
3) DQS_t/DQS_c is "don't care" prior to the start of t_{RPRE}. No transition of DQS is implied, as DQS_t/DQS_c can be HIGH, LOW, or HI-Z prior to t_{RPRE}.

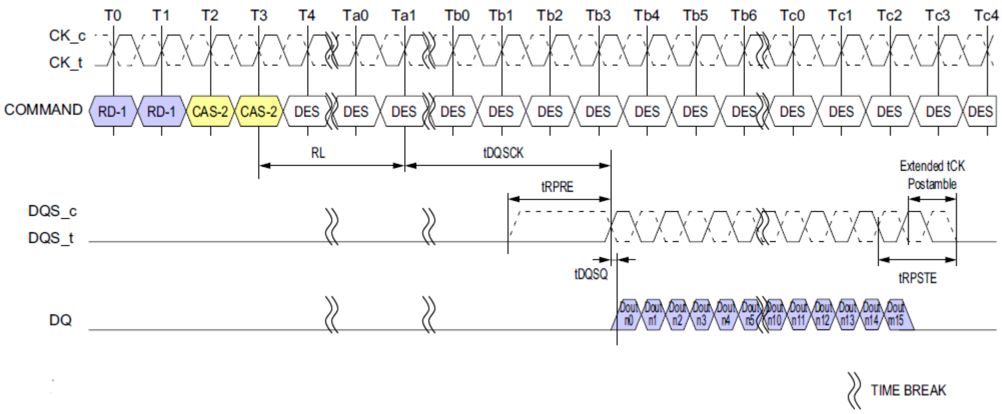


Figure 6: DQS READ Preamble Postamble : Static Preamble and 1.5nCK Postamble

NOTE :
1) BL = 16, Preamble = Static, Postamble = 1.5nCK (Extended).
2) DQS and DQ terminated V_{SSQ}.
3) DQS_t/DQS_c is "don't care" prior to the start of t_{RPRE}. No transition of DQS is implied, as DQS_t/DQS_c can be HIGH, LOW, or HI-Z prior to t_{RPRE}.

3.6 Burst Read Operation

A burst Read command is initiated with CS, and CA[5:0] asserted to the proper state at the rising edge of CK, as defined by the Command Truth Table. The command address bus inputs determine the starting column address for the burst. The two low-order address bits are not transmitted on the CA bus and are implied to be "0", so that the starting burst address is always a multiple of four (ex. 0x0, 0x4, 0x8, 0xC). The read latency (RL) is defined from the last rising edge of the clock that completes a read command (Ex: the second rising edge of the CAS-2 command) to the rising edge of the clock from which the t_{DQSCK} delay is measured. The first valid data is available RL × t_{CK} + t_{DQSCK} + t_{DQSQ} after the rising edge of Clock that completes a read command. The data strobe output is driven tRPRE before the first valid rising strobe edge. The first data-bit of the burst is synchronized with the first valid (i.e. post-preamble) rising edge of the data strobe. Each subsequent data-out appears on each DQ pin, edge-aligned with the data strobe. At the end of a burst the DQS signals are driven for another half cycle post-amble, or for a 1.5-cycle post-amble if the programmable post-amble bit is set in the mode register. The RL is programmed in the mode registers. Pin timings for the data strobe are measured relative to the cross-point of DQS_t and DQS_c.

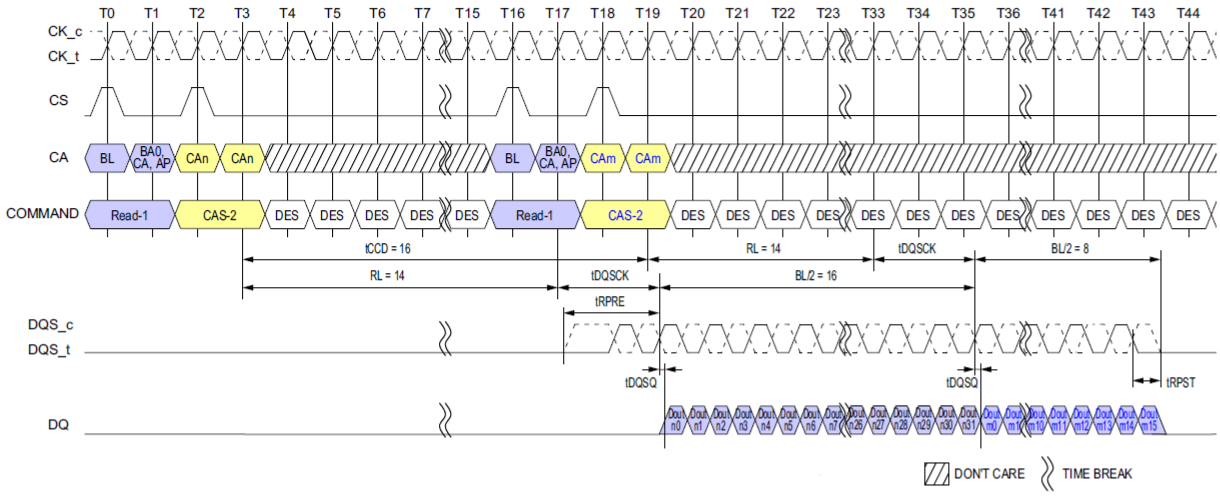


Figure 7: Burst Read Timing

NOTE :
1) BL = 32 for column n, BL = 16 for column m, RL = 14, Preamble = Toggle, Postamble = 0.5nCK, DQ/DQS: VSSQ termination.
2) Dout n/m = data-out from column n and column m.
3) DES commands are shown for ease of illustration; other commands may be valid at these times.

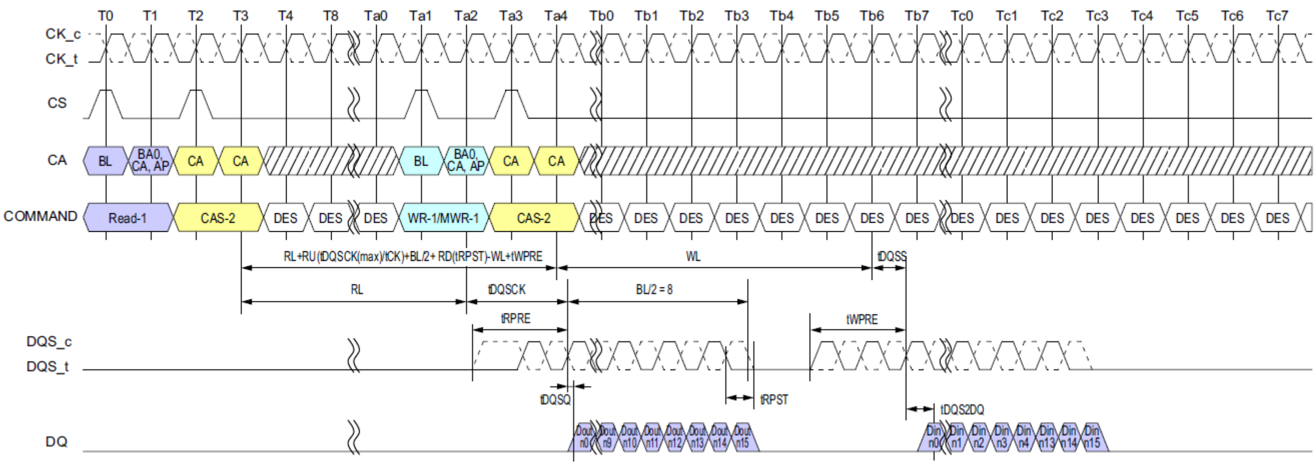


Figure 8: Burst Read followed by Burst WRITE or Burst Mask Write

NOTE :
1) BL=16, Read Preamble = Toggle, Read Postamble = 0.5nCK, Write Preamble = 2nCK, Write Postamble = 0.5nCK, DQ/DQS: V_{SSQ} termination.
2) Dout n = data-out from column n and Din n = data-in to columnm.n
3) DES commands are shown for ease of illustration; other commands may be valid at these times.

The minimum time from a burst read command to a Write or MASK WRITE command is defined by the Read Latency (RL) and the Burst Length (BL). Minimum READ-to-WRITE or MASK WRITE latency is RL + RU ($t_{DQSCK(MAX)}/t_{CK}$) + BL/2 + RD(t_{RPST}) - WL + t_{WPRE}.

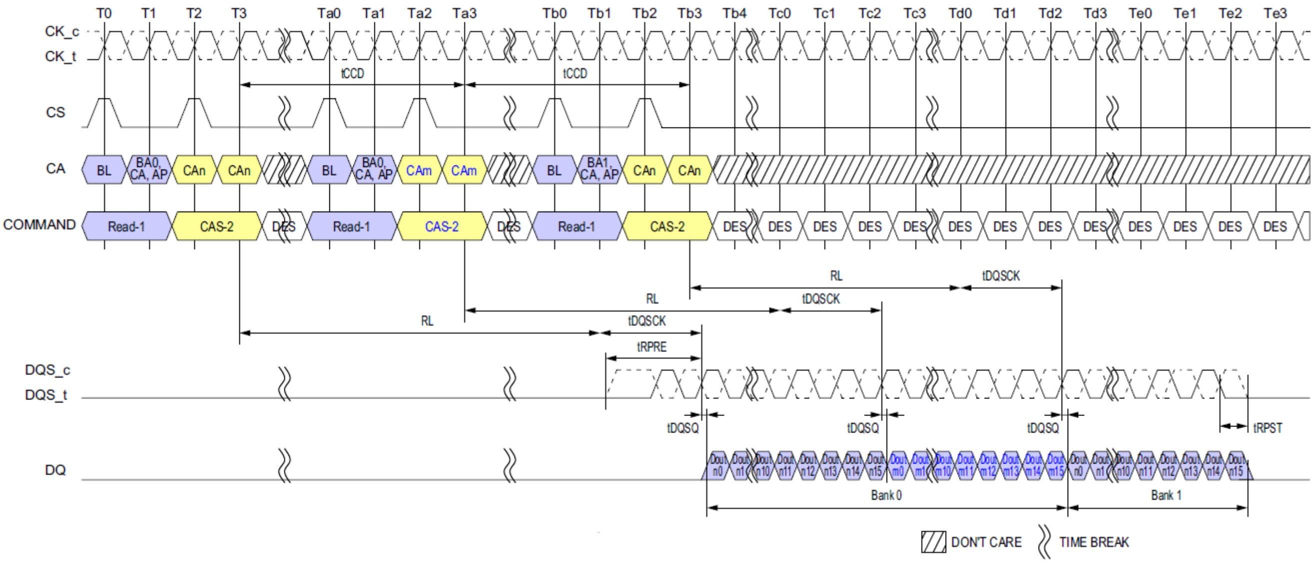


Figure 9: Seamless Burst Read

NOTE :
1) BL = 16, t_{CCD} = 8, Preamble = Toggle, Postamble = 0.5nCK, DQ/DQS: V_{SSQ} termination
2) Dout n/m = data-out from column n and column m.
3) DES commands are shown for ease of illustration; other commands may be valid at these times.

The seamless Burst READ operation is supported by placing a READ command at every $t_{CCD(Min)}$ interval for BL16 (or every 2 × $t_{CCD(Min)}$ for BL32). The seamless Burst READ can access any open bank.

3.7 Read Timing

The read timing is shown Figure 10.

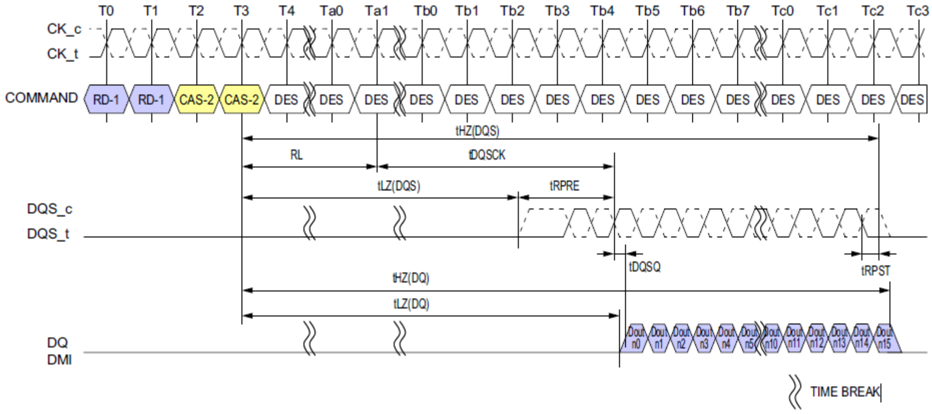


Figure 10: Read Timing

NOTE :
1) BL = 16, Preamble = Toggling, Postamble = 0.5nCK.
2) DQS, DQ and DMI terminated V_{SSQ}.
3) Output driver does not turn on before an end point of $t_{LZ(DQS)}$ and $t_{LZ(DQ)}$.
4) Output driver does not turn off before an end point of $t_{HZ(DQS)}$ and $t_{HZ(DQ)}$.

3.8 tLZ(DQS), tLZ(DQ), tHZ(DQS), tHZ(DQ) Calculation

tHZ and tLZ transitions occur in the same time window as valid data transitions. These parameters are referenced to a specific voltage level that specifies when the device output is no longer driving tHZ(DQS) and tHZ(DQ), or begins driving tLZ(DQS), tLZ(DQ).
This section shows a method to calculate the point when the device is no longer driving tHZ(DQS) and tHZ(DQ), or begins driving tLZ(DQS), tLZ(DQ), by measuring the signal at two different voltages. The actual voltage measurement points are not critical as long as the calculation is consistent. The parameters tLZ(DQS), tLZ(DQ), tHZ(DQS), and tHZ(DQ) are defined as single ended.

3.8.1 tLZ(DQS) and tHZ(DQS) Calculation for ATE (Automatic Test Equipment)

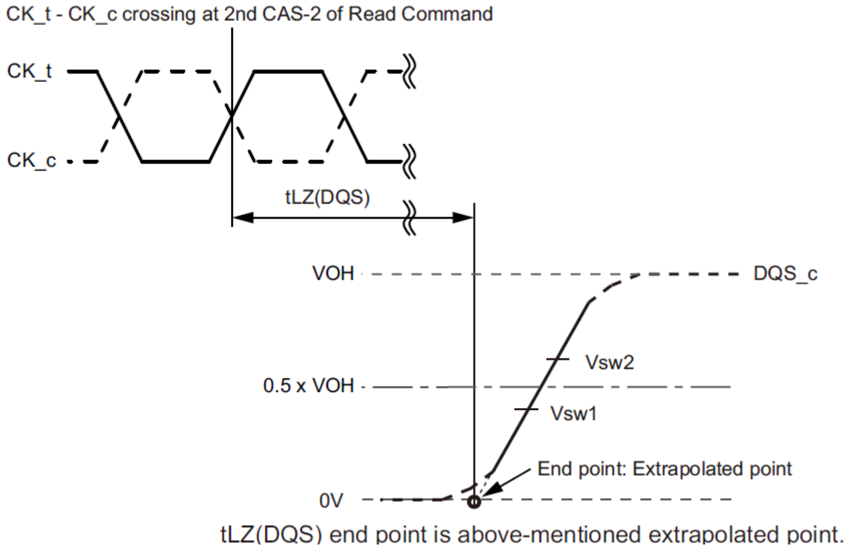


Figure 11: tLZ(DQS) method for calculating transitions and end point

NOTE :
1) Conditions for Calibration: Pull Down Driver Ron = 40ohm, $V_{OH} = V_{DDQ}/3$.
2) Termination condition for DQS_t and DQS_C = 50ohm to V_{SSQ}.
3) The V_{OH} level depends on MR22 OP[2:0] and MR3 OP[0] settings as well as device tolerances. Use the actual V_{OH} value for t_{HZ} and t_{LZ} measurements.

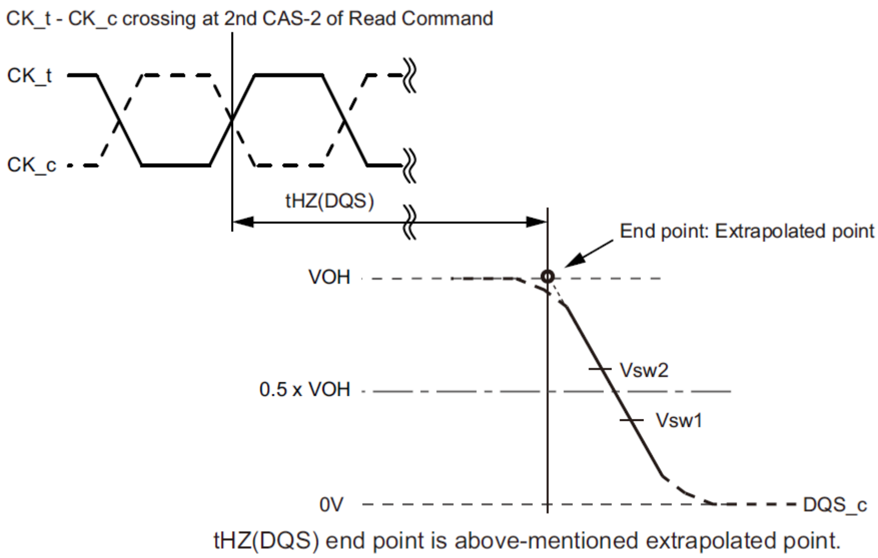


Figure 12: tHZ(DQS) method for calculating transitions and end point

NOTE :
1) Conditions for Calibration: Pull Down Driver Ron = 40ohm, $V_{OH} = V_{DDQ}/3$.
2) Termination condition for DQS_t and DQS_C = 50ohm to V_{SSQ}.
3) The V_{OH} level depends on MR22 OP[2:0] and MR3 OP[0] settings as well as device tolerances. Use the actual V_{OH} value for t_{HZ} and t_{LZ} measurements.

[Table 8] Reference Voltage for tLZ(DQS), tHZ(DQS) Timing Measurements

Measured Parameter	Measured Parameter Symbol	Vsw1[V]	Vsw2[V]
DQS_c low-impedance time from CK_t, CK_c	tLZ(DQS)	0.4 × VOH	0.6 × VOH
DQS_c high impedance time from CK_t, CK_c	tHZ(DQS)	0.4 × VOH	0.6 × VOH

3.8.2 tLZ(DQ) and tHZ(DQ) Calculation for ATE (Automatic Test Equipment)

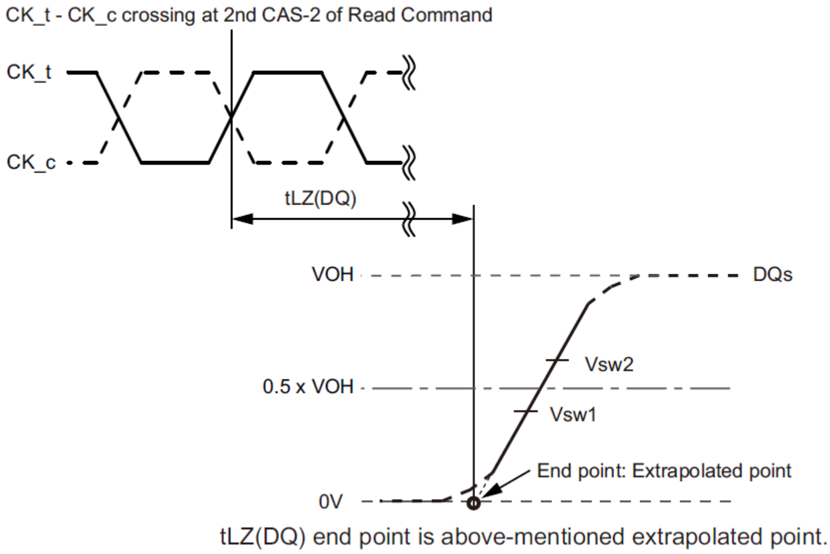


Figure 13: tLZ(DQ) method for calculating transitions and end point

NOTE :
1) Conditions for Calibration: Pull Down Driver Ron = 40ohm, $V_{OH} = V_{DDQ}/3$.
2) Termination condition for DQ and DMI = 50ohm to V_{SSQ}.
3) The V_{OH} level depends on MR22 OP[2:0] and MR3 OP[0] settings as well as device tolerances. Use the actual V_{OH} value for t_{HZ} and t_{LZ} measurements.

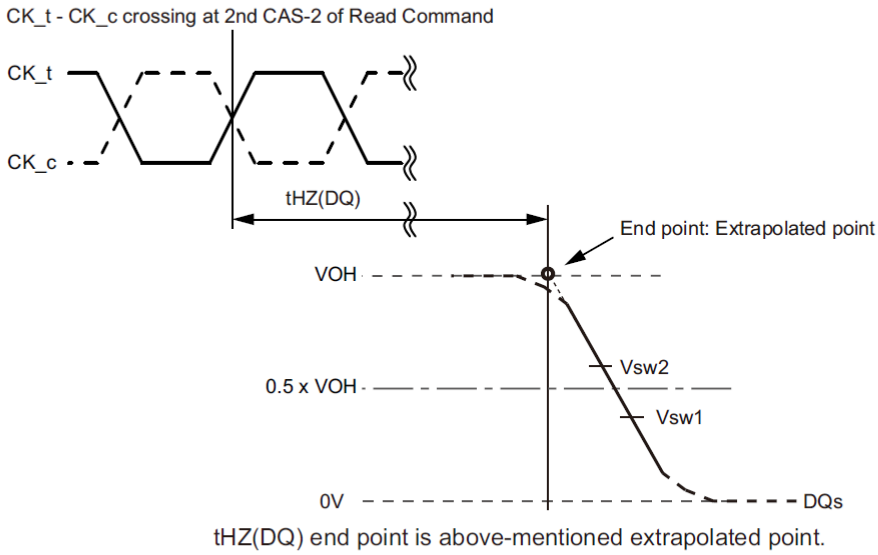


Figure 14: tHZ(DQ) method for calculating transitions and end point

NOTE :
1) Conditions for Calibration: Pull Down Driver Ron = 40ohm, $V_{OH} = V_{DDQ}/3$.
2) Termination condition for DQ and DMI = 50ohm to V_{SSQ}.
3) The V_{OH} level depends on MR22 OP[2:0] and MR3 OP[0] settings as well as device tolerances. Use the actual V_{OH} value for t_{HZ} and t_{LZ} measurements.

[Table 9] Reference Voltage for tLZ(DQ), tHZ(DQ) Timing Measurements

Measured Parameter	Measured Parameter Symbol	Vsw1[V]	Vsw2[V]
DQ low-impedance time from CK_t, CK_c	$t_{LZ(DQ)}$	0.4 × V_{OH}	0.6 × V_{OH}
DQ high impedance time from CK_t, CK_c	$t_{HZ(DQ)}$	0.4 × V_{OH}	0.6 × V_{OH}

3.8.3 tRPRE Calculation for ATE (Automatic Test Equipment)

The method for calculating differential pulse widths for tRPRE is shown in Figure 15 and Figure 16.

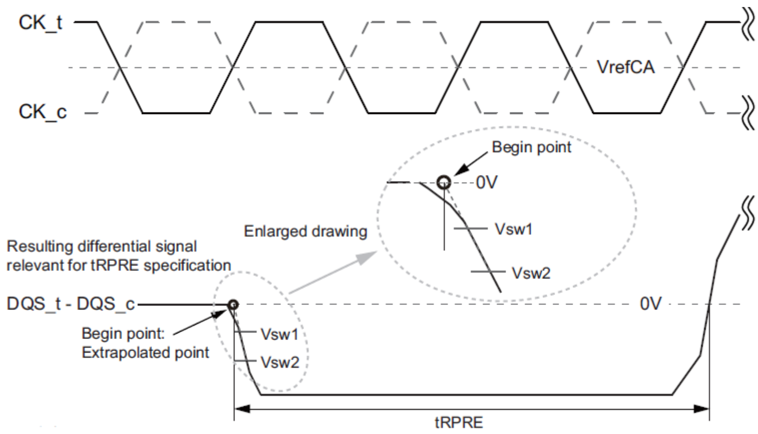


Figure 15: Method for calculating tRPRE transitions and endpoints

NOTE :
1) Conditions for Calibration: Pull Down Driver Ron = 40ohm, VOH = VDDQ/3.
2) Termination condition for DQS_t, DQS_c, DQ and DMI = 50ohm to VSSQ.
3) Preamble = Static.

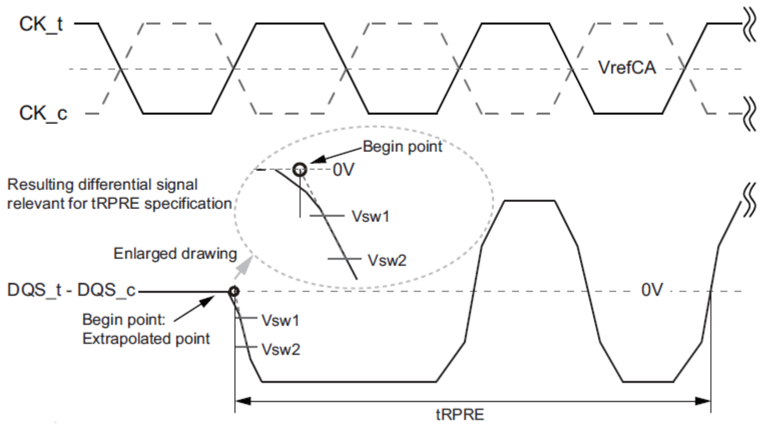


Figure 16: Method for calculating tRPRE transitions and endpoints

NOTE :
1) Conditions for Calibration: Pull Down Driver Ron = 40ohm, VOH = VDDQ/3.
2) Termination condition for DQS_t, DQS_c, DQ and DMI = 50ohm to VSSQ.
3) Preamble = Toggle.

[Table 10] Reference Voltage for tRPRE Timing Measurements

Measured Parameter	Measured Parameter Symbol	Vsw1[V]	Vsw2[V]	Note
DQS_t, DQS_c differential Read Preamble	t_{RPRE}	-(0.3 x VOH)	-(0.7 x VOH)	

3.8.4 tRPST Calculation for ATE(Automatic Test Equipment)

The method for calculating differential pulse widths for tRPST is shown in Figure 17.

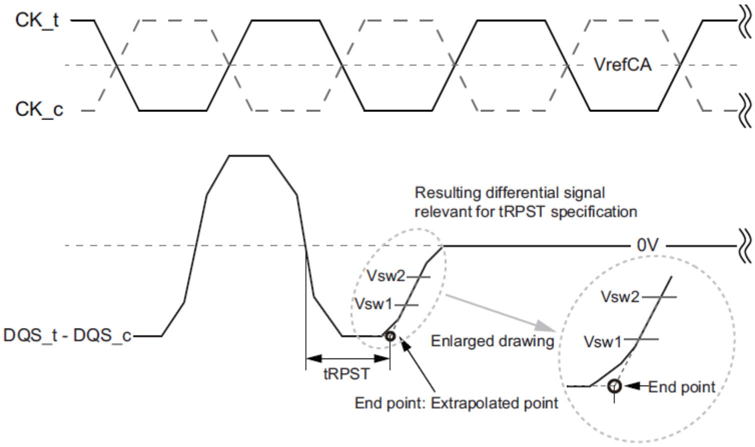


Figure 17: Method for calculating tRPST transitions and endpoints

NOTE :
1) Conditions for Calibration: Pull Down Driver Ron = 40ohm, VOH = VDDQ/3.
2) Termination condition for DQS_t, DQS_c, DQ and DMI = 50ohm to VSSQ.
3) Read Postamble: 0.5tCK.
4) The method for calculating differential pulse widths for 1.5 tCK Postamble is same as 0.5 tCK Postamble.

[Table 11] Reference Voltage for tRPST Timing Measurements

Measured Parameter	Measured Parameter Symbol	Vsw1[V]	Vsw2[V]	Note
DQS_t, DQS_c differential Read Postamble	tRPST	-(0.7 x VOH)	-(0.3 x VOH)	

3.9 Write Preamble And Postamble

The DQS strobe for the LPDDR4-SDRAM requires a pre-amble prior to the first latching edge (the rising edge of DQS_t with DATA "valid"), and it requires a post-amble after the last latching edge. The pre-amble and post-amble lengths are set via mode register writes (MRW).
For WRITE operations, a 2×t_{CK} pre-amble is required at all operating frequencies.
LPDDR4 will have a DQS Write post-amble of 0.5×tCK or extended to 1.5×t_{CK}. Standard DQS post-amble will be 0.5×t_{CK} driven by the memory controller for Writes. A mode register setting instructs the DRAM to drive an additional (extended) one cycle DQS Write post-amble. The drawings below show examples of DQS Write post-amble for both standard (t_{WPST}) and extended (t_{WPSTE}) post-amble operation.

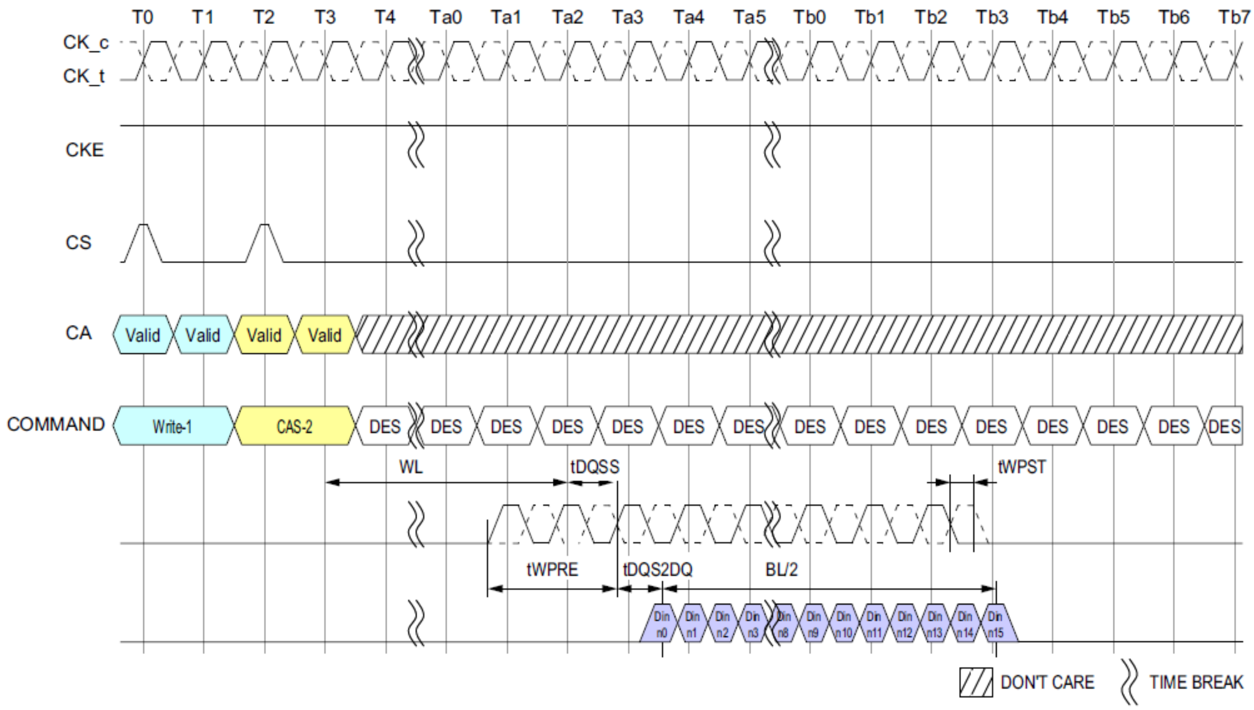


Figure 18: DQS Write Preamble and Postamble: 0.5nCK Postamble

NOTE :
1) BL = 16, Postamble = 0.5nCK.
2) DQS and DQ terminated V_{SSQ}.
3) DQS_t/DQS_c is “don’t care” prior to the start of t_{WPRE}. No transition of DQS is implied, as DQS_t/DQS_c can be HIGH, LOW, or HI-Z prior to t_{WPRE}.

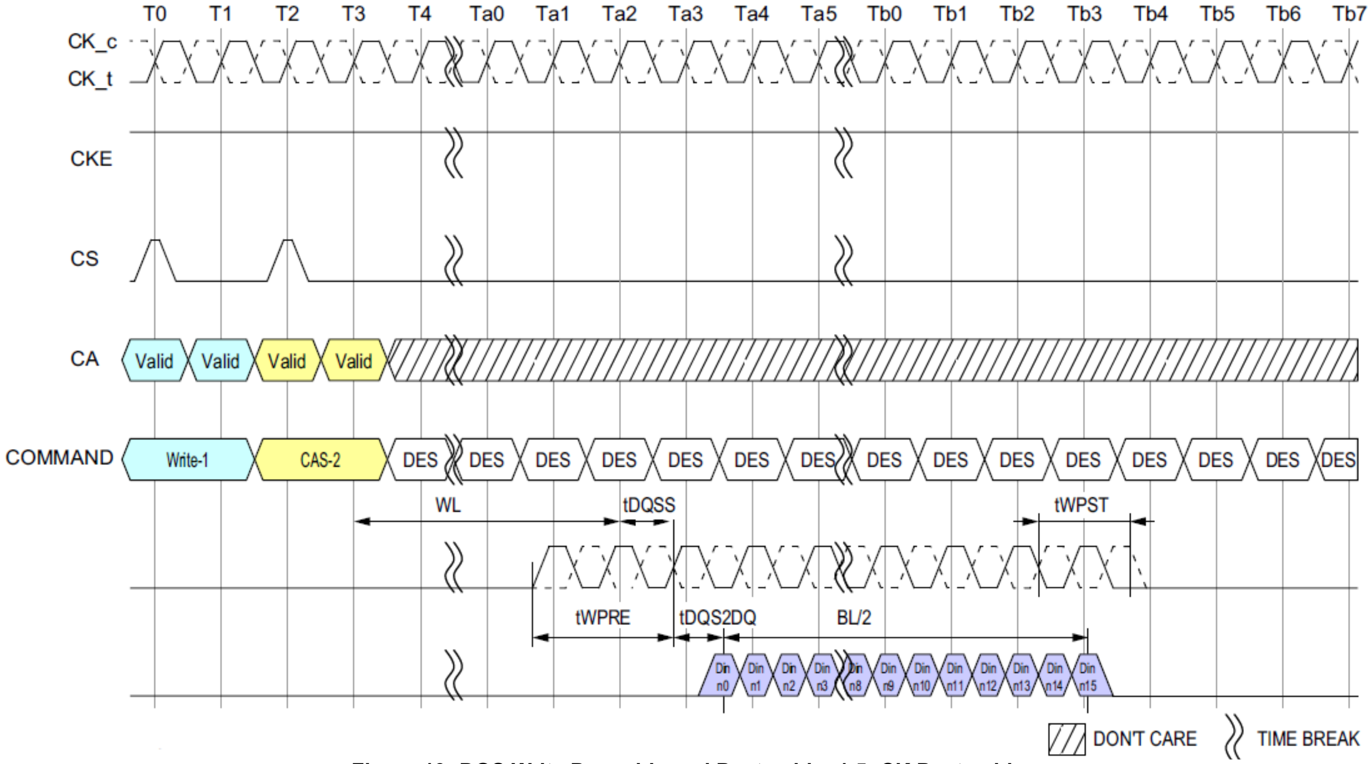


Figure 19: DQS Write Preamble and Postamble: 1.5nCK Postamble

NOTE :
1) BL = 16, Postamble = 1.5nCK.
2) DQS and DQ terminated V_{SSQ}.
3) DQS_t/DQS_c is “don’t care” prior to the start of t_{WPRE}. No transition of DQS is implied, as DQS_t/DQS_c can be HIGH, LOW, or HI-Z prior to t_{WPRE}.

3.10 Burst Write Operation

A burst WRITE command is initiated with CS, and CA[5:0] asserted to the proper state at the rising edge of CK, as defined by the Command Truth Table. Column addresses C[3:2] should be driven LOW for Burst WRITE commands, and column addresses C[1:0] are not transmitted on the CA bus (and are assumed to be zero), so that the starting column burst address is always aligned with a 32B boundary. The write latency (WL) is defined from the last rising edge of the clock that completes a write command (Ex: the second rising edge of the CAS-2 command) to the rising edge of the clock from which t_{DQSS} is measured. The first valid "latching" edge of DQS must be driven WL × t_{CK} + t_{DQSS} after the rising edge of Clock that completes a write command.

The LPDDR4-SDRAM uses an un-matched DQS-DQ path for lower power, so the DQS-strobe must arrive at the SDRAM ball prior to the DQ signal by the amount of t_{DQS2DQ}. The DQS-strobe output is driven t_{WPRE} before the first valid rising strobe edge. The t_{WPRE} pre-amble is required to be 2 × t_{CK}. The DQS-strobe must be trained to arrive at the DQ pad center-aligned with the DQ-data. The DQ-data must be held for t_{DIVW} (data input valid window) and the DQS must be periodically trained to stay centered in the t_{DIVW} window to compensate for timing changes due to temperature and voltage variation. Burst data is captured by the SDRAM on successive edges of DQS until the 16 or 32 bit data burst is complete. The DQS-strobe must remain active (toggling) for t_{WPST} (WRITE post-amble) after the completion of the burst WRITE.

After a burst WRITE operation, t_{WR} must be satisfied before a PRECHARGE command to the same bank can be issued. Pin input timings are measured relative to the crosspoint of DQS_t and DQS_c.

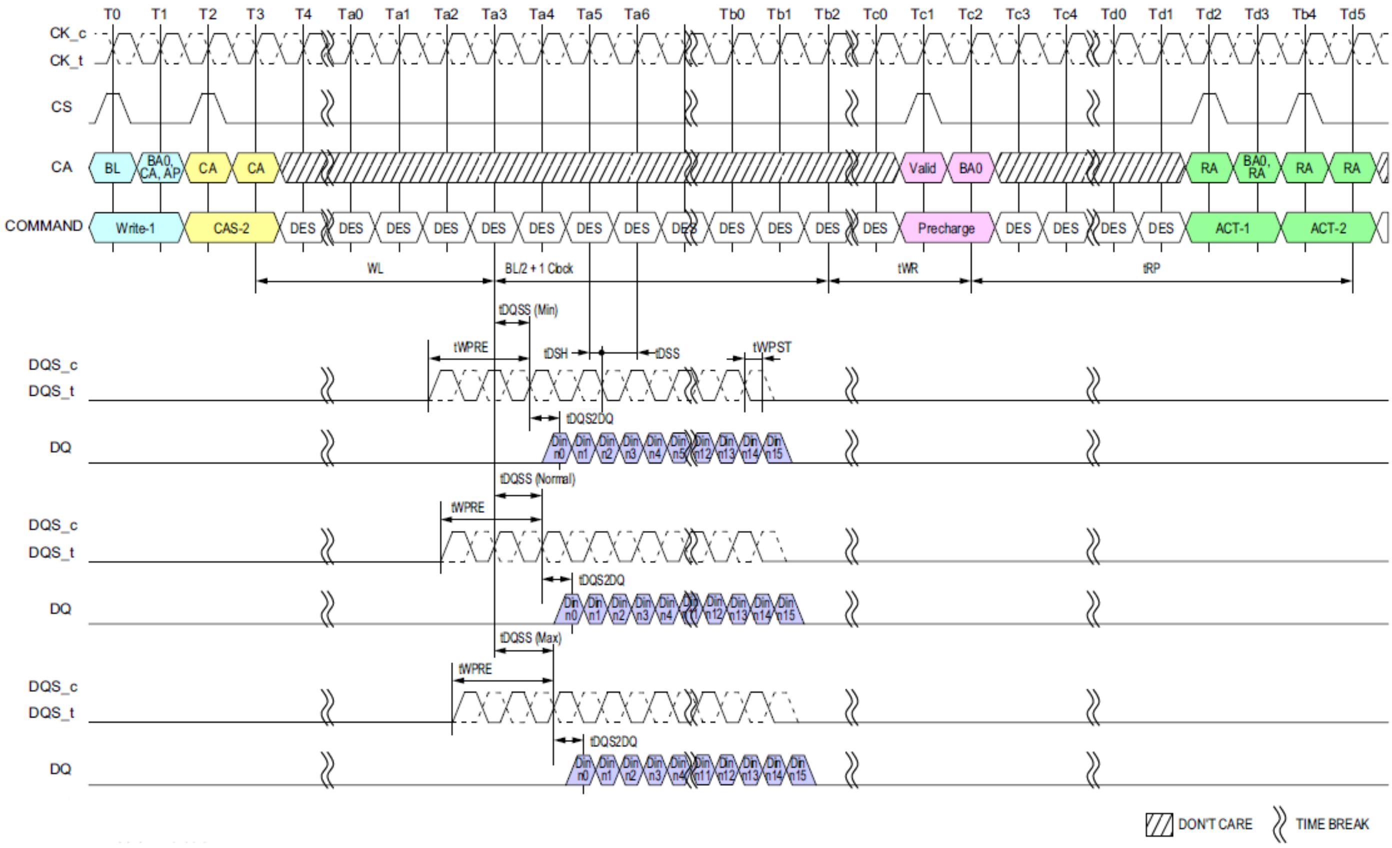


Figure 20: Burst Write Operation

NOTE :
1) BL=16, Write Postamble = 0.5nCK, DQ/DQS: V_{SSQ} termination.
2) Din n = data-in to columnm.n.
3) The minimum number of clock cycles from the burst write command to the burst read command for any bank is [WL + 1 + BL/2 + RU (t_{WR}/t_{CK})].
4) t_{WR} starts at the rising edge of CK after the last latching edge of DQS.
5) DES commands are shown for ease of illustration; other commands may be valid at these times.

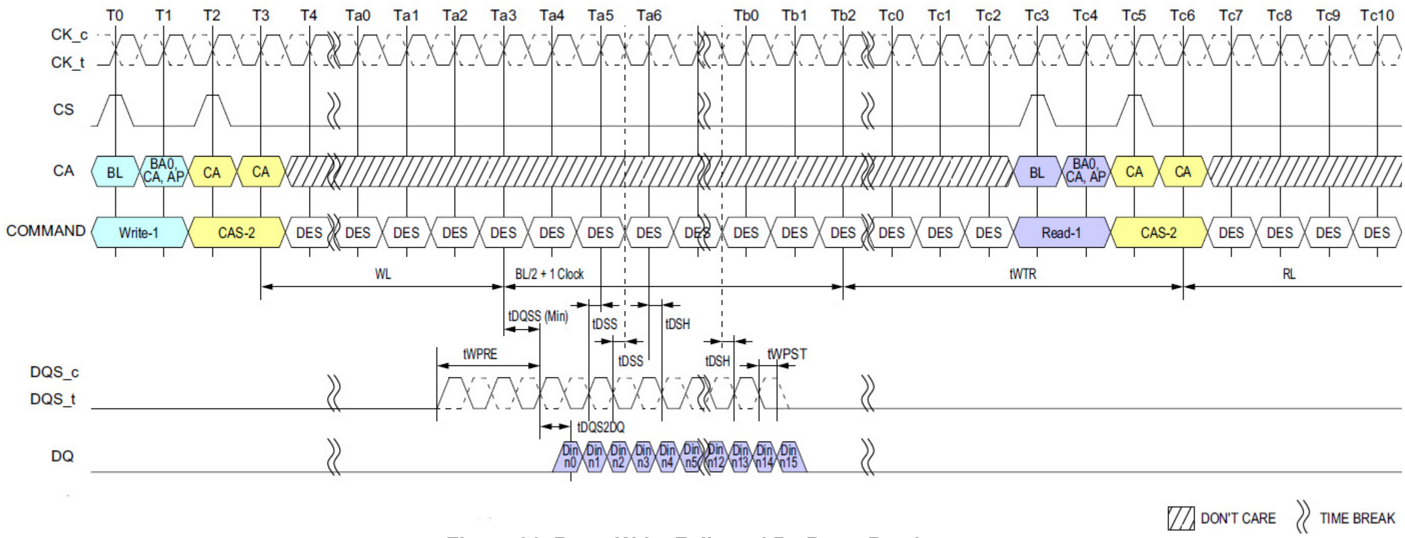


Figure 21: Burst Write Followed By Burst Read

NOTE :

1) BL=16, Write Postamble = 0.5nCK, DQ/DQS: VSSQ termination.
2) Din n = data-in to columnm.n.
3) The minimum number of clock cycles from the burst write command to the burst read command for any bank is [WL + 1 + BL/2 + RU (t_{WTR}/t_{CK})].
4) t_{WTR} starts at the rising edge of CK after the last latching edge of DQS.
5) DES commands are shown for ease of illustration; other commands may be valid at these times.

3.11 Write Timing

The write timing is shown Figure 22.

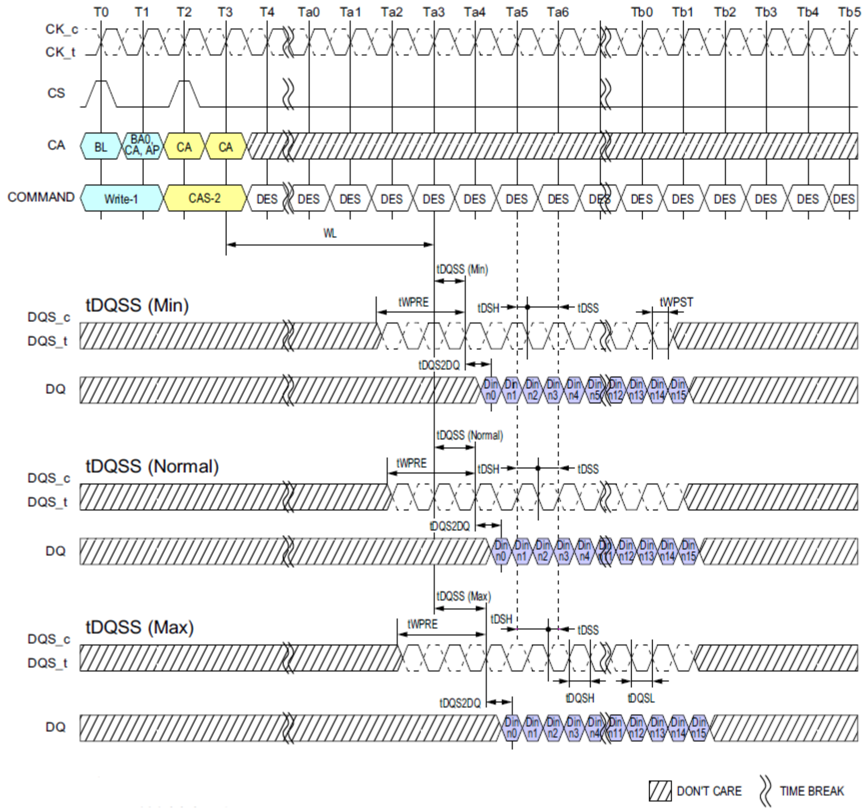


Figure 22: Write Timing

NOTE :
1) BL=16, Write Postamble = 0.5nCK.
2) Din n = data-in to columnm.n.
3) DES commands are shown for ease of illustration; other commands may be valid at these times.

3.11.1 tWPRE Calculation for ATE (Automatic Test Equipment)

The method for calculating differential pulse widths for t_{WPRE} is shown in Figure 23.

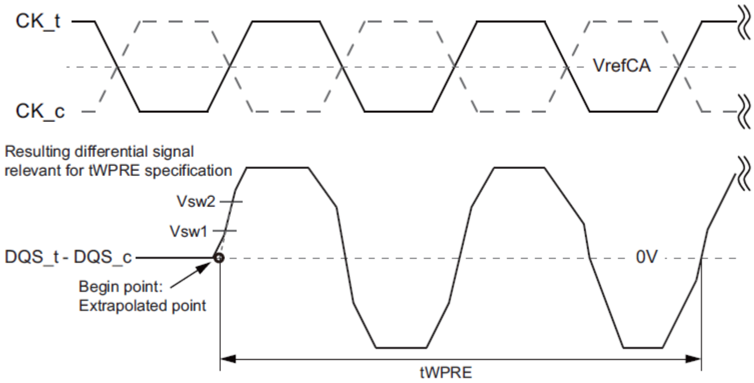


Figure 23: Method for calculating t_{WPRE} transitions and endpoints

NOTE :
1) Termination condition for DQS_t, DQS_c, DQ and DMI = 50ohm to V_{SSQ}.

[Table 12] Reference Voltage for t_{WPRE} Timing Measurements

Measured Parameter	Measured Parameter Symbol	Vsw1[V]	Vsw2[V]	Note
DQS_t, DQS_c differential WRITE Preamble	t_{WPRE}	VIHL_AC × 0.3	VIHL_AC × 0.7	

3.11.2 tWPST Calculation for ATE (Automatic Test Equipment)

The method for calculating differential pulse widths for tWPST is shown in Figure 24.

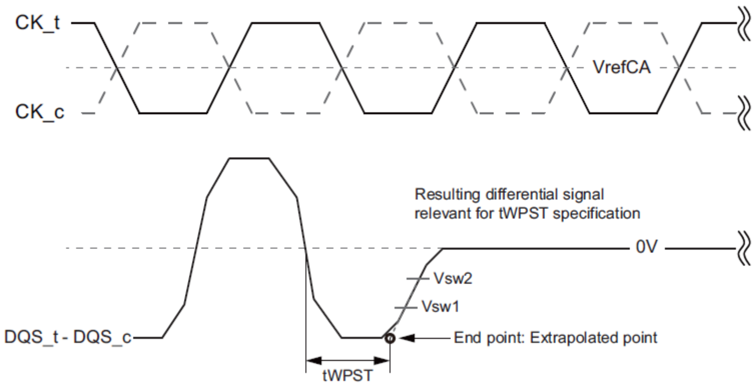


Figure 24: Method for calculating tWPST transitions and endpoints

NOTE :
1) Termination condition for DQS_t, DQS_c, DQ and DMI = 50ohm to V_{SSQ}.
2) Write Postamble: 0.5tCK
3) The method for calculating differential pulse widths for 1.5 tCK Postamble is same as 0.5 tCK Postamble.

[Table 13] Reference Voltage for tWPST Timing Measurements

Measured Parameter	Measured Parameter Symbol	Vsw1[V]	Vsw2[V]	Note
DQS_t, DQS_c differential WRITE Postamble	t_{WPST}	- (VIHL_AC x 0.7)	- (VIHL_AC x 0.3)	

3.12 Read And Write Latencies

[Table 14] Read and Write Latencies

Read Latency		Write Latency		nWR	nRTP	Lower Clock Frequency Limit (Greater than)	Upper Clock Frequency Limit (Same or less than)	Units	Notes
No DBI	w/ DBI	Set "A"	Set "B"						
6	6	4	4	6	8	10	266	MHz	1,2,3,4,5,6
10	12	6	8	10	8	266	533		
14	16	8	12	16	8	533	800		
20	22	10	18	20	8	800	1066		
24	28	12	22	24	10	1066	1333		
28	32	14	26	30	12	1333	1600		
32	36	16	30	34	14	1600	1866		
36	40	18	34	40	16	1866	2133		

NOTE :
1) The LPDDR4-SDRAM device should not be operated at a frequency above the Upper Frequency Limit, or below the Lower Frequency Limit, shown for each RL, WL, nRTP, or nWR value.
2) DBI for Read operations is enabled in MR3 OP[6]. When MR3 OP[6]=0_B, then the “No DBI” column should be used for Read Latency. When MR3 OP[6]=1_B, then the “w/DBI” column should be used for Read Latency.
3) Write Latency Set “A” and Set “B” is determined by MR2 OP[6]. When MR2 OP[6]=0_B, then Write Latency Set “A” should be used. When MR2 OP[6]=1_B, then Write Latency Set “B” should be used.
4) The programmed value of nWR is the number of clock cycles the LPDDR4-SDRAM device uses to determine the starting point of an internal Pre-charge operation after a Write burst with AP (auto-pre-charge). It is determined by RU(t_{WR}/t_{CK}).
5) The programmed value of nRTP is the number of clock cycles the LPDDR4-SDRAM device uses to determine the starting point of an internal Pre-charge operation after a Read burst with AP (auto-pre-charge). It is determined by RU(t_{RTP}/t_{CK}).
6) nRTP shown in this table is valid for BL16 only. For BL32, the SDRAM will add 8 clocks to the nRTP value before starting a precharge.

3.13 Write and Masked Write operation DQS controls (WDQS Control)

LPDDR4-SDRAMs support write and masked write operations with the following DQS controls. Before and after Write and Masked Write operations are issued, DQS_t/DQS_c is required to have a sufficient voltage gap to make sure the write buffers operating normally without any risk of metastability.

The LPDDR4-SDRAM is supported by either of two WDQS control modes below.

Mode 1: Read Based Control
Mode 2 : WDQS_on / WDQS_off definition based control

Regardless of ODT enable / disable, WDQS related timing described here does not allow any change of existing command timing constraints for all read / write operation. In case of any conflict or ambiguity on the command timing constraints caused by the spec here, the spec defined in table 74 in section 4.38 (or below) should have higher priority than WDQS control requirements.

Some legacy products may not provide WDQS control described below. However, in order to prevent the write preamble related failure, it is strongly recommended to support either of two WDQS controls to LPDDR4-SDRAMs. In the case of legacy SoC which may not provide WDQS control modes, it is required to consult DRAM vendors to guarantee the write / masked write operation appropriately.

3.13.1 WDQS Control Mode 1 - Read Based Control

The LPDDR4-SDRAM needs to be guaranteed the differential WDQS, but the differential WDQS can be controlled as described below. WDQS control requirements here can be ignored while differential read DQS is operated or while DQS hands over from Read to Write and vice versa.

1. At the time a write / masked write command is issued, SoC makes the transition from driving DQS_c high to driving differential DQS_t/DQS_c, followed by normal differential burst on DQS pins.
2. At the end of post amble of write /masked write burst, SoC resumes driving DQS_c high through the subsequent states except for DQS toggling and DQS turn around time of WT-RD and RD-WT as long as CKE is high.
3. When CKE is low, the state of DQS_t and DQS_c is allowed to be “Don’t Care”.

3.13.2 WDQS Control Mode 2 - WDQS_on/off

After write / masked write command is issued, DQS_t and DQS_c required to be differential from WDQS_on, and DQS_t and DQS_c can be "Don't Care" status from WDQS_off of write / masked write command. When ODT is enabled, WDQS_on and WDQS_off timing is located in the middle of the operations. When host disables ODT, WDQS_on and WDQS_off constraints conflict with tRTW. The timing does not conflict when ODT is enabled because WDQS_on and WDQS_off timing is covered in ODTLon and ODTLoff. However, regardless of ODT on/off, WDQS_on/off timing below does not change any command timing constraints for all read and write operations. In order to prevent the conflict, WDQS_on/off requirement can be ignored when WDQS_on/off timing is overlapped with read operation period including Read burst period and tRPST or overlapped with turn-around time (RD-WT or WT-RD). In addition, the period during DQS toggling caused by Read and Write can be counted as WDQS_on/off.

Parameters

- WDQS_on: the max delay from write / masked write command to differential DQS_t and DQS_c.
- WDQS_off : the min delay for DQS_t and DQS_c differential input after the last write / masked write command.
- WDQS_Exception : the period where WDQS_on and WDQS_off timing is overlapped with read operation or with DQS turn around (RD-WT, WT-RD).

- WDQS_Exception @ ODT disable = max (WL-WDQS_on+tDQSTA- tWPRE - n*tCK, 0 tCK) where RD to WT command gap = tRTW(min)@ODT disable + n*tCK
- WDQS_Exception @ ODT enable = tDQSTA

[Table 15] WDQS_on / WDQS_off Definition

WL		nWR	nRTP	WDQS_on (max)		WDQS_off (min)		Lower Clock Frequency Limit (>)	Upper Clock Frequency Limit (≤)
Set A	Set B			Set A	Set B	Set A	Set B		
4	4	6	8	0	0	15	15	10	266
6	8	10	8	0	0	18	20	266	533
8	12	16	8	0	6	21	25	533	800
10	18	20	8	4	12	24	32	800	1066
12	22	24	10	4	14	27	37	1066	1333
14	26	30	12	6	18	30	42	1333	1600
16	30	34	14	6	20	33	47	1600	1866
18	34	40	16	8	24	36	52	1866	2133
nCK	nCK	nCK	nCK	nCK	nCK	nCK	nCK	MHz	MHz

NOTE :
1) WDQS_on/off requirement can be ignored when WDQS_on/off timing is overlapped with read operation period including Read burst period and tRPST or overlapped with turn-around time (RD-WT or WT-RD).
2) The period during which DQS is toggling because of a Read or Write can be counted as part of the WDQS_on/off requirement.

[Table 16] WDQS_on / WDQS_off Allowable Variation Range

	Min	Max	Unit
WDQS_On	-0.25	0.25	tCK(avg)
WDQS_Off	-0.25	0.25	tCK(avg)

[Table 17] DQS turn around parameter

Parameter	Description	Value	Unit	Note
t_{DQSTA}	Turn-around time RDQS to WDQS for WDQS control case	TBD	-	1

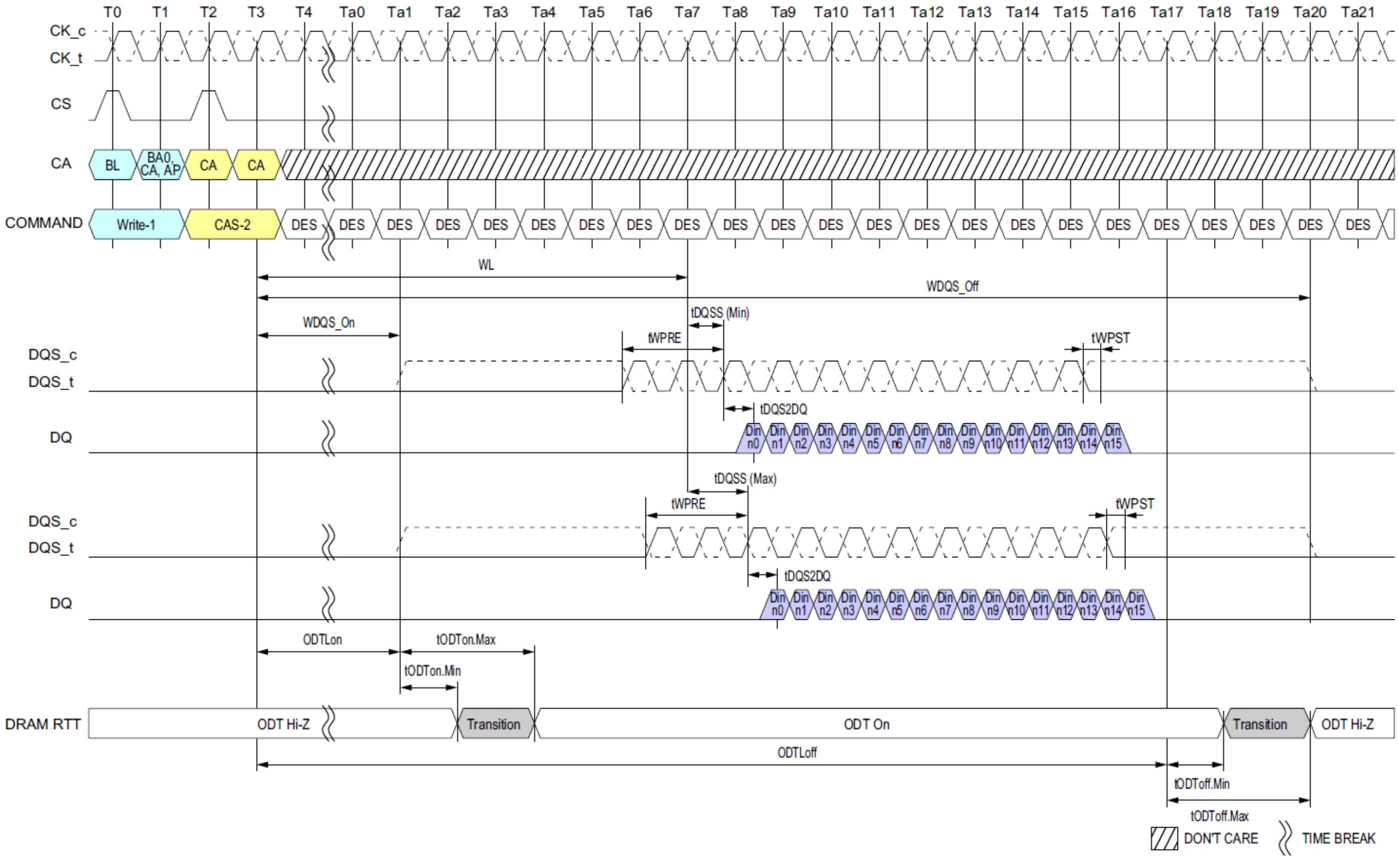


Figure 25: Burst Write Operation

NOTE :
1) BL=16, Write Postamble = 0.5nCK, DQ/DQS: VSSQ termination.
2) Din n = data-in to column n.
3) DES commands are shown for ease of illustration; other commands may be valid at these times.
4) DRAM RTT is only applied when ODT is enabled (MR11 OP[2:0] is not 000_B).

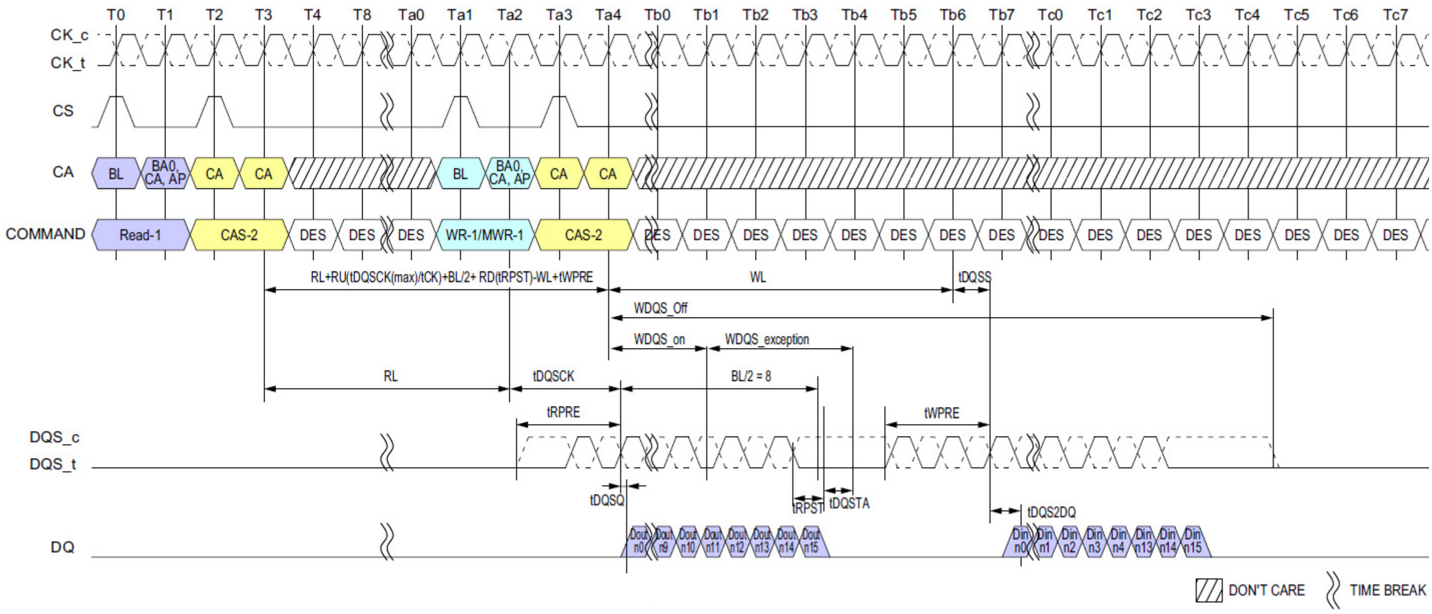


Figure 26: Burst Read followed by Burst Write or Burst Mask Write (ODT Disable)

NOTE :
1) BL=16, Read Preamble = Toggle, Read Postamble = 0.5nCK, Write Preamble = 2nCK, Write Postamble = 0.5nCK.
2) Dout n = data-out from column n and Din n = data-in to column n.
3) DES commands are shown for ease of illustration; other commands may be valid at these times.
4) WDQS_on and WDQS_off requirement can be ignored where WDQS_on/off timing is overlapped with read operation period including Read burst period and tRPST or overlapped with turn-around time (RD-WT or WT-RD).

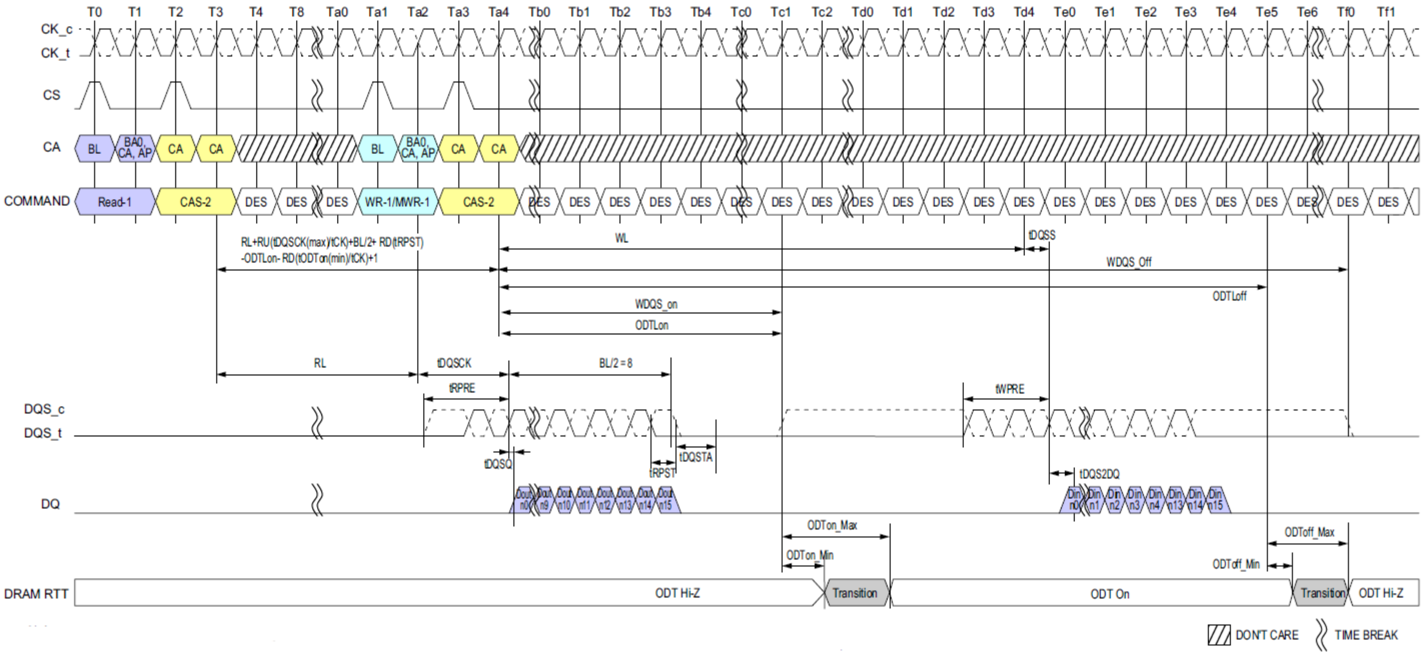


Figure 27: Burst Read followed by Burst Write or Burst Mask Write (ODT Enable)

NOTE :
1) BL=16, Read Preamble = Toggle, Read Postamble = 0.5nCK, Write Preamble = 2nCK, Write Postamble = 0.5nCK, DQ/DQS: VSSQ termination.
2) Dout n = data-out from column n and Din n = data-in to column n.
3) DES commands are shown for ease of illustration; other commands may be valid at these times.
4) WDQS_on and WDQS_off requirement can be ignored where WDQS_on/off timing is overlapped with read operation period including Read burst period and tRPST or over-lapped with turn-around time (RD-WT or WT-RD).

3.14 Postamble And Preamble Merging Behavior

The DQS strobe for the device requires a preamble prior to the first latching edge (the rising edge of DQS_t with data valid), and it requires a postamble after the last latching edge. The preamble and postamble options are set via MODE REGISTER WRITE commands.

In Read to Read or Write to Write operations with t_{CCD}=BL/2, postamble for 1st command and preamble for 2nd command will disappear to create consecutive DQS latching edge for seamless burst operations. But in the case of Read to Read or Write to Write operations with command interval of t_{CCD}+1,t_{CCD}+2, etc., they will not completely disappear because it's not seamless burst operations.

Timing diagrams in this material describe Postamble and Preamble merging behavior in Read to Read or Write to Write operations with t_{CCD}+n.

3.15 Read to Read operation

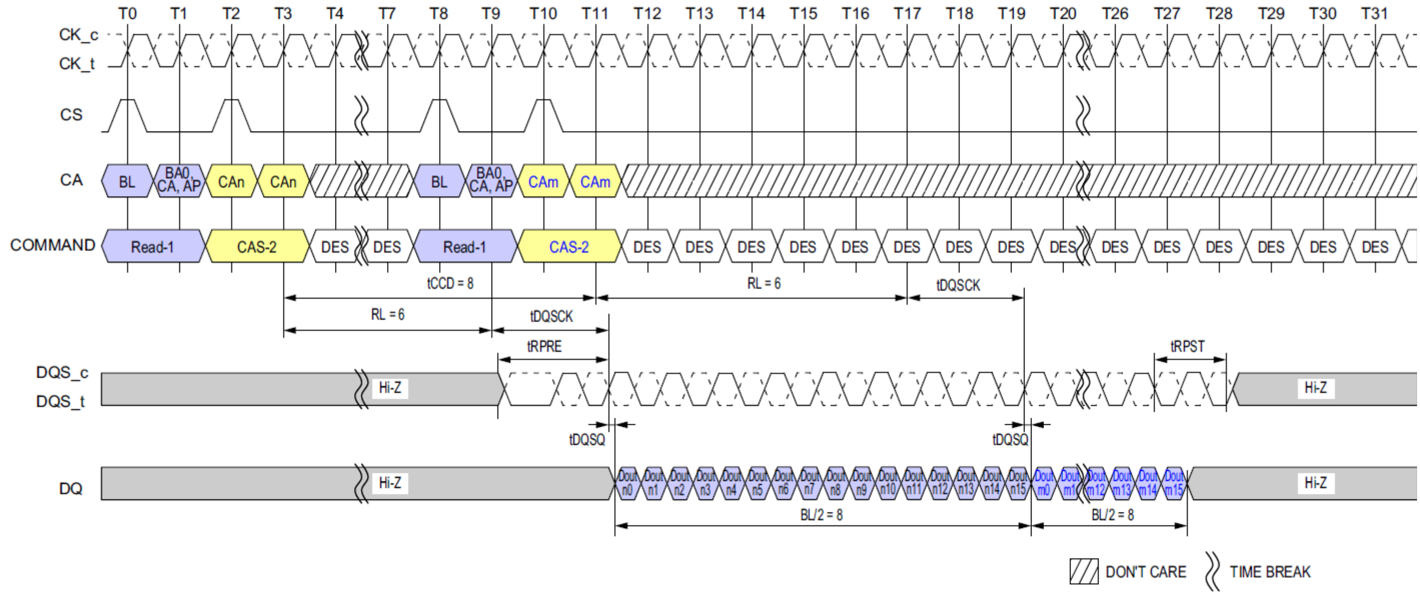


Figure 28: Seamless Reads Operation: t_{CCD} = Min, Preamble = Toggle, 1.5nCK Postamble

NOTE :
1) BL = 16 for column n and column m, RL = 6, Preamble = Toggle, Postamble = 1.5nCK.
2) Dout n/m = data-out from column n and column m.
3) DES commands are shown for ease of illustration; other commands may be valid at these times.

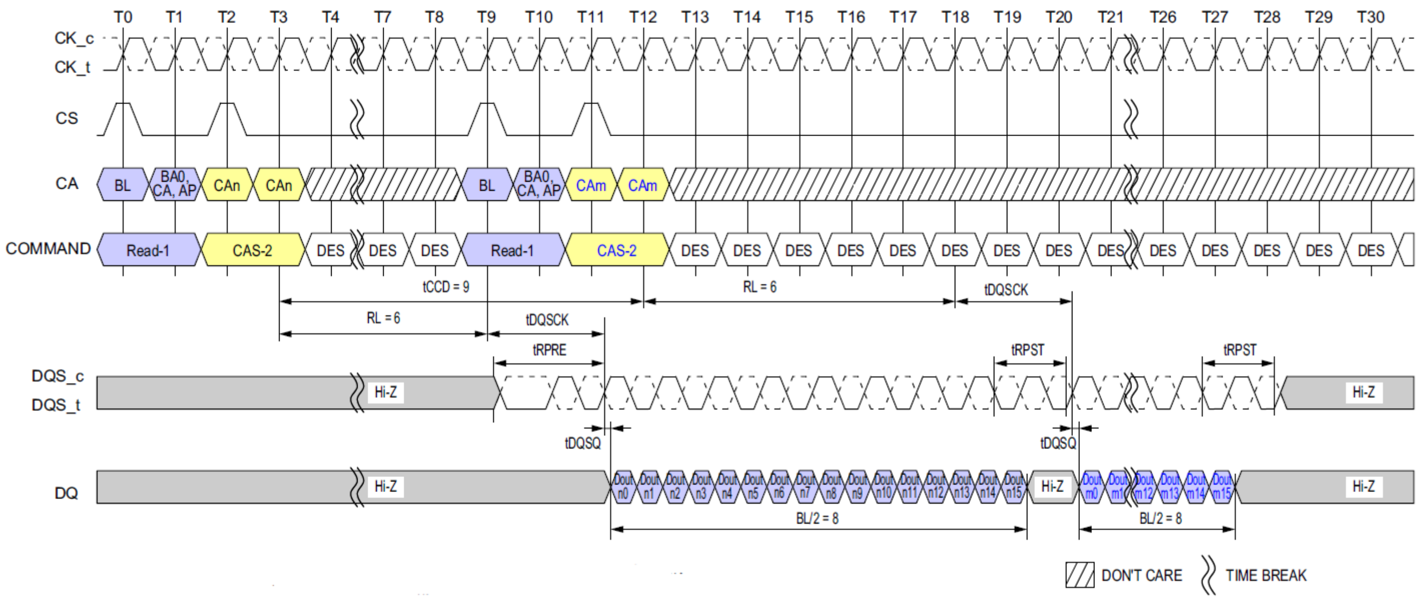


Figure 29: Consecutive Reads Operation: t_{CCD} = Min +1, Preamble = Toggle, 1.5nCK Postamble

NOTE :
1) BL = 16 for column n and column m, RL = 6, Preamble = Toggle, Postamble = 1.5nCK.
2) Dout n/m = data-out from column n and column m.
30 DES commands are shown for ease of illustration; other commands may be valid at these times.

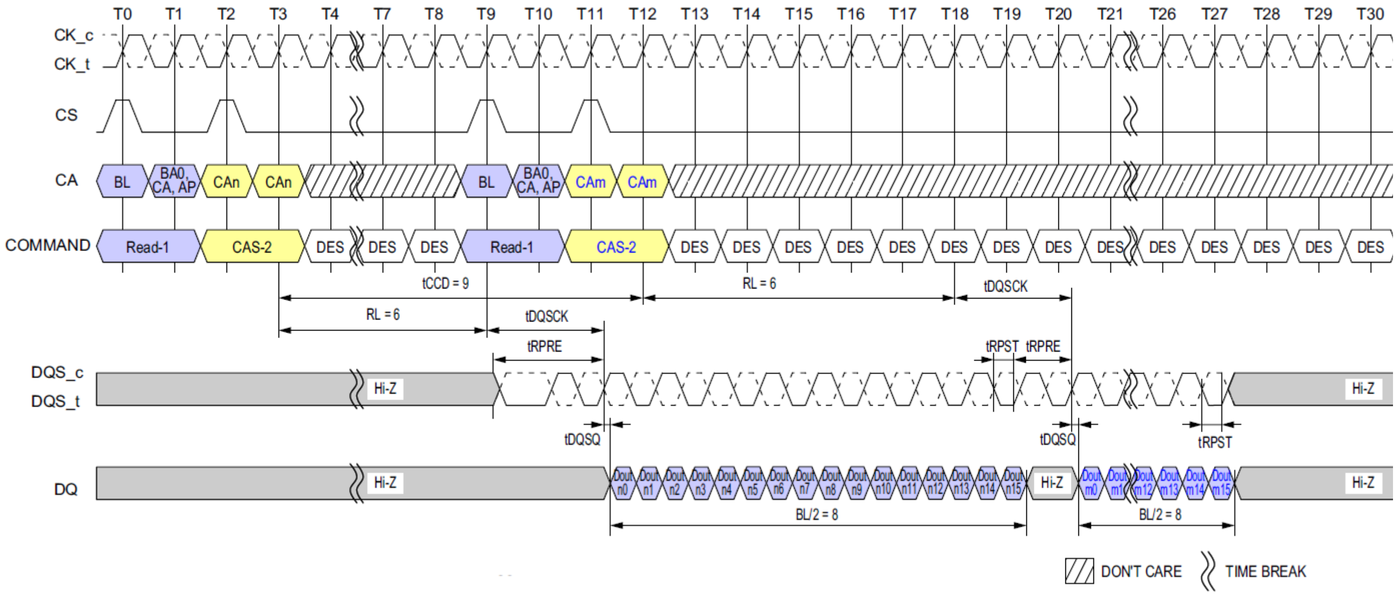


Figure 30: Consecutive Reads Operation: t_{CCD} = Min +1, Preamble = Toggle, 0.5nCK Postamble

NOTE :
1) BL = 16 for column n and column m, RL = 6, Preamble = Toggle, Postamble = 0.5nCK.
2) Dout n/m = data-out from column n and column m.
3) DES commands are shown for ease of illustration; other commands may be valid at these times.

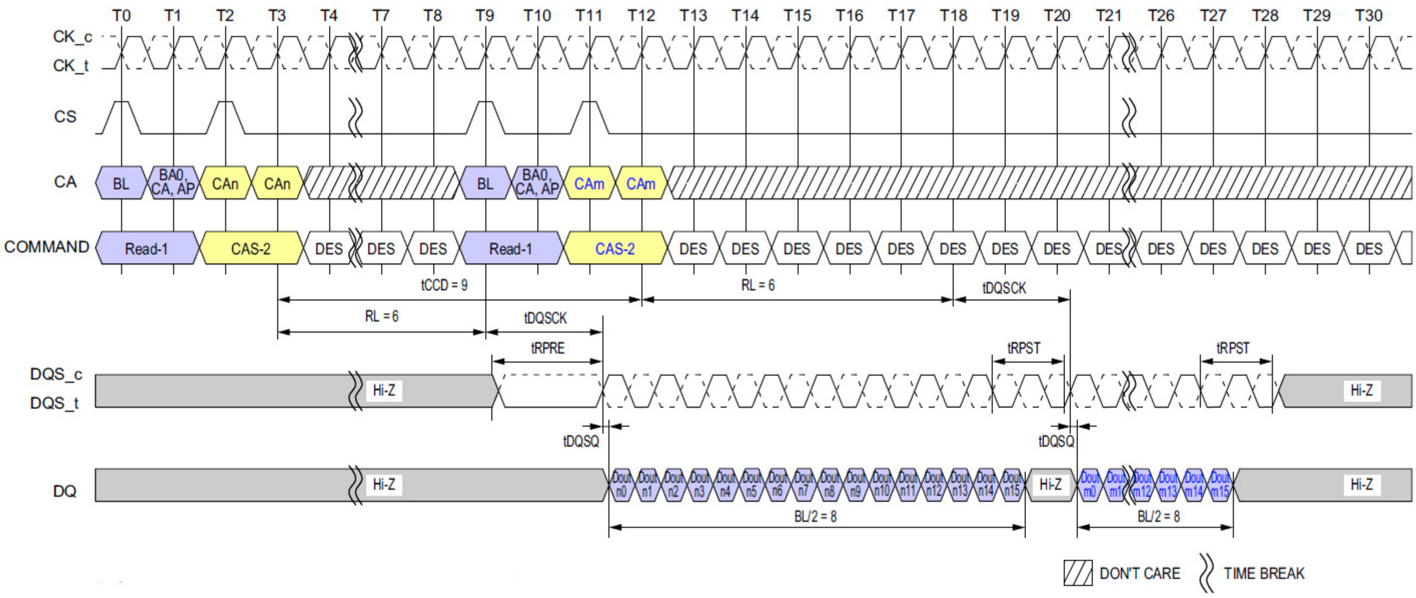


Figure 31: Consecutive Reads Operation: t_{CCD} = Min +1, Preamble = Static, 1.5nCK Postamble

NOTE :
1) BL = 16 for column n and column m, RL = 6, Preamble = Static, Postamble = 1.5nCK.
2) Dout n/m = data-out from column n and column m.
3) DES commands are shown for ease of illustration; other commands may be valid at these times.

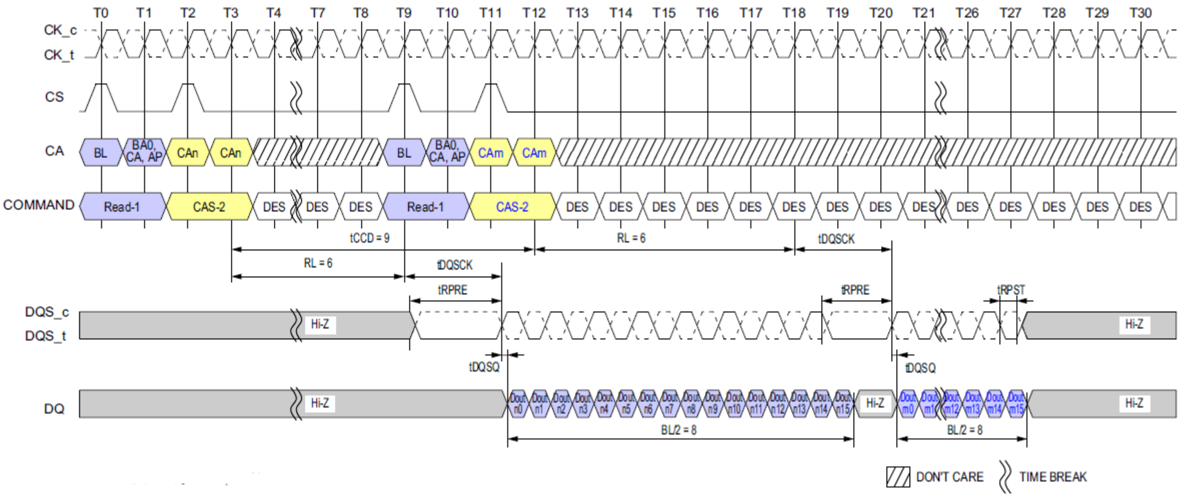


Figure 32: Consecutive Reads Operation: t_{CCD} = Min +1, Preamble = Static, 0.5nCK Postamble

NOTE :
1) BL = 16 for column n and column m, RL = 6, Preamble = Static, Postamble = 0.5nCK.
2) Dout n/m = data-out from column n and column m.
3) DES commands are shown for ease of illustration; other commands may be valid at these times.

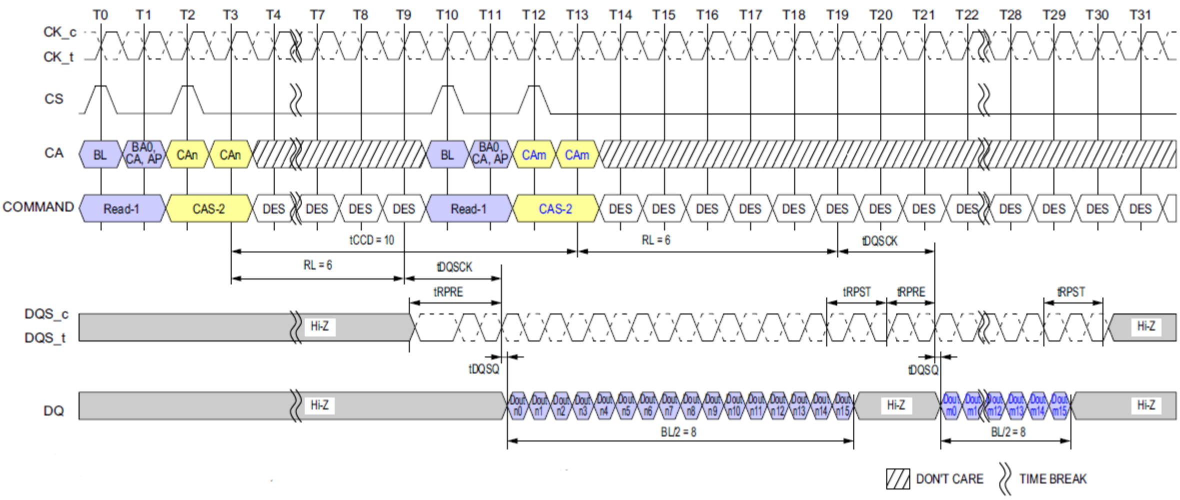


Figure 33: Consecutive Reads Operation: t_{CCD} = Min +2, Preamble = Toggle, 1.5nCK Postamble

NOTE :
1) BL = 16 for column n and column m, RL = 6, Preamble = Toggle, Postamble = 1.5nCK.
2) Dout n/m = data-out from column n and column m.
3) DES commands are shown for ease of illustration; other commands may be valid at these times.

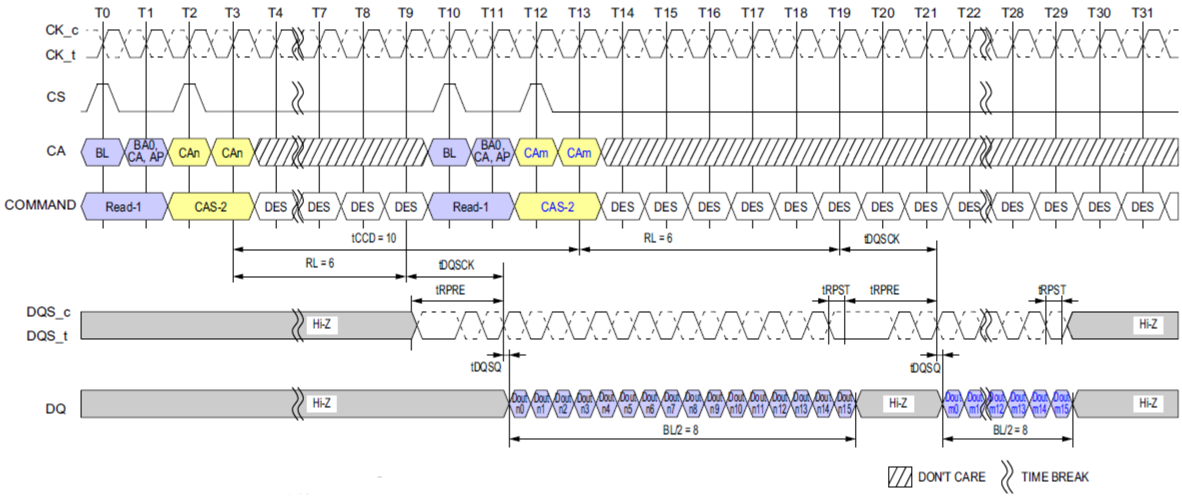


Figure 34: Consecutive Reads Operation: t_{CCD} = Min +2, Preamble = Toggle, 0.5nCK Postamble

NOTE :
1) BL = 16 for column n and column m, RL = 6, Preamble = Toggle, Postamble = 0.5nCK.
2) Dout n/m = data-out from column n and column m.
3) DES commands are shown for ease of illustration; other commands may be valid at these times.

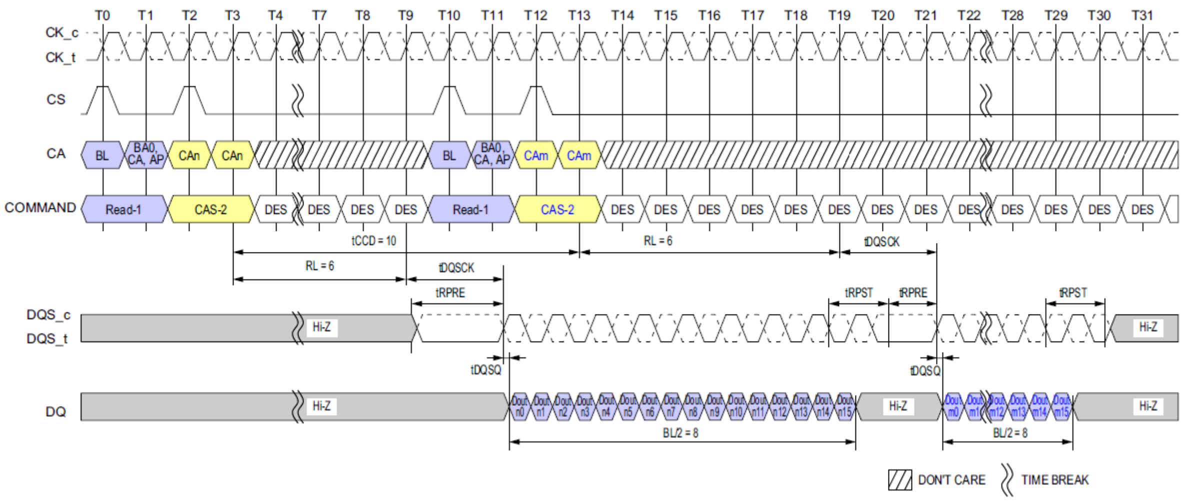


Figure 35: Consecutive Reads Operation: t_{CCD} = Min +2, Preamble = Static, 1.5nCK Postamble

NOTE :
1) BL = 16 for column n and column m, RL = 6, Preamble = Static, Postamble = 1.5nCK
2) Dout n/m = data-out from column n and column m.
3) DES commands are shown for ease of illustration; other commands may be valid at these times.

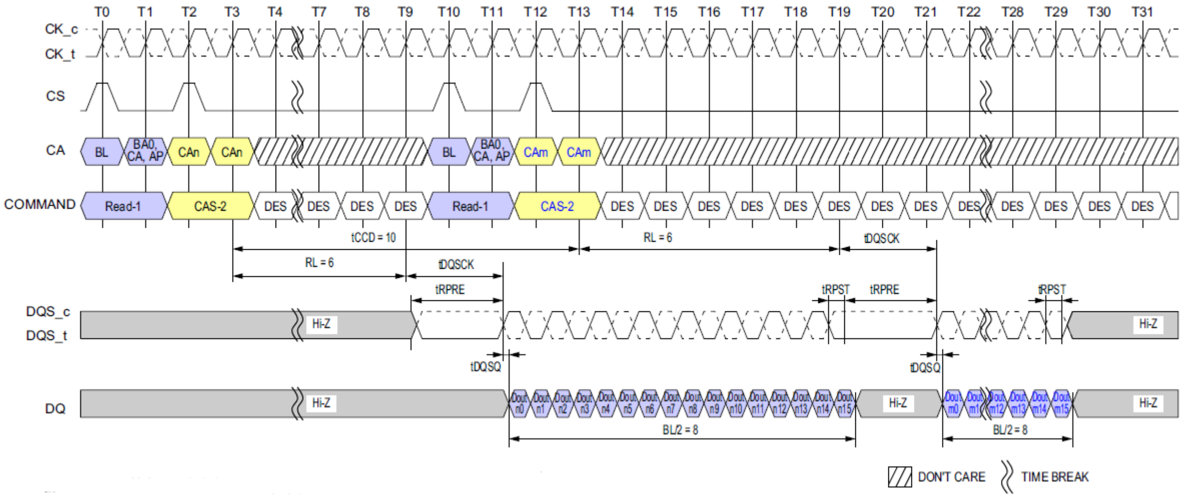


Figure 36: Consecutive Reads Operation: t_{CCD} = Min +2, Preamble = Static, 0.5nCK Postamble

NOTE :
1) BL = 16 for column n and column m, RL = 6, Preamble = Static, Postamble = 0.5nCK.
2) Dout n/m = data-out from column n and column m.
3) DES commands are shown for ease of illustration; other commands may be valid at these times.

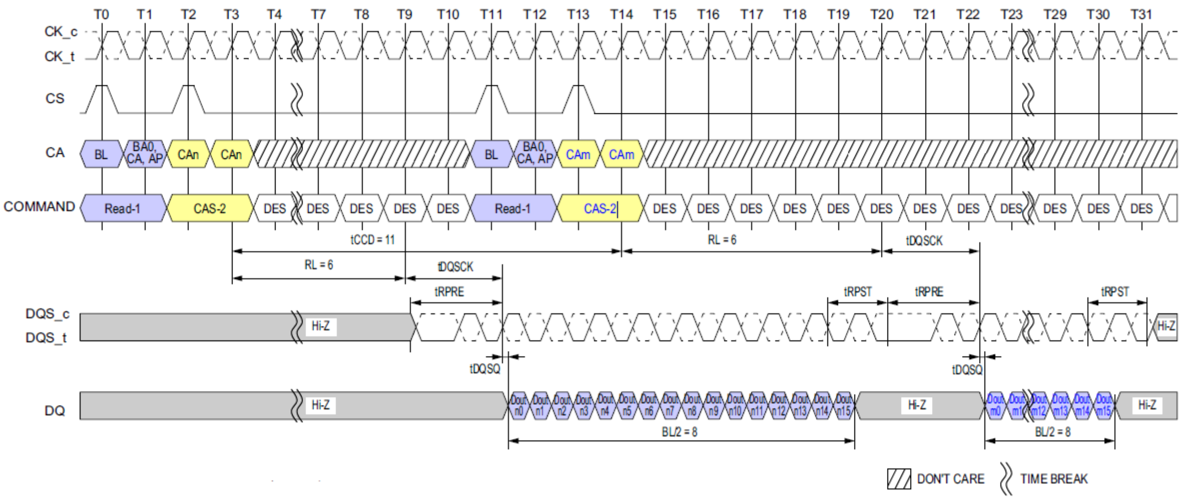


Figure 37: Consecutive Reads Operation: t_{CCD} = Min +3, Preamble = Toggle, 1.5nCK Postamble

NOTE :
1) BL = 16 for column n and column m, RL = 6, Preamble = Toggle, Postamble = 1.5nCK.
2) Dout n/m = data-out from column n and column m.
3) DES commands are shown for ease of illustration; other commands may be valid at these times.

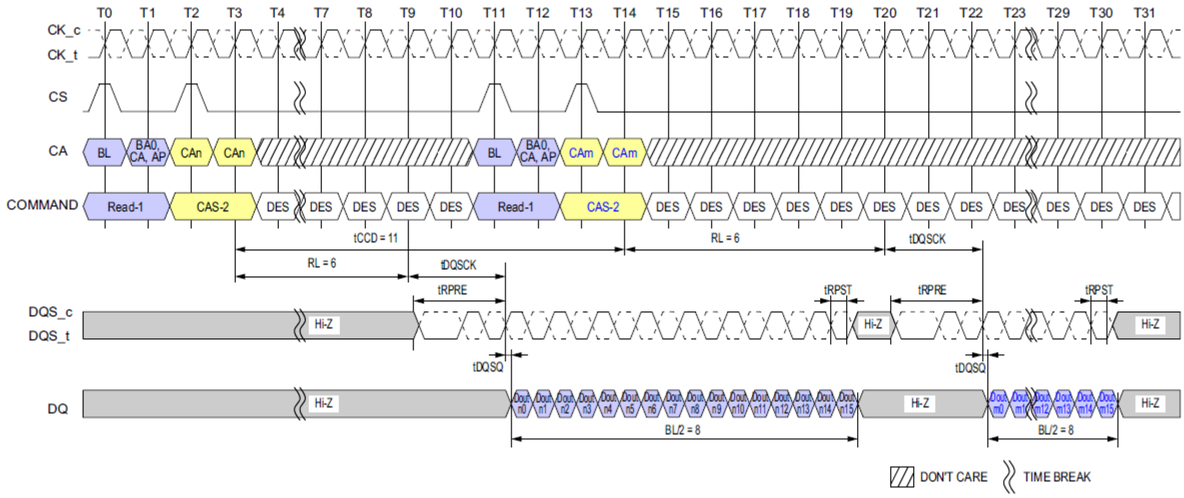


Figure 38: Consecutive Reads Operation: tCCD = Min +3, Preamble = Toggle, 0.5nCK Postamble

NOTE :
1) BL = 16 for column n and column m, RL = 6, Preamble = Toggle, Postamble = 0.5nCK.
2) Dout n/m = data-out from column n and column m.
3) DES commands are shown for ease of illustration; other commands may be valid at these times.

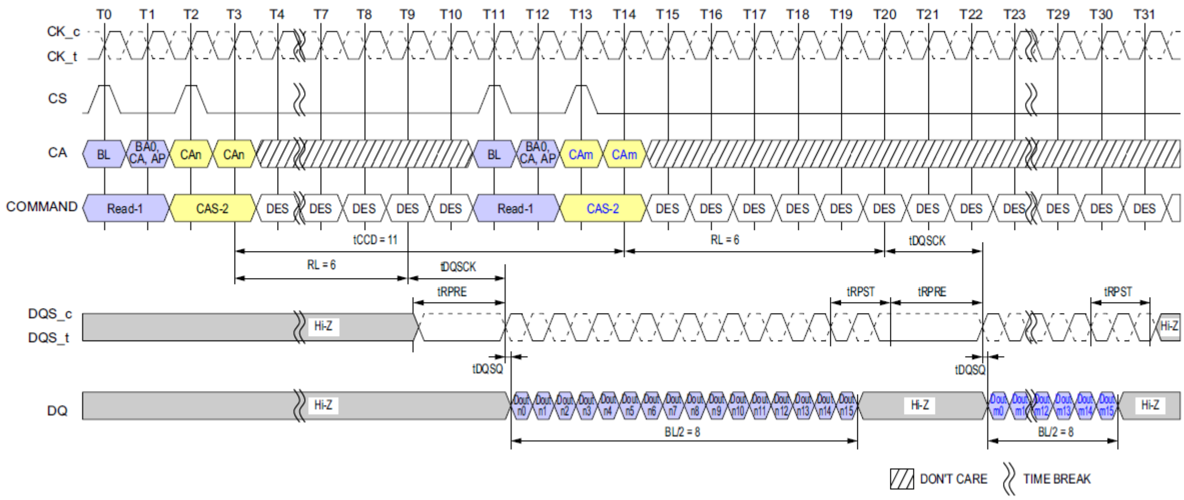


Figure 39: Consecutive Reads Operation: t_{CCD} = Min +3, Preamble = Static, 1.5nCK Postamble

NOTE :
1) BL = 16 for column n and column m, RL = 6, Preamble = Static, Postamble = 1.5nCK.
2) Dout n/m = data-out from column n and column m.
3) DES commands are shown for ease of illustration; other commands may be valid at these times.

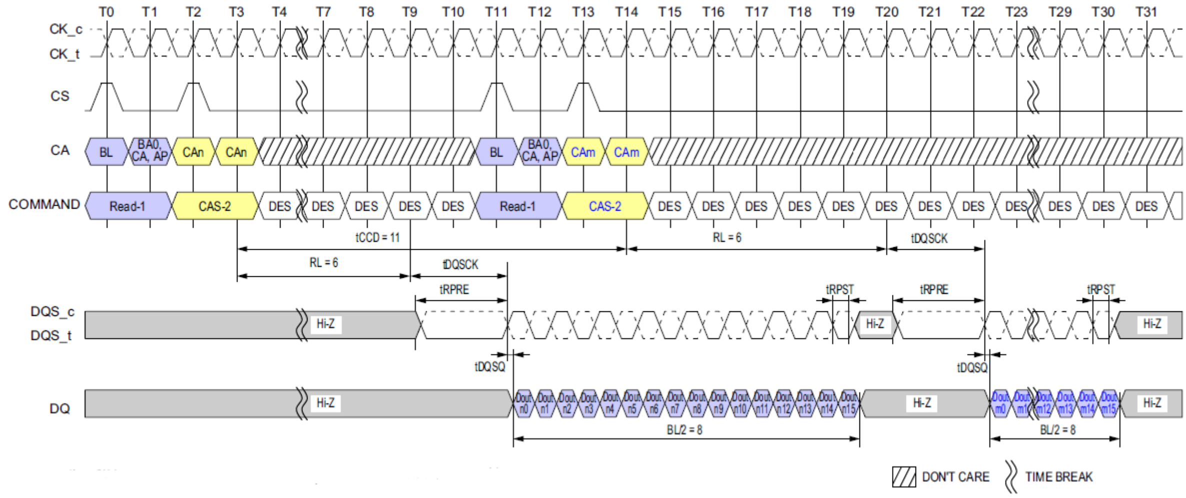


Figure 40: Consecutive Reads Operation: t_{CCD} = Min +3, Preamble = Static, 0.5nCK Postamble

NOTE :
1) BL = 16 for column n and column m, RL = 6, Preamble = Static, Postamble = 0.5nCK.
2) Dout n/m = data-out from column n and column m.
3) DES commands are shown for ease of illustration; other commands may be valid at these times.

3.16 Write to Write Operation

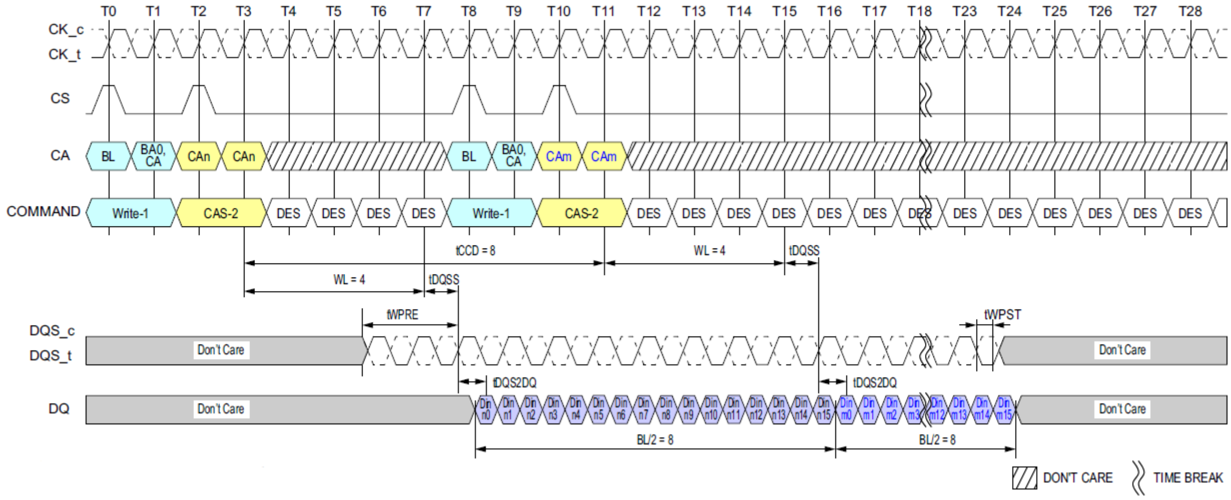


Figure 41: Seamless Writes Operation: t_{CCD} = Min, 0.5nCK Postamble

NOTE :
1) BL=16, Write Postamble = 0.5nCK.
2) Dout n/m = data-in to column n and column m.
3) The minimum number of clock cycles from the burst write command to the burst write command for any bank is BL/2.
4) DES commands are shown for ease of illustration; other commands may be valid at these times.

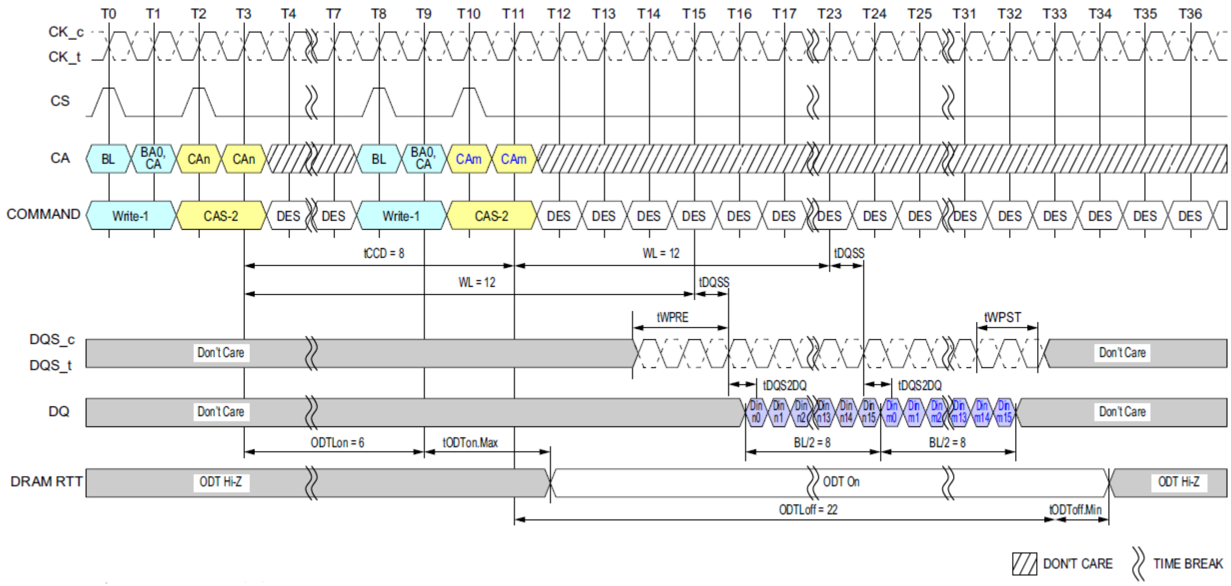


Figure 42: Seamless Writes Operation: t_{CCD} = Min, 1.5nCK Postamble, 533MHz < Clock Freq. ≤ 800MHz, ODT Worst Timing Case

NOTE :
1) Clock Frequency = 800MHz, $t_{CK(AVG)}$ = 1.25ns.
2) BL=16, Write Postamble = 1.5nCK.
3) Dout n/m = data-in to column n and column m.
4) The minimum number of clock cycles from the burst write command to the burst write command for any bank is BL/2.
5) DES commands are shown for ease of illustration; other commands may be valid at these times.

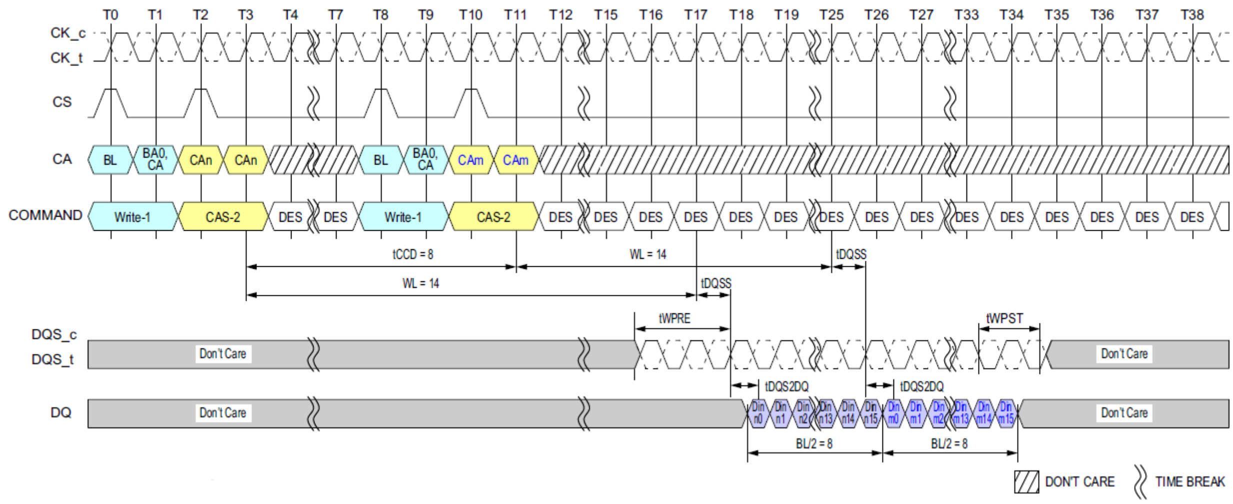


Figure 43: Seamless Writes Operation: t_{CCD} = Min, 1.5nCK Postamble

NOTE :
1) BL=16, Write Postamble = 1.5nCK.
2) Dout n/m = data-in to column n and column m.
3) The minimum number of clock cycles from the burst write command to the burst write command for any bank is BL/2.
4) DES commands are shown for ease of illustration; other commands may be valid at these times.

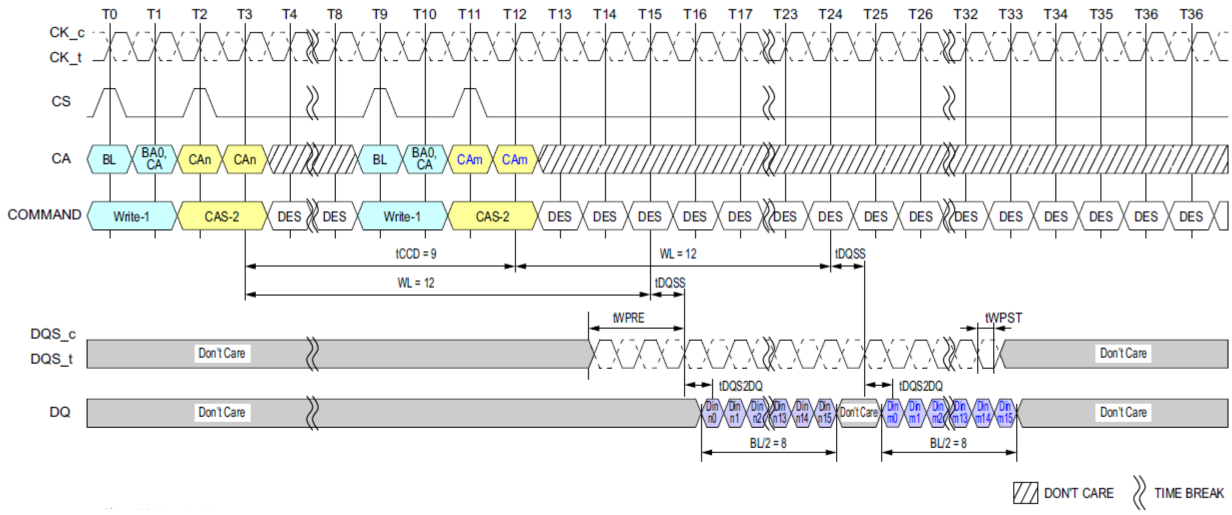


Figure 44: Consecutive Writes Operation: t_{CCD} = Min + 1, 0.5nCK Postamble

NOTE :
1) BL=16, Write Postamble = 0.5nCK.
2) Dout n/m = data-in to column n and column m.
3) DES commands are shown for ease of illustration; other commands may be valid at these times.

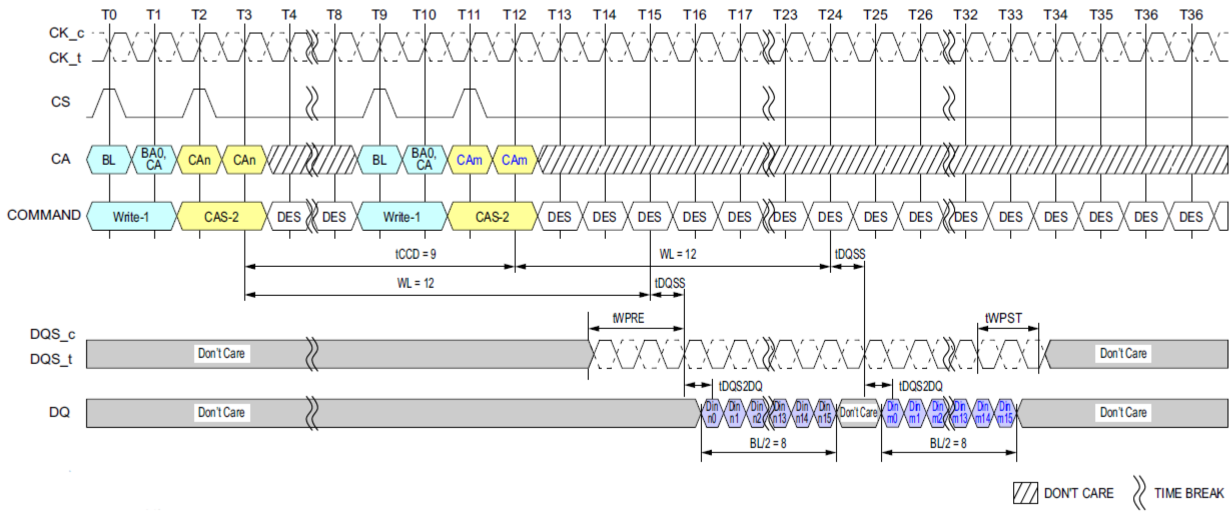


Figure 45: Consecutive Writes Operation: t_{CCD} = Min + 1, 1.5nCK Postamble

NOTE :
1) BL=16, Write Postamble = 1.5nCK.
2) Dout n/m = data-in to column n and column m.
3) DES commands are shown for ease of illustration; other commands may be valid at these times.

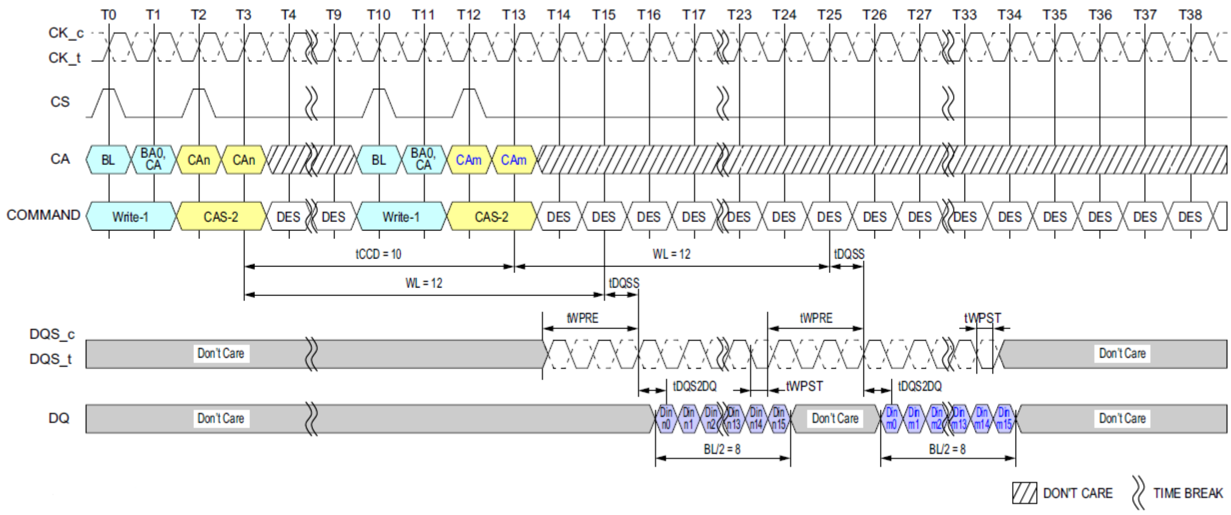


Figure 46: Consecutive Writes Operation: t_{CCD} = Min + 2, 0.5nCK Postamble

NOTE :
1) BL=16, Write Postamble = 0.5nCK.
2) Dout n/m = data-in to column n and column m.
3) DES commands are shown for ease of illustration; other commands may be valid at these times.

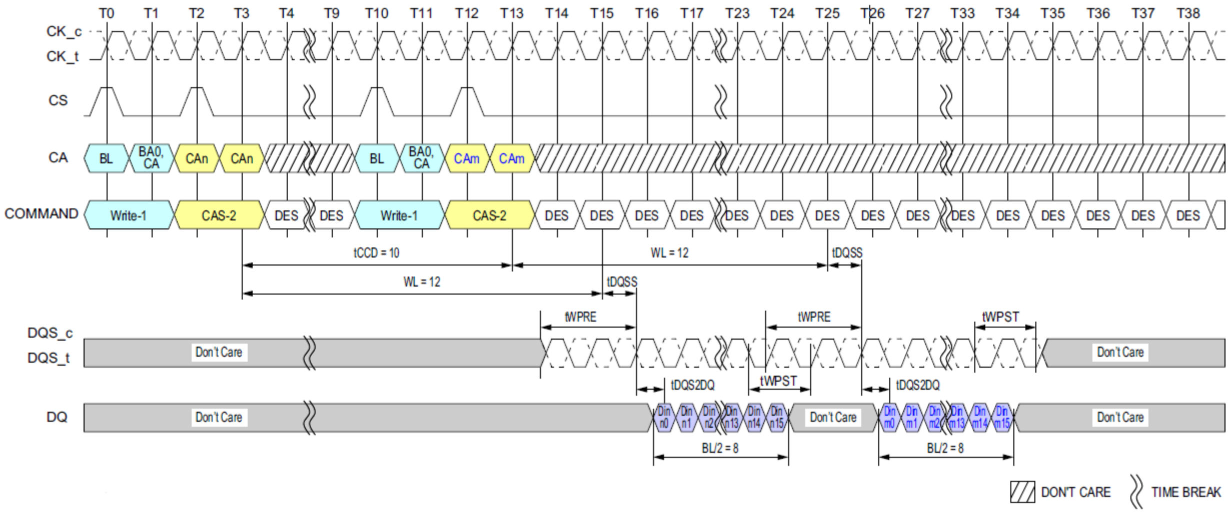


Figure 47: Consecutive Writes Operation: t_{CCD} = Min + 2, 1.5nCK Postamble

NOTE :

1) BL=16, Write Postamble = 1.5nCK.
2) Dout n/m = data-in to column n and column m.
3) DES commands are shown for ease of illustration; other commands may be valid at these times.

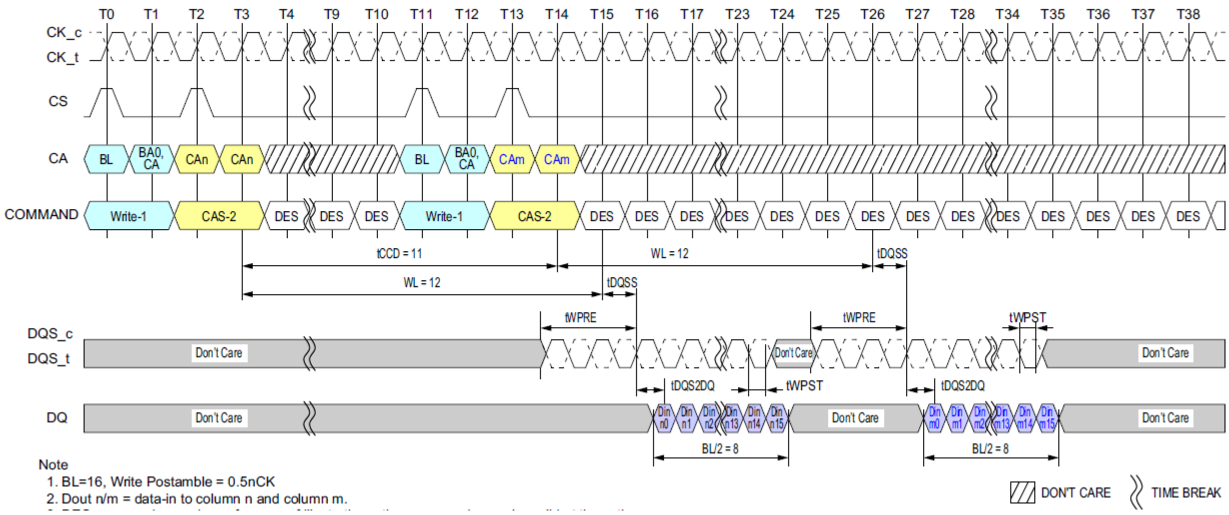


Figure 48: Consecutive Writes Operation: t_{CCD} = Min + 3, 0.5nCK Postamble

NOTE :

1) BL=16, Write Postamble = 0.5nCK.
2) Dout n/m = data-in to column n and column m.
3) DES commands are shown for ease of illustration; other commands may be valid at these times.

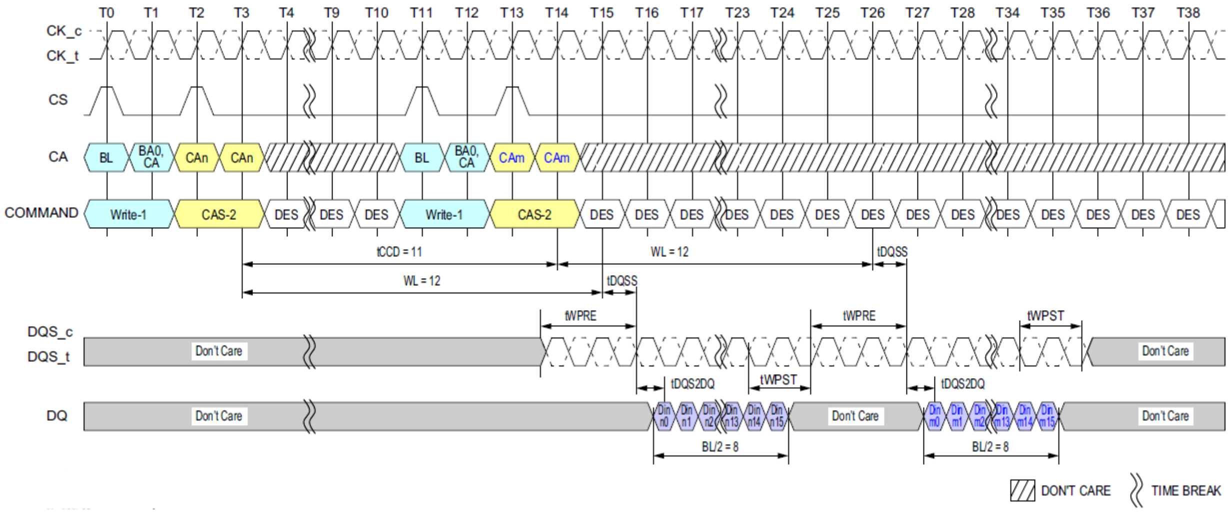


Figure 49: Consecutive Writes Operation: t_{CCD} = Min + 3, 1.5nCK Postamble

NOTE :
1) BL=16, Write Postamble = 1.5nCK.
2) Dout n/m = data-in to column n and column m.
3) DES commands are shown for ease of illustration; other commands may be valid at these times.

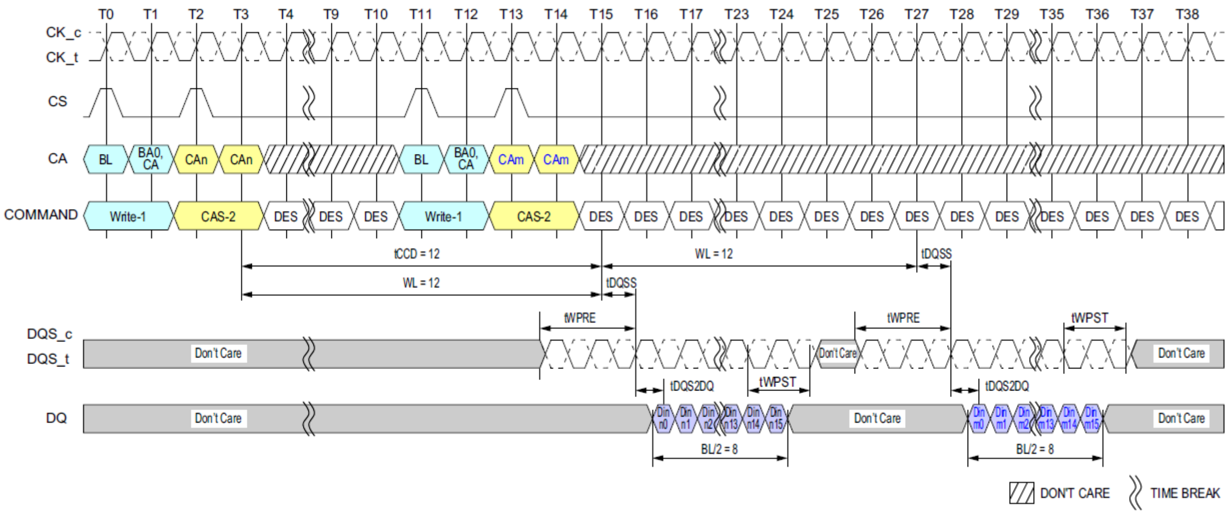


Figure 50: Consecutive Writes Operation: t_{CCD} = Min + 4, 1.5nCK Postamble

NOTE :
1) BL=16, Write Postamble = 1.5nCK.
2) Dout n/m = data-in to column n and column m.
3) DES commands are shown for ease of illustration; other commands may be valid at these times.

3.17 Masked Write Operation

The LPDDR4-SDRAM requires that Write operations which include a byte mask anywhere in the burst sequence must use the Masked Write command. This allows the DRAM to implement efficient data protection schemes based on larger data blocks. The Masked Write-1 command is used to begin the operation, followed by a CAS-2 command. A Masked Write command to the same bank cannot be issued until t_{CCDMW} later, to allow the LPDDR4-SDRAM to finish the internal Read-Modify-Write. One Data Mask-Invert (DMI) pin is provided per byte lane, and the Data Mask-Invert timings match data bit (DQ) timing. See the section on "Data Mask Invert" for more information on the use of the DMI signal.

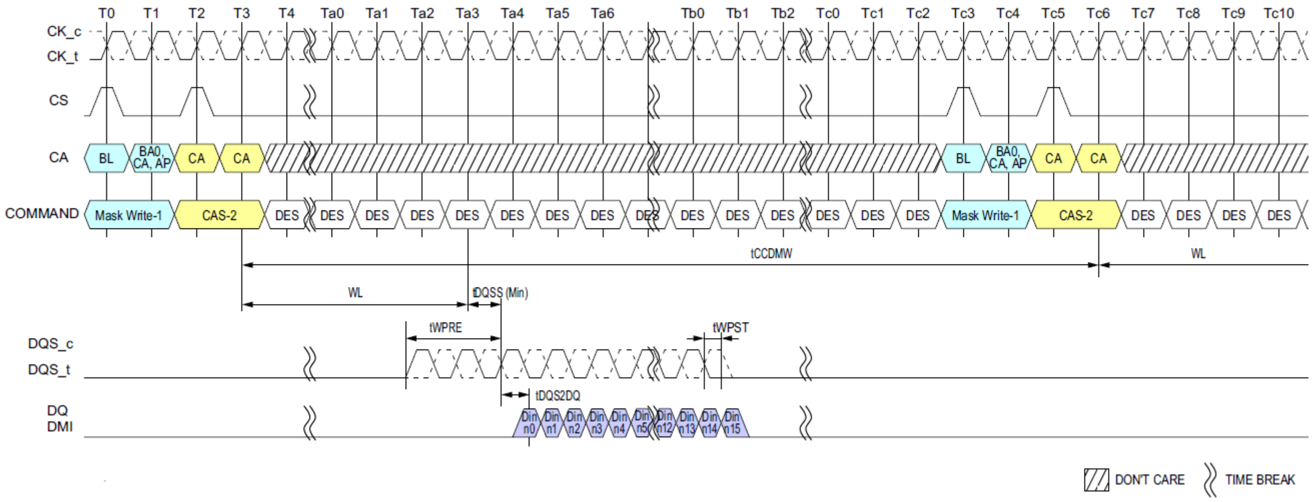


Figure 51: Masked Write Command - Same Bank

NOTE :
1) BL=16, Write Postamble = 0.5nCK, DQ/DQS: V_{SSQ} termination.
2) Din n = data-in to columnm.n.
3) Mask-Write supports only BL16 operations. For BL32 configuration, the system needs to insert only 16 bit wide data for masked write operation.
4) DES commands are shown for ease of illustration; other commands may be valid at these time.

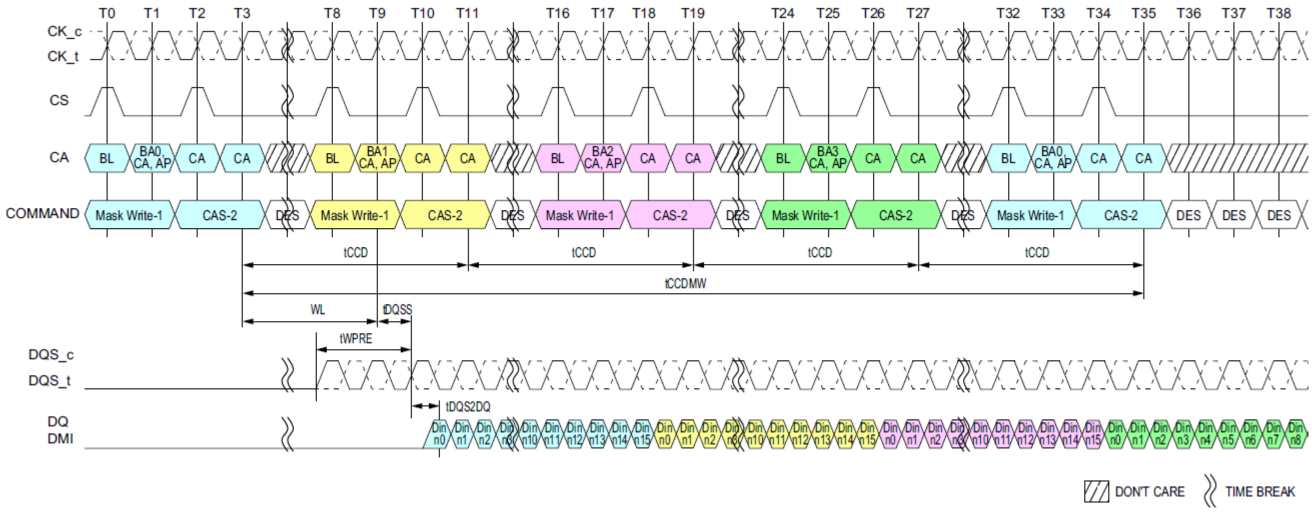


Figure 52: Masked Write Command - Different Bank

NOTE :
1) BL=16, DQ/DQS/DMI: V_{SSQ} termination.
2) Din n = data-in to columnm.n
3) Mask-Write supports only BL16 operations. For BL32 configuration, the system needs to insert only 16 bit wide data for masked write operation.
4) DES commands are shown for ease of illustration; other commands may be valid at these time.

3.18 Masked Write Timing Constraints For BL16

[Table 18] Timing constraints for Same bank: DQ ODT is Disabled

Next CMD / Current CMD	Active	Read (BL=16 or 32)	Write (BL=16 or 32)	Masked Write	Precharge
Active	illegal	RU (t_{RCD}/t_{CK})	RU (t_{RCD}/t_{CK})	RU (t_{RCD}/t_{CK})	RU (t_{RAS}/t_{CK})
Read with BL = 16	illegal	8 1)	RL+RU($t_{DQSCK(max)}/t_{CK}$)+BL/2-WL+t_{WPRE}+RD(t_{RPST})	RL+RU($t_{DQSCK(max)}/t_{CK}$)+BL/2-WL+t_{WPRE}+RD(t_{RPST})	BL/2+max{(8,RU(t_{RTP}/t_{CK})}-8
Read with BL = 32	illegal	16 2)	RL+RU($t_{DQSCK(max)}/t_{CK}$)+BL/2-WL+t_{WPRE}+RD(t_{RPST})	RL+RU($t_{DQSCK(max)}/t_{CK}$)+BL/2-WL+t_{WPRE}+RD(t_{RPST})	BL/2+max{(8,RU(t_{RTP}/t_{CK})}-8
Write with BL = 16	illegal	WL+1+BL/2+RU(t_{WTR}/t_{CK})	8 1)	t_{CCDMW} 3)	WL+ 1 + BL/2+RU(t_{WR}/t_{CK})
Write with BL = 32	illegal	WL+1+BL/2+RU(t_{WTR}/t_{CK})	16 2)	t_{CCDMW} + 8 4)	WL+ 1 + BL/2+RU(t_{WR}/t_{CK})
Masked Write	illegal	WL+1+BL/2+RU(t_{WTR}/t_{CK})	t_{CCD}	t_{CCDMW} 3)	WL+ 1 + BL/2+RU(t_{WR}/t_{CK})
Precharge	RU(t_{RP}/t_{CK}), RU(t_{RPab}/t_{CK})	illegal	illegal	illegal	4

NOTE :
1) In the case of BL = 16, t_{CCD} is 8 × t_{CK}.
2) In the case of BL = 32, t_{CCD} is 16 × t_{CK}.
3) t_{CCDMW} = 32 × t_{CK} (4×t_{CCD} at BL=16).
4) Write with BL=32 operation has 8 × tCK longer than BL =16.
5) tRPST values depend on MR1-OP[7] respectively.

[Table 19] Timing constraints for Same bank: DQ ODT is Enabled

Next CMD / Current CMD	Active	Read (BL=16 or 32)	Write (BL=16 or 32)	Masked Write	Precharge
Read with BL = 16	illegal	8 1)	RL+RU($t_{DQSCK(max)}/t_{CK}$)+BL/2+RD(t_{RPST})-ODTLon-RD($t_{ODTon,min}/t_{CK}$)	RL+RU($t_{DQSCK(max)}/t_{CK}$)+BL/2+RD(t_{RPST})-ODTLon-RD($t_{ODTon,min}/t_{CK}$)	BL/2+max{(8,RU(t_{RTP}/t_{CK})}-8
Read with BL = 32	illegal	16 2)	RL+RU($t_{DQSCK(max)}/t_{CK}$)+BL/2+RD(t_{RPST})-ODTLon-RD($t_{ODTon,min}/t_{CK}$)	RL+RU($t_{DQSCK(max)}/t_{CK}$)+BL/2+RD(t_{RPST})-ODTLon-RD($t_{ODTon,min}/t_{CK}$)	BL/2+max{(8,RU(t_{RTP}/t_{CK})}-8

NOTE :
1) In the case of BL = 16, t_{CCD} is 8×t_{CK}.
2) In the case of BL = 32, t_{CCD} is 16×t_{CK}.
3) The rest of the timing is same as DQ ODT is Disable case.
4) t_{RPST} values depend on MR1-OP[7] respectively.

[Table 20] Timing constraints for Different bank: DQ ODT is Disabled

Next CMD / Current CMD	Active	Read (BL=16 or 32)	Write (BL=16 or 32)	Masked Write	Precharge
Active	RU(t_{RRD}/t_{CK})	4	4	4	2
Read with BL = 16	4	8 1)	RL+RU($t_{DQSCK(max)}$/t_{CK})+BL/2-WL+t_{WPRE}+RD(t_{RPST})	RL+RU($t_{DQSCK(max)}$/t_{CK})+BL/2-WL+t_{WPRE}+RD(t_{RPST})	2
Read with BL = 32	4	16 2)	RL+RU($t_{DQSCK(max)}$/t_{CK})+BL/2-WL+t_{WPRE}+RD(t_{RPST})	RL+RU($t_{DQSCK(max)}$/t_{CK})+BL/2-WL+t_{WPRE}+RD(t_{RPST})	2
Write with BL = 16	4	WL+1+BL/2+RU(t_{WTR}/t_{CK})	8 1)	8 1)	2
Write with BL = 32	4	WL+1+BL/2+RU(t_{WTR}/t_{CK})	16 2)	16 2)	2
Masked Write	4	WL+1+BL/2+RU(t_{WTR}/t_{CK})	8 1)	8 1)	2
Precharge	4	4	4	4	4

NOTE :
1) In the case of BL = 16, t_{CCD} is 8 × tCK.
2) In the case of BL = 32, t_{CCD} is 16 × tCK.
3) tRPST values depend on MR1-OP[7] respectively.

[Table 21] Timing constraints for Different bank: DQ ODT is Enabled

Next CMD / Current CMD	Active	Read (BL=16 or 32)	Write (BL=16 or 32)	Masked Write	Precharge
Read with BL = 16	4	8 1)	RL+RU($t_{DQSCK(max)}$/t_{CK})+BL/2+RD(t_{RPST})-ODTLon-RD($t_{ODTon,min}$/t_{CK})	RL+RU($t_{DQSCK(max)}$/t_{CK})+BL/2+RD(t_{RPST})-ODTLon-RD($t_{ODTon,min}$/t_{CK})	2
Read with BL = 32	4	16 2)	RL+RU($t_{DQSCK(max)}$/t_{CK})+BL/2+RD(t_{RPST})-ODTLon-RD($t_{ODTon,min}$/t_{CK})	RL+RU($t_{DQSCK(max)}$/t_{CK})+BL/2+RD(t_{RPST})-ODTLon-RD($t_{ODTon,min}$/t_{CK})	2

NOTE :
1) In the case of BL = 16, t_{CCD} is 8×tCK.
2) In the case of BL = 32, t_{CCD} is 16×tCK.
3) The rest of the timing is same as DQ ODT is Disable case.
4) t_{RPST} values depend on MR1-OP[7] respectively.

3.19 Data Mask (DM) And Data Bus Inversion (DBIdc) Function

LPDDR4 SDRAM supports the function of Data Mask and Data Bus inversion. Its details are shown below.

- LPDDR4 device supports Data Mask (DM) function for Write operation.
- LPDDR4 device supports Data Bus Inversion (DBIdc) function for Write and Read operation.
- LPDDR4 supports DM and DBIdc function with a byte granularity.
- DBIdc function during Write or Masked Write can be enabled or disabled through MR3 OP[7].
- DBIdc function during Read can be enabled or disabled through MR3 OP[6].
- DM function during Masked Write can be enabled or disabled through MR13 OP[5].
- LPDDR4 device has one Data Mask Inversion (DMI) signal pin per byte; total of 2 DMI signals per channel.
- DMI signal is a bi-directional DDR signal and is sampled along with the DQ signals for Read and Write or Masked Write operation.

There are eight possible combinations for LPDDR4 device with DM and DBIdc function. Table 22 describes the functional behavior for all combinations.

[Table 22] Function Behavior of DMI Signal During Write, Masked Write and Read Operation

DM function	Write DBIdc Function	Read DBIdc Function	DMI Signal during Write Command	Signal during Masked Write Command	DMI Signal During Read	DMI Signal during MPC WR FIFO	DMI Signal during MPC RD FIFO	DMI Signal during MPC DQ Read Training	DMI Signal during MRR
Disable	Disable	Disable	Notes: 1	Notes: 1,3	Notes: 2	Note: 1	Note: 2	Note: 2	Note: 2
Disable	Enable	Disable	Notes: 4	Notes: 3	Notes: 2	Note: 9	Note: 10	Note: 11	Note: 2
Disable	Disable	Enable	Notes: 1	Notes: 3	Notes: 5	Note: 9	Note: 10	Note: 11	Note: 12
Disable	Enable	Enable	Notes: 4	Notes: 3	Notes: 5	Note: 9	Note: 10	Note: 11	Note: 12
Enable	Disable	Disable	Notes: 6	Notes: 7	Notes: 2	Note: 9	Note: 10	Note: 11	Note: 2
Enable	Enable	Disable	Notes: 4	Notes: 8	Notes: 2	Note: 9	Note: 10	Note: 11	Note: 2
Enable	Disable	Enable	Notes: 6	Notes: 7	Notes: 5	Note: 9	Note: 10	Note: 11	Note: 12
Enable	Enable	Enable	Notes: 4	Notes: 8	Notes: 5	Note: 9	Note: 10	Note: 11	Note: 12

NOTE :
1) DMI input signal is a don’t care. DMI input receivers are turned OFF.
2) DMI output drivers are turned OFF.
3) Masked Write Command is not allowed and is considered an illegal command as DM function is disabled.
4) DMI signal is treated as DBI signal and it indicates whether DRAM needs to invert the Write data received on DQs within a byte. The LPDDR4 device inverts Write data received on the DQ inputs in case DMI was sampled HIGH, or leaves the Write data non-inverted in case DMI was sampled LOW.
5) The LPDDR4 DRAM inverts Read data on its DQ outputs associated within a byte and drives DMI signal HIGH when the number of ‘1’ data bits within a given byte lane is greater than four; otherwise the DRAM does not invert the read data and drives DMI signal LOW.
6) The LPDDR4 DRAM does not perform any mask operation when it receives Write command. During the Write burst associated with Write command, DMI signal must be driven LOW.
7) The LPDDR4 DRAM requires an explicit Masked Write command for all masked write operations. DMI signal is treated as DM signal and it indicates which bit time within the burst is to be masked. When DMI signal is HIGH, DRAM masks that bit time across all DQs associated within a byte. All DQ input signals within a byte are don’t care (either HIGH or LOW) when DMI signal is HIGH. When DMI signal is LOW, the LPDDR4 DRAM does not perform mask operation and data received on DQ input is written to the array.
8) The LPDDR4 DRAM requires an explicit Masked Write command for all masked write operations. The LPDDR4 device masks the Write data received on the DQ inputs if the total count of ‘1’ data bits on DQ[2:7] or DQ[10:15] (for Lower Byte or Upper Byte respectively) is equal to or greater than five and DMI signal is LOW. Otherwise the LPDDR4 DRAM does not perform mask operation and treats it as a legal DBI pattern; DMI signal is treated as DBI signal and data received on DQ input is written to the array.
9) DMI signal is treated as a training pattern. The LPDDR4 SDRAM does not perform any mask operation and does not invert Write data received on the DQ inputs.
10) DMI signal is treated as a training pattern. The LPDDR4 SDRAM returns DMI pattern written in WR-FIFO.
11) DMI signal is treated as a training pattern. For more details, see RD DQ Calibration section.
12) DBI may apply or may not apply during normal MRR. It's vendor specific.
If read DBI is enable with MRS and vendor cannot support the DBI during MRR, DBI pin status should be low.
If read DBI is enable with MRS and vendor can support the DBI during MRR, the LPDDR4 DRAM inverts Mode Register Read data on its DQ outputs associated within a byte and drives DMI signal HIGH when the number of ‘1’ data bits within a given byte lane is greater than four; otherwise the DRAM does not invert the read data and drives DMI signal LOW.

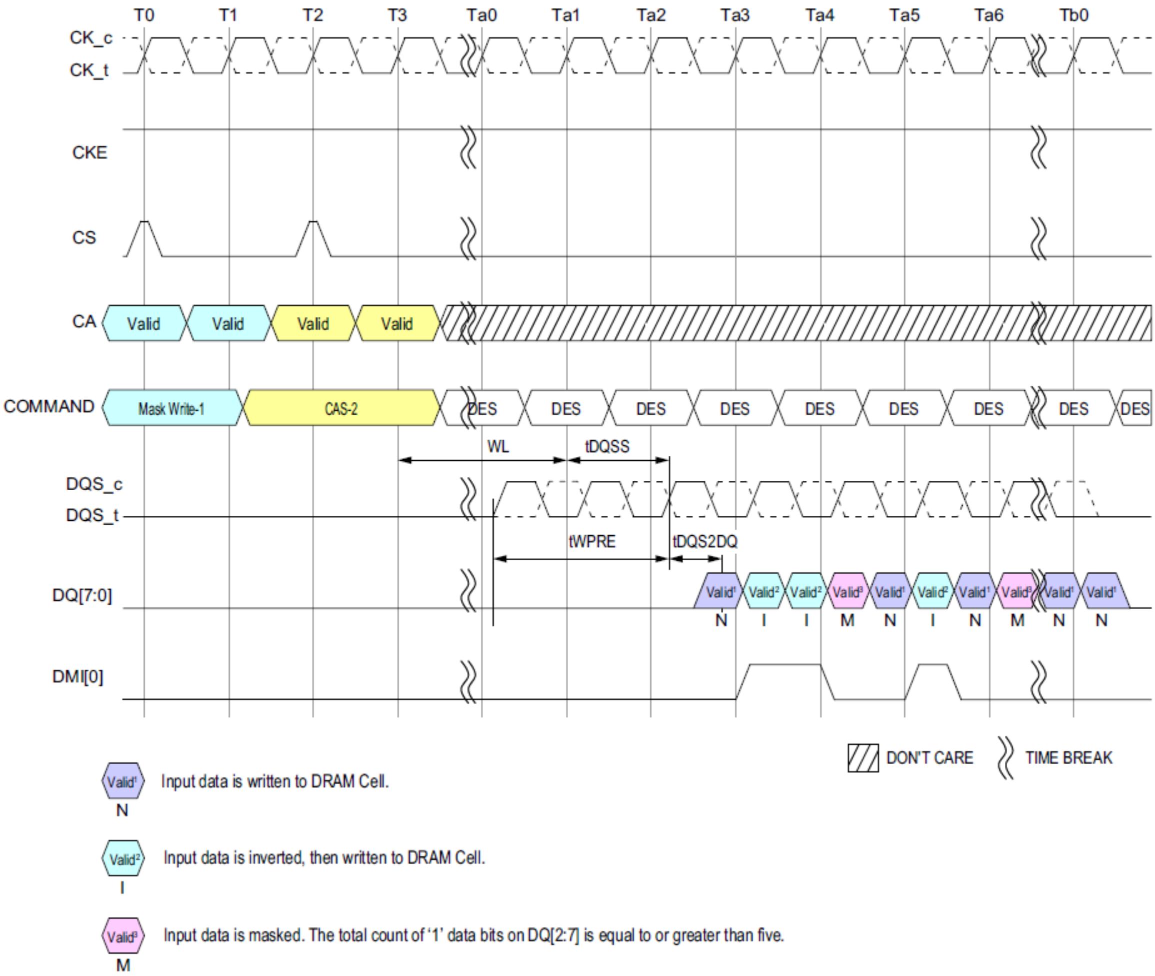


Figure 53: Masked Write Command w/ Write DBI Enabled; DM Enabled

NOTE :
1) Data Mask (DM) is Enable: MR13 OP [5] = 1_B, Data BUS Inversion (DBI) Write is Enable: MR3 OP[7] = 1_B.

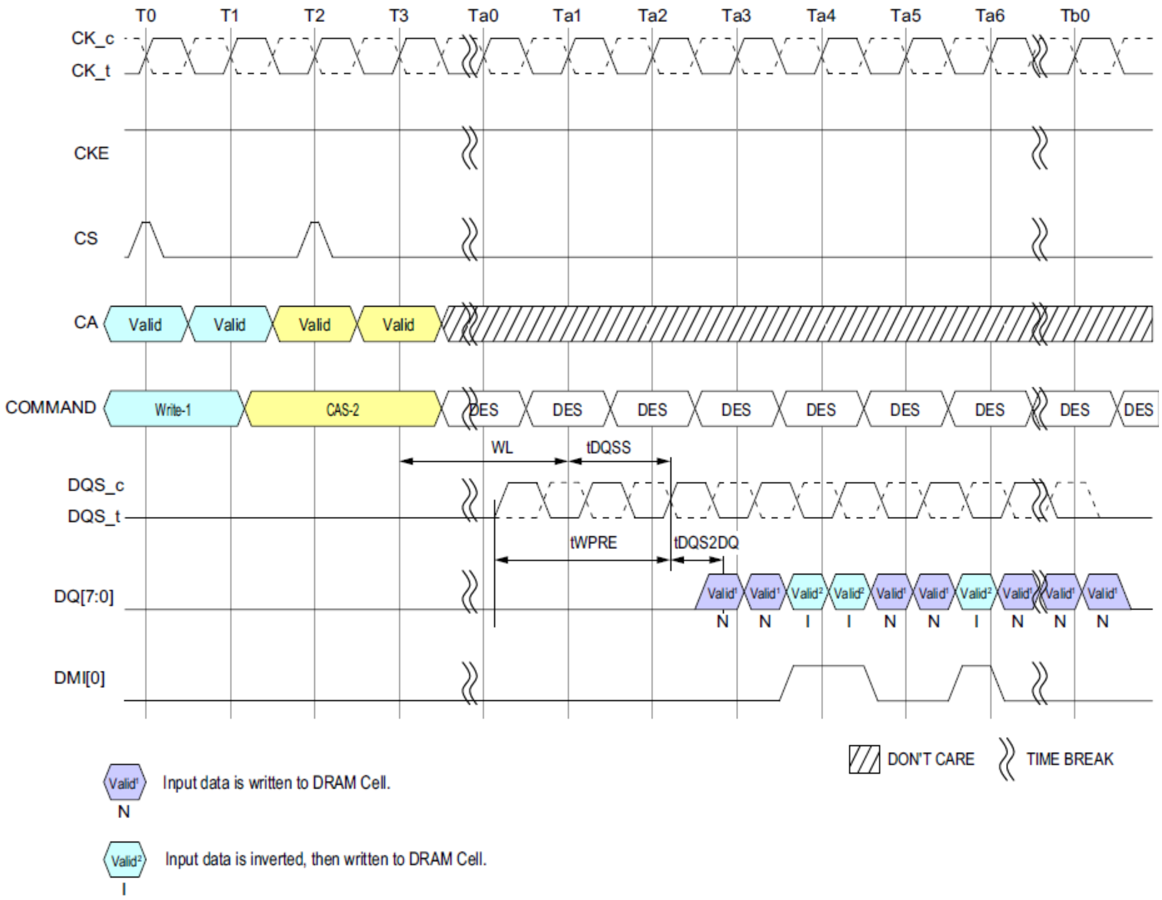


Figure 54: Write Command w/ Write DBI Enabled; DM Disabled

NOTE :
1) Data Mask (DM) is Disable: MR13 OP [5] = 0_B, Data BUS Inversion (DBI) Write is Enable: MR3 OP[7] = 1_B.

3.20 Precharge Operation

The PRECHARGE command is used to precharge or close a bank that has been activated. The PRECHARGE command is initiated with CS, and CA[5:0] in the proper state as defined by the Command Truth Table. The PRECHARGE command can be used to precharge each bank independently or all banks simultaneously. The AB flag and the bank address bit are used to determine which bank(s) to precharge. The precharged bank(s) will be available for subsequent row access t_{RPab} after an all-bank PRECHARGE command is issued, or t_{RPpb} after a single-bank PRECHARGE command is issued.
To ensure that LPDDR4 devices can meet the instantaneous current demands, the row-precharge time for an all-bank PRECHARGE (t_{RPab}) is longer than the per bank precharge time (t_{RPpb}).

[Table 23] Precharge Bank Selection

AB (CA[5], R1)	BA2 (CA[2], R2)	BA1 (CA[1], R2)	BA0 (CA[0], R2)	Precharged Bank(s)
0	0	0	0	Bank 0 only
0	0	0	1	Bank 1 only
0	0	1	0	Bank 2 only
0	0	1	1	Bank 3 only
0	1	0	0	Bank 4 only
0	1	0	1	Bank 5 only
0	1	1	0	Bank 6 only
0	1	1	1	Bank 7 only
1	Valid	Valid	Valid	All Banks

3.20.1 Burst READ Operation Followed By a PRECHARGE

The PRECHARGE command can be issued as early as BL/2 clock cycles after a READ command, but PRECHARGE cannot be issued until after t_{RAS} is satisfied. A new bank ACTIVATE command can be issued to the same bank after the row PRECHARGE time (t_{RP}) has elapsed. The minimum READ-to-PRECHARGE time must also satisfy a minimum analog time from the 2nd rising clock edge of the CAS-2 command. t_{RTP} begins BL/2-8 clock cycles after the READ command. For LPDDR4 READ-to-PRECHARGE timings see Table 24 Precharge & Auto Precharge clarification.

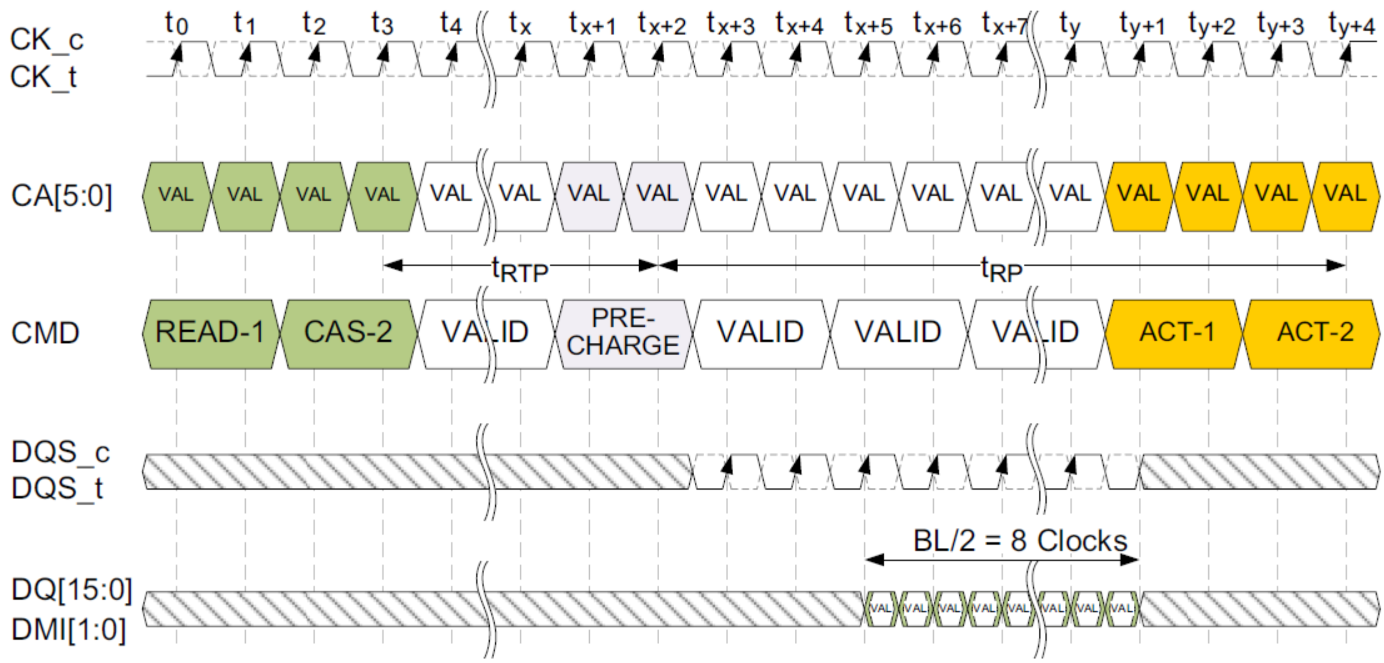


Figure 55: Burst READ Followed by PRECHARGE (Shown with BL16, 2tCK pre-amble)

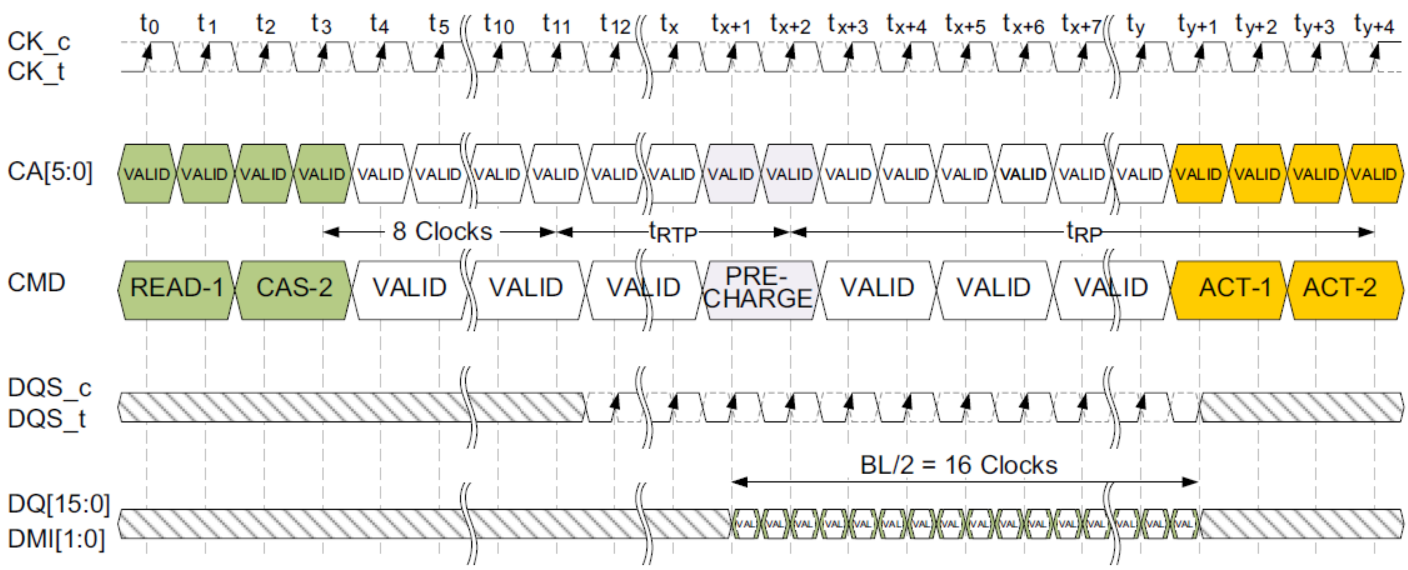


Figure 56: Burst READ followed by PRECHARGE (Shown with BL32, 2tCK pre-amble)

3.20.2 Burst WRITE Followed by PRECHARGE

A Write Recovery time (t_{WR}) must be provided before a PRECHARGE command may be issued. This delay is referenced from the next rising edge of CK_t after the last latching DQS clock of the burst. LPDDR4-SDRAM devices write data to the memory array in prefetch multiples (prefetch=16). An internal WRITE operation can only begin after a prefetch group has been clocked, so t_{WR} starts at the prefetch boundaries. The minimum WRITE-to-PRECHARGE time for commands to the same bank is WL + BL/2 + 1 + RU(t_{WR}/t_{CK}) clock cycles.

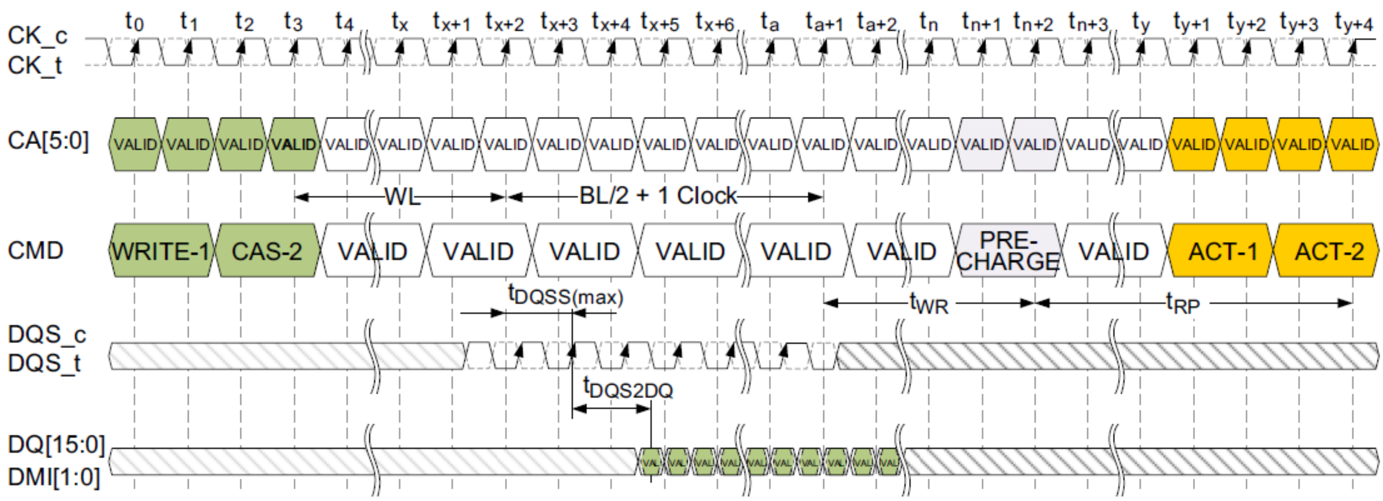


Figure 57: Burst WRITE Followed by PRECHARGE (Shown with BL16, 2tCK pre-amble)

3.20.3 Burst READ With Auto-PRECHARGE

If AP is HIGH when a READ command is issued, the READ with Auto-PRECHARGE function is engaged. An internal precharge procedure starts a following delay time after the READ command. And this delay time depends on BL setting.
BL = 16: nRTP
BL = 32: 8nCK + nRTP

For LPDDR4 Auto-PRECHARGE calculations, see Table 24. Following an Auto-PRECHARGE operation, an ACTIVATE command can be issued to the same bank if the following two conditions are both satisfied:

- The RAS precharge time (t_{RP}) has been satisfied from the clock at which the Auto-PRECHARGE begin, or
- The RAS cycle time (t_{RC}) from the previous bank activation has been satisfied.

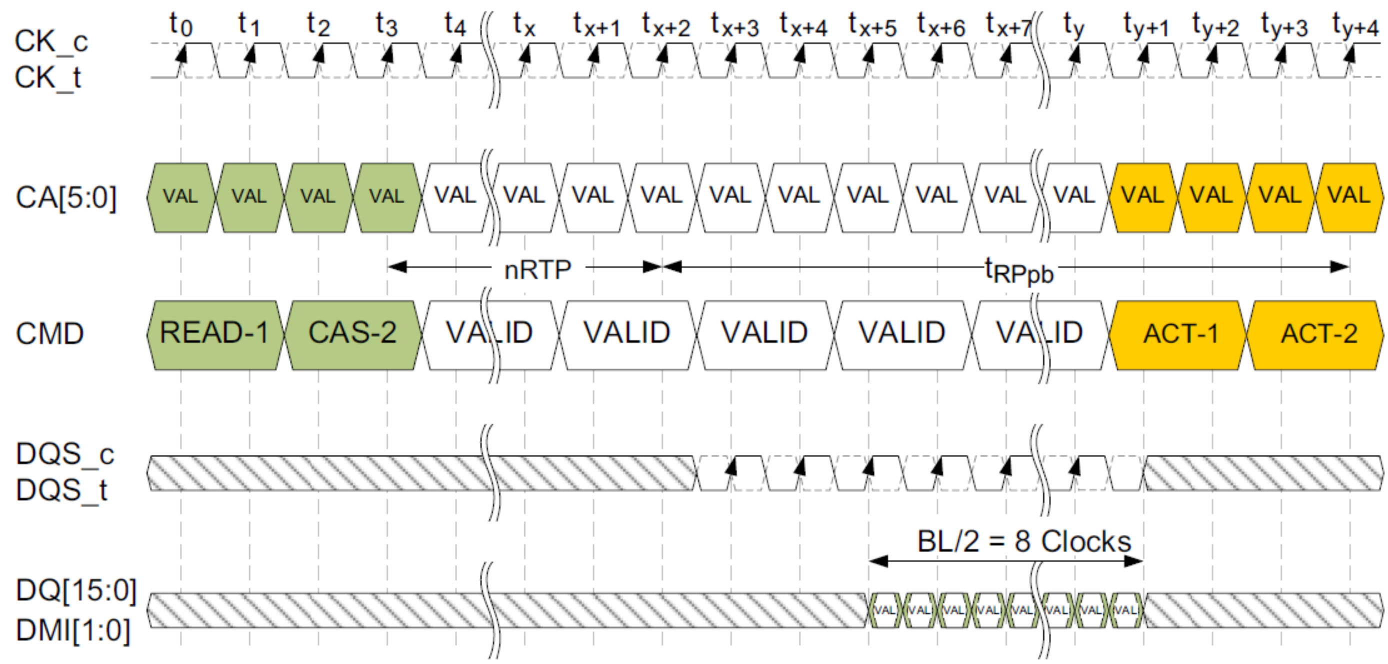


Figure 58: Burst READ with Auto-Precharge (Shown with BL16, 2tCK pre-amble)

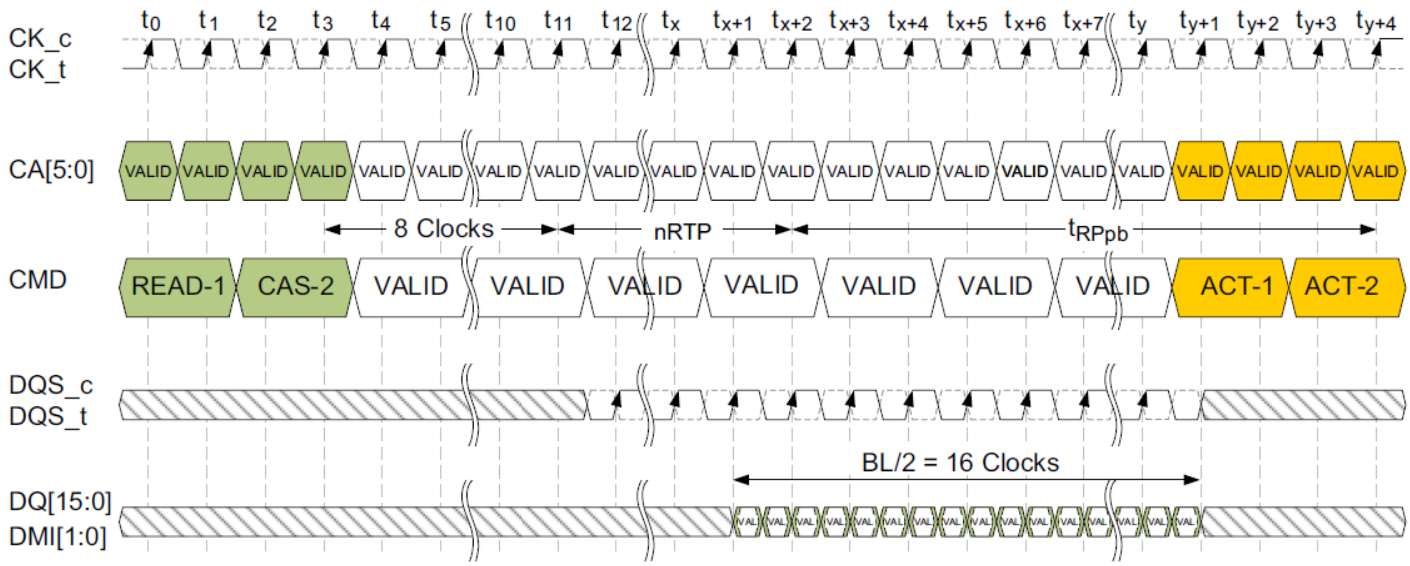


Figure 59: Burst READ with Auto-Precharge (Shown with BL32, 2tCK pre-amble)

3.20.4 Burst WRITE With Auto Precharge

If AP is HIGH when a WRITE command is issued, the WRITE with Auto-PRECHARGE function is engaged. The device starts an Auto-PRECHARGE on the rising edge t_{WR} cycles after the completion of the Burst WRITE.

Following a WRITE with Auto-PRECHARGE, an ACTIVATE command can be issued to the same bank if the following conditions are met:

- The RAS precharge time (t_{RP}) has been satisfied from the clock at which the Auto-PRECHARGE began, and
- The RAS cycle time (t_{RC}) from the previous bank activation has been satisfied.

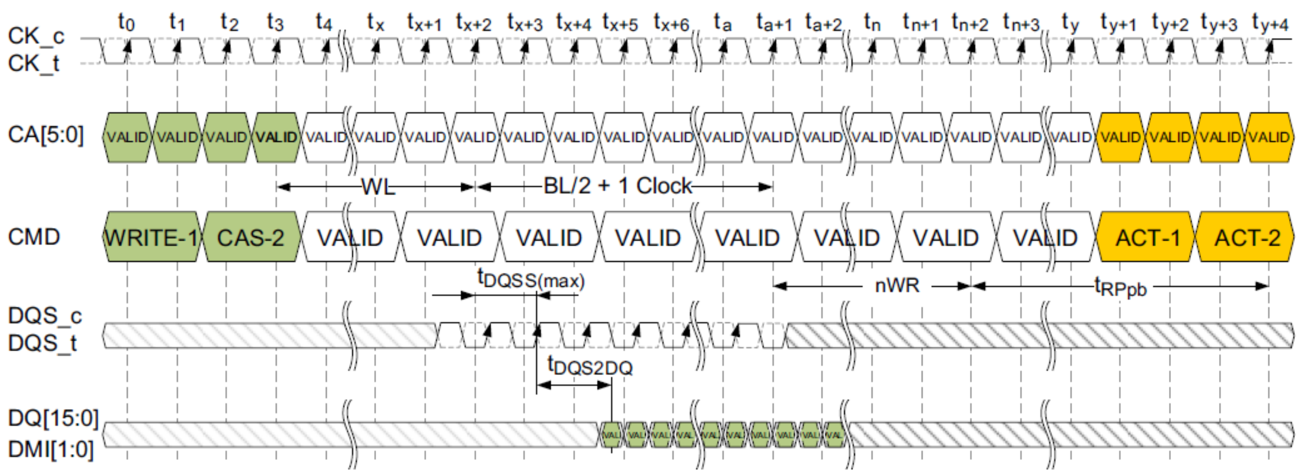


Figure 60: Burst WRITE with AutoPrecharge (Shown with BL16, 2tCK pre-amble)

3.21 Auto-Precharge Operation

Before a new row can be opened in an active bank, the active bank must be precharged using either the PRECHARGE command or the Auto-Precharge function. When a READ, a WRITE or Masked Write command is issued to the device, the AP bit (CA5) can be set to enable the active bank to automatically begin precharge at the earliest possible moment during the burst READ, WRITE or Masked Write cycle.

If AP is LOW when the READ or WRITE command is issued, then the normal READ, WRITE or Masked Write burst operation is executed and the bank remains active at the completion of the burst.

If AP is HIGH when the READ, WRITE or Masked Write command is issued, the Auto-Precharge function is engaged. This feature enables the PRECHARGE operation to be partially or completely hidden during burst READ cycles (dependent upon READ or WRITE latency), thus improving system performance for random data access.
Read with Auto-Precharge or Write/Mask Write with Auto-Precharge commands may be issued after tRCD has been satisfied. The LPDDR4 SDRAM RAS Lockout feature will schedule the internal precharge to assure that tRAS is satisfied.
tRC needs to be satisfied prior to issuing subsequent Activate commands to the same bank.
The figure below shows example of RAS lock function.

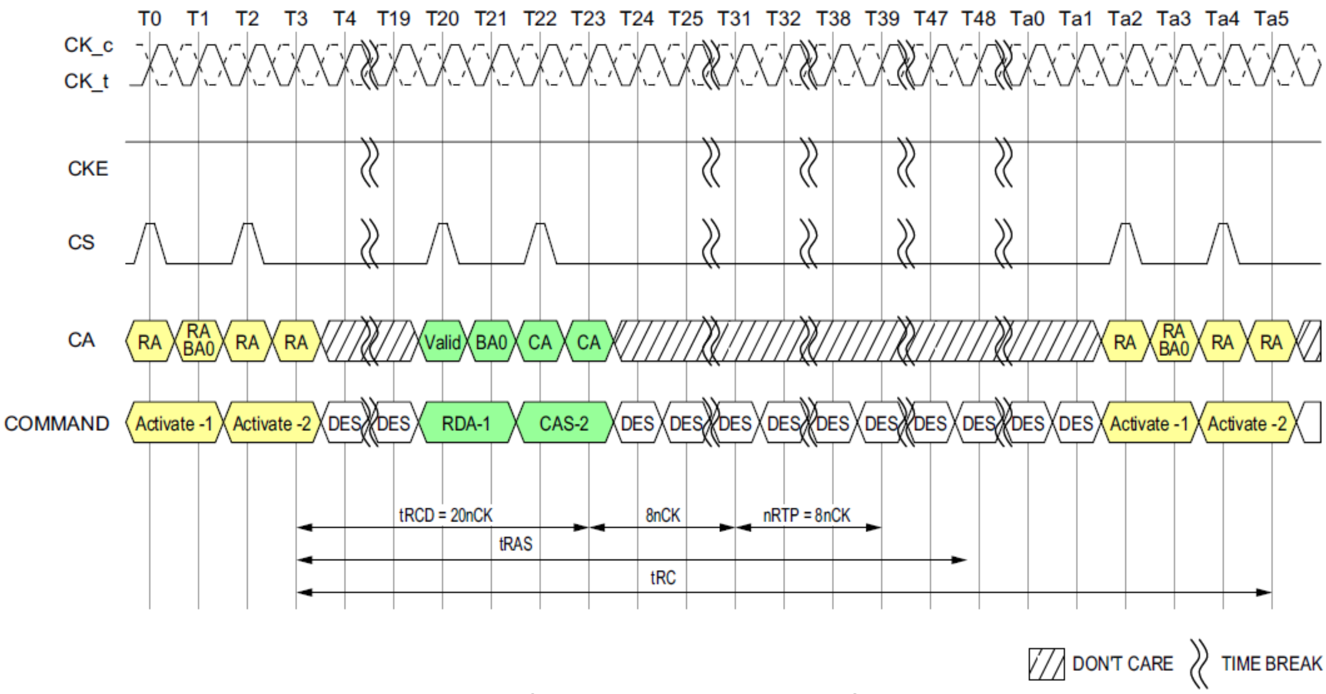


Figure 61: Command Input Timing with RAS lock

NOTE :
1) $t_{CK(AVG)}$ = 0.938ns, Data Rate = 2133Mbps, $t_{RCD(Min)}$ = Max(18ns, 4nCK), $t_{RAS(Min)}$ = Max(42ns, 3nCK), nRTP = 8nCK, BL = 32.
2) t_{RCD} = 20nCK comes from Roundup(18ns/0.938ns).
3) DES commands are shown for ease of illustration; other commands may be valid at these times.

3.21.1 Delay time from Write to Read with Auto-Precharge

In the case of write command followed by read with Auto-Precharge, controller must satisfy tWR for the write command before initiating the DRAM internal Auto-Precharge. It means that (t_{WTR} + nRTP) should be equal or longer than (t_{WR}) when BL setting is 16, as well as (t_{WTR} + nRTP +8nCK) should be equal or longer than (tWR) when BL setting is 32. Refer to the following figure for details.

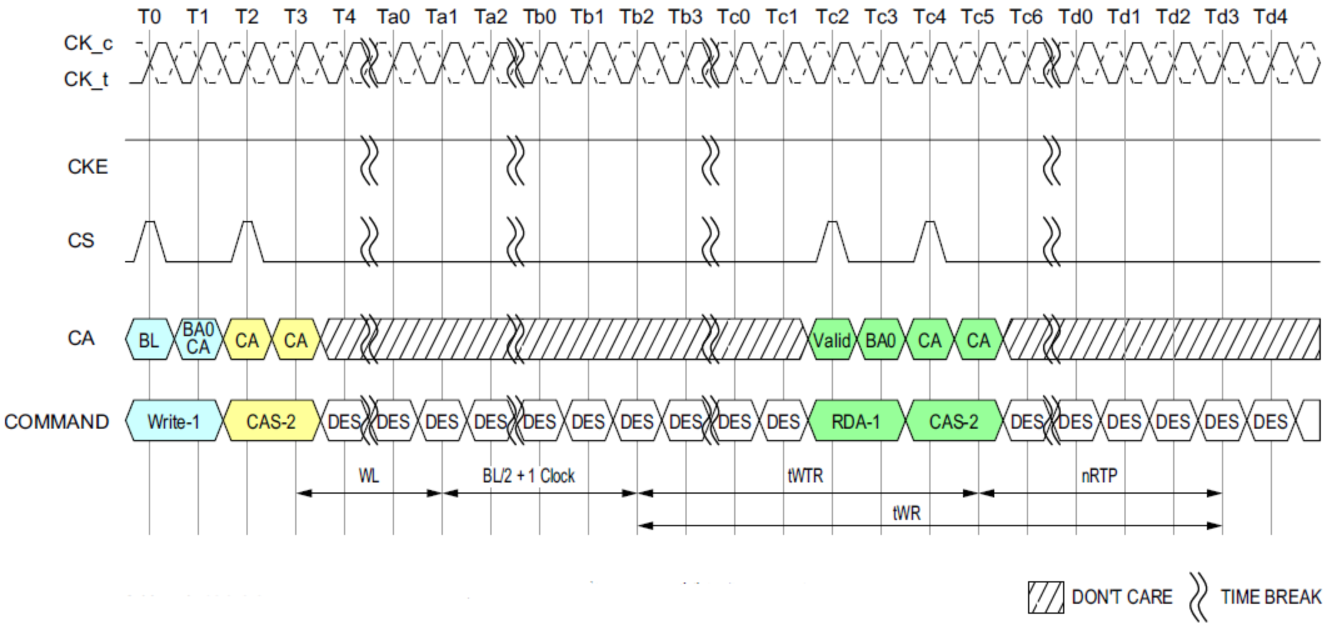


Figure 62: Delay time from Write to Read with Auto-Precharge

NOTE :
1) Burst Length at Read = 16.
2) DES commands are shown for ease of illustration; other commands may be valid at these times.

[Table 24] Timing Between Commands (PRECHARGE and Auto-PRECHARGE) : DQ ODT is Disable

"From" Command	"To" Command	Minimum Delay Between "From Command" and "To Command"	Unit	Notes
READ BL = 16	PRECHARGE (to same bank as READ)	tRTP	tCK	1,6
	PRECHARGE ALL	tRTP	tCK	1,6
READ BL = 32	PRECHARGE (to same bank as READ)	$8t_{CK} + t_{RTP}$	tCK	1,6
	PRECHARGE ALL	$8t_{CK} + t_{RTP}$	tCK	1,6
READ w/ AP BL=16	PRECHARGE (to same bank as READ w/ AP)	nRTP	tCK	1,10
	PRECHARGE ALL	nRTP	tCK	1,10
	ACTIVATE (to same bank as READ w/ AP)	nRTP + t_{RPpb}	tCK	1,8,10
	WRITE or WRITE w/ AP (same bank)	Illegal	-	
	MASK-WR or MASK-WR w/AP (same bank)	Illegal	-	
	WRITE or WRITE w/ AP (different bank)	RL+RU($t_{DQSCK(max)}/t_{CK}$)+BL/2+RD(t_{RPST})-WL+t_{WPRE}	tCK	3,4,5
	MASK-WR or MASK-WR w/AP (different bank)	RL+RU($t_{DQSCK(max)}/t_{CK}$)+BL/2+RD(t_{RPST})-WL+t_{WPRE}	tCK	3,4,5
	READ or READ w/ AP (same bank)	Illegal	-	
	READ or READ w/ AP (different bank)	BL/2	tCK	3
READ w/AP BL = 32	PRECHARGE (to same bank as READ w/AP)	8tCK + nRTP	tCK	1,10
	PRECHARGE All	8tCK + nRTP	tCK	1,10
	Activate (to same bank as READ w/AP)	8tCK + nRTP +t_{RPpb}	tCK	1,8,10
	WRITE or WRITE w/AP (same bank)	Illegal	-	
	MASK-WR or MASK-WR w/AP (same bank)	Illegal	-	
	WRITE or WRITE w/AP (different bank)	RL+RU($t_{DQSCK(max)}/t_{CK}$)+BL/2+RD(t_{RPST})-WL+t_{WPRE}	tCK	3,4,5
	MASK-WR or MASK-WR w/AP (different bank)	RL+RU($t_{DQSCK(max)}/t_{CK}$)+BL/2+RD(t_{RPST})-WL+t_{WPRE}	tCK	3,4,5
	READ or READ w/AP (same bank)	Illegal	-	
	READ or READ w/AP (different bank)	BL/2	tCK	3
WRITE BL = 16 & 32	PRECHARGE (to same bank as WRITE)	WL + BL/2 + t_{WR} + 1	tCK	1,7
	PRECHARGE All	WL + BL/2 + t_{WR} + 1	tCK	1,7
MASK-WR BL = 16	PRECHARGE (to same bank as MASK-WR)	WL + BL/2 + t_{WR} + 1	tCK	1,7
	PRECHARGE All	WL + BL/2 + t_{WR} + 1	tCK	1,7
WRITE w/ AP BL = 16 & 32	Precharge (to same Bank as Write w/ AP)	WL + BL/2 + nWR + 1	tCK	1,11
	PRECHARGE ALL	WL + BL/2 + nWR + 1	tCK	1,11
	ACTIVATE (to same bank as WRITE w/ AP)	WL + BL/2 +nWR + 1 + t_{RPpb}	tCK	1,8,11
	WRITE or WRITE w/ AP (same bank)	Illegal	-	
	READ or READ w/ AP (same bank)	Illegal	-	
	WRITE or WRITE w/ AP (different bank)	BL/2	tCK	3
	MASK-WR or MASK-WR w/AP (different bank)	BL/2	tCK	3
	READ or READ w/ AP (different bank)	WL + BL/2 + t_{WTR} + 1	tCK	3,9
MASK-WR w/AP BL = 16	PRECHARGE (to same bank as MASK-WR w/AP)	WL + BL/2 + nWR + 1	tCK	1,11
	PRECHARGE All	WL + BL/2 + nWR + 1	tCK	1,11
	ACTIVATE (to same bank as MASK-WR w/AP)	WL + BL/2 + nWR + 1 + t_{RPpb}	tCK	1,8,11
	WRITE or WRITE w/AP (same bank)	Illegal	-	3
	MASK-WR or MASK-WR w/AP (same bank)	Illegal	-	3
	WRITE or WRITE w/AP (different bank)	BL/2	tCK	3
	MASK-WR or MASK-WR w/AP (different bank)	BL/2	tCK	3
	READ or READ w/AP (same bank)	Illegal	-	3
	READ or READ w/AP (different bank)	WL + BL/2 + t_{WTR} + 1	tCK	3,9
PRECHARGE	PRECHARGE (to same bank as PRECHARGE)	4	tCK	1
	PRECHARGE ALL	4	tCK	1
PRECHARGE All	PRECHARGE	4	tCK	1
	PRECHARGE ALL	4	tCK	1

NOTE :
1) For a given bank, the precharge period should be counted from the latest precharge command, whether per-bank or all-bank, issued to that bank. The precharge period is satisfied t_{RP} after that latest precharge command.
2) Any command issued during the minimum delay time as specified in the table above is illegal.
3) After READ w/AP, seamless read operations to different banks are supported. After WRITE w/AP or MASK-WR w/AP, seamless write operations to different banks are supported. READ, WRITE, and MASK-WR operations may not be truncated or interrupted.
4) t_{RPST} values depend on MR1-OP[7] respectively.
5) t_{WPRE} values depend on MR1-OP[2] respectively.
6) Minimum Delay between "From Command" and "To Command" in clock cycle is calculated by dividing t_{RTP}(in ns) by t_{CK}(in ns) and rounding up to the next integer: Minimum Delay [cycles] = Roundup (t_{RTP}[ns] / t_{CK}[ns])
7) Minimum Delay between "From Command" and "To Command" in clock cycle is calculated by dividing t_{WR}(in ns) by t_{CK}(in ns) and rounding up to the next integer: Minimum Delay [cycles] = Roundup (t_{WR}[ns] / t_{CK}[ns])
8) Minimum Delay between "From Command" and "To Command" in clock cycle is calculated by dividing t_{RPpb}(in ns) by t_{CK}(in ns) and rounding up to the next integer: Minimum Delay [cycles] = Roundup (t_{RPpb}[ns] / t_{CK}[ns])
9) Minimum Delay between "From Command" and "To Command" in clock cycle is calculated by dividing t_{WTR}(in ns) by t_{CK}(in ns) and rounding up to the next integer: Minimum Delay [cycles] = Roundup (t_{WTR}[ns] / t_{CK}[ns])
10) For Read w/AP the value is nRTP which is defined in Mode Register 2.
11) For Write w/AP the value is nWR which is defined in Mode Register 1.

[Table 25] Timing Between Commands (read w/ AP and write command) : DQ ODT is Enable

"From" Command	"To" Command	Minimum Delay Between "From Command" and "To Command"	Unit	Notes
READ w/AP BL = 16	WRITE or WRITE w/AP (different bank)	RL+RU($t_{DQSCK(max)}$/t_{CK})+BL/2+RD(t_{RPST})-ODTLon-RD($t_{ODTon,min}$/t_{CK})+1	tCK	2, 3
	MASK-WR or MASK-WR w/AP (different bank)	RL+RU($t_{DQSCK(max)}$/t_{CK})+BL/2+RD(t_{RPST})-ODTLon-RD($t_{ODTon,min}$/t_{CK})+1	tCK	2, 3
READ w/AP BL = 32	WRITE or WRITE w/AP (different bank)	RL+RU($t_{DQSCK(max)}$/t_{CK})+BL/2+RD(t_{RPST})-ODTLon-RD($t_{ODTon,min}$/t_{CK})+1	tCK	2, 3
	MASK-WR or MASK-WR w/AP (different bank)	RL+RU($t_{DQSCK(max)}$/t_{CK})+BL/2+RD(t_{RPST})-ODTLon-RD($t_{ODTon,min}$/t_{CK})+1	tCK	2, 3

NOTE :
1) The rest of the timing about precharge and Auto-Precharge is same as DQ ODT is Disable case.
2) After READ w/AP, seamless read operations to different banks are supported. READ, WRITE, and MASK-WR operations may not be truncated or interrupted.
3) t_{RPST} values depend on MR1-OP[7] respectively.

3.22 Refresh Command

The REFRESH command is initiated with CS HIGH, CA0 LOW, CA1 LOW, CA2 LOW, CA3 HIGH and CA4 LOW at the first rising edge of the clock. Per-bank REFRESH is initiated with CA5 LOW at the first rising edge of the clock. All-bank REFRESH is initiated with CA5 HIGH at the first rising edge of the clock.

A per-bank REFRESH command (REFpb) is performed to the bank address as transferred on CA0, CA1 and CA2 at the second rising edge of the clock. Bank address BA0 is transferred on CA0, bank address BA1 is transferred on CA1 and bank address BA2 is transferred on CA2. A per-bank REFRESH command (REFpb) to the eight banks can be issued in any order. e.g. REFpb commands are issued in the following order: 1-3-0-2-4-7-5-6. After the eight banks have been refreshed using the per-bank REFRESH command the controller can send another set of per-bank REFRESH commands in the same order or a different order. e.g. REFpb commands are issued in the following order that is different from the previous order: 7-1-3-5-0-4-2-6. One of the possible order can also be a sequential round robin: 0-1-2-3-4-5-6-7. It is illegal to send a per-bank REFRESH command to the same bank unless all eight banks have been refreshed using the per-bank REFRESH command. The count of eight REFpb commands starts with the first REFpb command after a synchronization event.

The bank count is synchronized between the controller and the SDRAM by resetting the bank count to zero. Synchronization can occur upon asserting RESET_n or at every exit from self refresh. REFab command also synchronizes the counter between the controller and SDRAM to zero. The SDRAM device can be placed in self-refresh or a REFab command can be issued at any time without cycling through all eight banks using per-bank REFRESH command. After the bank count is synchronized to zero the controller can issue per-bank REFRESH commands in any order as described in the previous paragraph.

A REFab command issued when the bank counter is not zero will reset the bank counter to zero and the DRAM will perform refreshes to all banks as indicated by the row counter. If another refresh command (REFab or REFpb) is issued after the REFab command then it uses an incremented value of the row counter.
The table below shows examples of both bank and refresh counter increment behavior.

[Table 26] Bank and Refresh counter increment behavior

#	Command	BA0	BA1	BA2	Refresh Bank#	Bank Counter #	Ref Counter # (Row Address #)
0	Reset, SRX or REFab					To 0	-
1	REFpb	0	0	0	0	0 to 1	n
2	REFpb	0	0	1	1	1 to 2	
3	REFpb	0	1	0	2	2 to 3	
4	REFpb	0	1	1	3	3 to 4	
5	REFpb	1	0	0	4	4 to 5	
6	REFpb	1	0	1	5	5 to 6	
7	REFpb	1	1	0	6	6 to 7	
8	REFpb	1	1	1	7	7 to 0	
9	REFpb	1	1	0	6	0 to 1	n + 1
10	REFpb	1	1	1	7	1 to 2	
11	REFpb	0	0	1	1	2 to 3	
12	REFpb	0	1	1	3	3 to 4	
13	REFpb	1	0	1	5	4 to 5	
14	REFpb	0	1	0	2	5 to 6	
15	REFpb	0	0	0	0	6 to 7	
16	REFpb	1	0	0	4	7 to 0	
17	REFpb	0	0	0	0	0 to 1	n+2
18	REFpb	0	0	1	1	1 to 2	
19	REFpb	0	1	0	2	2 to 3	
20	***REFpb***	***V***	***V***	***V***	***0~7***	***To 0***	***n + 2***
21	REFpb	1	1	0	6	0 to 1	n + 3
22	REFpb	1	1	1	7	1 to 2	
Snip							

A bank must be idle before it can be refreshed. The controller must track the bank being refreshed by the per-bank REFRESH command.

The REFpb command must not be issued to the device until the following conditions are met:
- t_{RFCab} has been satisfied after the prior REFab command
- t_{RFCpb} has been satisfied after the prior REFpb command
- t_{RP} has been satisfied after the prior PRECHARGE command to that bank
- t_{RRD} has been satisfied after the prior ACTIVATE command (if applicable, for example after activating a row in a different bank than the one affected by the REFpb command).

The target bank is inaccessible during per-bank REFRESH cycle time (t_{RFCpb}), however, other banks within the device are accessible and can be addressed during the cycle. During the REFpb operation, any of the banks other than the one being refreshed can be maintained in an active state or accessed by a READ or a WRITE command. When the per-bank REFRESH cycle has completed, the affected bank will be in the idle state.
After issuing REFpb, these conditions must be met:
- t_{RFCpb} must be satisfied before issuing a REFab command
- t_{RFCpb} must be satisfied before issuing an ACTIVATE command to the same bank
- t_{RRD} must be satisfied before issuing an ACTIVATE command to a different bank
- t_{RFCpb} must be satisfied before issuing another REFpb command.

An all-bank REFRESH command (REFab) issues a REFRESH command to all banks. All banks must be idle when REFab is issued (for instance, by issuing a PRECHARGE-all command prior to issuing an all-bank REFRESH command). REFab also synchronizes the bank count between the controller and the SDRAM to zero. The REFab command must not be issued to the device until the following conditions have been met:
- t_{RFCab} has been satisfied following the prior REFab command
- t_{RFCpb} has been satisfied following the prior REFpb command
- t_{RP} has been satisfied following the prior PRECHARGE commands.

When an all-bank refresh cycle has completed, all banks will be idle. After issuing REFab:
- t_{RFCab} latency must be satisfied before issuing an ACTIVATE command
- t_{RFCab} latency must be satisfied before issuing a REFab or REFpb command.

[Table 27] REFRESH Command Scheduling Separation Requirements

Symbol	Minimum Delay From	To	Notes
t_{RFCab}	REFab	REFab	
		ACTIVATE command to any bank	
		REFpb	
t_{RFCpb}	REFpb	REFab	
		ACTIVATE command to same bank as REFpb	
		REFpb	
t_{RRD}	REFpb	ACTIVATE command to different bank than REFpb	
	ACTIVATE	REFpb	1
		ACTIVATE command to different bank than prior ACTIVATE command	

NOTE :
1) A bank must be in the Idle state before it is refreshed, so following an ACTIVATE command REFab is prohibited; REFpb is supported only if it affects a bank that is in the idle state.

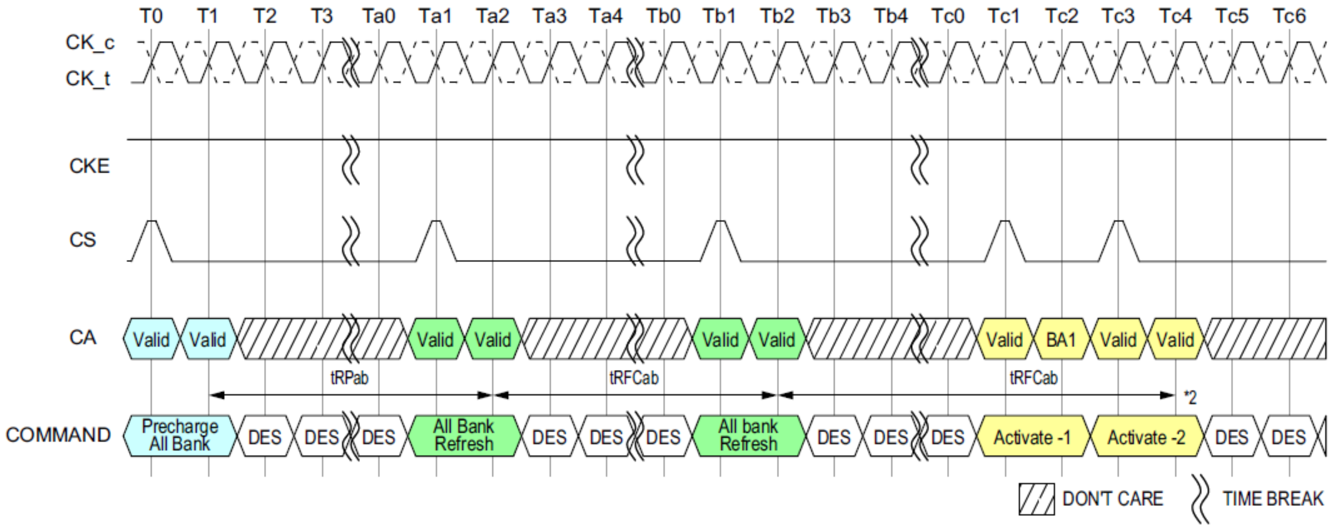


Figure 63: All-Bank Refresh Operation

NOTE :
1) DES commands are shown for ease of illustration; other commands may be valid at these times.
2) Activate Command is shown as an example. Other commands may be valid provided the timing specification is satisfied.

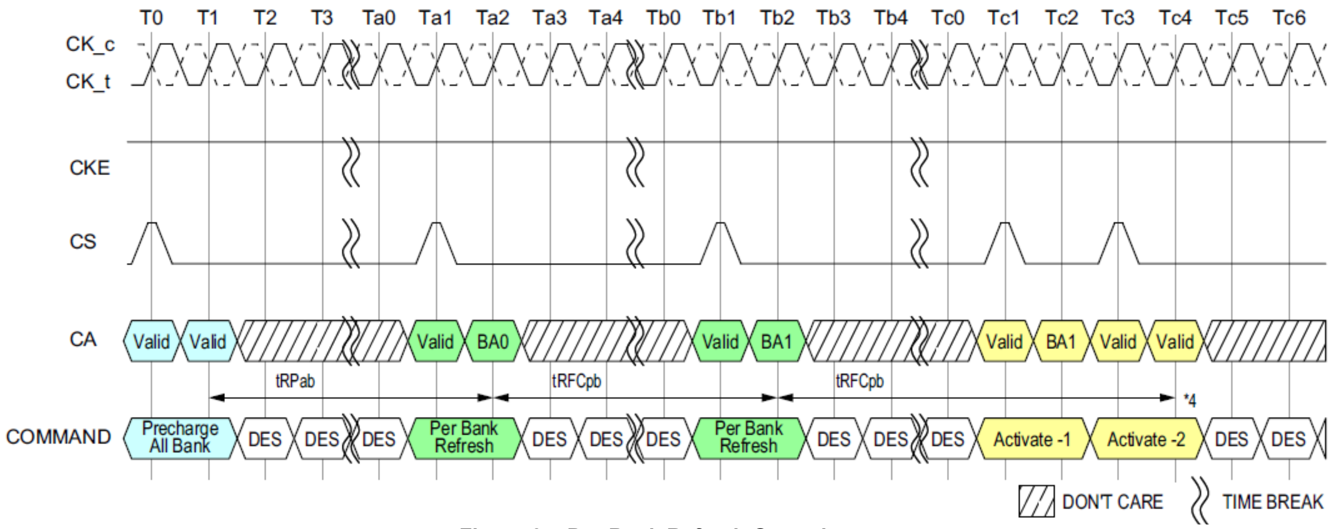


Figure 64: Per-Bank Refresh Operation

NOTE :
1) DES commands are shown for ease of illustration; other commands may be valid at these times.
2) In the beginning of this example, the REFpb bank is pointing to bank 0.
3) Operations to banks other than the bank being refreshed are supported during the t_{RFCpb} period.
4) Activate Command is shown as an example. Other commands may be valid provided the timing specification is satisfied.

In general, a Refresh command needs to be issued to the LPDDR4 SDRAM regularly every t_{REFI} × refresh rate multiplier interval. To allow for improved efficiency in scheduling and switching between tasks, some flexibility in the absolute refresh interval is provided. A maximum of 8 Refresh commands can be postponed during operation of the LPDDR4 SDRAM, meaning that at no point in time more than a total of 8 Refresh commands are allowed to be postponed and maximum number of pulled-in or postponed REF command is dependent on refresh rate. It is described in the table below. In case that 8 Refresh commands are postponed in a row, the resulting maximum interval between the surrounding Refresh commands is limited to 9 × t_{REFI} × refresh rate multiplier. A maximum of 8 additional Refresh commands can be issued in advance ("pulled in"), with each one reducing the number of regular Refresh commands required later by one. Note that pulling in more than 8 Refresh commands in advance does not further reduce the number of regular Refresh commands required later, so that the resulting maximum interval between two surrounding Refresh commands is limited to 9 × t_{REFI} × refresh rate multiplier. At any given time, a maximum of 16 REF commands can be issued within 2 × t_{REFI} × refresh rate multiplier. Self-Refresh Mode may be entered with a maximum of eight Refresh commands being postponed. After exiting Self-Refresh Mode with one or more Refresh commands postponed, additional Refresh commands may be postponed to the extent that the total number of postponed Refresh commands (before and after the Self-Refresh) will never exceed eight. During Self-Refresh Mode, the number of postponed or pulled-in REF commands does not change.

And for per bank refresh, a maximum 8 x 8 per bank refresh commands can be postponed or pulled in for scheduling efficiency. At any given time, a maximum of 2 x 8 x 8 per bank refresh commands can be issued within 2 x t_{REFI} × refresh rate multiplier.

[Table 28] Legacy REFRESH Command Timing Constraints

MR4 OP[2:0]	Refresh rate	Max. No. of pulled-in or postponed REFab	Max. Interval between two REFab	Max. No. of REFab within max(2×t_{REFI} × refresh rate multiplier, 16×t_{RFC})	Per-bank Refresh
000_B	Low Temp. Limit	N/A	N/A	N/A	N/A
001_B	4× t_{REFI}	8	9 × 4 × t_{REFI}	16	1/8 of REFab
010_B	2× t_{REFI}	8	9 × 2 × t_{REFI}	16	1/8 of REFab
011_B	1× t_{REFI}	8	9 × t_{REFI}	16	1/8 of REFab
100_B	0.5× t_{REFI}	8	9 × 0.5 × t_{REFI}	16	1/8 of REFab
101_B	0.25× t_{REFI}	8	9 × 0.25 × t_{REFI}	16	1/8 of REFab
110_B	0.25× t_{REFI}	8	9 × 0.25 × t_{REFI}	16	1/8 of REFab
111_B	High Temp. Limit	N/A	N/A	N/A	N/A

[Table 29] Modified REFRESH Command Timing Constraints

MR4 OP[2:0]	Refresh rate	Max. No. of pulled-in or postponed REFab	Max. Interval between two REFab	Max. No. of REFab within max(2×t_{REFI} × refresh rate multiplier, 16×t_{RFC})	Per-bank Refresh
000_B	Low Temp. Limit	N/A	N/A	N/A	N/A
001_B	4× t_{REFI}	2	3 × 4 × t_{REFI}	4	1/8 of REFab
010_B	2× t_{REFI}	4	5 × 2 × t_{REFI}	8	1/8 of REFab
011_B	1× t_{REFI}	8	9 × t_{REFI}	16	1/8 of REFab
100_B	0.5× t_{REFI}	8	9 × 0.5 × t_{REFI}	16	1/8 of REFab
101_B	0.25× t_{REFI}	8	9 × 0.25 × t_{REFI}	16	1/8 of REFab
110_B	0.25× t_{REFI}	8	9 × 0.25 × t_{REFI}	16	1/8 of REFab
111_B	High Temp. Limit	N/A	N/A	N/A	N/A

NOTE :

1) For any thermal transition phase where Refresh mode is transitioned to either 2× t_{REFI} or 4x t_{REFI}, DRAM will support the previous postponed refresh requirement provided the number of postponed refreshes is monotonically reduced to meet the new requirement. However, the pulled-in refresh commands in previous thermal phase are not applied in new thermal phase. Entering new thermal phase the controller must count the number of pulled-in refresh commands as zero, regardless of remaining pulled-in refresh commands in previous thermal phase.

2) LPDDR4 devices are refreshed properly if memory controller issues refresh commands with same or shorter refresh period than reported by MR4 OP[2:0]. If shorter refresh period is applied, the corresponding requirements from Table apply. For example, when MR4 OP[2:0]=001_B, controller can be in any refresh rate from 4 × t_{REFI} to 0.25x t_{REFI}. When MR4 OP[2:0]=010_B, the only prohibited refresh rate is 4x t_{REFI}.

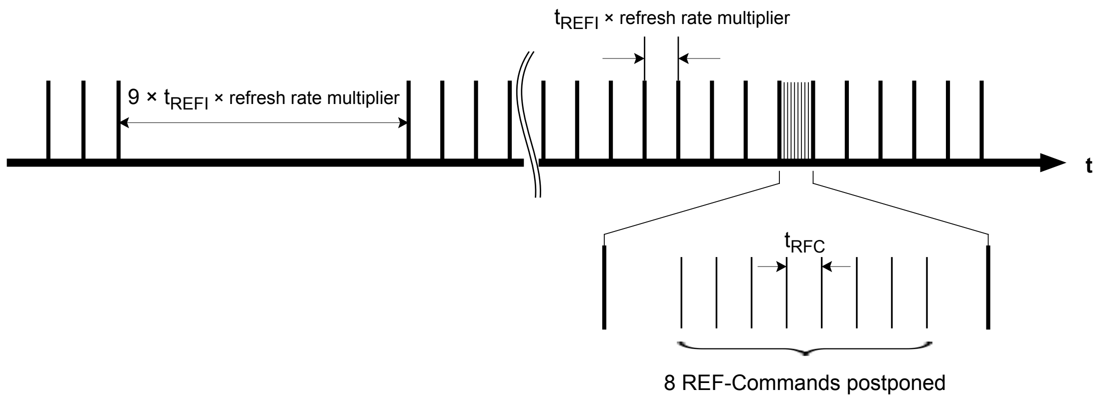


Figure 65: Postponing Refresh Commands (Example)

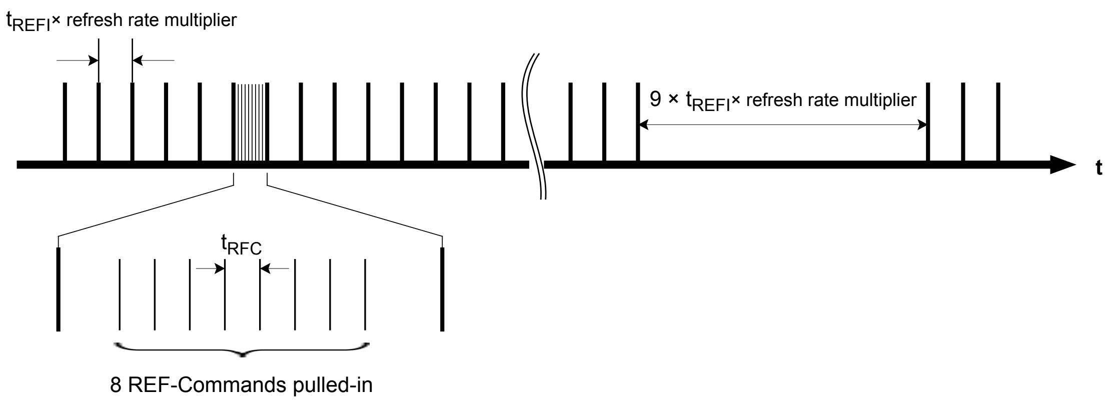


Figure 66: Pulling-in Refresh Commands (Example)

3.22.1 Burst Read operation followed by Per Bank Refresh

The Per Bank Refresh command can be issued after t_{RTP} + t_{RPpb} from Read command.

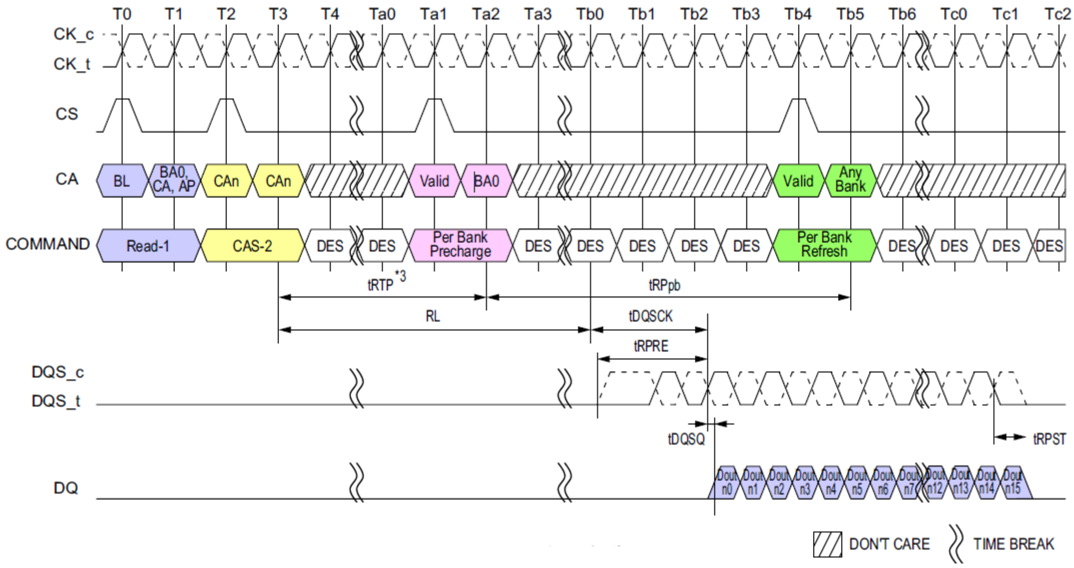


Figure 67: Burst Read operation followed by Per Bank Refresh

NOTE :
1) BL = 16, Preamble = Toggle, Postamble = 0.5nCK, DQ/DQS: VSSQ termination.
2) Dout n = data-out from column n.
3) In case of BL = 32, Delay time from Read to Per Bank Precharge is 8nCK + t_{RTP}.
4) DES commands are shown for ease of illustration; other commands may be valid at these times.

The Per Bank Refresh command can be issued after tRC from Read with Auto Precharge command.

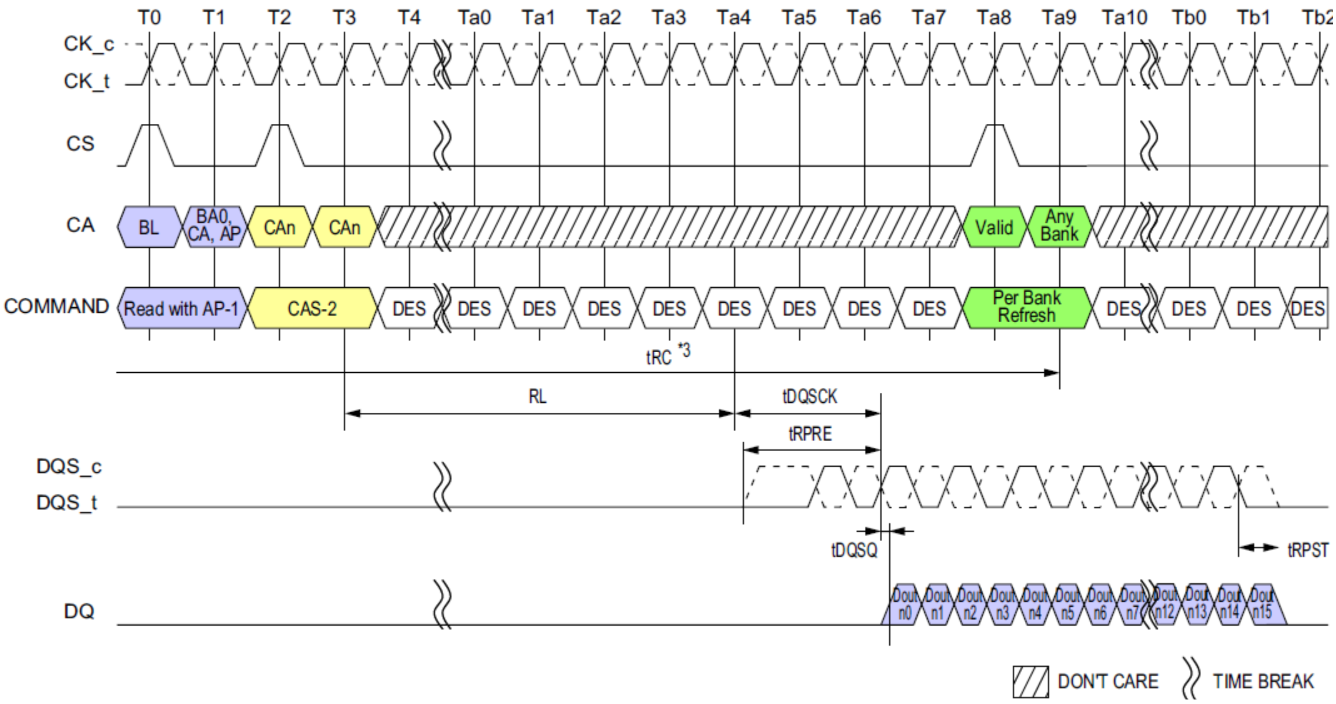


Figure 68: Burst Read with Auto-Precharge operation followed by Per Bank Refresh

NOTE :
1) BL = 16, Preamble = Toggle, Postamble = 0.5nCK, DQ/DQS: VSSQ termination.
2) Dout n = data-out from column n.
3) tRC needs to be satisfied prior to issuing subsequent Per Bank Refresh command.
4) DES commands are shown for ease of illustration; other commands may be valid at these times.

3.22.2 Refresh Requirements

[Table 30] Refresh Requirement Parameters (per density)

Parameter Requirements		Symbol	4 Gb	6Gb	8 Gb	12Gb	16Gb	24Gb	32Gb	Unit
Density per channel			2Gb	3Gb	4Gb	6Gb	8Gb	12Gb	16Gb	-
Number of Banks per channel			8							-
Refresh Window (t_{REFW}) Tcase ≤ 85°C		t_{REFW}	32							ms
Refresh Window (t_{REFW}) 1/2 Rate Refresh		t_{REFW}	16			16		TBD	TBD	ms
Refresh Window (t_{REFW}) 1/4 Rate Refresh		t_{REFW}	8			8		TBD	TBD	ms
Required number of REFRESH commands in a t_{REFW} window		R	8,192							-
Average refresh internal	REFab	t_{REFI}	3.904							us
	REFpb	t_{REFIpb}	488							ns
Refresh Cycle time (all banks)		t_{RFCab}	130	180		280		380		ns
Refresh Cycle time (Per Bank)		t_{RFCpb}	60	90		140		190		ns
Per-bank Refresh to Per-bank Refresh different bank Time		$t_{pbR2pbR}$	60	90						ns

NOTE:

1) Refresh for each channel is independent of the other channel on the die, or other channels in a package. Power delivery in the user's system should be verified to make sure the DC operating conditions are maintained when multiple channels are refreshed simultaneously.

2) Self Refresh abort feature is available for higher density devices starting with12Gb dual channel device and 6Gb single channel device and tXSR_abort(min) is defined as tRFCpb + 17.5ns.

3.23 Self Refresh Operation

3.23.1 Self Refresh Entry And Exit

The Self Refresh command can be used to retain data in the LPDDR4 SDRAM, the SDRAM retains data without external Refresh command. The device has a built-in timer to accommodate Self Refresh operation. The Self Refresh is entered by Self Refresh Entry Command defined by having CS High, CA0 Low, CA1 Low, CA2 Low; CA3 High; CA4 High, CA5 Valid (Valid that means it is Logic Level, High or Low) for the first rising edge and CS Low, CA0 Valid, CA1 Valid, CA2 Valid, CA3 Valid, CA4 Valid, CA5 Valid at the second rising edge of the clock. Self Refresh command is only allowed when read data burst is completed and SDRAM is idle state.

During Self Refresh mode, external clock input is needed and all input pin of SDRAM are activated. SDRAM can accept the following commands, MRR-1, CAS-2, DES, SRX, MPC, MRW-1, and MRW-2 except PASR Bank/Segment and SR Abort setting.

LPDDR4 SDRAM can operate in Self Refresh in both the standard or elevated temperature ranges. SDRAM will also manage Self Refresh power consumption when the operating temperature changes, lower at low temperature and higher at high temperatures.

For proper Self Refresh operation, power supply pins (V_{DD1}, V_{DD2} and V_{DDQ}) must be at valid levels. However V_{DDQ} may be turned off during Self-Refresh with Power Down after t_{CKELCK}(Max(5ns,5nCK))is satisfied (Refer to Figure 69 about t_{CKELCK}).

Prior to exiting Self-Refresh with Power Down, V_{DDQ} must be within specified limits. The minimum time that the SDRAM must remain in Self Refresh mode is tSR,min. Once Self Refresh Exit is registered, only MRR-1, CAS-2, DES, MPC, MRW-1 and MRW-2 except PASR Bank/Segment and SR Abort setting are allowed until tXSR is satisfied.

The use of Self Refresh mode introduces the possibility that an internally timed refresh event can be missed when Self Refresh Exit is registered. Upon exit from Self Refresh, it is required that at least one REFRESH command (8 per-bank or 1 all-bank) is issued before entry into a subsequent Self Refresh.

This REFRESH command is not included in the count of regular refresh commands required by the tREFI interval, and does not modify the postponed or pulled-in refresh counts; the REFRESH command does count toward the maximum refreshes permitted within 2 X tREFI× refresh rate multiplier.

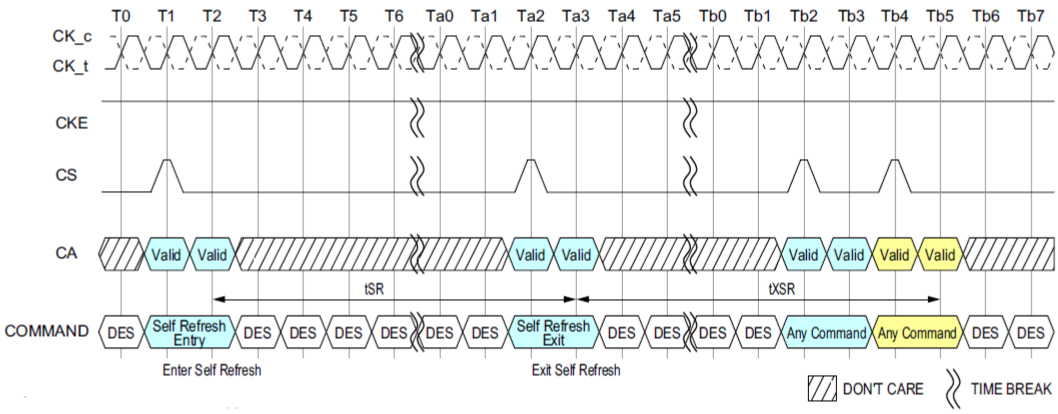


Figure 69: Self Refresh Entry/Exit Timing

NOTE :
1) MRR-1, CAS-2, DES, SRX, MPC, MRW-1 and MRW-2 except PASR Bank/Segment and SR Abort setting is allowed during Self Refresh.
2) DES commands are shown for ease of illustration; other commands may be valid at these times.

3.23.2 Power Down Entry And Exit During Self Refresh

Entering/Exiting Power Down Mode is allowed during Self Refresh mode in SDRAM. The related timing parameters between Self Refresh Entry/Exit and Power Down Entry/Exit are shown in Figure 70.

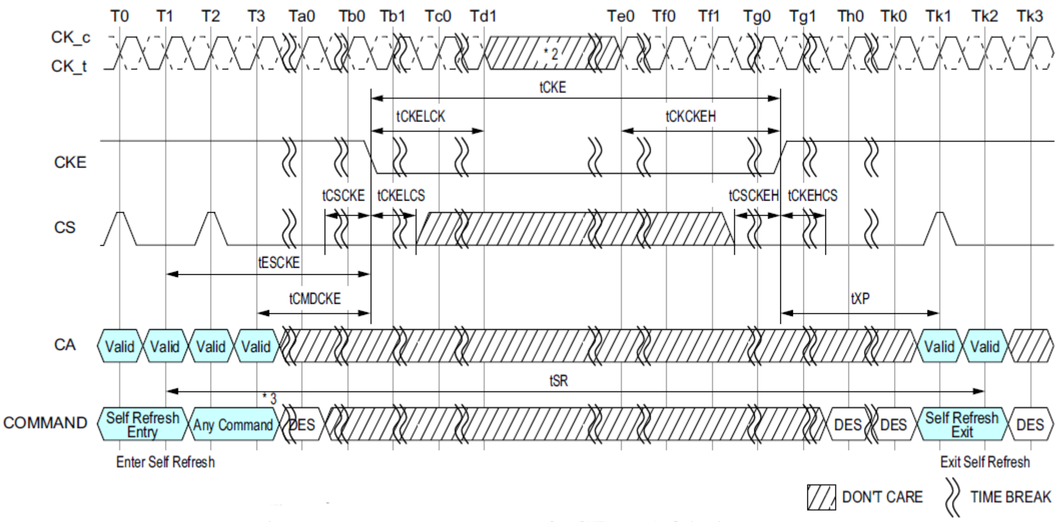


Figure 70: Self Refresh Entry/Exit Timing with Power Down Entry/Exit

NOTE :
1) MRR-1, CAS-2, DES, SRX, MPC, MRW-1 and MRW-2 except PASR Bank/Segment and SR Abort setting is allowed during Self Refresh.
2) Input clock frequency can be changed or the input clock can be stopped or floated after t_{CKELCK} satisfied and during power-down, provided that upon exiting power-down, the clock is stable and within specified limits for a minimum of t_{CKCKEH} of stable clock prior to power-down exit and the clock frequency is between the minimum and maximum specified frequency for the speed grade in use.
3) 2 Clock command for example.

3.23.3 Command Input Timing After Power Down Exit

Command input timings after Power Down Exit during Self Refresh mode are shown in Figure 71.

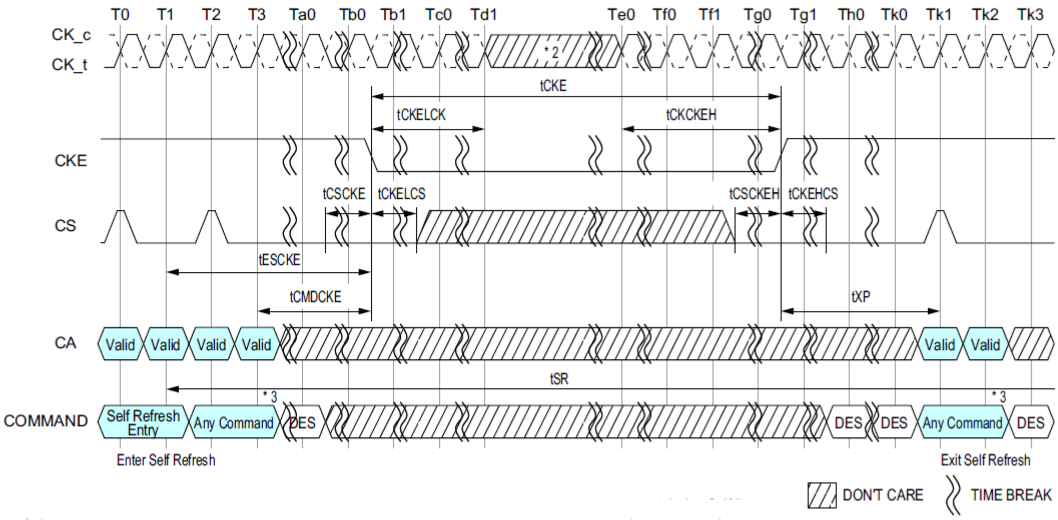


Figure 71: Command input timings after Power Down Exit during Self Refresh

NOTE:

1) MRR-1, CAS-2, DES, SRX, MPC, MRW-1 and MRW-2 except PASR Bank/Segment and SR Abort setting is allowed during Self Refresh.

2) Input clock frequency can be changed or the input clock can be stopped or floated after t_{CKELCK} satisfied and during power-down, provided that upon exiting power-down, the clock is stable and within specified limits for a minimum of t_{CKCKEH} of stable clock prior to power-down exit and the clock frequency is between the minimum and maximum specified frequency for the speed grade in use.

3) 2 Clock command for example.

3.23.4 Self Refresh AC Timing Table

[Table 31] Self Refresh AC Timing Table

Parameter	Symbol	Min/Max	Data Rate	Unit	Note
Self Refresh Timing					
Delay from SRE command to CKE Input low	t_{ESCKE}	Min	Max(1.75ns, 3t_{CK})	ns	1
Minimum Self Refresh Time	t_{SR}	Min	Max(15ns, 3t_{CK})	ns	1
Exit Self Refresh to Valid commands	t_{XSR}	Min	Max (t_{RFCab} + 7.5ns, 2t_{CK})	ns	1,2

NOTE :
1) Delay time has to satisfy both analog time(ns) and clock count(tCK).
It means that t_{ESCKE} will not expire until CK has toggled through at least 3 full cycles (3 *tCK) and 1.75ns has transpired.
The case which 3tCK is applied to is shown below.

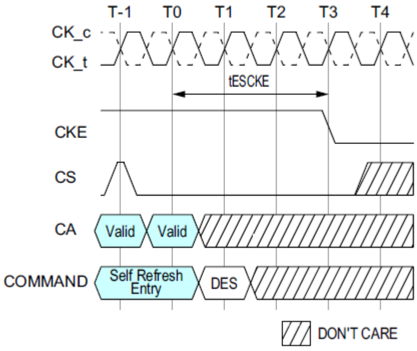


Figure 72: t_{ESCKE} Timing

2) MRR-1, CAS-2, DES, MPC, MRW-1 and MRW-2 except PASR Bank/Segment and SR Abort setting are only allowed during this period.

3.24 Self Refresh Abort

If MR4 OP[3] is enabled then DRAM aborts any ongoing refresh during Self Refresh exit and does not increment the internal refresh counter. Controller can issue a valid command after a delay of tXSR_abort instead of tXSR.

The value of tXSR_abort(min) is defined as tRFCpb + 17.5ns.

Upon exit from Self Refresh mode, the LPDDR4 SDRAM requires a minimum of one extra refresh (8 per bank or 1 all bank) before entry into a subsequent Self Refresh mode. This requirement remains the same irrespective of the setting of the MR bit for Self Refresh abort.

Self Refresh abort feature is available for higher density devices starting with12Gb dual channel device and 6Gb single channel device.

3.25 MRR, MRW, MPC Command During tXSR, tRFC

Mode Register Read (MRR), Mode Register Write (MRW) and Multi Purpose Command (MPC) can be issued during t_{XSR} period.

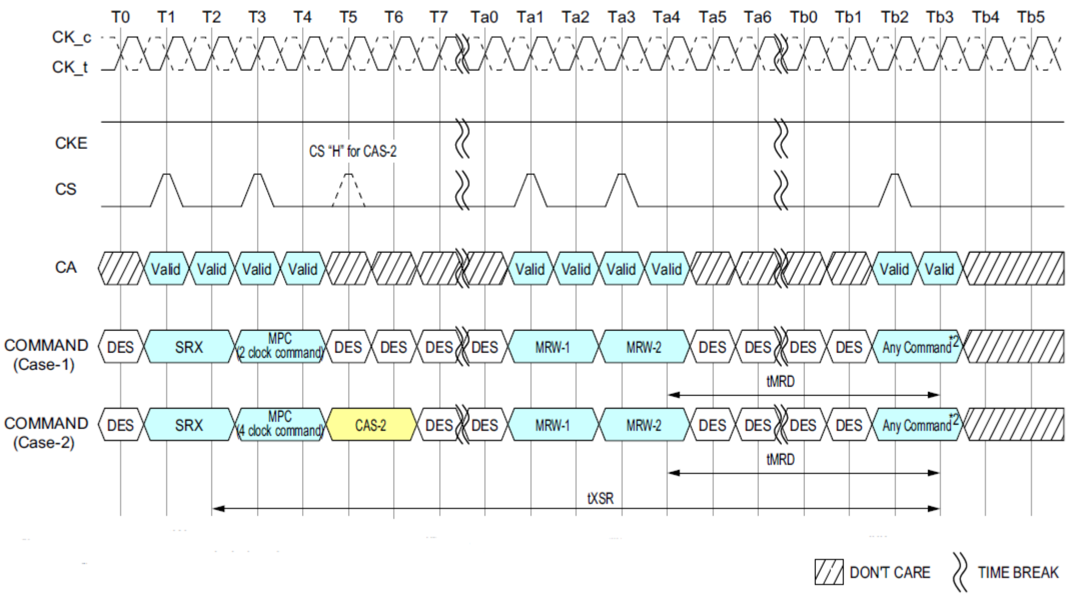


Figure 73: MRR, MRW and MPC Commands Issuing Timing during t_{XSR}

NOTE :
1) MPC and MRW command are shown in figure at this time, Any combination of MRR, MRW and MPC is allowed during t_{XSR} period.
2) Any command also includes MRR, MRW and all MPC command.

Mode Register Read (MRR), Mode Register Write (MRW) and Multi Purpose Command (MPC) can be issued during t_{RFC} period.

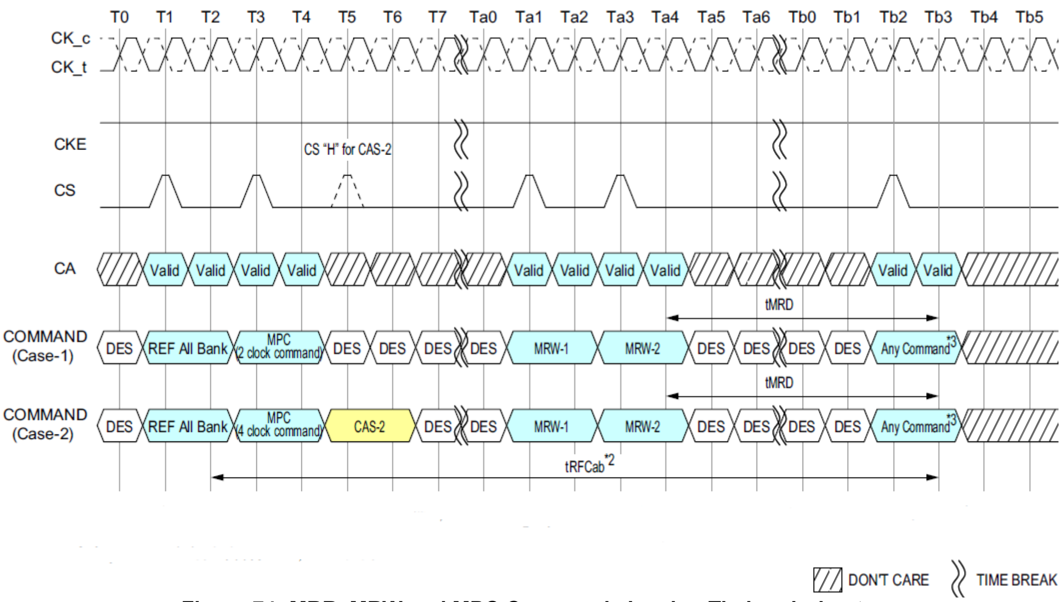


Figure 74: MRR, MRW and MPC Commands Issuing Timing during t_{RFC}

NOTE :
1) MPC and MRW command are shown in figure at this time, Any combination of MRR, MRW and MPC is allowed during t_{RFCab} or t_{RFCpb} period.
2) Refresh cycle time depends on Refresh command. In case of REF per Bank command issued, Refresh cycle time will be t_{RFCpb}.
3) Any command also includes MRR, MRW and all MPC command.

3.26 LPDDR4: Partial Array Self-Refresh: Bank Masking

LPDDR4 SDRAM has 8 banks (additional banks may be required for higher densities). Each bank of LPDDR4 SDRAM can be independently configured whether a self refresh operation is taking place. One mode register unit of 8 bits accessible via MRW command is assigned to program the bank masking status of each bank up to 8 banks. For bank masking bit assignments see Mode Register 16.

The mask bit to the bank controls a refresh operation of entire memory within the bank. If a bank is masked via MRW, a refresh operation to the entire bank is blocked and data retention by a bank is not guaranteed in self refresh mode. To enable a refresh operation to a bank, a coupled mask bit has to be programmed, “unmasked”. When a bank mask bit is unmasked, a refresh to a bank is determined by the programmed status of segment mask bits, which is described in the following chapter.

3.27 LPDDR4: Partial Array Self-Refresh: Segment Masking

Segment masking scheme may be used in lieu of or in combination with bank masking scheme in LPDDR4 SDRAM. The number of segments differ by the density and the setting of each segment mask bit is applied across all the banks. LPDDR4 devices utilize 8 segments per bank.For segment masking bit assignments, see Mode Register 17.

For those refresh-enabled banks, a refresh operation to the address range which is represented by a segment is blocked when the mask bit to this segment is programmed, “masked”. Programming of segment mask bits is similar to the one of bank mask bits. With LPDDR4, 8 segments are used as listed in Mode Register 17. One mode register unit is used for the programming of segment mask bits up to 8 bits. One more mode register unit may be reserved for future use. Programming of mask bits in the reserved registers has no effect on the device operation.

[Table 32] Example of Bank and Segment Masking use in LPDDR4 devices

	Segment Mask (MR17)	Bank 0	Bank 1	Bank 2	Bank 3	Bank 4	Bank 5	Bank 6	Bank 7
Bank Mask (MR16)		0	1	0	0	0	0	0	1
Segment 0	0		M						M
Segment 1	0		M						M
Segment 2	1	M	M	M	M	M	M	M	M
Segment 3	0		M						M
Segment 4	0		M						M
Segment 5	0		M						M
Segment 6	0		M						M
Segment 7	1	M	M	M	M	M	M	M	M

NOTE :
1) This table illustrates an example of an 8-bank LPDDR4 device, when a refresh operation to bank 1 and bank 7, as well as segment 2 and segment 7 are masked.

3.28 MODE REGISTER READ (MRR)

The Mode Register Read (MRR) command is used to read configuration and status data from the LPDDR4-SDRAM registers. The MRR command is initiated with CS and CA[5:0] in the proper state as defined by the Command Truth Table. The mode register address operands (MA[5:0]) allow the user to select one of 64 registers. The mode register contents are available on the first 4UI's data bits of DQ[7:0] after RL × t_{CK} + t_{DQSCK} + t_{DQSQ} following the MRR command. Subsequent data bits contain valid but undefined content. DQS is toggled for the duration of the Mode Register READ burst. The MRR has a command burst length 16. MRR operation must not be interrupted.

[Table 33] DQ output mapping

UI	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
DQ0	OP0				V											
DQ1	OP1				V											
DQ2	OP2				V											
DQ3	OP3				V											
DQ4	OP4				V											
DQ5	OP5				V											
DQ6	OP6				V											
DQ7	OP7				V											
DQ8-15	V															
DMI0-1	V															

NOTE :
1) MRR data are extended to first 4 UI's for DRAM controller to sample data easily.
2) DBI may apply or may not apply during normal MRR. It's vendor specific. If read DBI is enable with MRS and vendor cannot support the DBI during MRR, DMI pin status should be low.
3) The read pre-amble and post-amble of MRR are same as normal read.

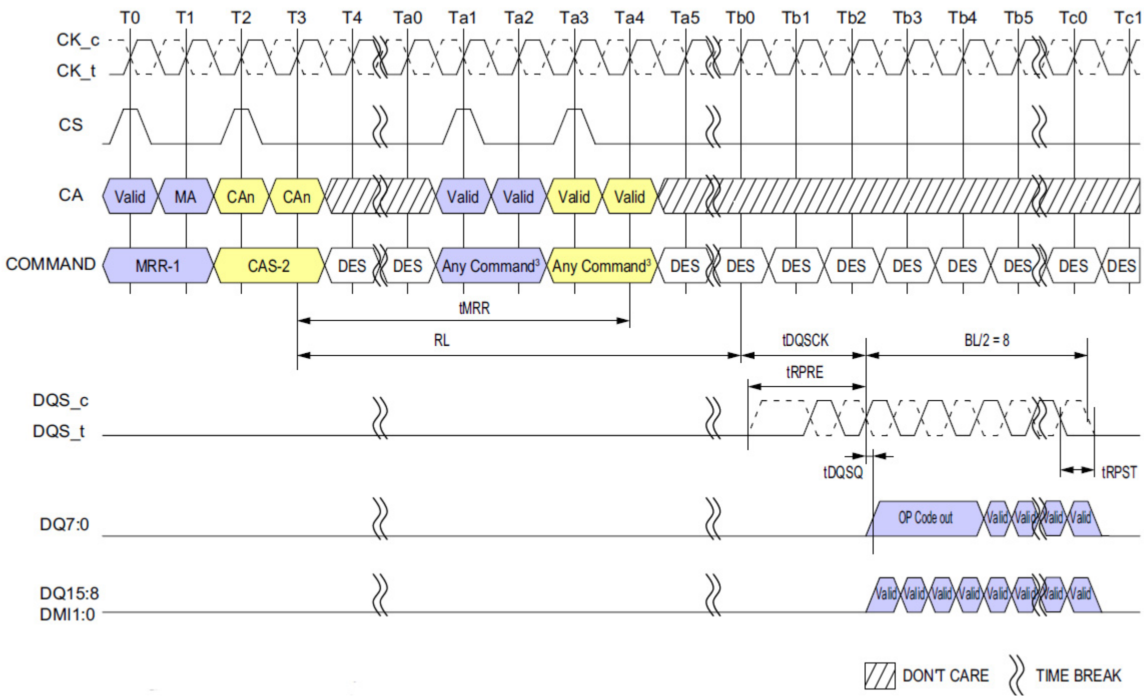


Figure 75: Mode Register Read Operation

NOTE :
1) Only BL=16 is supported.
2) Only DES is allowed during t_{MRR} period.
3) There are some exceptions about issuing commands after t_{MRR}. Refer to MRR/MRW Timing Constraints Table for detail.
4) DBI is Disable mode.
5) DES commands except t_{MRR} period are shown for ease of illustration; other commands may be valid at these times.
6) DQ/DQS: VSSQ termination.

3.28.1 MRR After Read And Write Command

After a prior READ command, the MRR command must not be issued earlier than BL/2 clock cycles, in a similar way WL + BL/2 + 1 + RU(t_{WTR}/t_{CK}) clock cycles after a prior Write, Write with AP, Mask Write, Mask Write with AP and MPC Write FIFO command in order to avoid the collision of Read and Write burst data on SDRAM's internal Data bus.

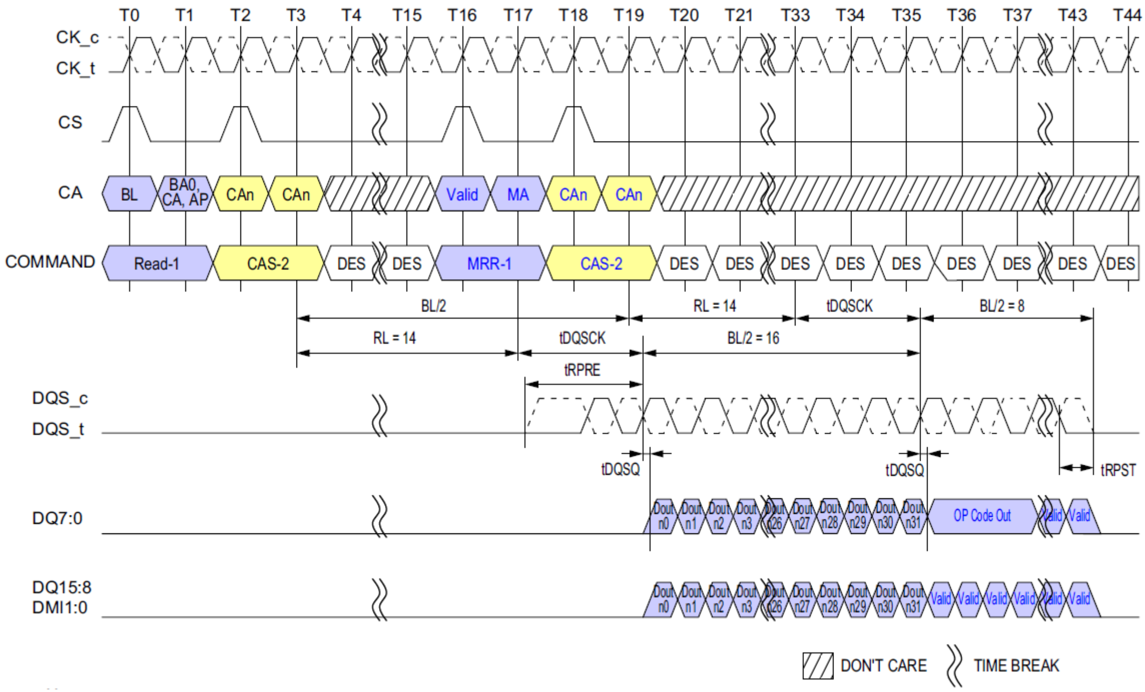


Figure 76: READ to MRR Timing

NOTE :
1) The minimum number of clock cycles from the burst READ command to the MRR command is BL/2.
2) Read BL = 32, MRR BL = 16, RL = 14, Preamble = Toggle, Postamble = 0.5nCK, DBI = Disable, DQ/DQS: VSSQ termination.
3) DES commands except t_{MRR} period are shown for ease of illustration; other commands may be valid at these times.

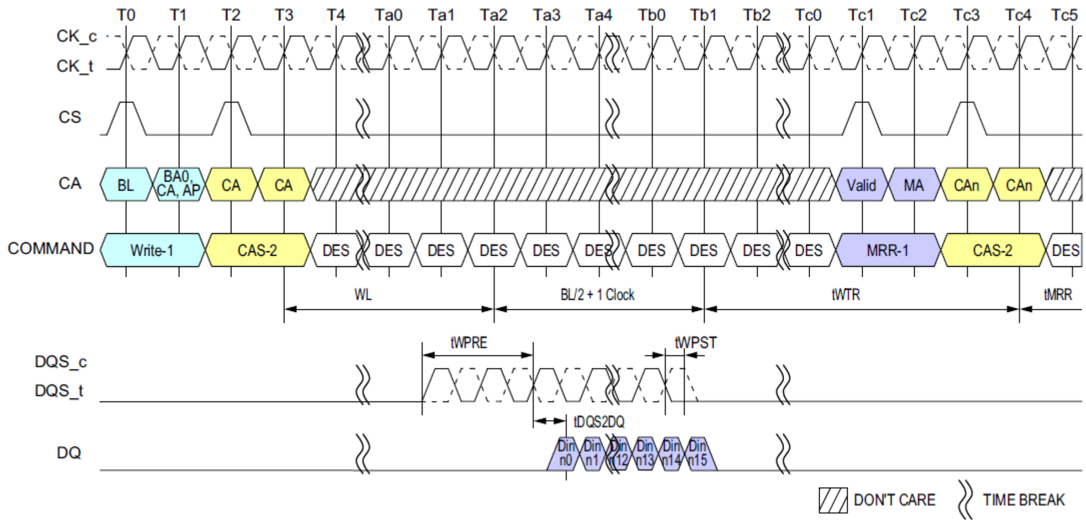


Figure 77: Write to MRR Timing

NOTE :
1) Write BL=16, Write Postamble = 0.5nCK, DQ/DQS: VSSQ termination.
2) Only DES is allowed during t_{MRR} period.
3) Din n = data-in to columnm.n.
4) The minimum number of clock cycles from the burst write command to MRR command is WL + BL/2 + 1 + RU(t_{WTR}/t_{CK}).
5) t_{WTR} starts at the rising edge of CK after the last latching edge of DQS.
6) DES commands except t_{MRR} period are shown for ease of illustration; other commands may be valid at these times.

3.28.2 MRR After Power-Down Exit

Following the power-down state, an additional time, t_{MRRI}, is required prior to issuing the mode register read (MRR) command. This additional time (equivalent to t_{RCD}) is required in order to be able to maximize power-down current savings by allowing more power-up time for the MRR data path after exit from power-down mode.

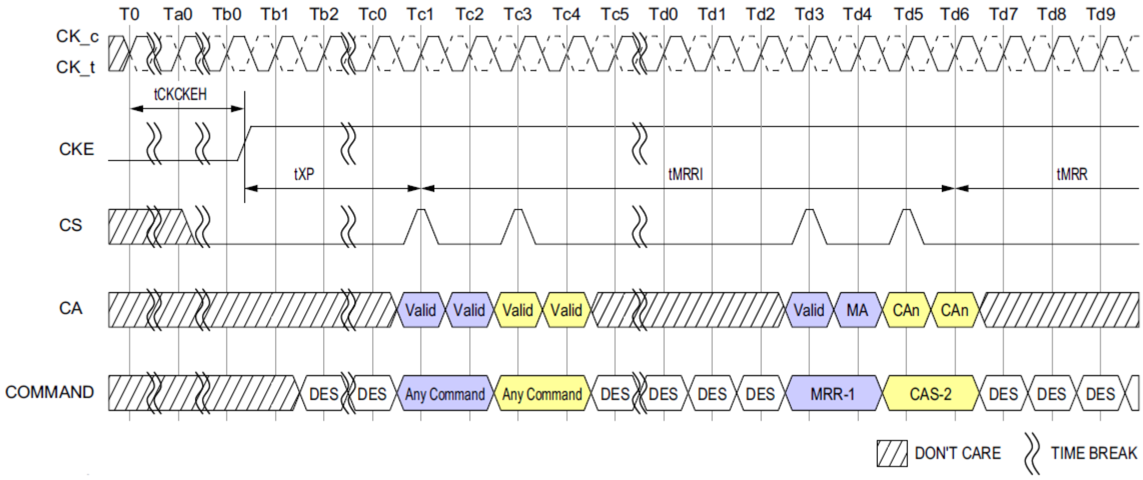


Figure 78: MRR Following Power-Down

NOTE :
1) Only DES is allowed during t_{MRR} period.
2) DES commands except t_{MRR} period are shown for ease of illustration; other commands may be valid at these times.

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3.29 Mode Register Write (MRW) Operation

The Mode Register Write (MRW) command is used to write configuration data to the mode registers. The MRW command is initiated by setting CKE, CS, and CA[5:0] to valid levels at a rising edge of the clock (see Command Truth Table). The mode register address and the data written to the mode registers is contained in CA[5:0] according to the Command Truth Table. The MRW command period is defined by t_{MRW}. Mode register Writes to read-only registers have no impact on the functionality of the device.

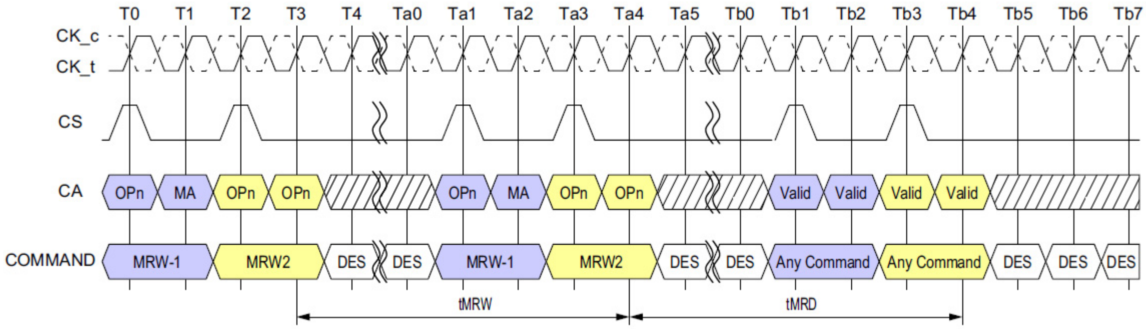


Figure 79: Mode Register Write Timing

NOTE :
1) Only DE-SELECT command is allowed during t_{MRW} and t_{MRD} periods.

3.29.1 Mode Register Write

MRW can be issued from either a Bank-Idle or Bank-Active state. Certain restrictions may apply for MRW from an Active state.

[Table 34] Truth Table for Mode Register Read (MRR) and Mode Register Write (MRW)

Current State	Command	Intermediate State	Next State
All Banks Idle	MRR	Mode Register Reading (All Banks Idle)	All Banks Idle
	MRW	Mode Register Writing (All Banks Idle)	All Banks Idle
Bank(s) Active	MRR	Mode Register Reading	Bank(s) Active
	MRW	Mode Register Writing	Bank(s) Active

[Table 35] MRR/MRW Timing Constraints : DQ ODT is Disable

From Command	To Command	Minimum Delay between "From Command" and "To Command"	Unit
MRR	MRR	t_{MRR}	-
	RD/RDA	t_{MRR}	-
	WR/WRA/MWR/MWRA	RL+RU($t_{DQSCK(max)}$/t_{CK})+BL/2+WL+t_{WPRE}+RD(t_{RPST})	nCK
	MRW	RL+RU($t_{DQSCK(max)}$/t_{CK})+BL/2+3	nCK
RD/RDA	MRR	BL/2	nCK
WR/WRA/MWR/MWRA		WL+1+BL/2+RU(t_{WTR}/t_{CK})	nCK
MRW		t_{MRD}	-
Power Down Exit		t_{XP}+t_{MRRI}	-
MRW	RD/RDA	t_{MRD}	-
	WR/WRA/MWR/MWRA	t_{MRD}	-
	MRW	t_{MRW}	-
RD/RD_FIFO/RD_DQ_CAL	MRW	RL+BL/2+RU($t_{DQSCKmax}$/t_{CK}) +RD(t_{RPST}) +max(RU(7.5ns/t_{CK}),8nCK)	nCK
RD with Auto-Precharge		RL+BL/2+RU($t_{DQSCKmax}$/t_{CK}) +RD(t_{RPST}) +max(RU(7.5ns/t_{CK}),8nCK)+nRTP-8	nCK
WR/MWR/WR_FIFO		WL+1+BL/2+max(RU(7.5ns/t_{CK}),8nCK)	nCK
WR/MWR with Auto-Precharge		WL+1+BL/2+max(RU(7.5ns/t_{CK}),8nCK)+nWR	nCK

[Table 36] MRR/MRW Timing Constraints : DQ ODT is Enable

From Command	To Command	Minimum Delay between "From Command" and "To Command"	Unit
MRR	MRR	t_{MRR}	-
	RD/RDA	t_{MRR}	-
	WR/WRA/MWR/MWRA	RL+RU($t_{DQSCK(max)}$/t_{CK})+BL/2 -ODTLon-RD($t_{ODTon(min)}$/t_{CK})+RD(t_{RPST})+1	nCK
	MRW	RL+RU($t_{DQSCK(max)}$/t_{CK})+BL/2+ 3	-
RD/RDA	MRR	BL/2	-
WR/WRA/MWR/MWRA		WL+1+BL/2+RU(t_{WTR}/t_{CK})	-
MRW		t_{MRD}	-
Power Down Exit		t_{XP}+t_{MRRI}	-
MRW	RD/RDA	t_{MRD}	-
	WR/WRA/MWR/MWRA	t_{MRD}	-
	MRW	t_{MRW}	-
RD/RD_FIFO/RD DQ CAL	MRW	RL+BL/2+RU($t_{DQSCKmax}$/t_{CK}) +RD(t_{RPST}) +max(RU(7.5ns/t_{CK}),8nCK)	nCK
RD with Auto-Precharge		RL+BL/2+RU($t_{DQSCKmax}$/t_{CK}) +RD(t_{RPST}) +max(RU(7.5ns/t_{CK}),8nCK)+nRTP-8	nCK
WR/ MWR/ WR FIFO		WL+1+BL/2+max(RU(7.5ns/tCK),8nCK)	nCK
WR/MWR with Auto-Precharge		WL+1+BL/2+max(RU(7.5ns/t_{CK}),8nCK)+nWR	nCK

3.30 V_{REF} Current Generator (VRCG)

LPDDR4 SDRAM Vref current generators (VRCG) incorporate a high current mode to reduce the settling time of the internal Vref(DQ) and Vref(CA) levels during training and when changing frequency set points during operation. The high current mode is enabled by setting MR13[OP3] = 1_B. Only Deselect commands may be issued until t_{VRCG_ENABLE} is satisfied. t_{VRCG_ENABLE} timing is shown in Figure 80.

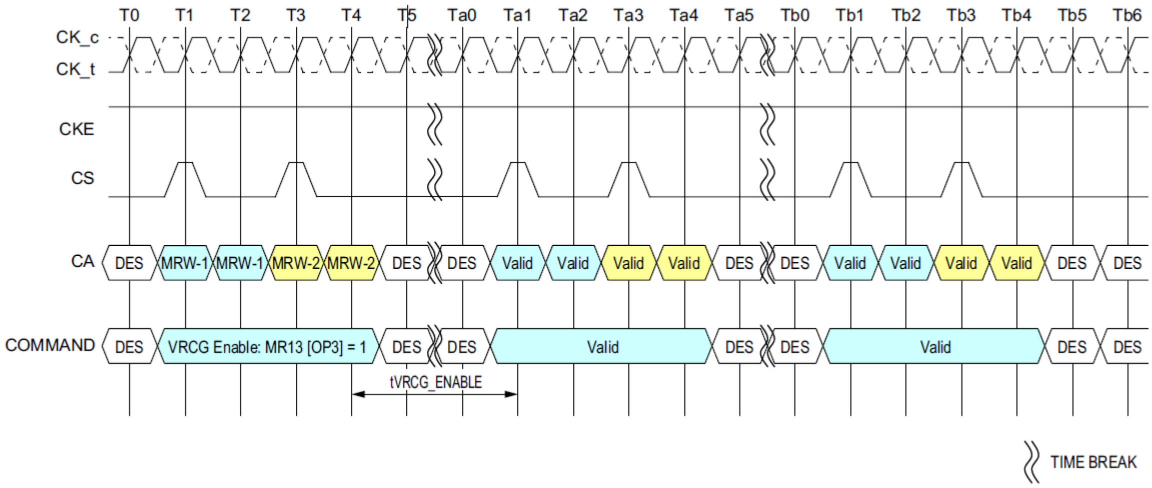


Figure 80: VRCG Enable timing

VRCG high current mode is disabled by setting MR13[OP3] = 0_B. Only Deselect commands may be issued until $t_{VRCG\_DISABLE}$ is satisfied. $t_{VRCG\_DISABLE}$ timing is shown in Figure 81.

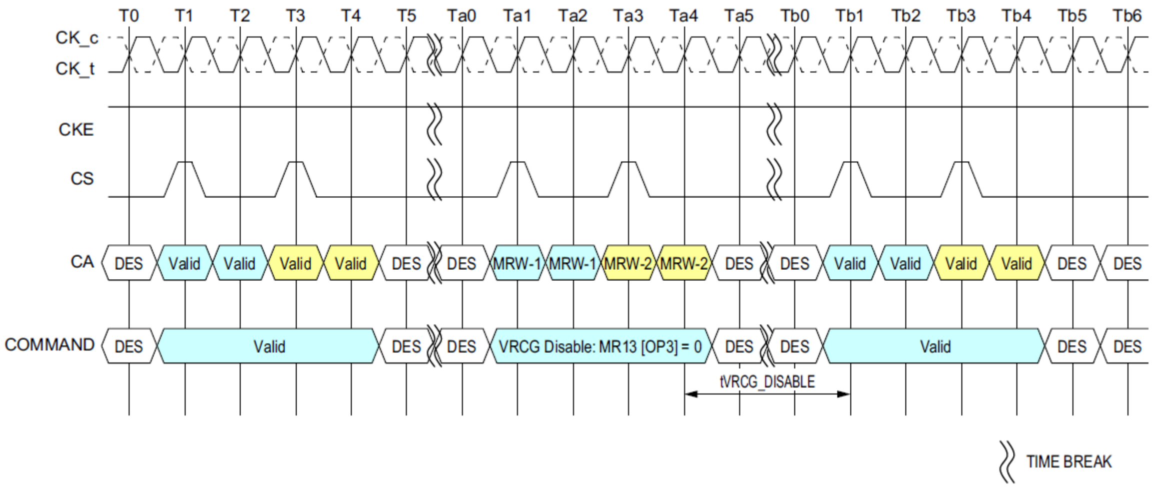
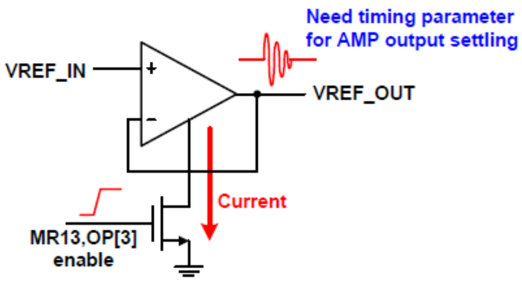


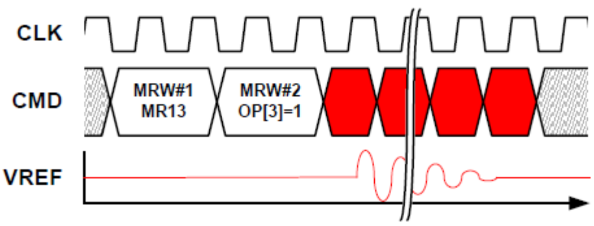
Figure 81: VRCG Disable timing

Note that LPDDR4 SDRAM devices support $V_{REF(CA)}$ and $V_{REF(DQ)}$ range and value changes without enabling VRCG high current mode.

3.31 VRCG(MR13,OP[3]) Operation Constraints

- While turning on/off OP-AMP with VRCG MRS, the OP-AMP switching noise can transfer to VREF_OUT node. Due to the VREF noise, DRAM could not sample “CMD” correctly
- Thus, DRAM requires the time for OP-AMP output to become stable.





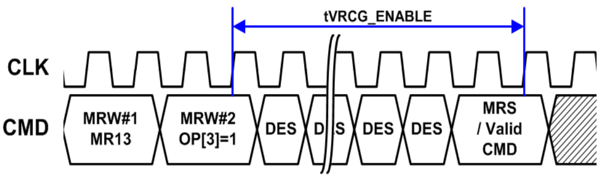
< Example : MR13 OP[3]= 0 → 1 >

MR13_Register Information (MA<5:0> = $0D_H$):

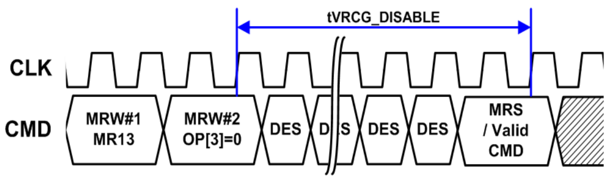
OP7	OP6	OP5	OP4	OP3	OP2	OP1	OP0
FSP-OP	FSP-WR	DM_DIS	(RFU)	VRCG	VRO	RPT	CBT

Function	Register Type	Operand	Data	Notes
VRCG (Vref Change)	Write	OP[3]	0_B: Normal operation (default) 1_B: VrefCA or VrefDQ fast response (high current) mode.	1

NOTE :
1) When OP[3]=1_B, the VREF circuit uses a high.current mode to improve VREF settling time. VRCG MRS may be issued when user want to change vrefDQ, vrefCA and operating frequency with around 200ns.



< MR13 OP[3]= 0 -> 1 >



< MR13 OP[3]= 1 -> 0 >

- To guarantee the OP-AMP settling, Samsung proposes t_{VRCG_ENABLE} /$t_{VRCG\_DISABLE}$ parameter
  - Each parameter indicates the time necessary for VREF settling after changing VRCG(MR13,OP[3]).

[Table 37] VRCG Enable/Disable Timing

Parameter	Symbol	min	max	Unit
VREF high current mode enable time	t_{VRCG_ENABLE}	-	100 1)	ns
VREF high current mode disable time	$t_{VRCG\_DISABLE}$	-	15 2)	ns

NOTE :
1) VREF high current mode enable setting time is max 100ns for DRAM. After tVRCG_Enable max time, controller can issue the command.
2) VREF high current mode disable setting time is max 15ns for DRAM. After tVRCG_DISABLE max time, controller can issue the command.

3.32 CA V_{REF} Training

The DRAM internal CA Vref specification parameters are voltage operating range, step size, Vref set tolerance, Vref step time and Vref valid level.
The voltage operating range specifies the minimum required Vref setting range for LPDDR4 DRAM devices. The minimum range is defined by Vrefmax and Vrefmin as depicted in Figure 82 below.

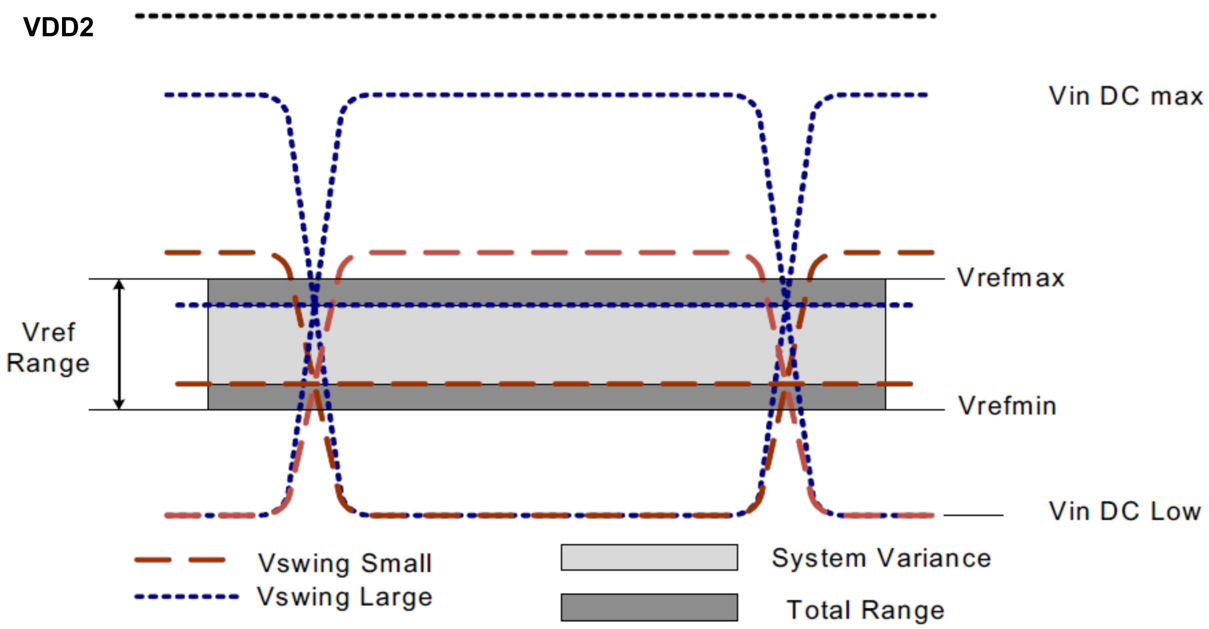


Figure 82: Vref operating range(Vrefmin, Vrefmax)

The Vref step size is defined as the step size between adjacent steps. However, for a given design, DRAM has one value for Vref step size that falls within the range.
The Vref set tolerance is the variation in the Vref voltage from the ideal setting. This accounts for accumulated error over multiple steps. There are two ranges for Vref set tolerance uncertainty. The range of Vref set tolerance uncertainty is a function of number of steps n.
The Vref set tolerance is measured with respect to the ideal line which is based on the two endpoints. Where the endpoints are at the min and max Vref values for a specified range. An illustration depicting an example of the step size and Vref set tolerance is below.

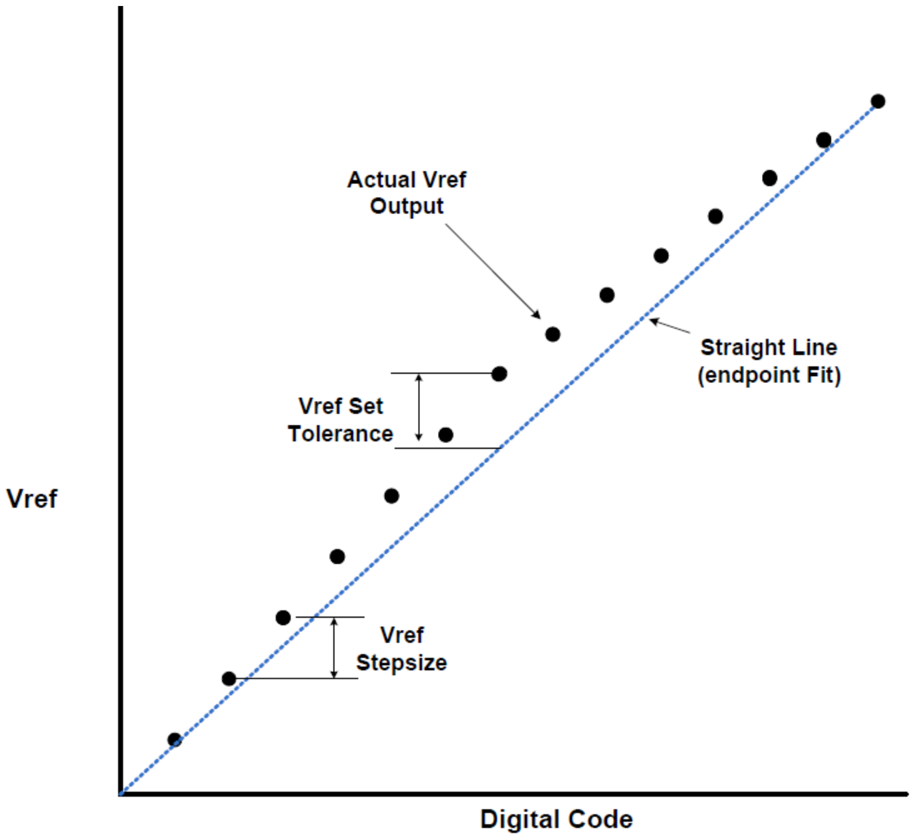


Figure 83: Example of Vref set tolerance(max case only shown) and step size

The Vref increment/decrement step times are define by Vref_time-short, Middle and long. The Vref_time-short, Vref_time-Middle and Vref_time-long is defined from TS to TE as shown in the Figure 84 below where TE is referenced to when the vref voltage is at the final DC level within the Vref valid tolerance(Vref_val_tol).
The Vref valid level is defined by Vref_val tolerance to qualify the step time TE as shown in Figure 85. This parameter is used to insure an adequate RC time constant behavior of the voltage level change after any Vref increment/decrement adjustment. This parameter is only applicable for DRAM component level validation/characterization.

Vref_time-Short is for a single step size increment/decrement change in Vref voltage.

Vref_time-Middle is at least 2 step sizes increment/decrement change within the same VrefCA range in Vref voltage.
Vref_time-Long is the time including up to Vrefmin to Vrefmax or Vrefmax to Vrefmin change across the VrefCA Range in Vref voltage.

TS - is referenced to MRS command clock
TE - is referenced to the Vref_val_tol

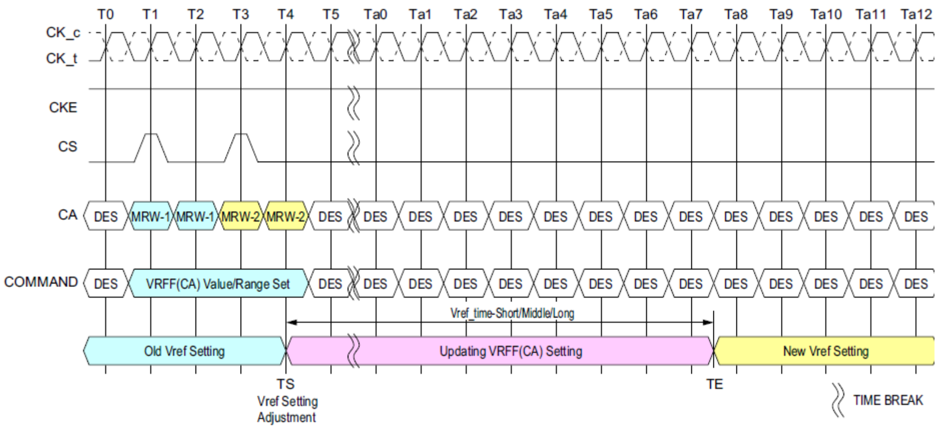


Figure 84: Vref_time for Short, Middle and Long Timing Diagram

The MRW command to the mode register bits are as follows.
MR12 OP[5:0] : $V_{REF(CA)}$ Setting
MR12 OP[6] : $V_{REF(CA)}$ Range

The minimum time required between two Vref MRS commands is Vref_time-short for single step and Vref_time-Middle for a full voltage range step.

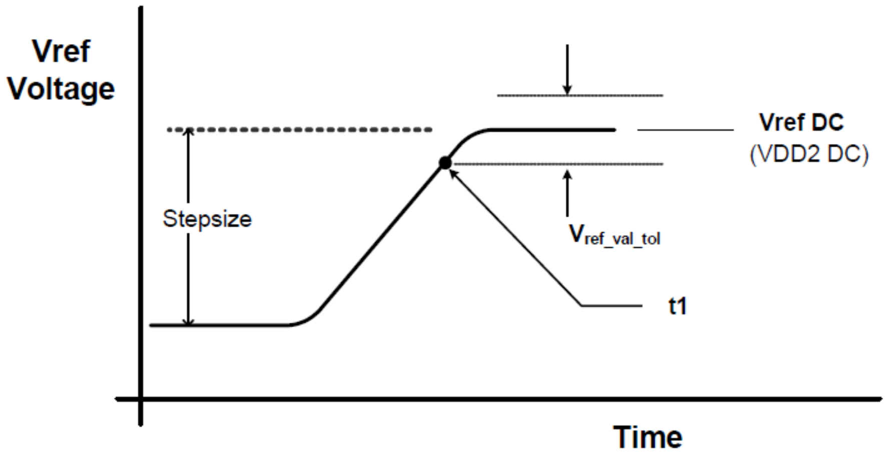


Figure 85: Vref step single step size increment case.

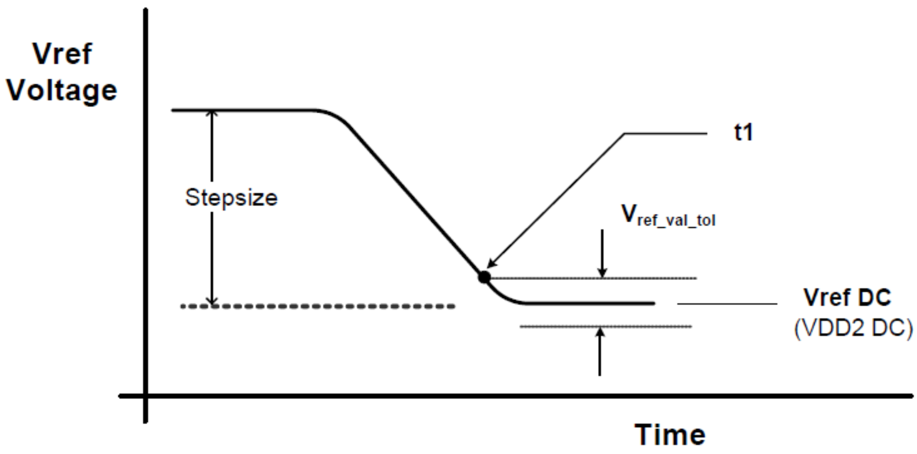


Figure 86: Vref step single step size decrement case

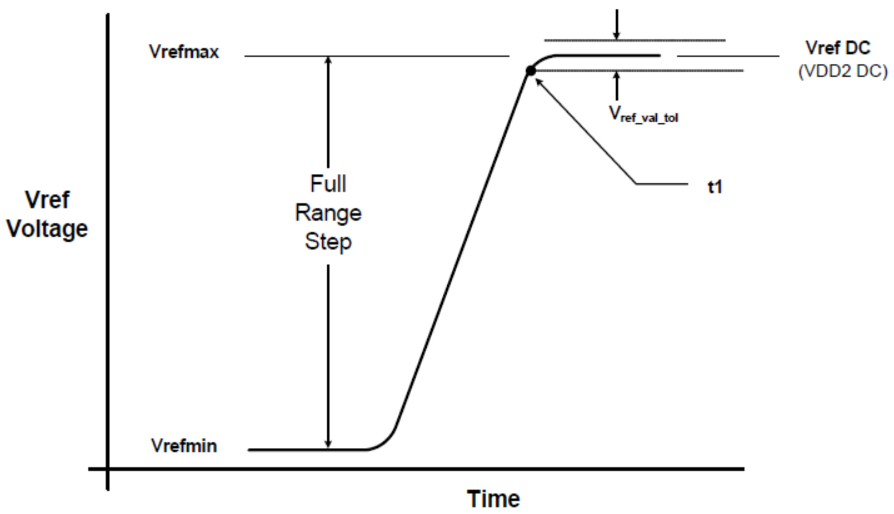


Figure 87: Vref full step from Vrefmin to Vrefmax case

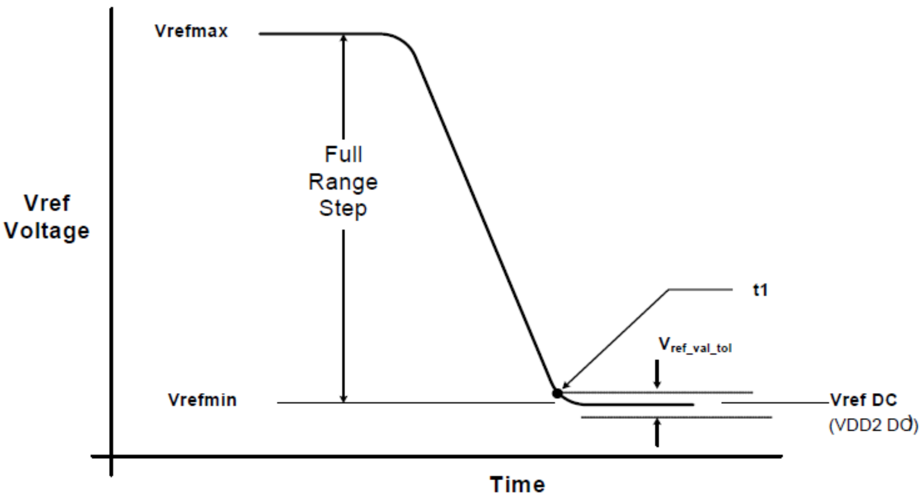


Figure 88: Vref full step from Vrefmax to Vrefmin case.

The table below contains the CA internal vref specifications that will be characterized at the component level for compliance. The component level characterization method is tbd.

[Table 38] CA Internal Vref Specifications

Parameter	Symbol	Min	Typ	Max	Unit	Notes
Vref Max operating point Range0	Vref_max_R0	-	-	30%	VDD2	1,11
Vref Min operating point Range0	Vref_min_R0	10%	-	-	VDD2	1,11
Vref Max operating Point Range1	Vref_max_R1	-	-	42%	VDD2	1,11
Vref Min operating point Range1	Vref_min_R1	22%	-	-	VDD2	1,11
Vref Step size	Vref_step	0.30%	0.40%	0.50%	VDD2	2
Vref Set Tolerance	Vref_set_tol	-1.00%	0.00%	1.00%	VDD2	3,4,6
		-0.10%	0.00%	0.10%	VDD2	3,5,7
Vref Step Time	Vref_time-Short	-	-	100	ns	8
	Vref_time_Middle	-	-	200	ns	12
	Vref_time-Long	-	-	250	ns	9
	Vref_time_weak	-	-	1	ms	13,14
Vref Valid tolerance	Vref_val_tol	-0.10%	0.00%	0.10%	VDD2	10

NOTE:
1) Vref DC voltage referenced to V_{DD2_DC}.
2) Vref step size increment/decrement range. Vref at DC level.
3) Vref_new = Vref_old + n×Vref_step; n= number of steps; if increment use "+"; If decrement use "-".
4) The minimum value of Vref setting tolerance = Vref_new - 1.0%×V_{DD2}. The maximum value of Vref setting tolerance = Vref_new + 1.0%×V_{DD2}. For n>4.
5) The minimum value of Vref setting tolerance = Vref_new - 0.10%×V_{DD2}. The maximum value of Vref setting tolerance = Vref_new + 0.10%×V_{DD2}. For n≤ 4.
6) Measured by recording the min and max values of the Vref output over the range, drawing a straight line between those points and comparing all other Vref output settings to that line.
7) Measured by recording the min and max values of the Vref output across 4 consecutive steps(n=4), drawing a straight line between those points and comparing all other Vref output settings to that line.
8) Time from MRS command to increment or decrement one step size for Vref.
9) Time from MRS command to increment or decrement Vrefmin to Vrefmax or Vrefmax to Vrefmin change across the VrefCA Range in Vref voltage.
10) Only applicable for DRAM component level test/characterization purpose. Not applicable for normal mode of operation. VREF valid is to qualify the step times which will be characterized at the component level.
11) DRAM range0 or 1 set by MR12 OP[6].
12) Time from MRS command to increment or decrement more than one step size up to a full range of Vref voltage within the same VrefCA range.
13) Applies when VRCG high current mode is not enabled, specified by MR13[OP3] = 0_B.
14) Vref_time_weak covers all Vref(CA) Range and Value change conditions are applied to Vref_time_Short/Middle/Long.

3.33 DQ V_{REF} Training

The DRAM internal DQ Vref specification parameters are voltage operating range, step size, Vref set tolerance, Vref step time and Vref valid level.

The voltage operating range specifies the minimum required Vref setting range for LPDDR4 DRAM devices. The minimum range is defined by Vrefmax and Vrefmin as depicted in Figure 89 below.

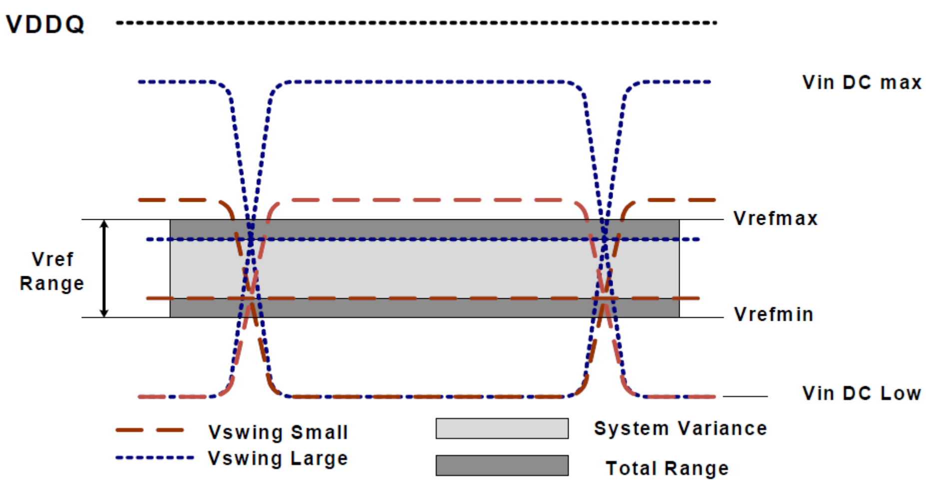


Figure 89: V_{REF} operating range (V_{REF}min, V_{REF}max)

The Vref step size is defined as the step size between adjacent steps. However, for a given design, DRAM has one value for Vref step size that falls within the range.

The Vref set tolerance is the variation in the Vref voltage from the ideal setting. This accounts for accumulated error over multiple steps. There are two ranges for Vref set tolerance uncertainty. The range of Vref set tolerance uncertainty is a function of number of steps n.

The Vref set tolerance is measured with respect to the ideal line which is based on the two endpoints. Where the endpoints are at the min and max Vref values for a specified range. An illustration depicting an example of the step size and Vref set tolerance is below.

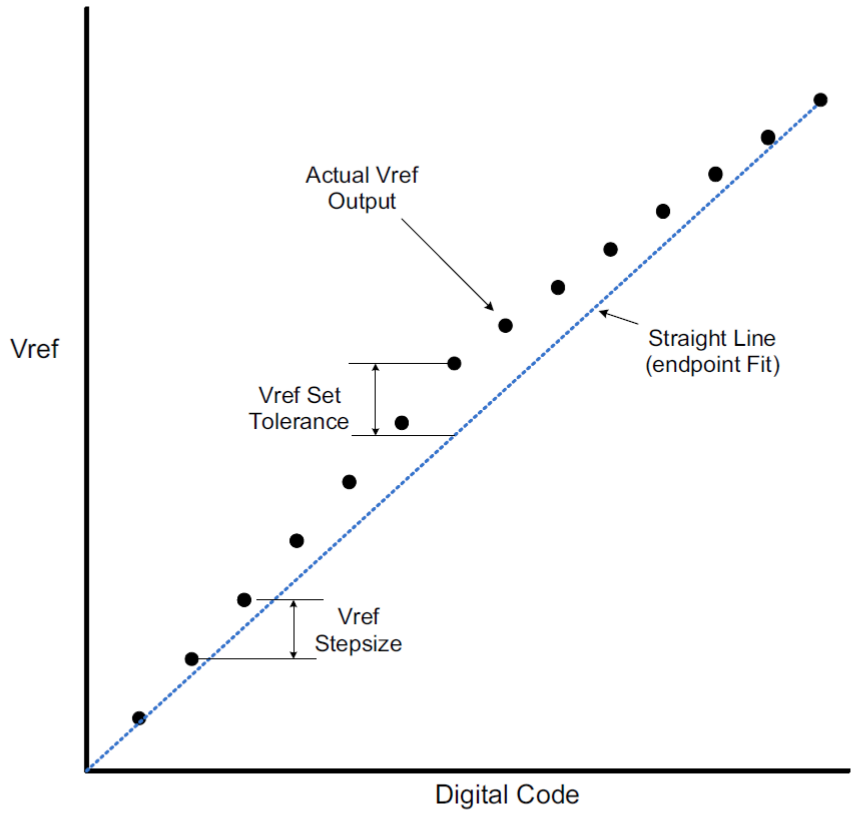


Figure 90: Example of V_{REF} set tolerance (max case only shown) and step size

The Vref increment/decrement step times are define by Vref_time-short, Middle and long. The Vref_time-short, Vref_time-Middle and Vref_time-long is defined from TS to TE as shown in the Figure 91 below where TE is referenced to when the vref voltage is at the final DC level within the Vref valid tolerance (Vref_val_tol).

The Vref valid level is defined by Vref_val tolerance to qualify the step time TE as shown in Figure 91. This parameter is used to insure an adequate RC time constant behavior of the voltage level change after any Vref increment/decrement adjustment. This parameter is only applicable for DRAM component level validation/characterization.

Vref_time-Short is for a single step size increment/decrement change in Vref voltage.
Vref_time-Middle is at least 2 step sizes increment/decrement change within the same VrefDQ range in Vref voltage.
Vref_time-Long is the time including up to Vrefmin to Vrefmax or Vrefmax to Vrefmin change across the VrefDQ Range in Vref voltage.
TS - is referenced to MRS command clock
TE - is referenced to the Vref_val_tol

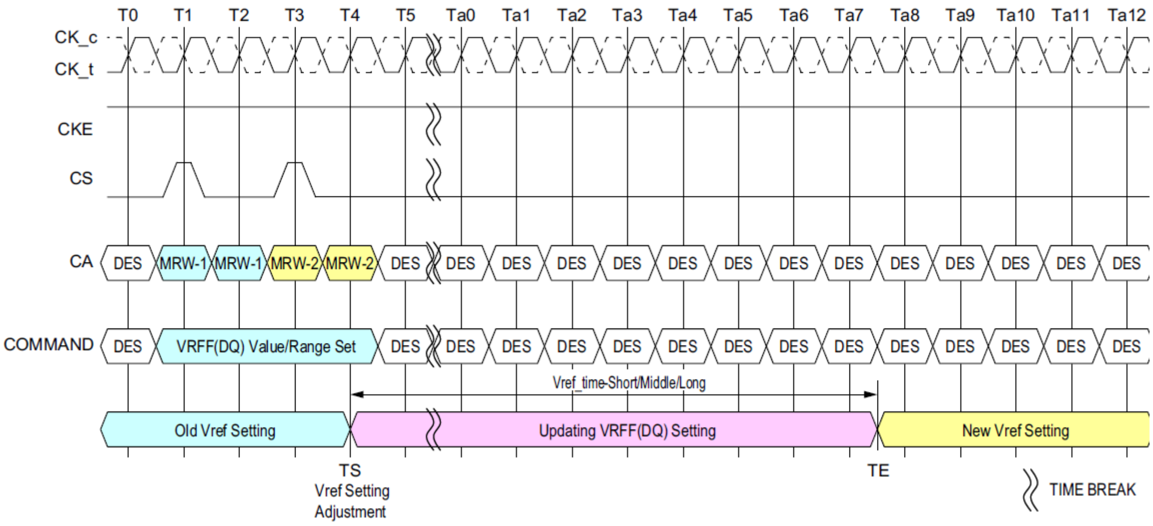


Figure 91: V_{REF}_time for short, Middle and long timing diagram

The MRW command to the mode register bits are as follows.
MR14 OP[5:0] : $V_{REF(DQ)}$ Setting
MR14 OP[6] : $V_{REF(DQ)}$ Range

The minimum time required between two Vref MRS commands is Vref_time-short for single step and Vref_time-Middle for a full voltage range step.

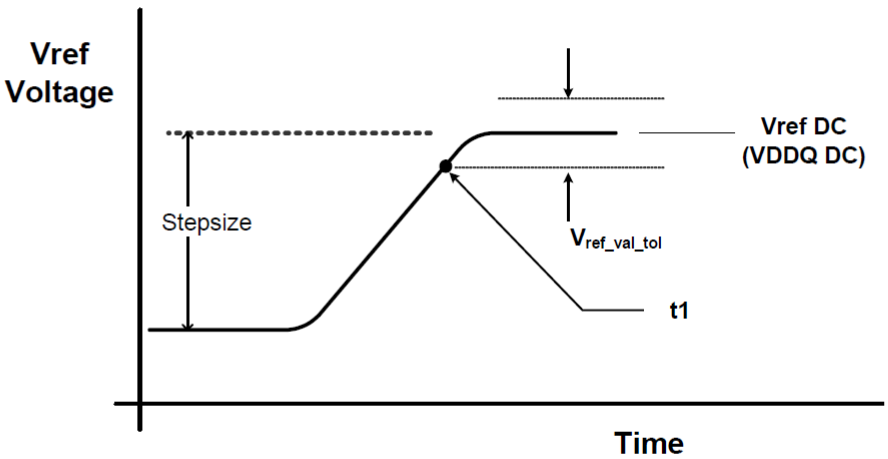


Figure 92: V_{REF} step single step size increment case

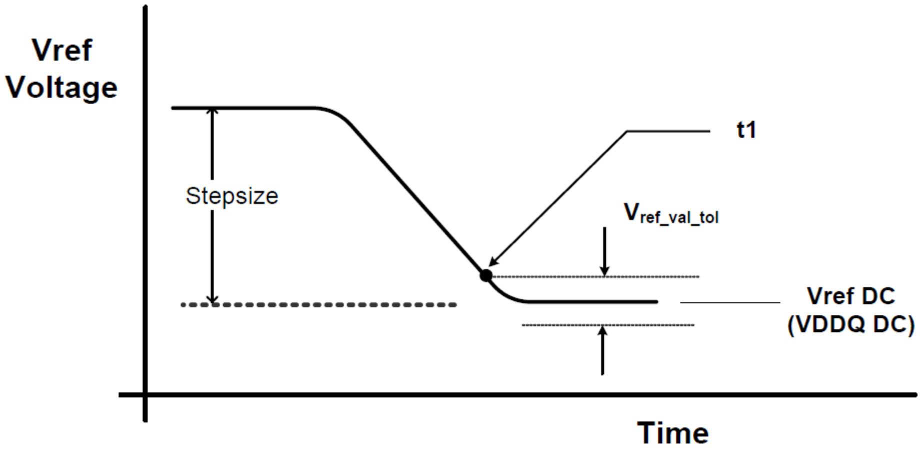


Figure 93: V_{REF} step single step size decrement case

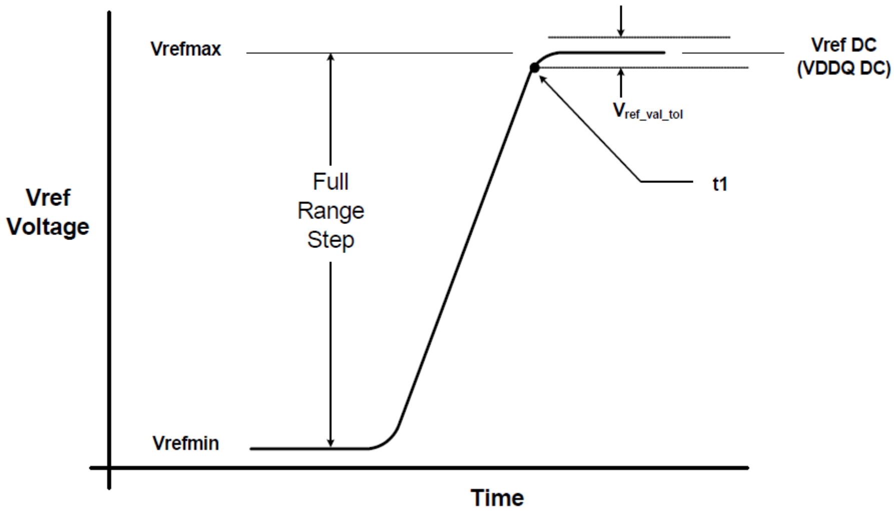


Figure 94: V_{REF} full step from V_{REF}min to V_{REF}max case

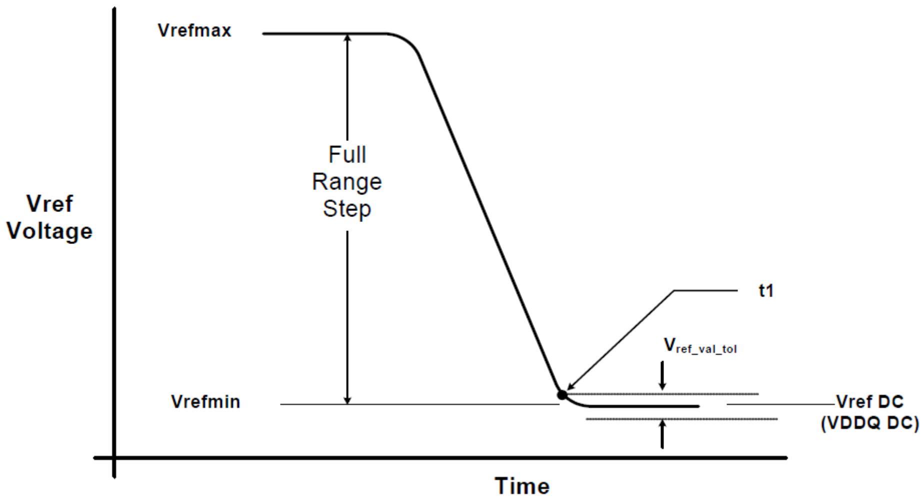


Figure 95: V_{REF} full step from V_{REF}max to V_{REF}min case

The table below contains the DQ internal vref specifications that will be characterized at the component level for compliance. The component level characterization method is tbd[A)].

[Table 39] DQ Internal V_{REF} Specifications

Parameter	Symbol	Min	Typ	Max	Unit	Notes
V_{REF} Max operating point Range0	V_{REF}_max_R0	-	-	30%	V_{DDQ}	1,11
V_{REF} Min operating point Range0	V_{REF}_min_R0	10%	-	-	V_{DDQ}	1,11
V_{REF} Max operating point Range1	V_{REF}_max_R1	-	-	42%	V_{DDQ}	1,11
V_{REF}Min operating point Range1	V_{REF}_min_R1	22%	-	-	V_{DDQ}	1,11
V_{REF} Step size	V_{REF}_step	0.30%	0.40%	0.50%	V_{DDQ}	2
V_{REF} Set Tolerance	V_{REF}_set_tol	-1.00%	0.00%	1.00%	V_{DDQ}	3,4,6
		-0.10%	0.00%	0.10%	V_{DDQ}	3,5,7
V_{REF} Step Time	V_{REF}_time-Short	-	-	100	ns	8
	V_{REF}_time-Middle	-	-	200	ns	12
	V_{REF}_time_Long	-	-	250	ns	9
	V_{REF}_time_weak	-	-	1	ms	13,14
V_{REF}Valid tolerance	v_{REF}_val_tol	-0.10%	0.00%	010%	V_{DDQ}	10

NOTE :
1) Vref DC voltage referenced to V_{DDQ_DC}.
2) Vref step size increment/decrement range. Vref at DC level.
3) Vref_new = Vref_old + n × Vref_step; n= number of steps; if increment use "+"; If decrement use "-".
4) The minimum value of Vref setting tolerance = Vref_new - 1.0% × V_{DDQ}. The maximum value of Vref setting tolerance = Vref_new + 1.0% × V_{DDQ}. For n>4.
5) The minimum value of Vref setting tolerance = Vref_new - 0.10% × V_{DDQ}. The maximum value of Vref setting tolerance = Vref_new + 0.10% × V_{DDQ}. For $n \leq 4$.
6) Measured by recording the min and max values of the Vref output over the range, drawing a straight line between those points and comparing all other Vref output settings to that line.
7) Measured by recording the min and max values of the Vref output across 4 consecutive steps(n=4), drawing a straight line between those points and comparing all other Vref output settings to that line.
8) Time from MRS command to increment or decrement one step size for Vref.
9) Time from MRS command to increment or decrement Vrefmin to Vrefmax or Vrefmax to Vrefmin change across the VrefDQ Range in Vref voltage.
10) Only applicable for DRAM component level test/characterization purpose. Not applicable for normal mode of operation. Vref valid is to qualify the step times which will be characterized at the component level.
11) DRAM range0 or 1 set by MR14 OP[6].
12) Time from MRS command to increment or decrement more than one step size up to a full range of Vref voltage within the same VrefDQ range.
13) Applies when VRCG high current mode is not enabled, specified by MR13[OP3] = 0.
14) Vref_time_weak covers all Vref(DQ) Range and Value change conditions are applied to Vref_time_Short/Middle/Long.

A) As of publication of this document, under discussion by the formulating committee.

3.34 Command Bus Training

The LPDDR4-SDRAM command bus must be trained before enabling termination for high-frequency operation. LPDDR4 provides an internal $V_{REF(CA)}$ that defaults to a level suitable for un-terminated, low-frequency operation, but the $V_{REF(CA)}$ must be trained to achieve suitable receiver voltage margin for terminated, high-frequency operation. The training mode described here centers the internal $V_{REF(CA)}$ in the CA data eye and at the same time allows for timing adjustments of the CS and CA signals to meet setup/hold requirements. Because it can be difficult to capture commands prior to training the CA inputs, the training mode described here uses a minimum of external commands to enter, train, and exit the Command Bus Training mode.
Note: it is up to the system designer to determine what constitutes “low-frequency” and “high-frequency” based on the capabilities of the system. Low-frequency should then be defined as an operating frequency in which the system can reliably communicate with the SDRAM before Command Bus Training is executed.

The LPDDR4-SDRAM die has a bond-pad (ODT-CA) for multi-rank operation. In a multi-rank system, the terminating rank should be trained first, followed by the nonterminating rank(s). See the ODT section for more information.

The LPDDR4-SDRAM uses Frequency Set-Points to enable multiple operating settings for the die. The LPDDR4-SDRAM defaults to FSP-OP[0] at power-up, which has the default settings to operate in un-terminated, low-frequency environments. Prior to training, the mode register settings should be configured by setting MR13 OP[6]=1_B (FSP-WR[1]) and setting all other mode register bits for FSP-OP[1] to the desired settings for high-frequency operation. Prior to entering Command Bus Training, the SDRAM will be operating from FSP-OP[x]. Upon Command Bus Training entry when CKE is driven LOW, the LPDDR4-SDRAM will automatically switch to the alternate FSP register set (FSP-OP[y]) and use the alternate register settings during training (See note 6 in Figure 96 for more information on FSP-OP register sets). Upon training exit when CKE is driven HIGH, the LPDDR4-SDRAM will automatically switch back to the original FSP register set (FSP-OP[x]), returning to the “known-good” state that was operating prior to training. The training values for $V_{REF(CA)}$ are not retained by the DRAM in FSP-OP[y] registers, and must be written to the registers after training exit.

• To enter Command Bus Training mode, issue a MRW-1 command followed by a MRW-2 command to set MR13 OP[0]=1_B (Command Bus Training Mode Enabled).

• After time t_{MRD}, CKE may be set LOW, causing the LPDDR4-SDRAM to switch from FSP-OP[x] to FSP-OP[y], and completing the entry into Command Bus Training mode.
A status of DQS_t, DQS_c, DQ and DMI are as follows, and DQ ODT state will be followed Frequency Set Point function except output pins.
- DQS_t[0], DQS_c[0] become input pins for capturing DQ[6:0] levels by its toggling.
- DQ[5:0] become input pins for setting $V_{REF(CA)}$ Level.
- DQ[6] becomes a input pin for setting $V_{REF(CA)}$ Range.
- DQ[7] and DMI[0] become input pins and their input level is Valid level or floating, either way is fine.
- DQ[13:8] become output pins to feedback its capturing value via command bus by CS signal.
- DQS_t[1], DQS_c[1],DMI[1] and DQ[15:14] become output pins or disable, it means that SDRAM may drive to a valid level or left floating.

• At time t_{CAENT} later, LPDDR4-SDRAM can accept to change its $V_{REF(CA)}$ Range and Value using input signals of DQS_t[0], DQS_c[0] and DQ[6:0] from existing value that’s setting via MR12 OP[6:0]. The mapping between MR12 OP code and DQs is shown in Table 40. At least one V_{REFCA} setting is required before proceed to next training steps.

[Table 40] Mapping of MR12 OP Code and DQ Numbers

	Mapping						
MR12 OP Code	OP6	OP5	OP4	OP3	OP2	OP1	OP0
DQ Number	DQ6	DQ5	DQ4	DQ3	DQ2	DQ1	DQ0

• The new $V_{REF(CA)}$ value must “settle” for time t_{VREF_LONG} before attempting to latch CA information.

• To verify that the receiver has the correct $V_{REF(CA)}$ setting and to further train the CA eye relative to clock (CK), values latched at the receiver on the CA bus are asynchronously output to the DQ bus.

• To exit Command Bus Training mode, drive CKE HIGH, and after time t_{VREF_LONG} issue the MRW-1 command followed by the MRW-2 command to set MR13 OP[0]=0_B. After time t_{MRW} the LPDDR4-SDRAM is ready for normal operation. After training exit the LPDDR4-SDRAM will automatically switch back to the FSP-OP registers that were in use prior to training.

Command Bus Training may executed from IDLE or self Refresh status. When executing CBT within the Self Refresh state, the SDRAM must not be a power down state (i.e. CKE must be HIGH prior to training entry). Command Bus Training entry and exit is the same, regardless of the SDRAM state from which CBT is initiated.

3.34.1 Training Sequence For Single-rank Systems:

Note that an example shown here is assuming an initial low-frequency, no-terminating operating point, training a high-frequency, terminating operating point. The green text is low-frequency, magenta text is high-frequency. Any operating point may be trained from any known good operating point

1. Set MR13 OP[6]=1_B to enable writing to Frequency Set Point ‘y’ (FSP-WR[y]) (or FSP-OP[x], See note).
2. Write FSP-WR[y] (or FSP-WR[x]) registers for all channels to set up high-frequency operating parameters.

3. Issue MRW-1 and MRW-2 commands to enter Command Bus Training mode.
4. Drive CKE LOW, and change CK frequency to the high-frequency operating point.
5. Perform Command Bus Training ($V_{REF(CA)}$, CS, and CA).
6. Exit training, a change CK frequency to the low-frequency operating point prior to driving CKE HIGH, then issue MRW-1 and MRW-2 commands. When CKE is driven HIGH, the SDRAM will automatically switch back to the FSP-OP registers that were in use prior to training (i.e. trained values are not retained by the SDRAM).
7. Write the trained values to FSP-WR[y] (or FSP-WR[x]) by issuing MRW-1 and MRW-2 commands to the SDRAM and setting all applicable mode register parameters.
8. Issue MRW-1 and MRW-2 commands to switch to FSP-OP[y] (or FSP-OP[x]), to turn on termination, and change CK frequency to the high frequency operating point. At this point the Command Bus is trained and you may proceed to other training or normal operation.

3.34.2 Training Sequence For Multi-rank Systems:

Note that the example shown here assumes an initial low-frequency operating point, training a high-frequency operating point. The green text is low-frequency, magenta text is high-frequency. Any operating point may be trained from any known good operating point

1. Set MR13 OP[6]=1_B to enable writing to Frequency Set Point 'y' (FSP-WR[y]) (or FSP-WR[x], See note).
2. Write FSP-WR[y] (or FSP-WR[x]) registers for all channels and ranks to set up high frequency operating parameters.
3. Read MR0 OP[7] on all channels and ranks to determine which die are terminating, signified by MR0 OP[7]=1_B.
4. Issue MRW-1 and MRW-2 commands to enter Command Bus Training mode on the terminating rank.
5. Drive CKE LOW on the terminating rank (or all ranks), and change CK frequency to the high frequency operating point.
6. Perform Command Bus Training on the terminating rank ($V_{REF(CA)}$, CS, and CA).
7. Exit training by driving CKE HIGH, change CK frequency to the low-frequency operating point, and issue MRW-1 and MRW-2 commands to write the trained values to FSP-WR[y] (or FSP-WR[x]). When CKE is driven HIGH, the SDRAM will automatically switch back to the FSP-OP registers that were in use prior to training (i.e. trained values are not retained by the SDRAM).
8. Issue MRW-1 and MRW-2 command to enter training mode on the non-terminating rank (but keep CKE HIGH)
9. Issue MRW-1 and MRW-2 commands to switch the terminating rank to FSP-OP[y] (or FSP-OP[x]), to turn on termination, and change CK frequency to the high frequency operating point.
10. Drive CKE LOW on the non-terminating (or all) ranks. The non-terminating rank(s) will now be using FSP-OP[y] (or FSP-OP[x]).
11. Perform Command Bus Training on the non-terminating rank ($V_{REF(CA)}$, CS, and CA).
12. Issue MRW-1 and MRW-2 commands to switch the terminating rank to FSP-OP[x] (or FSP-OP[y]) to turn off termination.
13. Exit training by driving CKE HIGH on the non-terminating rank, change CK frequency to the low frequency operating point, and issue MRW-1 and MRW-2 commands. When CKE is driven HIGH, the SDRAM will automatically switch back to the FSP-OP registers that were in use prior to training (i.e. trained values are not retained by the SDRAM).
14. Write the trained values to FSP-WR[y] (or FSP-WR[x]) by issuing MRW-1 and MRW- 2 commands to the SDRAM and setting all applicable mode register parameters.
15. Issue MRW-1 and MRW-2 commands to switch the terminating rank to FSP-OP[y] (or FSP-OP[x]), to turn on termination, and change CK frequency to the high frequency operating point. At this point the Command Bus is trained for both ranks and you may proceed to other training or normal operation.

3.34.3 Relation Between CA Input Pin DQ Output Pin

The relation between CA input pin DQ output pin is shown in Table 41.

[Table 41] Mapping of CA Input pin and DQ Output pin

	Mapping					
CA Number	CA5	CA4	CA3	CA2	CA1	CA0
DQ Number	DQ13	DQ12	DQ11	DQ10	DQ9	DQ8

3.34.4 Timing Diagram

The basic Timing diagrams of Command Bus Training are shown in following figures.

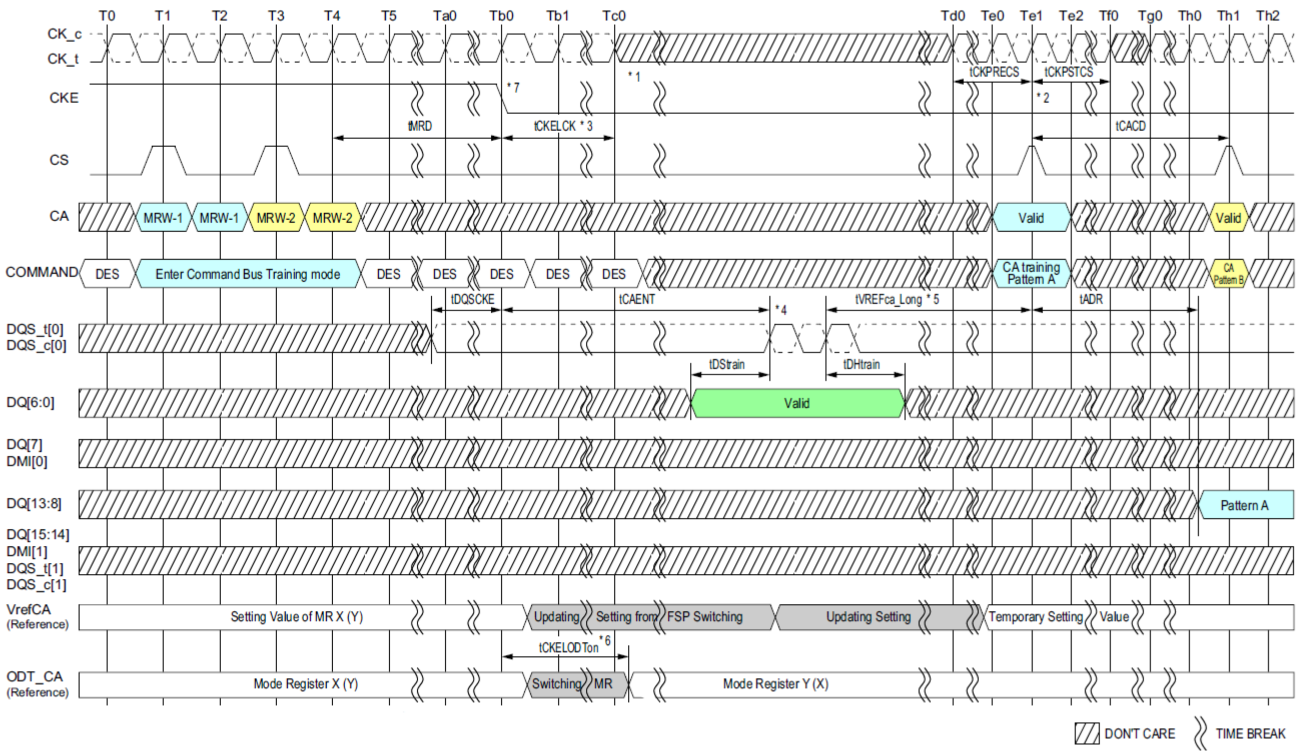


Figure 96: Entering Command Bus Training Mode and CA Training Pattern Input and Output with V_{REFCA} Value Update

NOTE :
1) After t_{CKELCK} clock can be stopped or frequency changed any time.
2) The input clock condition should be satisfied $t_{CKPRECS}$ and $t_{CKPSTCS}$.
3) Continue to Drive CK and Hold CA pins low until t_{CKELCK} after CKE is low (which disables command decoding).
4) DRAM may or may not capture first rising/falling edge of DQS_t/c due to an unstable first rising edge. Hence provide at least consecutive 2 pulses of DQS signal input is required in every DQS input signal at capturing DQ6:0 signals.
The captured value of DQ6:0 signal level by each DQS edges are overwritten at any time and the DRAM updates its VREFca setting of MR12 temporary after time t_{VREFca_Long}.
5) t_{VREF_LONG} may be reduced to t_{VREF_SHORT} if the following conditions are met: 1) The new Vref setting is a single step above or below the old Vref setting, and 2) The DQS pulses a single time, or the new Vref setting value on DQ[6:0] is static and meets $t_{DSTRAIN}/t_{DHTRAIN}$ for every DQS pulse applied.
6) When CKE is driven LOW, the SDRAM will switch its FSP-OP registers to use the alternate (i.e. non-active) set. Example: If the SDRAM is currently using FSP-OP[0], then it will switch to FSP-OP[1] when CKE is driven LOW. All operating parameters should be written to the alternate mode registers before entering Command Bus Training to ensure that ODT settings, RL/WL/nWR setting, etc., are set to the correct values. If the alternate FSP-OP has ODT_CA disabled then termination will not enable in CA Bus Training mode. If the ODT_CA pad is bonded to Vss, ODT_CA termination will never enable for that die.
7) When CKE is driven low in Command Bus Training mode, the LPDDR4-SDRAM will change operation to the alternate FSP, i.e. the inverse of the FSP programmed in the FSP-OP mode register.

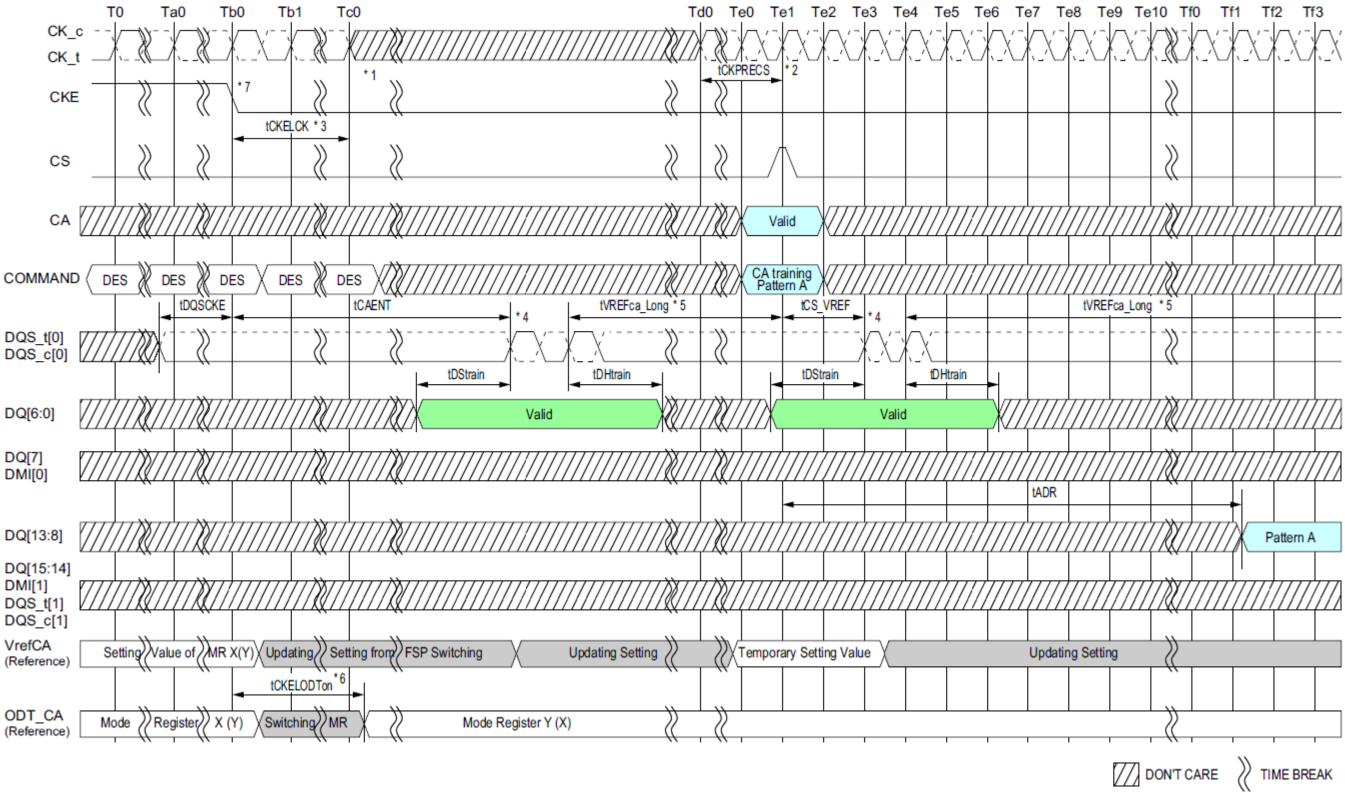


Figure 97: Consecutive V_{REFCA} Value Update

NOTE :

1) After t_{CKELCK} clock can be stopped or frequency changed any time.
2) The input clock condition should be satisfied $t_{CKPRECS}$.
3) Continue to Drive CK and Hold CA pins low until t_{CKELCK} after CKE is low (which disables command decoding).
4) DRAM may or may not capture first rising/falling edge of DQS_t/c due to an unstable first rising edge. Hence provide at least consecutive 2 pulses of DQS signal input is required in every DQS input signal at capturing DQ6:0 signals.
The captured value of DQ6:0 signal level by each DQS edges are overwritten at any time and the DRAM updates its V_{REFca} setting of MR12 temporary after time t_{VREFca_Long}.
5) t_{VREF_LONG} may be reduced to t_{VREF_SHORT} if the following conditions are met: 1) The new Vref setting is a single step above or below the old Vref setting, and 2) The DQS pulses a single time, or the new Vref setting value on DQ[6:0] is static and meets $t_{DSTRAIN}/t_{DHTRAIN}$ for every DQS pulse applied.
6) When CKE is driven LOW, the SDRAM will switch its FSP-OP registers to use the alternate (i.e. non-active) set. Example: If the SDRAM is currently using FSP-OP[0], then it will switch to FSP-OP[1] when CKE is driven LOW. All operating parameters should be written to the alternate mode registers before entering Command Bus Training to ensure that ODT settings, RL/WL/nWR setting, etc., are set to the correct values. If the alternate FSP-OP has ODT_CA disabled then termination will not enable in CA Bus Training mode. If the ODT_CA pad is bonded to Vss, ODT_CA termination will never enable for that die.
7) When CKE is driven low in Command Bus Training mode, the LPDDR4-SDRAM will change operation to the alternate FSP, i.e. the inverse of the FSP programmed in the FSP-OP mode register.

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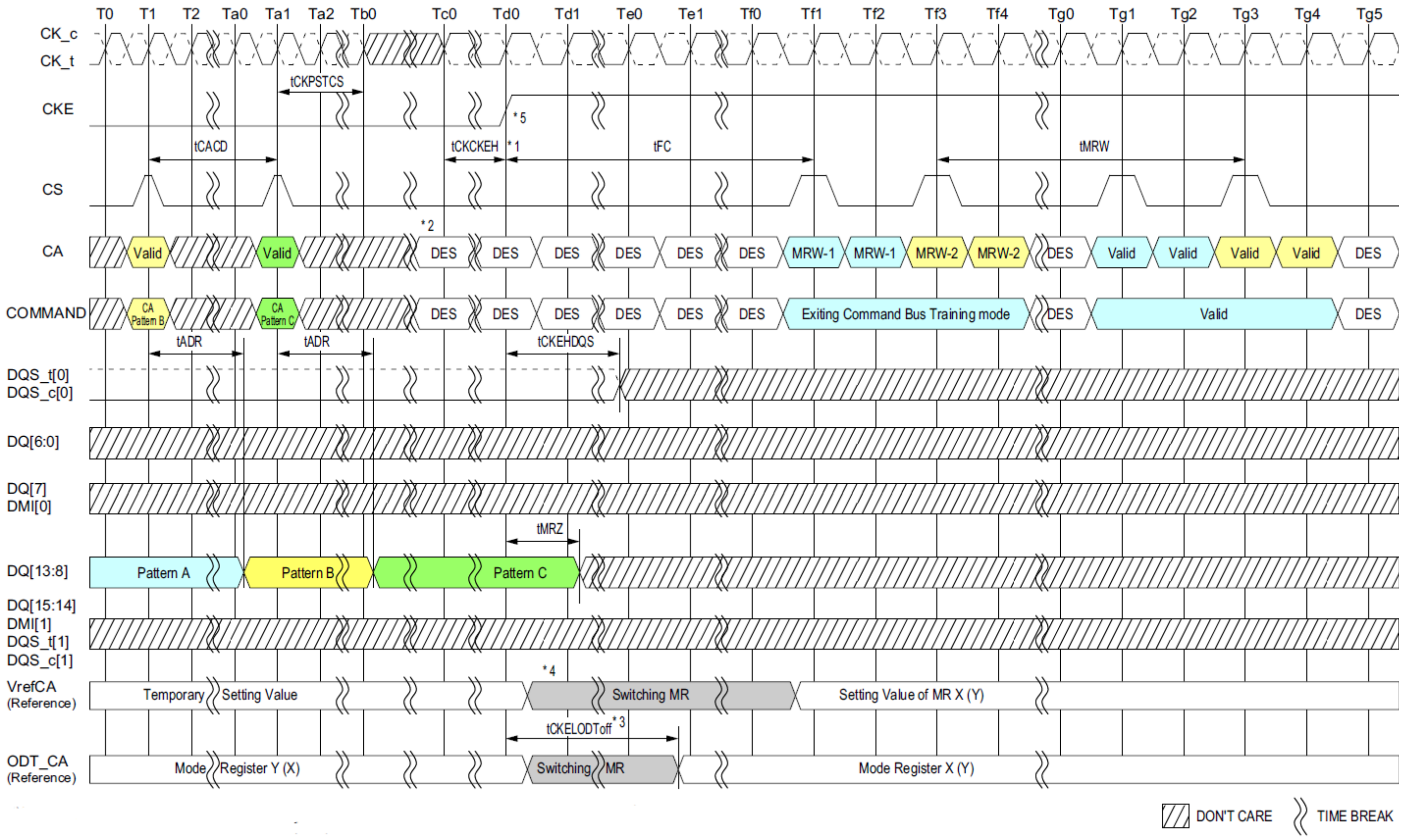

Figure 98: Exiting Command Bus Training Mode with Valid Command

NOTE :

1) Clock can be stopped or frequency changed any time before t_{CKCKEH}. CK must meet t_{CKCKEH} before CKE is driven high.
When CKE is driven high the clock frequency must be returned to the original frequency (the frequency corresponding to the FSP at which Command Bus Training mode was entered)
2) CS must be Deselect (low) t_{CKCKEH} before CKE is driven high.
3) When CKE is driven high, the SDRAM's ODT_CA will revert to the state/value defined by FSP-OP prior to Command Bus Training mode entry, i.e. the original frequency set point (FSP-OP, MR13-OP[7]).
Example: If the SDRAM was using FSP-OP[1] for training, then it will switch to FSP-OP[0] when CKE is driven HIGH.
4) Training values are not retained by the SDRAM, and must be written to the FSP-OP register set before returning to operation at the trained frequency.
Example: VREF(ca) will return to the value programmed in the original set point.
5) When CKE is driven high the LPDDR4-SDRAM will revert to the FSP in operation when Command Bus Training mode was entered.

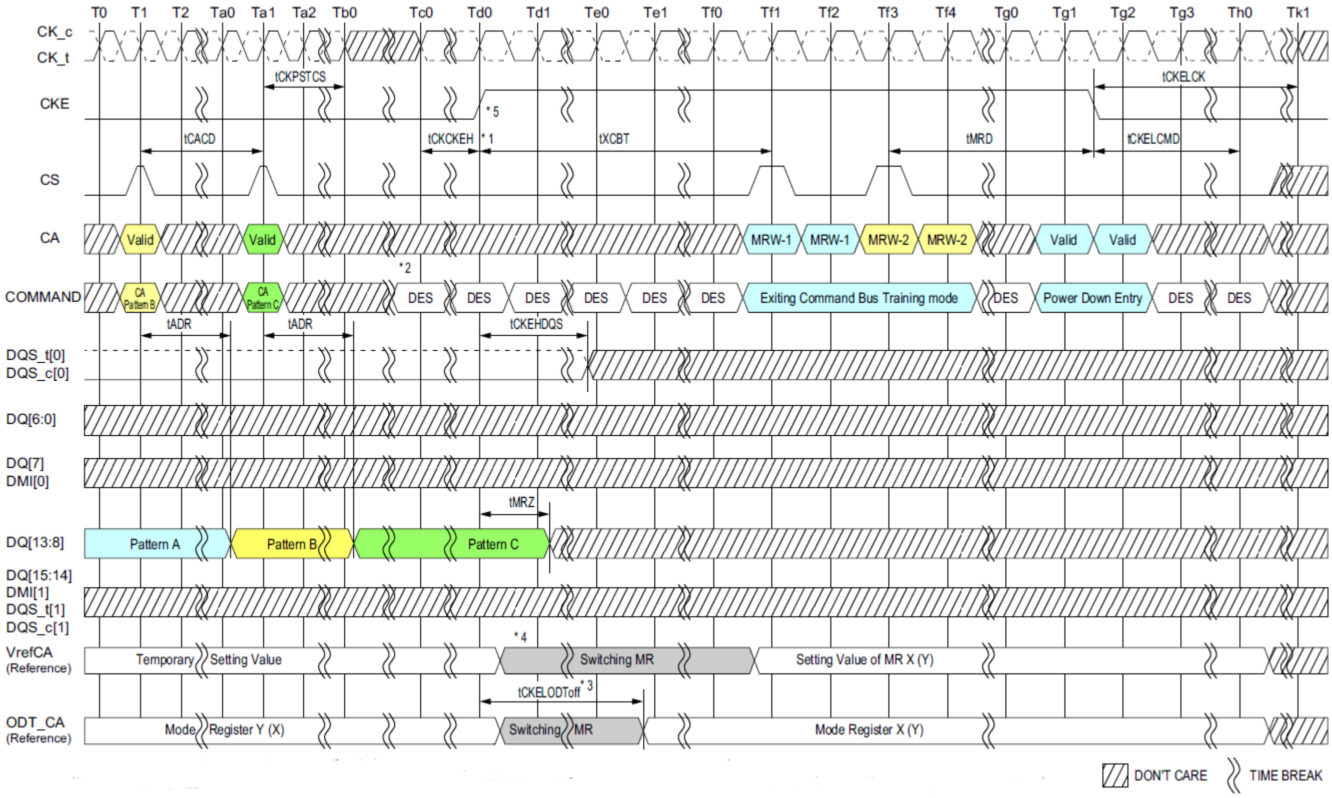

Figure 99: Exiting Command Bus Training Mode with Power Down Entry

NOTE :

1) Clock can be stopped or frequency changed any time before t_{CKCKEH}. CK must meet t_{CKCKEH} before CKE is driven high.
When CKE is driven high the clock frequency must be returned to the original frequency (the frequency corresponding to the FSP at which Command Bus Training mode was entered)

2) CS must be Deselect (low) t_{CKCKEH} before CKE is driven high.

3) When CKE is driven high, the SDRAM's ODT_CA will revert to the state/value defined by FSP-OP prior to Command Bus Training mode entry, i.e. the original frequency set point (FSP-OP, MR13-OP[7]).
Example: If the SDRAM was using FSP-OP[1] for training, then it will switch to FSP-OP[0] when CKE is driven HIGH.

4) Training values are not retained by the SDRAM, and must be written to the FSP-OP register set before returning to operation at the trained frequency.
Example: VREF(ca) will return to the value programmed in the original set point.

5) When CKE is driven high the LPDDR4-SDRAM will revert to the FSP in operation when Command Bus Training mode was entered.

3.34.5 Command Bus Training AC Timing Table

[Table 42] Command Bus Training AC Timing

Parameter	Symbol	Min/Max	Value	Units	Notes
Valid Clock Requirement after CKE Input low	t_{CKELCK}	Min	Max(5ns, 5nCK)	-	
Valid Clock Requirement before CKE Input High	t_{CKCKEH}	Min	Max(1.75ns, 3tCK)	-	
Data Setup for Vref Training Mode	$t_{DStrain}$	Min	2	ns	
Data Hold for Vref Training Mode	$t_{DHtrain}$	Min	2	ns	
Asynchronous Data Read	t_{ADR}	Max	20	ns	
Valid Strobe Requirement before CKE Low	t_{DQSCKE}	Min	10	ns	1
CA Bus Training Command to CA Bus Training Command Delay	t_{CACD}	Min	RU(t_{ADR}/t_{CK})	t_{CK}	2
First CA Bus Training Command Following CKE Low	t_{CAENT}	Min	250	ns	
CA Bus Training CKE High to DQ Tri-state	t_{MRZ}	Min	1.5	ns	
Vref Step Time - multiple steps	t_{VREFCA_LONG}	Max	250	ns	
Vref Step Time - one step	t_{VREFCA_SHORT}	Max	80	ns	
Valid Clock Requirement before CS High	$t_{CKPRECS}$	Min	2tck + t_{XP} (t_{XP} = Max(7.5ns, 5tCK))	-	
Valid Clock Requirement after CS High	$t_{CKPSTCS}$	Min	Max(7.5ns, 5tCK)	-	
Minimum delay from CKE High to Strobe High Impedance	$t_{CKEHDQS}$		10	ns	
Minimum Delay from CS to DQS toggle in command bus training	t_{CS_VREF}	Min	2	t_{CK}	
ODT turn-on Latency from CKE	$t_{CKELODTon}$	Min	20	ns	
ODT turn-off Latency from CKE	$t_{CKELODToff}$	Min	20	ns	
Exit Command Bus Training Mode to next valid command delay	t_{XCBT_Short}	MIN	Max(5nCK, 200ns)	-	3
	t_{XCBT_Middle}	MIN	Max(5nCK, 200ns)	-	3
	t_{XCBT_Long}	MIN	Max(5nCK, 250ns)	-	3

NOTE :
1) DQS_t has to retain a low level during t_{DQSCKE} period, as well as DQS_c has to retain a high level.
2) If t_{CACD} is violated, the data for samples which violate t_{CACD} will not be available, except for the last sample (where t_{CACD} after this sample is met). Valid data for the last sample will be available after t_{ADR}.
3) Exit Command Bus Training Mode to next valid command delay Time depends on value of V_{REF}(CA) setting: MR12 OP[5:0] and V_{REF}(CA) Range: MR12 OP[6] of FSP-OP 0 and 1. The details are shown in tFC value mapping table. Additionally exit Command Bus Training Mode to next valid command delay Time may affect V_{REF}(DQ) setting. Settling time of V_{REF}(DQ) level is same as V_{REF}(CA) level.

3.35 Frequency Set Point

Frequency Set-Points allow the LPDDR4-SDRAM CA Bus to be switched between two differing operating frequencies, with changes in voltage swings and termination values, without ever being in an un-trained state which could result in a loss of communication to the DRAM. This is accomplished by duplicating all CA Bus mode register parameters, as well as other mode register parameters commonly changed with operating frequency. These duplicated registers form two sets that use the same mode register addresses, with read/write access controlled by MR bit FSP-WR (Frequency Set-Point Write/Read) and the DRAM operating point controlled by another MR bit FSP-OP (Frequency Set-Point Operation). Changing the FSP-WR bit allows MR parameters to be changed for an alternate Frequency Set-Point without affecting the LPDDR4- SDRAM's current operation. Once all necessary parameters have been written to the alternate Set-Point, changing the FSP-OP bit will switch operation to use all of the new parameters simultaneously (within t_{FC}), eliminating the possibility of a loss of communication that could be caused by a partial configuration change.

Parameters which have two physical registers controlled by FSP-WR and FSP-OP are shown in the following table with the exception as outlined in note 1.

[Table 43] Mode Register Function with two physical registers

MR#	Operand	Function	Note
MR1	OP[2]	WR-PRE (WR Pre-amble Length)	1
	OP[3]	RD-PRE (RD Pre-amble Type)	
	OP[6:4]	nWR (Write-Recovery for Auto-Precharge commands)	
	OP[7]	PST (RD Post-Amble Length)	
MR2	OP[2:0]	RL (Read latency)	
	OP[5:3]	WL (Write latency)	
	OP[6]	WLS (Write Latency Set)	
MR3	OP[0]	PU-Cal (Pull-up Calibration Point)	2
	OP[1]	WR PST(WR Post-Amble Length)	
	OP[5:3]	PDDS (Pull-Down Drive Strength)	
	OP[6]	DBI-RD (DBI-Read Enable)	
	OP[7]	DBI-WR (DBI-Write Enable)	
MR11	OP[2:0]	DQ ODT (DQ Bus Receiver On-Die-Termination)	
	OP[6:4]	CA ODT (CA Bus Receiver On-Die-Termination)	
MR12	OP[5:0]	VREF(ca) (VREF(ca) Setting)	
	OP[6]	VR-CA (VREF(ca) Range)	
MR14	OP[5:0]	VREF(dq) (VREF(dq) Setting)	
	OP[6]	VR(dq) (VREF(dq) Range)	
MR22	OP[2:0]	SoC ODT (Controller ODT Value for VOH calibration)	
	OP[3]	ODTE-CK (CK ODT enabled for nonterminating rank)	
	OP[4]	ODTE-CS (CS ODT enable for non terminating rank)	
	OP[5]	ODTD-CA (CA ODT termination disable)	

NOTE:
1) Supporting the two physical registers for Burst Length: MR1 OP[1:0] is optional. Applications requiring support of both vendor options shall assure that both FSP-OP[0] and FSPOP[1] are set to the same code. Refer to vendor datasheets for detail.
2) For dual channel devices, PU-CAL setting is required as the same value for both Ch.A and Ch.B before issuing ZQ Cal start command. See Mode Register Definition for more details.

Table 44 shows how the two mode registers for each of the parameters above can be modified by setting the appropriate FSP-WR value, and how device operation can be switched between operating points by setting the appropriate FSP-OP value. The FSP-WR and FSP-OP functions operate completely independently.

[Table 44] Relation between MR Setting and DRAM Operation

Function	MR# & Operand	Data	Operation	Note
FSP-WR	MR13 OP[6]	0 (Default)	Data write to Mode Register N for FSP-OP[0] by MRW Command Data read from Mode Register N for FSP-OP[0] by MRR Command	1
		1	Data write to Mode Register N for FSP-OP[1] by MRW Command Data read from Mode Register N for FSP-OP[1] by MRR Command	
FSP-OP	MR13 OP[7]	0 (Default)	DRAM operates with Mode Register N for FSP-OP[0] setting.	2
		1	DRAM operates with Mode Register N for FSP-OP[1] setting.	

NOTE :
1) FSP-WR stands for Frequency Set Point Write/Read.
2) FSP-OP stands for Frequency Set Point Operating Point.

3.36 Frequency Set Point Update Timing

The Frequency set point update timing is shown in Figure 100. When changing the frequency set point via MR13 OP[7], the VRCG setting: MR13 OP[3] have to be changed into VREF Fast Response (high current) mode at the same time. After Frequency change time(tFC) is satisfied. VRCG can be changed into Normal Operation mode via MR13 OP[3].

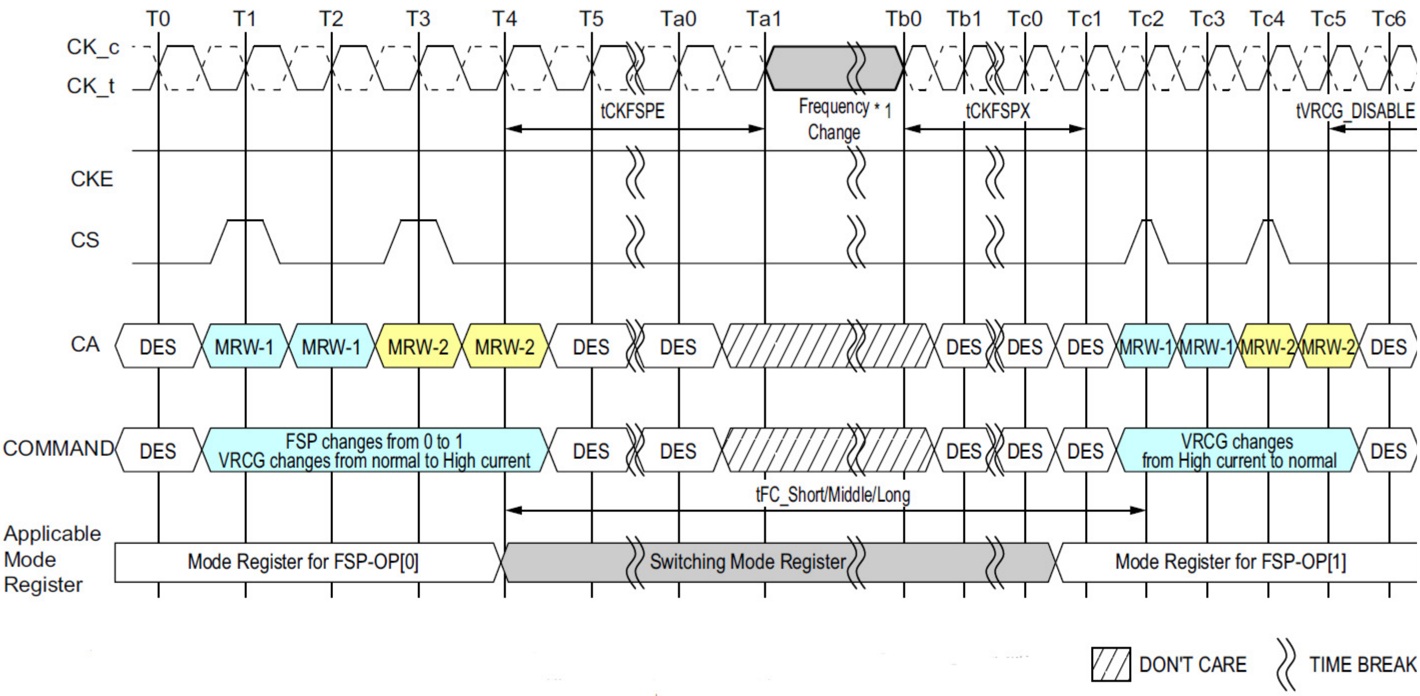

Figure 100: Frequency Set Point Switching Timing

NOTE :
1) The definition that is Clock frequency change during CKE HIGH should be followed at the frequency change operation.
For more information, refer to Section Input Clock Stop and Frequency Change.

[Table 45] Frequency Set Point AC Timing Table

Parameter	Symbol	Min/Max	Data Rate	Unit	Note
Frequency Set Point parameters					
Frequency Set Point Switching Time	t_{FC_short}	MIN	200	ns	1
	t_{FC_middle}	MIN	200	ns	1
	t_{FC_long}	MIN	250	ns	1
Valid Clock Requirement after Entering FSP Change	$t_{CKEFSPE}$	MIN	max(7.5ns, 4tCK)	-	
Valid Clock Requirement before 1st Valid Command after FSP change	t_{CKFSPX}	MIN	max(7.5ns, 4tCK)	-	

NOTE :
1) Frequency Set Point Switching Time depends on value of $V_{REF(CA)}$ setting: MR12 OP[5:0] and $V_{REF(CA)}$ Range: MR12 OP[6] of FSP-OP 0 and 1. The details are shown in Table 46. Additionally change of Frequency Set Point may affect $V_{REF(DQ)}$ setting. Settling time of $V_{REF(DQ)}$ level is same as $V_{REF(CA)}$ level.

[Table 46] tFC value mapping

Application	Step Size		Range	
	From FSP-OP0	To FSP-OP1	From FSP-OP0	To FSP-OP1
t_{FC_Short}	Base	A single step size increment/decrement	Base	No Change
t_{FC_Middle}	Base	Two or more step size increment/decrement	Base	No Change
t_{FC_Long}	-	-	Base	Change

NOTE :
1) As well as change from FSP-OP1 to FSP-OP0.

Table 47 provides an example of tFC value mapping when FSP-OP moves from OP0 to OP1.

[Table 47] tFC value mapping example

Case	From/To	FSP-OP: MR13 OP[7]	VREF(CA) Setting: MR12 OP[5:0]	VREF(CA) Range: MR12 OP[6]	Application	Note
1	From	0	001100	0	tRF_Short	1
	To	1	001101	0		
2	From	0	001100	0	tRF_Middle	2
	To	1	001110	0		
3	From	0	Don't Care	0	tRF_Long	3
	To	1	Don't Care	1		

NOTE :
1) A single step size increment/decrement for $V_{REF(CA)}$ Setting Value.
2) Two or more step size increment/decrement for $V_{REF(CA)}$ Setting Value.
3) $V_{REF(CA)}$ Range is changed. In this case changing $V_{REF(CA)}$ Setting doesn't affect t_{FC} value.

The LPDDR4-SDRAM defaults to FSP-OP[0] at power-up. Both Set-Points default to settings needed to operate in un-terminated, low-frequency environments. To enable the LPDDR4-SDRAM to operate at higher frequencies, Command Bus Training mode should be utilized to train the alternate Frequency Set-Point (Figure 101). See the section Command Bus Training for more details on this training mode.

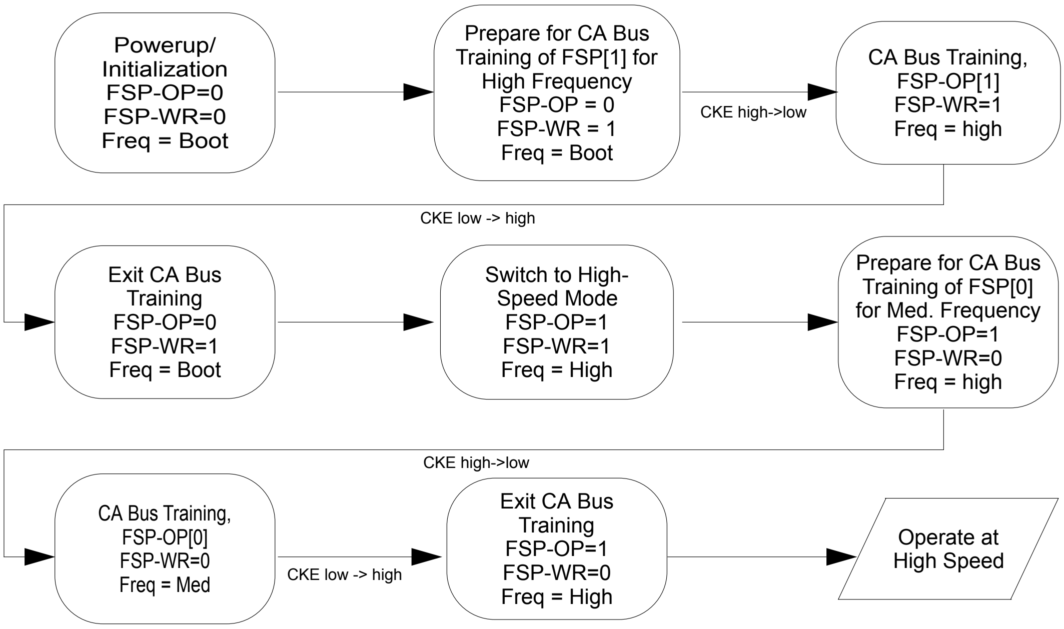

Figure 101: Training Two Frequency Set-Points

Once both Frequency Set-Points have been trained, switching between points can be performed by a single MRW followed by waiting for t_{FC} (Figure 102).

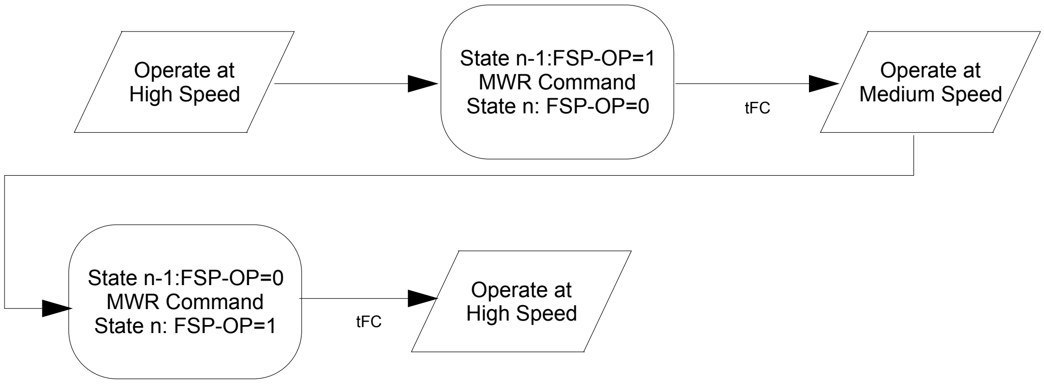

Figure 102: Switching Between Two Trained Frequency Set-Points

Switching to a third (or more) Set-Point can be accomplished if the memory controller has stored the previously-trained values (in particular the Vref-CA calibration value) and re-writes these to the alternate Set-Point before switching FSP-OP (Figure 103).

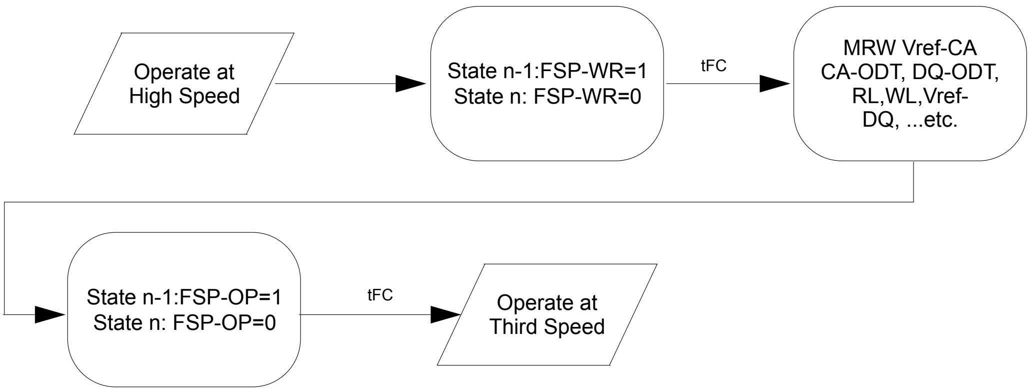

Figure 103: Switching to a Third Trained Frequency Set-Point

3.37 Mode Register WRITE-WR Leveling Mode

To improve signal-integrity performance, the LPDDR4 SDRAM provides a write-leveling feature to compensate CK-to-DQS timing skew affecting timing parameters such as t_{DQSS}, t_{DSS}, and t_{DSH}. The
DRAM samples the clock state with the rising edge of DQS signals, and asynchronously feeds back to the memory controller. The memory controller references this feedback to adjust the clock-to-data strobe signal relationship for each DQS_t/DQS_c signal pair.

All data bits (DQ[7:0] for DQS_t/DQS_c[0], and DQ[15:8] for DQS_t/DQS_c[1]) carry the training feedback to the controller. Both DQS signals in each channel must be leveled independently. Write-leveling entry/exit is independent between channels for dual channel devices.

The LPDDR4 SDRAM enters into write-leveling mode when mode register MR2-OP[7] is set HIGH. When entering write-leveling mode, the state of the DQ pins is undefined. During write-leveling mode, only DESELECT commands are allowed, or a MRW command to exit the write-leveling operation. Depending on the absolute values of t_{DQSL} and t_{DQSH} in the application, the value of t_{DQSS} may have to be better than the limits provided in the Write AC Timing Table in order to satisfy the t_{DSS} and t_{DSH} specifications. Upon completion of the write-leveling operation, the DRAM exits from write-leveling mode when MR2-OP[7] is reset LOW.

Write Leveling should be performed before Write Training (DQS2DQ Training).

3.37.1 Write Leveling Procedure:

1. Enter into Write-leveling mode by setting MR2-OP[7]=1_B.
2. Once entered into Write-leveling mode, DQS_t must be driven LOW and DQS_c HIGH after a delay of $t_{WLDQSEN}$.
3. Wait for a time t_{WLMRD} before providing the first DQS signal input. The delay time $t_{WLMRD(MAX)}$ is controller dependent.
4. DRAM may or may not capture first rising edge of DQS_t due to an unstable first rising edge. Hence provide at least consecutive 2 pulses of DQS signal input is required in every DQS input signal during Write Training Mode. The captured clock level by each DQS edges are overwritten at any time and the DRAM provides asynchronous feedback on all the DQ bits after time t_{WLO}.
5. The feedback provided by the DRAM is referenced by the controller to increment or decrement the DQS_t and/or DQS_c delay settings.
6. Repeat step 4 through step 5 until the proper DQS_t/DQS_c delay is established.
7. Exit from Write-leveling mode by setting MR2-OP[7]=0_B.

A Write Leveling timing example is shown in figure below.

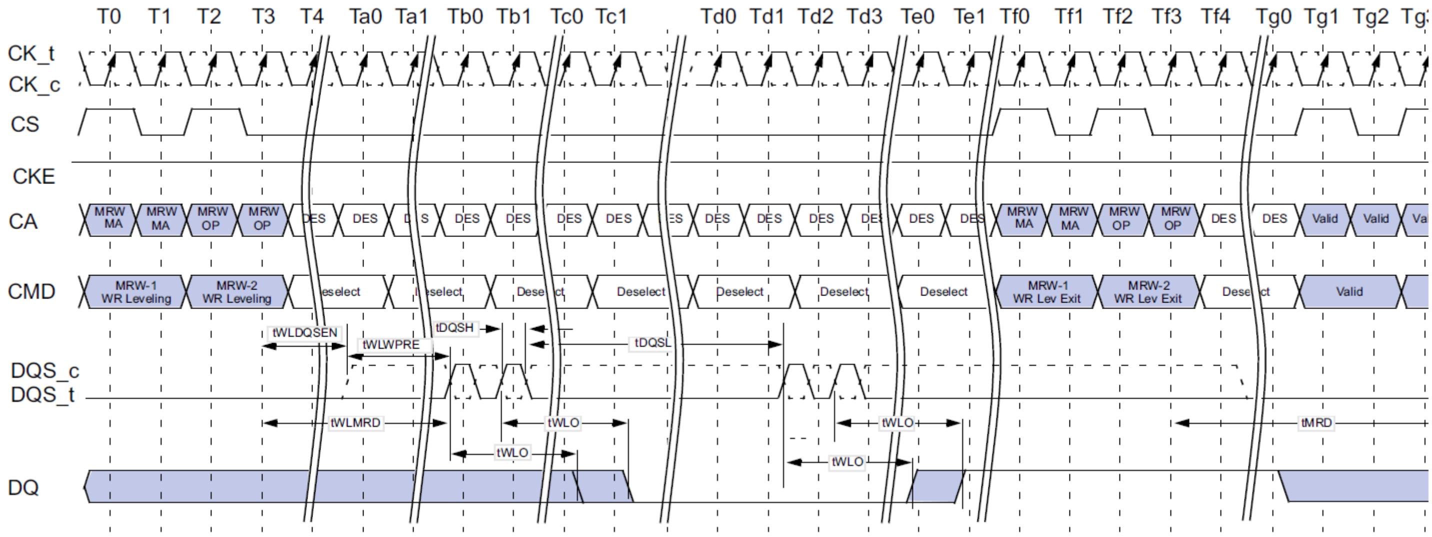

Figure 104: Write Leveling Timing, $t_{DQSL(max)}$

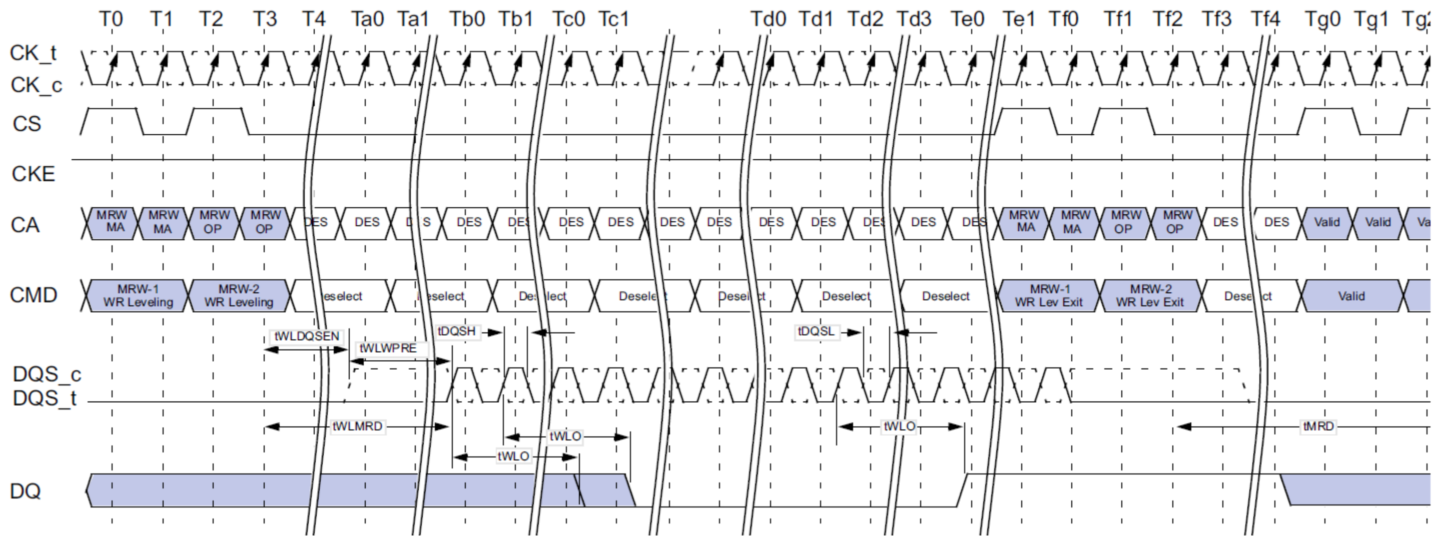

Figure 105: Write Leveling Timing, $t_{DQSL(min)}$

3.37.2 Input Clock Frequency Stop And Change

The input clock frequency can be stopped or changed from one stable clock rate to another stable clock rate during Write Leveling mode. The Frequency stop or change timing is shown in Figure 106.

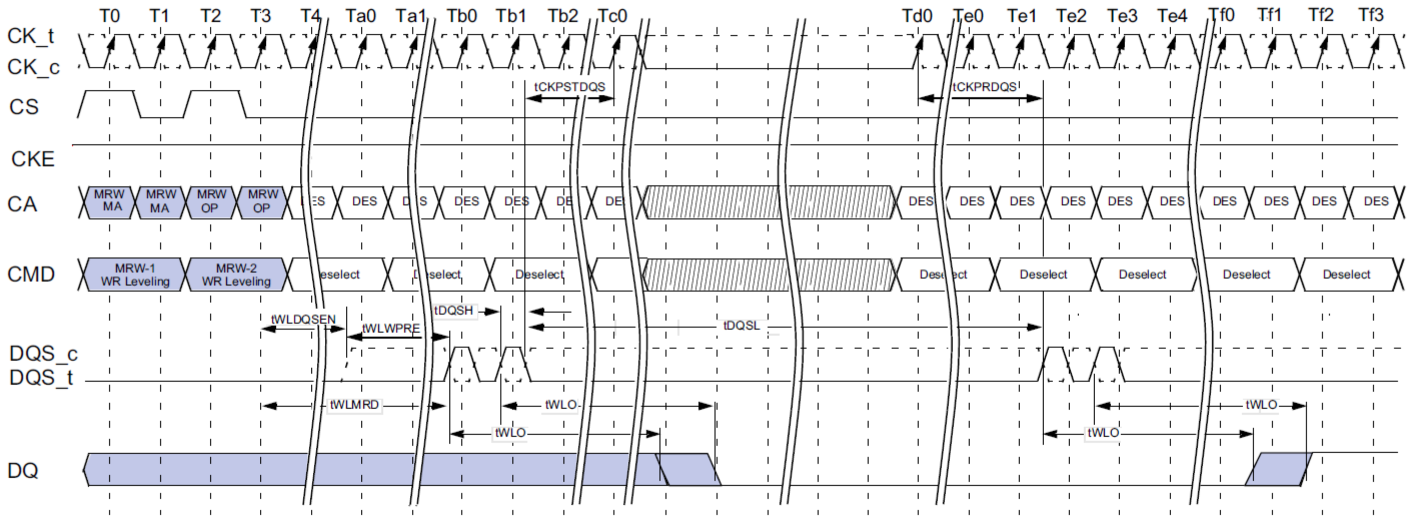

Figure 106: Clock Stop and Timing during Write Leveling

NOTE :
1) CK_t is held LOW and CK_c is held HIGH during clock stop.
2) CS shall be held LOW during clock stop.

[Table 48] Write Leveling Timing Parameters

Parameter	Symbol	Min/Max	Value	Units	Notes
DQS_t/DQS_c delay after write leveling mode is programmed	$t_{WLDQSEN}$	Min	20	t_{CK}	
		Max	-		
Write preamble for Write Leveling	t_{WLWPRE}	Min	20	t_{CK}	
		Max	-		
First DQS_t/DQS_c edge after write leveling mode is programmed	t_{WLMRD}	Min	40	t_{CK}	
		Max	-		
Write leveling output delay	t_{WLO}	Min	0	ns	
		Max	20		
Mode register set command delay	t_{MRD}	Min	max(14ns, 10nCK)	ns	
		Max	-		
Valid Clock Requirement before DQS Toggle	$t_{CKPRDQS}$	Min	max(7.5ns, 4nCK)	-	
		Max	-		
Valid Clock Requirement after DQS Toggle	$t_{CKPSTDQS}$	Min	max(7.5ns, 4nCK)	-	
		Max	-		

3.37.3 Write Leveling Setup And Hold Time

[Table 49] Write Leveling Setup and Hold Time

Parameter	Symbol	Min/ Max	Data Rate	Unit
			3733	
Write Leveling Parameters				
Write leveling hold time	t_{WLH}	MIN	60	ps
Write leveling setup time	t_{WLS}	MIN	60	ps
Write leveling input valid window	t_{WLIVW}	MIN	100	ps

NOTE :
1) In addition to the traditional setup and hold time specifications above, there is value in a input valid window based specification for write-leveling training. As the training is based on each device, worst case process skews for setup and hold do not make sense to close timing between CK and DQS.
2) t_{WLIVW} is defined in a similar manner to t_{dIVW_Total}, except that here it is a DQS input valid window with respect to CK. This would need to account for all VT (voltage and temperature) drift terms between CK and DQS within the DRAM that affect the write-leveling input valid window.

The DQS input mask for timing with respect to CK is shown in Figure 107. The "total" mask (t_{WLIVW}) defines the time the input signal must not encroach in order for the DQS input to be successfully captured by CK with a BER of lower than tbd. The mask is a receiver property and it is not the valid data-eye.

DQS_t/DQS_c and CK_t/CK_c at DRAM Latch

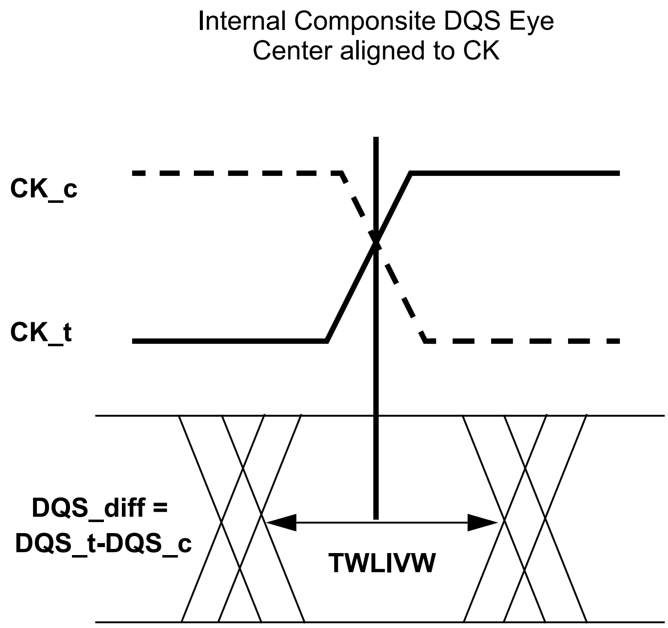

Figure 107: DQS_t/DQS_c to CK_t/CK_c timings at the DRAM pins referenced from the internal latch

3.38 RD DQ Calibration

LPDDR4 devices feature a RD DQ Calibration Training function that outputs a 16-bit user-defined pattern on the DQ pins. RD DQ Calibration is initiated by issuing a MPC-1 [RD DQ Calibration] command followed by a CAS-2 command, cause the LPDDR4-SDRAM to drive the contents of MR32 followed by the contents of MR40 on each of DQ[15:0] and DMI[1:0]. The pattern can be inverted on selected DQ pins according to user-defined invert masks written to MR15 and MR20.

3.38.1 RD DQ Calibration Training Procedure

The procedure for executing RD DQ Calibration is:

- Issue MRW commands to write MR32 (first eight bits), MR40 (second eight bits), MR15 (eight-bit invert mask for byte 0), and MR20 (eight-bit invert mask for byte 1)
  - * Optionally this step could be skipped to use the default patterns
    MR32 default = $5A_H$
    MR40 default = $3C_H$
    MR15 default = 55_H
    MR20 default = 55_H
- Issue an MPC-1 [RD DQ Calibration] command followed immediately by a CAS-2 command

* Each time an MPC-1 [RD DQ Calibration] command followed by a CAS-2 is received by the LPDDR4-SDRAM, a 16-bit data burst will, after the currently set RL, drive the eight bits programmed in MR32 followed by the eight bits programmed in MR40 on all I/O pins
* The data pattern will be inverted for I/O pins with a '1' programmed in the corresponding invert mask mode register bit (see Table)
* Note that the pattern is driven on the DMI pins, but no data bus inversion function is enabled, even if Read DBI is enabled in the DRAM mode register.
* The MPC-1 [RD DQ Calibration] command can be issued every t_{CCD} seamlessly, and t_{RTRRD} delay is required between array Read command and the MPC-1 [RD DQ Calibration] command as well the delay required between the MPC-1 [RD DQ Calibration] command and an array read.

- The operands received with the CAS-2 command must be driven LOW
- DQ Read Training can be performed with any or no banks active, during Refresh, or during SREF with CKE high.

[Table 50] Invert Mask Assignments

DQ Pin	0	1	2	3	DMI0	4	5	6	7
MR15 bit	0	1	2	3	N/A	4	5	6	7

DQ Pin	8	9	10	11	DMI1	12	13	14	15
MR20 bit	0	1	2	3	N/A	4	5	6	7

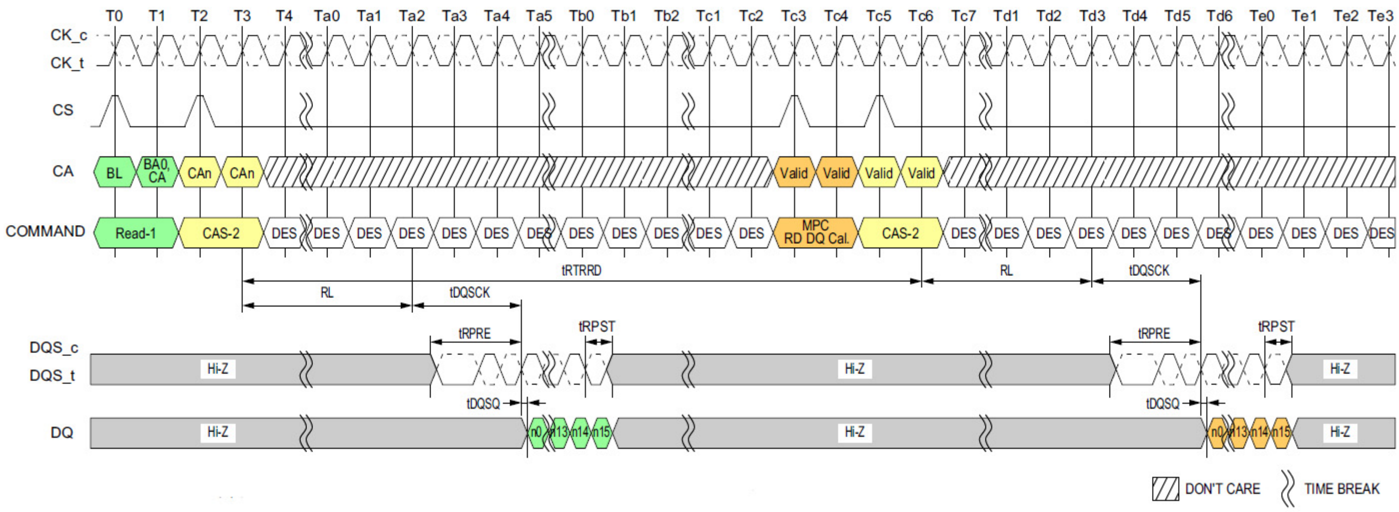

Figure 108: DQ Read Training Timing: Read to Read DQ Calibration

NOTE :
1) Read-1 to MPC [RD DQ Calibration] Operation is shown as an example of command-to-command timing.
Timing from Read-1 to MPC [RD DQ Calibration] command is t_{RTRRD}.
2) MPC [RD DQ Calibration] uses the same command-to-data timing relationship (RL, t_{DQSCK}, t_{DQSQ}) as a Read-1 command.
3) BL = 16, Read Preamble: Toggle, Read Postamble: 0.5nCK.
4) DES commands are shown for ease of illustration; other commands may be valid at these times.

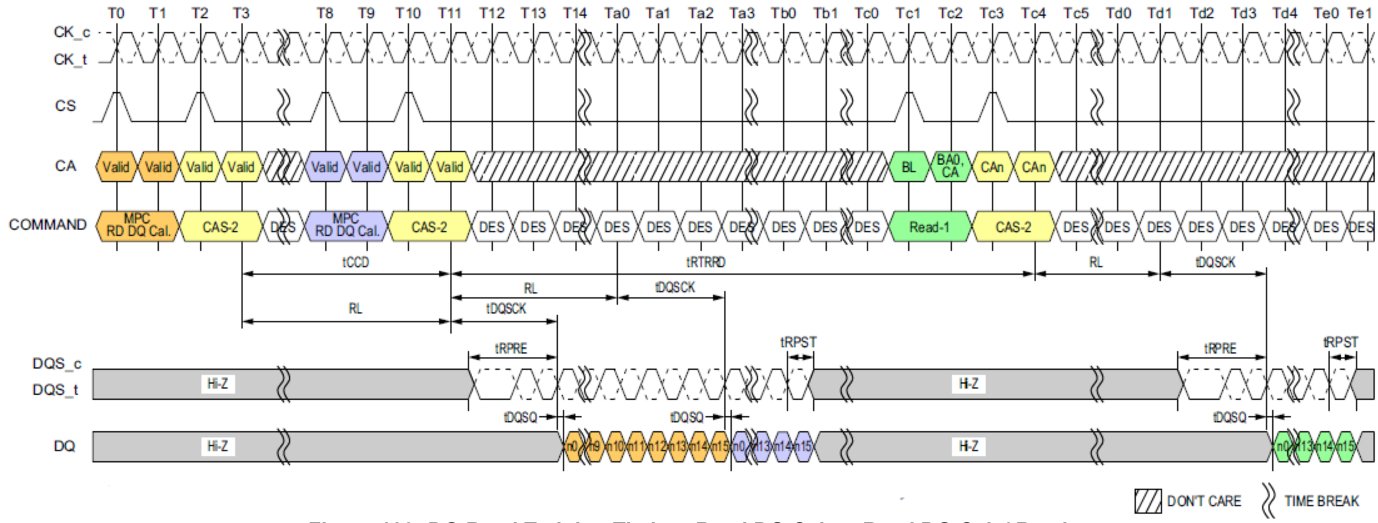

Figure 109: DQ Read Training Timing: Read DQ Cal. to Read DQ Cal. / Read

NOTE :
1) MPC [RD DQ Calibration] to MPC [RD DQ Calibration] Operation is shown as an example of command-to-command timing.
2) MPC [RD DQ Calibration] to Read-1 Operation is shown as an example of command-to-command timing.
3) MPC [RD DQ Calibration] uses the same command-to-data timing relationship (RL, t_{DQSCK}, t_{DQSQ}) as a Read-1 command.
4) Seamless MPC [RD DQ Calibration] commands may be executed by repeating the command every t_{CCD} time.
5) Timing from MPC [RD DQ Calibration] command to Read-1 is t_{RTRRD}.
6) BL = 16, Read Preamble: Toggle, Read Postamble: 0.5nCK.
7) DES commands are shown for ease of illustration; other commands may be valid at these times.

3.38.2 DQ Read Training Example

An example of DQ Read Training output is shown in Table 51. This shows the 16-bit data pattern that will be driven on each DQ in byte 0 when one DQ Read Training command is executed. This output assumes the following mode register values are used:

- MR32 = $1C_H$
- MR40 = 59_H
- MR15 = 55_H
- MR20 = 55_H

[Table 51] DQ Read Calibration Bit Ordering and Inversion Example

Pin		Bit Sequence ->															
	Invert	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
DQ0	Yes	1	0	1	0	0	1	1	0	1	1	1	0	0	0	1	1
DQ1	No	0	1	0	1	1	0	0	1	0	0	0	1	1	1	0	0
DQ2	Yes	1	0	1	0	0	1	1	0	1	1	1	0	0	0	1	1
DQ3	No	0	1	0	1	1	0	0	1	0	0	0	1	1	1	0	0
DMI0	Never	0	1	0	1	1	0	0	1	0	0	0	1	1	1	0	0
DQ4	Yes	1	0	1	0	0	1	1	0	1	1	1	0	0	0	1	1
DQ5	No	0	1	0	1	1	0	0	1	0	0	0	1	1	1	0	0
DQ6	Yes	1	0	1	0	0	1	1	0	1	1	1	0	0	0	1	1
DQ7	No	0	1	0	1	1	0	0	1	0	0	0	1	1	1	0	0
DQ8	Yes	1	0	1	0	0	1	1	0	1	1	1	0	0	0	1	1
DQ9	No	0	1	0	1	1	0	0	1	0	0	0	1	1	1	0	0
DQ10	Yes	1	0	1	0	0	1	1	0	1	1	1	0	0	0	1	1
DQ11	No	0	1	0	1	1	0	0	1	0	0	0	1	1	1	0	0
DMI1	Never	0	1	0	1	1	0	0	1	0	0	0	1	1	1	0	0
DQ12	Yes	1	0	1	0	0	1	1	0	1	1	1	0	0	0	1	1
DQ13	No	0	1	0	1	1	0	0	1	0	0	0	1	1	1	0	0
DQ14	Yes	1	0	1	0	0	1	1	0	1	1	1	0	0	0	1	1
DQ15	No	0	1	0	1	1	0	0	1	0	0	0	1	1	1	0	0

NOTE :
1) The patterns contained in MR32 and MR40 are transmitted on DQ[15:0] and DMI[1:0] when RD DQ Calibration is initiated via a MPC-1 [RD DQ Calibration] command. The pattern transmitted serially on each data lane, organized "little endian" such that the low-order bit in a byte is transmitted first. If the data pattern is 27_H, then the first bit transmitted with be a '1', followed by '1', '1', '0', '0', '1', '0', and '0'. The bit stream will be 00100111.
2) MR15 and MR20 may be used to invert the MR32/MR40 data pattern on the DQ pins. See MR15 and MR20 for more information. Data is never inverted on the DMI[1:0] pins.
3) DMI [1:0] outputs status follows Table 57.

[Table 52] MR Setting vs. DMI Status

DM Function MR13 OP[5]	Write DBIdc Function MR3 OP[7]	Read DBIdc Function MR3 OP[6]	DMI Status
1: Disable	0: Disable	0: Disable	Hi-Z
1: Disable	1: Enable	0: Disable	The data pattern is transmitted
1: Disable	0: Disable	1: Enable	The data pattern is transmitted
1: Disable	1: Enable	1: Enable	The data pattern is transmitted
0: Enable	0: Disable	0: Disable	The data pattern is transmitted
0: Enable	1: Enable	0: Disable	The data pattern is transmitted
0: Enable	0: Disable	1: Enable	The data pattern is transmitted
0: Enable	1: Enable	1: Enable	The data pattern is transmitted

4) No Data Bus Inversion (DBI) function is enacted during RD DQ Calibration, even if DBI is enabled in MR3-OP[6].

3.38.3 MPC of Read DQ Calibration After Power-Down Exit

Following the power-down state, an additional time, tMRRI, is required prior to issuing the MPC of Read DQ Calibration command. This additional time (equivalent to tRCD) is required in order to be able to maximize power-down current savings by allowing more power-up time for the Read DQ data in MR32 and MR40 data path after exit from standby, power-down mode.

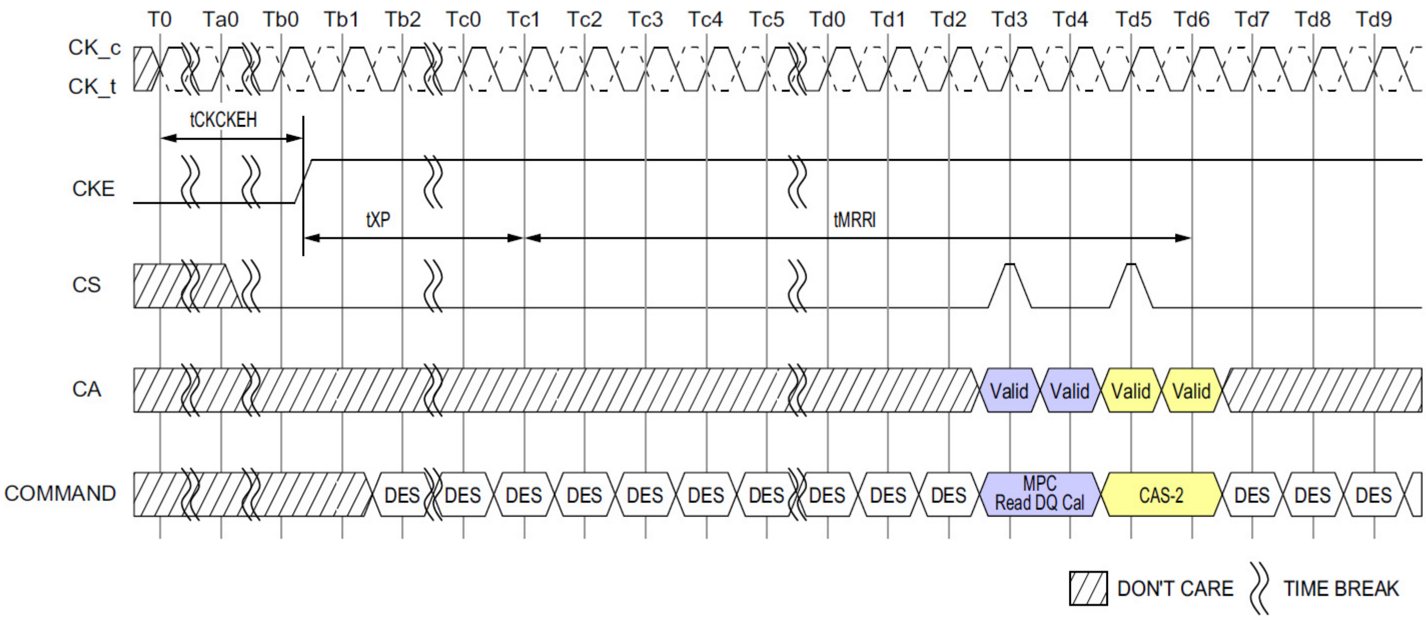

Figure 110: MPC Read DQ Calibration Following Power-Down State

3.39 DQS-DQ Training

The LPDDR4-SDRAM uses an un-matched DQS-DQ path to enable high speed performance and save power in the DRAM. As a result, the DQS strobe must be trained to arrive at the DQ latch center-aligned with the Data eye. The SDRAM DQ receiver is located at the DQ pad, and has a shorter internal delay in the SDRAM than does the DQS signal. The SDRAM DQ receiver will latch the data present on the DQ bus when DQS reaches the latch, and training is accomplished by delaying the DQ signals relative to DQS such that the Data eye arrives at the receiver latch centered on the DQS transition.

Two modes of training are available in LPDDR4:
- Command-based FIFO WR/RD with user patterns
- A internal DQS clock-tree oscillator, to determine the need for, and the magnitude of, required training.

The command-based FIFO WR/RD uses the MPC command with operands to enable this special mode of operation. When issuing the MPC command, if OP6 is set LOW then the DRAM will perform a NOP command. When OP6 is set HIGH, then OP[5:0] enable training functions or are reserved for future use (RFU). MPC commands that initiate a Read FIFO, READ DQ Calibration or Write FIFO to the SDRAM must be followed immediately by a CAS-2 command. See “Multi Purpose Command (MPC) Definition” for more information.

To perform Write training, the controller can issue a MPC [Write DQ FIFO] command with OP[6:0] set as described in the MPC Definition section, followed immediately by a CAS-2 command (CAS-2 operands should be driven LOW) to initiate a Write DQ FIFO. Timings for MPC [Write DQ FIFO] are identical to a Write command, with WL (Write Latency) timed from the 2nd rising clock edge of the CAS-2 command. Up to 5 consecutive MPC [Write DQ FIFO] commands with user defined patterns may be issued to the SDRAM to store up to 80 values (BL16 × 5) per pin that can be read back via the MPC [Read DQ FIFO] command. Write/Read FIFO Pointer operation is described later in this section.

After writing data to the SDRAM with the MPC [Write DQ FIFO] command, the data can be read back with the MPC [Read DQ FIFO] command and results compared with “expect” data to see if further training (DQ delay) is needed. MPC [Read DQ FIFO] is initiated by issuing a MPC command with OP[6:0] set as described in the MPC Definition section, followed immediately by a CAS-2 command (CAS-2 operands must be driven LOW). Timings for the MPC [Read DQ FIFO] command are identical to a Read command, with RL (Read Latency) timed from the 2nd rising clock edge of the CAS-2 command.

Read DQ FIFO is non-destructive to the data captured in the FIFO, so data may be read continuously until it is either overwritten by a Write DQ FIFO command or disturbed by CKE LOW or any of the following commands; Write, Masked Write, Read, Read DQ Calibration and a MRR. If fewer than 5 Write DQ FIFO commands were executed, then unwritten registers will have un-defined (but valid) data when read back.

The following command about MRW is only allowed from MPC [Write DQ FIFO] command to MPC [Read DQ FIFO].
Allowing MRW command is for OP[7]:FSP-OP, OP[6]:FSP-WR and OP[3]:VRCG of MR13 and MR14. And the rest of MRW command is prohibited.

For example: If 5 Write DQ FIFO commands are executed sequentially, then a series of Read DQ FIFO commands will read valid data from FIFO[0], FIFO[1]….FIFO[4], and will then wrap back to FIFO[0] on the next Read DQ FIFO.

On the other hand, if fewer than 5 Write DQ FIFO commands are executed sequentially (example=3), then a series of Read DQ FIFO commands will return valid data for FIFO[0], FIFO[1], and FIFO[2], but the next two Read DQ FIFO commands will return un.defined data for FIFO[3] and FIFO[4] before wrapping back to the valid data in FIFO[0].

3.39.1 FIFO Pointer Reset And Synchronism

The Read and Write DQ FIFO pointers are reset under the following conditions:
- Power-up initialization
- RESET_n asserted
- Power-down entry
- Self Refresh power down entry

The MPC [Write DQ FIFO] command advances the WR-FIFO pointer, and the MPC [Read DQ FIFO] advances the RD-FIFO pointer. Also any normal (non-FIFO) Read Operation (RD, RDA) advances both
WR-FIFO pointer and RD-FIFO pointer. Issuing (non-FIFO) Read Operation command is inhibited during Write training period. To keep the pointers aligned, the SoC memory controller must adhere to the following restriction at the end of Write training period:

$$b = a + (n*c)$$

Where:
‘a’ is the number of MPC [Write DQ FIFO] commands
‘b’ is the number of MPC [Read DQ FIFO] commands
‘c’ is the FIFO depth (=5 for LPDDR4)
‘n’ is a positive integer, ≥ 0

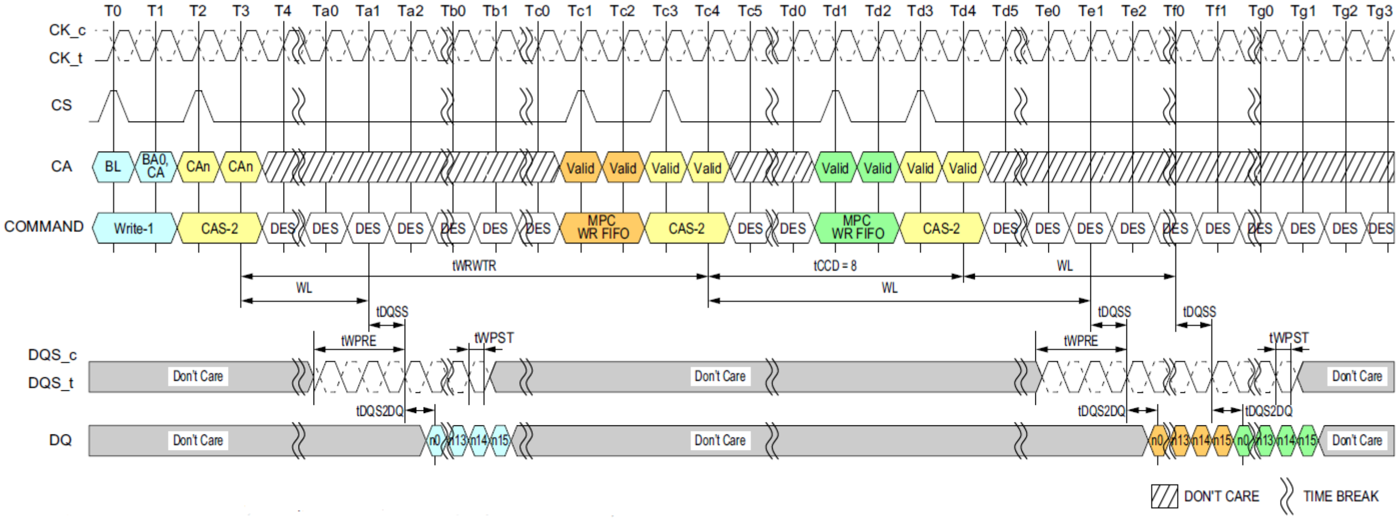

Figure 111: Write to MPC [Write FIFO] Operation Timing

NOTE :
1) MPC [WR FIFO] can be executed with a single bank or multiple banks active, during Refresh, or during SREF with CKE HIGH.
2) Write-1 to MPC is shown as an example of command-to-command timing for MPC. Timing from Write-1 to MPC [WR-FIFO] is t_{WRWTR}.
3) Seamless MPC [WR-FIFO] commands may be executed by repeating the command every t_{CCD} time.
4) MPC [WR-FIFO] uses the same command-to-data timing relationship (WL, t_{DQSS}, t_{DQS2DQ}) as a Write-1 command.
5) A maximum of 5 MPC [WR-FIFO] commands may be executed consecutively without corrupting FIFO data. The 6th MPC [WR-FIFO] command will overwrite the FIFO data from the first command. If fewer than 5 MPC [WR-FIFO] commands are executed, then the remaining FIFO locations will contain undefined data.
6) For the CAS-2 command following a MPC command, the CAS-2 operands must be driven "LOW."
7) To avoid corrupting the FIFO contents, MPC [RD-FIFO] must immediately follow MPC [WR-FIFO]/CAS-2 without any other command disturbing FIFO pointers in-between. FIFO pointers are disturbed by CKE Low, Write, Masked Write, Read, Read DQ Calibration and MRR.
8) BL = 16, Write Postamble = 0.5nCK.
9) DES commands are shown for ease of illustration; other commands may be valid at these times.

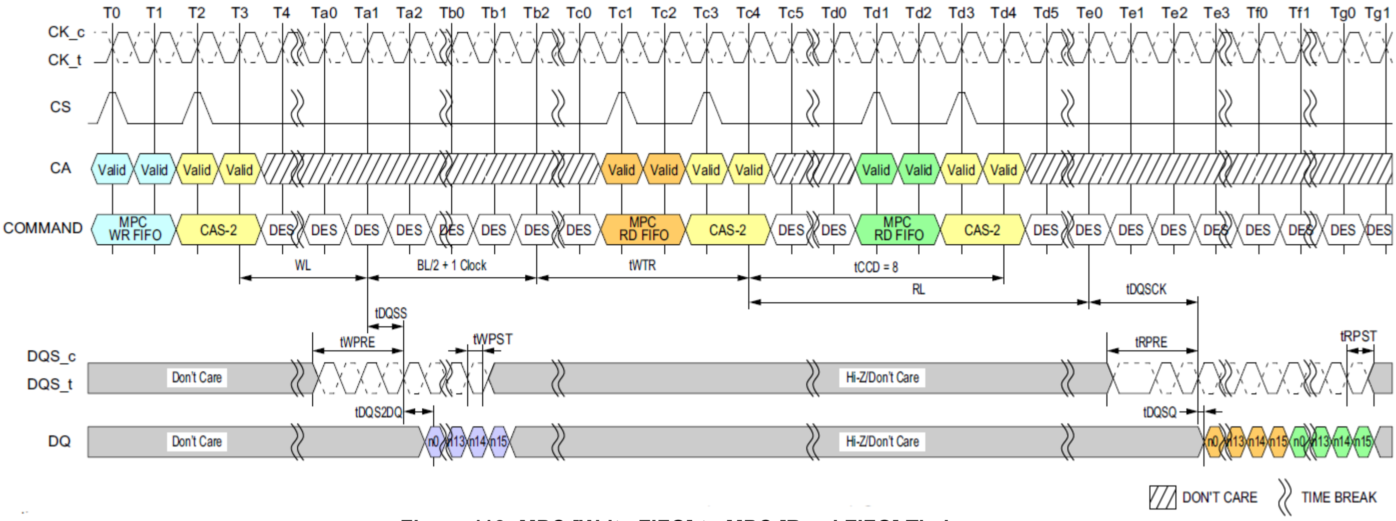

Figure 112: MPC [Write FIFO] to MPC [Read FIFO] Timing

NOTE :
1) MPC [RD-FIFO] can be executed with a single bank or multiple banks active, during Refresh, or during SREF with CKE HIGH.
2) MPC [WR-FIFO] to MPC [RD-FIFO] is shown as an example of command-to-command timing for MPC. Timing from MPC [WR-FIFO] to MPC [RD-FIFO] is specified in the command-to-command timing table.
3) Seamless MPC [RD-FIFO] commands may be executed by repeating the command every t_{CCD} time.
4) MPC [RD-FIFO] uses the same command-to-data timing relationship (RL, t_{DQSCK}, t_{DQSQ}) as a Read-1 command.
5) Data may be continuously read from the FIFO without any data corruption. After 5 MPC [RD-FIFO] commands the FIFO pointer will wrap back to the 1st FIFO and continue advancing. If fewer than 5 MPC [WR-FIFO] commands were executed, then the MPC [RD-FIFO] commands to those FIFO locations will return undefined data. See the Write Training section for more information on the FIFO pointer behavior.
6) For the CAS-2 command immediately following a MPC command, the CAS-2 operands must be driven "LOW."
7) DMI[1:0] signals will be driven if any of WR-DBI, RD-DBI, or DM is enabled in the mode registers. See Write Training section for more information on DMI behavior.
8) BL = 16, Write Postamble = 0.5nCK, Read Preamble: Toggle, Read Postamble: 0.5nCK.
9) DES commands are shown for ease of illustration; other commands may be valid at these times.

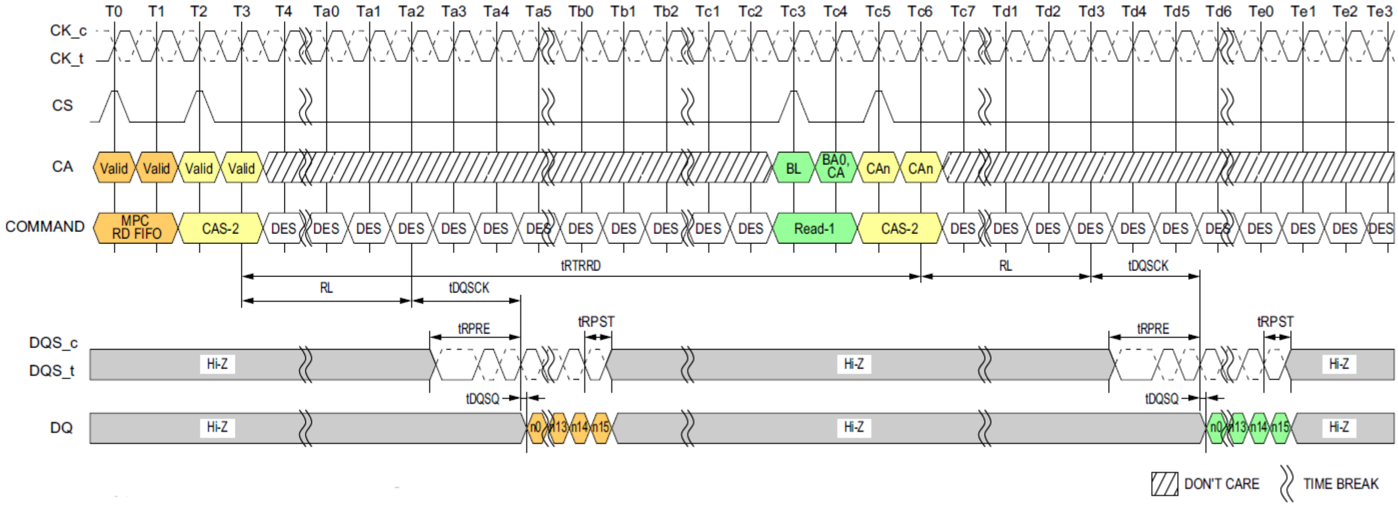

Figure 113: MPC [Read FIFO] to Read Timing

NOTE :
1) MPC [RD-FIFO] can be executed with a single bank or multiple banks active, during Refresh, or during SREF with CKE HIGH.
2) MPC [RD-FIFO] to Read-1 Operation is shown as an example of command-to-command timing for MPC. Timing from MPC [RD-FIFO] command to Read is t_{RTRRD}.
3) Seamless MPC [RD-FIFO] commands may be executed by repeating the command every t_{CCD} time.
4) MPC [RD-FIFO] uses the same command-to-data timing relationship (RL, t_{DQSCK}, t_{DQSQ}) as a Read-1 command.
5) Data may be continuously read from the FIFO without any data corruption. After 5 MPC [RD-FIFO] commands the FIFO pointer will wrap back to the 1st FIFO and continue advancing. If fewer than 5 MPC [WR-FIFO] commands were executed, then the MPC [RD-FIFO] commands to those FIFO locations will return undefined data. See the Write Training section for more information on the FIFO pointer behavior.
6) For the CAS-2 command immediately following a MPC command, the CAS-2 operands must be driven “LOW.”
7) DMI[1:0] signals will be driven if any of WR-DBI, RD-DBI, or DM is enabled in the mode registers. See Write Training section for more information on DMI behavior.
8) BL = 16, Read Preamble: Toggle, Read Postamble: 0.5nCK.
9) DES commands are shown for ease of illustration; other commands may be valid at these times.

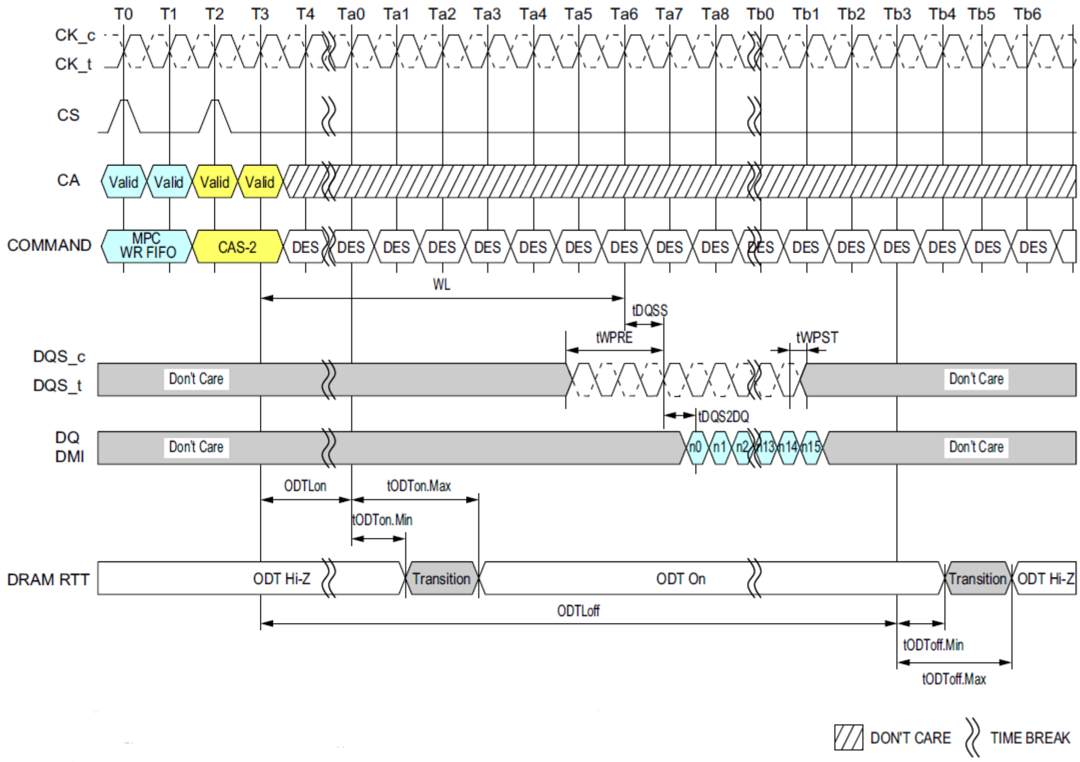

Figure 114: MPC [Write FIFO] with DQ ODT Timing

NOTE :
1) MPC [WR FIFO] can be executed with a single bank or multiple banks active, during Refresh, or during SREF with CKE HIGH.
2) MPC [WR-FIFO] uses the same command-to-data/ODT timing relationship (WL, t_{DQSS}, t_{DQS2DQ}, ODTLon, ODTLoff, t_{ODTon}, t_{ODToff}) as a Write-1 command.
3) For the CAS-2 command immediately following a MPC command, the CAS-2 operands must be driven “LOW.”
4) BL = 16, Write Postamble = 0.5nCK
5) DES commands are shown for ease of illustration; other commands may be valid at these times.

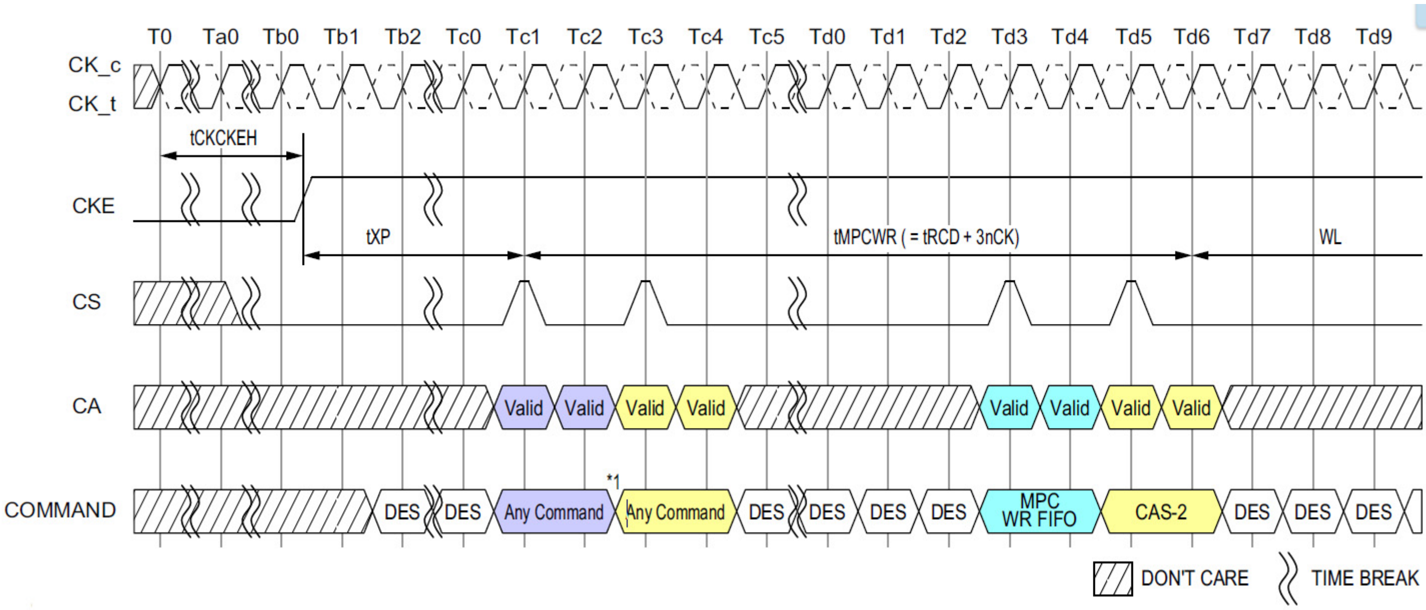

Figure 115: Power Down Exit to MPC [Write FIFO] Timing

NOTE :
1) Any commands except MPC WR FIFO and other exception commands defined other section in this document (i.e. MPC Read DQ Cal).
2) DES commands are shown for ease of illustration; other commands may be valid at these times.

[Table 53] MPC [Write FIFO] AC Timing

Parameter	Symbol	Min/Max	Data Rate	Unit	Note
MPC Write FIFO Timing					
Additional time after tXP has expired until MPC [Write FIFO] command may be issued	t_{MPCWR}	Min	t_{RCD} + 3nCK		

SAMSUNG

3.40 DQS Interval Oscillator

As voltage and temperature change on the SDRAM die, the DQS clock tree delay will shift and may require re-training. The LPDDR4-SDRAM includes an internal DQS clock.tree oscillator to measure the amount of delay over a given time interval (determined by the controller), allowing the controller to compare the trained delay value to the delay value seen at a later time. The DQS Oscillator will provide the controller with important information regarding the need to re.train, and the magnitude of potential error.

The DQS Interval Oscillator is started by issuing a MPC [Start DQS Osc] command with OP[6:0] set as described in the MPC Operation section, which will start an internal ring oscillator that counts the number of time a signal propagates through a copy of the DQS clock tree.

The DQS Oscillator may be stopped by issuing a MPC [Stop DQS Osc] command with OP[6:0] set as described in the MPC Operation section, or the controller may instruct the SDRAM to count for a specific number of clocks and then stop automatically (See MR23 for more information). If MR23 is set to automatically stop the DQS Oscillator, then the MPC [Stop DQS Osc] command should not be used (illegal). When the DQS Oscillator is stopped by either method, the result of the oscillator counter is automatically stored in MR18 and MR19.

The controller may adjust the accuracy of the result by running the DQS Interval Oscillator for shorter (less accurate) or longer (more accurate) duration. The accuracy of the result for a given temperature and voltage is determined by the following equation:

$$\textit{DQS Oscillator Granularity Error} = \frac{2 * (\textit{DQS delay})}{\textit{Run Time}}$$

Where:
Run Time = total time between start and stop commands
DQS delay = the value of the DQS clock tree delay (tDQS2DQ min/max)

Additional matching error must be included, which is the difference between DQS training circuit and the actual DQS clock tree across voltage and temperature. The matching error is vendor specific.
Therefore, the total accuracy of the DQS Oscillator counter is given by:

$$\textit{DQS oscillator accuracy} = 1 - \textit{granularity error} - \textit{matching error}$$

Example: If the total time between start and stop commands is 100ns, and the maximum DQS clock tree delay is 800ps (tDQS2DQ max), then the DQS Oscillator Granularity Error is:

$$\textit{DQS Oscillator Granularity Error} = \frac{2 * (0.8ns)}{100ns} = 1.6\%$$

This equates to a granularity timing error or 12.8ps.
Assuming a circuit Matching Error of 5.5ps across voltage and temperature, then the accuracy is:

$$\textit{DQS Oscillator Accuracy} = 1 - \frac{12.8 + 5.5}{800} = 97.7\%$$

Example: Running the DQS Oscillator for a longer period improves the accuracy. If the total time between start and stop commands is 500ns, and the maximum DQS clock tree delay is 800ps (tDQS2DQ max), then the DQS Oscillator Granularity Error is:

$$\textit{DQS Oscillator Granularity Error} = \frac{2 * (0.8ns)}{500ns} = 0.32\%$$

This equates to a granularity timing error or 2.56ps.
Assuming a circuit Matching Error of 5.5ps across voltage and temperature, then the accuracy is:

$$\textit{DQS Oscillator Accuracy} = 1 - \frac{2.56 + 5.5}{800} = 99.0\%$$

The result of the DQS Interval Oscillator is defined as the number of DQS Clock Tree Delays that can be counted within the “run time,” determined by the controller. The result is stored in MR18-OP[7:0] and MR19-OP[7:0]. MR18 contains the least significant bits (LSB) of the result, and MR19 contains the most significant bits (MSB) of the result. MR18 and MR19 are overwritten by the SDRAM when a MPC-1 [Stop DQS Osc] command is received. The SDRAM counter will count to its maximum value (=2^16) and stop. If the maximum value is read from the mode registers, then the memory controller must assume that the counter overflowed the register and discard the result. The longest “run time” for the oscillator that will not overflow the counter registers can be calculated as follows:

$$\textit{Longest runtime interval} = 2^{16} \times t_{DQS2DQ(MIN)} = 2^{16} \times 0.2ns = 13.1us$$

3.40.1 Interval Oscillator Matching Error

The interval oscillator matching error is defined as the difference between the DQS training ckt(interval oscillator) and the actual DQS clock tree across voltage and temperature.

- Parameters:
  - tDQS2DQ: Actual DQS clock tree delay
  - tDQSOSC: Training ckt(interval oscillator) delay
  - OSCOffset: Average delay difference over voltage and temp(shown in Figure 116)
  - OSCMatch: DQS oscillator matching error

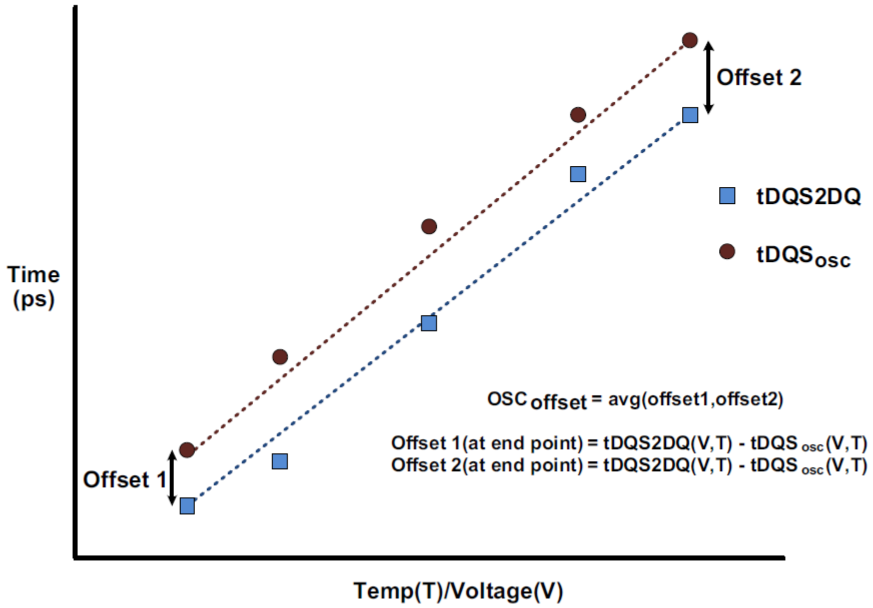

Figure 116: Interval oscillator offset OSC offset

- OSC_{Match} :

$$OSC_{Match} = [tDQS2DQ_{(V,T)} - tDQS_{OSC(V,T)} - OSC_{offset}]$$

- $tDQS_{OSC}$:

$$tDQS_{OSC(V,T)} = \frac{Runtime}{2 * Count}$$

[Table 54] DQS Oscillator Matching Error Specification

Parameter	Symbol	Min	Max	Units	Notes
DQS Oscillator Matching Error	OSC_{Match}	-20	20	ps	1,2,3,4,5,6,7
DQS Oscillator Offset	OSC_{offset}	-100	100	ps	2,4,7

NOTE :
1) The OSC_{Match} is the matching error per between the actual DQS and DQS interval oscillator over voltage and temp.
2) This parameter will be characterized or guaranteed by design.
3) The OSC_{Match} is defined as the following:
$OSC_{Match} = [t_{DQS2DQ}(V,T) - t_{DQSOSC}(V,T) - OSC_{offset}]$
Where $t_{DQS2DQ(V,T)}$ and $t_{DQSOSC(V,T)}$ are determined over the same voltage and temp conditions.
4) The runtime of the oscillator must be at least 200ns for determining $t_{DQSOSC(V,T)}$

$$tDQS_{OSC(V,T)} = \frac{Runtime}{2 * Count}$$

5) The input stimulus for t_{DQS2DQ} will be consistent over voltage and temp conditions.
6) The OSC_{offset} is the average difference of the endpoints across voltage and temp.
7) These parameters are defined per channel.
8) $t_{DQS2DQ(V,T)}$ delay will be the average of DQS to DQ delay over the runtime period.

3.40.2 DQS Interval Oscillator Readout Timing

OSC Stop to its counting value readout timing is shown in Figure 117 and Figure 118.

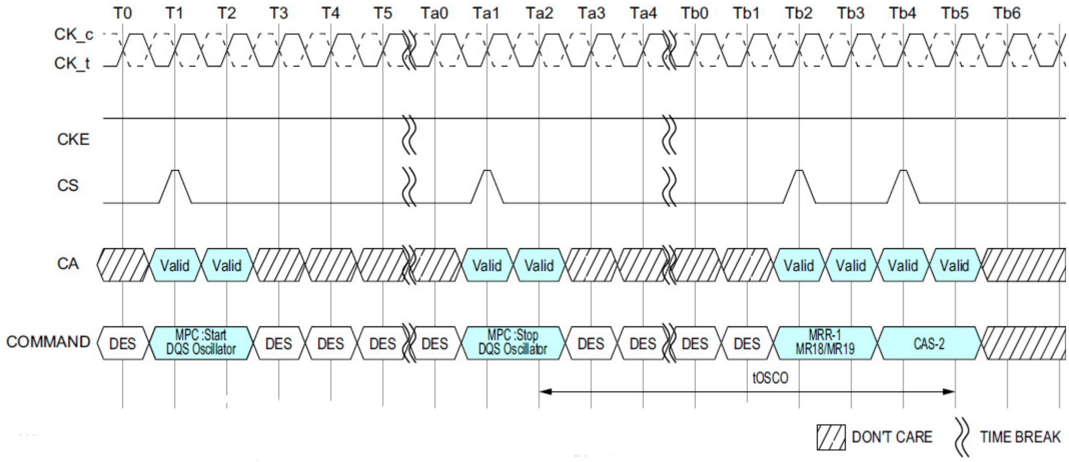

Figure 117: In case of DQS Interval Oscillator is stopped by MPC Command

NOTE :
1) DQS interval timer run time setting : MR23 OP[7:0] = 00000000_B.
2) DES commands are shown for ease of illustration; other commands may be valid at these times.

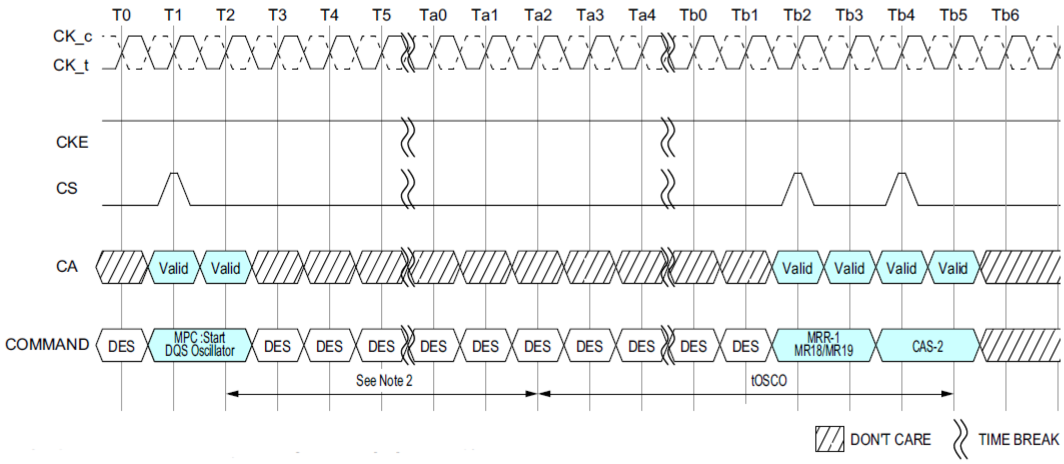

Figure 118: In case of DQS Interval Oscillator is stopped by DQS interval timer

NOTE :
1) DQS interval timer run time setting : MR23 OP[7:0] ≠ 00000000_B.
2) Setting counts of MR23.
3) DES commands are shown for ease of illustration; other commands may be valid at these times.

[Table 55] DQS Interval Oscillator AC Timing

Parameter	Symbol	Min/Max	Value	Units	Notes
Delay time from OSC stop to Mode Register Readout	t_{OSCO}	Min	Max(40ns,8tCK)	ns	

NOTE :
1) Start DQS OSC command is prohibited until $t_{OSCO(Min)}$ is satisfied.

3.41 Read Preamble Training

LPDDR4 READ Preamble Training is supported through the MPC function.

This mode can be used to train or read level the DQS receivers. Once READ Preamble Training is enabled by MR13[OP1] = 1, the LPDDR4 DRAM will drive DQS_t LOW, DQS_c HIGH within tSDO and remain at these levels until an MPC DQ READ Calibration command is issued.

During READ Preamble Training the DQS preamble provided during normal operation will not be driven by the DRAM. Once the MPC DQ READ Calibration command is issued, the DRAM will drive DQS_t/DQD_c and DQ like a normal READ burst after RL and t_{DQSCK}. Prior to the MPC DQ READ Calibration command, the DRAM may or may not drive DQ[15:0] in this mode.

While in READ Preamble Training Mode, only READ DQ Calibration commands may be issued.
- Issue an MPC [RD DQ Calibration] command followed immediately by a CAS-2 command.
- Each time an MPC [RD DQ Calibration] command followed by a CAS-2 is received by the LPDDR4 SDRAM, a 16-bit data burst will after the currently set RL, drive the eight bits programmed in MR32 followed by the eight bits programmed in MR40 on all I/O pins.
- The data pattern will be inverted for I/O pins with a '1' programmed in the corresponding invert mask mode register bit.
- Note that the pattern is driven on the DMI pins, but no data bus inversion function is enabled, even if Read DBI is enabled in the DRAM mode register.
- This command can be issued every t_{CCD} seamlessly.
- The operands received with the CAS-2 command must be driven LOW.

READ Preamble Training is exited within tSDO after setting MR13[OP1] = 0.
LPDDR4 supports the READ Preamble Training as optional feature. Refer to vendor specific datasheets.

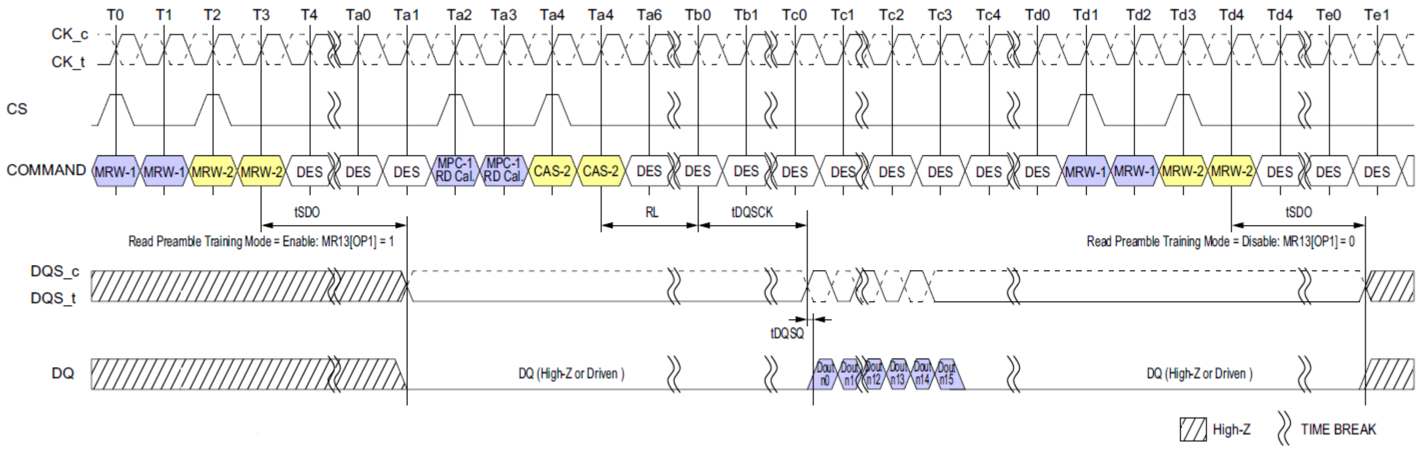

Figure 119: Read Preamble Training

NOTE :
1) Read DQ Calibration supports only BL16 operation.

[Table 56] Timing Parameters

Parameter	Symbol	Min	Max	Unit	Notes
Delay from MRW command to DQS Driven	t_{SDO}	-	Max(12nCK, 20ns)	-	

3.42 Multi-Purpose Command (MPC)

LPDDR4-SDRAMs use the MPC command to issue a NOP and to access various training modes. The MPC command is initiated with CS, and CA[5:0] asserted to the proper state at the rising edge of CK, as defined by the Command Truth Table. The MPC command has seven operands (OP[6:0]) that are decoded to execute specific commands in the SDRAM. OP[6] is a special bit that is decoded on the first rising CK edge of the MPC command. When OP[6]=0_B then the SDRAM executes a NOP (no operation) command, and when OP[6]=1_B then the SDRAM further decodes one of several training commands.

When OP[6]=1_B and when the training command includes a Read or Write operation, the MPC command must be followed immediately by a CAS-2 command. For training commands that Read or Write the SDRAM, read latency (RL) and write latency (WL) are counted from the second rising CK edge of the CAS-2 command with the same timing relationship as any normal Read or Write command. The operands of the CAS-2 command following a MPC Read/Write command must be driven LOW. The following MPC commands must be followed by a CAS-2 command:
- Write FIFO
- Read FIFO
- Read DQ Calibration

All other MPC-1 commands do not require a CAS-2 command, including:
- NOP
- Start DQS Interval Oscillator
- Stop DQS Interval Oscillator
- Start ZQ Calibration
- Latch ZQ Calibration

[Table 57] MPC Command Definition

SDRAM Command	SDR Command Pins			SDR CA Pins						CK_t EDGE	Notes
	CKE		CS	CA0	CA1	CA2	CA3	CA4	CA5		
	CK_t(n-1)	CK_t(n)									
MPC (Train, NOP)	H	H	H	L	L	L	L	L	OP6	R1	1,2
			L	OP0	OP1	OP2	OP3	OP4	OP5	R2	

[Table 58] MPC Command Definition for OP[6:0]

Function	Operand	Data	Notes
Training Modes	OP[6:0]	**0XXXXXX$_B$**: NOP **1000001$_B$**: RD FIFO : RD FIFO supports only BL16 operation **1000011$_B$**: RD DQ Calibration (MR32/MR40) **1000101$_B$**: RFU **1000111$_B$**: WR FIFO : WR FIFO supports only BL16 operation **1001001$_B$**: RFU **1001011$_B$**: Start DQS Osc **1001101$_B$**: Stop DQS Osc **1001111$_B$**: ZQCal Start **1010001$_B$**: ZQCal Latch **All Others**: Reserved	1,2,3

NOTE :
1) See command truth table for more information.
2) MPC commands for Read or Write training operations must be immediately followed by CAS-2 command consecutively without any other commands in-between. MPC command must be issued first before issuing the CAS-2 command.
3) Write FIFO and Read FIFO commands will only operate as BL16, ignoring the burst length selected by MR1 OP[1:0].

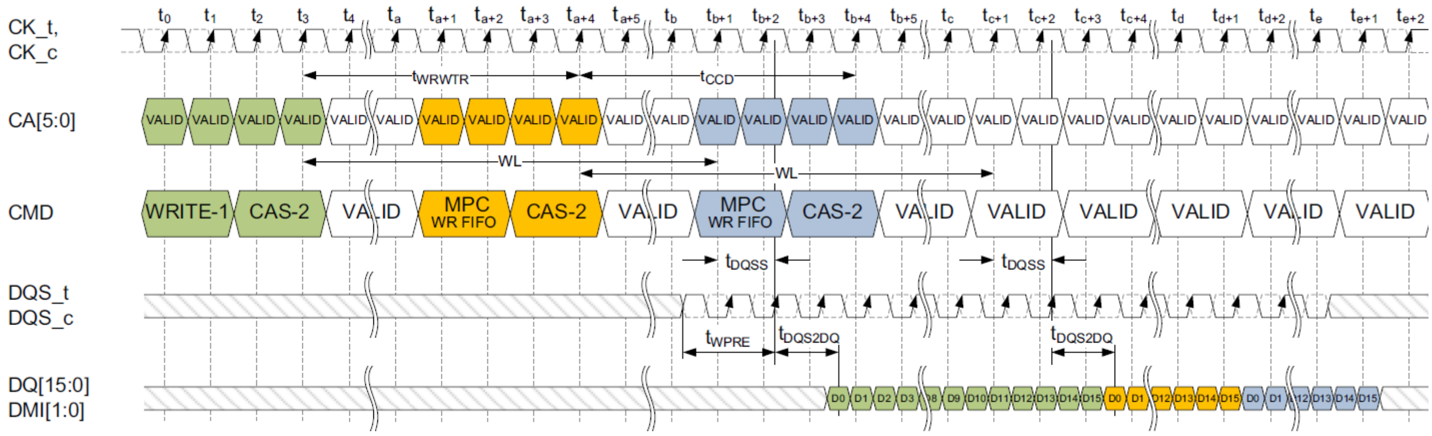

Figure 120: MPC [WRITE FIFO] Operation: tWPRE=2nCK, tWPST=0.5nCK

NOTE :
1) MPC [WR FIFO] can be executed with a single bank or multiple banks active, during Refresh, or during SREF with CKE HIGH.
2) Write-1 to MPC is shown as an example of command-to-command timing for MPC. Timing from Write-1 to MPC [WR-FIFO] is t_{WRWTR}.
3) Seamless MPC [WR-FIFO] commands may be executed by repeating the command every t_{CCD} time.
4) MPC [WR-FIFO] uses the same command-to-data timing relationship (WL, t_{DQSS}, t_{DQS2DQ}) as a Write-1 command.
5) A maximum of 5 MPC [WR-FIFO] commands may be executed consecutively without corrupting FIFO data. The 6th MPC [WR-FIFO] command will overwrite the FIFO data from the first command. If fewer than 5 MPC [WR-FIFO] commands are executed, then the remaining FIFO locations will contain undefined data.
6) For the CAS-2 command following a MPC command, the CAS-2 operands must be driven "LOW."
7) To avoid corrupting the FIFO contents, MPC-1 [RD-FIFO] must immediately follow MPC-1 [WR-FIFO]/CAS-2 without any other command disturbing FIFO pointers in-between. FIFO pointers are disturbed by CKE Low, Write, Masked Write, Read, Read DQ Calibration and MRR. See Write Training section for more information on FIFO pointer behavior.

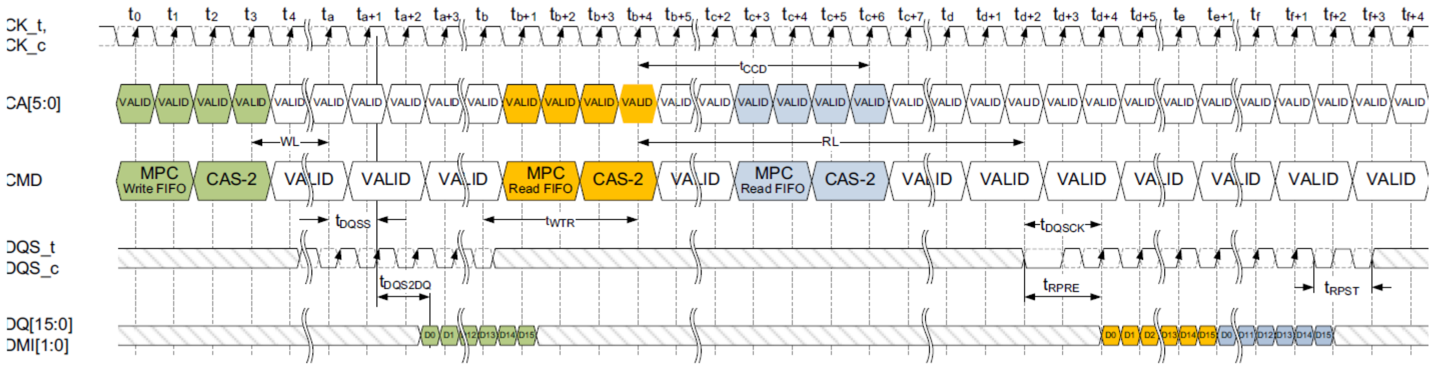

Figure 121: MPC [RD FIFO] Read Operation: tWPRE=2nCK, tWPST=0.5nCK, tRPRE=toggling, tRPST=1.5nCK

NOTE:
1) MPC [RD FIFO] can be executed with a single bank or multiple banks active, during Refresh, or during SREF with CKE HIGH.
2) Write-1 to MPC is shown as an example of command-to-command timing for MPC. Timing from Write-1 to MPC-1 [WR-FIFO] is t_{WRWTR}.
3) Seamless MPC [RD-FIFO] commands may be executed by repeating the command every t_{CCD} time.
4) MPC [RD-FIFO] uses the same command-to-data timing relationship (RL, t_{DQSCK}) as a Read-1 command.
5) Data may be continuously read from the FIFO without any data corruption. After 5 MPC [RD-FIFO] commands the FIFO pointer will wrap back to the 1st FIFO and continue advancing. If fewer than 5 MPC [WR-FIFO] commands were executed, then the MPC [RD-FIFO] commands to those FIFO locations will return undefined data. See the Write Training section for more information on the FIFO pointer behavior.
6) For the CAS-2 command immediately following a MPC command, the CAS-2 operands must be driven "LOW."
7) DMI[1:0] signals will be driven if any of WR-DBI, RD-DBI, or DM is enabled in the mode registers. See Write Training section for more information on DMI behavior.

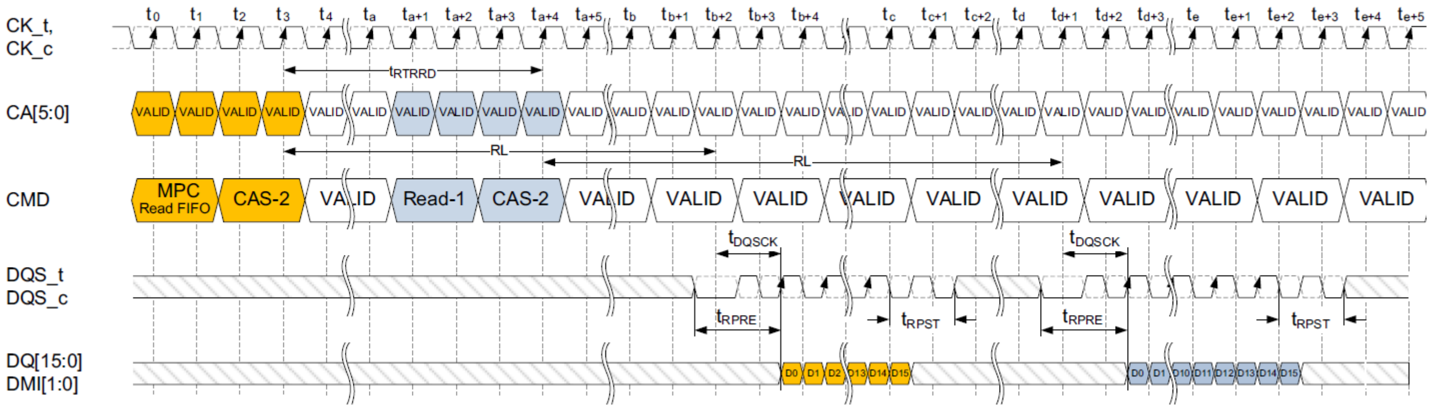

Figure 122: MPC [RD FIFO] Operation : tRPRE=toggling, tRPST=1.5nCK

NOTE :
1) MPC [RD FIFO] can be executed with a single bank or multiple banks active, during Refresh, or during SREF with CKE HIGH.
2) MPC [RD-FIFO] to Read-1 Operation is shown as an example of command-to-command timing for MPC. Timing from MPC-1 [RD-FIFO] command to Read is t_{RTRRD}.
3) Seamless MPC [RD-FIFO] commands may be executed by repeating the command every t_{CCD} time.
4) MPC [RD-FIFO] uses the same command-to-data timing relationship (RL, t_{DQSCK}) as a Read-1 command.
5) Data may be continuously read from the FIFO without any data corruption. After 5 MPC [RD-FIFO] commands the FIFO pointer will wrap back to the 1st FIFO and continue advancing. If fewer than 5 MPC [WR-FIFO] commands were executed, then the MPC [RD-FIFO] commands to those FIFO locations will return undefined data. See the Write Training section for more information on the FIFO pointer behavior.
6) For the CAS-2 command immediately following a MPC command, the CAS-2 operands must be driven “LOW.”
7) DMI[1:0] signals will be driven if any of WR-DBI, RD-DBI, or DM is enabled in the mode registers. See Write Training section for more information on DMI behavior.

[Table 59] Timing Constraints for Training Commands

Previous Command	Next command	Minimum Delay	Unit	Notes
WR/MWR	MPC [WR FIFO]	t_{WRWTR}	nCK	1
	MPC [RD FIFO]	Not Allowed	-	2
	MPC [RD DQ Calibration]	$WL+RU(t_{DQSS(max)}/t_{CK}) +BL/2+RU(t_{WTR}/t_{CK})$	nCK	
RD/MRR	MPC [WR FIFO]	t_{RTRRD}	nCK	3
	MPC [RD FIFO]	Not Allowed	-	2
	MPC [RD DQ Calibration]	t_{RTRRD}	nCK	3
MPC [WR FIFO]	WR/MWR	Not Allowed	-	2
	MPC [WR FIFO]	t_{CCD}	nCK	
	RD/MRR	Not Allowed	-	2
	MPC [RD FIFO]	$WL+RU(t_{DQSS(max)}/t_{CK}) +BL/2+RU(t_{WTR}/t_{CK})$	nCK	
	MPC [RD DQ Calibration]	Not Allowed	-	2
MPC [RD FIFO]	WR/MWR	t_{RTRRD}	nCK	3
	MPC [WR FIFO]	t_{RTW}	nCK	4
	RD/MRR	t_{RTRRD}	nCK	3
	MPC [RD FIFO]	t_{CCD}	nCK	
	MPC [RD DQ Calibration]	t_{RTRRD}	nCK	3
MPC [RD DQ Calibration]	WR/MWR	t_{RTRRD}	nCK	3
	MPC [WR FIFO]	t_{RTRRD}	nCK	3
	RD/MRR	t_{RTRRD}	nCK	3
	MPC [RD FIFO]	Not Allowed	-	2
	MPC [RD DQ Calibration]	t_{CCD}	nCK	

NOTE :
1) $t_{WRWTR} = WL + BL/2 + RU(t_{DQSS(max)} / t_{CK}) + max(RU(7.5ns / t_{CK}), 8nCK)$.
2) No commands are allowed between MPC [WR FIFO] and MPC-1 [RD FIFO] except MRW commands related to training parameters.
3) $t_{RTRRD} = RL + RU(t_{DQSCK(max)}/t_{CK}) + BL/2 + RD(t_{RPST}) + max(RU(7.5ns/t_{CK}),8nCK)$.
4) t_{RTW} = In Case of DQ ODT Disable MR11 OP[2:0] = 000_B:
$RL+RU(t_{DQSCK(max)}/ t_{CK}) +BL/2- WL+t_{WPRE}+RD(t_{RPST})$
In Case of DQ ODT Enable MR11 OP[2:0] ≠ 000_B:
$RL + RU(t_{DQSCK(max)}/t_{CK}) + BL/2 + RD(t_{RPST}) - ODTLon - RD(t_{ODTon,min}/t_{CK}) + 1$.

3.43 Thermal Offset

Because of their tight thermal coupling with the LPDDR4 device, hot spots on an SOC can induce thermal gradients across the LPDDR4 device. As these hot spots may not be located near the device thermal sensor, the devices' temperature compensated self-refresh circuit may not generate enough refresh cycles to guarantee memory retention. To address this shortcoming, the controller can provide a thermal offset that the memory uses to adjust its TCSR circuit to ensure reliable operation.

This offset is provided through MR4(6:5) to either or to both the channels (dual-channel devices). This temperature offset may modify refresh behavior for the channel to which the offset is provided. It will take a max of 200us to have the change reflected in MR4(2:0) for the channel to which the offset is provided. If the induced thermal gradient from the device temperature sensor location to the hot spot location of the controller is larger than 15 degrees C, then self-refresh mode will not reliably maintain memory contents.

To accurately determine the temperature gradient between the memory thermal sensor and the induced hot spot, the memory thermal sensor location must be provided to the LPDDR4 memory controller.

Support of thermal offset function is optional. Please refer to vendor datasheet to figure out if the function is supported or not.

3.44 Temperature Sensor

LPDDR4 devices feature a temperature sensor whose status can be read from MR4. This sensor can be used to determine an appropriate refresh rate, determine whether AC timing de-rating is required in the elevated temperature range, and/or monitor the operating temperature. Either the temperature sensor or the device T_{OPER} may be used to determine whether operating temperature requirements are being met.

LPDDR4 devices shall monitor device temperature and update MR4 according to t_{TSI}. Upon assertion of CKE (Low to High transition), the device temperature status bits shall be no older than t_{TSI}. MR4 will be updated even when device is in self refresh state with CKE HIGH.

When using the temperature sensor, the actual device case temperature may be higher than the TOPER specification that applies for the Standard or elevated Temperature Ranges. For example, TCASE may be above 85°C when MR4[2:0] equals 011_B.

LPDDR4 devices shall allow for 2°C temperature margin between the point at which the device updates the MR4 value and the point at which the controller reconfigures the system accordingly. In the case of tight thermal coupling of the memory device to external hot spots, the maximum device temperature might be higher than what is indicated by MR4.

To assure proper operation using the temperature sensor, applications should consider the following factors:

- Temp Gradient is the maximum temperature gradient experienced by the memory device at the temperature of interest over a range of 2°C.
- ReadInterval is the time period between MR4 reads from the system.
- Temp Sensor Interval (t_{TSI}) is maximum delay between internal updates of MR4.
- Sys Resp Delay is the maximum time between a read of MR4 and the response by the system.

In order to determine the required frequency of polling MR4, the system shall use the maximum Temp Gradient and the maximum response time of the system using the following equation:

$$\mathit{TempGradient} \times (\mathit{ReadInterval} + tTSI + \mathit{SysRespDelay}) <= 2°C$$

[Table 60] Temperature Sensor

Parameter	Symbol	Max/Min	Value	Unit
System Temperature Gradient	Temp Gradient	Max	System Dependent	°C/s
MR4 ReadInterval	ReadInterval	Max	System Dependent	ms
Temperature Sensor Interval	t_{TSI}	Max	32	ms
System Response Delay	Sys Resp Delay	Max	System Dependent	ms
Device Temperature Margin	Temp Margin	Max	2	°C

For example, if Temp Gradient is 10°C/s and the Sys Resp Delay is 1 ms:

$$(10°C/s) \times (\mathit{ReadInterval} + 32ms + 1ms) <= 2°C$$

In this case, ReadInterval shall be no greater than 167 ms.

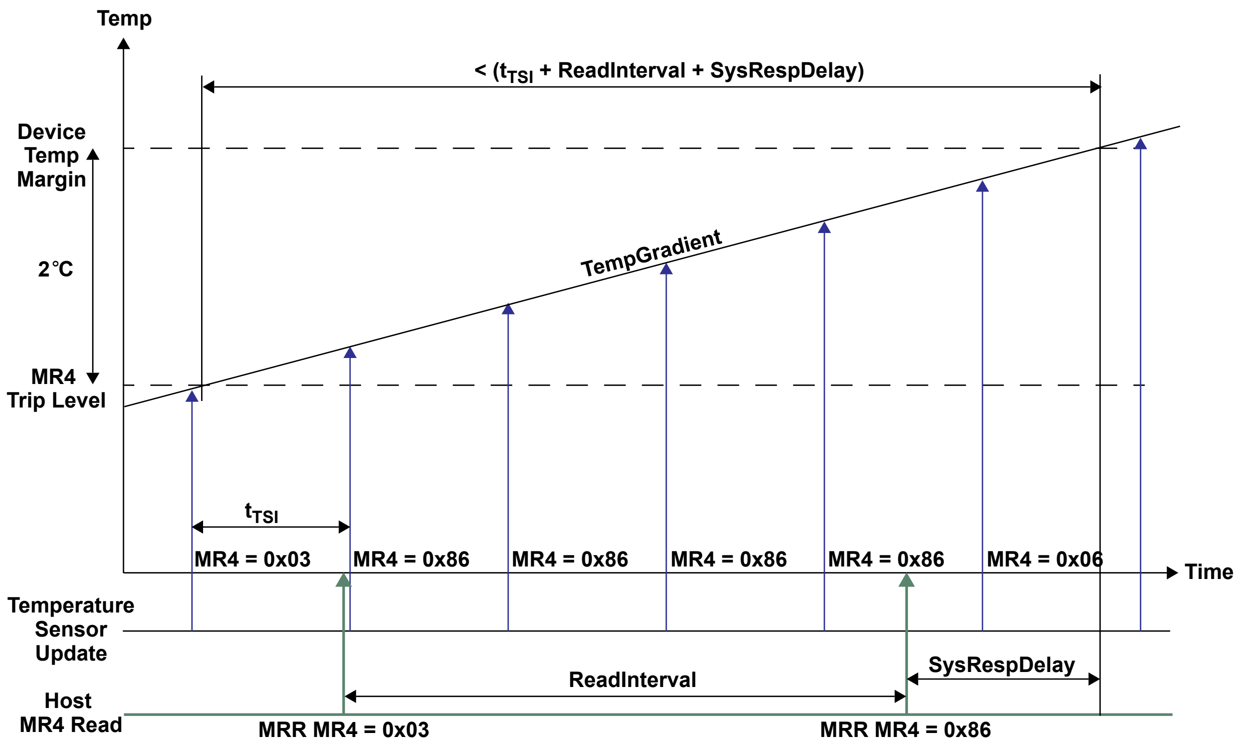

Figure 123: Temp Sensor Timing

3.45 ZQ Calibration

The MPC command is used to initiate ZQ Calibration, which calibrates the output driver impedance across process, temperature, and voltage. ZQ Calibration occurs in the background of device operation, Dual channel devices share common ZQ circuitry between channels which follow a protocol (See Section section 3.45.2) to allow for channel independence.

There are two ZQ Calibration modes initiated with the MPC command: ZQCal Start, and ZQCal Latch. ZQCal Start initiates the SDRAM's calibration procedure, and ZQCal Latch captures the result and loads it into the SDRAM's drivers.

A ZQCal Start command may be issued anytime the LPDDR4-SDRAM is not in a power-down state. A ZQCal Latch Command may be issued anytime outside of power-down after t_{ZQCAL} has expired and all DQ bus operations have completed. The CA Bus must maintain a Deselect state during t_{ZQLAT} to allow CA ODT calibration settings to be updated. The following mode register fields that modify I/O parameters cannot be changed following a ZQCal Start command and before t_{ZQCAL} has expired:

- PU-Cal (Pull-up Calibration VOH Point)
- PDDS (Pull Down Drive Strength and Rx Termination)
- DQ-ODT (DQ ODT Value)
- CA-ODT (CA ODT Value)

3.45.1 ZQCal Reset

The ZQCal Reset command resets the output impedance calibration to a default accuracy of +/- 30% across process, voltage, and temperature. This command is used to ensure output impedance accuracy to +/- 30% when ZQCal Start and ZQCal Latch commands are not used.

The ZQCal Reset command is executed by writing MR10-OP[0]=1_B.

[Table 61] ZQCal Timing Parameters

Parameter	Symbol	Min/Max	Value	Unit
ZQ Calibration Time	t_{ZQCAL}	MIN	1	us
ZQ Calibration Latch Time	t_{ZQLAT}	MIN	max(30ns, 8tCK)	ns
ZQ Calibration Reset Time	$t_{ZQRESET}$	MIN	max(50ns, 3tCK)	ns

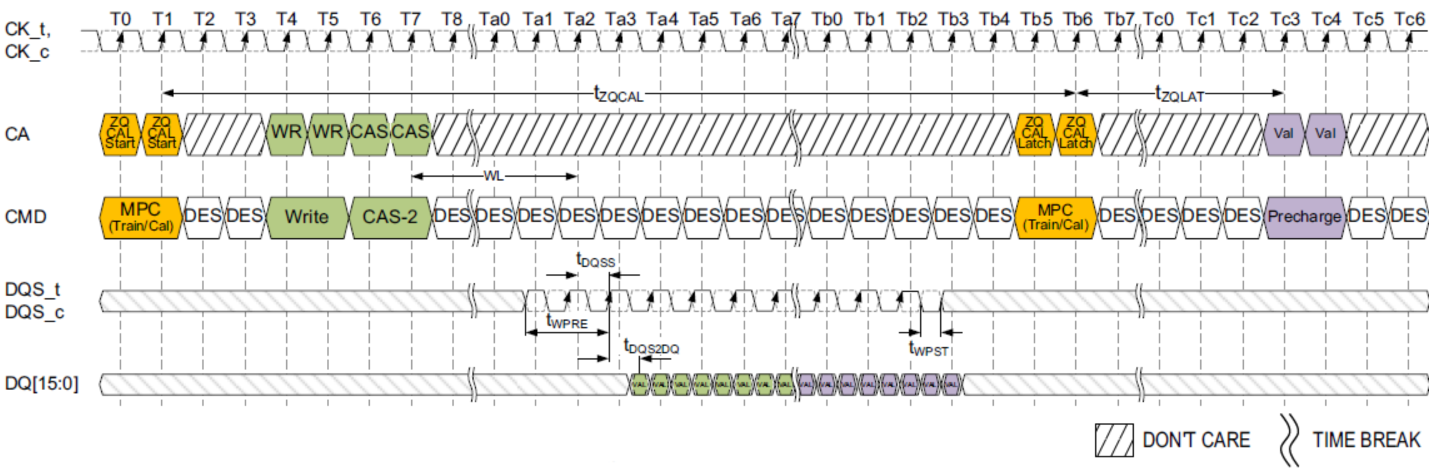

Figure 124: ZQCal Timing

NOTE :
1) Write and Precharge operations shown for illustrative purposes. Any single or multiple valid commands may be executed within the t_{ZQCAL} time and prior to latching the results.
2) Before the ZQ-Latch command can be executed, any prior commands utilizing the DQ bus must have completed. Write commands with DQ Termination must be given enough time to turn off the DQ-ODT before issuing the ZQ-Latch command. See the ODT section for ODT timing.

3.45.2 Multi-Channel Considerations for Dual Channel Devices

The LPDDR4-SDRAM includes a single ZQ pin and associated ZQ Calibration circuitry.
Calibration values from this circuit will be used by both channels according to the following protocol:

1. ZQCal Start commands may be issued to either or both channels.
2. ZQCal Start commands may be issued when either or both channels are executing other commands and other commands may be issued during t_{ZQCAL}.
3. ZQCal Start commands may be issued to both channels simultaneously.
4. The ZQCal Start command will begin the calibration unless a previously requested ZQ calibration is in progress.
5. If a ZQCal Start command is received while a ZQ calibration is in progress on the SDRAM, the ZQCal Start command will be ignored and the in-progress calibration will not be interrupted.
6. ZQCal Latch commands are required for each channel.
7. ZQCal Latch commands may be issued to both channels simultaneously.
8. ZQCal Latch commands will latch results of the most recent ZQCal Start command provided t_{ZQCAL} has been met.
9. ZQCal Latch commands which do not meet t_{ZQCAL} will latch the results of the most recently completed ZQ calibration.
10. ZQ Reset MRW commands will only reset the calibration values for the channel issuing the command.

In compliance with complete channel independence, either channel may issue ZQCal Start and ZQCal Latch commands as needed without regard to the state of the other channel.

3.45.3 ZQ External Resistor, Tolerance, And Capacitive Loading

To use the ZQ calibration function, a 240 ohm +/- 1% tolerance external resistor must be connected between the ZQ pin and V_{DDQ}.

If the system configuration shares the CA bus to form a x32 (or wider) channel, the ZQ pin of each die's x16 channel shall use a separate ZQCal resistor.

If the system configuration has more than one rank, and if the ZQ pins of both ranks are attached to a single resistor, then the SDRAM controller must ensure that the ZQCal's don't overlap.

The total capacitive loading on the ZQ pin must be limited to 25pF.

Example: If a system configuration shares a CA bus between 'n' channels to form a n × 16 wide bus, and no means are available to control the ZQCal separately for each channel (i.e. separate CS, CKE, or CK), then each x16 channel must have a separate ZQCal resistor.

Example: For a x32, two rank system, each x16 channel must have its own ZQCal resistor, but the ZQCal resistor can be shared between ranks on each x16 channel. In this configuration, the CS signal can be used to ensure that the ZQCal commands for Rank[0] and Rank[1] don't overlap.

3.46 On Die Termination For Command/Address Bus

ODT (On-Die Termination) is a feature of the LPDDR4 SDRAM that allows the SDRAM to turn on/off termination resistance for CK_t, CK_c, CS and CA[5:0] signals without the ODT control pin. The ODT feature is designed to improve signal integrity of the memory channel by allowing the DRAM controller to turn on and off termination resistance for any target DRAM devices via Mode Register setting.
A simple functional representation of the DRAM ODT feature is shown in Figure 125.

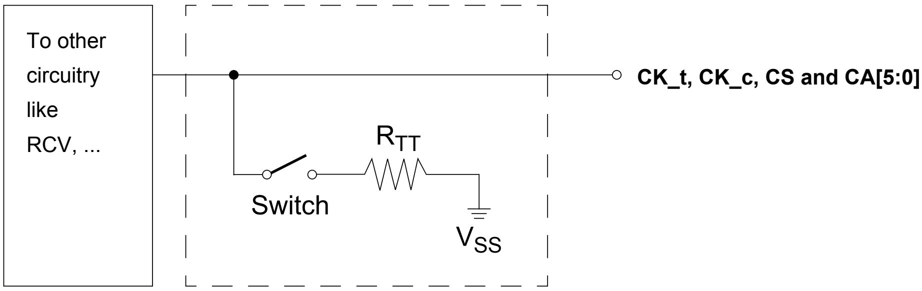

Figure 125: Functional Representation of CA ODT

3.46.1 ODT Mode Register And ODT State Table

ODT termination values are set and enabled via MR11. The CA bus (CK_t, CK_C, CS, CA[5:0]) ODT resistance values are set by MR11 OP[6:4]. The default state for the CA is ODT disabled.
ODT is applied on the CA bus to the CK_t, CK_c, CS and CA[5:0] signals. The CA ODT of the device is designed to enable one rank to terminate the entire command bus in a multi rank system, so only one termination load will be present even if multiple devices are sharing the command signals. For this reason, CA ODT remains on even when the device is in the power-down or self-refresh power-down states.
The die has a bond-pad (ODT_CA) for multi rank operations. When the ODT_CA pad is LOW, the die will not terminate the CA bus regardless of the state of the mode register CA ODT bits (MR11 OP[6:4]). If, however, the ODT_CA bond-pad is HIGH, and the mode register CA ODT bits are enabled, the die will terminate the CA bus with the ODT values found in MR11 OP[6:4]. In a multi rank system, the terminating rank should be trained first, followed by the non-terminating rank(s).

[Table 62] Command Bus ODT State

ODTE-CA MR11[6:4]	ODT_CA bond pad	ODTD-CA MR22[5]	ODTF-CK MR22[3]	ODTF-CS MR22[4]	ODT State for CA	ODT State for CK_t/CK_c	ODT State for CS
Disabled [1]	Valid [2]	Valid [3]	Valid [3]	Valid [3]	Off	Off	Off
Valid [3]	0	Valid [3]	0	0	Off	Off	Off
Valid [3]	0	Valid [3]	0	1	Off	Off	On
Valid [3]	0	Valid [3]	1	0	Off	On	Off
Valid [3]	0	Valid [3]	1	1	Off	On	On
Valid [3]	1	0	Valid [3]	Valid [3]	On	On	On
Valid [3]	1	1	Valid [3]	Valid [3]	Off	On	On

NOTE :
1) Default Value.
2) “Valid” means “H or L (but a defined logic level)”.
3) “Valid” means “0 or 1”.
4) The state of ODT_CA is not changed when the DRAM enters power-down mode. This maintains termination for alternate ranks in multi-rank systems.

3.46.2 ODT Mode Register And ODT Characteristics

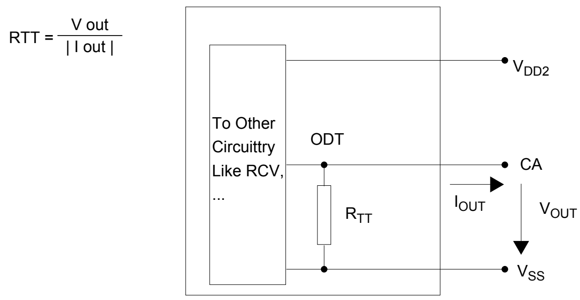

Figure 126: On Die Termination for CA

[Table 63] ODT DC Electrical Characteristics, assuming RZQ = 240 ohm +/-1% over the entire operating temperature range after a proper ZQ calibration up to 3200Mbps

MR11 [6:4]	RTT	Vout	Min	Nom	Max	Unit	Note
001	240ohm	VOLdc= 0.1 × VDD2	0.8	1.0	1.1	RZQ	1,2,3
		VOMdc= 0.33 × VDD2	0.9	1.0	1.1	RZQ	1,2,3
		VOHdc= 0.5 × VDD2	0.9	1.0	1.2	RZQ	1,2,3
010	120ohm	VOLdc= 0.1 × VDD2	0.8	1.0	1.1	RZQ/2	1,2,3
		VOMdc= 0.33 × VDD2	0.9	1.0	1.1	RZQ/2	1,2,3
		VOHdc= 0.5 × VDD2	0.9	1.0	1.2	RZQ/2	1,2,3
011	80ohm	VOLdc= 0.1 × VDD2	0.8	1.0	1.1	RZQ/3	1,2,3
		VOMdc= 0.33 × VDD2	0.9	1.0	1.1	RZQ/3	1,2,3
		VOHdc= 0.5 × VDD2	0.9	1.0	1.2	RZQ/3	1,2,3
100	60ohm	VOLdc= 0.1 × VDD2	0.8	1.0	1.1	RZQ/4	1,2,3
		VOMdc= 0.33 × VDD2	0.9	1.0	1.1	RZQ/4	1,2,3
		VOHdc= 0.5 × VDD2	0.9	1.0	1.2	RZQ/4	1,2,3
101	48ohm	VOLdc= 0.1 × VDD2	0.8	1.0	1.1	RZQ/5	1,2,3
		VOMdc= 0.33 × VDD2	0.9	1.0	1.1	RZQ/5	1,2,3
		VOHdc= 0.5 × VDD2	0.9	1.0	1.2	RZQ/5	1,2,3
110	40ohm	VOLdc= 0.1 × VDD2	0.8	1.0	1.1	RZQ/6	1,2,3
		VOMdc= 0.33 × VDD2	0.9	1.0	1.1	RZQ/6	1,2,3
		VOHdc= 0.5 × VDD2	0.9	1.0	1.2	RZQ/6	1,2,3
Mismatch CA-CA within clk group		0.33× VDD2	-		TBD	%	1,2,4

NOTE :
1) The tolerance limits are specified after calibration with stable voltage and temperature. For the behavior of the tolerance limits if temperature or voltage changes after calibration, see following section on voltage and temperature sensitivity[A].
2) Pull-dn ODT resistors are recommended to be calibrated at 0.33×V_{DD2}. Other calibration schemes may be used to achieve the linearity spec shown above, e.g. calibration at 0.5×V_{DD2} and 0.1×V_{DD2}.
3) Measurement definition for RTT:TBD
4) CA to CA mismatch within clock group (CA,CS) variation for a given component including CK_t and CK_c (characterized).

$$\text{CA - CA}_{\text{Mismatch}} = \frac{\text{RODT(max) - RODT(min)}}{\text{RODT(avg)}}$$

NOTE:
A. As of publication of this document, under discussion by the formulating committee.

[Table 64] ODT DC Electrical Characteristics, assuming RZQ = 240ohm +/-1% over the entire operating temperature range after a proper ZQ calibration for beyond 3200Mbps

MR11 [6:4]	RTT	Vout	Min	Nom	Max	Unit	Note
001	240ohm	VOLdc= 0.1 × VDD2	0.8	1.0	1.1	RZQ	1,2,3
		VOMdc= 0.33 × VDD2	0.9	1.0	1.1	RZQ	1,2,3
		VOHdc= 0.5 × VDD2	0.9	1.0	1.3	RZQ	1,2,3
010	120ohm	VOLdc= 0.1 × VDD2	0.8	1.0	1.1	RZQ/2	1,2,3
		VOMdc= 0.33 × VDD2	0.9	1.0	1.1	RZQ/2	1,2,3
		VOHdc= 0.5 × VDD2	0.9	1.0	1.3	RZQ/2	1,2,3
011	80ohm	VOLdc= 0.1 × VDD2	0.8	1.0	1.1	RZQ/3	1,2,3
		VOMdc= 0.33 × VDD2	0.9	1.0	1.1	RZQ/3	1,2,3
		VOHdc= 0.5 × VDD2	0.9	1.0	1.3	RZQ/3	1,2,3
100	60ohm	VOLdc= 0.1 × VDD2	0.8	1.0	1.1	RZQ/4	1,2,3
		VOMdc= 0.33 × VDD2	0.9	1.0	1.1	RZQ/4	1,2,3
		VOHdc= 0.5 × VDD2	0.9	1.0	1.3	RZQ/4	1,2,3
101	48ohm	VOLdc= 0.1 × VDD2	0.8	1.0	1.1	RZQ/5	1,2,3
		VOMdc= 0.33 × VDD2	0.9	1.0	1.1	RZQ/5	1,2,3
		VOHdc= 0.5 × VDD2	0.9	1.0	1.3	RZQ/5	1,2,3
110	40ohm	VOLdc= 0.1 × VDD2	0.8	1.0	1.1	RZQ/6	1,2,3
		VOMdc= 0.33 × VDD2	0.9	1.0	1.1	RZQ/6	1,2,3
		VOHdc= 0.5 × VDD2	0.9	1.0	1.3	RZQ/6	1,2,3
Mismatch CA-CA within clk group		0.33× VDD2	-		TBD	%	1,2,4

NOTE :
1) The tolerance limits are specified after calibration with stable voltage and temperature. For the behavior of the tolerance limits if temperature or voltage changes after calibration, see following section on voltage and temperature sensitivity1.
2) Pull-dn ODT resistors are recommended to be calibrated at 0.33×VDD2. Other calibration schemes may be used to achieve the linearity spec shown above, e.g. calibration at 0.5×VDD2 and 0.1×VDD2.
3) Measurement definition for RTT:TBD
4) CA to CA mismatch within clock group (CA,CS) variation for a given component including CK_t and CK_c (characterized).

$$\text{CA - CA}_{\text{Mismatch}} = \frac{\text{RODT(max) - RODT(min)}}{\text{RODT(avg)}}$$

NOTE :
A. As of publication of this document, under discussion by the formulating committee.

3.46.3 ODT for Command/Address update time

ODT for Command/Address update time after Mode Register set are shown in Figure 127.

ODT for Command/Address update time after Mode Register set are shown in Figure 118.

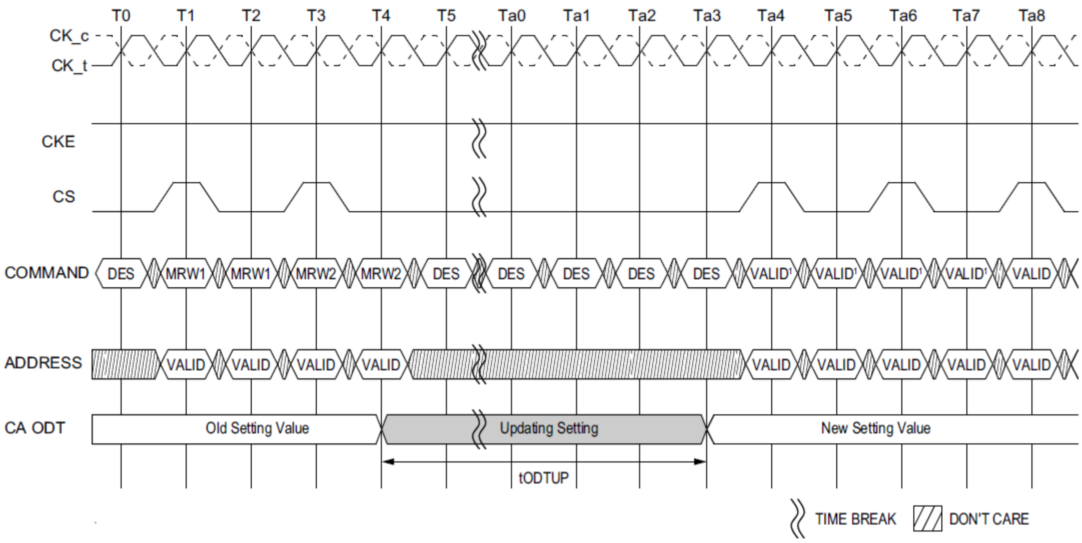

Figure 127: ODT for Command/Address setting update timing in 4 Clock Cycle Command

NOTE :
1) 4 Clock Cycle Command.

[Table 65] ODT CA AC Timing

Speed		LPDDR4-1600/1866/2133/2400/3200/4266		Unit	NOTE
Parameter	Symbol	MIN	MAX		
ODT CA Value Update Time	t_{ODTUP}	RU(TBDns/$t_{CK(avg)}$)	-		

3.47 On-Die Termination

ODT (On-Die Termination) is a feature of the LPDDR4 SDRAM that allows the DRAM to turn on/off termination resistance for each DQ, DQS_t, DQS_c and DMI signals without the ODT control pin. The ODT feature is designed to improve signal integrity of the memory channel by allowing the DRAM controller to turn on and off termination resistance for any target DRAM devices during Write or Mask Write operation.
The ODT feature is off and cannot be supported in Power Down and Self-Refresh modes.
A simple functional representation of the DRAM ODT feature is shown in Figure 128.

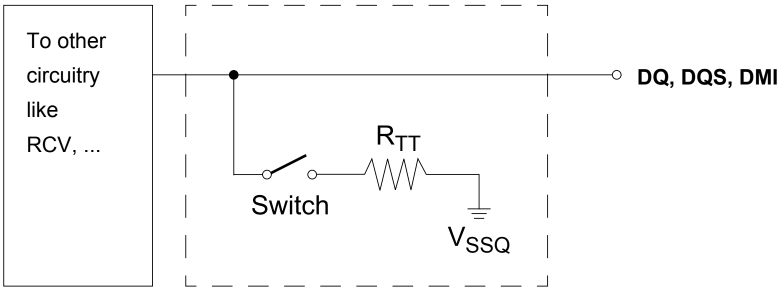

Figure 128: Functional Representation of ODT

The switch is enabled by the internal ODT control logic, which uses the Write-1 or Mask Write-1 command and other mode register control information. The value of RTT is determined by the settings of Mode Register bits.

3.47.1 ODT Mode Register

The ODT Mode is enabled if MR11 OP[3:0] are non zero. In this case, the value of RTT is determined by the settings of those bits. The ODT Mode is disabled if MR11 OP[3] = 0.

3.47.2 Asynchronous ODT

When ODT Mode is enabled in MR11 OP[3:0], DRAM ODT is always Hi-Z. DRAM ODT feature is automatically turned ON asynchronously based on the Write-1 or Mask Write-1 command that DRAM samples. After the write burst is complete, DRAM ODT featured is automatically turned OFF asynchronously.
Following timing parameters apply when DRAM ODT mode is enabled:

- ODTLon, $t_{ODTon,min}$, $t_{ODTon,max}$
- ODTLoff, $t_{ODToff,min}$, $t_{ODToff,max}$

ODTLon is a synchronous parameter and it is the latency from CAS-2 command to t_{ODTon} reference. ODTLon latency is a fixed latency value for each speed bin. Each speed bin has a different ODTLon latency.
Minimum RTT turn-on time ($t_{ODTon,min}$) is the point in time when the device termination circuit leaves high impedance state and ODT resistance begins to turn on.
Maximum RTT turn on time ($t_{ODTon,max}$) is the point in time when the ODT resistance is fully on.
$t_{ODTon,min}$ and tODTon,max are measured once ODTLon latency is satisfied from CAS-2 command.
ODTLoff is a synchronous parameter and it is the latency from CAS-2 command to t_{ODToff} reference. ODTLoff latency is a fixed latency value for each speed bin. Each speed bin has a different ODTLoff latency.
Minimum RTT turn-off time ($t_{ODToff,min}$) is the point in time when the device termination circuit starts to turn off the ODT resistance.
Maximum ODT turn off time ($t_{ODToff,max}$) is the point in time when the on-die termination has reached high impedance.
$t_{ODToff,min}$ and $t_{ODToff,max}$ are measured once ODTLoff latency is satisfied from CAS-2 command.

[Table 66] ODTLon and ODTLoff Latency Values

ODTLon Latency[1] t_{WPRE} = 2 tCK		ODTLoff Latency[2]		Lower Clock Frequency Limit (>)	Upper Clock Frequency Limit(≤)
WL Set “A”	WL Set “B”	WL Set “A”	WL Set “B”		
N/A	N/A	N/A	N/A	10	266
N/A	N/A	N/A	N/A	266	533
N/A	6	N/A	22	533	800
4	12	20	28	800	1066
4	14	22	32	1066	1333
6	18	24	36	1333	1600
6	20	26	40	1600	1866
8	24	28	44	1866	2133
nCK	nCK	nCK	nCK	MHz	MHz

NOTE:
1) ODTLon is referenced from CAS-2 command. See Figure 129.
2) ODTLoff as shown in table assumes BL=16. For BL32, 8 t_{CK} should be added.

[Table 67] Asynchronous ODT Turn On and Turn Off Timing

Parameter	800 - 2133MHz	Unit
$t_{ODTon, min}$	1.5	ns
$t_{ODTon, max}$	3.5	ns
$t_{ODToff, min}$	1.5	ns
$t_{ODToff, max}$	3.5	ns

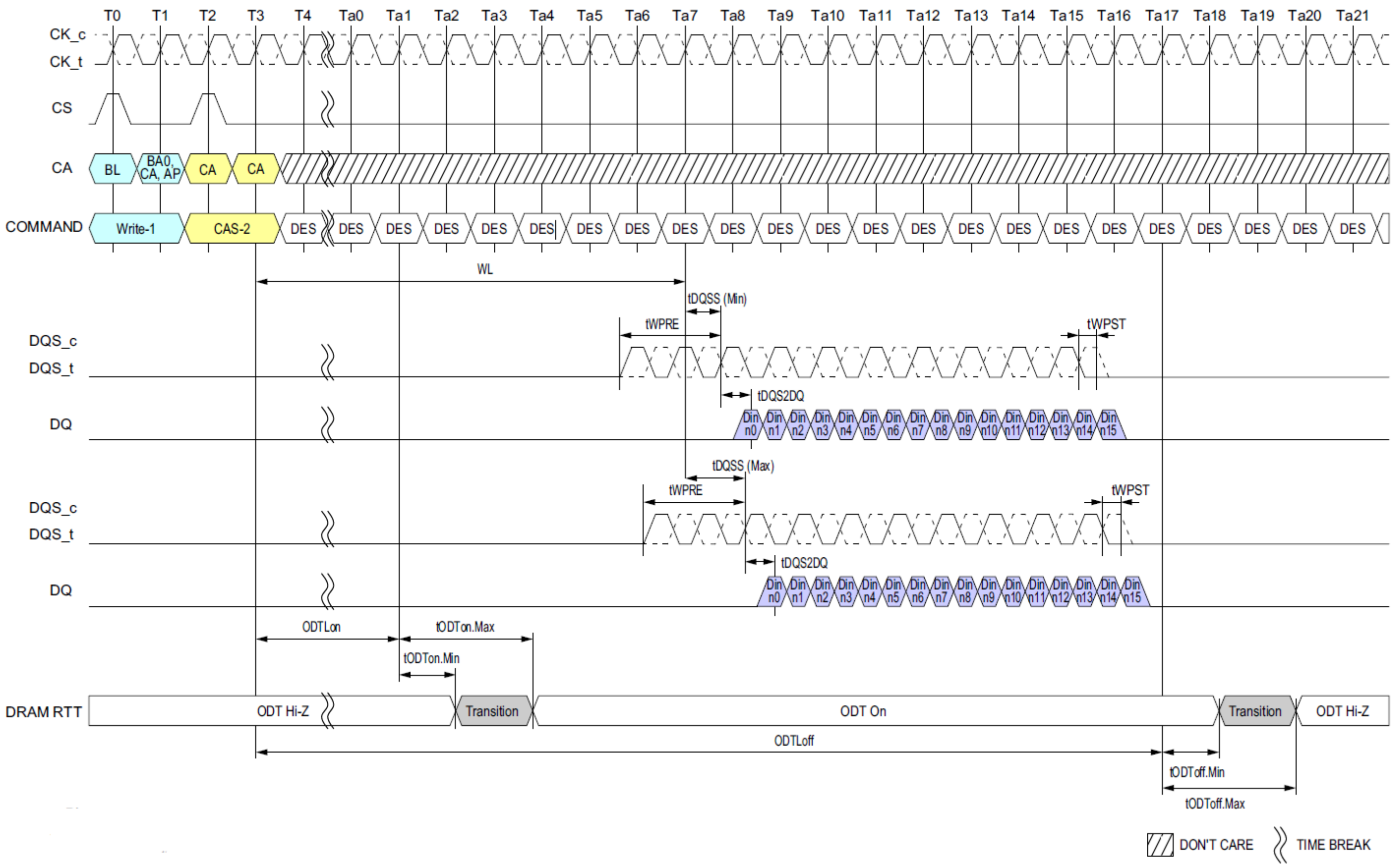

Figure 129: Asynchronous ODTon/ODToff Timing

NOTE :
1) BL=16, Write Postamble = 0.5nCK, DQ/DQS: VSSQ termination.
2) Din n = data-in to columnm.n.
3) DES commands are shown for ease of illustration; other commands may be valid at these times.

3.47.3 ODT during Write Leveling

If ODT is enabled in MR11 OP[3:0], in Write Leveling mode, DRAM always provides the termination on DQS_t/DQS_c signals. DQ termination is always off in Write Leveling mode regardless.

[Table 68] DRAM Termination Function In Write Leveling Mode

ODT Enabled in MR11	DQS_t/DQS_c termination	DQ termination
Disabled	OFF	OFF
Enabled	ON	OFF

3.48 On Die Termination For DQ, DQS And DMI

On-Die Termination effective resistance RTT is defined by MR11 OP[2:0].
ODT is applied to the DQ, DMI, DQS_t and DQS_c pins.
A functional representation of the on-die termination is shown in the figure below.

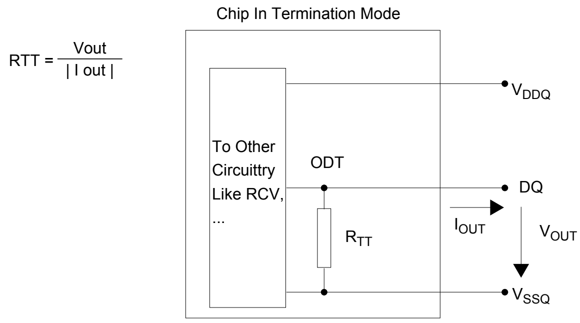

Figure 130: On Die Termination

[Table 69] ODT DC Electrical Characteristics, assuming RZQ = 240 ohm +/-1% over the entire operating temperature range after a proper ZQ calibration for up to 3200Mbps.

MR11 [2:0]	RTT	Vout	Min	Nom	Max	Unit	Note
001	240ohm	VOLdc= 0.1 × VDDQ	0.8	1.0	1.1	RZQ	1,2,3
		VOMdc= 0.33 × VDDQ	0.9	1.0	1.1	RZQ	1,2,3
		VOHdc= 0.5 × VDDQ	0.9	1.0	1.2	RZQ	1,2,3
010	120ohm	VOLdc= 0.1 × VDDQ	0.8	1.0	1.1	RZQ/2	1,2,3
		VOMdc= 0.33 × VDDQ	0.9	1.0	1.1	RZQ/2	1,2,3
		VOHdc= 0.5 × VDDQ	0.9	1.0	1.2	RZQ/2	1,2,3
011	80ohm	VOLdc= 0.1 × VDDQ	0.8	1.0	1.1	RZQ/3	1,2,3
		VOMdc= 0.33 × VDDQ	0.9	1.0	1.1	RZQ/3	1,2,3
		VOHdc= 0.5 × VDDQ	0.9	1.0	1.2	RZQ/3	1,2,3
100	60ohm	VOLdc= 0.1 × VDDQ	0.8	1.0	1.1	RZQ/4	1,2,3
		VOMdc= 0.33 × VDDQ	0.9	1.0	1.1	RZQ/4	1,2,3
		VOHdc= 0.5 × VDDQ	0.9	1.0	1.2	RZQ/4	1,2,3
101	48ohm	VOLdc= 0.1 × VDDQ	0.8	1.0	1.1	RZQ/5	1,2,3
		VOMdc= 0.33 × VDDQ	0.9	1.0	1.1	RZQ/5	1,2,3
		VOHdc= 0.5 × VDDQ	0.9	1.0	1.2	RZQ/5	1,2,3
110	40ohm	VOLdc= 0.1 × VDDQ	0.8	1.0	1.1	RZQ/6	1,2,3
		VOMdc= 0.33 × VDDQ	0.9	1.0	1.1	RZQ/6	1,2,3
		VOHdc= 0.5 × VDDQ	0.9	1.0	1.2	RZQ/6	1,2,3
Mismatch DQ-DQ within byte		0.33 × VDDQ	-		2	%	1,2,4

NOTE :
1) The tolerance limits are specified after calibration with stable voltage and temperature. For the behavior of the tolerance limits if temperature or voltage changes after calibration, see following section on voltage and temperature sensitivity[A].
2) Pull-dn ODT resistors are recommended to be calibrated at 0.33×V_{DDQ}. Other calibration schemes may be used to achieve the linearity spec shown above, e.g. calibration at 0.5×VDDQ and 0.1×VDDQ.
3) Measurement definition for RTT:TBD
4) DQ to DQ mismatch within byte variation for a given component including DQS_t and DQS_c (characterized).

$$\text{DQ - DQmismatch} = \frac{\text{RODT(max) - RODT(min)}}{\text{RODT(avg)}}$$

NOTE :
A) As of publication of this document, under discussion by the formulating committee.

[Table 70] ODT DC Electrical Characteristics, assuming RZQ = 240 ohm +/-1% over the entire operating temperature range after a proper ZQ calibration for beyond 3200Mbps.

MR11 [2:0]	RTT	Vout	Min	Nom	Max	Unit	Note
001	240ohm	VOLdc= 0.1 × V_{DDQ}	0.8	1.0	1.1	RZQ	1,2,3
		VOMdc= 0.33 × V_{DDQ}	0.9	1.0	1.1	RZQ	1,2,3
		VOHdc= 0.5 × V_{DDQ}	0.9	1.0	1.3	RZQ	1,2,3
010	120ohm	VOLdc= 0.1 × V_{DDQ}	0.8	1.0	1.1	RZQ/2	1,2,3
		VOMdc= 0.33 × V_{DDQ}	0.9	1.0	1.1	RZQ/2	1,2,3
		VOHdc= 0.5 × V_{DDQ}	0.9	1.0	1.3	RZQ/2	1,2,3
011	80ohm	VOLdc= 0.1 × V_{DDQ}	0.8	1.0	1.1	RZQ/3	1,2,3
		VOMdc= 0.33 × V_{DDQ}	0.9	1.0	1.1	RZQ/3	1,2,3
		VOHdc= 0.5 × V_{DDQ}	0.9	1.0	1.3	RZQ/3	1,2,3
100	60ohm	VOLdc= 0.1 × V_{DDQ}	0.8	1.0	1.1	RZQ/4	1,2,3
		VOMdc= 0.33 × V_{DDQ}	0.9	1.0	1.1	RZQ/4	1,2,3
		VOHdc= 0.5 × V_{DDQ}	0.9	1.0	1.3	RZQ/4	1,2,3
101	48ohm	VOLdc= 0.1 × V_{DDQ}	0.8	1.0	1.1	RZQ/5	1,2,3
		VOMdc= 0.33 × V_{DDQ}	0.9	1.0	1.1	RZQ/5	1,2,3
		VOHdc= 0.5 × V_{DDQ}	0.9	1.0	1.3	RZQ/5	1,2,3
110	40ohm	VOLdc= 0.1 × V_{DDQ}	0.8	1.0	1.1	RZQ/6	1,2,3
		VOMdc= 0.33 × V_{DDQ}	0.9	1.0	1.1	RZQ/6	1,2,3
		VOHdc= 0.5 × V_{DDQ}	0.9	1.0	1.3	RZQ/6	1,2,3
Mismatch DQ-DQ within byte		0.33 × V_{DDQ}	-		2	%	1,2,4

NOTE :
1) The tolerance limits are specified after calibration with stable voltage and temperature. For the behavior of the tolerance limits if temperature or voltage changes after calibration, see following section on voltage and temperature sensitivity[A)].
2) Pull-dn ODT resistors are recommended to be calibrated at 0.33*VDDQ. Other calibration schemes may be used to achieve the linearity spec shown above, e.g. calibration at 0.5×VDDQ and 0.1×VDDQ.
3) Measurement definition for RTT: TBD
4) DQ to DQ mismatch within byte variation for a given component including DQS_t and DQS_c (characterized).

$$DQ\text{ - }DQmismatch = \frac{RODT(max) - RODT(min)}{RODT(avg)}$$

NOTE :
A) As of publication of this document, under discussion by the formulating committee.

3.49 ODT Timing Constraints

3.49.1 Same bank in same rank (ODT On)

[Units : tCK]

Next CMD / Current CMD	Active	Read	Write	Masked Write	Precharge
Active	illegal	$RU(t_{RCD}/t_{CK})$	$RU(t_{RCD}/t_{CK})$	$RU(t_{RCD}/t_{CK})$	$RU(t_{RAS}/t_{CK})$
Read	illegal	BL/2	$RL+RU(t_{DQSCK(max)}/t_{CK})+BL/2+RD(t_{RPST})-ODTLon-RD(t_{ODTon(min)}/t_{CK})$	$RL+RU(t_{DQSCK(max)}/t_{CK})+BL/2+RD(t_{RPST})-ODTLon-RD(t_{ODTon(min)}/t_{CK})$	$BL/2+max\{(8,RU(t_{RTP}/t_{CK})\}-8$
Write	illegal	$WL+1+BL/2+RU(t_{WTR}/t_{CK})$	BL/2	"t_{CCDMW}" for BL16 Write "$t_{CCDMW}+8$" for BL32 Write	$WL+1+BL/2+RU(t_{WR}/t_{CK})$
Masked Write	illegal	$WL+1+BL/2+RU(t_{WTR}/t_{CK})$	BL/2	t_{CCDMW} 1)	$WL+1+BL/2+RU(t_{WR})/t_{CK})$
Precharge	$RU(t_{RP}/t_{CK})$ or $RU(t_{RPab}/t_{CK})$	illegal	illegal	illegal	$4(t_{PPD})$

NOTE :
1) $t_{CCDMW} = 4 \times t_{CCD}$ ($=32t_{CK}$).

3.49.2 Different bank in same rank (ODT On)

[Units : tCK]

Next CMD / Current CMD	Active	Read	Write	Masked Write	Precharge
Active	$RU(t_{RRD}/t_{CK})$	4	4	4	2
Read	4	BL/2	$RL+RU(t_{DQSCK(max)}/t_{CK})+BL/2+RD(t_{RPST})-ODTLon-RD(t_{ODTon(min)}/t_{CK})$	$RL+RU(t_{DQSCK(max)}/t_{CK})+BL/2+RD(t_{RPST})-ODTLon-RD(t_{ODTon(min)}/t_{CK})$	2
Write	4	$WL+1+BL/2+RU(t_{WTR}/t_{CK})$	BL/2	BL/2	2
Masked Write	4	$WL+1+BL/2+RU(t_{WTR}/t_{CK})$	BL/2	BL/2	2
Precharge	4	4	4	4	$4(t_{PPD})$

NOTE :
1) RD=Round Down.

3.49.3 Different rank (ODT On)

[Units : tCK]

Next CMD / Current CMD	Active	Read	Write	Masked Write	Precharge
Active	4	4	4	4	2
Read	4	BL/2 + RU[(t_{DQSCK}(max) - t_{DQSCK}(min))/t_{CK}] + RD(t_{RPST}+t_{RPRE}	RL+RU($t_{DQSCK(max)}$/t_{CK}) +BL/2+RD(t_{RPST})-ODT-Lon-RD($t_{ODTon(min)}$/t_{CK})	RL+RU($t_{DQSCK(max)}$ / t_{CK}) +BL/2+RD(t_{RPST})-ODT-Lon-RD($t_{ODTon(min)}$ /t_{CK})	2
Write	4	WL+BL/2 + 2 + RU($t_{OD-Toff, (max)}$ /t_{CK})-RD($t_{DQSCK(min)}$/t_{CK}) - RL + t_{RPRE}	BL/2 + 1 + RU($t_{OD-Toff(max)}$ /t_{CK}) + t_{WPRE}	BL/2 + 1 +RU(t_{ODToff} (max) / t_{CK}) + t_{WPRE}	2
Masked Write	4	WL+BL/2 + 2 + RU($t_{OD-Toff(max)}$ /t_{CK}) - RD($t_{DQSCK(min)}$ / t_{CK}) - RL + t_{RPRE}	BL/2 + 1 + RU($t_{OD-Toff(max)}$ /t_{CK}) + t_{WPRE}	BL/2 + 1 + RU(t_{ODToff} (max)/t_{CK}) + t_{WPRE}	2
Precharge	4	4	4	4	2

NOTE :
1) RD=Round Down.

3.49.4 Same bank in same rank (ODT Off)

[Units : tCK]

Next CMD / Current CMD	Active	Read	Write	Masked Write	Precharge
Active	illegal	RU(t_{RCD}/t_{CK})	RU(t_{RCD}/t_{CK})	RU(t_{RCD}/t_{CK})	RU(t_{RAS}/t_{CK})
Read	illegal	BL/2	RL+RU($t_{DQSCK(max)}$/t_{CK}) +BL/2+RD(t_{RPST}) - WL+ t_{WPRE}	RL+RU($t_{DQSCK(max)}$/t_{CK}) +BL/2+RD(t_{RPST}) - WL+ t_{WPRE}	BL/2+max[(8,RU(t_{RTP}/ t_{CK})]-8
Write	illegal	WL+1+BL/2 +RU(t_{WTR}/t_{CK})	BL/2	"t_{CCDMW}" for BL16 Write "t_{CCDMW}+8" for BL32 Write	WL+ 1 + BL/2 +RU(t_{WR}/t_{CK})
Masked Write	illegal	WL+1+BL/2 +RU(t_{WTR}/t_{CK})	BL/2	t_{CCDMW} 2)	WL+ 1 + BL/2 +RU(t_{WR})/t_{CK})
Precharge	RU(t_{RP}/t_{CK}) or RU(t_{RPab}/t_{CK})	illegal	illegal	illegal	4(t_{PPD})

NOTE :
1) RD=Round Down.
2) t_{CCDMW} = 4×t_{CCD} (=32t_{CK}).

3.49.5 Different bank in same rank (ODT Off)

[Units : tCK]

Next CMD / Current CMD	Active	Read	Write	Masked Write	Precharge
Active	RU(t_{RRD}/t_{CK})	4	4	4	2
Read	4	BL/2	RL+RU($t_{DQSCK(max)}$/t_{CK}) +BL/2+RD(t_{RPST})- WL+ t_{WPRE}	RL+RU($t_{DQSCK(max)}$/t_{CK}) +BL/2+RD(t_{RPST})- WL+ t_{WPRE}	2
Write	4	WL+1+BL/2 +RU(t_{WTR}/t_{CK})	BL/2	BL/2	2
Masked Write	4	WL+1+BL/2 +RU(t_{WTR}/t_{CK})	BL/2	BL/2	2
Precharge	4	4	4	4	4 (t_{PPD})

NOTE :
1) RD=Round Down.

3.49.6 Different rank (ODT Off)

[Units : tCK]

Next CMD / Current CMD	Active	Read	Write	Masked Write	Precharge
Active	4	4	4	4	2
Read	4	BL/2 + RU[($t_{DQSCK(max)}$ - $t_{DQSCK(min)}$)/t_{CK}] + RD[t_{RPST}] + t_{RPRE}	RL+ RU[$t_{DQSCK(max)}$/t_{CK}] +BL/2 + RD[t_{RPST}] - WL+t_{WPRE}	RL+ RU[$t_{DQSCK(max)}$/t_{CK}] +BL/2 + RD[t_{RPST}] - WL+ t_{WPRE}	2
Write	4	WL+BL/2 + 1 + RD[t_{WPST}] - RL - RD[$t_{DQSCK(min)}$/t_{CK}] + t_{RPRE}	BL/2+1+RD[t_{WPST}]	BL/2+1+RD[t_{RPST}]	2
Masked Write	4	WL+BL/2 + 1 + RD[t_{WPST}] - RL - RD[$t_{DQSCK(min)}$/t_{CK}] + t_{RPRE}	BL/2+1+RD[t_{WPST}]	BL/2+1+RD[t_{RPST}]	2
Precharge	4	4	4	4	2

NOTE :
1) RD=Round Down.
2) Write (or Masked Write) - Write (or Masked Write) may need one clock cycle bubble due to t_{DQS2DQ} rank to rank variation.
3) 1.5 tCK write postamble increases gap when ODT is off, due to variation in t_{DQS2DQ}.

3.50 Output Driver and Termination Register Temperature and Voltage Sensitivity

If temperature and/or voltage change after calibration, the tolerance limits widen according to the Tables shown below.

[Table 71] Output Driver and Termination Register Sensitivity Definition

Resistor	Definition Point	Min	Max	Unit	Notes
R_{ONPD}	$0.33 \times V_{DDQ}$	$90-(dR_{ON}dT \times \|\Delta T\|)-(dR_{ON}dV \times \|\Delta V\|)$	$110+(dR_{ON}dT \times \|\Delta T\|)+(dR_{ON}dV \times \|\Delta V\|)$	%	1,2
VOH_{PU}	$0.33 \times V_{DDQ}$	$90-(dV_{OH}dT \times \|\Delta T\|)-(dV_{OH}dV \times \|\Delta V\|)$	$110+(dV_{OH}dT \times \|\Delta T\|)+(dV_{OH}dV \times \|\Delta V\|)$	%	1,2,5
$R_{TT(I/O)}$	$0.33 \times V_{DDQ}$	$90-(dR_{ON}dT \times \|\Delta T\|)-(dR_{ON}dV \times \|\Delta V\|)$	$110+(dR_{ON}dT \times \|\Delta T\|)+(dR_{ON}dV \times \|\Delta V\|)$	%	1,2,3
$R_{TT(In)}$	$0.33 \times V_{DD2}$	$90-(dR_{ON}dT \times \|\Delta T\|)-(dR_{ON}dV \times \|\Delta V\|)$	$110+(dR_{ON}dT \times \|\Delta T\|)+(dR_{ON}dV \times \|\Delta V\|)$	%	1,2,4

NOTE :
1) $\Delta T = T - T(@ \text{Calibration})$, $\Delta V = V - V(@ \text{Calibration})$.
2) $dR_{ON}dT$, dRONdV, $dV_{OH}dT$, $dV_{OH}dV$, $dR_{TT}dV$, and dRTTdT are not subject to production test but are verified by design and characterization.
3) This parameter applies to Input/Output pin such as DQS, DQ and DMI.
4) This parameter applies to Input pin such as CK, CA and CS.
5) Refer to Pull Up/Pull Down Driver Characteristics for VOH_{PU}.

[Table 72] Output Driver and Termination Register Temperature and Voltage Sensitivity

Symbol	Parameter	Min	Max	Unit
$dR_{ON}dT$	R_{ON} Temperature Sensitivity	0.00	0.75	%/°C
$dR_{ON}dV$	R_{ON} Voltage Sensitivity	0.00	0.20	%/mV
$dV_{OH}dT$	V_{OH} Temperature Sensitivity	0.00	0.75	%/°C
$dV_{OH}dV$	V_{OH} Voltage Sensitivity	0.00	0.35	%/mV
$dR_{TT}dT$	R_{TT} Temperature Sensitivity	0.00	0.75	%/°C
$dR_{TT}dV$	R_{TT} Voltage Sensitivity	0.00	0.20	%/mV

3.51 Power-Down Mode

3.51.1 Power-Down Entry And Exit

Power-down is asynchronously entered when CKE is driven LOW. CKE must not go LOW while the following operations are in progress.

- Mode Register Read
- Mode Register Write
- Read
- Write
- Vref(CA) Range and Value setting via MRW
- Vref(DQ) Range and Value setting via MRW
- Command Bus Training mode Entering/Exiting via MRW
- VRCG High Current mode Entering/Exiting via MRW

And the LPDDR4 DRAM cannot be placed in power-down state during “Start DQS Interval Oscillator” operation.

CKE can go LOW while any other operations such as row activation, Precharge, Auto Precharge, or Refresh are in progress. The power-down IDD specification will not be applied until such operations are complete. Power-down entry and exit are shown in Figure 131.

Entering power-down deactivates the input and output buffers, excluding CKE and Reset_n. To ensure that there is enough time to account for internal delay on the CKE signal path, CS input is required stable Low level and CA input level is don’t care after CKE is driven LOW, this timing period is defined as t_{CKELCS}. Clock input is required after CKE is driven LOW, this timing period is defined as t_{CKELCK}. CKE LOW will result in deactivation of all input receivers except Reset_n after t_{CKELCK} has expired. In power-down mode, CKE must be held LOW; all other input signals except Reset_n are “Don’t Care.” CKE LOW must be maintained until $t_{CKE,min}$ is satisfied.

No refresh operations are performed in power-down mode except Self-Refresh power-down. The maximum duration in non-Self-Refresh power-down mode is only limited by the refresh requirements outlined in the Refresh command section.

The power-down state is asynchronously exited when CKE is driven HIGH. CKE HIGH must be maintained until t_{CKE},min is satisfied. A valid, executable command can be applied with power-down exit latency t_{XP} after CKE goes HIGH. Power-down exit latency is defined in the AC timing parameter table.

Clock frequency change or Clock Stop is inhibited during t_{CMDCKE}, t_{CKELCK}, t_{CKCKEH}, t_{XP}, $t_{MRWCKEL}$ and t_{ZQCKE} periods.

If power-down occurs when all banks are idle, this mode is referred to as idle power-down; if power-down occurs when there is a row active in any bank, this mode is referred to as active power-down. And If power-down occurs when Self Refresh is in progress, this mode is referred to as Self Refresh power-down in which the internal refresh is continuing in the same way as Self Refresh mode.

VDDQ may be turned off during power-down after tCKELCK(Max(5ns,5nCK)) is satisfied (Refer to Figure 137 about t_{CKELCK}). Prior to exiting power-down, VDDQ must be within its minimum/maximum operating range.

When CA, CK and/or CS ODT is enabled via MR11 OP[6:4] and also via MR22 or CA-ODT pad setting, the rank providing ODT will continue to terminate the command bus in all DRAM states including power-down when VDDQ is stable and within its minimum/maximum operating range.

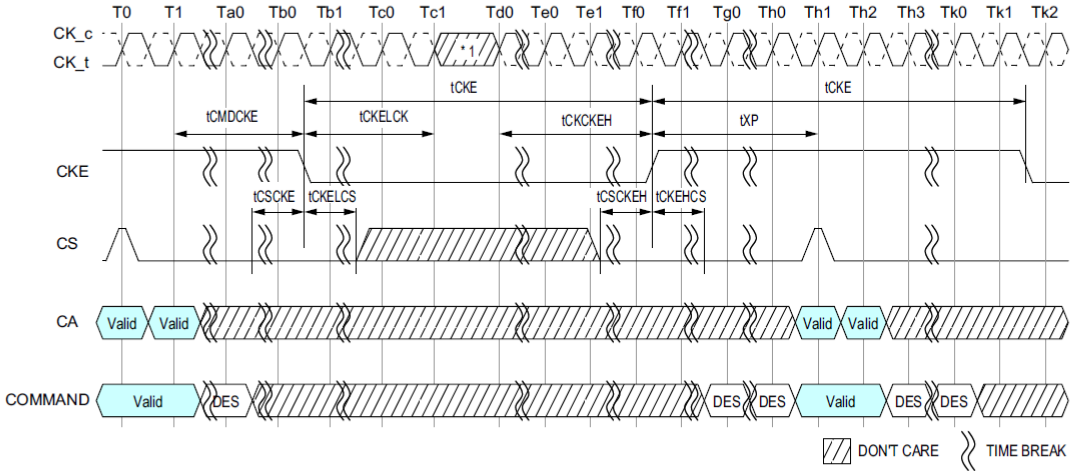

Figure 131: Basic power-down entry and exit timing

NOTE :
1) Input clock frequency can be changed or the input clock can be stopped or floated during power-down, provided that upon exiting power-down, the clock is stable and within specified limits for a minimum of t_{CKCKEH} of stable clock prior to power-down exit and the clock frequency is between the minimum and maximum specified frequency for the speed grade in use.

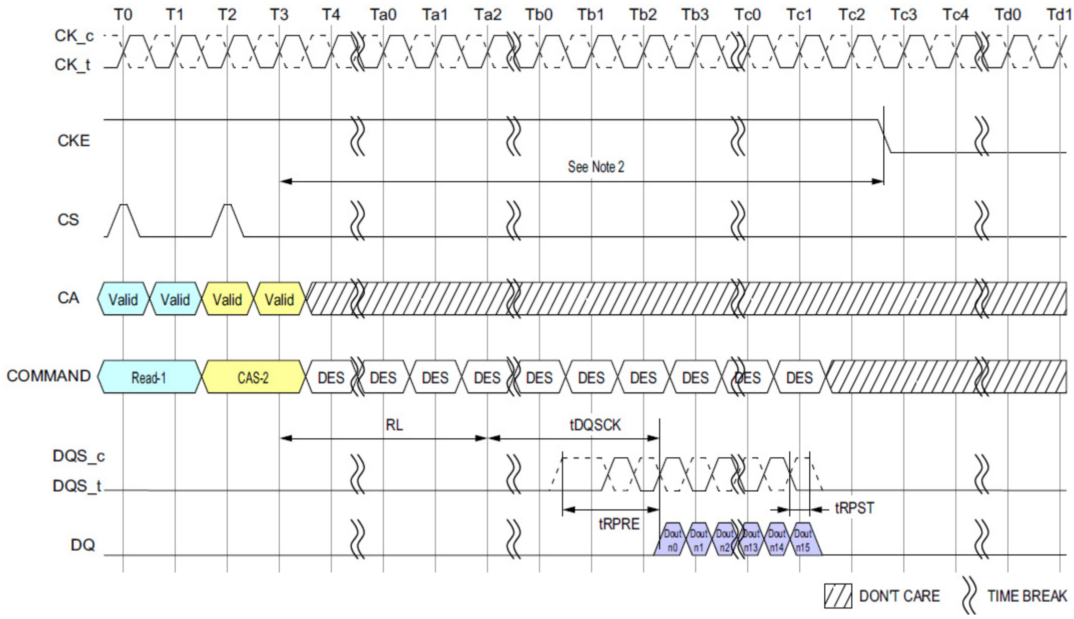

Figure 132: Read and Read with Auto Precharge to Power-Down Entry

NOTE :
1) CKE must be held HIGH until the end of the burst operation.
2) Minimum Delay time from Read Command or Read with Auto Precharge Command to falling edge of CKE signal is as follows.
Read Post-amble = 0.5nCK : MR1 OP[7]=0_B : (RL × t_{CK}) + $t_{DQSCK(Max)}$ + ((BL/2) × t_{CK}) + $1t_{CK}$.
Read Post-amble = 1.5nCK : MR1 OP[7]=1_B : (RL × t_{CK}) + $t_{DQSCK(Max)}$ + ((BL/2) × t_{CK}) + $2t_{CK}$.

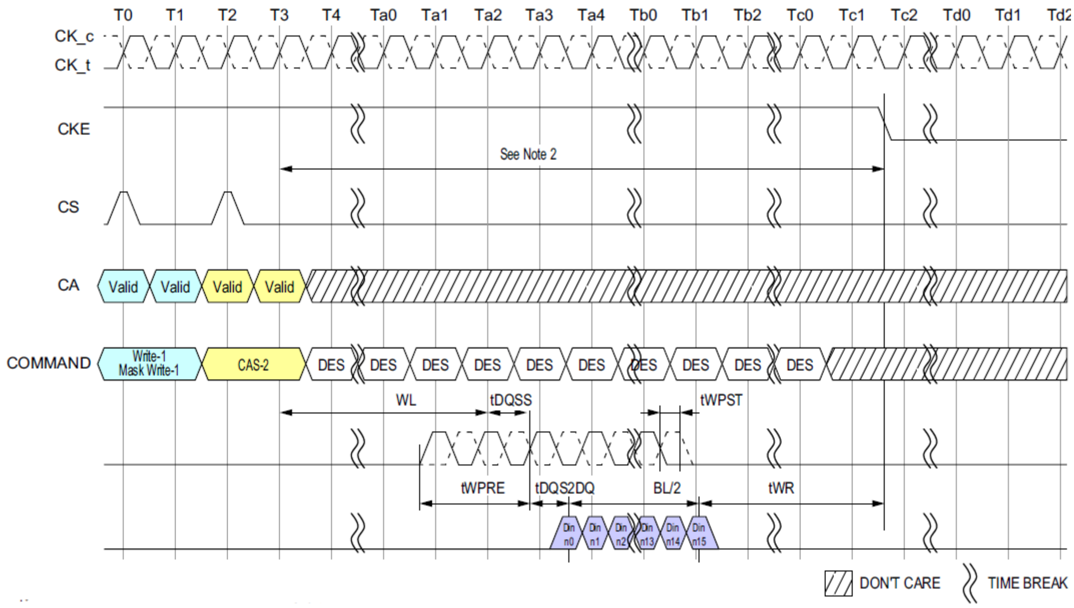

Figure 133: Write and Mask Write to Power-Down Entry

NOTE :

1) CKE must be held HIGH until the end of the burst operation.
2) Minimum Delay time from Write Command or Mask Write Command to falling edge of CKE signal is as follows.
$(WL \times t_{CK}) + t_{DQSS(Max)} + t_{DQS2DQ(Max)} + ((BL/2) \times t_{CK}) + t_{WR}$.
3) This timing is applied regardless of DQ ODT Disable/Enable setting: MR11[OP2:0].
4) This timing diagram only applies to the Write and Mask Write Commands without Auto Precharge.

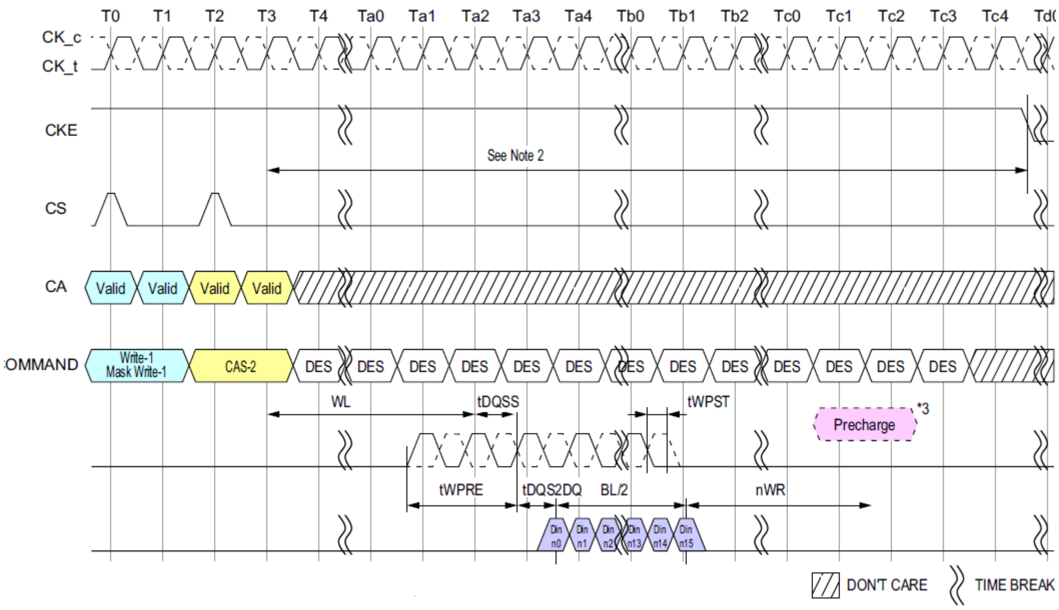

Figure 134: Write with Auto Precharge and Mask Write with Auto Precharge to Power-Down Entry

NOTE :

1) CKE must be held HIGH until the end of the burst operation.
2) Delay time from Write with Auto Precharge Command or Mask Write with Auto Precharge Command to falling edge of CKE signal is more than
$(WL \times t_{CK}) + t_{DQSS(Max)} + t_{DQS2DQ(Max)} + ((BL/2) \times t_{CK}) + (nWR \times t_{CK}) + (2 \times t_{CK})$.
3) Internal Precharge Command.
4) This timing is applied regardless of DQ ODT Disable/Enable setting: MR11[OP2:0].

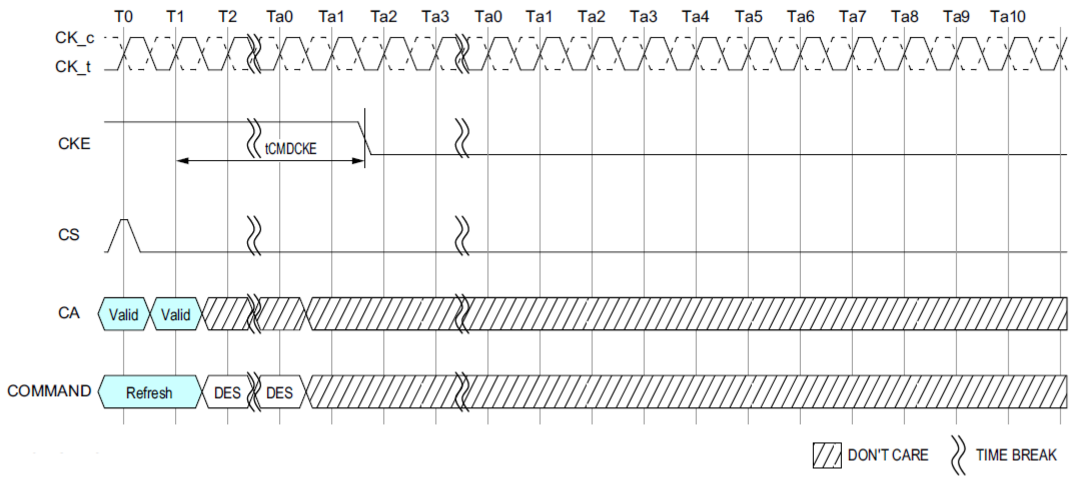

Figure 135: Refresh entry to Power-Down Entry

NOTE :
1) CKE must be held HIGH until t_{CMDCKE} is satisfied.

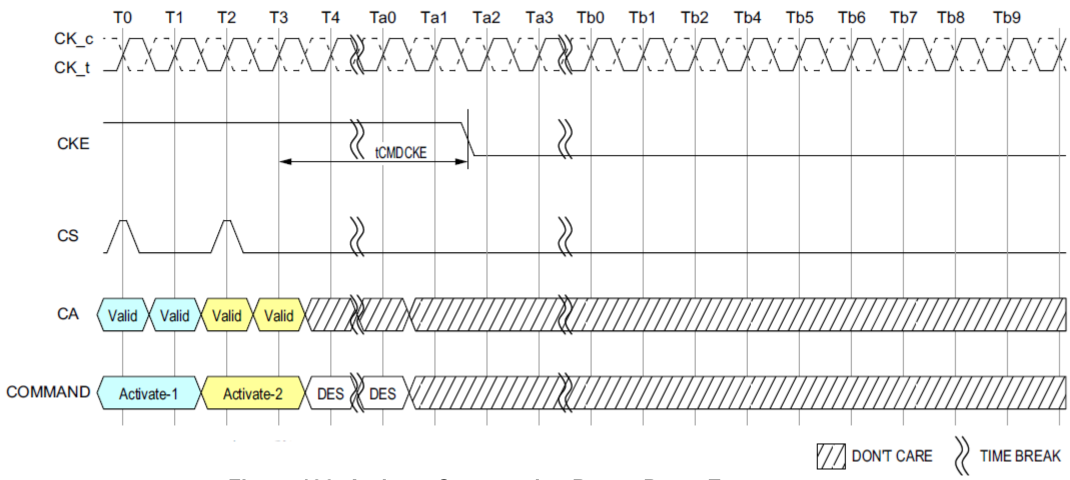

Figure 136: Activate Command to Power-Down Entry

NOTE :
1) CKE must be held HIGH until t_{CMDCKE} is satisfied.

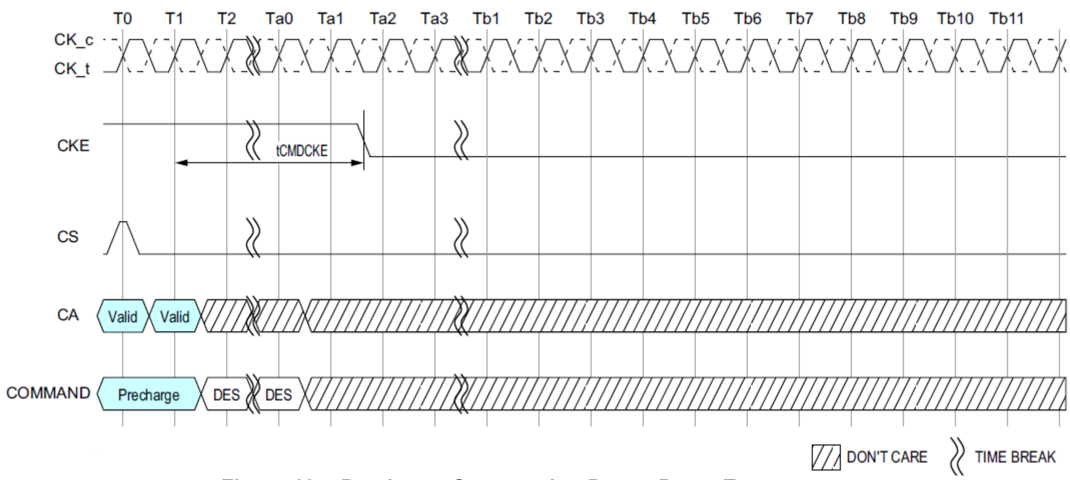

Figure 137: Precharge Command to Power-Down Entry

NOTE :
1) CKE must be held HIGH until t_{CMDCKE} is satisfied.

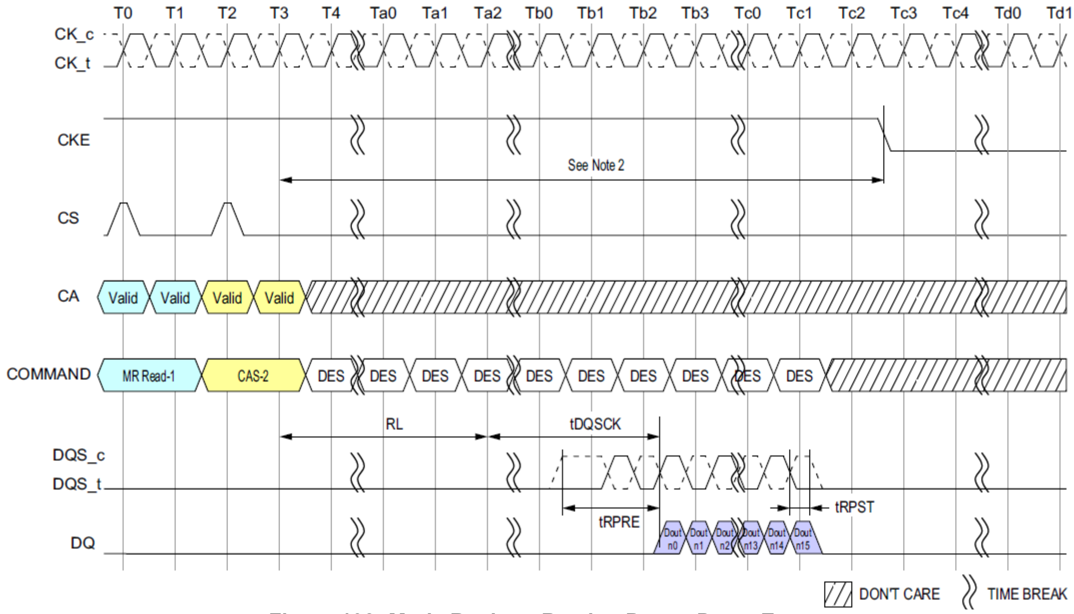

Figure 138: Mode Register Read to Power-Down Entry

NOTE :
1) CKE must be held HIGH until the end of the burst operation.
2) Minimum Delay time from Mode Register Read Command to falling edge of CKE signal is as follows:
Read Post-amble = 0.5nCK : MR1 OP[7]=0_B : (RL × t_{CK}) + $t_{DQSCK(Max)}$ + ((BL/2) × t_{CK}) + 1t_{CK}.
Read Post-amble = 1.5nCK : MR1 OP[7]=1_B : (RL × t_{CK}) + $t_{DQSCK(Max)}$ + ((BL/2) × t_{CK}) + 2t_{CK}.

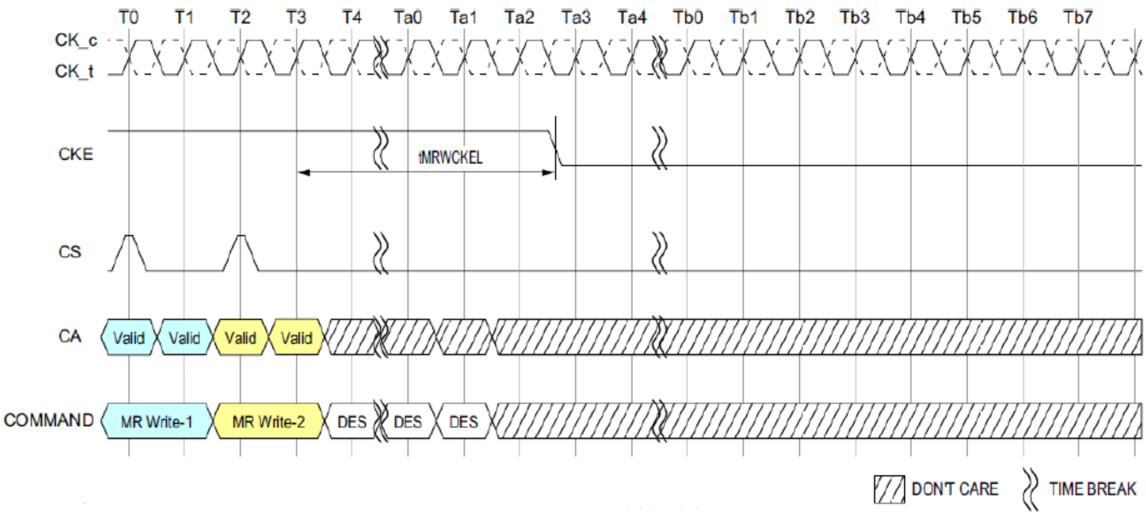

Figure 139: Mode Register Write to Power-Down Entry

NOTE :
1) CKE must be held HIGH until $t_{MRWCKEL}$ is satisfied.
2) This timing is the general definition for Power Down Entry after Mode Register Write Command.
When a Mode Register Write Command changes a parameter or starts an operation that requires special timing longer than $t_{MRWCKEL}$, that timing must be satisfied before CKE is driven low.
Changing the Vref(DQ) value is one example, in this case the appropriate Vref_time-Short/Middle/Long must be satisfied.

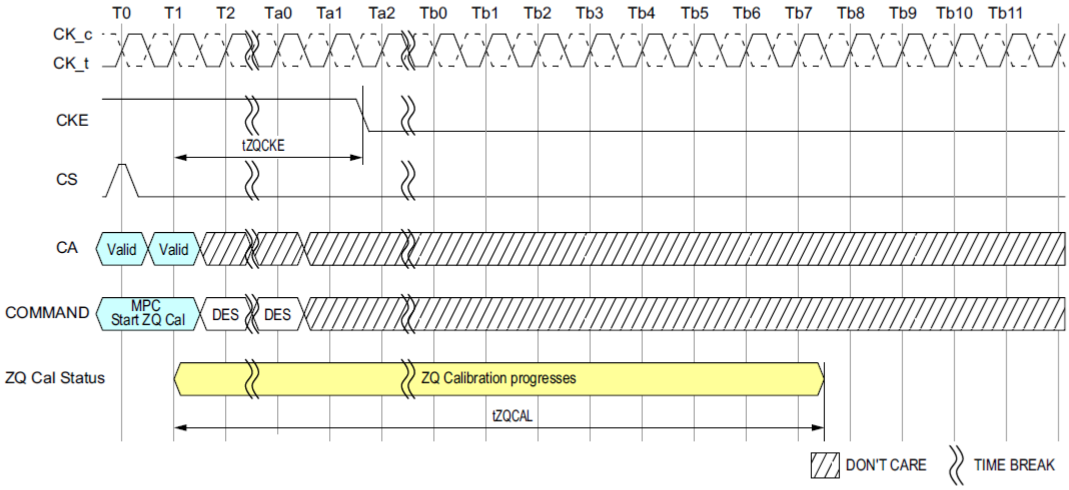

Figure 140: Multi purpose Command for Start ZQ Calibration to Power-Down Entry

NOTE :
1) ZQ Calibration continues if CKE goes low after t_{ZQCKE} is satisfied.

3.52 Input Clock Stop And Frequency Change

LPDDR4 SDRAMs support input clock frequency change during CKE LOW under the following conditions:
- $t_{CK(abs)}$min is met for each clock cycle;
- Refresh requirements apply during clock frequency change;
- During clock frequency change, only REFab or REFpb commands may be executing;
- Any Activate or Precharge commands have executed to completion prior to changing the frequency;
- The related timing conditions (t_{RCD}, t_{RP}) have been met prior to changing the frequency;
- The initial clock frequency shall be maintained for a minimum of tCKELCK after CKE goes LOW;
- The clock satisfies $t_{CH(abs)}$ and $t_{CL(abs)}$ for a minimum of tCKCKEH prior to CKE going HIGH.

After the input clock frequency is changed and CKE is held HIGH, additional MRW commands may be required to set the WR, RL etc. These settings may need to be adjusted to meet minimum timing requirements at the target clock frequency.

LPDDR4 devices support clock stop during CKE LOW under the following conditions:
- CK_t and CK_c are don’t care during clock stop;;
- Refresh requirements apply during clock stop;
- During clock stop, only REFab or REFpb commands may be executing;
- Any Activate or Precharge commands have executed to completion prior to stopping the clock;
- The related timing conditions (t_{RCD}, t_{RP}) have been met prior to stopping the clock;
- The initial clock frequency shall be maintained for a minimum of tCKELCK after CKE goes LOW;
- The clock satisfies $t_{CH(abs)}$ and $t_{CL(abs)}$ for a minimum of tCKCKEH prior to CKE going HIGH.

LPDDR4 devices support input clock frequency change during CKE HIGH under the following conditions:
- $t_{CK(abs)}$min is met for each clock cycle;
- Refresh requirements apply during clock frequency change;
- Any Activate, Read, Write, Precharge, Mode Register Write, or Mode Register Read commands must have executed to completion, including any associated data bursts prior to changing the frequency;
- The related timing conditions (t_{RCD}, t_{WR}, t_{WRA}, t_{RP}, t_{MRW}, t_{MRR}, etc.) have been met prior to changing the frequency;
- CS shall be held LOW during clock frequency change;
- During clock frequency change, only REFab or REFpb commands may be executing;
- The LPDDR4 SDRAM is ready for normal operation after the clock satisfies $t_{CH(abs)}$ and $t_{CL(abs)}$ for a minimum of 2 × t_{CK}+t_{XP}.

After the input clock frequency is changed, additional MRW commands may be required to set the WR, RL etc.
These settings may need to be adjusted to meet minimum timing requirements at the target clock frequency.

LPDDR4 devices support clock stop during CKE HIGH under the following conditions:
- CK_t is held LOW and CK_c is held HIGH during clock stop;
- CS shall be held LOW during clock stop;
- Refresh requirements apply during clock stop;
- During clock stop, only REFab or REFpb commands may be executing;
- Any Activate, Read, Write, MPC(WRFIFO, RDFIFO, RDDQCAL), Precharge, Mode Register Write or Mode Register Read commands must have executed to completion, including any associated data bursts and extra 4 clock cycles must be provided prior to stopping the clock;
- The related timing conditions (t_{RCD}, t_{WR}, t_{RP}, t_{MRW}, t_{MRR}, t_{ZQLAT} etc.) have been met prior to stopping the clock;
- Read with auto pre-charge and write with auto pre-charge commands need extra 4 clock cycles in addition to the related timing constraints, nWR and nRTP, to complete the operations.
- REFab, REFpb, SRE, SRX and MPC(Zqcal Start)commands are required to have 4 additional clocks prior to stopping the clock same as CKE=L case.
- The LPDDR4 SDRAM is ready for normal operation after the clock is restarted and satisfies $t_{CH(abs)}$ and $t_{CL(abs)}$ for a minimum of 2×t_{CK}+t_{XP}

3.53 Post Package Repair (PPR)

LPDDR4 supports Fail Row address repair as optional feature and it is readable through MR25 OP[7:0] PPR provides repair method in the system and Fail Row address can be repaired by the electrical programming of Electrical-fuse scheme.
With PPR, LPDDR4 can correct 1Row per Bank.
Electrical-fuse cannot be switched back to un-fused states once it is programmed. The controller should prevent unintended the PPR mode entry and repair.

3.53.1 Fail Row Address Repair

The following is procedure of PPR.
1. Before entering 'PPR' mode, All banks must be Precharged.
2. Enable PPR using MR4 bit "OP4=1" and wait t_{MRD}.
3. Issue ACT command with Fail Row address.
4. Wait t_{PGM} to allow DRAM repair target Row Address internally then issue PRE.
5. Wait t_{PGM_Exit} after PRE which allow DRAM to recognize repaired Row address RAn.
6. Exit PPR with setting MR4 bit "OP4=0".
7. Issue RESET command after t_{PGMPST}.
8. Repeat steps in 'Reset Initialization with Stable Power' section.
9. In More than one fail address repair case, Repeat Step 2 to 8.
Once PPR mode is exited, to confirm if target row is repaired correctly, host can verify by writing data into the target row and reading it back after PPR exit with MR4 [OP4=0] and t_{PGMPST}.

The following Timing diagram shows PPR operation.

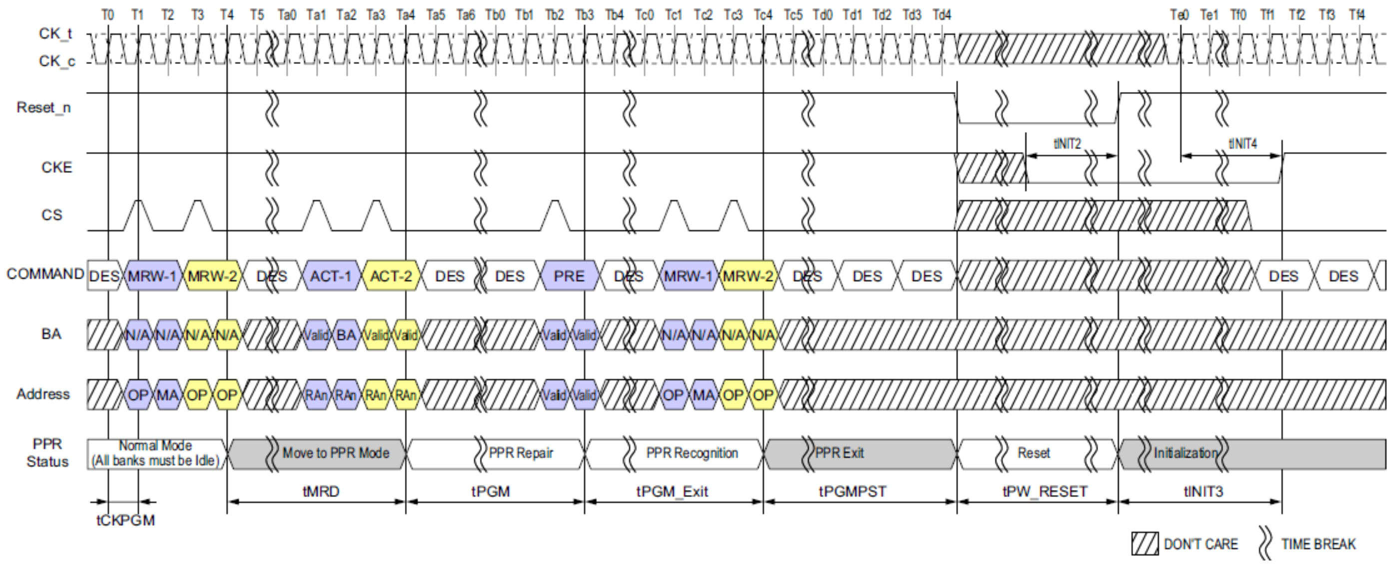

Figure 141: PPR Timing

NOTE :
1) During t_{PGM}, any other commands (including refresh) are not allowed on each die.
2) With one PPR command, only one row can be repaired at one time per die.
3) RESET command is required at the end of every PPR procedure.
4) During PPR, memory contents is not refreshed and may be lost.
5) Assert Reset_n below 0.2 X VDD2. Reset_n needs to be maintained LOW for minimum tPW_RESET. CKE must be pulled LOW at least 10ns before de-assserting Reset_n.
6) After RESET command, follow steps 4 to 10 in 'Voltage Ramp and Device Initialization' section.

[Table 73] PPR Timing Parameters

Parameter	Symbol	Min	Max	Unit	Note
PPR Programming Time	t_{PGM}	1000	-	ms	
PPR Exit Time	t_{PGM_Exit}	15	-	ns	
New Address Setting time	t_{PGMPST}	50	-	us	
PPR Programming Clock	t_{CKPGM}	1.25	-	ns	